

## 45. Management Data Input/Output (MDIO) Interface

### 45.1 Overview

This clause defines the logical and electrical characteristics of an extension to the two signal Management Data Input/Output (MDIO) Interface specified in Clause 22.

The purpose of this extension is to provide the ability to access more device registers while still retaining logical compatibility with the MDIO interface defined in Clause 22. Clause 22 specifies the MDIO frame format and uses an ST code of 01 to access registers. In this clause, additional registers are added to the address space by defining MDIO frames that use an ST code of 00.

This extension to the MDIO interface is applicable to the following:

- Implementations that operate at speeds of 2.5 Gb/s and above.
- Implementations of 10PASS-TS and 2BASE-TL subscriber network Physical Layer devices.
- Implementations of 10, 100, or 1000 Mb/s with additional management functions beyond those defined in Clause 22.

The MDIO electrical interface is optional. Where no physical embodiment of the MDIO exists, provision of an equivalent mechanism to access the registers is recommended.

Throughout this clause, an “a.b.c” format is used to identify register bits, where “a” is the device address, “b” is the register address, and “c” is the bit number within the register.

#### 45.1.1 Summary of major concepts

The following are major concepts of the MDIO Interface:

- a) Preserve the management frame structure defined in 22.2.4.5.
- b) Define a mechanism to address more registers than specified in 22.2.4.5.
- c) Define ST and OP codes to identify and control the extended access functions.
- d) Provide an electrical interface specification that is compatible with common digital CMOS ASIC processes.

#### 45.1.2 Application

This clause defines a management interface between Station Management (STA) and the sublayers that form a Physical Layer device (PHY) entity. Where a sublayer, or grouping of sublayers, is an individually manageable entity, it is known as an MDIO Manageable Device (MMD). This clause allows a single STA, through a single MDIO interface, to access up to 32 PHYs (defined as PRTAD in the frame format defined in 45.3) consisting of up to 32 MMDs as shown in Figure 45–1. The MDIO interface can support up to a maximum of 65 536 registers in each MMD.

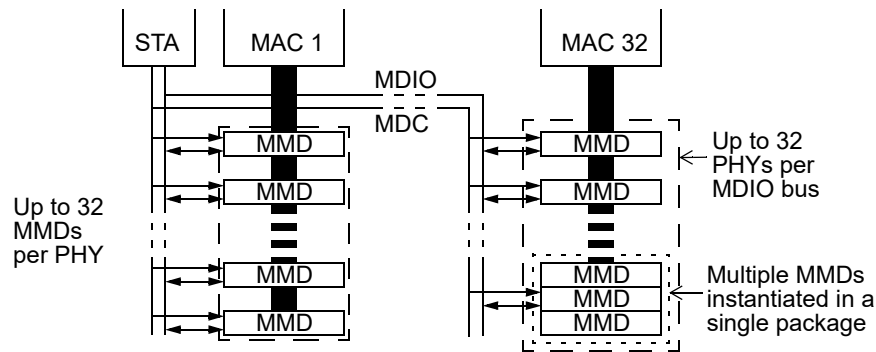


Figure 45–1—DTE and MMD devices

## 45.2 MDIO Interface registers

The management interface specified in Clause 22 provides a simple, two signal, serial interface to connect a Station Management entity and a managed PHY for providing access to management parameters and services. The interface is referred to as the MII management interface.

The MDIO interface is based on the MII management interface, but differs from it in several ways. The MDIO interface uses indirect addressing to create an extended address space allowing a much larger number of registers to be accessed within each MMD. The MDIO address space is orthogonal to the MII management interface address space. The mechanism for the addressing is defined in 45.3. The MDIO electrical interface operates at lower voltages than those specified for the MII management interface. The electrical interface is specified in 45.4. For cases where a single entity combines Clause 45 MMDs with Clause 22 registers, then the Clause 22 registers may be accessed using the Clause 45 electrical interface and the Clause 22 management frame structure. The list of possible MMDs is shown in Table 45–1. The PHY XS and DTE XS devices are the two partner devices used to extend the interface that sits immediately below the Reconciliation Sublayer. For 10 Gigabit Ethernet, the interface extenders are defined as the XGXS devices. For 10PASS-TS and 2BASE-TL, control and monitoring of the TC sublayer is defined in the TC MMD. For 10, 100 and 1000 Mb/s PHYs, further management capability is defined in the Clause 22 extension MMD.

Table 45–1—MDIO Manageable Device addresses

| Device address | MMD name |
|----------------|----------|
| 0              | Reserved |
| 1              | PMA/PMD  |
| 2              | WIS      |
| 3              | PCS      |
| 4              | PHY XS   |
| 5              | DTE XS   |
| 6              | TC       |

**Table 45–1—MDIO Manageable Device addresses (*continued*)**

| Device address | MMD name            |
|----------------|---------------------|
| 7              | Auto-Negotiation    |
| 8              | Separated PMA (1)   |
| 9              | Separated PMA (2)   |
| 10             | Separated PMA (3)   |
| 11             | Separated PMA (4)   |
| 12             | OFDM PMA/PMD        |
| 13             | Power Unit          |
| 14 through 28  | Reserved            |
| 29             | Clause 22 extension |
| 30             | Vendor specific 1   |
| 31             | Vendor specific 2   |

10PASS-TS and 2BASE-TL each have two port subtypes, 10PASS-TS-O, 10PASS-TS-R, 2BASE-TL-O and 2BASE-TL-R. Hereafter, referred to generically as -O and -R. The -O subtype corresponds to the port located at the service provider end of a subscriber link (the central office end). The -R subtype corresponds to the port located at the subscriber end of a subscriber link (the remote end). See 61.1 for more information.

Some register behavior may differ based on the port subtype. In the case where a register’s behavior or definition differs between port subtypes, it is noted in the register description and in the bit definition tables (denoted by “O:” and “R:” in the R/W column).

The Clause 22 extension MMD allows new features to be added to 10, 100, and 1000 Mb/s PHYs beyond those already defined in Clause 22.

If a device supports the MDIO interface it shall respond to all possible register addresses for the device and return a value of zero for undefined and unsupported registers. Writes to undefined registers and read-only registers shall have no effect. The operation of an MMD shall not be affected by writes to reserved and unsupported register bits, and such register bits shall return a value of zero when read.

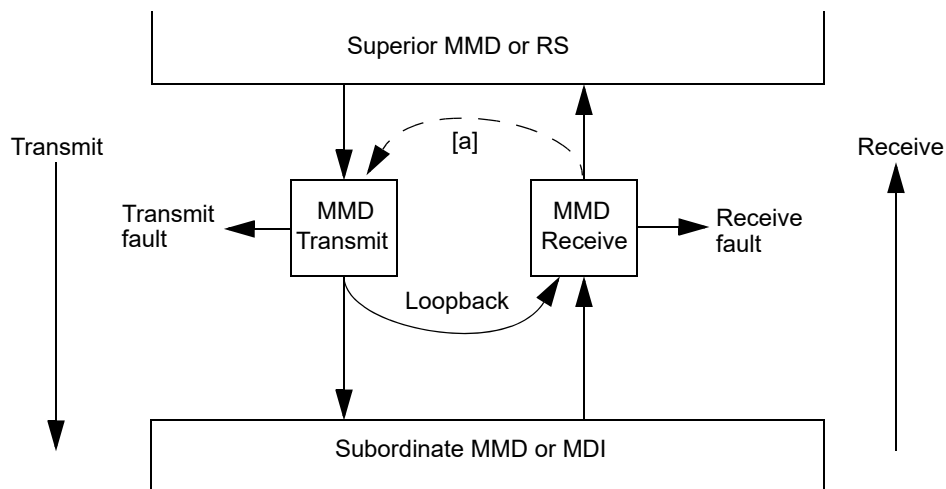
In the case of two registers that together form a 32-bit counter, whenever the most significant 16-bit register of the counter is read, the 32-bit counter value is latched into the register pair, the value being latched before the contents of the most significant 16 bits are driven on the MDIO interface and the contents of both registers are cleared to all zeros. A subsequent read from the least significant 16-bit register will return the least significant 16 bits of the latched value, but will not change the contents of the register pair. Writing to these registers has no effect. Counters that adhere to this behavior are marked in their bit definition tables with the tag “MW = Multi-word”.

To ensure compatibility with future use of reserved bits and registers, the Management Entity should write to reserved bits with a value of zero and ignore reserved bits on read.

Some of the bits within MMD registers are defined as latching low (LL) or latching high (LH). When a bit is defined as latching low and the condition for the bit to be low has occurred, the bit shall remain low until after it has been read via the management interface. Once such a read has occurred, the bit shall assume a value based on the current state of the condition it monitors. When a bit is defined as latching high and the condition for the bit to be high has occurred, the bit shall remain high until after it has been read via the management interface. Once such a read has occurred, the bit shall assume a value based on the current state of the condition it monitors.

For multi-bit fields, the lowest numbered bit of the field in the register corresponds to the least significant bit of the field.

Figure 45–2 describes the signal terminology used for the MMDs.



**Figure 45–2—MMD signal terminology**

[a] Direction of the optional PHY XS loopback

Each MMD contains registers 5 and 6, as defined in Table 45–2. Bits read as a one in this register indicate which MMDs are instantiated within the same package as the MMD being accessed. Bit 5.0 is used to indicate that Clause 22 functionality has been implemented within a Clause 45 electrical interface device. Bit 6.13 indicates that Clause 22 functionality is extended using the Clause 45 electrical interface through MMD 29. The definition of the term package is vendor specific and could be a chip, module, or other similar entity.

**Table 45–2—Devices in package registers bit definitions**

| Bit(s) <sup>a</sup> | Name                             | Description                                                                                            | R/W <sup>b</sup> |
|---------------------|----------------------------------|--------------------------------------------------------------------------------------------------------|------------------|
| m.6.15              | Vendor-specific device 2 present | 1 = Vendor-specific device 2 present in package<br>0 = Vendor-specific device 2 not present in package | RO               |
| m.6.14              | Vendor-specific device 1 present | 1 = Vendor-specific device 1 present in package<br>0 = Vendor-specific device 1 not present in package | RO               |
| m.6.13              | Clause 22 extension present      | 1 = Clause 22 extension present in package<br>0 = Clause 22 extension not present in package           | RO               |
| m.6.12:0            | Reserved                         | Value always 0                                                                                         | RO               |
| m.5.15:14           | Reserved                         | Value always 0                                                                                         | RO               |
| m.5.13              | Power Unit present               | 1 = Power Unit present in package<br>0 = Power Unit not present in package                             | RO               |
| m.5.12              | OFDM                             | 1 = OFDM present in package<br>0 = OFDM not present in package                                         | RO               |

**Table 45–2—Devices in package registers bit definitions (*continued*)**

| Bit(s) <sup>a</sup> | Name                        | Description                                                                                           | R/W <sup>b</sup> |
|---------------------|-----------------------------|-------------------------------------------------------------------------------------------------------|------------------|
| m.5.11              | Separated PMA (4) present   | 1 = Separated PMA <sup>c</sup> (4) present in package<br>0 = Separated PMA (4) not present in package | RO               |
| m.5.10              | Separated PMA (3) present   | 1 = Separated PMA (3) present in package<br>0 = Separated PMA (3) not present in package              | RO               |
| m.5.9               | Separated PMA (2) present   | 1 = Separated PMA (2) present in package<br>0 = Separated PMA (2) not present in package              | RO               |
| m.5.8               | Separated PMA (1) present   | 1 = Separated PMA (1) present in package<br>0 = Separated PMA (1) not present in package              | RO               |
| m.5.7               | Auto-Negotiation present    | 1 = Auto-Negotiation present in package<br>0 = Auto-Negotiation not present in package                | RO               |
| m.5.6               | TC present                  | 1 = TC present in package<br>0 = TC not present in package                                            | RO               |
| m.5.5               | DTE XS present              | 1 = DTE XS present in package<br>0 = DTE XS not present in package                                    | RO               |
| m.5.4               | PHY XS present              | 1 = PHY XS present in package<br>0 = PHY XS not present in package                                    | RO               |
| m.5.3               | PCS present                 | 1 = PCS present in package<br>0 = PCS not present in package                                          | RO               |
| m.5.2               | WIS present                 | 1 = WIS present in package<br>0 = WIS not present in package                                          | RO               |
| m.5.1               | PMD/PMA present             | 1 = PMA/PMD present in package<br>0 = PMA/PMD not present in package                                  | RO               |
| m.5.0               | Clause 22 registers present | 1 = Clause 22 registers present in package<br>0 = Clause 22 registers not present in package          | RO               |

<sup>a</sup>m = address of MMD accessed (see Table 45–1)

<sup>b</sup>RO = Read only

<sup>c</sup>Separated PMAs are defined in 45.2.1

#### 45.2.1 PMA/PMD registers

For devices operating at 25 Gb/s or higher speeds, the PMA may be instantiated as multiple sublayers (see 83.1.4, 109.1.4, and 120.1.4 for how MMD addresses are allocated to multiple PMA sublayers for the respective speeds). A PMA sublayer that is packaged with the PMD is addressed as MMD 1. More addressable instances of PMA sublayers, each one separated from lower addressable instances, may be implemented and addressed as MMD 8, 9, 10, and 11 where MMD 8 is the closest to the PMD and MMD 11 is the furthest from the PMD. The addresses and functions of all registers in MMD 8, 9, 10, and 11 are defined identically to MMD 1, except registers m.5 and m.6 as defined in Table 45–2.

The assignment of registers in the PMA/PMD is shown in Table 45–3.

**Table 45–3—PMA/PMD registers**

| Register address | Register name                                    | Subclause |
|------------------|--------------------------------------------------|-----------|
| 1.0              | PMA/PMD control 1                                | 45.2.1.1  |
| 1.1              | PMA/PMD status 1                                 | 45.2.1.2  |
| 1.2, 1.3         | PMA/PMD device identifier                        | 45.2.1.3  |
| 1.4              | PMA/PMD speed ability                            | 45.2.1.4  |
| 1.5, 1.6         | PMA/PMD devices in package                       | 45.2.1.5  |
| 1.7              | PMA/PMD control 2                                | 45.2.1.6  |
| 1.8              | PMA/PMD status 2                                 | 45.2.1.7  |
| 1.9              | PMA/PMD transmit disable                         | 45.2.1.8  |
| 1.10             | PMD receive signal detect                        | 45.2.1.9  |
| 1.11             | PMA/PMD extended ability                         | 45.2.1.10 |
| 1.12             | 10G-EPON PMA/PMD ability                         | 45.2.1.11 |
| 1.13             | 40G/100G PMA/PMD extended ability                | 45.2.1.12 |
| 1.14, 1.15       | PMA/PMD package identifier                       | 45.2.1.13 |
| 1.16             | EEE capability                                   | 45.2.1.14 |
| 1.17             | EPoC PMA/PMD ability                             | 45.2.1.15 |
| 1.18             | BASE-T1 PMA/PMD extended ability                 | 45.2.1.16 |
| 1.19             | 25G PMA/PMD extended ability                     | 45.2.1.17 |
| 1.20             | 50G PMA/PMD extended ability                     | 45.2.1.18 |
| 1.21             | 2.5G/5G PMA/PMD extended ability                 | 45.2.1.19 |
| 1.22             | BASE-H PMA/PMD extended ability                  | 45.2.1.20 |
| 1.23             | 200G PMA/PMD extended ability                    | 45.2.1.21 |
| 1.24             | 400G PMA/PMD extended ability                    | 45.2.1.22 |
| 1.25             | PMA/PMD extended ability 2                       | 45.2.1.23 |
| 1.26             | 40G/100G PMA/PMD extended ability 2              | 45.2.1.24 |
| 1.27             | PMD transmit disable extension                   | 45.2.1.25 |
| 1.28             | PMD receive signal detect extension              | 45.2.1.26 |
| 1.29             | PMA/PMD control 3                                | 45.2.1.27 |
| 1.30             | 10P/2B PMA/PMD control                           | 45.2.1.28 |
| 1.31             | 10P/2B PMA/PMD status                            | 45.2.1.29 |
| 1.32             | 10P/2B link partner PMA/PMD control <sup>a</sup> | 45.2.1.30 |
| 1.33             | 10P/2B link partner PMA/PMD status <sup>a</sup>  | 45.2.1.31 |
| 1.34             | BiDi PMA/PMD extended ability 1                  | 45.2.1.32 |
| 1.35             | BiDi PMA/PMD extended ability 2                  | 45.2.1.33 |
| 1.36             | 10P/2B link loss counter                         | 45.2.1.34 |
| 1.37             | 10P/2B RX SNR margin                             | 45.2.1.35 |

**Table 45–3—PMA/PMD registers (*continued*)**

| Register address             | Register name                                          | Subclause |
|------------------------------|--------------------------------------------------------|-----------|
| 1.38                         | 10P/2B link partner RX SNR margin <sup>a</sup>         | 45.2.1.36 |
| 1.39                         | 10P/2B line attenuation                                | 45.2.1.37 |
| 1.40                         | 10P/2B link partner line attenuation <sup>a</sup>      | 45.2.1.38 |
| 1.41                         | 10P/2B line quality thresholds                         | 45.2.1.39 |
| 1.42                         | 2B link partner line quality thresholds <sup>a</sup>   | 45.2.1.40 |
| 1.43                         | 10P FEC correctable errors counter                     | 45.2.1.41 |
| 1.44                         | 10P FEC uncorrectable errors counter                   | 45.2.1.42 |
| 1.45                         | 10P link partner FEC correctable errors <sup>a</sup>   | 45.2.1.43 |
| 1.46                         | 10P link partner FEC uncorrectable errors <sup>a</sup> | 45.2.1.44 |
| 1.47                         | 10P electrical length                                  | 45.2.1.45 |
| 1.48                         | 10P link partner electrical length <sup>a</sup>        | 45.2.1.46 |
| 1.49                         | 10P PMA/PMD general configuration <sup>a</sup>         | 45.2.1.47 |
| 1.50                         | 10P PSD configuration <sup>a</sup>                     | 45.2.1.48 |
| 1.51, 1.52                   | 10P downstream data rate configuration <sup>a</sup>    | 45.2.1.49 |
| 1.53                         | 10P downstream Reed-Solomon configuration <sup>a</sup> | 45.2.1.50 |
| 1.54, 1.55                   | 10P upstream data rate <sup>a</sup>                    | 45.2.1.51 |
| 1.56                         | 10P upstream Reed-Solomon configuration <sup>a</sup>   | 45.2.1.52 |
| 1.57, 1.58                   | 10P tone group                                         | 45.2.1.53 |
| 1.59, 1.60, 1.61, 1.62, 1.63 | 10P tone control parameters <sup>a</sup>               | 45.2.1.54 |
| 1.64                         | 10P tone control action <sup>a</sup>                   | 45.2.1.55 |
| 1.65, 1.66, 1.67             | 10P tone status                                        | 45.2.1.56 |
| 1.68                         | 10P outgoing indicator bits                            | 45.2.1.57 |
| 1.69                         | 10P incoming indicator bits                            | 45.2.1.58 |
| 1.70                         | 10P cyclic extension configuration                     | 45.2.1.59 |
| 1.71                         | 10P attainable downstream data rate                    | 45.2.1.60 |
| 1.72 through 1.79            | Reserved                                               |           |
| 1.80                         | 2B general parameter                                   | 45.2.1.61 |
| 1.81 through 1.88            | 2B PMD parameters                                      | 45.2.1.62 |
| 1.89                         | 2B code violation errors counter                       | 45.2.1.63 |
| 1.90                         | 2B link partner code violation errors <sup>a</sup>     | 45.2.1.64 |
| 1.91                         | 2B errored seconds counter                             | 45.2.1.65 |
| 1.92                         | 2B link partner errored seconds <sup>a</sup>           | 45.2.1.66 |
| 1.93                         | 2B severely errored seconds counter                    | 45.2.1.67 |
| 1.94                         | 2B link partner severely errored seconds <sup>a</sup>  | 45.2.1.68 |
| 1.95                         | 2B LOSW counter                                        | 45.2.1.69 |
| 1.96                         | 2B link partner LOSW <sup>a</sup>                      | 45.2.1.70 |
| 1.97                         | 2B unavailable seconds counter                         | 45.2.1.71 |

**Table 45–3—PMA/PMD registers (*continued*)**

| Register address    | Register name                                             | Subclause  |
|---------------------|-----------------------------------------------------------|------------|
| 1.98                | 2B link partner unavailable seconds <sup>a</sup>          | 45.2.1.72  |
| 1.99                | 2B state defects                                          | 45.2.1.73  |
| 1.100               | 2B link partner state defects <sup>a</sup>                | 45.2.1.74  |
| 1.101               | 2B negotiated constellation                               | 45.2.1.75  |
| 1.102 through 1.109 | 2B extended PMD parameters                                | 45.2.1.76  |
| 1.110 through 1.128 | Reserved                                                  |            |
| 1.129               | MultiGBASE-T status                                       | 45.2.1.77  |
| 1.130               | MultiGBASE-T pair swap and polarity                       | 45.2.1.78  |
| 1.131               | MultiGBASE-T TX power backoff and PHY short reach setting | 45.2.1.79  |
| 1.132               | MultiGBASE-T test mode                                    | 45.2.1.80  |
| 1.133               | SNR operating margin channel A                            | 45.2.1.81  |
| 1.134               | SNR operating margin channel B                            | 45.2.1.82  |
| 1.135               | SNR operating margin channel C                            | 45.2.1.83  |
| 1.136               | SNR operating margin channel D                            | 45.2.1.84  |
| 1.137               | Minimum margin channel A                                  | 45.2.1.85  |
| 1.138               | Minimum margin channel B                                  | 45.2.1.86  |
| 1.139               | Minimum margin channel C                                  | 45.2.1.87  |
| 1.140               | Minimum margin channel D                                  | 45.2.1.88  |
| 1.141               | RX signal power channel A                                 | 45.2.1.89  |
| 1.142               | RX signal power channel B                                 | 45.2.1.90  |
| 1.143               | RX signal power channel C                                 | 45.2.1.91  |
| 1.144               | RX signal power channel D                                 | 45.2.1.92  |
| 1.145 through 1.146 | MultiGBASE-T skew delay                                   | 45.2.1.93  |
| 1.147               | MultiGBASE-T fast retrain status and control              | 45.2.1.94  |
| 1.148 through 1.149 | Reserved                                                  |            |
| 1.150               | BASE-R PMD control                                        | 45.2.1.95  |
| 1.151               | BASE-R PMD status                                         | 45.2.1.96  |
| 1.152               | BASE-R LP coefficient update, lane 0                      | 45.2.1.97  |
| 1.153               | BASE-R LP status report, lane 0                           | 45.2.1.98  |
| 1.154               | BASE-R LD coefficient update, lane 0                      | 45.2.1.99  |
| 1.155               | BASE-R LD status report, lane 0                           | 45.2.1.100 |
| 1.156               | BASE-R PMD status 2                                       | 45.2.1.101 |
| 1.157               | BASE-R PMD status 3                                       | 45.2.1.102 |
| 1.158 through 1.159 | Reserved                                                  |            |
| 1.160               | 1000BASE-KX/2.5GBASE-KX control                           | 45.2.1.103 |
| 1.161               | 1000BASE-KX/2.5GBASE-KX status                            | 45.2.1.104 |
| 1.162 through 1.164 | PMA overhead control 1, 2, and 3                          | 45.2.1.105 |



**Table 45–3—PMA/PMD registers (*continued*)**

| Register address    | Register name                                                                           | Subclause                   |
|---------------------|-----------------------------------------------------------------------------------------|-----------------------------|
| 1.165, 1.166        | PMA overhead status 1 and 2                                                             | 45.2.1.106                  |
| 1.167 through 1.169 | Reserved                                                                                |                             |
| 1.170               | BASE-R FEC ability                                                                      | 45.2.1.107                  |
| 1.171               | BASE-R FEC control                                                                      | 45.2.1.108                  |
| 1.172 through 1.173 | Single-lane PHY BASE-R FEC corrected blocks counter                                     | 45.2.1.109                  |
| 1.174 through 1.175 | Single-lane PHY BASE-R FEC uncorrected blocks counter                                   | 45.2.1.110                  |
| 1.176 through 1.178 | Reserved                                                                                |                             |
| 1.179               | CAUI-4 C2M and 25GAUI C2M recommended CTLE                                              | 45.2.1.111                  |
| 1.180               | 25GAUI C2C and lane 0 CAUI-4 C2C transmitter equalization, receive direction            | 45.2.1.112                  |
| 1.181 through 1.183 | CAUI-4 chip-to-chip transmitter equalization, receive direction, lane 1 through lane 3  | 45.2.1.113                  |
| 1.184               | 25GAUI C2C and lane 0 CAUI-4 C2C transmitter equalization, transmit direction           | 45.2.1.114                  |
| 1.185 through 1.187 | CAUI-4 chip-to-chip transmitter equalization, transmit direction, lane 1 through lane 3 | 45.2.1.115                  |
| 1.188 through 1.199 | Reserved                                                                                |                             |
| 1.200               | RS-FEC control                                                                          | 45.2.1.116                  |
| 1.201               | RS-FEC status                                                                           | 45.2.1.117                  |
| 1.202, 1.203        | RS-FEC corrected codewords counter                                                      | 45.2.1.118                  |
| 1.204, 1.205        | RS-FEC uncorrected codewords counter                                                    | 45.2.1.119                  |
| 1.206               | RS-FEC lane mapping                                                                     | 45.2.1.120                  |
| 1.207 through 1.209 | Reserved                                                                                |                             |
| 1.210 through 1.217 | RS-FEC symbol error counter, lane 0 to 3                                                | 45.2.1.121,<br>45.2.1.122   |
| 1.218 through 1.229 | Reserved                                                                                |                             |
| 1.230 through 1.249 | RS-FEC BIP error counter, lane 0 to 19                                                  | 45.2.1.123,<br>45.2.1.124   |
| 1.250 through 1.269 | RS-FEC PCS lane mapping, lane 0 to 19                                                   | 45.2.1.125,<br>45.2.1.126   |
| 1.270 through 1.279 | Reserved                                                                                |                             |
| 1.280 through 1.283 | RS-FEC PCS alignment status 1 through 4                                                 | 45.2.1.127 to<br>45.2.1.130 |
| 1.284 through 1.299 | Reserved                                                                                |                             |
| 1.300 through 1.339 | BASE-R FEC corrected blocks counter, lanes 0 through 19                                 | 45.2.1.131                  |
| 1.340 through 1.399 | Reserved                                                                                |                             |
| 1.400 through 1.407 | 200GAUI-8 and 400GAUI-16 chip-to-module recommended CTLE, lane 0 through lane 7         | 45.2.1.132,<br>45.2.1.133   |
| 1.408 through 1.415 | 400GAUI-16 chip-to-module recommended CTLE, lane 8 through lane 15                      | 45.2.1.134                  |
| 1.416 through 1.499 | Reserved                                                                                |                             |

**Table 45–3—PMA/PMD registers (*continued*)**

| Register address      | Register name                                                                                                                   | Subclause                 |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| 1.500 through 1.515   | 50GAUI-n, 100GAUI-2, 200GAUI-n, and 400GAUI-n chip-to-chip transmitter equalization, receive direction, lane 0 through lane 15  | 45.2.1.135,<br>45.2.1.136 |
| 1.516 through 1.531   | 50GAUI-n, 100GAUI-2, 200GAUI-n, and 400GAUI-n chip-to-chip transmitter equalization, transmit direction, lane 0 through lane 15 | 45.2.1.137,<br>45.2.1.138 |
| 1.532 through 1.599   | Reserved                                                                                                                        |                           |
| 1.600                 | PMA precoder control Tx output                                                                                                  | 45.2.1.139                |
| 1.601                 | PMA precoder control Rx input                                                                                                   | 45.2.1.140                |
| 1.602                 | PMA precoder control Rx output                                                                                                  | 45.2.1.141                |
| 1.603                 | PMA precoder control Tx input                                                                                                   | 45.2.1.142                |
| 1.604                 | PMA precoder request flag                                                                                                       | 45.2.1.143                |
| 1.605                 | PMA precoder request Rx input status                                                                                            | 45.2.1.144                |
| 1.606                 | PMA precoder request Tx input status                                                                                            | 45.2.1.145                |
| 1.607 through 1.649   | Reserved                                                                                                                        |                           |
| 1.650, 1.651          | RS-FEC degraded SER activate threshold                                                                                          | 45.2.1.146                |
| 1.652, 1.653          | RS-FEC degraded SER deactivate threshold                                                                                        | 45.2.1.147                |
| 1.654, 1.655          | RS-FEC degraded SER interval                                                                                                    | 45.2.1.148                |
| 1.656 through 1.699   | Reserved                                                                                                                        |                           |
| 1.700 through 1.739   | BASE-R FEC uncorrected blocks counter, lanes 0 through 19                                                                       | 45.2.1.149                |
| 1.740 through 1.799   | Reserved                                                                                                                        |                           |
| 1.800                 | Tx optical channel control                                                                                                      | 45.2.1.150                |
| 1.801                 | Tx optical channel ability 1                                                                                                    | 45.2.1.151                |
| 1.802                 | Tx optical channel ability 2                                                                                                    | 45.2.1.152                |
| 1.803                 | Tx optical channel ability 3                                                                                                    | 45.2.1.153                |
| 1.804 through 1.819   | Reserved                                                                                                                        |                           |
| 1.820                 | Rx optical channel control                                                                                                      | 45.2.1.154                |
| 1.821                 | Rx optical channel ability 1                                                                                                    | 45.2.1.155                |
| 1.822                 | Rx optical channel ability 2                                                                                                    | 45.2.1.156                |
| 1.823                 | Rx optical channel ability 3                                                                                                    | 45.2.1.157                |
| 1.824 through 1.899   | Reserved                                                                                                                        |                           |
| 1.900                 | BASE-H PMA/PMD control                                                                                                          | 45.2.1.158                |
| 1.901 through 1.999   | Reserved                                                                                                                        |                           |
| 1.1000 through 1.1002 | Nx25G-EPON PMA/PMD extended ability                                                                                             | 45.2.1.159                |
| 1.1003 through 1.1099 | Reserved                                                                                                                        |                           |
| 1.1100                | BASE-R LP coefficient update, lane 0 (copy)                                                                                     | 45.2.1.97                 |
| 1.1101 through 1.1109 | BASE-R LP coefficient update, lanes 1 through 9                                                                                 | 45.2.1.160                |
| 1.1110 through 1.1119 | Reserved                                                                                                                        |                           |
| 1.1120 through 1.1123 | BASE-R PAM4 PMD training LP control, lane 0 through lane 3                                                                      | 45.2.1.161                |
| 1.1124 through 1.1199 | Reserved                                                                                                                        |                           |

**Table 45–3—PMA/PMD registers (*continued*)**

| Register address      | Register name                                              | Subclause  |
|-----------------------|------------------------------------------------------------|------------|
| 1.1200                | BASE-R LP status report, lane 0 (copy)                     | 45.2.1.98  |
| 1.1201 through 1.1209 | BASE-R LP status report, lanes 1 through 9                 | 45.2.1.162 |
| 1.1210 through 1.1219 | Reserved                                                   |            |
| 1.1220 through 1.1223 | BASE-R PAM4 PMD training LP status, lane 0 through lane 3  | 45.2.1.163 |
| 1.1224 through 1.1299 | Reserved                                                   |            |
| 1.1300                | BASE-R LD coefficient update, lane 0 (copy)                | 45.2.1.99  |
| 1.1301 through 1.1309 | BASE-R LD coefficient update, lanes 1 through 9            | 45.2.1.164 |
| 1.1310 through 1.1319 | Reserved                                                   |            |
| 1.1320 through 1.1323 | BASE-R PAM4 PMD training LD control, lane 0 through lane 3 | 45.2.1.165 |
| 1.1324 through 1.1399 | Reserved                                                   |            |
| 1.1400                | BASE-R LD status report, lane 0 (copy)                     | 45.2.1.100 |
| 1.1401 through 1.1409 | BASE-R LD status report, lanes 1 through 9                 | 45.2.1.166 |
| 1.1410 through 1.1419 | Reserved                                                   |            |
| 1.1420 through 1.1423 | BASE-R PAM4 PMD training LD status, lane 0 through lane 3  | 45.2.1.167 |
| 1.1424 through 1.1449 | Reserved                                                   |            |
| 1.1450 through 1.1453 | PMD training pattern, lanes 0 to 3                         | 45.2.1.168 |
| 1.1454 through 1.1499 | Reserved                                                   |            |
| 1.1500                | Test-pattern ability                                       | 45.2.1.169 |
| 1.1501                | PRBS pattern testing control                               | 45.2.1.170 |
| 1.1502 through 1.1509 | Reserved                                                   |            |
| 1.1510                | Square wave testing control                                | 45.2.1.171 |
| 1.1511                | Reserved                                                   |            |
| 1.1512                | PRBS13Q testing control                                    | 45.2.1.172 |
| 1.1513 through 1.1599 | Reserved                                                   |            |
| 1.1600 through 1.1615 | PRBS Tx error counters, lane 0 through lane 15             | 45.2.1.173 |
| 1.1616 through 1.1699 | Reserved                                                   |            |
| 1.1700 through 1.1715 | PRBS Rx error counters, lane 0 through lane 15             | 45.2.1.174 |
| 1.1716 through 1.1799 | Reserved                                                   |            |
| 1.1800                | TimeSync PMA/PMD capability                                | 45.2.1.175 |
| 1.1801 through 1.1804 | TimeSync PMA/PMD transmit path data delay                  | 45.2.1.176 |
| 1.1805 through 1.1808 | TimeSync PMA/PMD receive path data delay                   | 45.2.1.177 |
| 1.1809 through 1.1899 | Reserved                                                   |            |
| 1.1900                | 10GPASS-XR control and status                              | 45.2.1.178 |
| 1.1901                | DS OFDM control                                            | 45.2.1.179 |
| 1.1902 through 1.1906 | DS OFDM channel frequency control                          | 45.2.1.180 |
| 1.1907                | US OFDM control                                            | 45.2.1.181 |
| 1.1908                | US OFDM channel frequency control                          | 45.2.1.182 |

**Table 45–3—PMA/PMD registers (*continued*)**

| Register address      | Register name                           | Subclause  |
|-----------------------|-----------------------------------------|------------|
| 1.1909                | US OFDMA pilot pattern                  | 45.2.1.183 |
| 1.1910                | Profile control                         | 45.2.1.184 |
| 1.1911                | DS PHY Link control                     | 45.2.1.185 |
| 1.1912                | US PHY Link control                     | 45.2.1.186 |
| 1.1913 and 1.1914     | PHY Discovery control                   | 45.2.1.187 |
| 1.1915                | New CNU control                         | 45.2.1.188 |
| 1.1916 through 1.1920 | New CNU info                            | 45.2.1.189 |
| 1.1921                | DS PHY Link frame counter               | 45.2.1.190 |
| 1.1922 and 1.1923     | PMA/PMD timing offset                   | 45.2.1.191 |
| 1.1924                | PMA/PMD power offset                    | 45.2.1.192 |
| 1.1925 and 1.1926     | PMA/PMD ranging offset                  | 45.2.1.193 |
| 1.1927 through 1.1929 | DS PMA/PMD data rate                    | 45.2.1.194 |
| 1.1930 through 1.1932 | US PMA/PMD data rate                    | 45.2.1.195 |
| 1.1933 and 1.1934     | 10GPASS-XR FEC codeword counter         | 45.2.1.196 |
| 1.1935 and 1.1936     | 10GPASS-XR FEC codeword success counter | 45.2.1.197 |
| 1.1937 and 1.1938     | 10GPASS-XR FEC codeword fail counter    | 45.2.1.198 |
| 1.1939                | PHY Link EPFH counter                   | 45.2.1.199 |
| 1.1940                | PHY Link EPFH error counter             | 45.2.1.200 |
| 1.1941                | PHY Link EPCH counter                   | 45.2.1.201 |
| 1.1942                | PHY Link EPCH error counter             | 45.2.1.202 |
| 1.1943                | PHY Link EMB counter                    | 45.2.1.203 |
| 1.1944                | PHY Link EMB error counter              | 45.2.1.204 |
| 1.1945                | PHY Link FPMB counter                   | 45.2.1.205 |
| 1.1946                | PHY Link FPMB error counter             | 45.2.1.206 |
| 1.1947                | US PHY Link response time               | 45.2.1.207 |
| 1.1948                | 10GPASS-XR modulation ability           | 45.2.1.208 |
| 1.1949                | PHY Discovery Response power control    | 45.2.1.209 |
| 1.1950                | US target receive power                 | 45.2.1.210 |
| 1.1951 through 1.1955 | DS transmit power                       | 45.2.1.211 |
| 1.1956 and 1.1957     | US receive power measurement            | 45.2.1.212 |
| 1.1958                | Reported power                          | 45.2.1.213 |
| 1.1959 through 1.2099 | Reserved                                |            |
| 1.2100                | BASE-T1 PMA/PMD control                 | 45.2.1.214 |
| 1.2101                | Reserved                                |            |
| 1.2102                | 100BASE-T1 PMA/PMD test control         | 45.2.1.215 |
| 1.2103 through 1.2199 | Reserved                                |            |
| 1.2200                | IFEC control                            | 45.2.1.216 |

**Table 45–3—PMA/PMD registers (*continued*)**

| Register address      | Register name                          | Subclause                 |
|-----------------------|----------------------------------------|---------------------------|
| 1.2201                | IFEC status                            | 45.2.1.217                |
| 1.2202, 1.2203        | IFEC corrected codewords counter       | 45.2.1.218                |
| 1.2204, 1.2205        | IFEC uncorrected codewords counter     | 45.2.1.219                |
| 1.2206                | IFEC lane mapping                      | 45.2.1.220                |
| 1.2207 through 1.2209 | Reserved                               |                           |
| 1.2210 through 1.2217 | IFEC symbol error counter, lane 0 to 3 | 45.2.1.221,<br>45.2.1.222 |
| 1.2218 through 1.2245 | Reserved                               |                           |
| 1.2246                | SC-FEC alignment status 1              | 45.2.1.223                |
| 1.2247                | SC-FEC alignment status 2              | 45.2.1.224                |
| 1.2248, 1.2249        | Reserved                               |                           |
| 1.2250 through 1.2269 | SC-FEC lane mapping, lane 0 through 19 | 45.2.1.225,<br>45.2.1.226 |
| 1.2270 through 1.2275 | Reserved                               |                           |
| 1.2276, 1.2277        | SC-FEC corrected codewords counter     | 45.2.1.227                |
| 1.2278, 1.2279        | SC-FEC uncorrected codewords counter   | 45.2.1.228                |
| 1.2280 through 1.2283 | SC-FEC total bits                      | 45.2.1.229                |
| 1.2284 through 1.2287 | SC-FEC corrected bits                  | 45.2.1.230                |
| 1.2288 through 1.2293 | Reserved                               |                           |
| 1.2294                | 10BASE-T1L PMA control                 | 45.2.1.231                |
| 1.2295                | 10BASE-T1L PMA status                  | 45.2.1.232                |
| 1.2296                | 10BASE-T1L test mode control           | 45.2.1.233                |
| 1.2297                | 10BASE-T1S PMA control                 | 45.2.1.234                |
| 1.2298                | 10BASE-T1S PMA status                  | 45.2.1.235                |
| 1.2299                | 10BASE-T1S test mode control           | 45.2.1.236                |
| 1.2300 through 1.2303 | Reserved                               |                           |
| 1.2304                | 1000BASE-T1 PMA control                | 45.2.1.237                |
| 1.2305                | 1000BASE-T1 PMA status                 | 45.2.1.238                |
| 1.2306                | 1000BASE-T1 training                   | 45.2.1.239                |
| 1.2307                | 1000BASE-T1 link partner training      | 45.2.1.240                |
| 1.2308                | 1000BASE-T1 test mode control          | 45.2.1.241                |
| 1.2309                | MultiGBASE-T1 PMA control              | 45.2.1.242                |
| 1.2310                | MultiGBASE-T1 PMA status               | 45.2.1.243                |
| 1.2311                | MultiGBASE-T1 training                 | 45.2.1.244                |
| 1.2312                | MultiGBASE-T1 link partner training    | 45.2.1.245                |
| 1.2313                | MultiGBASE-T1 test mode control        | 45.2.1.246                |
| 1.2314                | MultiGBASE-T1 SNR operating margin     | 45.2.1.247                |
| 1.2315                | MultiGBASE-T1 minimum SNR margin       | 45.2.1.248                |

**Table 45–3—PMA/PMD registers (*continued*)**

| Register address          | Register name                                | Subclause  |
|---------------------------|----------------------------------------------|------------|
| 1.2316                    | MultiGBASE-T1 user defined data              | 45.2.1.249 |
| 1.2317                    | MultiGBASE-T1 link partner user defined data | 45.2.1.250 |
| 1.2318 through 1.32767    | Reserved                                     |            |
| 1.32 768 through 1.65 535 | Vendor specific                              |            |

<sup>a</sup>Register is defined only for -O port types and is reserved for -R ports.

#### 45.2.1.1 PMA/PMD control 1 register (Register 1.0)

The assignment of bits in the PMA/PMD control 1 register is shown in Table 45–4. The default value for each bit of the PMA/PMD control 1 register has been chosen so that the initial state of the device upon power up or completion of reset is a normal operational state without management intervention.

**Table 45–4—PMA/PMD control 1 register bit definitions**

| Bit(s)   | Name                  | Description                                                                                       | R/W <sup>a</sup> |
|----------|-----------------------|---------------------------------------------------------------------------------------------------|------------------|
| 1.0.15   | Reset                 | 1 = PMA/PMD reset<br>0 = Normal operation                                                         | R/W<br>SC        |
| 1.0.14   | Reserved              | Value always 0                                                                                    | RO               |
| 1.0.13   | Speed selection (LSB) | 1.0.6 1.0.13<br>1 1 = bits 5:2 select speed<br>1 0 = 1000 Mb/s<br>0 1 = 100 Mb/s<br>0 0 = 10 Mb/s | R/W              |
| 1.0.12   | Reserved              | Value always 0                                                                                    | RO               |
| 1.0.11   | Low power             | 1 = Low-power mode<br>0 = Normal operation                                                        | R/W              |
| 1.0.10:7 | Reserved              | Value always 0                                                                                    | RO               |
| 1.0.6    | Speed selection (MSB) | 1.0.6 1.0.13<br>1 1 = bits 5:2 select speed<br>1 0 = 1000 Mb/s<br>0 1 = 100 Mb/s<br>0 0 = 10 Mb/s | R/W              |

**Table 45–4—PMA/PMD control 1 register bit definitions (*continued*)**

| Bit(s)  | Name                | Description                                                                                                                                                                                                                                                                         | R/W <sup>a</sup> |
|---------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.0.5:2 | Speed selection     | 5 4 3 2<br>1 1 x x = Reserved<br>1 0 1 x = Reserved<br>1 0 0 1 = 400 Gb/s<br>1 0 0 0 = 200 Gb/s<br>0 1 1 1 = 5 Gb/s<br>0 1 1 0 = 2.5 Gb/s<br>0 1 0 1 = 50 Gb/s<br>0 1 0 0 = 25 Gb/s<br>0 0 1 1 = 100 Gb/s<br>0 0 1 0 = 40 Gb/s<br>0 0 0 1 = 10PASS-TS/2BASE-TL<br>0 0 0 0 = 10 Gb/s | R/W              |
| 1.0.1   | PMA remote loopback | 1 = Enable PMA remote loopback mode<br>0 = Disable PMA remote loopback mode                                                                                                                                                                                                         | R/W              |
| 1.0.0   | PMA local loopback  | 1 = Enable PMA local loopback mode<br>0 = Disable PMA local loopback mode                                                                                                                                                                                                           | R/W              |

<sup>a</sup>R/W = Read/Write, SC = Self-clearing, RO = Read only

#### 45.2.1.1.1 Reset (1.0.15)

Resetting a PMA/PMD is accomplished by setting bit 1.0.15 to a one. This action shall set all PMA/PMD registers to their default states. As a consequence, this action may change the internal state of the PMA/PMD and the state of the physical link. This action may also initiate a reset in any other MMDs that are instantiated in the same package. This bit is self-clearing, and a PMA/PMD shall return a value of one in bit 1.0.15 when a reset is in progress; otherwise, it shall return a value of zero. A PMA/PMD is not required to accept a write transaction to any of its registers until the reset process is completed. The control and management interface shall be restored to operation within 0.5 s from the setting of bit 1.0.15. During a reset, a PMD/PMA shall respond to reads from register bits 1.0.15 and 1.8.15:14. All other register bits should be ignored.

NOTE—This operation may interrupt data communication. The data path of a PMD, depending on type and temperature, may take many seconds to run at optimum error ratio after exiting from reset or low-power mode.

#### 45.2.1.1.2 Low power (1.0.11)

A PMA/PMD may be placed into a low-power mode by setting bit 1.0.11 to a one. This action may also initiate a low-power mode in any other MMDs that are instantiated in the same package. The low-power mode is exited by resetting the PMA/PMD. The behavior of the PMA/PMD in transition to and from the low-power mode is implementation specific and any interface signals should not be relied upon. While in the low-power mode, the device shall, as a minimum, respond to management transactions necessary to exit the low-power mode. The default value of bit 1.0.11 is zero.

NOTE—This operation will interrupt data communication. The data path of a PMD, depending on type and temperature, may take many seconds to run at optimum error ratio after exiting from reset or low-power mode.

#### **45.2.1.1.3 Speed selection (1.0.13, 1.0.6, 1.0.5:2)**

For devices operating at 10 Mb/s, 100 Mb/s, or 1000 Mb/s the speed of the PMA/PMD may be selected using bits 13 and 6. The speed abilities of the PMA/PMD are advertised in the PMA/PMD speed ability register. These two bits use the same definition as the speed selection bits defined in Clause 22.

For devices not operating at 10 Mb/s, 100 Mb/s, or 1000 Mb/s, the speed of the PMA/PMD may be selected using bits 5 through 2. When bits 5 through 2 are set to 0000 the use of a 10G PMA/PMD is selected. More specific selection is performed using the PMA/PMD control 2 register (register 1.7) (see 45.2.1.6). The speed abilities of the PMA/PMD are advertised in the PMA/PMD speed ability register. A PMA/PMD may ignore writes to the PMA/PMD speed selection bits that select speeds it has not advertised in the PMA/PMD speed ability register. It is the responsibility of the STA entity to ensure that mutually acceptable speeds are applied consistently across all the MMDs on a particular PHY.

The PMA/PMD speed selection defaults to a supported ability.

When set to 0001, bits 5:2 select the use of the 10PASS-TS or 2BASE-TL PMA/PMD. More specific mode selection is performed using the 10P/2B PMA control register (45.2.1.28).

When bits 5 through 2 are set to 0010 the use of a 40G PMA/PMD is selected; when set to 0011 the use of a 100G PMA/PMD is selected; when set to 0100 the use of a 25G PMA/PMD is selected; when set to 0101 the use of a 50G PMA/PMD is selected; when set to 0110 the use of a 2.5G PMA/PMD is selected; when set to 0111 the use of a 5G PMA/PMD is selected; when set to 1000 the use of a 200G PMA/PMD is selected; when set to 1001 the use of a 400G PMA/PMD is selected. More specific selection is performed using the PMA/PMD control 2 register (register 1.7) (see 45.2.1.6.3).

#### **45.2.1.1.4 PMA remote loopback (1.0.1)**

The PMA shall be placed in a remote loopback mode of operation when bit 1.0.1 is set to a one. When bit 1.0.1 is set to a one, the PMA shall accept data on the receive path and return it on the transmit path.

The remote loopback function is optional for all port types, except 2BASE-TL and 10PASS-TS, which do not support loopback. A device's ability to perform the remote loopback function is advertised in the remote loopback ability bit of the related speed-dependent status register. A PMA that is unable to perform the remote loopback function shall ignore writes to this bit and shall return a value of zero when read. For 25/40/100 Gb/s operation, the remote loopback ability bit is specified in register 1.13. For 200 Gb/s and 400 Gb/s operation, the remote loopback ability bit is specified in registers 1.23 and 1.24, respectively.

The default value of bit 1.0.1 is zero.

#### **45.2.1.1.5 PMA local loopback (1.0.0)**

The PMA shall be placed in a local loopback mode of operation when bit 1.0.0 is set to a one. When bit 1.0.0 is set to a one, the PMA shall accept data on the transmit path and return it on the receive path.

The local loopback function is mandatory for the 1000BASE-KX, 2.5GBASE-X, 5GBASE-R, 10GBASE-KR, 10GBASE-X, 40GBASE-KR4, 40GBASE-CR4, and 100GBASE-CR10 port type and optional for all other port types, except 2BASE-TL, 10PASS-TS, and 10/1GBASE-PRX, which do not support loopback. A PMA that is unable to perform the local loopback function shall ignore writes to this bit and shall return a value of zero when read. The local loopback functionality is detailed in the relevant PMA clause. For 10/25/40/100/200/400 Gb/s operation, the local loopback ability bit is specified in the PMA/PMD status 2 register.

The default value of bit 1.0.0 is zero.



NOTE—The signal path through the PMA that is exercised in the loopback mode of operation is implementation specific, but it is recommended that the signal path encompass as much of the PMA circuitry as is practical. The intention of providing this loopback mode of operation is to permit a diagnostic or self-test function to perform the transmission and reception of a PDU, thus testing the transmit and receive data paths. Other loopback signal paths may be enabled using loopback controls within other MMDs.

#### 45.2.1.2 PMA/PMD status 1 register (Register 1.1)

The assignment of bits in the PMA/PMD status 1 register is shown in Table 45–5. All the bits in the PMA/PMD status 1 register are read only; therefore, a write to the PMA/PMD status 1 register shall have no effect.

**Table 45–5—PMA/PMD status 1 register bit definitions**

| Bit(s)    | Name                | Description                                                                        | R/W <sup>a</sup> |
|-----------|---------------------|------------------------------------------------------------------------------------|------------------|
| 1.1.15:10 | Reserved            | Value always 0                                                                     | RO               |
| 1.1.9     | PIASA               | PMA ingress AUI stop ability                                                       | RO               |
| 1.1.8     | PEASA               | PMA egress AUI stop ability                                                        | RO               |
| 1.1.7     | Fault               | 1 = Fault condition detected<br>0 = Fault condition not detected                   | RO               |
| 1.1.6:3   | Reserved            | Value always 0                                                                     | RO               |
| 1.1.2     | Receive link status | 1 = PMA/PMD receive link up<br>0 = PMA/PMD receive link down                       | RO/LL            |
| 1.1.1     | Low-power ability   | 1 = PMA/PMD supports low-power mode<br>0 = PMA/PMD does not support low-power mode | RO               |
| 1.1.0     | Reserved            | Value always 0                                                                     | RO               |

<sup>a</sup>RO = Read only, LL = Latching low

##### 45.2.1.2.1 PMA ingress AUI stop ability (1.1.9)

If bit 1.1.9 is set to one, then the PMA is indicating that the PMA sublayer attached by the ingress AUI is permitted to stop signaling during LPI. If the bit is set to zero, then the PMA is indicating that the PMA sublayer attached by the ingress AUI is not permitted to stop signaling during LPI. If the PMA sublayer attached by the ingress AUI does not support EEE capability or is not capable to stop signaling, then this bit has no effect.

##### 45.2.1.2.2 PMA egress AUI stop ability (1.1.8)

If bit 1.1.8 is set to one, then the PMA is indicating that the PMA sublayer attached by the egress AUI is permitted to stop signaling during LPI. If the bit is set to zero, then the PMA is indicating that the PMA sublayer attached by the egress AUI is not permitted to stop signaling during LPI. If the PMA sublayer attached by the egress AUI does not support EEE capability or is not capable to stop signaling, then this bit has no effect.

#### **45.2.1.2.3 Fault (1.1.7)**

Fault is a global PMA/PMD variable. When read as a one, bit 1.1.7 indicates that either (or both) the PMA or the PMD has detected a fault condition on either the transmit or receive paths. When read as a zero, bit 1.1.7 indicates that neither the PMA nor the PMD has detected a fault condition. For 10/25/40/50/100/200/400 Gb/s operation, bit 1.1.7 is set to a one when either of the fault bits (1.8.11, 1.8.10) located in register 1.8 are set to a one. For 10PASS-TS or 2BASE-TL operations, when read as a one, a fault has been detected and more detailed information is conveyed in 45.2.1.34, 45.2.1.57, 45.2.1.58, and 45.2.1.73.

#### **45.2.1.2.4 Receive link status (1.1.2)**

When read as a one, bit 1.1.2 indicates that the PMA/PMD receive link is up. When read as a zero, bit 1.1.2 indicates that the PMA/PMD receive link is down. The receive link status bit shall be implemented with latching low behavior.

While a 10PASS-TS or 2BASE-TL PMA/PMD is initializing, this bit shall indicate receive link down (see 45.2.1.29).

#### **45.2.1.2.5 Low-power ability (1.1.1)**

When read as a one, bit 1.1.1 indicates that the PMA/PMD supports the low-power feature. When read as a zero, bit 1.1.1 indicates that the PMA/PMD does not support the low-power feature. If a PMA/PMD supports the low-power feature, then it is controlled using the low-power bit 1.0.11.

#### **45.2.1.3 PMA/PMD device identifier (Registers 1.2 and 1.3)**

Registers 1.2 and 1.3 provide a 32-bit value, which may constitute a unique identifier for a particular type of PMA/PMD. The identifier shall be composed of the 3rd through 24th bits of the Organizationally Unique Identifier (OUI) assigned to the device manufacturer by the IEEE, plus a six-bit model number, plus a four-bit revision number.

A PMA/PMD may return a value of zero in each of the 32 bits of the PMA/PMD device identifier to indicate that a unique identifier as described above is not provided.

The format of the PMA/PMD device identifier is specified in 22.2.4.3.1.

NOTE—The use of only 22 bits of the OUI as described here has been deprecated by the IEEE Registration Authority. In this case, Company ID (CID) is not an acceptable alternative to OUI due to the possibility that a CID and OUI could be identical in the 22-bit subset. The definition of vendor-specific device identifiers for other applications is expected to use the full 24 bits to accommodate the use of either an OUI or CID.

#### **45.2.1.4 PMA/PMD speed ability (Register 1.4)**

The assignment of bits in the PMA/PMD speed ability register is shown in Table 45–6.

##### **45.2.1.4.1 400G capable (1.4.15)**

When read as a one, bit 1.4.15 indicates that the PMA/PMD is able to operate at a data rate of 400 Gb/s. When read as a zero, bit 1.4.15 indicates that the PMA/PMD is not able to operate at a data rate of 400 Gb/s.

**Table 45–6—PMA/PMD speed ability register bit definitions**

| Bit(s) | Name               | Description                                                                                                                                                         | R/W <sup>a</sup> |
|--------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.4.15 | 400G capable       | 1 = PMA/PMD is capable of operating at 400 Gb/s<br>0 = PMA/PMD is not capable of operating at 400 Gb/s                                                              | RO               |
| 1.4.14 | 5G capable         | 1 = PMA/PMD is capable of operating at 5 Gb/s<br>0 = PMA/PMD is not capable of operating as 5 Gb/s                                                                  | RO               |
| 1.4.13 | 2.5G capable       | 1 = PMA/PMD is capable of operating at 2.5 Gb/s<br>0 = PMA/PMD is not capable of operating as 2.5 Gb/s                                                              | RO               |
| 1.4.12 | 200G capable       | 1 = PMA/PMD is capable of operating at 200 Gb/s<br>0 = PMA/PMD is not capable of operating at 200 Gb/s                                                              | RO               |
| 1.4.11 | 25G capable        | 1 = PMA/PMD is capable of operating at 25 Gb/s<br>0 = PMA/PMD is not capable of operating at 25 Gb/s                                                                | RO               |
| 1.4.10 | 10GPASS-XR capable | 1 = PMA/PMD is capable of operating as 10GPASS-XR<br>0 = PMA/PMD is not capable of operating as 10GPASS-XR                                                          | RO               |
| 1.4.9  | 100G capable       | 1 = PMA/PMD is capable of operating at 100 Gb/s<br>0 = PMA/PMD is not capable of operating as 100 Gb/s                                                              | RO               |
| 1.4.8  | 40G capable        | 1 = PMA/PMD is capable of operating at 40 Gb/s<br>0 = PMA/PMD is not capable of operating as 40 Gb/s                                                                | RO               |
| 1.4.7  | 10/1G capable      | 1 = PMA/PMD is capable of operating at 10 Gb/s downstream and 1 Gb/s upstream<br>0 = PMA/PMD is not capable of operating at 10 Gb/s downstream and 1 Gb/s upstream. | RO               |
| 1.4.6  | 10M capable        | 1 = PMA/PMD is capable of operating at 10 Mb/s<br>0 = PMA/PMD is not capable of operating as 10 Mb/s                                                                | RO               |
| 1.4.5  | 100M capable       | 1 = PMA/PMD is capable of operating at 100 Mb/s<br>0 = PMA/PMD is not capable of operating at 100 Mb/s                                                              | RO               |
| 1.4.4  | 1000M capable      | 1 = PMA/PMD is capable of operating at 1000 Mb/s<br>0 = PMA/PMD is not capable of operating at 1000 Mb/s                                                            | RO               |
| 1.4.3  | 50G capable        | 1 = PMA/PMD is capable of operating at 50 Gb/s<br>0 = PMA/PMD is not capable of operating at 50 Gb/s                                                                | RO               |
| 1.4.2  | 10PASS-TS capable  | 1 = PMA/PMD is capable of operating as 10PASS-TS<br>0 = PMA/PMD is not capable of operating as 10PASS-TS                                                            | RO               |
| 1.4.1  | 2BASE-TL capable   | 1 = PMA/PMD is capable of operating as 2BASE-TL<br>0 = PMA/PMD is not capable of operating as 2BASE-TL                                                              | RO               |
| 1.4.0  | 10G capable        | 1 = PMA/PMD is capable of operating at 10 Gb/s<br>0 = PMA/PMD is not capable of operating at 10 Gb/s                                                                | RO               |

<sup>a</sup>RO = Read only

#### 45.2.1.4.2 5G capable (1.4.14)

When read as a one, bit 1.4.14 indicates that the PMA/PMD is able to operate at a data rate of 5 Gb/s. When read as a zero, bit 1.4.14 indicates that the PMA/PMD is not able to operate at a data rate of 5 Gb/s.

#### **45.2.1.4.3 2.5G capable (1.4.13)**

When read as a one, bit 1.4.13 indicates that the PMA/PMD is able to operate at a data rate of 2.5 Gb/s. When read as a zero, bit 1.4.13 indicates that the PMA/PMD is not able to operate at a data rate of 2.5 Gb/s.

#### **45.2.1.4.4 200G capable (1.4.12)**

When read as a one, bit 1.4.12 indicates that the PMA/PMD is able to operate at a data rate of 200 Gb/s. When read as a zero, bit 1.4.12 indicates that the PMA/PMD is not able to operate at a data rate of 200 Gb/s.

#### **45.2.1.4.5 25G capable (1.4.11)**

When read as a one, bit 1.4.11 indicates that the PMA/PMD is able to operate at a data rate of 25 Gb/s. When read as a zero, bit 1.4.11 indicates that the PMA/PMD is not able to operate at a data rate of 25 Gb/s.

#### **45.2.1.4.6 10GPASS-XR capable (1.4.10)**

When read as one, bit 1.4.10 indicates that the PMA/PMD is able to operate as 10GPASS-XR. When read as zero, bit 1.4.10 indicates that the PMA/PMD is not able to operate as 10GPASS-XR.

#### **45.2.1.4.7 100G capable (1.4.9)**

When read as a one, bit 1.4.9 indicates that the PMA/PMD is able to operate at a data rate of 100 Gb/s. When read as a zero, bit 1.4.9 indicates that the PMA/PMD is not able to operate at a data rate of 100 Gb/s.

#### **45.2.1.4.8 40G capable (1.4.8)**

When read as a one, bit 1.4.8 indicates that the PMA/PMD is able to operate at a data rate of 40 Gb/s. When read as a zero, bit 1.4.8 indicates that the PMA/PMD is not able to operate at a data rate of 40 Gb/s.

#### **45.2.1.4.9 10/1G capable (1.4.7)**

When read as a one, bit 1.4.7 indicates that the PMA/PMD is able to operate at a data rate of 10 Gb/s in the downstream direction and 1 Gb/s in the upstream direction. When read as a zero, bit 1.4.7 indicates that the PMA/PMD is not able to operate at a data rate of 10 Gb/s in the downstream direction and 1 Gb/s in the upstream direction.

#### **45.2.1.4.10 10M capable (1.4.6)**

When read as a one, bit 1.4.6 indicates that the PMA/PMD is able to operate at a data rate of 10 Mb/s. When read as a zero, bit 1.4.6 indicates that the PMA/PMD is not able to operate at a data rate of 10 Mb/s.

#### **45.2.1.4.11 100M capable (1.4.5)**

When read as a one, bit 1.4.5 indicates that the PMA/PMD is able to operate at a data rate of 100 Mb/s. When read as a zero, bit 1.4.5 indicates that the PMA/PMD is not able to operate at a data rate of 100 Mb/s.

#### **45.2.1.4.12 1000M capable (1.4.4)**

When read as a one, bit 1.4.4 indicates that the PMA/PMD is able to operate at a data rate of 1000 Mb/s. When read as a zero, bit 1.4.4 indicates that the PMA/PMD is not able to operate at a data rate of 1000 Mb/s.

#### **45.2.1.4.13 50G capable (1.4.3)**

When read as a one, bit 1.4.3 indicates that the PMA/PMD is able to operate at a data rate of 50 Gb/s. When read as a zero, bit 1.4.3 indicates that the PMA/PMD is not able to operate at a data rate of 50 Gb/s.

#### **45.2.1.4.14 10PASS-TS capable (1.4.2)**

When read as a one, bit 1.4.2 indicates that the PMA/PMD is able to operate as 10PASS-TS. When read as a zero, bit 1.4.2 indicates that the PMA/PMD is not able to operate as 10PASS-TS.

#### **45.2.1.4.15 2BASE-TL capable (1.4.1)**

When read as a one, bit 1.4.1 indicates that the PMA/PMD is able to operate as 2BASE-TL. When read as a zero, bit 1.4.1 indicates that the PMA/PMD is not able to operate as 2BASE-TL.

#### **45.2.1.4.16 10G capable (1.4.0)**

When read as a one, bit 1.4.0 indicates that the PMA/PMD is able to operate at a data rate of 10 Gb/s. When read as a zero, bit 1.4.0 indicates that the PMA/PMD is not able to operate at a data rate of 10 Gb/s.

### **45.2.1.5 PMA/PMD devices in package (Registers 1.5 and 1.6)**

The PMA/PMD devices in package registers are defined in Table 45–2.

#### **45.2.1.6 PMA/PMD control 2 register (Register 1.7)**

The assignment of bits in the PMA/PMD control 2 register is shown in Table 45–7.

##### **45.2.1.6.1 PMA ingress AUI stop enable (1.7.9)**

If bit 1.7.9 is set to one then the PMA may stop the ingress direction AUI signaling during LPI otherwise it shall keep active signaling on that AUI. If the PMA does not support EEE capability or is not able to stop the ingress direction AUI signaling (see 45.2.1.2.1) then this bit has no effect.

##### **45.2.1.6.2 PMA egress AUI stop enable (1.7.8)**

If bit 1.7.8 is set to one then the PMA may stop the egress direction AUI signaling during LPI otherwise it shall keep active signaling on that AUI. If the PMA does not support EEE capability or is not able to stop the egress direction AUI signaling (see 45.2.1.2.2) then this bit has no effect.

##### **45.2.1.6.3 PMA/PMD type selection (1.7.6:0)**

The PMA/PMD type of the PMA/PMD shall be selected using bits 6 to 0. The PMA/PMD type abilities of the PMA/PMD are advertised in bits 9 and 7 through 0 of the PMA/PMD status 2 register; the PMA/PMD extended ability register; the 40G/100G PMA/PMD extended ability register; the 50G PMA/PMD extended ability register; the 200G PMA/PMD extended ability register; and the 400G PMA/PMD extended ability register. A PMA/PMD shall ignore writes to the PMA/PMD type selection bits that select PMA/PMD types it has not advertised. It is the responsibility of the STA entity to ensure that mutually acceptable MMD types are applied consistently across all the MMDs on a particular PHY.

The PMA/PMD type selection defaults to a supported ability.

**Table 45–7—PMA/PMD control 2 register bit definitions**

| Bit(s)    | Name                   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | R/W <sup>a</sup> |
|-----------|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.7.15:10 | Reserved               | Value always 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | RO               |
| 1.7.9     | PIASE                  | PMA ingress AUI stop enable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | R/W              |
| 1.7.8     | PEASE                  | PMA egress AUI stop enable                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | R/W              |
| 1.7.7     | Reserved               | Value always 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | RO               |
| 1.7.6:0   | PMA/PMD type selection | <p>6 5 4 3 2 1 0</p> <p>1 1 1 1 1 x x = reserved</p> <p>1 1 1 1 0 1 x = reserved</p> <p>1 1 1 1 0 0 1 = reserved</p> <p>1 1 1 1 0 0 0 = 50GBASE-BR40-U PMA/PMD</p> <p>1 1 1 0 1 1 1 = 50GBASE-BR20-U PMA/PMD</p> <p>1 1 1 0 1 1 0 = 50GBASE-BR10-U PMA/PMD</p> <p>1 1 1 0 1 0 1 = 50GBASE-BR40-D PMA/PMD</p> <p>1 1 1 0 1 0 0 = 50GBASE-BR20-D PMA/PMD</p> <p>1 1 1 0 0 1 1 = 50GBASE-BR10-D PMA/PMD</p> <p>1 1 1 0 0 1 0 = 25GBASE-BR40-U PMA/PMD</p> <p>1 1 1 0 0 0 1 = 25GBASE-BR20-U PMA/PMD</p> <p>1 1 1 0 0 0 0 = 25GBASE-BR10-U PMA/PMD</p> <p>1 1 0 1 1 1 1 = 25GBASE-BR40-D PMA/PMD</p> <p>1 1 0 1 1 1 0 = 25GBASE-BR20-D PMA/PMD</p> <p>1 1 0 1 1 0 1 = 25GBASE-BR10-D PMA/PMD</p> <p>1 1 0 1 1 0 0 = 10GBASE-BR40-U PMA/PMD</p> <p>1 1 0 1 0 1 1 = 10GBASE-BR20-U PMA/PMD</p> <p>1 1 0 1 0 1 0 = 10GBASE-BR10-U PMA/PMD</p> <p>1 1 0 1 0 0 1 = 10GBASE-BR40-D PMA/PMD</p> <p>1 1 0 1 0 0 0 = 10GBASE-BR20-D PMA/PMD</p> <p>1 1 0 0 1 1 1 = 10GBASE-BR10-D PMA/PMD</p> <p>1 1 0 0 1 1 0 = reserved</p> <p>1 1 0 0 1 0 x = reserved</p> <p>1 1 0 0 0 1 1 = 400GBASE-ER8 PMA/PMD</p> <p>1 1 0 0 0 1 0 = 400GBASE-LR4-6 PMA/PMD</p> <p>1 1 0 0 0 0 1 = 400GBASE-FR4 PMA/PMD</p> <p>1 1 0 0 0 0 0 = 400GBASE-SR4.2 PMA/PMD</p> <p>1 0 1 1 1 1 1 = 400GBASE-SR8 PMA/PMD</p> <p>1 0 1 1 1 1 0 = reserved</p> <p>1 0 1 1 1 0 1 = reserved</p> <p>1 0 1 1 1 0 0 = 400GBASE-LR8 PMA/PMD</p> <p>1 0 1 1 0 1 1 = 400GBASE-FR8 PMA/PMD</p> <p>1 0 1 1 0 1 0 = 400GBASE-DR4 PMA/PMD</p> <p>1 0 1 1 0 0 1 = 400GBASE-SR16 PMA/PMD</p> <p>1 0 1 1 0 0 0 = 200GBASE-ER4 PMA/PMD</p> <p>1 0 1 0 1 1 x = reserved</p> <p>1 0 1 0 1 0 1 = 200GBASE-LR4 PMA/PMD</p> <p>1 0 1 0 1 0 0 = 200GBASE-FR4 PMA/PMD</p> <p>1 0 1 0 0 1 1 = 200GBASE-DR4 PMA/PMD</p> <p>1 0 1 0 0 1 0 = 200GBASE-SR4 PMA/PMD</p> <p>1 0 1 0 0 0 1 = 200GBASE-CR4 PMA/PMD</p> <p>1 0 1 0 0 0 0 = 200GBASE-KR4 PMA/PMD</p> <p>1 0 0 1 1 1 1 = reserved</p> <p>1 0 0 1 1 1 0 = 100GBASE-ZR PMA/PMD</p> <p>1 0 0 1 1 0 1 = 100GBASE-LR1 PMA/PMD</p> <p>1 0 0 1 1 0 0 = 100GBASE-FR1 PMA/PMD</p> <p>1 0 0 1 0 1 1 = 100GBASE-DR PMA/PMD</p> <p>1 0 0 1 0 1 0 = 100GBASE-SR2 PMA/PMD</p> <p>1 0 0 1 0 0 1 = 100GBASE-CR2 PMA/PMD</p> | R/W              |

**Table 45–7—PMA/PMD control 2 register bit definitions (*continued*)**

| Bit(s)                          | Name                                           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | R/W <sup>a</sup> |
|---------------------------------|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.7.6:0<br>( <i>continued</i> ) | PMA/PMD type selection<br>( <i>continued</i> ) | <p>1 0 0 1 0 0 0 = 100GBASE-KR2 PMA/PMD</p> <p>1 0 0 0 1 1 x = reserved</p> <p>1 0 0 0 1 0 1 = 50GBASE-ER PMA/PMD</p> <p>1 0 0 0 1 0 0 = 50GBASE-LR PMA/PMD</p> <p>1 0 0 0 0 1 1 = 50GBASE-FR PMA/PMD</p> <p>1 0 0 0 0 1 0 = 50GBASE-SR PMA/PMD</p> <p>1 0 0 0 0 0 1 = 50GBASE-CR PMA/PMD</p> <p>1 0 0 0 0 0 0 = 50GBASE-KR PMA/PMD</p> <p>0 1 1 1 1 1 x = reserved</p> <p>0 1 1 1 1 0 1 = BASE-T1 PMA/PMD<sup>b</sup></p> <p>0 1 1 1 1 0 0 = 5GBASE-KR PMA/PMD</p> <p>0 1 1 1 0 1 1 = 2.5GBASE-KX PMA/PMD</p> <p>0 1 1 1 0 1 0 = 25GBASE-SR PMA/PMD</p> <p>0 1 1 1 0 0 1 = 25GBASE-KR or 25GBASE-KR-S PMA/PMD</p> <p>0 1 1 1 0 0 0 = 25GBASE-CR or 25GBASE-CR-S PMA/PMD</p> <p>0 1 1 0 1 1 1 = 25GBASE-T PMA</p> <p>0 1 1 0 1 1 0 = 25GBASE-ER PMA/PMD</p> <p>0 1 1 0 1 0 1 = 25GBASE-LR PMA/PMD</p> <p>0 1 1 0 1 0 0 = BASE-H PMA/PMD<sup>c</sup></p> <p>0 1 1 0 0 1 1 = 10GPASS-XR-U PMA/PMD</p> <p>0 1 1 0 0 1 0 = 10GPASS-XR-D PMA/PMD</p> <p>0 1 1 0 0 0 1 = 5GBASE-T PMA</p> <p>0 1 1 0 0 0 0 = 2.5GBASE-T PMA</p> <p>0 1 0 1 1 1 1 = 100GBASE-SR4 PMA/PMD</p> <p>0 1 0 1 1 1 0 = 100GBASE-CR4 PMA/PMD</p> <p>0 1 0 1 1 0 1 = 100GBASE-KR4 PMA/PMD</p> <p>0 1 0 1 1 0 0 = 100GBASE-KP4 PMA/PMD</p> <p>0 1 0 1 0 1 1 = 100GBASE-ER4 PMA/PMD</p> <p>0 1 0 1 0 1 0 = 100GBASE-LR4 PMA/PMD</p> <p>0 1 0 1 0 0 1 = 100GBASE-SR10 PMA/PMD</p> <p>0 1 0 1 0 0 0 = 100GBASE-CR10 PMA/PMD</p> <p>0 1 0 0 1 1 1 = reserved</p> <p>0 1 0 0 1 1 0 = 40GBASE-T PMA</p> <p>0 1 0 0 1 0 1 = 40GBASE-ER4 PMA/PMD</p> <p>0 1 0 0 1 0 0 = 40GBASE-FR PMA/PMD</p> <p>0 1 0 0 0 1 1 = 40GBASE-LR4 PMA/PMD</p> <p>0 1 0 0 0 1 0 = 40GBASE-SR4 PMA/PMD</p> <p>0 1 0 0 0 0 1 = 40GBASE-CR4 PMA/PMD</p> <p>0 1 0 0 0 0 0 = 40GBASE-KR4 PMA/PMD</p> <p>0 0 1 1 1 1 1 = 10/1GBASE-PRX-U4</p> <p>0 0 1 1 1 1 0 = 10GBASE-PR-U4</p> <p>0 0 1 1 1 0 1 = 10/1GBASE-PRX-D4</p> <p>0 0 1 1 1 0 0 = 10GBASE-PR-D4</p> <p>0 0 1 1 0 1 1 = reserved</p> <p>0 0 1 1 0 1 0 = 10GBASE-PR-U3</p> <p>0 0 1 1 0 0 1 = 10GBASE-PR-U1</p> <p>0 0 1 1 0 0 0 = 10/1GBASE-PRX-U3</p> <p>0 0 1 0 1 1 1 = 10/1GBASE-PRX-U2</p> <p>0 0 1 0 1 1 0 = 10/1GBASE-PRX-U1</p> <p>0 0 1 0 1 0 1 = 10GBASE-PR-D3</p> <p>0 0 1 0 1 0 0 = 10GBASE-PR-D2</p> <p>0 0 1 0 0 1 1 = 10GBASE-PR-D1</p> <p>0 0 1 0 0 1 0 = 10/1GBASE-PRX-D3</p> <p>0 0 1 0 0 0 1 = 10/1GBASE-PRX-D2</p> <p>0 0 1 0 0 0 0 = 10/1GBASE-PRX-D1</p> <p>0 0 0 1 1 1 1 = 10BASE-T PMA/PMD</p> <p>0 0 0 1 1 1 0 = 100BASE-TX PMA/PMD</p> <p>0 0 0 1 1 0 1 = 1000BASE-KX PMA/PMD</p> <p>0 0 0 1 1 0 0 = 1000BASE-T PMA/PMD</p> | R/W              |

**Table 45–7—PMA/PMD control 2 register bit definitions (*continued*)**

| Bit(s)                          | Name                                           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                         | R/W <sup>a</sup> |
|---------------------------------|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.7.6:0<br>( <i>continued</i> ) | PMA/PMD type selection<br>( <i>continued</i> ) | 0 0 0 1 0 1 1 = 10GBASE-KR PMA/PMD<br>0 0 0 1 0 1 0 = 10GBASE-KX4 PMA/PMD<br>0 0 0 1 0 0 1 = 10GBASE-T PMA<br>0 0 0 1 0 0 0 = 10GBASE-LRM PMA/PMD<br>0 0 0 0 1 1 1 = 10GBASE-SR PMA/PMD<br>0 0 0 0 1 1 0 = 10GBASE-LR PMA/PMD<br>0 0 0 0 1 0 1 = 10GBASE-ER PMA/PMD<br>0 0 0 0 1 0 0 = 10GBASE-LX4 PMA/PMD<br>0 0 0 0 0 1 1 = 10GBASE-SW PMA/PMD<br>0 0 0 0 0 1 0 = 10GBASE-LW PMA/PMD<br>0 0 0 0 0 0 1 = 10GBASE-EW PMA/PMD<br>0 0 0 0 0 0 0 = 10GBASE-CX4 PMA/PMD |                  |

<sup>a</sup>R/W = Read/Write, RO = Read only

<sup>b</sup>If BASE-T1 is selected, bits 1.2100.3:0 are used to differentiate which BASE-T1 PMA/PMD is selected.

<sup>c</sup>If BASE-H PMA/PMD is selected, register 1.900 is used to differentiate which BASE-H PMA/PMD is selected.

#### 45.2.1.7 PMA/PMD status 2 register (Register 1.8)

The assignment of bits in the PMA/PMD status 2 register is shown in Table 45–8. All the bits in the PMA/PMD status 2 register are read only; a write to the PMA/PMD status 2 register shall have no effect.

**Table 45–8—PMA/PMD status 2 register bit definitions**

| Bit(s)    | Name                         | Description                                                                                                                                                                                | R/W <sup>a</sup> |
|-----------|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.8.15:14 | Device present               | 15 14<br>1 0 = Device responding at this address<br>1 1 = No device responding at this address<br>0 1 = No device responding at this address<br>0 0 = No device responding at this address | RO               |
| 1.8.13    | Transmit fault ability       | 1 = PMA/PMD has the ability to detect a fault condition on the transmit path<br>0 = PMA/PMD does not have the ability to detect a fault condition on the transmit path                     | RO               |
| 1.8.12    | Receive fault ability        | 1 = PMA/PMD has the ability to detect a fault condition on the receive path<br>0 = PMA/PMD does not have the ability to detect a fault condition on the receive path                       | RO               |
| 1.8.11    | Transmit fault               | 1 = Fault condition on transmit path<br>0 = No fault condition on transmit path                                                                                                            | RO/LH            |
| 1.8.10    | Receive fault                | 1 = Fault condition on receive path<br>0 = No fault condition on receive path                                                                                                              | RO/LH            |
| 1.8.9     | Extended abilities           | 1 = PMA/PMD has extended abilities listed in register 1.11<br>0 = PMA/PMD does not have extended abilities                                                                                 | RO               |
| 1.8.8     | PMD transmit disable ability | 1 = PMD has the ability to disable the transmit path<br>0 = PMD does not have the ability to disable the transmit path                                                                     | RO               |
| 1.8.7     | 10GBASE-SR ability           | 1 = PMA/PMD is able to perform 10GBASE-SR<br>0 = PMA/PMD is not able to perform 10GBASE-SR                                                                                                 | RO               |



**Table 45–8—PMA/PMD status 2 register bit definitions (*continued*)**

| Bit(s) | Name                       | Description                                                                                                                            | R/W <sup>a</sup> |
|--------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.8.6  | 10GBASE-LR ability         | 1 = PMA/PMD is able to perform 10GBASE-LR<br>0 = PMA/PMD is not able to perform 10GBASE-LR                                             | RO               |
| 1.8.5  | 10GBASE-ER ability         | 1 = PMA/PMD is able to perform 10GBASE-ER<br>0 = PMA/PMD is not able to perform 10GBASE-ER                                             | RO               |
| 1.8.4  | 10GBASE-LX4 ability        | 1 = PMA/PMD is able to perform 10GBASE-LX4<br>0 = PMA/PMD is not able to perform 10GBASE-LX4                                           | RO               |
| 1.8.3  | 10GBASE-SW ability         | 1 = PMA/PMD is able to perform 10GBASE-SW<br>0 = PMA/PMD is not able to perform 10GBASE-SW                                             | RO               |
| 1.8.2  | 10GBASE-LW ability         | 1 = PMA/PMD is able to perform 10GBASE-LW<br>0 = PMA/PMD is not able to perform 10GBASE-LW                                             | RO               |
| 1.8.1  | 10GBASE-EW ability         | 1 = PMA/PMD is able to perform 10GBASE-EW<br>0 = PMA/PMD is not able to perform 10GBASE-EW                                             | RO               |
| 1.8.0  | PMA local loopback ability | 1 = PMA has the ability to perform a local loopback function<br>0 = PMA does not have the ability to perform a local loopback function | RO               |

<sup>a</sup>RO = Read only, LH = Latching high

#### 45.2.1.7.1 Device present (1.8.15:14)

When read as <10>, bits 1.8.15:14 indicate that a device is present and responding at this register address. When read as anything other than <10>, bits 1.8.15:14 indicate that no device is present at this register address or that the device is not functioning properly.

#### 45.2.1.7.2 Transmit fault ability (1.8.13)

When read as a one, bit 1.8.13 indicates that the PMA/PMD has the ability to detect a fault condition on the transmit path. When read as a zero, bit 1.8.13 indicates that the PMA/PMD does not have the ability to detect a fault condition on the transmit path.

#### 45.2.1.7.3 Receive fault ability (1.8.12)

When read as a one, bit 1.8.12 indicates that the PMA/PMD has the ability to detect a fault condition on the receive path. When read as a zero, bit 1.8.12 indicates that the PMA/PMD does not have the ability to detect a fault condition on the receive path.

#### 45.2.1.7.4 Transmit fault (1.8.11)

When read as a one, bit 1.8.11 indicates that the PMA/PMD has detected a fault condition on the transmit path. When read as a zero, bit 1.8.11 indicates that the PMA/PMD has not detected a fault condition on the transmit path. Detection of a fault condition on the transmit path is optional and the ability to detect such a condition is advertised by bit 1.8.13. A PMA/PMD that is unable to detect a fault condition on the transmit path shall return a value of zero for this bit. The description of the transmit fault function for the various PMA/PMDs is given in Table 45–9.

The transmit fault bit shall be implemented with latching high behavior.

The default value of bit 1.8.11 is zero.

**Table 45–9—Transmit fault description location**

| PMA/PMD                                              | Description location |
|------------------------------------------------------|----------------------|
| 10BASE-T1L                                           | 146.4.2              |
| 100BASE-T1                                           | 96.4.2               |
| 2.5GBASE-KX                                          | 128.6.8              |
| 2.5GBASE-T, 5GBASE-T                                 | 126.4.2.2            |
| 2.5GBASE-T1, 5GBASE-T1, 10GBASE-T1                   | 149.4.2.2            |
| 5GBASE-KR                                            | 130.6.8              |
| 10GBASE-BR10, 10GBASE-BR20, 10GBASE-BR40             | 158.5.8              |
| 10GBASE-KR                                           | 72.6.8               |
| 10GBASE-LRM                                          | 68.4.8               |
| 10GBASE-S, 10GBASE-L, 10GBASE-E                      | 52.4.8               |
| 10GBASE-LX4                                          | 53.4.10              |
| 10GBASE-CX4                                          | 54.5.10              |
| 10GBASE-T                                            | 55.4.2.2             |
| 10GBASE-KX4                                          | 71.6.10              |
| 25GBASE-KR, 25GBASE-KR-S                             | 111.7.8              |
| 25GBASE-CR, 25GBASE-CR-S                             | 110.7.8              |
| 25GBASE-SR                                           | 112.5.8              |
| 25GBASE-LR, 25GBASE-ER                               | 114.5.8              |
| 25GBASE-BR10, 25GBASE-BR20, 25GBASE-BR40             | 159.5.8              |
| 25GBASE-T                                            | 113.4.2.2            |
| 40GBASE-KR4                                          | 84.7.10              |
| 40GBASE-CR4, 100GBASE-CR10                           | 85.7.10              |
| 40GBASE-SR4, 100GBASE-SR10                           | 86.5.10              |
| 40GBASE-LR4, 40GBASE-ER4                             | 87.5.10              |
| 40GBASE-FR                                           | 89.5.8               |
| 40GBASE-T                                            | 113.4.2.2            |
| 50GBASE-KR, 100GBASE-KR2, 200GBASE-KR4               | 137.8.9              |
| 50GBASE-CR, 100GBASE-CR2, 200GBASE-CR4               | 136.8.9              |
| 50GBASE-SR, 100GBASE-SR2, 200GBASE-SR4, 400GBASE-SR8 | 138.5.10             |
| 50GBASE-FR, 50GBASE-LR, 50GBASE-ER                   | 139.5.8              |
| 50GBASE-BR10, 50GBASE-BR20, 50GBASE-BR40             | 160.5.8              |
| 100GBASE-KP4                                         | 94.3.8               |
| 100GBASE-KR4                                         | 93.7.10              |

**Table 45–9—Transmit fault description location (*continued*)**

| PMA/PMD                                                                               | Description location |
|---------------------------------------------------------------------------------------|----------------------|
| 100GBASE-CR4                                                                          | 92.7.10              |
| 100GBASE-SR4                                                                          | 95.5.10              |
| 100GBASE-DR, 100GBASE-FR1, 100GBASE-LR1                                               | 140.5.8              |
| 100GBASE-LR4, 100GBASE-ER4                                                            | 88.5.10              |
| 100GBASE-ZR                                                                           | 154.5.8              |
| 200GBASE-DR4                                                                          | 121.5.10             |
| 200GBASE-FR4, 200GBASE-LR4, 200GBASE-ER4,<br>400GBASE-FR8, 400GBASE-LR8, 400GBASE-ER8 | 122.5.10             |
| 400GBASE-SR16                                                                         | 123.5.10             |
| 400GBASE-SR4.2                                                                        | 150.5.10             |
| 400GBASE-DR4                                                                          | 124.5.10             |
| 400GBASE-FR4, 400GBASE-LR4-6                                                          | 151.5.10             |

#### 45.2.1.7.5 Receive fault (1.8.10)

When read as a one, bit 1.8.10 indicates that the PMA/PMD has detected a fault condition on the receive path. When read as a zero, bit 1.8.10 indicates that the PMA/PMD has not detected a fault condition on the receive path. Detection of a fault condition on the receive path is optional and the ability to detect such a condition is advertised by bit 1.8.12. A PMA/PMD that is unable to detect a fault condition on the receive path shall return a value of zero for this bit. The description of the receive fault function for the various PMA/PMDs is given in Table 45–10.

**Table 45–10—Receive fault description location**

| PMA/PMD                                  | Description location |
|------------------------------------------|----------------------|
| 10BASE-T1L                               | 146.4.3              |
| 100BASE-T1                               | 96.4.3               |
| 2.5GBASE-KX                              | 128.6.9              |
| 2.5GBASE-T, 5GBASE-T                     | 126.4.2.4            |
| 2.5GBASE-T1, 5GBASE-T1, 10GBASE-T1       | 149.4.2.3            |
| 5GBASE-KR                                | 130.6.9              |
| 10GBASE-BR10, 10GBASE-BR20, 10GBASE-BR40 | 158.5.9              |
| 10GBASE-KR                               | 72.6.9               |
| 10GBASE-LRM                              | 68.4.9               |
| 10GBASE-S, 10GBASE-L, 10GBASE-E          | 52.4.9               |
| 10GBASE-LX4                              | 53.4.11              |

**Table 45–10—Receive fault description location (*continued*)**

| PMA/PMD                                                                               | Description location |
|---------------------------------------------------------------------------------------|----------------------|
| 10GBASE-CX4                                                                           | 54.5.11              |
| 10GBASE-T                                                                             | 55.4.2.4             |
| 10GBASE-KX4                                                                           | 71.6.11              |
| 25GBASE-KR, 25GBASE-KR-S                                                              | 111.7.9              |
| 25GBASE-CR, 25GBASE-CR-S                                                              | 110.7.9              |
| 25GBASE-SR                                                                            | 112.5.9              |
| 25GBASE-LR, 25GBASE-ER                                                                | 114.5.9              |
| 25GBASE-BR10, 25GBASE-BR20, 25GBASE-BR40                                              | 159.5.9              |
| 25GBASE-T                                                                             | 113.4.2.4            |
| 40GBASE-KR4                                                                           | 84.7.11              |
| 40GBASE-CR4, 100GBASE-CR10                                                            | 85.7.11              |
| 40GBASE-SR4, 100GBASE-SR10                                                            | 86.5.11              |
| 40GBASE-LR4, 40GBASE-ER4                                                              | 87.5.11              |
| 40GBASE-FR                                                                            | 89.5.9               |
| 40GBASE-T                                                                             | 113.4.2.4            |
| 50GBASE-KR, 100GBASE-KR2, 200GBASE-KR4                                                | 137.8.10             |
| 50GBASE-CR, 100GBASE-CR2, 200GBASE-CR4                                                | 136.8.10             |
| 50GBASE-SR, 100GBASE-SR2, 200GBASE-SR4, 400GBASE-SR8                                  | 138.5.11             |
| 50GBASE-FR, 50GBASE-LR, 50GBASE-ER                                                    | 139.5.9              |
| 50GBASE-BR10, 50GBASE-BR20, 50GBASE-BR40                                              | 160.5.9              |
| 100GBASE-KP4                                                                          | 94.3.9               |
| 100GBASE-KR4                                                                          | 93.7.11              |
| 100GBASE-CR4                                                                          | 92.7.11              |
| 100GBASE-SR4                                                                          | 95.5.11              |
| 100GBASE-DR, 100GBASE-FR1, 100GBASE-LR1                                               | 140.5.9              |
| 100GBASE-LR4, 100GBASE-ER4                                                            | 88.5.11              |
| 100GBASE-ZR                                                                           | 154.5.9              |
| 200GBASE-DR4                                                                          | 121.5.11             |
| 200GBASE-FR4, 200GBASE-LR4, 200GBASE-ER4,<br>400GBASE-FR8, 400GBASE-LR8, 400GBASE-ER8 | 122.5.11             |
| 400GBASE-SR16                                                                         | 123.5.11             |
| 400GBASE-SR4.2                                                                        | 150.5.11             |
| 400GBASE-DR4                                                                          | 124.5.11             |
| 400GBASE-FR4, 400GBASE-LR4-6                                                          | 151.5.11             |

The receive fault bit shall be implemented with latching high behavior.

#### **45.2.1.7.6 PMA/PMD extended abilities (1.8.9)**

When read as a one, bit 1.8.9 indicates that the PMA/PMD has extended abilities listed in register 1.11. When read as a zero, bit 1.8.9 indicates that the PMA/PMD does not have extended abilities.

#### **45.2.1.7.7 PMD transmit disable ability (1.8.8)**

When read as a one, bit 1.8.8 indicates that the PMD is able to perform the transmit disable function. When read as a zero, bit 1.8.8 indicates that the PMD is not able to perform the transmit disable function. If a PMD is able to perform the transmit disable function, then it is controlled using the PMD transmit disable register.

#### **45.2.1.7.8 10GBASE-SR ability (1.8.7)**

When read as a one, bit 1.8.7 indicates that the PMA/PMD is able to support a 10GBASE-SR PMA/PMD type. When read as a zero, bit 1.8.7 indicates that the PMA/PMD is not able to support a 10GBASE-SR PMA/PMD type.

#### **45.2.1.7.9 10GBASE-LR ability (1.8.6)**

When read as a one, bit 1.8.6 indicates that the PMA/PMD is able to support a 10GBASE-LR PMA/PMD type. When read as a zero, bit 1.8.6 indicates that the PMA/PMD is not able to support a 10GBASE-LR PMA/PMD type.

#### **45.2.1.7.10 10GBASE-ER ability (1.8.5)**

When read as a one, bit 1.8.5 indicates that the PMA/PMD is able to support a 10GBASE-ER PMA/PMD type. When read as a zero, bit 1.8.5 indicates that the PMA/PMD is not able to support a 10GBASE-ER PMA/PMD type.

#### **45.2.1.7.11 10GBASE-LX4 ability (1.8.4)**

When read as a one, bit 1.8.4 indicates that the PMA/PMD is able to support a 10GBASE-LX4 PMA/PMD type. When read as a zero, bit 1.8.4 indicates that the PMA/PMD is not able to support a 10GBASE-LX4 PMA/PMD type.

#### **45.2.1.7.12 10GBASE-SW ability (1.8.3)**

When read as a one, bit 1.8.3 indicates that the PMA/PMD is able to support a 10GBASE-SW PMA/PMD type. When read as a zero, bit 1.8.3 indicates that the PMA/PMD is not able to support a 10GBASE-SW PMA/PMD type.

#### **45.2.1.7.13 10GBASE-LW ability (1.8.2)**

When read as a one, bit 1.8.2 indicates that the PMA/PMD is able to support a 10GBASE-LW PMA/PMD type. When read as a zero, bit 1.8.2 indicates that the PMA/PMD is not able to support a 10GBASE-LW PMA/PMD type.

#### **45.2.1.7.14 10GBASE-EW ability (1.8.1)**

When read as a one, bit 1.8.1 indicates that the PMA/PMD is able to support a 10GBASE-EW PMA/PMD type. When read as a zero, bit 1.8.1 indicates that the PMA/PMD is not able to support a 10GBASE-EW PMA/PMD type.

#### 45.2.1.7.15 PMA local loopback ability (1.8.0)

When read as a one, bit 1.8.0 indicates that the PMA is able to perform the local loopback function. When read as a zero, bit 1.8.0 indicates that the PMA is not able to perform the local loopback function. If a PMA is able to perform the local loopback function, then it is controlled using the PMA local loopback bit 1.0.0.

#### 45.2.1.8 PMD transmit disable register (Register 1.9)

The assignment of bits in the PMD transmit disable register is shown in Table 45–11. The transmit disable functionality is optional and a PMD's ability to perform the transmit disable functionality is advertised in the PMD transmit disable ability bit 1.8.8. A PMD that does not implement the transmit disable functionality shall ignore writes to the PMD transmit disable register and may return a value of zero for all bits. A PMD device that operates using a single lane and has implemented the transmit disable function shall use bit 1.9.0 to control the function. Such devices shall ignore writes to bits 1.9.15:1 and return a value of zero for those bits when they are read. The description of the transmit disable function for the various PMA/PMDs is given in Table 45–12.

NOTE 1—This register is extended by the PMD transmit disable extension register (see 45.2.1.25).

NOTE 2—Disabling the transmitter on one or more lanes stops the entire link from carrying data.

**Table 45–11—PMD transmit disable register bit definitions**

| Bit(s) | Name                    | Description                                                                     | R/W <sup>a</sup> |
|--------|-------------------------|---------------------------------------------------------------------------------|------------------|
| 1.9.15 | PMD transmit disable 14 | 1 = Disable output on transmit lane 14<br>0 = Enable output on transmit lane 14 | R/W              |
| 1.9.14 | PMD transmit disable 13 | 1 = Disable output on transmit lane 13<br>0 = Enable output on transmit lane 13 | R/W              |
| 1.9.13 | PMD transmit disable 12 | 1 = Disable output on transmit lane 12<br>0 = Enable output on transmit lane 12 | R/W              |
| 1.9.12 | PMD transmit disable 11 | 1 = Disable output on transmit lane 11<br>0 = Enable output on transmit lane 11 | R/W              |
| 1.9.11 | PMD transmit disable 10 | 1 = Disable output on transmit lane 10<br>0 = Enable output on transmit lane 10 | R/W              |
| 1.9.10 | PMD transmit disable 9  | 1 = Disable output on transmit lane 9<br>0 = Enable output on transmit lane 9   | R/W              |
| 1.9.9  | PMD transmit disable 8  | 1 = Disable output on transmit lane 8<br>0 = Enable output on transmit lane 8   | R/W              |
| 1.9.8  | PMD transmit disable 7  | 1 = Disable output on transmit lane 7<br>0 = Enable output on transmit lane 7   | R/W              |
| 1.9.7  | PMD transmit disable 6  | 1 = Disable output on transmit lane 6<br>0 = Enable output on transmit lane 6   | R/W              |
| 1.9.6  | PMD transmit disable 5  | 1 = Disable output on transmit lane 5<br>0 = Enable output on transmit lane 5   | R/W              |
| 1.9.5  | PMD transmit disable 4  | 1 = Disable output on transmit lane 4<br>0 = Enable output on transmit lane 4   | R/W              |
| 1.9.4  | PMD transmit disable 3  | 1 = Disable output on transmit lane 3<br>0 = Enable output on transmit lane 3   | R/W              |

**Table 45–11—PMD transmit disable register bit definitions (*continued*)**

| Bit(s) | Name                        | Description                                                                   | R/W <sup>a</sup> |
|--------|-----------------------------|-------------------------------------------------------------------------------|------------------|
| 1.9.3  | PMD transmit disable 2      | 1 = Disable output on transmit lane 2<br>0 = Enable output on transmit lane 2 | R/W              |
| 1.9.2  | PMD transmit disable 1      | 1 = Disable output on transmit lane 1<br>0 = Enable output on transmit lane 1 | R/W              |
| 1.9.1  | PMD transmit disable 0      | 1 = Disable output on transmit lane 0<br>0 = Enable output on transmit lane 0 | R/W              |
| 1.9.0  | Global PMD transmit disable | 1 = Disable transmitter output<br>0 = Enable transmitter output               | R/W              |

<sup>a</sup>R/W = Read/Write

#### 45.2.1.8.1 PMD transmit disable 14 (1.9.15)

When bit 1.9.15 is set to a one, the PMD shall disable output on lane 14 of the transmit path. When bit 1.9.15 is set to zero, the PMD shall enable output on lane 14 of the transmit path.

The default value for bit 1.9.15 is zero.

NOTE—Transmission will not be enabled when this bit is set to zero unless the global PMD transmit disable bit is also zero.

**Table 45–12—Transmit disable description location**

| PMA/PMD                                      | Description location |
|----------------------------------------------|----------------------|
| 2.5GBASE-KX                                  | 128.6.5              |
| 2.5GBASE-T and 5GBASE-T                      | 126.4.2.3            |
| 5GBASE-KR                                    | 130.6.5              |
| 10GBASE-BR10, 10GBASE-BR20, and 10GBASE-BR40 | 158.5.6              |
| 10GBASE-KR                                   | 72.6.5               |
| 10GBASE-LRM                                  | 68.4.7               |
| Other 10GBASE-R                              | 52.4.7               |
| 10GBASE-LX4                                  | 53.4.7               |
| 10GBASE-CX4                                  | 54.5.6               |
| 10GBASE-T                                    | 55.4.2.3             |
| 10GBASE-KX4                                  | 71.6.6               |
| 25GBASE-KR and 25GBASE-KR-S                  | 111.7.5              |
| 25GBASE-CR and 25GBASE-CR-S                  | 110.7.5              |
| 25GBASE-SR                                   | 112.5.6              |
| 25GBASE-LR and 25GBASE-ER                    | 114.5.6              |

**Table 45–12—Transmit disable description location (*continued*)**

| PMA/PMD                                                                                | Description location |
|----------------------------------------------------------------------------------------|----------------------|
| 25GBASE-BR10, 25GBASE-BR20, and 25GBASE-BR40                                           | 159.5.6              |
| 25GBASE-T                                                                              | 113.4.2.3            |
| 40GBASE-KR4                                                                            | 84.7.6               |
| 40GBASE-CR4 and 100GBASE-CR10                                                          | 85.7.6               |
| 40GBASE-SR4 and 100GBASE-SR10                                                          | 86.5.7               |
| 40GBASE-LR4 and 40GBASE-ER4                                                            | 87.5.7               |
| 40GBASE-FR                                                                             | 89.5.6               |
| 40GBASE-T                                                                              | 113.4.2.3            |
| 50GBASE-KR, 100GBASE-KR2, and 200GBASE-KR4                                             | 137.8.6              |
| 50GBASE-CR, 100GBASE-CR2, and 200GBASE-CR4                                             | 136.8.6              |
| 50GBASE-SR, 100GBASE-SR2, 200GBASE-SR4, and 400GBASE-SR8                               | 138.5.7              |
| 50GBASE-FR, 50GBASE-LR, and 50GBASE-ER                                                 | 139.5.6              |
| 50GBASE-BR10, 50GBASE-BR20, and 50GBASE-BR40                                           | 160.5.6              |
| 100GBASE-KP4                                                                           | 94.3.6.6             |
| 100GBASE-KR4                                                                           | 93.7.6               |
| 100GBASE-CR4                                                                           | 92.7.6               |
| 100GBASE-SR4                                                                           | 95.5.7               |
| 100GBASE-DR, 100GBASE-FR1, and 100GBASE-LR1                                            | 140.5.6              |
| 100GBASE-LR4 and 100GBASE-ER4                                                          | 88.5.7               |
| 100GBASE-ZR                                                                            | 154.5.6              |
| 200GBASE-DR4                                                                           | 121.5.7              |
| 200GBASE-FR4, 200GBASE-LR4, 200GBASE-ER4, 400GBASE-FR8, 400GBASE-LR8, and 400GBASE-ER8 | 122.5.7              |
| 400GBASE-SR16                                                                          | 123.5.7              |
| 400GBASE-SR4.2                                                                         | 150.5.7              |
| 400GBASE-DR4                                                                           | 124.5.7              |
| 400GBASE-FR4 and 400GBASE-LR4-6                                                        | 151.5.7              |

#### 45.2.1.8.2 PMD transmit disable 4 through 14 (1.9.5 through 1.9.14)

Bits 1.9.5 through 1.9.14 are defined similarly to bit 1.9.15 for lanes 4 through 13, respectively.

#### 45.2.1.8.3 PMD transmit disable 3 (1.9.4)

When bit 1.9.4 is set to a one, the PMD shall disable output on lane 3 of the transmit path. When bit 1.9.4 is set to a zero, the PMD shall enable output on lane 3 of the transmit path.



The default value for bit 1.9.4 is zero.

NOTE—Transmission will not be enabled when this bit is set to a zero unless the global PMD transmit disable bit is also zero.

#### **45.2.1.8.4 PMD transmit disable 2 (1.9.3)**

When bit 1.9.3 is set to a one, the PMD shall disable output on lane 2 of the transmit path. When bit 1.9.3 is set to a zero, the PMD shall enable output on lane 2 of the transmit path.

The default value for bit 1.9.3 is zero.

NOTE—Transmission will not be enabled when this bit is set to a zero unless the global PMD transmit disable bit is also zero.

#### **45.2.1.8.5 PMD transmit disable 1 (1.9.2)**

When bit 1.9.2 is set to a one, the PMD shall disable output on lane 1 of the transmit path. When bit 1.9.2 is set to a zero, the PMD shall enable output on lane 1 of the transmit path.

The default value for bit 1.9.2 is zero.

NOTE—Transmission will not be enabled when this bit is set to a zero unless the global PMD transmit disable bit is also zero.

#### **45.2.1.8.6 PMD transmit disable 0 (1.9.1)**

When bit 1.9.1 is set to a one, the PMD shall disable output on lane 0 of the transmit path. When bit 1.9.1 is set to a zero, the PMD shall enable output on lane 0 of the transmit path.

The default value for bit 1.9.1 is zero.

NOTE—Transmission will not be enabled when this bit is set to a zero unless the global PMD transmit disable bit is also zero.

#### **45.2.1.8.7 Global PMD transmit disable (1.9.0)**

When bit 1.9.0 is set to a one, the PMD shall disable output on the transmit path. When bit 1.9.0 is set to a zero, the PMD shall enable output on the transmit path.

For single wavelength PMD types, transmission will be disabled when this bit is set to one. When this bit is set to zero, transmission is enabled.

For multiple wavelength or lane PMD types, transmission will be disabled on all lanes when this bit is set to one. When this bit is set to zero, the lanes are individually controlled by their corresponding transmit disable bits 1.9.4:1.

The default value for bit 1.9.0 is zero.

#### **45.2.1.9 PMD receive signal detect register (Register 1.10)**

The assignment of bits in the PMD receive signal detect register is shown in Table 45–13. The PMD receive signal detect register is mandatory. PMD types that use only a single lane indicate the status of the receive signal detect using bit 1.10.0 and return a value of zero for bits 1.10.15:1. PMD types that use multiple

wavelengths or lanes indicate the status of each lane in bits 1.10.15:1 and the logical AND of those bits in bit 1.10.0.

NOTE—This register is extended by the PMD receive signal detect extension register (see 45.2.1.26).

**Table 45–13—PMD receive signal detect register bit definitions**

| Bit(s)  | Name                             | Description                                                                          | R/W <sup>a</sup> |
|---------|----------------------------------|--------------------------------------------------------------------------------------|------------------|
| 1.10.15 | PMD receive signal detect 14     | 1 = Signal detected on receive lane 14<br>0 = Signal not detected on receive lane 14 | RO               |
| 1.10.14 | PMD receive signal detect 13     | 1 = Signal detected on receive lane 13<br>0 = Signal not detected on receive lane 13 | RO               |
| 1.10.13 | PMD receive signal detect 12     | 1 = Signal detected on receive lane 12<br>0 = Signal not detected on receive lane 12 | RO               |
| 1.10.12 | PMD receive signal detect 11     | 1 = Signal detected on receive lane 11<br>0 = Signal not detected on receive lane 11 | RO               |
| 1.10.11 | PMD receive signal detect 10     | 1 = Signal detected on receive lane 10<br>0 = Signal not detected on receive lane 10 | RO               |
| 1.10.10 | PMD receive signal detect 9      | 1 = Signal detected on receive lane 9<br>0 = Signal not detected on receive lane 9   | RO               |
| 1.10.9  | PMD receive signal detect 8      | 1 = Signal detected on receive lane 8<br>0 = Signal not detected on receive lane 8   | RO               |
| 1.10.8  | PMD receive signal detect 7      | 1 = Signal detected on receive lane 7<br>0 = Signal not detected on receive lane 7   | RO               |
| 1.10.7  | PMD receive signal detect 6      | 1 = Signal detected on receive lane 6<br>0 = Signal not detected on receive lane 6   | RO               |
| 1.10.6  | PMD receive signal detect 5      | 1 = Signal detected on receive lane 5<br>0 = Signal not detected on receive lane 5   | RO               |
| 1.10.5  | PMD receive signal detect 4      | 1 = Signal detected on receive lane 4<br>0 = Signal not detected on receive lane 4   | RO               |
| 1.10.4  | PMD receive signal detect 3      | 1 = Signal detected on receive lane 3<br>0 = Signal not detected on receive lane 3   | RO               |
| 1.10.3  | PMD receive signal detect 2      | 1 = Signal detected on receive lane 2<br>0 = Signal not detected on receive lane 2   | RO               |
| 1.10.2  | PMD receive signal detect 1      | 1 = Signal detected on receive lane 1<br>0 = Signal not detected on receive lane 1   | RO               |
| 1.10.1  | PMD receive signal detect 0      | 1 = Signal detected on receive lane 0<br>0 = Signal not detected on receive lane 0   | RO               |
| 1.10.0  | Global PMD receive signal detect | 1 = Signal detected on receive<br>0 = Signal not detected on receive                 | RO               |

<sup>a</sup>RO = Read only

#### **45.2.1.9.1 PMD receive signal detect 14 (1.10.15)**

When bit 1.10.15 is read as a one, a signal has been detected on lane 14 of the PMD receive path. When bit 1.10.15 is read as a zero, a signal has not been detected on lane 14 of the PMD receive path.

#### **45.2.1.9.2 PMD receive signal detect 4 through 13 (1.10.5 through 1.10.14)**

Bits 1.10.5 through 1.10.14 are defined similarly to bit 1.10.15 for lanes 4 through 13, respectively.

#### **45.2.1.9.3 PMD receive signal detect 3 (1.10.4)**

When bit 1.10.4 is read as a one, a signal has been detected on lane 3 of the PMD receive path. When bit 1.10.4 is read as a zero, a signal has not been detected on lane 3 of the PMD receive path.

#### **45.2.1.9.4 PMD receive signal detect 2 (1.10.3)**

When bit 1.10.3 is read as a one, a signal has been detected on lane 2 of the PMD receive path. When bit 1.10.3 is read as a zero, a signal has not been detected on lane 2 of the PMD receive path.

#### **45.2.1.9.5 PMD receive signal detect 1 (1.10.2)**

When bit 1.10.2 is read as a one, a signal has been detected on lane 1 of the PMD receive path. When bit 1.10.2 is read as a zero, a signal has not been detected on lane 1 of the PMD receive path.

#### **45.2.1.9.6 PMD receive signal detect 0 (1.10.1)**

When bit 1.10.1 is read as a one, a signal has been detected on lane 0 of the PMD receive path. When bit 1.10.1 is read as a zero, a signal has not been detected on lane 0 of the PMD receive path.

#### **45.2.1.9.7 Global PMD receive signal detect (1.10.0)**

When bit 1.10.0 is read as a one, a signal has been detected on all the PMD receive paths. When bit 1.10.0 is read as a zero, a signal has not been detected on at least one of the PMD receive paths.

Single wavelength PMD types indicate the status of their receive path signal using this bit.

Multiple wavelength or multiple lane PMD types indicate the global status of the lane-by-lane signal detect indications using this bit. This bit is read as a one when all the lane signal detect indications are one; otherwise, this bit is read as a zero.

#### 45.2.1.10 PMA/PMD extended ability register (Register 1.11)

The assignment of bits in the PMA/PMD extended ability register is shown in Table 45–14. All of the bits in the PMA/PMD extended ability register are read only; a write to the PMA/PMD extended ability register shall have no effect.

**Table 45–14—PMA/PMD Extended Ability register bit definitions**

| Bit(s)  | Name                         | Description                                                                                                                                     | R/W <sup>a</sup> |
|---------|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.11.15 | BASE-H extended abilities    | 1 = PMA/PMD has BASE-H extended abilities listed in register 1.22<br>0 = PMA/PMD does not have BASE-H extended abilities                        | RO               |
| 1.11.14 | 2.5G/5G extended abilities   | 1 = PMA/PMD has 2.5G/5G extended abilities listed in register 1.21<br>0 = PMA/PMD does not have 2.5G/5G extended abilities                      | RO               |
| 1.11.13 | 200G/400G extended abilities | 1 = PMA/PMD has 200G/400G extended abilities listed in register 1.23 or register 1.24<br>0 = PMA/PMD does not have 200G/400G extended abilities | RO               |
| 1.11.12 | 25G extended abilities       | 1 = PMA/PMD has 25G extended abilities listed in register 1.19<br>0 = PMA/PMD does not have 25G extended abilities                              | RO               |
| 1.11.11 | BASE-T1 extended abilities   | 1 = PMA/PMD has BASE-T1 extended abilities listed in register 1.18<br>0 = PMA/PMD does not have BASE-T1 extended abilities                      | RO               |
| 1.11.10 | 40G/100G extended abilities  | 1 = PMA/PMD has 40G/100G extended abilities listed in register 1.13<br>0 = PMA/PMD does not have 40G/100G extended abilities                    | RO               |
| 1.11.9  | P2MP ability                 | 1 = PMA/PMD has P2MP abilities listed in register 1.12<br>0 = PMA/PMD does not have P2MP abilities                                              | RO               |
| 1.11.8  | 10BASE-T ability             | 1 = PMA/PMD is able to perform 10BASE-T<br>0 = PMA/PMD is not able to perform 10BASE-T                                                          | RO               |
| 1.11.7  | 100BASE-TX ability           | 1 = PMA/PMD is able to perform 100BASE-TX<br>0 = PMA/PMD is not able to perform 100BASE-TX                                                      | RO               |
| 1.11.6  | 1000BASE-KX ability          | 1 = PMA/PMD is able to perform 1000BASE-KX<br>0 = PMA/PMD is not able to perform 1000BASE-KX                                                    | RO               |
| 1.11.5  | 1000BASE-T ability           | 1 = PMA/PMD is able to perform 1000BASE-T<br>0 = PMA/PMD is not able to perform 1000BASE-T                                                      | RO               |
| 1.11.4  | 10GBASE-KR ability           | 1 = PMA/PMD is able to perform 10GBASE-KR<br>0 = PMA/PMD is not able to perform 10GBASE-KR                                                      | RO               |
| 1.11.3  | 10GBASE-KX4 ability          | 1 = PMA/PMD is able to perform 10GBASE-KX4<br>0 = PMA/PMD is not able to perform 10GBASE-KX4                                                    | RO               |

**Table 45–14—PMA/PMD Extended Ability register bit definitions (*continued*)**

| Bit(s) | Name                | Description                                                                                  | R/W <sup>a</sup> |
|--------|---------------------|----------------------------------------------------------------------------------------------|------------------|
| 1.11.2 | 10GBASE-T ability   | 1 = PMA/PMD is able to perform 10GBASE-T<br>0 = PMA/PMD is not able to perform 10GBASE-T     | RO               |
| 1.11.1 | 10GBASE-LRM ability | 1 = PMA/PMD is able to perform 10GBASE-LRM<br>0 = PMA/PMD is not able to perform 10GBASE-LRM | RO               |
| 1.11.0 | 10GBASE-CX4 ability | 1 = PMA/PMD is able to perform 10GBASE-CX4<br>0 = PMA/PMD is not able to perform 10GBASE-CX4 | RO               |

<sup>a</sup>RO = Read only

#### **45.2.1.10.1 BASE-H extended abilities (1.11.15)**

When read as one, bit 1.11.15 indicates that the PMA/PMD has BASE-H extended abilities listed in register 1.22. When read as zero, bit 1.11.15 indicates that the PMA/PMD does not have BASE-H extended abilities.

#### **45.2.1.10.2 2.5G/5G extended abilities (1.11.14)**

When read as a one, bit 1.11.14 indicates that the PMA/PMD has 2.5G/5G extended abilities listed in register 1.21. When read as a zero, bit 1.11.14 indicates that the PMA/PMD does not have 2.5G/5G extended abilities.

#### **45.2.1.10.3 200G/400G extended abilities (1.11.13)**

When read as a one, bit 1.11.13 indicates that the PMA/PMD has 200G extended abilities listed in register 1.23 or 400G extended abilities listed in register 1.24. When read as a zero, bit 1.11.13 indicates that the PMA/PMD does not have 200G or 400G extended abilities.

#### **45.2.1.10.4 25G extended abilities (1.11.12)**

When read as a one, bit 1.11.12 indicates that the PMA/PMD has 25G extended abilities listed in register 1.19. When read as a zero, bit 1.11.12 indicates that the PMA/PMD does not have 25G extended abilities.

#### **45.2.1.10.5 BASE-T1 extended abilities (1.11.11)**

When read as a one, bit 1.11.11 indicates that the PMA/PMD has BASE-T1 extended abilities listed in register 1.18. When read as a zero, bit 1.11.11 indicates that the PMA/PMD does not have BASE-T1 extended abilities.

#### **45.2.1.10.6 40G/100G extended abilities (1.11.10)**

When read as a one, bit 1.11.10 indicates that the PMA/PMD has 40G/100G extended abilities listed in register 1.13. When read as a zero, bit 1.11.10 indicates that the PMA/PMD does not have 40G/100G extended abilities.

#### **45.2.1.10.7 P2MP ability (1.11.9)**

When read as a one, bit 1.11.9 indicates that the PMA/PMD has P2MP abilities listed in register 1.12. When read as a zero, bit 1.11.9 indicates that the PMA/PMD does not have P2MP abilities.

#### **45.2.1.10.8 10BASE-T ability (1.11.8)**

When read as a one, bit 1.11.8 indicates that the PMA/PMD is able to operate as a 10BASE-T PMA/PMD type. When read as a zero, bit 1.11.8 indicates that the PMA/PMD is not able to operate as a 10BASE-T PMA/PMD type.

#### **45.2.1.10.9 100BASE-TX ability (1.11.7)**

When read as a one, bit 1.11.7 indicates that the PMA/PMD is able to operate as a 100BASE-TX PMA/PMD type. When read as a zero, bit 1.11.7 indicates that the PMA/PMD is not able to operate as a 100BASE-TX PMA/PMD type.

#### **45.2.1.10.10 1000BASE-KX ability (1.11.6)**

When read as a one, bit 1.11.6 indicates that the PMA/PMD is able to operate as 1000BASE-KX. When read as a zero, bit 1.11.6 indicates that the PMA/PMD is not able to operate as 1000BASE-KX.

#### **45.2.1.10.11 1000BASE-T ability (1.11.5)**

When read as a one, bit 1.11.5 indicates that the PMA/PMD is able to operate as a 1000BASE-T PMA/PMD type. When read as a zero, bit 1.11.5 indicates that the PMA/PMD is not able to operate as a 1000BASE-T PMA/PMD type.

#### **45.2.1.10.12 10GBASE-KR ability (1.11.4)**

When read as a one, bit 1.11.4 indicates that the PMA/PMD is able to operate as 10GBASE-KR. When read as a zero, bit 1.11.4 indicates that the PMA/PMD is not able to operate as 10GBASE-KR.

#### **45.2.1.10.13 10GBASE-KX4 ability (1.11.3)**

When read as a one, bit 1.11.3 indicates that the PMA/PMD is able to operate as 10GBASE-KX4. When read as a zero, bit 1.11.3 indicates that the PMA/PMD is not able to operate as 10GBASE-KX4.

#### **45.2.1.10.14 10GBASE-T ability (1.11.2)**

When read as a one, bit 1.11.2 indicates that the PMA/PMD is able to operate as a 10GBASE-T PMA/PMD type. When read as a zero, bit 1.11.2 indicates that the PMA/PMD is not able to operate as a 10GBASE-T PMA/PMD type.

#### **45.2.1.10.15 10GBASE-LRM ability (1.11.1)**

When read as a one, bit 1.11.1 indicates that the PMA/PMD is able to operate as 10GBASE-LRM. When read as a zero, bit 1.11.1 indicates that the PMA/PMD is not able to operate as 10GBASE-LRM.

#### **45.2.1.10.16 10GBASE-CX4 ability (1.11.0)**

When read as a one, bit 1.11.0 indicates that the PMA/PMD is able to operate as a 10GBASE-CX4 PMA/PMD type. When read as a zero, bit 1.11.0 indicates that the PMA/PMD is not able to operate as a 10GBASE-CX4 PMA/PMD type.

#### 45.2.1.11 10G-EPON PMA/PMD ability register (Register 1.12)

The assignment of bits in the 10G-EPON PMA/PMD ability register is shown in Table 45–15. All of the bits in the 10G-EPON PMA/PMD ability register are read only; a write to the 10G-EPON PMA/PMD ability register shall have no effect.

**Table 45–15—10G-EPON PMA/PMD ability register bit definitions**

| Bit(s)  | Name                     | Description                                                                                            | R/W <sup>a</sup> |
|---------|--------------------------|--------------------------------------------------------------------------------------------------------|------------------|
| 1.12.15 | Reserved                 | Value always 0                                                                                         | RO               |
| 1.12.14 | 10GBASE-PR-D4 ability    | 1 = PMA/PMD is able to perform 10GBASE-PR-D4<br>0 = PMA/PMD is not able to perform 10GBASE-PR-D4       | RO               |
| 1.12.13 | 10GBASE-PR-U4 ability    | 1 = PMA/PMD is able to perform 10GBASE-PR-U4<br>0 = PMA/PMD is not able to perform 10GBASE-PR-U4       | RO               |
| 1.12.12 | 10/1GBASE-PRX-D4 ability | 1 = PMA/PMD is able to perform 10/1GBASE-PRX-D4<br>0 = PMA/PMD is not able to perform 10/1GBASE-PRX-D4 | RO               |
| 1.12.11 | 10/1GBASE-PRX-U4 ability | 1 = PMA/PMD is able to perform 10/1GBASE-PRX-U4<br>0 = PMA/PMD is not able to perform 10/1GBASE-PRX-U4 | RO               |
| 1.12.10 | 10/1GBASE-PRX-D1 ability | 1 = PMA/PMD is able to perform 10/1GBASE-PRX-D1<br>0 = PMA/PMD is not able to perform 10/1GBASE-PRX-D1 | RO               |
| 1.12.9  | 10/1GBASE-PRX-D2 ability | 1 = PMA/PMD is able to perform 10/1GBASE-PRX-D2<br>0 = PMA/PMD is not able to perform 10/1GBASE-PRX-D2 | RO               |
| 1.12.8  | 10/1GBASE-PRX-D3 ability | 1 = PMA/PMD is able to perform 10/1GBASE-PRX-D3<br>0 = PMA/PMD is not able to perform 10/1GBASE-PRX-D3 | RO               |
| 1.12.7  | 10GBASE-PR-D1 ability    | 1 = PMA/PMD is able to perform 10GBASE-PR-D1<br>0 = PMA/PMD is not able to perform 10GBASE-PR-D1       | RO               |
| 1.12.6  | 10GBASE-PR-D2 ability    | 1 = PMA/PMD is able to perform 10GBASE-PR-D2<br>0 = PMA/PMD is not able to perform 10GBASE-PR-D2       | RO               |
| 1.12.5  | 10GBASE-PR-D3 ability    | 1 = PMA/PMD is able to perform 10GBASE-PR-D3<br>0 = PMA/PMD is not able to perform 10GBASE-PR-D3       | RO               |
| 1.12.4  | 10/1GBASE-PRX-U1 ability | 1 = PMA/PMD is able to perform 10/1GBASE-PRX-U1<br>0 = PMA/PMD is not able to perform 10/1GBASE-PRX-U1 | RO               |
| 1.12.3  | 10/1GBASE-PRX-U2 ability | 1 = PMA/PMD is able to perform 10/1GBASE-PRX-U2<br>0 = PMA/PMD is not able to perform 10/1GBASE-PRX-U2 | RO               |
| 1.12.2  | 10/1GBASE-PRX-U3 ability | 1 = PMA/PMD is able to perform 10/1GBASE-PRX-U3<br>0 = PMA/PMD is not able to perform 10/1GBASE-PRX-U3 | RO               |
| 1.12.1  | 10GBASE-PR-U1 ability    | 1 = PMA/PMD is able to perform 10GBASE-PR-U1<br>0 = PMA/PMD is not able to perform 10GBASE-PR-U1       | RO               |
| 1.12.0  | 10GBASE-PR-U3 ability    | 1 = PMA/PMD is able to perform 10GBASE-PR-U3<br>0 = PMA/PMD is not able to perform 10GBASE-PR-U3       | RO               |

<sup>a</sup>RO = Read only

##### 45.2.1.11.1 10GBASE-PR-D4 ability (1.12.14)

When read as a one, bit 1.12.14 indicates that the PMA/PMD is able to operate as a 10GBASE-PR-D4 PMA/PMD type. When read as a zero, bit 1.12.14 indicates that the PMA/PMD is not able to operate as a 10GBASE-PR-D4 PMA/PMD type.

#### **45.2.1.11.2 10GBASE-PR-U4 ability (1.12.13)**

When read as a one, bit 1.12.13 indicates that the PMA/PMD is able to operate as a 10GBASE-PR-U4 PMA/PMD type. When read as a zero, bit 1.12.13 indicates that the PMA/PMD is not able to operate as a 10GBASE-PR-U4 PMA/PMD type.

#### **45.2.1.11.3 10/1GBASE-PRX-D4 ability (1.12.12)**

When read as a one, bit 1.12.12 indicates that the PMA/PMD is able to operate as a 10/1GBASE-PRX-D4 PMA/PMD type. When read as a zero, bit 1.12.12 indicates that the PMA/PMD is not able to operate as a 10/1GBASE-PRX-D4 PMA/PMD type.

#### **45.2.1.11.4 10/1GBASE-PRX-U4 ability (1.12.11)**

When read as a one, bit 1.12.11 indicates that the PMA/PMD is able to operate as a 10/1GBASE-PRX-U4 PMA/PMD type. When read as a zero, bit 1.12.11 indicates that the PMA/PMD is not able to operate as a 10/1GBASE-PRX-U4 PMA/PMD type.

#### **45.2.1.11.5 10/1GBASE-PRX-D1 ability (1.12.10)**

When read as a one, bit 1.12.10 indicates that the PMA/PMD is able to operate as a 10/1GBASE-PRX-D1 PMA/PMD type. When read as a zero, bit 1.12.10 indicates that the PMA/PMD is not able to operate as a 10/1GBASE-PRX-D1 PMA/PMD type.

#### **45.2.1.11.6 10/1GBASE-PRX-D2 ability (1.12.9)**

When read as a one, bit 1.12.9 indicates that the PMA/PMD is able to operate as a 10/1GBASE-PRX-D2 PMA/PMD type. When read as a zero, bit 1.12.9 indicates that the PMA/PMD is not able to operate as a 10/1GBASE-PRX-D2 PMA/PMD type.

#### **45.2.1.11.7 10/1GBASE-PRX-D3 ability (1.12.8)**

When read as a one, bit 1.12.8 indicates that the PMA/PMD is able to operate as a 10/1GBASE-PRX-D3 PMA/PMD type. When read as a zero, bit 1.12.8 indicates that the PMA/PMD is not able to operate as a 10/1GBASE-PRX-D3 PMA/PMD type.

#### **45.2.1.11.8 10GBASE-PR-D1 ability (1.12.7)**

When read as a one, bit 1.12.7 indicates that the PMA/PMD is able to operate as a 10GBASE-PR-D1 PMA/PMD type. When read as a zero, bit 1.12.7 indicates that the PMA/PMD is not able to operate as a 10GBASE-PR-D1 PMA/PMD type.

#### **45.2.1.11.9 10GBASE-PR-D2 ability (1.12.6)**

When read as a one, bit 1.12.6 indicates that the PMA/PMD is able to operate as a 10GBASE-PR-D2 PMA/PMD type. When read as a zero, bit 1.12.6 indicates that the PMA/PMD is not able to operate as a 10GBASE-PR-D2 PMA/PMD type.

#### **45.2.1.11.10 10GBASE-PR-D3 ability (1.12.5)**

When read as a one, bit 1.12.5 indicates that the PMA/PMD is able to operate as a 10GBASE-PR-D3 PMA/PMD type. When read as a zero, bit 1.12.5 indicates that the PMA/PMD is not able to operate as a 10GBASE-PR-D3 PMA/PMD type.



#### 45.2.1.11.11 10/1GBASE-PRX-U1 ability (1.12.4)

When read as a one, bit 1.12.4 indicates that the PMA/PMD is able to operate as a 10/1GBASE-PRX-U1 PMA/PMD type. When read as a zero, bit 1.12.4 indicates that the PMA/PMD is not able to operate as a 10/1GBASE-PRX-U1 PMA/PMD type.

#### 45.2.1.11.12 10/1GBASE-PRX-U2 ability (1.12.3)

When read as a one, bit 1.12.3 indicates that the PMA/PMD is able to operate as a 10/1GBASE-PRX-U2 PMA/PMD type. When read as a zero, bit 1.12.3 indicates that the PMA/PMD is not able to operate as a 10/1GBASE-PRX-U2 PMA/PMD type.

#### 45.2.1.11.13 10/1GBASE-PRX-U3 ability (1.12.2)

When read as a one, bit 1.12.2 indicates that the PMA/PMD is able to operate as a 10/1GBASE-PRX-U3 PMA/PMD type. When read as a zero, bit 1.12.2 indicates that the PMA/PMD is not able to operate as a 10/1GBASE-PRX-U3 PMA/PMD type.

#### 45.2.1.11.14 10GBASE-PR-U1 ability (1.12.1)

When read as a one, bit 1.12.1 indicates that the PMA/PMD is able to operate as a 10GBASE-PR-U1 PMA/PMD type. When read as a zero, bit 1.12.1 indicates that the PMA/PMD is not able to operate as a 10GBASE-PR-U1 PMA/PMD type.

#### 45.2.1.11.15 10GBASE-PR-U3 ability (1.12.0)

When read as a one, bit 1.12.0 indicates that the PMA/PMD is able to operate as a 10GBASE-PR-U3 PMA/PMD type. When read as a zero, bit 1.12.0 indicates that the PMA/PMD is not able to operate as a 10GBASE-PR-U3 PMA/PMD type.

#### 45.2.1.12 40G/100G PMA/PMD extended ability register (Register 1.13)

The assignment of bits in the 40G/100G PMA/PMD extended ability register is shown in Table 45–16. All of the bits in the PMA/PMD extended ability register are read only; a write to the PMA/PMD extended ability register shall have no effect.

**Table 45–16—40G/100G PMA/PMD extended ability register bit definitions**

| Bit(s)  | Name                        | Description                                                                                                                              | R/W <sup>a</sup> |
|---------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.13.15 | PMA remote loopback ability | 1 = PMA has the ability to perform a remote loopback function<br>0 = PMA does not have the ability to perform a remote loopback function | RO               |
| 1.13.14 | 100GBASE-CR4 ability        | 1 = PMA/PMD is able to perform 100GBASE-CR4<br>0 = PMA/PMD is not able to perform 100GBASE-CR4                                           | RO               |
| 1.13.13 | 100GBASE-KR4 ability        | 1 = PMA/PMD is able to perform 100GBASE-KR4<br>0 = PMA/PMD is not able to perform 100GBASE-KR4                                           | RO               |
| 1.13.12 | 100GBASE-KP4 ability        | 1 = PMA/PMD is able to perform 100GBASE-KP4<br>0 = PMA/PMD is not able to perform 100GBASE-KP4                                           | RO               |
| 1.13.11 | 100GBASE-ER4 ability        | 1 = PMA/PMD is able to perform 100GBASE-ER4<br>0 = PMA/PMD is not able to perform 100GBASE-ER4                                           | RO               |

**Table 45–16—40G/100G PMA/PMD extended ability register bit definitions (*continued*)**

| Bit(s)  | Name                  | Description                                                                                      | R/W <sup>a</sup> |
|---------|-----------------------|--------------------------------------------------------------------------------------------------|------------------|
| 1.13.10 | 100GBASE-LR4 ability  | 1 = PMA/PMD is able to perform 100GBASE-LR4<br>0 = PMA/PMD is not able to perform 100GBASE-LR4   | RO               |
| 1.13.9  | 100GBASE-SR10 ability | 1 = PMA/PMD is able to perform 100GBASE-SR10<br>0 = PMA/PMD is not able to perform 100GBASE-SR10 | RO               |
| 1.13.8  | 100GBASE-CR10 ability | 1 = PMA/PMD is able to perform 100GBASE-CR10<br>0 = PMA/PMD is not able to perform 100GBASE-CR10 | RO               |
| 1.13.7  | 100GBASE-SR4 ability  | 1 = PMA/PMD is able to perform 100GBASE-SR4<br>0 = PMA/PMD is not able to perform 100GBASE-SR4   | RO               |
| 1.13.6  | 40GBASE-T ability     | 1 = PMA/PMD is able to perform 40GBASE-T<br>0 = PMA/PMD is not able to perform 40GBASE-T         | RO               |
| 1.13.5  | 40GBASE-ER4 ability   | 1 = PMA/PMD is able to perform 40GBASE-ER4<br>0 = PMA/PMD is not able to perform 40GBASE-ER4     | RO               |
| 1.13.4  | 40GBASE-FR ability    | 1 = PMA/PMD is able to perform 40GBASE-FR<br>0 = PMA/PMD is not able to perform 40GBASE-FR       | RO               |
| 1.13.3  | 40GBASE-LR4 ability   | 1 = PMA/PMD is able to perform 40GBASE-LR4<br>0 = PMA/PMD is not able to perform 40GBASE-LR4     | RO               |
| 1.13.2  | 40GBASE-SR4 ability   | 1 = PMA/PMD is able to perform 40GBASE-SR4<br>0 = PMA/PMD is not able to perform 40GBASE-SR4     | RO               |
| 1.13.1  | 40GBASE-CR4 ability   | 1 = PMA/PMD is able to perform 40GBASE-CR4<br>0 = PMA/PMD is not able to perform 40GBASE-CR4     | RO               |
| 1.13.0  | 40GBASE-KR4 ability   | 1 = PMA/PMD is able to perform 40GBASE-KR4<br>0 = PMA/PMD is not able to perform 40GBASE-KR4     | RO               |

<sup>a</sup>RO = Read only

#### 45.2.1.12.1 PMA remote loopback ability (1.13.15)

When read as a one, bit 1.13.15 indicates that the PMA is able to perform the remote loopback function. When read as a zero, bit 1.13.15 indicates that the PMA is not able to perform the remote loopback function. If a PMA is able to perform the remote loopback function, then it is controlled using the PMA remote loopback bit 1.0.1 (see 45.2.1.1.4).

#### 45.2.1.12.2 100GBASE-CR4 ability (1.13.14)

When read as a one, bit 1.13.14 indicates that the PMA/PMD is able to operate as a 100GBASE-CR4 PMA/PMD type. When read as a zero, bit 1.13.14 indicates that the PMA/PMD is not able to operate as a 100GBASE-CR4 PMA/PMD type.

#### 45.2.1.12.3 100GBASE-KR4 ability (1.13.13)

When read as a one, bit 1.13.13 indicates that the PMA/PMD is able to operate as a 100GBASE-KR4 PMA/PMD type. When read as a zero, bit 1.13.13 indicates that the PMA/PMD is not able to operate as a 100GBASE-KR4 PMA/PMD type.

#### **45.2.1.12.4 100GBASE-KP4 ability (1.13.12)**

When read as a one, bit 1.13.12 indicates that the PMA/PMD is able to operate as a 100GBASE-KP4 PMA/PMD type. When read as a zero, bit 1.13.12 indicates that the PMA/PMD is not able to operate as a 100GBASE-KP4 PMA/PMD type.

#### **45.2.1.12.5 100GBASE-ER4 ability (1.13.11)**

When read as a one, bit 1.13.11 indicates that the PMA/PMD is able to operate as a 100GBASE-ER4 PMA/PMD type. When read as a zero, bit 1.13.11 indicates that the PMA/PMD is not able to operate as a 100GBASE-ER4 PMA/PMD type.

#### **45.2.1.12.6 100GBASE-LR4 ability (1.13.10)**

When read as a one, bit 1.13.10 indicates that the PMA/PMD is able to operate as a 100GBASE-LR4 PMA/PMD type. When read as a zero, bit 1.13.10 indicates that the PMA/PMD is not able to operate as a 100GBASE-LR4 PMA/PMD type.

#### **45.2.1.12.7 100GBASE-SR10 ability (1.13.9)**

When read as a one, bit 1.13.9 indicates that the PMA/PMD is able to operate as a 100GBASE-SR10 PMA/PMD type. When read as a zero, bit 1.13.9 indicates that the PMA/PMD is not able to operate as a 100GBASE-SR10 PMA/PMD type.

#### **45.2.1.12.8 100GBASE-CR10 ability (1.13.8)**

When read as a one, bit 1.13.8 indicates that the PMA/PMD is able to operate as a 100GBASE-CR10 PMA/PMD type. When read as a zero, bit 1.13.8 indicates that the PMA/PMD is not able to operate as a 100GBASE-CR10 PMA/PMD type.

#### **45.2.1.12.9 100GBASE-SR4 ability (1.13.7)**

When read as a one, bit 1.13.7 indicates that the PMA/PMD is able to operate as a 100GBASE-SR4 PMA/PMD type. When read as a zero, bit 1.13.7 indicates that the PMA/PMD is not able to operate as a 100GBASE-SR4 PMA/PMD type.

#### **45.2.1.12.10 40GBASE-T ability (1.13.6)**

When read as a one, bit 1.13.6 indicates that the PMA/PMD is able to operate as a 40GBASE-T PMA type. When read as a zero, bit 1.13.6 indicates that the PMA/PMD is not able to operate as a 40GBASE-T PMA type.

#### **45.2.1.12.11 40GBASE-ER4 ability (1.13.5)**

When read as a one, bit 1.13.5 indicates that the PMA/PMD is able to operate as a 40GBASE-ER4 PMA/PMD type. When read as a zero, bit 1.13.5 indicates that the PMA/PMD is not able to operate as a 40GBASE-ER4 PMA/PMD type.

#### **45.2.1.12.12 40GBASE-FR ability (1.13.4)**

When read as a one, bit 1.13.4 indicates that the PMA/PMD is able to operate as a 40GBASE-FR PMA/PMD type. When read as a zero, bit 1.13.4 indicates that the PMA/PMD is not able to operate as a 40GBASE-FR PMA/PMD type.

#### 45.2.1.12.13 40GBASE-LR4 ability (1.13.3)

When read as a one, bit 1.13.3 indicates that the PMA/PMD is able to operate as a 40GBASE-LR4 PMA/PMD type. When read as a zero, bit 1.13.3 indicates that the PMA/PMD is not able to operate as a 40GBASE-LR4 PMA/PMD type.

#### 45.2.1.12.14 40GBASE-SR4 ability (1.13.2)

When read as a one, bit 1.13.2 indicates that the PMA/PMD is able to operate as a 40GBASE-SR4 PMA/PMD type. When read as a zero, bit 1.13.2 indicates that the PMA/PMD is not able to operate as a 40GBASE-SR4 PMA/PMD type.

#### 45.2.1.12.15 40GBASE-CR4 ability (1.13.1)

When read as a one, bit 1.13.1 indicates that the PMA/PMD is able to operate as a 40GBASE-CR4 PMA/PMD type. When read as a zero, bit 1.13.1 indicates that the PMA/PMD is not able to operate as a 40GBASE-CR4 PMA/PMD type.

#### 45.2.1.12.16 40GBASE-KR4 ability (1.13.0)

When read as a one, bit 1.13.0 indicates that the PMA/PMD is able to operate as a 40GBASE-KR4 PMA/PMD type. When read as a zero, bit 1.13.0 indicates that the PMA/PMD is not able to operate as a 40GBASE-KR4 PMA/PMD type.

#### 45.2.1.13 PMA/PMD package identifier (Registers 1.14 and 1.15)

Registers 1.14 and 1.15 provide a 32-bit value, which may constitute a unique identifier for a particular type of package that the PMA/PMD is instantiated within. The identifier shall be composed of the 3rd through 24th bits of the Organizationally Unique Identifier (OUI) assigned to the package manufacturer by the IEEE, plus a six-bit model number, plus a four-bit revision number. A PMA/PMD may return a value of zero in each of the 32 bits of the package identifier.

A non-zero package identifier may be returned by one or more MMDs in the same package. The package identifier may be the same as the device identifier.

The format of the package identifier is specified in 22.2.4.3.1.

#### 45.2.1.14 EEE capability (Register 1.16)

This register is used to indicate the capability of the PMA/PMD to support EEE functions for each PHY type. The assignment of bits in the EEE capability register is shown in Table 45–17.

**Table 45–17—EEE capability register bit definitions**

| Bit(s)     | Name                    | Description                                                                                              | R/W <sup>a</sup> |
|------------|-------------------------|----------------------------------------------------------------------------------------------------------|------------------|
| 1.16.15:12 | Reserved                | Value always 0                                                                                           | RO               |
| 1.16.11    | 100GBASE-CR4 deep sleep | 1 = EEE deep sleep is supported for 100GBASE-CR4<br>0 = EEE deep sleep is not supported for 100GBASE-CR4 | RO               |
| 1.16.10    | 100GBASE-KR4 deep sleep | 1 = EEE deep sleep is supported for 100GBASE-KR4<br>0 = EEE deep sleep is not supported for 100GBASE-KR4 | RO               |

**Table 45–17—EEE capability register bit definitions (*continued*)**

| Bit(s)   | Name                     | Description                                                                                                | R/W <sup>a</sup> |
|----------|--------------------------|------------------------------------------------------------------------------------------------------------|------------------|
| 1.16.9   | 100GBASE-KP4 deep sleep  | 1 = EEE deep sleep is supported for 100GBASE-KP4<br>0 = EEE deep sleep is not supported for 100GBASE-KP4   | RO               |
| 1.16.8   | 100GBASE-CR10 deep sleep | 1 = EEE deep sleep is supported for 100GBASE-CR10<br>0 = EEE deep sleep is not supported for 100GBASE-CR10 | RO               |
| 1.16.7:3 | Reserved                 | Value always 0                                                                                             | RO               |
| 1.16.2   | 25GBASE-R deep sleep     | 1 = EEE deep sleep is supported for 25GBASE-R<br>0 = EEE deep sleep is not supported for 25GBASE-R         | RO               |
| 1.16.1   | 40GBASE-CR4 deep sleep   | 1 = EEE deep sleep is supported for 40GBASE-CR4<br>0 = EEE deep sleep is not supported for 40GBASE-CR4     | RO               |
| 1.16.0   | 40GBASE-KR4 deep sleep   | 1 = EEE deep sleep is supported for 40GBASE-KR4<br>0 = EEE deep sleep is not supported for 40GBASE-KR4     | RO               |

<sup>a</sup> RO = Read only

#### **45.2.1.14.1 100GBASE-CR4 EEE deep sleep supported (1.16.11)**

If the device supports EEE deep sleep operation for 100GBASE-CR4 as defined in 92.1, this bit shall be set to a one; otherwise this bit shall be set to a zero.

#### **45.2.1.14.2 100GBASE-KR4 EEE deep sleep supported (1.16.10)**

If the device supports EEE deep sleep operation for 100GBASE-KR4 as defined in 93.1, this bit shall be set to a one; otherwise this bit shall be set to a zero.

#### **45.2.1.14.3 100GBASE-KP4 EEE deep sleep supported (1.16.9)**

If the device supports EEE deep sleep operation for 100GBASE-KP4 as defined in 94.1, this bit shall be set to a one; otherwise this bit shall be set to a zero.

#### **45.2.1.14.4 100GBASE-CR10 EEE deep sleep supported (1.16.8)**

If the device supports EEE deep sleep operation for 100GBASE-CR10 as defined in 85.1, this bit shall be set to a one; otherwise this bit shall be set to a zero.

#### **45.2.1.14.5 25GBASE-R deep sleep (1.16.2)**

If the device supports EEE deep sleep operation for 25GBASE-R, bit 1.16.2 shall be set to a one; otherwise this bit is set to a zero.

#### **45.2.1.14.6 40GBASE-CR4 EEE deep sleep supported (1.16.1)**

If the device supports EEE deep sleep operation for 40GBASE-CR4 as defined in 85.1, this bit shall be set to a one; otherwise this bit shall be set to a zero.

#### 45.2.1.14.7 40GBASE-KR4 EEE deep sleep supported (1.16.0)

If the device supports EEE deep sleep operation for 40GBASE-KR4 as defined in 84.1, this bit shall be set to a one; otherwise this bit shall be set to a zero.

#### 45.2.1.15 EPoC PMA/PMD ability register (Register 1.17)

The assignment of bits in the EPoC PMA/PMD ability register is shown in Table 45–18. All of the bits in the EPoC PMA/PMD ability register are read only; a write to the EPoC PMA/PMD ability register shall have no effect.

**Table 45–18—EPoC PMA/PMD ability register bit definitions**

| Bit(s)    | Name                 | Description                                                                                    | R/W <sup>a</sup> |
|-----------|----------------------|------------------------------------------------------------------------------------------------|------------------|
| 1.17.15:2 | Reserved             | Value always 0                                                                                 | RO               |
| 1.17.1    | 10GPASS-XR-D ability | 1 = PMA/PMD is able to perform 10GPASS-XR-D<br>0 = PMA/PMD is not able to perform 10GPASS-XR-D | RO               |
| 1.17.0    | 10GPASS-XR-U ability | 1 = PMA/PMD is able to perform 10GPASS-XR-U<br>0 = PMA/PMD is not able to perform 10GPASS-XR-U | RO               |

<sup>a</sup>RO = Read only

##### 45.2.1.15.1 10GPASS-XR-D ability (1.17.1)

When read as one, bit 1.17.1 indicates that the PMA/PMD is able to operate as a 10GPASS-XR-D PMA/PMD type. When read as zero, bit 1.17.1 indicates that the PMA/PMD is not able to operate as a 10GPASS-XR-D PMA/PMD type.

##### 45.2.1.15.2 10GPASS-XR-U ability (1.17.0)

When read as one, bit 1.17.0 indicates that the PMA/PMD is able to operate as a 10GPASS-XR-U PMA/PMD type. When read as zero, bit 1.17.0 indicates that the PMA/PMD is not able to operate as a 10GPASS-XR-U PMA/PMD type.

#### 45.2.1.16 BASE-T1 PMA/PMD extended ability register (1.18)

The assignment of bits in the BASE-T1 PMA/PMD extended ability register is shown in Table 45–19. All of the bits in the PMA/PMD extended ability register are read only; a write to the PMA/PMD extended ability register shall have no effect.

##### 45.2.1.16.1 10GBASE-T1 ability (1.18.6)

When read as a one, bit 1.18.6 indicates that the PMA/PMD is able to operate as a 10GBASE-T1 PMA type. When read as a zero, bit 1.18.6 indicates that the PMA/PMD is not able to operate as a 10GBASE-T1 PMA type.

##### 45.2.1.16.2 5GBASE-T1 ability (1.18.5)

When read as a one, bit 1.18.5 indicates that the PMA/PMD is able to operate as a 5GBASE-T1 PMA type. When read as a zero, bit 1.18.5 indicates that the PMA/PMD is not able to operate as a 5GBASE-T1 PMA type.

**Table 45–19—BASE-T1 PMA/PMD extended ability register bit definitions**

| Bit(s)    | Name                | Description                                                                                  | R/W <sup>a</sup> |
|-----------|---------------------|----------------------------------------------------------------------------------------------|------------------|
| 1.18.15:7 | Reserved            | Value always 0                                                                               | RO               |
| 1.18.6    | 10GBASE-T1 ability  | 1 = PMA/PMD is able to perform 10GBASE-T1<br>0 = PMA/PMD is not able to perform 10GBASE-T1   | RO               |
| 1.18.5    | 5GBASE-T1 ability   | 1 = PMA/PMD is able to perform 5GBASE-T1<br>0 = PMA/PMD is not able to perform 5GBASE-T1     | RO               |
| 1.18.4    | 2.5GBASE-T1 ability | 1 = PMA/PMD is able to perform 2.5GBASE-T1<br>0 = PMA/PMD is not able to perform 2.5GBASE-T1 | RO               |
| 1.18.3    | 10BASE-T1S ability  | 1 = PMA/PMD is able to perform 10BASE-T1S<br>0 = PMA/PMD is not able to perform 10BASE-T1S   | RO               |
| 1.18.2    | 10BASE-T1L ability  | 1 = PMA/PMD is able to perform 10BASE-T1L<br>0 = PMA/PMD is not able to perform 10BASE-T1L   | RO               |
| 1.18.1    | 1000BASE-T1 ability | 1 = PMA/PMD is able to perform 1000BASE-T1<br>0 = PMA/PMD is not able to perform 1000BASE-T1 | RO               |
| 1.18.0    | 100BASE-T1 ability  | 1 = PMA/PMD is able to perform 100BASE-T1<br>0 = PMA/PMD is not able to perform 100BASE-T1   | RO               |

<sup>a</sup>RO = Read only

#### **45.2.1.16.3 2.5GBASE-T1 ability (1.18.4)**

When read as a one, bit 1.18.4 indicates that the PMA/PMD is able to operate as a 2.5GBASE-T1 PMA type. When read as a zero, bit 1.18.4 indicates that the PMA/PMD is not able to operate as a 2.5GBASE-T1 PMA type.

#### **45.2.1.16.4 10BASE-T1S ability (1.18.3)**

When read as a one, bit 1.18.3 indicates that the PMA/PMD is able to operate as a 10BASE-T1S PMA type. When read as a zero, bit 1.18.3 indicates that the PMA/PMD is not able to operate as a 10BASE-T1S PMA type.

#### **45.2.1.16.5 10BASE-T1L ability (1.18.2)**

When read as a one, bit 1.18.2 indicates that the PMA/PMD is able to operate as a 10BASE-T1L PMA type. When read as a zero, bit 1.18.2 indicates that the PMA/PMD is not able to operate as a 10BASE-T1L PMA type.

#### **45.2.1.16.6 1000BASE-T1 ability (1.18.1)**

When read as a one, bit 1.18.1 indicates that the PMA/PMD is able to operate as a 1000BASE-T1 PMA type. When read as a zero, bit 1.18.1 indicates that the PMA/PMD is not able to operate as a 1000BASE-T1 PMA type.

#### 45.2.1.16.7 100BASE-T1 ability (1.18.0)

When read as a one, bit 1.18.0 indicates that the PMA/PMD is able to operate as a 100BASE-T1 PMA type. When read as a zero, bit 1.18.0 indicates that the PMA/PMD is not able to operate as a 100BASE-T1 PMA type.

#### 45.2.1.17 25G PMA/PMD extended ability register (Register 1.19)

The assignment of bits in the 25G PMA/PMD extended ability register is shown in Table 45–20.

**Table 45–20—25G PMA/PMD extended ability register bit definitions**

| Bit(s)    | Name                 | Description                                                                                    | R/W <sup>a</sup> |
|-----------|----------------------|------------------------------------------------------------------------------------------------|------------------|
| 1.19.15:8 | Reserved             | Value always 0                                                                                 | RO               |
| 1.19.7    | 25GBASE-ER ability   | 1 = PMA/PMD is able to perform 25GBASE-ER<br>0 = PMA/PMD is not able to perform 25GBASE-ER     | RO               |
| 1.19.6    | 25GBASE-LR ability   | 1 = PMA/PMD is able to perform 25GBASE-LR<br>0 = PMA/PMD is not able to perform 25GBASE-LR     | RO               |
| 1.19.5    | 25GBASE-T ability    | 1 = PMA/PMD is able to perform 25GBASE-T<br>0 = PMA/PMD is not able to perform 25GBASE-T       | RO               |
| 1.19.4    | 25GBASE-SR ability   | 1 = PMA/PMD is able to perform 25GBASE-SR<br>0 = PMA/PMD is not able to perform 25GBASE-SR     | RO               |
| 1.19.3    | 25GBASE-CR ability   | 1 = PMA/PMD is able to perform 25GBASE-CR<br>0 = PMA/PMD is not able to perform 25GBASE-CR     | RO               |
| 1.19.2    | 25GBASE-CR-S ability | 1 = PMA/PMD is able to perform 25GBASE-CR-S<br>0 = PMA/PMD is not able to perform 25GBASE-CR-S | RO               |
| 1.19.1    | 25GBASE-KR ability   | 1 = PMA/PMD is able to perform 25GBASE-KR<br>0 = PMA/PMD is not able to perform 25GBASE-KR     | RO               |
| 1.19.0    | 25GBASE-KR-S ability | 1 = PMA/PMD is able to perform 25GBASE-KR-S<br>0 = PMA/PMD is not able to perform 25GBASE-KR-S | RO               |

<sup>a</sup>RO = Read only

##### 45.2.1.17.1 25GBASE-ER ability (1.19.7)

When read as a one, bit 1.19.7 indicates that the PMA/PMD is able to operate as a 25GBASE-ER PMA/PMD type. When read as a zero, bit 1.19.7 indicates that the PMA/PMD is not able to operate as a 25GBASE-ER PMA/PMD type.

##### 45.2.1.17.2 25GBASE-LR ability (1.19.6)

When read as a one, bit 1.19.6 indicates that the PMA/PMD is able to operate as a 25GBASE-LR PMA/PMD type. When read as a zero, bit 1.19.6 indicates that the PMA/PMD is not able to operate as a 25GBASE-LR PMA/PMD type.



#### 45.2.1.17.3 25GBASE-T ability (1.19.5)

When read as a one, bit 1.19.5 indicates that the PMA/PMD is able to operate as a 25GBASE-T PMA type. When read as a zero, bit 1.19.5 indicates that the PMA/PMD is not able to operate as a 25GBASE-T PMA type.

#### 45.2.1.17.4 25GBASE-SR ability (1.19.4)

When read as a one, bit 1.19.4 indicates that the PMA/PMD is able to operate as a 25GBASE-SR PMA/PMD type. When read as a zero, bit 1.19.4 indicates that the PMA/PMD is not able to operate as a 25GBASE-SR PMA/PMD type.

#### 45.2.1.17.5 25GBASE-CR ability (1.19.3)

When read as a one, bit 1.19.3 indicates that the PMA/PMD is able to operate as a 25GBASE-CR PMA/PMD type. When read as a zero, bit 1.19.3 indicates that the PMA/PMD is not able to operate as a 25GBASE-CR PMA/PMD type.

#### 45.2.1.17.6 25GBASE-CR-S ability (1.19.2)

When read as a one, bit 1.19.2 indicates that the PMA/PMD is able to operate as a 25GBASE-CR-S PMA/PMD type. When read as a zero, bit 1.19.2 indicates that the PMA/PMD is not able to operate as a 25GBASE-CR-S PMA/PMD type.

#### 45.2.1.17.7 25GBASE-KR ability (1.19.1)

When read as a one, bit 1.19.1 indicates that the PMA/PMD is able to operate as a 25GBASE-KR PMA/PMD type. When read as a zero, bit 1.19.1 indicates that the PMA/PMD is not able to operate as a 25GBASE-KR PMA/PMD type.

#### 45.2.1.17.8 25GBASE-KR-S ability (1.19.0)

When read as a one, bit 1.19.0 indicates that the PMA/PMD is able to operate as a 25GBASE-KR-S PMA/PMD type. When read as a zero, bit 1.19.0 indicates that the PMA/PMD is not able to operate as a 25GBASE-KR-S PMA/PMD type.

#### 45.2.1.18 50G PMA/PMD extended ability (Register 1.20)

The assignment of bits in the 50G PMA/PMD extended ability register is shown in Table 45–21.

**Table 45–21—50G PMA/PMD extended ability register bit definitions**

| Bit(s)    | Name                            | Description                                                                                                                                      | R/W <sup>a</sup> |
|-----------|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.20.15   | 50G PMA remote loopback ability | 1 = 50G PMA has the ability to perform a remote loopback function<br>0 = 50G PMA does not have the ability to perform a remote loopback function | RO               |
| 1.20.14:6 | Reserved                        | Value always 0                                                                                                                                   | RO               |
| 1.20.5    | 50GBASE-ER ability              | 1 = PMA/PMD is able to perform 50GBASE-ER<br>0 = PMA/PMD is not able to perform 50GBASE-ER                                                       | RO               |

**Table 45–21—50G PMA/PMD extended ability register bit definitions (*continued*)**

| Bit(s) | Name               | Description                                                                                | R/W <sup>a</sup> |
|--------|--------------------|--------------------------------------------------------------------------------------------|------------------|
| 1.20.4 | 50GBASE-LR ability | 1 = PMA/PMD is able to perform 50GBASE-LR<br>0 = PMA/PMD is not able to perform 50GBASE-LR | RO               |
| 1.20.3 | 50GBASE-FR ability | 1 = PMA/PMD is able to perform 50GBASE-FR<br>0 = PMA/PMD is not able to perform 50GBASE-FR | RO               |
| 1.20.2 | 50GBASE-SR ability | 1 = PMA/PMD is able to perform 50GBASE-SR<br>0 = PMA/PMD is not able to perform 50GBASE-SR | RO               |
| 1.20.1 | 50GBASE-CR ability | 1 = PMA/PMD is able to perform 50GBASE-CR<br>0 = PMA/PMD is not able to perform 50GBASE-CR | RO               |
| 1.20.0 | 50GBASE-KR ability | 1 = PMA/PMD is able to perform 50GBASE-KR<br>0 = PMA/PMD is not able to perform 50GBASE-KR | RO               |

<sup>a</sup>RO = Read only

#### **45.2.1.18.1 50G PMA remote loopback ability (1.20.15)**

When read as a one, bit 1.20.15 indicates that the 50G PMA is able to perform the remote loopback function. When read as a zero, bit 1.20.15 indicates that the 50G PMA is not able to perform the remote loopback function. If a PMA is able to perform the remote loopback function, then it is controlled using the PMA remote loopback bit 1.0.1 (see 45.2.1.1.4).

#### **45.2.1.18.2 50GBASE-ER ability (1.20.5)**

When read as a one, bit 1.20.5 indicates that the PMA/PMD is able to operate as a 50GBASE-ER PMA/PMD type. When read as a zero, bit 1.20.5 indicates that the PMA/PMD is not able to operate as a 50GBASE-ER PMA/PMD type.

#### **45.2.1.18.3 50GBASE-LR ability (1.20.4)**

When read as a one, bit 1.20.4 indicates that the PMA/PMD is able to operate as a 50GBASE-LR PMA/PMD type. When read as a zero, bit 1.20.4 indicates that the PMA/PMD is not able to operate as a 50GBASE-LR PMA/PMD type.

#### **45.2.1.18.4 50GBASE-FR ability (1.20.3)**

When read as a one, bit 1.20.3 indicates that the PMA/PMD is able to operate as a 50GBASE-FR PMA/PMD type. When read as a zero, bit 1.20.3 indicates that the PMA/PMD is not able to operate as a 50GBASE-FR PMA/PMD type.

#### **45.2.1.18.5 50GBASE-SR ability (1.20.2)**

When read as a one, bit 1.20.2 indicates that the PMA/PMD is able to operate as a 50GBASE-SR PMA/PMD type. When read as a zero, bit 1.20.2 indicates that the PMA/PMD is not able to operate as a 50GBASE-SR PMA/PMD type.

#### 45.2.1.18.6 50GBASE-CR ability (1.20.1)

When read as a one, bit 1.20.1 indicates that the PMA/PMD is able to operate as a 50GBASE-CR PMA/PMD type. When read as a zero, bit 1.20.1 indicates that the PMA/PMD is not able to operate as a 50GBASE-CR PMA/PMD type.

#### 45.2.1.18.7 50GBASE-KR ability (1.20.0)

When read as a one, bit 1.20.0 indicates that the PMA/PMD is able to operate as a 50GBASE-KR PMA/PMD type. When read as a zero, bit 1.20.0 indicates that the PMA/PMD is not able to operate as a 50GBASE-KR PMA/PMD type.

#### 45.2.1.19 2.5G/5G PMA/PMD extended ability register (Register 1.21)

The assignment of bits in the 2.5G/5G PMA/PMD extended ability register is shown in Table 45–22. All of the bits in the PMA/PMD extended ability register are read only; a write to the PMA/PMD extended ability register shall have no effect.

**Table 45–22—2.5G/5G PMA/PMD extended ability register bit definitions**

| Bit       | Name                | Description                                                                                  | R/W <sup>a</sup> |
|-----------|---------------------|----------------------------------------------------------------------------------------------|------------------|
| 1.21.15:4 | Reserved            | Value always 0                                                                               | RO               |
| 1.21.3    | 5GBASE-KR ability   | 1 = PMA/PMD is able to perform 5GBASE-KR<br>0 = PMA/PMD is not able to perform 5GBASE-KR     | RO               |
| 1.21.2    | 2.5GBASE-KX ability | 1 = PMA/PMD is able to perform 2.5GBASE-KX<br>0 = PMA/PMD is not able to perform 2.5GBASE-KX | RO               |
| 1.21.1    | 5GBASE-T ability    | 1 = PMA/PMD is able to perform 5GBASE-T<br>0 = PMA/PMD is not able to perform 5GBASE-T       | RO               |
| 1.21.0    | 2.5GBASE-T ability  | 1 = PMA/PMD is able to perform 2.5GBASE-T<br>0 = PMA/PMD is not able to perform 2.5GBASE-T   | RO               |

<sup>a</sup>RO = Read only

NOTE—2.5GBASE-T1 and 5GBASE-T1 PMA/PMD extended abilities can be found in register 1.18, see Table 45–19.

##### 45.2.1.19.1 5GBASE-KR ability (1.21.3)

When read as a one, bit 1.21.3 indicates that the PMA/PMD is able to operate as a 5GBASE-KR PMA/PMD type. When read as a zero, bit 1.21.3 indicates that the PMA/PMD is not able to operate as a 5GBASE-KR PMA/PMD type.

##### 45.2.1.19.2 2.5GBASE-KX ability (1.21.2)

When read as a one, bit 1.21.2 indicates that the PMA/PMD is able to operate as a 2.5GBASE-KX PMA/PMD type. When read as a zero, bit 1.21.2 indicates that the PMA/PMD is not able to operate as a 2.5GBASE-KX PMA/PMD type.

#### 45.2.1.19.3 5GBASE-T ability (1.21.1)

When read as a one, bit 1.21.1 indicates that the PMA/PMD is able to operate as a 5GBASE-T PMA type.  
When read as a zero, bit 1.21.1 indicates that the PMA is not able to operate as a 5GBASE-T PMA type.

#### 45.2.1.19.4 2.5GBASE-T ability (1.21.0)

When read as a one, bit 1.21.0 indicates that the PMA/PMD is able to operate as a 2.5GBASE-T PMA type.  
When read as a zero, bit 1.21.0 indicates that the PMA/PMD is not able to operate as a 2.5GBASE-T PMA type.

#### 45.2.1.20 BASE-H PMA/PMD extended ability register (Register 1.22)

The assignment of bits in the BASE-H PMA/PMD extended ability register is shown in Table 45–23. All of the bits in the BASE-H PMA/PMD extended ability register are read only; a write to the BASE-H PMA/PMD extended ability register shall have no effect.

**Table 45–23—BASE-H PMA/PMD extended ability register bit definitions**

| Bit(s)    | Name                 | Description                                                                                    | R/W <sup>a</sup> |
|-----------|----------------------|------------------------------------------------------------------------------------------------|------------------|
| 1.22.15:3 | Reserved             | Value always 0                                                                                 | RO               |
| 1.22.2    | 1000BASE-RHC ability | 1 = PMA/PMD is able to perform 1000BASE-RHC<br>0 = PMA/PMD is not able to perform 1000BASE-RHC | RO               |
| 1.22.1    | 1000BASE-RHB ability | 1 = PMA/PMD is able to perform 1000BASE-RHB<br>0 = PMA/PMD is not able to perform 1000BASE-RHB | RO               |
| 1.22.0    | 1000BASE-RHA ability | 1 = PMA/PMD is able to perform 1000BASE-RHA<br>0 = PMA/PMD is not able to perform 1000BASE-RHA | RO               |

<sup>a</sup>RO = Read Only

#### 45.2.1.21 200G PMA/PMD extended ability register (Register 1.23)

The assignment of bits in the 200G PMA/PMD extended ability register is shown in Table 45–24.

**Table 45–24—200G PMA/PMD extended ability register bit definitions**

| Bit(s)    | Name                             | Description                                                                                                                                        | R/W <sup>a</sup> |
|-----------|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.23.15   | 200G PMA remote loopback ability | 1 = 200G PMA has the ability to perform a remote loopback function<br>0 = 200G PMA does not have the ability to perform a remote loopback function | RO               |
| 1.23.14:7 | Reserved                         | Value always 0                                                                                                                                     | RO               |
| 1.23.6    | 200GBASE-ER4 ability             | 1 = PMA/PMD is able to perform 200GBASE-ER4<br>0 = PMA/PMD is not able to perform 200GBASE-ER4                                                     | RO               |
| 1.23.5    | 200GBASE-LR4 ability             | 1 = PMA/PMD is able to perform 200GBASE-LR4<br>0 = PMA/PMD is not able to perform 200GBASE-LR4                                                     | RO               |
| 1.23.4    | 200GBASE-FR4 ability             | 1 = PMA/PMD is able to perform 200GBASE-FR4<br>0 = PMA/PMD is not able to perform 200GBASE-FR4                                                     | RO               |

**Table 45–24—200G PMA/PMD extended ability register bit definitions (*continued*)**

| Bit(s) | Name                 | Description                                                                                    | R/W <sup>a</sup> |
|--------|----------------------|------------------------------------------------------------------------------------------------|------------------|
| 1.23.3 | 200GBASE-DR4 ability | 1 = PMA/PMD is able to perform 200GBASE-DR4<br>0 = PMA/PMD is not able to perform 200GBASE-DR4 | RO               |
| 1.23.2 | 200GBASE-SR4 ability | 1 = PMA/PMD is able to perform 200GBASE-SR4<br>0 = PMA/PMD is not able to perform 200GBASE-SR4 | RO               |
| 1.23.1 | 200GBASE-CR4 ability | 1 = PMA/PMD is able to perform 200GBASE-CR4<br>0 = PMA/PMD is not able to perform 200GBASE-CR4 | RO               |
| 1.23.0 | 200GBASE-KR4 ability | 1 = PMA/PMD is able to perform 200GBASE-KR4<br>0 = PMA/PMD is not able to perform 200GBASE-KR4 | RO               |

<sup>a</sup>RO = Read only

#### **45.2.1.21.1 200G PMA remote loopback ability (1.23.15)**

When read as a one, bit 1.23.15 indicates that the 200G PMA is able to perform the remote loopback function. When read as a zero, bit 1.23.15 indicates that the 200G PMA is not able to perform the remote loopback function. If a PMA is able to perform the remote loopback function, then it is controlled using the PMA remote loopback bit 1.0.1 (see 45.2.1.1.4).

#### **45.2.1.21.2 200GBASE-ER4 ability (1.23.6)**

When read as a one, bit 1.23.6 indicates that the PMA/PMD is able to operate as a 200GBASE-ER4 PMA/PMD type. When read as a zero, bit 1.23.6 indicates that the PMA/PMD is not able to operate as a 200GBASE-ER4 PMA/PMD type.

#### **45.2.1.21.3 200GBASE-LR4 ability (1.23.5)**

When read as a one, bit 1.23.5 indicates that the PMA/PMD is able to operate as a 200GBASE-LR4 PMA/PMD type. When read as a zero, bit 1.23.5 indicates that the PMA/PMD is not able to operate as a 200GBASE-LR4 PMA/PMD type.

#### **45.2.1.21.4 200GBASE-FR4 ability (1.23.4)**

When read as a one, bit 1.23.4 indicates that the PMA/PMD is able to operate as a 200GBASE-FR4 PMA/PMD type. When read as a zero, bit 1.23.4 indicates that the PMA/PMD is not able to operate as a 200GBASE-FR4 PMA/PMD type.

#### **45.2.1.21.5 200GBASE-DR4 ability (1.23.3)**

When read as a one, bit 1.23.3 indicates that the PMA/PMD is able to operate as a 200GBASE-DR4 PMA/PMD type. When read as a zero, bit 1.23.3 indicates that the PMA/PMD is not able to operate as a 200GBASE-DR4 PMA/PMD type.

#### **45.2.1.21.6 200GBASE-SR4 ability (1.23.2)**

When read as a one, bit 1.23.2 indicates that the PMA/PMD is able to operate as a 200GBASE-SR4 PMA/PMD type. When read as a zero, bit 1.23.2 indicates that the PMA/PMD is not able to operate as a 200GBASE-SR4 PMA/PMD type.

#### 45.2.1.21.7 200GBASE-CR4 ability (1.23.1)

When read as a one, bit 1.23.1 indicates that the PMA/PMD is able to operate as a 200GBASE-CR4 PMA/PMD type. When read as a zero, bit 1.23.1 indicates that the PMA/PMD is not able to operate as a 200GBASE-CR4 PMA/PMD type.

#### 45.2.1.21.8 200GBASE-KR4 ability (1.23.0)

When read as a one, bit 1.23.0 indicates that the PMA/PMD is able to operate as a 200GBASE-KR4 PMA/PMD type. When read as a zero, bit 1.23.0 indicates that the PMA/PMD is not able to operate as a 200GBASE-KR4 PMA/PMD type.

#### 45.2.1.22 400G PMA/PMD extended ability register (Register 1.24)

The assignment of bits in the 400G PMA/PMD extended ability register is shown in Table 45–25.

**Table 45–25—400G PMA/PMD extended ability register bit definitions**

| Bit(s)     | Name                             | Description                                                                                                                                        | R/W <sup>a</sup> |
|------------|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.24.15    | 400G PMA remote loopback ability | 1 = 400G PMA has the ability to perform a remote loopback function<br>0 = 400G PMA does not have the ability to perform a remote loopback function | RO               |
| 1.24.14:11 | Reserved                         | Value always 0                                                                                                                                     | RO               |
| 1.24.10    | 400GBASE-ER8 ability             | 1 = PMA/PMD is able to perform 400GBASE-ER8<br>0 = PMA/PMD is not able to perform 400GBASE-ER8                                                     | RO               |
| 1.24.9     | 400GBASE-LR4-6 ability           | 1 = PMA/PMD is able to perform 400GBASE-LR4-6<br>0 = PMA/PMD is not able to perform 400GBASE-LR4-6                                                 | RO               |
| 1.24.8     | 400GBASE-FR4 ability             | 1 = PMA/PMD is able to perform 400GBASE-FR4<br>0 = PMA/PMD is not able to perform 400GBASE-FR4                                                     | RO               |
| 1.24.7     | 400GBASE-SR4.2 ability           | 1 = PMA/PMD is able to perform 400GBASE-SR4.2<br>0 = PMA/PMD is not able to perform 400GBASE-SR4.2                                                 | RO               |
| 1.24.6     | 400GBASE-SR8 ability             | 1 = PMA/PMD is able to perform 400GBASE-SR8<br>0 = PMA/PMD is not able to perform 400GBASE-SR8                                                     | RO               |
| 1.24.5     | 400GBASE-LR8 ability             | 1 = PMA/PMD is able to perform 400GBASE-LR8<br>0 = PMA/PMD is not able to perform 400GBASE-LR8                                                     | RO               |
| 1.24.4     | 400GBASE-FR8 ability             | 1 = PMA/PMD is able to perform 400GBASE-FR8<br>0 = PMA/PMD is not able to perform 400GBASE-FR8                                                     | RO               |
| 1.24.3     | 400GBASE-DR4 ability             | 1 = PMA/PMD is able to perform 400GBASE-DR4<br>0 = PMA/PMD is not able to perform 400GBASE-DR4                                                     | RO               |
| 1.24.2     | 400GBASE-SR16 ability            | 1 = PMA/PMD is able to perform 400GBASE-SR16<br>0 = PMA/PMD is not able to perform 400GBASE-SR16                                                   | RO               |
| 1.24.1:0   | Reserved                         | Value always 0                                                                                                                                     | RO               |

<sup>a</sup>RO = Read only

##### 45.2.1.22.1 400G PMA remote loopback ability (1.24.15)

When read as a one, bit 1.24.15 indicates that the 400G PMA is able to perform the remote loopback function. When read as a zero, bit 1.24.15 indicates that the 400G PMA is not able to perform the remote

loopback function. If a PMA is able to perform the remote loopback function, then it is controlled using the PMA remote loopback bit 1.0.1 (see 45.2.1.1.4).

#### **45.2.1.22.2 400GBASE-ER8 ability (1.24.10)**

When read as a one, bit 1.24.10 indicates that the PMA/PMD is able to operate as a 400GBASE-ER8 PMA/PMD type. When read as a zero, bit 1.24.10 indicates that the PMA/PMD is not able to operate as a 400GBASE-ER8 PMA/PMD type.

#### **45.2.1.22.3 400GBASE-LR4-6 ability (1.24.9)**

When read as a one, bit 1.24.9 indicates that the PMA/PMD is able to operate as a 400GBASE-LR4-6 PMA/PMD type. When read as a zero, bit 1.24.9 indicates that the PMA/PMD is not able to operate as a 400GBASE-LR4-6 PMA/PMD type.

#### **45.2.1.22.4 400GBASE-FR4 ability (1.24.8)**

When read as a one, bit 1.24.8 indicates that the PMA/PMD is able to operate as a 400GBASE-FR4 PMA/PMD type. When read as a zero, bit 1.24.8 indicates that the PMA/PMD is not able to operate as a 400GBASE-FR4 PMA/PMD type.

#### **45.2.1.22.5 400GBASE-SR4.2 ability (1.24.7)**

When read as a one, bit 1.24.7 indicates that the PMA/PMD is able to operate as a 400GBASE-SR4.2 PMA/PMD type. When read as a zero, bit 1.24.7 indicates that the PMA/PMD is not able to operate as a 400GBASE-SR4.2 PMA/PMD type.

#### **45.2.1.22.6 400GBASE-SR8 ability (1.24.6)**

When read as a one, bit 1.24.6 indicates that the PMA/PMD is able to operate as a 400GBASE-SR8 PMA/PMD type. When read as a zero, bit 1.24.6 indicates that the PMA/PMD is not able to operate as a 400GBASE-SR8 PMA/PMD type.

#### **45.2.1.22.7 400GBASE-LR8 ability (1.24.5)**

When read as a one, bit 1.24.5 indicates that the PMA/PMD is able to operate as a 400GBASE-LR8 PMA/PMD type. When read as a zero, bit 1.24.5 indicates that the PMA/PMD is not able to operate as a 400GBASE-LR8 PMA/PMD type.

#### **45.2.1.22.8 400GBASE-FR8 ability (1.24.4)**

When read as a one, bit 1.24.4 indicates that the PMA/PMD is able to operate as a 400GBASE-FR8 PMA/PMD type. When read as a zero, bit 1.24.4 indicates that the PMA/PMD is not able to operate as a 400GBASE-FR8 PMA/PMD type.

#### **45.2.1.22.9 400GBASE-DR4 ability (1.24.3)**

When read as a one, bit 1.24.3 indicates that the PMA/PMD is able to operate as a 400GBASE-DR4 PMA/PMD type. When read as a zero, bit 1.24.3 indicates that the PMA/PMD is not able to operate as a 400GBASE-DR4 PMA/PMD type.

#### 45.2.1.22.10 400GBASE-SR16 ability (1.24.2)

When read as a one, bit 1.24.2 indicates that the PMA/PMD is able to operate as a 400GBASE-SR16 PMA/PMD type. When read as a zero, bit 1.24.5 indicates that the PMA/PMD is not able to operate as a 400GBASE-SR16 PMA/PMD type.

#### 45.2.1.23 PMA/PMD extended ability 2 (Register 1.25)

The assignment of bits in the PMA/PMD extended ability 2 register is shown in Table 45–26.

**Table 45–26—PMA/PMD extended ability 2 register bit definitions**

| Bit(s)    | Name                   | Description                                                                                                        | R/W <sup>a</sup> |
|-----------|------------------------|--------------------------------------------------------------------------------------------------------------------|------------------|
| 1.25.15:1 | Reserved               | Value always 0                                                                                                     | RO               |
| 1.25.0    | 50G extended abilities | 1 = PMA/PMD has 50G extended abilities listed in register 1.20<br>0 = PMA/PMD does not have 50G extended abilities | RO               |

<sup>a</sup>RO = Read only

##### 45.2.1.23.1 50G extended abilities (1.25.0)

When read as a one, bit 1.25.0 indicates that the PMA/PMD has 50G extended abilities listed in register 1.20. When read as a zero, bit 1.25.0 indicates that the PMA/PMD does not have 50G extended abilities.

#### 45.2.1.24 40G/100G PMA/PMD extended ability 2 (Register 1.26)

The assignment of bits in the 40G/100G PMA/PMD extended ability 2 register is shown in Table 45–27.

**Table 45–27—40G/100G PMA/PMD extended ability 2 register bit definitions**

| Bit(s)     | Name                 | Description                                                                                    | R/W <sup>a</sup> |
|------------|----------------------|------------------------------------------------------------------------------------------------|------------------|
| 1.26.15:10 | Reserved             | Value always 0                                                                                 | RO               |
| 1.26.9     | 100GBASE-SR2 ability | 1 = PMA/PMD is able to perform 100GBASE-SR2<br>0 = PMA/PMD is not able to perform 100GBASE-SR2 | RO               |
| 1.26.8     | 100GBASE-CR2 ability | 1 = PMA/PMD is able to perform 100GBASE-CR2<br>0 = PMA/PMD is not able to perform 100GBASE-CR2 | RO               |
| 1.26.7     | 100GBASE-KR2 ability | 1 = PMA/PMD is able to perform 100GBASE-KR2<br>0 = PMA/PMD is not able to perform 100GBASE-KR2 | RO               |
| 1.26.6     | 100GBASE-ZR ability  | 1 = PMA/PMD is able to perform 100GBASE-ZR<br>0 = PMA/PMD is not able to perform 100GBASE-ZR   | RO               |
| 1.26.5     | 100GBASE-LR1 ability | 1 = PMA/PMD is able to perform 100GBASE-LR1<br>0 = PMA/PMD is not able to perform 100GBASE-LR1 | RO               |
| 1.26.4     | 100GBASE-FR1 ability | 1 = PMA/PMD is able to perform 100GBASE-FR1<br>0 = PMA/PMD is not able to perform 100GBASE-FR1 | RO               |



**Table 45–27—40G/100G PMA/PMD extended ability 2 register bit definitions (*continued*)**

| Bit(s)   | Name                | Description                                                                                  | R/W <sup>a</sup> |
|----------|---------------------|----------------------------------------------------------------------------------------------|------------------|
| 1.26.3   | 100GBASE-DR ability | 1 = PMA/PMD is able to perform 100GBASE-DR<br>0 = PMA/PMD is not able to perform 100GBASE-DR | RO               |
| 1.26.2:0 | Reserved            | Value always 0                                                                               | RO               |

<sup>a</sup> RO = Read only

#### **45.2.1.24.1 100GBASE-SR2 ability (1.26.9)**

When read as a one, bit 1.26.9 indicates that the PMA/PMD is able to operate as a 100GBASE-SR2 PMA/PMD type. When read as a zero, bit 1.26.9 indicates that the PMA/PMD is not able to operate as a 100GBASE-SR2 PMA/PMD type.

#### **45.2.1.24.2 100GBASE-CR2 ability (1.26.8)**

When read as a one, bit 1.26.8 indicates that the PMA/PMD is able to operate as a 100GBASE-CR2 PMA/PMD type. When read as a zero, bit 1.26.8 indicates that the PMA/PMD is not able to operate as a 100GBASE-CR2 PMA/PMD type.

#### **45.2.1.24.3 100GBASE-KR2 ability (1.26.7)**

When read as a one, bit 1.26.7 indicates that the PMA/PMD is able to operate as a 100GBASE-KR2 PMA/PMD type. When read as a zero, bit 1.26.7 indicates that the PMA/PMD is not able to operate as a 100GBASE-KR2 PMA/PMD type.

#### **45.2.1.24.4 100GBASE-ZR ability (1.26.6)**

When read as a one, bit 1.26.6 indicates that the PMA/PMD is able to operate as a 100GBASE-ZR PMA/PMD type. When read as a zero, bit 1.26.6 indicates that the PMA/PMD is not able to operate as a 100GBASE-ZR PMA/PMD type.

#### **45.2.1.24.5 100GBASE-LR1 ability (1.26.5)**

When read as a one, bit 1.26.5 indicates that the PMA/PMD is able to operate as a 100GBASE-LR1 PMA/PMD type. When read as a zero, bit 1.26.5 indicates that the PMA/PMD is not able to operate as a 100GBASE-LR1 PMA/PMD type.

#### **45.2.1.24.6 100GBASE-FR1 ability (1.26.4)**

When read as a one, bit 1.26.4 indicates that the PMA/PMD is able to operate as a 100GBASE-FR1 PMA/PMD type. When read as a zero, bit 1.26.4 indicates that the PMA/PMD is not able to operate as a 100GBASE-FR1 PMA/PMD type.

#### **45.2.1.24.7 100GBASE-DR ability (1.26.3)**

When read as a one, bit 1.26.3 indicates that the PMA/PMD is able to operate as a 100GBASE-DR PMA/PMD type. When read as a zero, bit 1.26.3 indicates that the PMA/PMD is not able to operate as a 100GBASE-DR PMA/PMD type.

#### 45.2.1.25 PMD transmit disable extension register (Register 1.27)

The assignment of bits in the PMD transmit disable extension register is shown in Table 45–28. The transmit disable functionality is optional and a PMD’s ability to perform the transmit disable functionality is advertised in the PMD transmit disable ability bit 1.8.8. A PMD that does not implement the transmit disable functionality shall ignore writes to the PMD transmit disable extension register and may return a value of zero for all bits. A PMD device that operates using a single lane and has implemented the transmit disable function shall use bit 1.9.0 to control the function. Such devices shall ignore writes to bit 1.27.0 and return a value of zero for that bit when read.

NOTE 1—This register is an extension of the PMD transmit disable register (see 45.2.1.8).

NOTE 2—Disabling the transmitter on one or more lanes stops the entire link from carrying data.

**Table 45–28—PMD transmit disable extension register bit definitions**

| Bit(s)    | Name                    | Description                                                                     | R/W <sup>a</sup> |
|-----------|-------------------------|---------------------------------------------------------------------------------|------------------|
| 1.27.15:1 | Reserved                | Value always 0                                                                  | RO               |
| 1.27.0    | PMD transmit disable 15 | 1 = Disable output on transmit lane 15<br>0 = Enable output on transmit lane 15 | R/W              |

<sup>a</sup>RO = Read only, R/W = Read/Write

##### 45.2.1.25.1 PMD transmit disable 15 (1.27.0)

When bit 1.27.0 is set to a one, the PMD shall disable output on lane 15 of the transmit path. When bit 1.27.0 is set to a zero, the PMD shall enable output on lane 15 of the transmit path.

The default value for bit 1.27.0 is zero.

NOTE—Transmission will not be enabled when this bit is set to a zero unless the global PMD transmit disable bit is also zero.

#### 45.2.1.26 PMD receive signal detect extension register (Register 1.28)

The assignment of bits in the PMD receive signal detect extension register is shown in Table 45–29. PMD types that use only a single lane indicate the status of the receive signal detect using bit 1.10.0 and return a value of zero for bit 1.28.0. PMD types that use multiple wavelengths or lanes indicate the status of each lane in bits 1.10.15:1 and 1.28.0 (if needed) and the logical AND of those bits in bit 1.10.0.

NOTE—This register is an extension of the PMD receive signal detect register (see 45.2.1.9).

**Table 45–29—PMD receive signal detect extension register bit definitions**

| Bit(s)    | Name                         | Description                                                                          | R/W <sup>a</sup> |
|-----------|------------------------------|--------------------------------------------------------------------------------------|------------------|
| 1.28.15:1 | Reserved                     | Value always 0                                                                       | RO               |
| 1.28.0    | PMD receive signal detect 15 | 1 = Signal detected on receive lane 15<br>0 = Signal not detected on receive lane 15 | RO               |

<sup>a</sup>RO = Read only

#### **45.2.1.26.1 PMD receive signal detect 15 (1.28.0)**

When bit 1.28.0 is read as a one, a signal has been detected on lane 15 of the PMD receive path. When bit 1.28.0 is read as a zero, a signal has not been detected on lane 15 of the PMD receive path.

#### **45.2.1.27 PMA/PMD control 3 register (Register 1.29)**

The assignment of bits in the PMA/PMD control 3 register is shown in Table 45–30.

##### **45.2.1.27.1 Downstream differential encoding (1.29.15)**

Downstream differential encoding is selected using bit 1.29.15. This bit is read/write in the OLT and read only in the ONU with the default value of 0 (indicating that downstream differential encoding is not enabled).

In the OLT, this bit controls whether downstream differential encoding is selected for the transmit PMA output.

In the ONU, this bit indicates whether the downstream differential decoding is enabled in the ONU receive PMA.

##### **45.2.1.27.2 PMA/PMD type selection (1.29.5:0)**

The PMA/PMD type of the PMA/PMD is selected using bits 5 to 0. The PMA/PMD type abilities of the PMA/PMD are advertised in the Nx25G-EPON PMA/PMD extended ability registers (registers 1.1000 through 1.1002, see 45.2.1.159). A PMA/PMD shall ignore writes to the PMA/PMD type selection bits that select PMA/PMD types it has not advertised. It is the responsibility of the STA entity to ensure that mutually acceptable MMD types are applied consistently across all the MMDs on a particular PHY.

#### **45.2.1.28 10P/2B PMA/PMD control register (Register 1.30)**

The assignment of bits in the 10P/2B PMA control register is shown in Table 45–31.

##### **45.2.1.28.1 PMA/PMD link control (1.30.15)**

The STA may enable the PMA/PMD link and initiate link initialization by writing this bit to a one. While link is initializing or up, this bit shall remain a one and writing a one to this bit shall be ignored. The STA may force the link down by writing a zero to this bit. While this bit is set to zero, the PHY shall not send G.994.1 handshake tones. For -O subtypes, upon link drop or MMD reset, the PMA/PMD shall set these bits to zero. For -R subtypes, upon link drop or MMD reset, the PMA/PMD shall set these bits to one.

##### **45.2.1.28.2 STFU (1.30.14)**

When this bit is set to a one, the PMA/PMD sends a message to the link partner instructing it to be silent for the silence time (see 45.2.1.28.3). Writing to this bit is valid only when the PMA/PMD link status bits in the PMA/PMD status register (see 45.2.1.29.4) are set to “link is down (ready)”. Writes are otherwise ignored.

This bit clears to zero when the silence command is sent, or upon the execution of an MMD reset.

##### **45.2.1.28.3 Silence time (1.30.13:8)**

The value of these bits sets the silence time conveyed in a STFU operation (see 45.2.1.28.2). The silence time is encoded according to Equation (45–1), where  $x$  is the decimal value of the bits:

**Table 45–30—PMA/PMD control 3 register bit definitions**

| Bit(s)    | Name                             | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | R/W <sup>a</sup>        |
|-----------|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| 1.29.15   | Downstream differential encoding | 1 = Downstream differential encoding enabled<br>0 = Downstream differential encoding disabled                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | R/W in OLT<br>RO in ONU |
| 1.29.14:6 | Reserved                         | Value always 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | RO                      |
| 1.29.5:0  | PMA/PMD type selection           | 5 4 3 2 1 0<br>1 1 x x x x = Reserved<br>1 0 1 x x x = Reserved<br>1 0 0 1 1 1 = 50GBASE-PQX-U3<br>1 0 0 1 1 0 = 50GBASE-PQX-U2<br>1 0 0 1 0 1 = 50GBASE-PQX-D3<br>1 0 0 1 0 0 = 50GBASE-PQX-D2<br>1 0 0 0 1 1 = 50GBASE-PQG-U3<br>1 0 0 0 1 0 = 50GBASE-PQG-U2<br>1 0 0 0 0 1 = 50GBASE-PQG-D3<br>1 0 0 0 0 0 = 50GBASE-PQG-D2<br>0 1 1 1 1 1 = 50/25GBASE-PQX-U3<br>0 1 1 1 1 0 = 50/25GBASE-PQX-U2<br>0 1 1 1 0 1 = 50/25GBASE-PQX-D3<br>0 1 1 1 0 0 = 50/25GBASE-PQX-D2<br>0 1 1 0 1 1 = 50/25GBASE-PQG-U3<br>0 1 1 0 1 0 = 50/25GBASE-PQG-U2<br>0 1 1 0 0 1 = 50/25GBASE-PQG-D3<br>0 1 1 0 0 0 = 50/25GBASE-PQG-D2<br>0 1 0 1 1 1 = 50/10GBASE-PQX-U3<br>0 1 0 1 1 0 = 50/10GBASE-PQX-U2<br>0 1 0 1 0 1 = 50/10GBASE-PQX-D3<br>0 1 0 1 0 0 = 50/10GBASE-PQX-D2<br>0 1 0 0 1 1 = 50/10GBASE-PQG-U3<br>0 1 0 0 1 0 = 50/10GBASE-PQG-U2<br>0 1 0 0 0 1 = 50/10GBASE-PQG-D3<br>0 1 0 0 0 0 = 50/10GBASE-PQG-D2<br>0 0 1 1 1 1 = 25GBASE-PQX-U3<br>0 0 1 1 1 0 = 25GBASE-PQX-U2<br>0 0 1 1 0 1 = 25GBASE-PQX-D3<br>0 0 1 1 0 0 = 25GBASE-PQX-D2<br>0 0 1 0 1 1 = 25GBASE-PQG-U3<br>0 0 1 0 1 0 = 25GBASE-PQG-U2<br>0 0 1 0 0 1 = 25GBASE-PQG-D3<br>0 0 1 0 0 0 = 25GBASE-PQG-D2<br>0 0 0 1 1 1 = 25/10GBASE-PQX-U3<br>0 0 0 1 1 0 = 25/10GBASE-PQX-U2<br>0 0 0 1 0 1 = 25/10GBASE-PQX-D3<br>0 0 0 1 0 0 = 25/10GBASE-PQX-D2<br>0 0 0 0 1 1 = 25/10GBASE-PQG-U3<br>0 0 0 0 1 0 = 25/10GBASE-PQG-U2<br>0 0 0 0 0 1 = 25/10GBASE-PQG-D3<br>0 0 0 0 0 0 = 25/10GBASE-PQG-D2 | R/W                     |

<sup>a</sup>R/W = Read/Write, RO = Read only

$$\text{time} = 10 \times (x + 1) \text{ seconds} \quad (45-1)$$

#### 45.2.1.28.4 Port subtype select (1.30.7)

This register bit selects the port subtype for PMA/PMD operation. The bit defaults to a supported mode. The PHY shall ignore writes that select an unsupported mode (see 45.2.1.28).

**Table 45–31—10P/2B PMA control register bit definitions**

| Bit(s)    | Name                      | Description                                                                                                                                                                                                                                                                              | R/W <sup>a</sup> |
|-----------|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.30.15   | PMA/PMD link control      | 1 = begin initialization, enable link (-R default)<br>0 = force link down (-O default)                                                                                                                                                                                                   | R/W              |
| 1.30.14   | STFU                      | Silence the far unit<br>1 = send silence command<br>0 = silence command inactive (default)                                                                                                                                                                                               | R/W              |
| 1.30.13:8 | Silence time              | Silence time = 10 × (value of bits + 1) seconds                                                                                                                                                                                                                                          | R/W              |
| 1.30.7    | Port subtype select       | 1 = port operates as an -O subtype<br>0 = port operates as a -R subtype                                                                                                                                                                                                                  | R/W              |
| 1.30.6    | Handshake cleardown       | 1 = send cleardown command<br>0 = idle (default)                                                                                                                                                                                                                                         | R/W, SC          |
| 1.30.5    | Ignore incoming handshake | 1 = PMA/PMD does not respond to handshake tones<br>0 = PMA/PMD responds to handshake tones                                                                                                                                                                                               | R/W              |
| 1.30.4:0  | PMA/PMD type selection    | 4 3 2 1 0<br>0 0 0 0 0 = 10PASS-TS PMA/PMD type<br>0 0 0 0 1 = 2BASE-TL PMA/PMD type<br>0 0 0 1 0 = 2BASE-TL or 10PASS-TS (-R only)<br>0 0 0 1 1 = 2BASE-TL preferred, or 10PASS-TS (-O only)<br>0 0 1 0 0 = 10PASS-TS preferred, or 2BASE-TL (-O only)<br>all other values are reserved | R/W              |

<sup>a</sup>R/W = Read/Write, SC = Self-clearing

Changing this bit alters the fundamental operation of the PMA/PMD, therefore, writes to change this bit shall be ignored if the link is up or initializing (see 45.2.1.29.4).

#### **45.2.1.28.5 Handshake cleardown (1.30.6)**

Setting this bit to a one shall cause the PMA/PMD to issue a G.994.1 cleardown command to the link partner (see 61.4.3). The PMA/PMD shall clear this bit to zero after the cleardown command has been sent or upon MMD reset. If the PMA/PMD link is not in the “link down (ready)” state (see 45.2.1.29.4), writes to this register shall be ignored.

#### **45.2.1.28.6 Ignore incoming handshake (1.30.5)**

When set to a one, the PMA/PMD shall not respond to received handshake tones (see 61.4.3). When set to a zero, the PMA/PMD shall respond to received handshake tones normally, according to 61.4.3 and G.994.1. Upon MMD reset, this bit shall be cleared to zero.

#### **45.2.1.28.7 PMA/PMD type selection (1.30.4:0)**

The PMA/PMD type of a 10P/2B PHY may be selected using bits 4 through 0. A PHY shall ignore writes to the type selection bits that select PMA/PMD types it has not advertised in the speed ability register. It is the responsibility of the STA entity to ensure that mutually acceptable port types are applied consistently across all the MMDs on a particular PHY.

A value of 00010 may be set in -R subtype PMA/PMDs that have both 2BASE-TL and 10PASS-TS capability set in the PMA/PMD speed ability register. The PMA/PMD type of the -R is set upon link initialization by the -O.

Values of 00011 and 00100 may be set in -O subtype PMA/PMDs that have both 2BASE-TL and 10PASS-TS capabilities set in the PMA/PMD speed ability register. These values indicate whether the -R is set to 10PASS-TS or 2BASE-TL respectively. If the -R is not capable of the “preferred” mode, the -R is set to 10PASS-TS or 2BASE-TL respectively.

The selection is advertised during link initialization G.994 handshake.

The PMA/PMD type selection defaults to a supported ability.

#### 45.2.1.29 10P/2B PMA/PMD status register (Register 1.31)

The assignment of bits in the 10P/2B PMA/PMD status register is shown in Table 45–32.

**Table 45–32—10P/2B PMA/PMD status register bit definitions**

| Bit(s)    | Name                | Description                                                                                                                                                                                             | R/W <sup>a</sup> |
|-----------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.31.15:5 | Data rate           | Current operating bit rate of the PMD<br>$n$ = the value of the bits<br>Data rate = $64n$ kb/s                                                                                                          | RO               |
| 1.31.4    | CO supported        | 1 = -O subtype operation supported<br>0 = -O subtype operation not supported                                                                                                                            | RO               |
| 1.31.3    | CPE supported       | 1 = -R subtype operation supported<br>0 = -R subtype operation not supported                                                                                                                            | RO               |
| 1.31.2:0  | PMA/PMD link status | 2 1 0<br>0 0 0 = link is down (not ready)<br>1 0 0 = link is down (ready)<br>0 0 1 = link is initializing<br>0 1 0 = link is up, 10PASS-TS<br>0 1 1 = link is up, 2BASE-TL<br>all other values reserved | RO               |

<sup>a</sup>RO = Read only

##### 45.2.1.29.1 Data rate (1.31.15:5)

These bits indicate the current bit rate of an operational PMA/PMD link. These bits shall be set to all zeros when the link is down or initializing.

##### 45.2.1.29.2 CO supported (1.31.4)

This bit indicates that the PMA/PMD supports operation as a -O subtype. This bit is set to a one when the capability is supported and zero otherwise. This bit reflects the signal PMA\_PMD\_type in 61.3.2.1.

##### 45.2.1.29.3 CPE supported (1.31.3)

This bit indicates that the PMA/PMD supports operation as a -R subtype. This bit is set to a one when the capability is supported and zero otherwise. This bit reflects the signal PMA\_PMD\_type in 61.3.2.1.

#### 45.2.1.29.4 PMA/PMD link status (1.31.2:0)

The overall state of the PMA/PMD link is reflected in bits 2:0. After the PMA/PMD is linked to the remote PHY, the PHY shall set these bits to indicate the PMA/PMD port type that is linked (010 for 10PASS-TS and 011 for 2BASE-TL). The corresponding signal, PMA\_received\_synchronized, is defined in 61.3.2.1.

While the link is initializing, these bits shall be set to 001.

When read as 000, these bits shall indicate that PMA/PMD link is down and the PMA/PMD is not detecting handshake tones from a link partner. This state is known as “not ready”.

When read as 100, these bits shall indicate that the PMA/PMD link is down and the PMA/PMD is detecting handshake tones from a link partner. This state is known as “ready”.

#### 45.2.1.30 Link partner PMA/PMD control register (Register 1.32)

The 10PASS-TS-O and 2BASE-TL-O PMA/PMDs allow access to certain register values of their link partner via the local MDIO interface. A summary of link partner parameters that may be sent or retrieved is provided in Table 45–33.

**Table 45–33—Link partner PMA/PMD registers and PMA/PMD register duals**

|                                             | Register type | Link partner register                | Local register counterpart |         |
|---------------------------------------------|---------------|--------------------------------------|----------------------------|---------|
|                                             | PHY subtype   | -O                                   | -O                         | -R      |
| Link partner register name                  |               | Address and access type <sup>a</sup> |                            |         |
| 10P/2B link partner RX SNR margin           |               | 1.38 RO                              | 1.37 RO                    | 1.37 RO |
| 10P/2B link partner line attenuation        |               | 1.40 RO                              | 1.39 RO                    | 1.39 RO |
| 10P/2B link partner line quality thresholds |               | 1.42 R/W                             | 1.41 R/W                   | n/a     |
| 10P link partner FEC correctable errors     |               | 1.45 RO                              | 1.43 RO                    | 1.43 RO |
| 10P link partner FEC uncorrectable errors   |               | 1.46 RO                              | 1.44 RO                    | 1.44 RO |
| 10P link partner electrical length          |               | 1.48 RO                              | 1.47 RO                    | 1.47 RO |
| 2B link partner code violation errors       |               | 1.90 RO                              | 1.89 RO                    | 1.89 RO |
| 2B link partner errored seconds             |               | 1.92 RO                              | 1.91 RO                    | 1.91 RO |
| 2B link partner severely errored seconds    |               | 1.94 RO                              | 1.93 RO                    | 1.93 RO |
| 2B link partner LOSW                        |               | 1.96 RO                              | 1.95 RO                    | 1.95 RO |
| 2B link partner unavailable seconds         |               | 1.98 RO                              | 1.97 RO                    | 1.97 RO |
| 2B link partner state defects register      |               | 1.100 RO                             | 1.99 RO                    | 1.99 RO |

<sup>a</sup>R/W = Read/Write, RO = read only, n/a = undefined

The Link partner PMA/PMD control register allows the -O STA to control the transmission and retrieval of parameters from its -R link partner.

The -R STA may read values exchanged by the -O STA in the local register counterpart to the link partner register. For example, the -O 10P link partner electrical length register will be populated with the contents of the -R 10P electrical length register upon a successful “Get link partner parameters” command. Similarly, the -R 10P/2B line quality thresholds register will contain the values sent by the -O in the 10P/2B link partner line quality thresholds register, after a successful “Send link partner parameters” command.

The link partner registers listed in Table 45–33 have the same behavior upon being read or reset as their local register counterparts.

This register is defined for -O port subtypes only.

Bit definitions for the Link partner PMA/PMD control register are found in Table 45–34.

**Table 45–34—Link partner PMA/PMD control register bit definitions**

| Bit(s)    | Name                         | Description                                                       | R/W <sup>a</sup> |
|-----------|------------------------------|-------------------------------------------------------------------|------------------|
| 1.32.15   | Get link partner parameters  | 1 = get link partner parameters<br>0 = operation complete, ready  | R/W, SC          |
| 1.32.14   | Reserved                     | Value always 0                                                    | RO               |
| 1.32.13   | Send link partner parameters | 1 = send link partner parameters<br>0 = operation complete, ready | R/W, SC          |
| 1.32.12:0 | Reserved                     | Value always 0                                                    | RO               |

<sup>a</sup>R/W = Read/Write, SC = Self-clearing, RO = read only

#### 45.2.1.30.1 Get link partner parameters (1.32.15)

When this bit is set to a one, the -O PHY updates its link partner registers shown in Table 45–33 with values from the link partner. While the operation is in progress, the PHY shall keep the bit set as one. If the “Get link partner parameters” operation does not complete within 10 seconds, its result shall be marked as “failed” (see 45.2.1.31) and the operation marked as “complete”. After completion of the operation or upon reset, the PHY shall reset the bit to zero. A write to this bit when link is down shall cause the result to be marked as “failed” and the operation marked as “complete”.

#### 45.2.1.30.2 Send link partner parameters (1.32.13)

When this bit is set to a one, the -O PHY sends the contents of the 2B link partner line quality thresholds register (see 45.2.1.39) to the link partner. While the operation is in progress, the PHY shall keep the bit set as one. The “Send link partner parameters” operation has to complete within 10 seconds, or its result shall be marked as “failed” (see 45.2.1.30) and the operation marked as “complete”. After completion of the operation or upon reset, the PHY shall reset the bit to zero. A write to this bit when link is down shall cause the result to be marked as “failed” and the operation marked as “complete”.

#### 45.2.1.31 Link partner PMA/PMD status register (Register 1.33)

The Link partner PMA/PMD status register reflects the result of the operations that are performed using the Link Partner PMA/PMD control register (1.32).

This register is defined for -O port subtypes only.

Bit definitions for the Link partner PMA/PMD status register are found in Table 45–35.



**Table 45–35—Link Partner PMA/PMD status register bit definitions**

| Bit(s)    | Name                     | Description                                      | R/W <sup>a</sup> |
|-----------|--------------------------|--------------------------------------------------|------------------|
| 1.33.15   | Reserved                 | Value always 0                                   | RO               |
| 1.33.14   | Get link partner result  | 1 = operation failed<br>0 = operation successful | RO, LH           |
| 1.33.13   | Reserved                 | Value always 0                                   | RO               |
| 1.33.12   | Send link partner result | 1 = operation failed<br>0 = operation successful | RO, LH           |
| 1.33.11:0 | Reserved                 | Value always 0                                   | RO               |

<sup>a</sup>RO = Read only, LH = Latching high

#### 45.2.1.31.1 Get link partner result (1.33.14)

After a “Get link partner parameters” operation terminates, this bit reflects the result of the operation. If the operation did not complete successfully, the PHY shall set this bit to a one. Upon being read or a reset, the PHY shall set the bit to zero.

The definition of an unsuccessful “Get link partner parameters” operation is unspecified and left to the implementation.

#### 45.2.1.31.2 Send link partner result (1.33.12)

After a “Send link partner parameters” operation terminates, this bit reflects the result of the operation. If the operation did not complete successfully, the PHY shall set this bit to a one. Upon being read or a reset, the PHY resets the bit to zero.

The definition of an unsuccessful “Send link partner parameters” operation is unspecified and left to the implementation.

#### 45.2.1.32 BiDi PMA/PMD extended ability 1 (Register 1.34)

The assignment of bits in the BiDi PMA/PMD extended ability 1 register is shown in Table 45–36.

**Table 45–36—BiDi PMA/PMD extended ability 1 register bit definitions**

| Bit(s)     | Name                   | Description                                                                                        | R/W <sup>a</sup> |
|------------|------------------------|----------------------------------------------------------------------------------------------------|------------------|
| 1.34.15:12 | Reserved               | Value always 0                                                                                     | RO               |
| 1.34.11    | 25GBASE-BR40-U ability | 1 = PMA/PMD is able to perform 25GBASE-BR40-U<br>0 = PMA/PMD is not able to perform 25GBASE-BR40-U | RO               |
| 1.34.10    | 25GBASE-BR40-D ability | 1 = PMA/PMD is able to perform 25GBASE-BR40-D<br>0 = PMA/PMD is not able to perform 25GBASE-BR40-D | RO               |
| 1.34.9     | 25GBASE-BR20-U ability | 1 = PMA/PMD is able to perform 25GBASE-BR20-U<br>0 = PMA/PMD is not able to perform 25GBASE-BR20-U | RO               |
| 1.34.8     | 25GBASE-BR20-D ability | 1 = PMA/PMD is able to perform 25GBASE-BR20-D<br>0 = PMA/PMD is not able to perform 25GBASE-BR20-D | RO               |

**Table 45–36—BiDi PMA/PMD extended ability 1 register bit definitions (*continued*)**

| Bit(s) | Name                   | Description                                                                                        | R/W <sup>a</sup> |
|--------|------------------------|----------------------------------------------------------------------------------------------------|------------------|
| 1.34.7 | 25GBASE-BR10-U ability | 1 = PMA/PMD is able to perform 25GBASE-BR10-U<br>0 = PMA/PMD is not able to perform 25GBASE-BR10-U | RO               |
| 1.34.6 | 25GBASE-BR10-D ability | 1 = PMA/PMD is able to perform 25GBASE-BR10-D<br>0 = PMA/PMD is not able to perform 25GBASE-BR10-D | RO               |
| 1.34.5 | 10GBASE-BR40-U ability | 1 = PMA/PMD is able to perform 10GBASE-BR40-U<br>0 = PMA/PMD is not able to perform 10GBASE-BR40-U | RO               |
| 1.34.4 | 10GBASE-BR40-D ability | 1 = PMA/PMD is able to perform 10GBASE-BR40-D<br>0 = PMA/PMD is not able to perform 10GBASE-BR40-D | RO               |
| 1.34.3 | 10GBASE-BR20-U ability | 1 = PMA/PMD is able to perform 10GBASE-BR20-U<br>0 = PMA/PMD is not able to perform 10GBASE-BR20-U | RO               |
| 1.34.2 | 10GBASE-BR20-D ability | 1 = PMA/PMD is able to perform 10GBASE-BR20-D<br>0 = PMA/PMD is not able to perform 10GBASE-BR20-D | RO               |
| 1.34.1 | 10GBASE-BR10-U ability | 1 = PMA/PMD is able to perform 10GBASE-BR10-U<br>0 = PMA/PMD is not able to perform 10GBASE-BR10-U | RO               |
| 1.34.0 | 10GBASE-BR10-D ability | 1 = PMA/PMD is able to perform 10GBASE-BR10-D<br>0 = PMA/PMD is not able to perform 10GBASE-BR10-D | RO               |

<sup>a</sup>RO = Read only

#### **45.2.1.32.1 25GBASE-BR40-U ability (1.34.11)**

When read as a one, bit 1.34.11 indicates that the PMA/PMD is able to operate as a 25GBASE-BR40-U PMA/PMD type. When read as a zero, bit 1.34.11 indicates that the PMA/PMD is not able to operate as a 25GBASE-BR40-U PMA/PMD type.

#### **45.2.1.32.2 25GBASE-BR40-D ability (1.34.10)**

When read as a one, bit 1.34.10 indicates that the PMA/PMD is able to operate as a 25GBASE-BR40-D PMA/PMD type. When read as a zero, bit 1.34.10 indicates that the PMA/PMD is not able to operate as a 25GBASE-BR40-D PMA/PMD type.

#### **45.2.1.32.3 25GBASE-BR20-U ability (1.34.9)**

When read as a one, bit 1.34.9 indicates that the PMA/PMD is able to operate as a 25GBASE-BR20-U PMA/PMD type. When read as a zero, bit 1.34.9 indicates that the PMA/PMD is not able to operate as a 25GBASE-BR20-U PMA/PMD type.

#### **45.2.1.32.4 25GBASE-BR20-D ability (1.34.8)**

When read as a one, bit 1.34.8 indicates that the PMA/PMD is able to operate as a 25GBASE-BR20-D PMA/PMD type. When read as a zero, bit 1.34.8 indicates that the PMA/PMD is not able to operate as a 25GBASE-BR20-D PMA/PMD type.

#### **45.2.1.32.5 25GBASE-BR10-U ability (1.34.7)**

When read as a one, bit 1.34.7 indicates that the PMA/PMD is able to operate as a 25GBASE-BR10-U PMA/PMD type. When read as a zero, bit 1.34.7 indicates that the PMA/PMD is not able to operate as a 25GBASE-BR10-U PMA/PMD type.

#### **45.2.1.32.6 25GBASE-BR10-D ability (1.34.6)**

When read as a one, bit 1.34.6 indicates that the PMA/PMD is able to operate as a 25GBASE-BR10-D PMA/PMD type. When read as a zero, bit 1.34.6 indicates that the PMA/PMD is not able to operate as a 25GBASE-BR10-D PMA/PMD type.

#### **45.2.1.32.7 10GBASE-BR40-U ability (1.34.5)**

When read as a one, bit 1.34.5 indicates that the PMA/PMD is able to operate as a 10GBASE-BR40-U PMA/PMD type. When read as a zero, bit 1.34.5 indicates that the PMA/PMD is not able to operate as a 10GBASE-BR40-U PMA/PMD type.

#### **45.2.1.32.8 10GBASE-BR40-D ability (1.34.4)**

When read as a one, bit 1.34.4 indicates that the PMA/PMD is able to operate as a 10GBASE-BR40-D PMA/PMD type. When read as a zero, bit 1.34.4 indicates that the PMA/PMD is not able to operate as a 10GBASE-BR40-D PMA/PMD type.

#### **45.2.1.32.9 10GBASE-BR20-U ability (1.34.3)**

When read as a one, bit 1.34.3 indicates that the PMA/PMD is able to operate as a 10GBASE-BR20-U PMA/PMD type. When read as a zero, bit 1.34.3 indicates that the PMA/PMD is not able to operate as a 10GBASE-BR20-U PMA/PMD type.

#### **45.2.1.32.10 10GBASE-BR20-D ability (1.34.2)**

When read as a one, bit 1.34.2 indicates that the PMA/PMD is able to operate as a 10GBASE-BR20-D PMA/PMD type. When read as a zero, bit 1.34.2 indicates that the PMA/PMD is not able to operate as a 10GBASE-BR20-D PMA/PMD type.

#### **45.2.1.32.11 10GBASE-BR10-U ability (1.34.1)**

When read as a one, bit 1.34.1 indicates that the PMA/PMD is able to operate as a 10GBASE-BR10-U PMA/PMD type. When read as a zero, bit 1.34.1 indicates that the PMA/PMD is not able to operate as a 10GBASE-BR10-U PMA/PMD type.

#### **45.2.1.32.12 10GBASE-BR10-D ability (1.34.0)**

When read as a one, bit 1.34.0 indicates that the PMA/PMD is able to operate as a 10GBASE-BR10-D PMA/PMD type. When read as a zero, bit 1.34.0 indicates that the PMA/PMD is not able to operate as a 10GBASE-BR10-D PMA/PMD type.

#### 45.2.1.33 BiDi PMA/PMD extended ability 2 (Register 1.35)

The assignment of bits in the BiDi PMA/PMD extended ability 2 register is shown in Table 45–37.

**Table 45–37—BiDi PMA/PMD extended ability 2 register bit definitions**

| Bit(s)    | Name                   | Description                                                                                        | R/W <sup>a</sup> |
|-----------|------------------------|----------------------------------------------------------------------------------------------------|------------------|
| 1.35.15:6 | Reserved               | Value always 0                                                                                     | RO               |
| 1.35.5    | 50GBASE-BR40-U ability | 1 = PMA/PMD is able to perform 50GBASE-BR40-U<br>0 = PMA/PMD is not able to perform 50GBASE-BR40-U | RO               |
| 1.35.4    | 50GBASE-BR40-D ability | 1 = PMA/PMD is able to perform 50GBASE-BR40-D<br>0 = PMA/PMD is not able to perform 50GBASE-BR40-D | RO               |
| 1.35.3    | 50GBASE-BR20-U ability | 1 = PMA/PMD is able to perform 50GBASE-BR20-U<br>0 = PMA/PMD is not able to perform 50GBASE-BR20-U | RO               |
| 1.35.2    | 50GBASE-BR20-D ability | 1 = PMA/PMD is able to perform 50GBASE-BR20-D<br>0 = PMA/PMD is not able to perform 50GBASE-BR20-D | RO               |
| 1.35.1    | 50GBASE-BR10-U ability | 1 = PMA/PMD is able to perform 50GBASE-BR10-U<br>0 = PMA/PMD is not able to perform 50GBASE-BR10-U | RO               |
| 1.35.0    | 50GBASE-BR10-D ability | 1 = PMA/PMD is able to perform 50GBASE-BR10-D<br>0 = PMA/PMD is not able to perform 50GBASE-BR10-D | RO               |

<sup>a</sup>RO = Read only

##### 45.2.1.33.1 50GBASE-BR40-U ability (1.35.5)

When read as a one, bit 1.35.5 indicates that the PMA/PMD is able to operate as a 50GBASE-BR40-U PMA/PMD type. When read as a zero, bit 1.35.5 indicates that the PMA/PMD is not able to operate as a 50GBASE-BR40-U PMA/PMD type.

##### 45.2.1.33.2 50GBASE-BR40-D ability (1.35.4)

When read as a one, bit 1.35.4 indicates that the PMA/PMD is able to operate as a 50GBASE-BR40-D PMA/PMD type. When read as a zero, bit 1.35.4 indicates that the PMA/PMD is not able to operate as a 50GBASE-BR40-D PMA/PMD type.

##### 45.2.1.33.3 50GBASE-BR20-U ability (1.35.3)

When read as a one, bit 1.35.3 indicates that the PMA/PMD is able to operate as a 50GBASE-BR20-U PMA/PMD type. When read as a zero, bit 1.35.3 indicates that the PMA/PMD is not able to operate as a 50GBASE-BR20-U PMA/PMD type.

##### 45.2.1.33.4 50GBASE-BR20-D ability (1.35.2)

When read as a one, bit 1.35.2 indicates that the PMA/PMD is able to operate as a 50GBASE-BR20-D PMA/PMD type. When read as a zero, bit 1.35.2 indicates that the PMA/PMD is not able to operate as a 50GBASE-BR20-D PMA/PMD type.

#### 45.2.1.33.5 50GBASE-BR10-U ability (1.35.1)

When read as a one, bit 1.35.1 indicates that the PMA/PMD is able to operate as a 50GBASE-BR10-U PMA/PMD type. When read as a zero, bit 1.35.1 indicates that the PMA/PMD is not able to operate as a 50GBASE-BR10-U PMA/PMD type.

#### 45.2.1.33.6 50GBASE-BR10-D ability (1.35.0)

When read as a one, bit 1.35.0 indicates that the PMA/PMD is able to operate as a 50GBASE-BR10-D PMA/PMD type. When read as a zero, bit 1.35.0 indicates that the PMA/PMD is not able to operate as a 50GBASE-BR10-D PMA/PMD type.

#### 45.2.1.34 10P/2B PMA/PMD link loss register (Register 1.36)

The 10P/2B PMA/PMD link loss register is a 16 bit counter that contains the number of times the PMA/PMD link is lost. Link is considered lost when the PMA\_receive\_synchronized signal (see 61.3.2.1) transitions from up to down. These bits shall be reset to all zeros when the register is read by the management function or upon execution of the MMD reset. These bits shall be held at all ones in the case of overflow. The assignment of bits in the 10P/2B PMA/PMD link loss register is shown in Table 45–38.

**Table 45–38—10P/2B PMA/PMD link loss register bit definitions**

| Bit(s)    | Name             | Description              | R/W <sup>a</sup> |
|-----------|------------------|--------------------------|------------------|
| 1.36.15:0 | Link lost events | The bytes of the counter | RO, NR           |

<sup>a</sup>RO = Read only, NR = Non Roll-over

#### 45.2.1.35 10P/2B RX SNR margin register (Register 1.37)

For further information on 2BASE-TL SNR margin, see 63.3. For 10PASS-TS SNR margin, see 62.3.

The bit definitions for the 10P/2B RX SNR margin register are found in Table 45–39.

**Table 45–39—10P/2B RX SNR margin register bit definition**

| Bit(s)    | Name          | Description               | R/W <sup>a</sup> |
|-----------|---------------|---------------------------|------------------|
| 1.37.15:8 | Reserved      | Value always 0            | RO               |
| 1.37.7:0  | RX SNR margin | Value of SNR margin in dB | RO               |

<sup>a</sup>RO = Read only

#### 45.2.1.36 10P/2B link partner RX SNR margin register (Register 1.38)

The 10P/2B link partner RX SNR margin register provides access to the link partner’s receive SNR margin. The contents of this register are refreshed when the STA activates the “Get link partner parameter” command in the Link partner PMA/PMD control register (see 45.2.1.30) and the command completes successfully (see 45.2.1.31.1).

This register is defined for -O port subtypes only.

The bit definitions for this register are found in Table 45–39.

#### 45.2.1.37 10P/2B line attenuation register (Register 1.39)

This register reports the line attenuation as measured by the PMA/PMD. For more information, see the reference documents in 63.3 and 62.3.

The bit definitions for the 10P/2B line attenuation register are found in Table 45–40.

**Table 45–40—10P/2B line attenuation register bit definitions**

| Bit(s)    | Name             | Description                                                             | R/W <sup>a</sup> |
|-----------|------------------|-------------------------------------------------------------------------|------------------|
| 1.39.15:0 | Line attenuation | The value of the line attenuation in dB, as perceived by the local PMD. | RO               |

<sup>a</sup>RO = Read only

#### 45.2.1.38 10P/2B link partner line attenuation register (Register 1.40)

The 10P/2B link partner line attenuation register provides access to the link partner’s perceived line attenuation margin. The contents of this register are refreshed when the STA activates the “Get link partner parameter” command in the Link partner PMA/PMD control register (see 45.2.1.30) and the command completes successfully (see 45.2.1.31.1).

This register is defined for -O port subtypes only.

The bit definitions for this register are found in Table 45–40.

#### 45.2.1.39 10P/2B line quality thresholds register (Register 1.41)

The 10P/2B line quality thresholds register sets the target environment for the 10PASS-TS/2BASE-TL connection. The line quality is defined by the SNR margin and line attenuation values.

Bit definitions for the 10P/2B line quality threshold register are found in Table 45–41.

**Table 45–41—10P/2B line quality thresholds register bit definition**

| Bit(s)    | Name                       | Description                 | R/W <sup>a</sup> |
|-----------|----------------------------|-----------------------------|------------------|
| 1.41.15:8 | Loop attenuation threshold | Attenuation threshold in dB | O: R/W<br>R: RO  |
| 1.41.7:4  | SNR margin threshold       | SNR margin threshold in dB  | O: R/W<br>R: RO  |
| 1.41.3:0  | Reserved                   | Value always 0              | RO               |

<sup>a</sup>R/W = Read/Write, RO = Read only

##### 45.2.1.39.1 Loop attenuation threshold (1.41.15:8)

These bits set the loop attenuation threshold for 2BASE-TL PMA/PMDs. Writing to these bits on a 10PASS-TS PMA/PMD shall have no effect. The threshold value is in units of dB. For more information on the loop attenuation threshold, see 63.2.2.3.

#### 45.2.1.39.2 SNR margin threshold (1.41.7:4)

These bits set the SNR margin threshold for 10PASS-TS and 2BASE-TL PMA/PMDs. The threshold is expressed in units of dB. For more information of the SNR margin threshold, see 63.2.2.3 for 2BASE-TL and Section 10 of the document referenced in 62.1.3 for 10PASS-TS.

#### 45.2.1.40 2B link partner line quality thresholds register (Register 1.42)

The 2B link partner line quality thresholds register allows the -O STA to set its -R link partner's line quality thresholds. The contents of this register are transmitted to the -R when the STA activates the "Send link partner parameter" command in the Link Partner PMA/PMD control register (see 45.2.1.30) and the command completes successfully (see 45.2.1.31.2).

This register is defined for -O port subtypes only.

The bit definitions for this register are found in Table 45–41.

#### 45.2.1.41 10P FEC correctable errors counter (Register 1.43)

The 10P FEC correctable errors counter is a 16 bit counter that contains the number of FEC codewords that have been received and corrected. For more information on 10PASS-TS FEC, see 62.2.4.2. These bits shall be reset to all zeros upon execution of the MMD reset and upon being read. The assignment of bits in the 10P FEC correctable error counter is shown in Table 45–42.

**Table 45–42—10P FEC correctable errors counter register bit definitions**

| Bit(s)    | Name                         | Description              | R/W <sup>a</sup> |
|-----------|------------------------------|--------------------------|------------------|
| 1.43.15:0 | Correctable codewords [15:0] | The bytes of the counter | RO               |

<sup>a</sup>RO = Read only

#### 45.2.1.42 10P FEC uncorrectable errors counter (Register 1.44)

The 10P FEC uncorrectable errors counter is a 16 bit counter that contains the number of FEC codewords that have been received and are uncorrectable. For more information on 10PASS-TS FEC, see 62.2.4.2. These bits shall be reset to all zeros upon execution of the MMD reset and upon being read. The assignment of bits in the 10P FEC uncorrectable error counter is shown in Table 45–43.

**Table 45–43—10P FEC uncorrectable errors counter bit definitions**

| Bit(s)    | Name                           | Description              | R/W <sup>a</sup> |
|-----------|--------------------------------|--------------------------|------------------|
| 1.44.15:0 | Uncorrectable codewords [15:0] | The bytes of the counter | RO               |

<sup>a</sup>RO = Read only

#### 45.2.1.43 10P link partner FEC correctable errors register (Register 1.45)

The 10P link partner FEC correctable errors register provides the -O STA with a snapshot of the -R link partner's FEC correctable errors counter. Because this register is not a counter, its value will increment only

when refreshed. The contents of this register are refreshed when the STA activates the “Get link partner parameter” command in the Link Partner PMA/PMD control register (see 45.2.1.30) and the command completes successfully (see 45.2.1.31.1).

This register is defined for -O port subtypes only.

The bit definitions for this register are found in Table 45–42 and 45.2.1.41.

#### 45.2.1.44 10P link partner FEC uncorrectable errors register (Register 1.46)

The 10P link partner FEC uncorrectable errors register provides the -O STA a snapshot of the -R link partner’s FEC uncorrectable errors counter. Because this register is not a counter, its value will increment only when refreshed. The contents of this register are refreshed when the STA activates the “Get link partner parameter” command in the Link Partner PMA/PMD control register (see 45.2.1.30) and the command completes successfully (see 45.2.1.31.1).

This register is defined for -O port subtypes only.

The bit definitions for this register are found in Table 45–43 and 45.2.1.42.

#### 45.2.1.45 10P electrical length register (Register 1.47)

The bit definitions for the 10P electrical length register are found in Table 45–44.

**Table 45–44—10P electrical length register bit definitions**

| Bit(s)    | Name              | Description                                                                    | R/W <sup>a</sup> |
|-----------|-------------------|--------------------------------------------------------------------------------|------------------|
| 1.47.15:0 | Electrical length | The electrical length of the medium (in meters), as perceived at the local PMD | RO               |

<sup>a</sup>RO = Read only

##### 45.2.1.45.1 Electrical length (1.47.15:0)

After the link is established, these bits contain the measured electrical length (in meters) of the medium as measured by the PMD. If the link is down or the PMD is unable to determine the electrical length, these bits shall be set to all ones (see 62.3.2).

#### 45.2.1.46 10P link partner electrical length register (Register 1.48)

The 10P link partner electrical length register provides access to the link partner’s electrical length measurement. The contents of this register are refreshed when the STA activates the “Get link partner parameter” command in the Link Partner PMA/PMD control register (see 45.2.1.30) and the command completes successfully (see 45.2.1.31.1).

This register is defined for -O port subtypes only.

The bit definitions for this register are found in Table 45–44 and 45.2.1.45.1.

#### 45.2.1.47 10P PMA/PMD general configuration register (Register 1.49)

The 10P PMA/PMD general configuration register is defined for -O port types only.



The 10P PMA/PMD general configuration register bit definitions are found in Table 45–45.

**Table 45–45—10P PMA/PMD general configuration register bit definitions**

| Bit(s)    | Name             | Description            | R/W <sup>a</sup>       |
|-----------|------------------|------------------------|------------------------|
| 1.49.15:8 | Reserved         | Value always 0         | RO                     |
| 1.49.7:0  | TX window length | Transmit window length | O: R/W<br>R: undefined |

<sup>a</sup>RO = Read only, R/W = Read/Write

#### 45.2.1.47.1 TX window length (1.49.7:0)

Bits 7:0 control the PMD transmit window length within the cyclic prefix and suffix in units of number of samples, as defined in 62.3.2.

#### 45.2.1.48 10P PSD configuration register (Register 1.50)

This register is defined for -O port subtypes only.

The 10P PSD configuration register bit definitions may be found in Table 45–46.

**Table 45–46—10P PSD configuration register bit definitions**

| Bit(s)    | Name        | Description                                  | R/W <sup>a</sup>       |
|-----------|-------------|----------------------------------------------|------------------------|
| 1.50.15:9 | Reserved    | Value always 0                               | RO                     |
| 1.50.8    | PBO disable | 1 = PBO disabled<br>0 = PBO normal operation | O: R/W<br>R: undefined |
| 1.50.7:0  | Reserved    | Value always 0                               | RO                     |

<sup>a</sup>RO = Read only, R/W = Read/Write

#### 45.2.1.48.1 PBO disable (1.50.8)

Setting this bit to a one disables UPBO for performance testing purposes. Refer to 62.3.4.4.

#### 45.2.1.49 10P downstream data rate configuration (Registers 1.51, 1.52)

These registers are defined for -O port subtypes only.

The bit definitions for the 10P downstream data rate configuration registers are found in Table 45–47.

**Table 45–47—10P downstream data rate configuration register bit definitions**

| Bit(s)    | Name                         | Description                                                                                                       | R/W <sup>a</sup>       |
|-----------|------------------------------|-------------------------------------------------------------------------------------------------------------------|------------------------|
| 1.51.15:0 | Minimum downstream data rate | Sets the minimum required downstream payload data rate<br>$M$ = value of bits<br>Data rate = $M \times 64000$ b/s | O: R/W<br>R: undefined |
| 1.52.15:0 | Maximum downstream data rate | Sets the maximum downstream payload data rate<br>$M$ = value of bits<br>Data rate = $M \times 64000$ b/s          | O: R/W<br>R: undefined |

<sup>a</sup>R/W = Read/Write

#### 45.2.1.50 10P downstream Reed-Solomon configuration (Register 1.53)

This register is defined for -O port subtypes only.

The bit definitions for 10P downstream Reed-Solomon configuration are found in Table 45–48.

**Table 45–48—10P downstream Reed-Solomon configuration register bit definitions**

| Bit(s)    | Name               | Description                                              | R/W <sup>a</sup>       |
|-----------|--------------------|----------------------------------------------------------|------------------------|
| 1.53.15:1 | Reserved           | Value always 0                                           | RO                     |
| 1.53.0    | RS codeword length | 1 = codeword length of 144<br>0 = codeword length of 240 | O: R/W<br>R: undefined |

<sup>a</sup>RO = Read only, R/W = Read/Write

##### 45.2.1.50.1 RS codeword length (1.53.0)

This bit selects the Reed-Solomon forward error correction codeword length used in the downstream direction. For more information, see 62.2.4.2.

#### 45.2.1.51 10P upstream data rate configuration (Registers 1.54, 1.55)

These registers are defined for -O port subtypes only.

The bit definitions for 10P upstream data rate configuration are found in Table 45–49.

**Table 45–49—10P upstream data rate configuration register bit definitions**

| Bit(s)    | Name                       | Description                                                                                             | R/W <sup>a</sup>       |
|-----------|----------------------------|---------------------------------------------------------------------------------------------------------|------------------------|
| 1.54.15:0 | Minimum upstream data rate | Sets the required upstream payload data rate<br>$M$ = value of bits<br>Data rate = $M \times 64000$ b/s | O: R/W<br>R: undefined |
| 1.55.15:0 | Maximum upstream data rate | Sets the maximum upstream payload data rate<br>$M$ = value of bits<br>Data rate = $M \times 64000$ b/s  | O: R/W<br>R: undefined |

<sup>a</sup>R/W = Read/Write

#### 45.2.1.52 10P upstream 10P upstream Reed-Solomon configuration register (Register 1.56)

This register is defined for -O port subtypes only.

The bit definitions for the 10P upstream Reed-Solomon configuration are found in Table 45–50.

**Table 45–50—10P upstream Reed-Solomon configuration register bit definitions**

| Bit(s)    | Name               | Description                                            | R/W <sup>a</sup>       |
|-----------|--------------------|--------------------------------------------------------|------------------------|
| 1.56.15:1 | Reserved           | Value always 0                                         | RO                     |
| 1.56.0    | RS codeword length | 1 = codeword length = 144<br>0 = codeword length = 240 | O: R/W<br>R: undefined |

<sup>a</sup>RO = Read only, R/W = Read/Write

##### 45.2.1.52.1 RS codeword length (1.56.0)

This bit selects the Reed-Solomon forward error correction codeword length used in the upstream direction. For more information, see 62.2.4.2.

#### 45.2.1.53 10P tone group registers (Registers 1.57, 1.58)

10PASS-TS operates by modulating 4096 individual tones across the transmission spectrum. Each tone can be assigned a PSD level, desired SNR margin and transmission direction (downstream or upstream). To reduce the complexity of addressing individual tones, tones are addressed by group. The STA sets the lower and upper tones in a group, sets the parameters for that group, and issues a command to activate those parameters for that group. See 62.3.4.7 for details on the mechanism that transfers tone information across the link to and from the -R link partner.

This register allows the STA to specify the range of tones to control. The bit definitions for the 10P tone group register are defined in Table 45–51.

**Table 45–51—10P tone group register bit definitions**

| Bit(s)    | Name       | Description                                                                                | R/W <sup>a</sup> |
|-----------|------------|--------------------------------------------------------------------------------------------|------------------|
| 1.57.15:0 | Lower tone | The number of the lower frequency tone in the group.<br>Valid when $\leq$ the Upper tone.  | R/W              |
| 1.58.15:0 | Upper tone | The number of the higher frequency tone in the group.<br>Valid when $\geq$ the Lower tone. | R/W              |

<sup>a</sup>R/W = Read/Write

#### 45.2.1.54 10P tone control parameters (Registers 1.59, 1.60, 1.61, 1.62, 1.63)

These registers allow the STA to specify parameters for the tones selected in the 10P tone group registers. These values do not take effect until the corresponding activation commands are issued in the 10P tone control action register. The bit definitions for the 10P tone control parameters are shown in Table 45–52.

These registers are defined for -O port subtypes only.

**Table 45–52—10P tone control parameters register bit definitions**

| Bit(s)    | Name              | Description                                                                                                                                                        | R/W <sup>a</sup> |
|-----------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.59.15   | Tone active       | 1 = selected tones are active<br>0 = selected tones are disabled                                                                                                   | R/W              |
| 1.59.14   | Tone direction    | 1 = selected tones assigned to upstream communication<br>0 = selected tones assigned to downstream communication                                                   | R/W              |
| 1.59.13:5 | Max SNR margin    | Assigns the maximum SNR margin the PMD may achieve<br>$M$ = value of bits<br>Max SNR margin = $M/4$ dB                                                             | R/W              |
| 1.59.4:0  | Reserved          | Value always 0                                                                                                                                                     | RO               |
| 1.60.15:9 | Reserved          | Value always 0                                                                                                                                                     | RO               |
| 1.60.8:0  | Target SNR margin | Assigns the target SNR margin for the selected tones<br>$M$ = value of bits<br>Target SNR margin = $M/4$ dB                                                        | R/W              |
| 1.61.15:9 | Reserved          | Value always 0                                                                                                                                                     | RO               |
| 1.61.8:0  | Min SNR margin    | Assigns the minimum SNR margin for the selected tones<br>$M$ = value of bits<br>Min SNR margin = $M/4$ dB                                                          | R/W              |
| 1.62.15:9 | Reserved          | Value always 0                                                                                                                                                     | RO               |
| 1.62.8:0  | PSD level         | Assigns a TX PSD level for the selected tones in dBm/Hz<br>$P$ = value of bits (2's complement)<br>PSD level = $P/4 - 100$ dBm/Hz                                  | R/W              |
| 1.63.15:9 | Reserved          | Value always 0                                                                                                                                                     | RO               |
| 1.63.8:0  | USPBO reference   | Assigns the level of the USPBO reference at the points represented by the selected tones<br>$P$ = value of bits (2's complement)<br>PSD level = $P/4 - 100$ dBm/Hz | R/W              |

<sup>a</sup>R/W = Read/Write, RO = Read only

#### 45.2.1.54.1 Tone active (1.59.15)

These bits are used to control the activity of the selected tones. When the “Change tone activity” command is issued (1.64.4), the selected tones will be either activated or deactivated based on the value set in these bits.

#### 45.2.1.54.2 Tone direction (1.59.14)

These bits are used to control the direction of the selected tones. When the “Change tone direction” command is issued (1.64.3), the selected tones will adopt the direction set in these bits.

#### 45.2.1.54.3 Max SNR margin (1.59.13:5)

These bits control the maximum SNR margin for the selected tones. When the “Change SNR margin” command is issued (1.64.2), the PMA/PMD will use the value set in these bits in calculations related to maximum SNR margin. The SNR margin is in units of dB, derived by dividing the value of bits 13:5 by 4.

#### 45.2.1.54.4 Target SNR margin (1.60.8:0)

These bits control the target SNR margin for the selected tones. When the “Change SNR margin” command is issued (1.64.2), the PMA/PMD will use the value set in these bits in calculations related to target SNR margin. The target SNR margin is in units of dB, derived by dividing the value of bits 8:0 by 4.

#### 45.2.1.54.5 Minimum SNR margin (1.61.8:0)

These bits control the minimum SNR margin for the selected tones. When the “Change SNR margin” command is issued (1.64.2), the PMA/PMD will use the value set in these bits in calculations related to minimum SNR margin. The minimum SNR margin is in units of dB, derived by dividing the value of bits 8:0 by 4.

#### 45.2.1.54.6 PSD level (1.62.8:0)

These bits control the transmit PSD level of the selected tones. When the “Change PSD level” command is issued (1.64.1), the PMA/PMD will set the PSD level of the selected tones to according to Equation (45–2), where  $x$  is the value of bits 8:0:

$$power = \frac{x}{4} - 100 \frac{\text{dBm}}{\text{Hz}} \quad (45-2)$$

#### 45.2.1.54.7 USPBO reference (1.63.8:0)

These bits control the reference level for the upstream power back-off function of the PMA/PMD. When the “Change USPBO reference PSD” command (1.64.0) is issued, the portion of the USPBO reference curve that corresponds to the selected tones is changed to the value specified by these bits. The USPBO reference level is specified according to Equation (45–3), where  $x$  is the value of bits 8:0:

$$power = \frac{x}{4} - 100 \frac{\text{dBm}}{\text{Hz}} \quad (45-3)$$

#### 45.2.1.55 10P tone control action register (Register 1.64)

The operations in this register apply to the tones selected in the 10P tone group registers (1.57, 1.58).

This register is defined for -O port subtypes only.

The bit definitions for the 10P tone control action register are shown in Table 45–53.

**Table 45–53—10P tone control action register bit definitions**

| Bit(s)    | Name                      | Description                                                                                                                 | R/W <sup>a</sup> |
|-----------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.64.15:6 | Reserved                  | Value always 0                                                                                                              | RO               |
| 1.64.5    | Refresh tone status       | 1 = refresh selected tones for the 10P tone status registers<br>0 = ready, operation complete                               | R/W, SC          |
| 1.64.4    | Change tone activity      | 1 = activate tone active setting as in tone control parameter register<br>0 = ready, operation complete                     | R/W, SC          |
| 1.64.3    | Change tone direction     | 1 = activate tone direction setting as in tone control parameter register<br>0 = ready, operation complete                  | R/W, SC          |
| 1.64.2    | Change SNR margin         | 1 = activate min, max and target SNR margin settings as in tone control parameter register<br>0 = ready, operation complete | R/W, SC          |
| 1.64.1    | Change PSD level          | 1 = activate PSD level setting as in tone control parameter register<br>0 = ready, operation complete                       | R/W, SC          |
| 1.64.0    | Change UPBO reference PSD | 1 = activate UPBO reference PSD settings as in tone control parameter register<br>0 = ready, operation complete             | R/W, SC          |

<sup>a</sup>RO = Read only, R/W = Read/Write, SC = Self-clearing

#### 45.2.1.55.1 Refresh tone status (1.64.5)

When this bit is set to a one, the tone status information from the local and link partner is gathered so that it may be read using the 10P tone status registers (1.65, 1.66, and 1.67). While the tones are being refreshed, this bit shall remain set as one. This bit shall be reset to zero when the refresh operation is over or upon reset.

NOTE—Refreshing a large number of tones may take a long time to complete.

#### 45.2.1.55.2 Change tone activity (1.64.4)

When this bit is set to a one, the selected tones are enabled or disabled according to the assignment in the tone active bit of the 10P tone control parameters register (1.59.15). While the tones are being activated/deactivated, this bit shall remain set as one. This bit shall be reset to zero when the operation is over or upon reset.

#### 45.2.1.55.3 Change tone direction (1.64.3)

When this bit is set to a one, the transmission direction of selected tones is changed according to the assignment in the tone direction bit of the 10P tone control parameters register (1.59.14). While the tones are being assigned, this bit shall remain set as one. This bit shall be reset to zero when the operation is over or upon reset.

#### 45.2.1.55.4 Change SNR margin (1.64.2)

When this bit is set to a one, the SNR margin parameters for the selected tones are loaded according the assignment in the Minimum, Target and Maximum SNR margin bits of the 10P tone control parameters register (1.59.13:5, 1.60.8:0, 1.61.8:0). While the parameters are being loaded, this bit shall remain set as one. This bit shall be reset to zero when the operation is over or upon reset.

#### 45.2.1.55.5 Change PSD level (1.64.1)

When this bit is set to a one, the PSD level for the selected tones is set according to the value in the PSD level bits of the 10P tone control parameters register (1.62.8:0). While the PSD is being set, this bit shall remain set as one. This bit shall be reset to zero when the operation is over or upon reset.

#### 45.2.1.55.6 Change USPBO reference PSD (1.64.0)

When this bit is set to a one, the upstream power back-off reference PSD level for the selected tones is set according to the value in the USPBO PSD reference bits of the 10P tone control parameters register (1.63.8:0). While the reference PSD is being set, this bit shall remain set as one. This bit shall be reset to zero when the operation is over or upon reset.

#### 45.2.1.56 10P tone status registers (Registers 1.65, 1.66, 1.67)

The 10P tone status registers allow the STA to query the status of any individual tone in the link. The values read in the 10P tone status register correspond to the tone whose number is set in the “Lower tone” of the 10P tone group registers (see 45.2.1.53).

The status of some tones is read from the link partner. Because the constant update of these values would be a strain on channel resources, these values are only updated for selected tones when the “Refresh tone table” command is issued in the 10P tone control action register (1.64).

The 10P tone status registers bit definitions are given in Table 45–54.

**Table 45–54—10P tone status registers bit definitions**

| Bit(s)     | Name           | Description                                                                                        | R/W <sup>a</sup> |
|------------|----------------|----------------------------------------------------------------------------------------------------|------------------|
| 1.65.15    | Refresh status | 1 = tone entry has been refreshed<br>0 = tone entry has not been refreshed since last read         | RO               |
| 1.65.14    | Active         | 1 = tone is disabled<br>0 = tone is active                                                         | RO               |
| 1.65.13    | Direction      | 1 = tone is assigned to upstream communication<br>0 = tone is assigned to downstream communication | RO               |
| 1.65.12:8  | Reserved       | Value always 0                                                                                     | RO               |
| 1.65.7:0   | RX PSD         | PSD of the tone at the receiver in dBm/Hz                                                          | RO               |
| 1.66.15:8  | TX PSD         | PSD of the tone at the transmitter in dBm/Hz                                                       | RO               |
| 1.66.7:3   | Bit load       | The number of bits currently loaded on the tone                                                    | RO               |
| 1.66.2:0   | Reserved       | Value always 0                                                                                     | RO               |
| 1.67.15:10 | Reserved       | Value always 0                                                                                     | RO               |
| 1.67.9:0   | SNR margin     | Current SNR margin for the tone<br>$R$ = value of bits<br>SNR Margin = $R/4$ dB                    | RO               |

<sup>a</sup>RO = Read only

#### 45.2.1.56.1 Refresh status (1.65.15)

When read as a one, bit 1.65.15 indicates that the values for this tone table have not been read since the last “Refresh tone status” command was issued by the 10P tone control action register (1.64). Upon reading this bit or upon reset, the bit shall be reset to zero.

#### 45.2.1.56.2 Active (1.65.14)

When read as a one, this bit indicates that the selected tone is disabled (i.e., powered off and not carrying data).

#### 45.2.1.56.3 Direction (1.65.13)

When read as a one, this bit indicates that the selected tone is assigned to upstream communication. When read as a zero, the tone is assigned to downstream communication.

#### 45.2.1.56.4 RX PSD (1.65.7:0)

These bits report the PSD of the selected tone as perceived at the receiver in units of dBm/Hz.

#### 45.2.1.56.5 TX PSD (1.66.15:8)

These bits report the PSD of the selected tone as output by the transmitter in units of dBm/Hz.

#### 45.2.1.56.6 Bit load (1.66.7:3)

These bits report the number of bits encoded on the selected tone.

#### 45.2.1.56.7 SNR margin (1.67.9:0)

These bits report the current SNR margin for the selected tone, as perceived by the receiver, in units of dB. The value of the SNR margin is obtained by dividing the decimal value of bits 9:0 by 4.

#### 45.2.1.57 10P outgoing indicator bits status register (Register 1.68)

The 10P outgoing indicator bits status register conveys the current state of the indicator bits being sent to the link partner (See 62.3.4.7). The bit definitions for the 10P indicator bits status register are shown in Table 45–55.

**Table 45–55—10P outgoing indicator bits status register bit definition**

| Bit(s)    | Name     | Description                                                        | R/W <sup>a</sup> |
|-----------|----------|--------------------------------------------------------------------|------------------|
| 1.68.15:9 | Reserved | Value always 0                                                     | RO               |
| 1.68.8    | LoM      | 1 = received signal below SNR margin threshold<br>0 = normal state | RO               |
| 1.68.7    | lpr      | 1 = power supply voltage invalid<br>0 = normal state               | RO               |
| 1.68.6    | po       | 1 = PMA/PMD is being powered off<br>0 = normal state               | RO               |
| 1.68.5    | Rdi      | 1 = severely errored frames have been received<br>0 = normal state | RO               |
| 1.68.4    | los      | 1 = signal power is lower than the threshold<br>0 = normal state   | RO               |



**Table 45–55—10P outgoing indicator bits status register bit definition (*continued*)**

| Bit(s)   | Name     | Description                                                                                        | R/W <sup>a</sup> |
|----------|----------|----------------------------------------------------------------------------------------------------|------------------|
| 1.68.3:2 | Reserved | Value always 0                                                                                     | RO               |
| 1.68.1   | fec-s    | 1 = corrected errors have been detected in the received FEC block of slow data<br>0 = normal state | RO               |
| 1.68.0   | be-s     | 1 = non-corrected errors have been detected in the received block of slow data<br>0 = normal state | RO               |

<sup>a</sup>RO = Read only

NOTE—These bit refer to “slow” data. 10PASS-TS uses the slow data channel as referenced in ATIS-0600424. The name is kept here to simplify a comparison between the two documents.

#### **45.2.1.57.1 LoM (1.68.8)**

When read as a one, this bit indicates that the PMA/PMD is receiving a signal whose SNR margin is below the set threshold (see 45.2.1.39). The status of this condition is reported to the link partner PMA/PMD via the 10PASS-TS indicator bits.

#### **45.2.1.57.2 lpr (1.68.7)**

When read as a one, this bit indicates that the PMA/PMD is not receiving sufficient power supply input for proper operation. The specific conditions that cause this bit to be set are implementation specific. The status of this condition is reported to the link partner PMA/PMD via the 10PASS-TS indicator bits.

#### **45.2.1.57.3 po (1.68.6)**

When read as a one, this bit indicates that the PMA/PMD has been instructed to power off. The specific conditions that cause this bit to be set are implementation specific. The status of this condition is reported to the link partner PMA/PMD via the 10PASS-TS indicator bits.

#### **45.2.1.57.4 Rdi (1.68.5)**

When read as a one, this bit indicates that the PMA/PMD has received PMA/PMD frames containing severe errors. The status of this condition is reported to the link partner PMA/PMD via the 10PASS-TS indicator bits.

#### **45.2.1.57.5 los (1.68.4)**

When read as a one, this bit indicates that the PMA/PMD is not receiving a valid signal. The status of this condition is reported to the link partner PMA/PMD via the 10PASS-TS indicator bits.

#### **45.2.1.57.6 fec-s (1.68.1)**

When read as a one, this bit indicates that the PMA/PMD is receiving FEC blocks with one or more correctable errors. The status of this condition is reported to the link partner PMA/PMD via the 10PASS-TS indicator bits.

#### 45.2.1.57.7 be-s (1.68.0)

When read as a one, this bit indicates that the PMA/PMD is receiving FEC blocks with one or more uncorrectable errors. The status of this condition is reported to the link partner PMA/PMD via the 10PASS-TS indicator bits.

#### 45.2.1.58 10P incoming indicator bits status register (Register 1.69)

The 10P indicator bits status register conveys the current state of the indicator bits being received from the link partner's PMA (see 62.3.4.7). The bit definitions for the 10P incoming indicator bits status register are shown in Table 45–56.

**Table 45–56—10P incoming indicator bits status register bit definition**

| Bit(s)    | Name     | Description                                                                                        | R/W <sup>a</sup> |
|-----------|----------|----------------------------------------------------------------------------------------------------|------------------|
| 1.69.15:9 | Reserved | Value always 0                                                                                     | RO               |
| 1.69.8    | LoM      | 1 = received signal below SNR margin threshold<br>0 = normal state                                 | RO               |
| 1.69.7    | Flpr     | 1 = power supply voltage invalid<br>0 = normal state                                               | RO               |
| 1.69.6    | Fpo      | 1 = PMA/PMD is being powered off<br>0 = normal state                                               | RO               |
| 1.69.5    | Rdi      | 1 = severely errored frames have been received<br>0 = normal state                                 | RO               |
| 1.69.4    | Flos     | 1 = signal power is lower than the threshold<br>0 = normal state                                   | RO               |
| 1.69.3:2  | Reserved | Value always 0                                                                                     | RO               |
| 1.69.1    | Ffec-s   | 1 = corrected errors have been detected in the received FEC block of slow data<br>0 = normal state | RO               |
| 1.69.0    | Febe-s   | 1 = non-corrected errors have been detected in the received block of slow data<br>0 = normal state | RO               |

<sup>a</sup>RO = Read only

NOTE—These bit refer to “slow” data. 10PASS-TS uses the slow data channel as referenced in ATIS-0600424. The name is kept here to simplify a comparison between the two documents.

##### 45.2.1.58.1 LoM (1.69.8)

When read as a one, this bit indicates that the link partner PMA/PMD is receiving a signal whose SNR margin is below the set threshold (see 45.2.1.39). The status of this condition is reported to the local PMA/PMD via the 10PASS-TS indicator bits.

##### 45.2.1.58.2 Flpr (1.69.7)

When read as a one, this bit indicates that the link partner PMA/PMD is not receiving sufficient power supply input for proper operation. The specific conditions that cause this bit to be set are implementation specific. The status of this condition is reported to the local PMA/PMD via the 10PASS-TS indicator bits.

#### 45.2.1.58.3 Fpo (1.69.6)

When read as a one, this bit indicates that the link partner PMA/PMD has been instructed to power off. The specific conditions that cause this bit to be set are implementation specific. The status of this condition is reported to the local PMA/PMD via the 10PASS-TS indicator bits.

#### 45.2.1.58.4 Rdi (1.69.5)

When read as a one, this bit indicates that the link partner PMA/PMD has received PMA/PMD frames containing severe errors. The status of this condition is reported to the local PMA/PMD via the 10PASS-TS indicator bits.

#### 45.2.1.58.5 Flos (1.69.4)

When read as a one, this bit indicates that the link partner PMA/PMD has is not receiving a valid signal. The status of this condition is reported to the local PMA/PMD via the 10PASS-TS indicator bits.

#### 45.2.1.58.6 Ffec-s (1.69.1)

When read as a one, this bit indicates that the link partner PMA/PMD is receiving FEC blocks with one or more correctable errors. The status of this condition is reported to the local PMA/PMD via the 10PASS-TS indicator bits.

#### 45.2.1.58.7 Febe-s (1.69.0)

When read as a one, this bit indicates that the link partner PMA/PMD is receiving FEC blocks with one or more uncorrectable errors. The status of this condition is reported to the local PMA/PMD via the 10PASS-TS indicator bits.

#### 45.2.1.59 10P cyclic extension configuration register (Register 1.70)

The 10P cyclic extension configuration register controls the length of the cyclic extension for the 10P PMD. For more information, see 62.3.4.2. The value of the cyclic extension is equal to the decimal value set in bits 15:0. Values of decimal 10, 20 and 40 are valid. Writes to set any other values shall be ignored. Upon reset, the PMD shall set these bits to a decimal value of 20.

The bit definitions for this register are shown in Table 45–57.

**Table 45–57—10P cyclic extension configuration register bit definitions**

| Bit(s)    | Name             | Description                       | R/W <sup>a</sup> |
|-----------|------------------|-----------------------------------|------------------|
| 1.70.15:0 | Cyclic extension | The value of the cyclic extension | O: R/W<br>R: RO  |

<sup>a</sup>R/W = Read/Write, RO = Read only

#### 45.2.1.60 10P attainable downstream data rate register (Register 1.71)

The 10P attainable downstream data rate register reports the data rate that the -R link partner measures to be the highest data rate for downstream transmission. The data rate is encoded as 1 kb/s times the decimal value of the register bits 15:0. The value of the register bits are not valid until after link is “up” (see 45.2.1.29.4).

The bit definitions for this register are found in Table 45–58.

**Table 45–58—10P attainable downstream data rate register bit definitions**

| Bit(s)    | Name                            | Description                    | R/W <sup>a</sup> |
|-----------|---------------------------------|--------------------------------|------------------|
| 1.71.15:0 | Attainable downstream data rate | Data rate in 1 kb/s increments | RO               |

<sup>a</sup>RO = Read only

#### 45.2.1.61 2B general parameter register (Register 1.80)

The 2B general parameter register controls various parameters for the operation of the 2BASE-TL PMA/PMD.

This register is read only for -R ports which may be read so the STA may know the mode selected by the -O port. The selected parameters on the -O are sent to the -R link partner on link initialization. For more information on how these parameters are passed across the physical link using G.994.1 signaling, see 61B.3.2.

The bit definitions for the 2B general parameter register are found in Table 45–59.

**Table 45–59— 2B general parameter register bit definition**

| Bit(s)     | Name                 | Description                                                                                                                  | R/W <sup>a</sup> |
|------------|----------------------|------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.80.15    | Reserved             | Value always 0                                                                                                               | RO               |
| 1.80.14:10 | PMMS target margin   | margin = 14:10 - 10dB                                                                                                        | R/W              |
| 1.80.9     | Line probing control | 1 = use line probing<br>0 = do not use line probing (default)                                                                | R/W              |
| 1.80.8     | Noise environment    | 1 = current condition<br>0 = worst case (default)                                                                            | R/W              |
| 1.80.7:2   | Reserved             | Value always 0                                                                                                               | RO               |
| 1.80.1:0   | Region               | Selects the regional annex to operate under<br>00 = Annex A<br>01 = Annex B<br>10 = Annex C<br>11 = reserved, writes ignored | O: R/W<br>R: RO  |

<sup>a</sup>R/W = Read/Write, RO = Read only

##### 45.2.1.61.1 PMMS target margin (1.80.14:10)

The PMMS target margin specified in bits 14:10 specifies the noise margin that the PMMS procedure tries to attain. The margin is expressed in dB as the decimal value of bits 14:10 minus 10 dB. The margin specified is measured either against either worst case or current line conditions, based on the value set in bit 1.80.8.

The PMMS margin value is transferred during 2BASE-TL initialization via the worst case PMMS margin bits in Table 61B-57 and Table 61B-43, or the current condition PMMS margin bits in Table 61B-48 and Table 61B-44.

#### **45.2.1.61.2 Line probing control (1.80.9)**

When set to a one, this bit tells the PMA/PMD to perform line probing the next time link is initialized. When set to a zero, the PMA/PMD does not use line probing. Line probing causes the PMA/PMD to select probe duration and the link data rate. For more information, see the documents referenced in 63.3.

#### **45.2.1.61.3 Noise environment (1.80.8)**

This bit controls the reference noise used during line probing. When set to a one, the noise environment is based on the current line conditions. When set to a zero, the noise environment is based on worst case models. For more information, see the documents referenced in 63.3.

#### **45.2.1.61.4 Region (1.80.1:0)**

These bits select the regional annex that is used for the operation of the 2BASE-TL PMA/PMD. These annexes refer to clauses in documents referenced by the 2BASE-TL PMA/PMD specification. These are not annexes in IEEE Std 802.3. For details on each annex, see the document referenced in 63.1.3.

#### **45.2.1.62 2B PMD parameters registers (Registers 1.81 through 1.88)**

The 2B PMD parameters registers set the transmission parameters for an individual 2BASE-TL PMA/PMD link. When the link is initialized, these parameters are used by the link partner PMA/PMDs in an attempt to achieve specified settings.

These registers allow one to specify a single fixed data rate or up to four data rate ranges at the -O PMA/PMD. An additional set of four data ranges are found in the 2B extended PMD parameters registers (1.102 through 1.109). Bit descriptions for the 2B extended PMD parameters registers are found in 45.2.1.76. Together these sets allow up to eight data rate ranges to be specified.

If at least one data rate range is specified with different Minimum and Maximum data rates, the link is trained with the highest attainable rate. If line probing is enabled, the highest rate is determined by the result of line probing and the “Data rate step” value is ignored. If line probing is disabled, the minimum and maximum rate, “Data rate step” and “Power” values are used to determine the highest attainable rate.

In the case of a single fixed rate specified (Minimum data rate<sub>1</sub> equals Maximum data rate<sub>1</sub>, Data rate step[1:8] set to zero, Minimum/Maximum data rate[2:8] set to zero), the link is trained at the specified rate.

When multiple ranges are specified, the PMD selects the first attainable range, starting sequentially from the first range.

Since writing to this register does not have an immediate effect, reading this register returns the desired parameters, which are not necessarily the current operating parameters.

For more information on how these parameters are passed across the physical link using G.994.1 signaling (see 61B.3.2).

The bit definitions for the 2B PMD parameters register are found in Table 45–60.

**Table 45–60— 2B PMD parameters registers bit definition**

| Bit(s)     | Name            | Description                                                                                                | R/W <sup>a</sup> |
|------------|-----------------|------------------------------------------------------------------------------------------------------------|------------------|
| 1.81.15    | Reserved        | Value always 0                                                                                             | RO               |
| 1.81.14:8  | Min data rate1  | Min data rate of the first range<br>$n$ = value of the bits, $n$ valid 3 to 89<br>Data Rate = $64n$ kb/s   | O: R/W<br>R: RO  |
| 1.81.7     | Reserved        | Value always 0                                                                                             | RO               |
| 1.81.6:0   | Max data rate1  | Max data rate of the first range<br>$n$ = value of the bits, $n$ valid 3 to 89<br>Data Rate = $64n$ kb/s   | O: R/W<br>R: RO  |
| 1.82.15:14 | Reserved        | Value always 0                                                                                             | RO               |
| 1.82.13:7  | Data rate step1 | Data rate step of the first range<br>$n$ = value of the bits, $n$ valid 1 to 86<br>Data Rate = $64n$ kb/s  | O: R/W<br>R: RO  |
| 1.82.6:2   | Power1          | $x$ = multiple of 0.5 dBm to add to 5 dBm offset<br>Power = $(5 + 0.5x)$ dBm                               | O: R/W<br>R: RO  |
| 1.82.1:0   | Constellation1  | 11 = reserved<br>10 = 16-TCPAM<br>01 = 32-TCPAM<br>00 = automatically set by the PHY                       | O: R/W<br>R: RO  |
| 1.83.15    | Reserved        | Value always 0                                                                                             | RO               |
| 1.83.14:8  | Min data rate2  | Min data rate of the second range<br>$n$ = value of the bits, $n$ valid 3 to 89<br>Data Rate = $64n$ kb/s  | O: R/W<br>R: RO  |
| 1.83.7     | Reserved        | Value always 0                                                                                             | RO               |
| 1.83.6:0   | Max data rate2  | Max data rate of the second range<br>$n$ = value of the bits, $n$ valid 3 to 89<br>Data Rate = $64n$ kb/s  | O: R/W<br>R: RO  |
| 1.84.15:14 | Reserved        | Value always 0                                                                                             | RO               |
| 1.84.13:7  | Data rate step2 | Data rate step of the second range<br>$n$ = value of the bits, $n$ valid 1 to 86<br>Data Rate = $64n$ kb/s | O: R/W<br>R: RO  |
| 1.84.6:2   | Power2          | $x$ = multiple of 0.5 dBm to add to 5 dBm offset<br>Power = $(5 + 0.5x)$ dBm                               | O: R/W<br>R: RO  |
| 1.84.1:0   | Constellation2  | 11 = reserved<br>10 = 16-TCPAM<br>01 = 32-TCPAM<br>00 = automatically set by the PHY                       | O: R/W<br>R: RO  |
| 1.85.15    | Reserved        | Value always 0                                                                                             | RO               |
| 1.85.14:8  | Min data rate3  | Min data rate of the third range<br>$n$ = value of the bits, $n$ valid 3 to 89<br>Data Rate = $64n$ kb/s   | O: R/W<br>R: RO  |
| 1.85.7     | Reserved        | Value always 0                                                                                             | RO               |

**Table 45–60— 2B PMD parameters registers bit definition (*continued*)**

| Bit(s)     | Name            | Description                                                                                                | R/W <sup>a</sup> |
|------------|-----------------|------------------------------------------------------------------------------------------------------------|------------------|
| 1.85.6:0   | Max data rate3  | Max data rate of the third range<br>$n$ = value of the bits, $n$ valid 3 to 89<br>Data Rate = $64n$ kb/s   | O: R/W<br>R: RO  |
| 1.86.15:14 | Reserved        | Value always 0                                                                                             | RO               |
| 1.86.13:7  | Data rate step3 | Data rate step of the third range<br>$n$ = value of the bits, $n$ valid 1 to 86<br>Data Rate = $64n$ kb/s  | O: R/W<br>R: RO  |
| 1.86.6:2   | Power3          | $x$ = multiple of 0.5 dBm to add to 5 dBm offset<br>Power = $(5 + 0.5x)$ dBm                               | O: R/W<br>R: RO  |
| 1.86.1:0   | Constellation3  | 11 = reserved<br>10 = 16-TCPAM<br>01 = 32-TCPAM<br>00 = automatically set by the PHY                       | O: R/W<br>R: RO  |
| 1.87.15    | Reserved        | Value always 0                                                                                             | RO               |
| 1.87.14:8  | Min data rate4  | Min data rate of the fourth range<br>$n$ = value of the bits, $n$ valid 3 to 89<br>Data Rate = $64n$ kb/s  | O: R/W<br>R: RO  |
| 1.87.7     | Reserved        | Value always 0                                                                                             | RO               |
| 1.87.6:0   | Max data rate4  | Max data rate of the fourth range<br>$n$ = value of the bits, $n$ valid 3 to 89<br>Data Rate = $64n$ kb/s  | O: R/W<br>R: RO  |
| 1.88.15:14 | Reserved        | Value always 0                                                                                             | RO               |
| 1.88.13:7  | Data rate step4 | Data rate step of the fourth range<br>$n$ = value of the bits, $n$ valid 1 to 86<br>Data Rate = $64n$ kb/s | O: R/W<br>R: RO  |
| 1.88.6:2   | Power4          | $x$ = multiple of 0.5 dBm to add to 5 dBm offset<br>Power = $(5 + 0.5x)$ dBm                               | O: R/W<br>R: RO  |
| 1.88.1:0   | Constellation4  | 11 = reserved<br>10 = 16-TCPAM<br>01 = 32-TCPAM<br>00 = automatically set by the PHY                       | O: R/W<br>R: RO  |

<sup>a</sup>R/W = Read/Write, RO = Read only

#### 45.2.1.62.1 Minimum data rate (1.81, 1.83, 1.85, 1.87. Bits 14:8)

Bits 14:8 in registers 1.81 through 1.87 set the minimum data rate for each of the four ranges. Valid values for these bits are decimal 3 through 89, writes to set an invalid value shall be ignored. The rate is expressed in units of kb/s and is derived by multiplying the decimal value of bits 14:8 by 64.

#### 45.2.1.62.2 Max data rate (1.81, 1.83, 1.85, 1.87. Bits 6:0)

Bits 6:0 in registers 1.81 through 1.87 set the maximum data rate for each of the four ranges. Valid values for these bits are decimal 3 through 89, writes to set an invalid value shall be ignored. The rate is expressed in units of kb/s and is derived by multiplying the decimal value of bits 6:0 by 64.

#### 45.2.1.62.3 Data rate step (1.82, 1.84, 1.86, 1.88. Bits 13:7)

Bits 13:7 in registers 1.82 through 1.88 set the granularity used by the PMA/PMD when determining the line rate. Valid values for these bits are decimal 1 through 86, writes to set an invalid value shall be ignored. The data rate step is expressed in units of kb/s and is derived by multiplying the decimal value of bits 13:7 by 64.

#### 45.2.1.62.4 Power (1.82, 1.84, 1.86, 1.88. Bits 6:2)

Bits 6:2 in registers 1.82 through 1.88 set the allowed power level for each data rate range. The power levels set in these bits override those of the annex selected in the 2B general parameter register (1.80). The power level is expressed in units of dBm and is derived by Equation (45–4), where  $x$  equals the value of bits 6:2.

$$power = \left(5 + \frac{x}{2}\right) \text{ dBm} \quad (45-4)$$

#### 45.2.1.62.5 Constellation (1.82, 1.84, 1.86, 1.88. Bits 1:0)

Bits 1:0 in registers 1.82 through 1.88 set the allowed constellation for each data rate range. Setting a value of 10 or 01 restricts the constellation to 16- or 32-TCPAM respectively. When set to a value of 00, the PMD automatically determines the constellation during initialization. Attempts to set a value of 11 shall be ignored.

#### 45.2.1.63 2B code violation errors counter (Register 1.89)

The 2B code violation errors counter is a 16-bit counter that contains the number of the 2BASE-TL CRC anomalies. See 63.2.2.3 for more information. These bits shall be set to all zeros upon an MMD reset and upon being read.

Bit definitions for the 2B code violation errors counter are found in Table 45–61.

**Table 45–61—2B code violation errors counter bit definitions**

| Bit(s)    | Name                   | Description              | R/W <sup>a</sup> |
|-----------|------------------------|--------------------------|------------------|
| 1.89.15:0 | Code violations [15:0] | The bytes of the counter | RO               |

<sup>a</sup>RO = Read only

#### 45.2.1.64 2B link partner code violations register (Register 1.90)

The 2B link partner code violations register provides the -O STA with a snapshot of the -R link partner’s 2B code violations counter. Because this register is not a counter, its value will be updated only when refreshed. The contents of this register are refreshed when the STA activates the “Get link partner parameter” command in the Link Partner PMA/PMD control register (see 45.2.1.30) and the command completes successfully (see 45.2.1.31.1).

This register is defined for -O port subtypes only.

The bit definitions for this register are found in Table 45–61.



#### 45.2.1.65 2B errored seconds counter (Register 1.91)

This 8-bit counter contains the number of errored seconds (see 63.2.2.3) These bits shall be set to all zeros when the register is read by management or upon an MMD reset.

Bit definitions for the 2B errored seconds counter are found in Table 45–62.

**Table 45–62—2B errored seconds counter bit definitions**

| Bit(s)    | Name                  | Description             | R/W <sup>a</sup> |
|-----------|-----------------------|-------------------------|------------------|
| 1.91.15:8 | Reserved              | Value always 0          | RO               |
| 1.91.7:0  | Errored seconds [7:0] | The byte of the counter | RO               |

<sup>a</sup>RO = Read only

#### 45.2.1.66 2B link partner errored seconds register (Register 1.92)

The 2B link partner errored seconds register provides the -O STA with a snapshot of the -R link partner’s 2B errored seconds counter. Because this register is not a counter, its value will be updated only when refreshed. The contents of this register are refreshed when the STA activates the “Get link partner parameter” command in the Link Partner PMA/PMD control register (see 45.2.1.30) and the command completes successfully (see 45.2.1.31.1).

This register is defined for -O port subtypes only.

The bit definitions for this register are found in Table 45–62.

#### 45.2.1.67 2B severely errored seconds counter (Register 1.93)

This 8-bit counter contains the number severely errored seconds (see 63.2.2.3). These bits shall be set to all zeros when the register is read by management or upon an MMD reset.

Bit definitions for the 2B severely errored seconds register are found in Table 45–63.

**Table 45–63—2B severely errored counter register bit definitions**

| Bit(s)    | Name                           | Description             | R/W <sup>a</sup> |
|-----------|--------------------------------|-------------------------|------------------|
| 1.93.15:8 | Reserved                       | Value always 0          | RO               |
| 1.93.7:0  | Severely errored seconds [7:0] | The byte of the counter | RO               |

<sup>a</sup>RO = Read only

#### 45.2.1.68 2B link partner severely errored seconds register (Register 1.94)

The 2B link partner severely errored seconds register provides the -O STA with a snapshot of the -R link partner’s 2B severely errored seconds counter. Because this register is not a counter, its value will be updated only when refreshed. The contents of this register are refreshed when the STA activates the “Get link partner parameter” command in the Link Partner PMA/PMD control register (see 45.2.1.30) and the command completes successfully (see 45.2.1.31.1).

This register is defined for -O port subtypes only.

The bit definitions for this register are found in Table 45–63.

#### 45.2.1.69 2B LOSW counter (Register 1.95)

This 8-bit counter contains the number of loss of sync seconds (see 63.2.2.3). These bits shall be set to all zeros when the register is read by management or upon an MMD reset.

Bit definitions for the 2B LOSW counter are found in Table 45–64.

**Table 45–64—2B LOSW counter bit definitions**

| Bit(s)    | Name                       | Description             | R/W <sup>a</sup> |
|-----------|----------------------------|-------------------------|------------------|
| 1.95.15:8 | Reserved                   | Value always 0          | RO               |
| 1.95.7:0  | Loss of sync seconds [7:0] | The byte of the counter | RO               |

<sup>a</sup>RO = Read only

#### 45.2.1.70 2B link partner LOSW register (Register 1.96)

The 2B link partner LOSW register provides the -O STA with a snapshot of the -R link partner’s 2B LOSW counter. Because this register is not a counter, its value will be updated only when refreshed. The contents of this register are refreshed when the STA activates the “Get link partner parameter” command in the Link partner PMA/PMD control register (see 45.2.1.30) and the command completes successfully (see 45.2.1.31.1).

This register is defined for -O port subtypes only.

The bit definitions for this register are found in Table 45–64.

#### 45.2.1.71 2B unavailable seconds counter (Register 1.97)

This 8-bit counter contains the number of unavailable seconds (see 63.2.2.3). These bits shall be set to all zeros when the register is read by management or upon an MMD reset.

Bit definitions for the 2B unavailable seconds counter are found in Table 45–65.

**Table 45–65—2B unavailable seconds counter bit definitions**

| Bit(s)    | Name                      | Description             | R/W <sup>a</sup> |
|-----------|---------------------------|-------------------------|------------------|
| 1.97.15:8 | Reserved                  | Value always 0          | RO               |
| 1.97.7:0  | unavailable seconds [7:0] | The byte of the counter | RO               |

<sup>a</sup>RO = Read only

#### 45.2.1.72 2B link partner unavailable seconds register (Register 1.98)

The 2B link partner unavailable seconds register provides the -O STA with a snapshot of the -R link partner's 2B unavailable seconds counter. Because this register is not a counter, its value will be updated only when refreshed. The contents of this register are refreshed when the STA activates the “Get link partner parameter” command in the Link Partner PMA/PMD control register (see 45.2.1.30) and the command completes successfully (see 45.2.1.31.1).

This register is defined for -O port subtypes only.

The bit definitions for this register are found in Table 45–65.

#### 45.2.1.73 2B state defects register (Register 1.99)

The 2B state defects register is used to communicate defect states from the 2BASE-TL PMD (see 63.2.2.3). The thresholds for these defects are set using the 2B line quality threshold register (see 45.2.1.39). The register bits are cleared to zero when read by the STA or upon MMD reset. On a -R PMA/PMD, these bits are also cleared to zero upon the successful reception of a “Get link partner parameters” command (see 45.2.1.29.1).

Bit definitions for the 2B state defects register are found in Table 45–66.

**Table 45–66—2B state defects register bit definitions**

| Bit(s)    | Name                    | Description                                                  | R/W <sup>a</sup> |
|-----------|-------------------------|--------------------------------------------------------------|------------------|
| 1.99.15   | Segment defect          | 1 = segment defect detected<br>0 = normal condition          | RO, LH           |
| 1.99.14   | SNR margin defect       | 1 = SNR margin defect detected<br>0 = normal condition       | RO, LH           |
| 1.99.13   | Loop attenuation defect | 1 = loop attenuation defect detected<br>0 = normal condition | RO, LH           |
| 1.99.12   | Loss of sync word       | 1 = loss of sync word detected<br>0 = normal condition       | RO, LH           |
| 1.99.11:0 | Reserved                | Value always 0                                               | RO               |

<sup>a</sup>RO = Read only, LH = Latching high

##### 45.2.1.73.1 Segment defect (1.99.15)

When read as a one, this bit indicates that the local PMA/PMD has detected a segment defect.

##### 45.2.1.73.2 SNR margin defect (1.99.14)

When read as a one, this bit indicates that the local PMA/PMD has received a signal whose SNR is below the set threshold (see 45.2.1.39).

##### 45.2.1.73.3 Loop attenuation defect (1.99.13)

When read as a one, this bit indicates that the PMA/PMD has detected that the loop attenuation is below the set threshold (see 45.2.1.39).

#### 45.2.1.73.4 Loss of sync word (1.99.12)

When read as a one, this bit indicates that the PMA/PMD has lost PMA/PMD frame sync.

#### 45.2.1.74 2B link partner state defects register (Register 1.100)

The 2B link partner state defects register provides the -O STA with a snapshot of the -R link partner's 2B state defects register. The contents of this register are refreshed when the STA activates the "Get link partner parameter" command in the Link Partner PMA/PMD control register (see 45.2.1.30) and the command completes successfully (see 45.2.1.31.1).

This register is defined for -O port subtypes only.

The bit definitions for this register are found in Table 45–66.

#### 45.2.1.75 2B negotiated constellation register (Register 1.101)

The bit definitions for this register are shown in Table 45–67.

**Table 45–67—2B register bit definition**

| Bit(s)     | Name                     | Description                                                          | R/W <sup>a</sup> |
|------------|--------------------------|----------------------------------------------------------------------|------------------|
| 1.101.15:2 | Reserved                 | Value always 0                                                       | RO               |
| 1.101.1:0  | Negotiated constellation | 11 = reserved<br>10 = 16-TCPAM<br>01 = 32-TCPAM<br>00 = undetermined | RO               |

<sup>a</sup>RO = Read only

##### 45.2.1.75.1 Negotiated constellation (1.101.1:0)

These bits report the resulting constellation that was obtained after initialization. For more information on configuring 2BASE-TL PMA/PMD link initialization, see the 2B PMD parameter registers (see 45.2.1.62). When read as 10 or 01, the constellation has been set as either 16- or 32-TCPAM respectively. When read as 00, the local PMD has not arrived at a constellation with its link partner (as may be the case while link is down or initializing, after reset or upon a failed initialization).

#### 45.2.1.76 2B extended PMD parameters registers (Registers 1.102 through 1.109)

The 2B extended PMD parameters registers define four additional data range sets to be used in conjunction with the 2B PMD parameters registers when additional PMD configuration detail is desired. For a complete description of the use of these registers, see 45.2.1.62.

Bit definitions for these registers can be found in Table 45–68.

**Table 45–68— 2B extended PMD parameters registers bit definition**

| Bit(s)      | Name            | Description                                                                                                | R/W <sup>a</sup> |
|-------------|-----------------|------------------------------------------------------------------------------------------------------------|------------------|
| 1.102.15    | Reserved        | Value always 0                                                                                             | RO               |
| 1.102.14:8  | Min data rate5  | Min data rate of the fifth range<br>$n$ = value of the bits, $n$ valid 3 to 89<br>Data Rate = $64n$ kb/s   | O: R/W<br>R: RO  |
| 1.102.7     | Reserved        | Value always 0                                                                                             | RO               |
| 1.102.6:0   | Max data rate5  | Max data rate of the fifth range<br>$n$ = value of the bits, $n$ valid 3 to 89<br>Data Rate = $64n$ kb/s   | O: R/W<br>R: RO  |
| 1.103.15:14 | Reserved        | Value always 0                                                                                             | RO               |
| 1.103.13:7  | Data rate step5 | Data rate step of the fifth range<br>$n$ = value of the bits, $n$ valid 1 to 86<br>Data Rate = $64n$ kb/s  | O: R/W<br>R: RO  |
| 1.103.6:2   | Power5          | $x$ = multiple of 0.5 dBm to add to 5 dBm offset<br>Power = $(5 + 0.5x)$ dBm                               | O: R/W<br>R: RO  |
| 1.103.1:0   | Constellation5  | 11 = reserved<br>10 = 16-TCPAM<br>01 = 32-TCPAM<br>00 = automatically set by the PHY                       | O: R/W<br>R: RO  |
| 1.104.15    | Reserved        | Value always 0                                                                                             | RO               |
| 1.104.14:8  | Min data rate6  | Min data rate of the sixth range<br>$n$ = value of the bits, $n$ valid 3 to 89<br>Data Rate = $64n$ kb/s   | O: R/W<br>R: RO  |
| 1.104.7     | Reserved        | Value always 0                                                                                             | RO               |
| 1.104.6:0   | Max data rate6  | Max data rate of the sixth range<br>$n$ = value of the bits, $n$ valid 3 to 89<br>Data Rate = $64n$ kb/s   | O: R/W<br>R: RO  |
| 1.105.15:14 | Reserved        | Value always 0                                                                                             | RO               |
| 1.105.13:7  | Data rate step6 | Data rate step of the sixth range<br>$n$ = value of the bits, $n$ valid 1 to 86<br>Data Rate = $64n$ kb/s  | O: R/W<br>R: RO  |
| 1.105.6:2   | Power6          | $x$ = multiple of 0.5 dBm to add to 5 dBm offset<br>Power = $(5 + 0.5x)$ dBm                               | O: R/W<br>R: RO  |
| 1.105.1:0   | Constellation6  | 11 = reserved<br>10 = 16-TCPAM<br>01 = 32-TCPAM<br>00 = automatically set by the PHY                       | O: R/W<br>R: RO  |
| 1.106.15    | Reserved        | Value always 0                                                                                             | RO               |
| 1.106.14:8  | Min data rate7  | Min data rate of the seventh range<br>$n$ = value of the bits, $n$ valid 3 to 89<br>Data Rate = $64n$ kb/s | O: R/W<br>R: RO  |
| 1.106.7     | Reserved        | Value always 0                                                                                             | RO               |

**Table 45–68— 2B extended PMD parameters registers bit definition (*continued*)**

| Bit(s)      | Name                        | Description                                                                                                 | R/W <sup>a</sup> |
|-------------|-----------------------------|-------------------------------------------------------------------------------------------------------------|------------------|
| 1.106.6:0   | Max data rate <sup>7</sup>  | Max data rate of the seventh range<br>$n$ = value of the bits, $n$ valid 3 to 89<br>Data Rate = $64n$ kb/s  | O: R/W<br>R: RO  |
| 1.107.15:14 | Reserved                    | Value always 0                                                                                              | RO               |
| 1.107.13:7  | Data rate step <sup>7</sup> | Data rate step of the seventh range<br>$n$ = value of the bits, $n$ valid 1 to 86<br>Data Rate = $64n$ kb/s | O: R/W<br>R: RO  |
| 1.107.6:2   | Power <sup>7</sup>          | $x$ = multiple of 0.5 dBm to add to 5 dBm offset<br>Power = $(5 + 0.5x)$ dBm                                | O: R/W<br>R: RO  |
| 1.107.1:0   | Constellation <sup>7</sup>  | 11 = reserved<br>10 = 16-TCPAM<br>01 = 32-TCPAM<br>00 = automatically set by the PHY                        | O: R/W<br>R: RO  |
| 1.108.15    | Reserved                    | Value always 0                                                                                              | RO               |
| 1.108.14:8  | Min data rate <sup>8</sup>  | Min data rate of the eighth range<br>$n$ = value of the bits, $n$ valid 3 to 89<br>Data Rate = $64n$ kb/s   | O: R/W<br>R: RO  |
| 1.108.7     | Reserved                    | Value always 0                                                                                              | RO               |
| 1.108.6:0   | Max data rate <sup>8</sup>  | Max data rate of the eighth range<br>$n$ = value of the bits, $n$ valid 3 to 89<br>Data Rate = $64n$ kb/s   | O: R/W<br>R: RO  |
| 1.109.15:14 | Reserved                    | Value always 0                                                                                              | RO               |
| 1.109.13:7  | Data rate step <sup>8</sup> | Data rate step of the eighth range<br>$n$ = value of the bits, $n$ valid 1 to 86<br>Data Rate = $64n$ kb/s  | O: R/W<br>R: RO  |
| 1.109.6:2   | Power <sup>8</sup>          | $x$ = multiple of 0.5 dBm to add to 5 dBm offset<br>Power = $(5 + 0.5x)$ dBm                                | O: R/W<br>R: RO  |
| 1.109.1:0   | Constellation <sup>8</sup>  | 11 = reserved<br>10 = 16-TCPAM<br>01 = 32-TCPAM<br>00 = automatically set by the PHY                        | O: R/W<br>R: RO  |

<sup>a</sup>R/W = Read/Write, RO = Read only

#### 45.2.1.76.1 Minimum data rate (1.102, 1.104, 1.106, 1.108. Bits 14:8)

Bits 14:8 in registers 1.102, 1.104, 1.106, and 1.108 set the minimum data rate for each of the four extended ranges. Valid values for these bits are decimal 3 through 89. writes to set an invalid value shall be ignored. The rate is expressed in units of kb/s and is derived by multiplying the decimal value of bits 14:8 by 64.

#### 45.2.1.76.2 Max data rate (1.102, 1.104, 1.106, 1.108. Bits 6:0)

Bits 6:0 in registers 1.102, 1.104, 1.106, and 1.108 set the maximum data rate for each of the four extended ranges. Valid values for these bits are decimal 3 through 89, writes to set an invalid value shall be ignored. The rate is expressed in units of kb/s and is derived by multiplying the decimal value of bits 6:0 by 64.

#### 45.2.1.76.3 Data rate step (1.103, 1.105, 1.107, 1.109. Bits 13:7)

Bits 13:7 in registers 1.103, 1.105, 1.107 and 1.109 set the granularity used by the PMA/PMD when determining the line rate. Valid values for these bits are decimal 1 through 86, writes to set an invalid value shall be ignored. The data rate step is expressed in units of kb/s and is derived by multiplying the decimal value of bits 13:7 by 64.

#### 45.2.1.76.4 Power (1.103, 1.105, 1.107, 1.109. Bits 6:2)

Bits 6:2 in registers 1.103, 1.105, 1.107 and 1.109 set the allowed power level for each extended data rate range. The power levels set in these bits override those of the annex selected in the 2B general parameter register (1.80). The power level is expressed in units of dBm and is derived by Equation (45–5), where  $x$  equals the value of bits 6:2.

$$\text{power} = \left( 5 + \frac{x}{2} \right) \text{ dBm} \quad (45-5)$$

#### 45.2.1.76.5 Constellation (1.103, 1.105, 1.107, 1.109. Bits 1:0)

Bits 1:0 in registers 1.103, 1.105, 1.107 and 1.109 set the allowed constellation for each extended data rate range. Setting a value of 10 or 01 restricts the constellation to 16- or 32-TCPAM respectively. When set to a value of 00, the PMD automatically determines the constellation during initialization. Attempts to set a value of 11 shall be ignored.

#### 45.2.1.77 MultiGBASE-T status (Register 1.129)

The assignment of bits in the MultiGBASE-T status register is shown in Table 45–69.

**Table 45–69—MultiGBASE-T status register bit definitions**

| Bit(s)     | Name                 | Description                                                                      | R/W <sup>a</sup> |
|------------|----------------------|----------------------------------------------------------------------------------|------------------|
| 1.129.15:1 | Reserved             | Value always 0                                                                   | RO               |
| 1.129.0    | LP information valid | 1 = Link partner information is valid<br>0 = Link partner information is invalid | RO               |

<sup>a</sup>RO = Read only

##### 45.2.1.77.1 LP information valid (1.129.0)

When read as a one, bit 1.129.0 indicates that the startup protocol defined in 55.4.2.5 (for 10GBASE-T), 113.4.2.5 (for 25G/40GBASE-T), or 126.4.2.5 (for 2.5G/5GBASE-T) has been completed, and that the contents of bits 1.130.11:0, 1.131.15:10, 1.145.14:8, 1.146.14:8, and 1.146.6:0, which are established during the startup protocol, are valid. When read as a zero, bit 1.129.0 indicates that the startup process has not been completed, and that the contents of these bits that are established during the startup protocol are invalid. A PMA in the MultiGBASE-T set shall return a value of zero in bit 1.129.0 if PMA link\_status=FAIL.

#### 45.2.1.78 MultiGBASE-T pair swap and polarity register (Register 1.130)

Register 1.130 reflects the status of the pair-to-pair connectivity between the local device and the link partner (see Table 45–70). The polarity of each individual pair is also indicated.

**Table 45–70—MultiGBASE-T pair swap and polarity register bit definitions**

| Bit(s)      | Name                 | Description                                                                                            | R/W <sup>a</sup> |
|-------------|----------------------|--------------------------------------------------------------------------------------------------------|------------------|
| 1.130.15:12 | Reserved             | Value always 0                                                                                         | RO               |
| 1.130.11    | Pair D polarity      | 1 = Polarity of pair D is reversed<br>0 = Polarity of pair D is not reversed                           | RO               |
| 1.130.10    | Pair C polarity      | 1 = Polarity of pair C is reversed<br>0 = Polarity of pair C is not reversed                           | RO               |
| 1.130.9     | Pair B polarity      | 1 = Polarity of pair B is reversed<br>0 = Polarity of pair B is not reversed                           | RO               |
| 1.130.8     | Pair A polarity      | 1 = Polarity of pair A is reversed<br>0 = Polarity of pair A is not reversed                           | RO               |
| 1.130.7:2   | Reserved             | Value always 0                                                                                         | RO               |
| 1.130.1:0   | MDI/MDI-X connection | 1 0<br>1 1 = No crossover<br>1 0 = Reserved<br>0 1 = Reserved<br>0 0 = Pair A/B and pair C/D crossover | RO               |

<sup>a</sup>RO = Read only

#### 45.2.1.78.1 Pair D polarity (1.130.11)

When read as zero, bit 1.130.11 indicates that the polarity within pair D is not reversed. When read as one, bit 1.130.11 indicates that the polarity of pair D is reversed.

#### 45.2.1.78.2 Pair C polarity (1.130.10)

When read as zero, bit 1.130.10 indicates that the polarity within pair C is not reversed. When read as one, bit 1.130.10 indicates that the polarity of pair C is reversed.

#### 45.2.1.78.3 Pair B polarity (1.130.9)

When read as zero, bit 1.130.9 indicates that the polarity within pair B is not reversed. When read as one, bit 1.130.9 indicates that the polarity of pair B is reversed.

#### 45.2.1.78.4 Pair A polarity (1.130.8)

When read as zero, bit 1.130.8 indicates that the polarity within pair A is not reversed. When read as one, bit 1.130.8 indicates that the polarity of pair A is reversed.

#### 45.2.1.78.5 MDI/MDI-X connection (1.130.1:0)

Bits 1.130.1:0 indicate the resolution of the automatic MDI/MDI-X configuration defined in 55.4.4. This function is intended to eliminate the need for crossover cables between similar devices. The automatic configuration method used shall comply with 40.4.4.1 and 40.4.4.2.

#### 45.2.1.79 MultiGBASE-T TX power backoff and PHY short reach setting (Register 1.131)

The complete assignment of bits in the MultiGBASE-T TX power backoff and short reach mode settings register is shown in Table 45–71.



**Table 45–71—MultiGBASE-T TX power backoff and PHY short reach setting register bit definitions**

| Bit(s)      | Name                                  | Description                                                                                                                                 | R/W <sup>a</sup> |
|-------------|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.131.15:13 | Link partner TX power backoff setting | 15 14 13<br>1 1 1 = 14 dB<br>1 1 0 = 12 dB<br>1 0 1 = 10 dB<br>1 0 0 = 8 dB<br>0 1 1 = 6 dB<br>0 1 0 = 4 dB<br>0 0 1 = 2 dB<br>0 0 0 = 0 dB | RO               |
| 1.131.12:10 | TX power backoff setting              | 12 11 10<br>1 1 1 = 14 dB<br>1 1 0 = 12 dB<br>1 0 1 = 10 dB<br>1 0 0 = 8 dB<br>0 1 1 = 6 dB<br>0 1 0 = 4 dB<br>0 0 1 = 2 dB<br>0 0 0 = 0 dB | RO               |
| 1.131.9:1   | Reserved                              | Value always 0                                                                                                                              | RO               |
| 1.131.0     | Short reach mode                      | 1 = PHY is operating in short reach mode<br>0 = PHY is not operating in short reach mode                                                    | R/W              |

<sup>a</sup>RO = Read only, R/W = Read/Write

#### 45.2.1.79.1 MultiGBASE-T TX power backoff settings (1.131.15:10)

The MultiGBASE-T TX power backoff settings reflects the TX power backoff selected during the startup negotiation process. The 10GBASE-T startup negotiation process and all TX power backoff settings are defined in 55.4.2.5 and 55.4.5.1. The 25GBASE-T and 40GBASE-T startup negotiation process and all TX power backoff settings are defined in 113.4.2.5 and 113.4.5.1. For 2.5GBASE-T and 5GBASE-T, startup negotiation process and all TX power backoff settings are defined in 126.4.2.5 and 126.4.5.1. If LP information valid bit, 1.129.0, is set to one then bits 1.131.15:13 indicates the TX power backoff setting of the link partner and bits 1.131.12:10 indicates the TX power backoff setting of the local device.

#### 45.2.1.79.2 PHY short reach mode (1.131.0)

The short reach mode of the 10GBASE-T PHY provides a means for operation on a cable plant that has parametric performance equivalent to 30 m of Class F and Class E<sub>A</sub> cabling as defined in 55.5.4.5. The short reach mode of the 25GBASE-T and 40GBASE-T PHYs provides for operation on a direct attach link segment that has parametric performance defined in 113.7.4. Setting bit 1.131.0 to a one puts the PHY in short reach mode, and setting bit 1.131.0 to a zero puts the PHY into normal (non-short reach) mode. The default value for this bit is zero.

#### 45.2.1.80 MultiGBASE-T test mode register (Register 1.132)

The assignment of bits in the MultiGBASE-T test mode register is shown in Table 45–72. The default values for each bit should be chosen so that the initial state of the device upon power up or reset is a normal operational state without management intervention.

**Table 45–72—MultiGBASE-T test mode register bit definitions**

| Bit(s)      | Name                         | Description                                                                                                                                                                                           | R/W <sup>a</sup> |
|-------------|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.132.15:13 | Test mode control            | 15 14 13<br>1 1 1 = Test mode 7<br>1 1 0 = Test mode 6<br>1 0 1 = Test mode 5<br>1 0 0 = Test mode 4<br>0 1 1 = Test mode 3<br>0 1 0 = Test mode 2<br>0 0 1 = Test mode 1<br>0 0 0 = Normal operation | R/W              |
| 1.132.12:10 | Transmitter test frequencies | 12 11 10<br>1 1 1 = Reserved<br>1 1 0 = Dual tone 5<br>1 0 1 = Dual tone 4<br>1 0 0 = Dual tone 3<br>0 1 1 = Reserved<br>0 1 0 = Dual tone 2<br>0 0 1 = Dual tone 1<br>0 0 0 = Reserved               | R/W              |
| 1.132.9:0   | Reserved                     | Value always 0                                                                                                                                                                                        | RO               |

<sup>a</sup>R/W = Read/Write, RO = Read only

#### 45.2.1.80.1 Test mode control (1.132.15:13)

Transmitter test mode operations defined by bits 1.132.15:13 are described for 10GBASE-T in 55.5.2 and Table 55–12, for 2.5G/5GBASE-T in 126.5.2 and Table 126–13, and for 25G/40GBASE-T in 113.5.2 and Table 113–16. The default value for bits 1.132.15:13 is zero.

#### 45.2.1.80.2 Transmitter test frequencies (1.132.12:10)

When test mode 4 is selected by setting bits 1.132.15:13 to one, zero, zero respectively, bits 1.132.12:10 select the transmit test frequency as shown in Table 45–72. Detailed use and operation of these transmitter test frequencies is described in 55.5.2, 113.5.2, and 126.5.2.

#### 45.2.1.81 SNR operating margin channel A register (Register 1.133)

Register 1.133 contains the current SNR operating margin measured at the slicer input for channel A for the PMAs in the MultiGBASE-T set. It is reported with 0.1 dB of resolution to an accuracy of 0.5 dB within the range of –12.7 dB to 12.7 dB. The number is in offset binary notation, with 0.0 dB represented by 0x8000. Implementation of this register is optional.

#### 45.2.1.82 SNR operating margin channel B register (Register 1.134)

Register 1.134 contains the current SNR operating margin measured at the slicer input for channel B for the PMAs in the MultiGBASE-T set. It is reported in units of 0.1 dB to an accuracy of 0.5 dB within the range of –12.7 dB to 12.7 dB. The number is in offset binary notation, with 0.0 dB represented by 0x8000. Implementation of this register is optional.

#### **45.2.1.83 SNR operating margin channel C register (Register 1.135)**

Register 1.135 contains the current SNR operating margin measured at the slicer input for channel C for the PMAs in the MultiGBASE-T set. It is reported in units of 0.1 dB to an accuracy of 0.5 dB within the range of –12.7 dB to 12.7 dB. The number is in offset binary notation, with 0.0 dB represented by 0x8000. Implementation of this register is optional.

#### **45.2.1.84 SNR operating margin channel D register (Register 1.136)**

Register 1.136 contains the current SNR operating margin measured at the slicer input for channel D for the PMAs in the MultiGBASE-T set. It is reported in units of 0.1 dB to an accuracy of 0.5 dB within the range of –12.7 dB to 12.7 dB. The number is in offset binary notation, with 0.0 dB represented by 0x8000. Implementation of this register is optional.

#### **45.2.1.85 Minimum margin channel A register (Register 1.137)**

The Minimum margin channel A register contains a latched copy of the lowest value observed in the SNR operating margin channel A register (1.133) since the last read. It is reported in units of 0.1 dB to an accuracy of 0.5 dB within the range of –12.7 dB to 12.7 dB. The number is in offset binary notation, with 0.0 dB represented by 0x8000. Implementation of this register is optional.

#### **45.2.1.86 Minimum margin channel B register (Register 1.138)**

The Minimum margin channel B register contains a latched copy of the lowest value observed in the SNR operating margin channel B register (1.134) since the last read. It is reported in units of 0.1 dB to an accuracy of 0.5 dB within the range of –12.7 dB to 12.7 dB. The number is in offset binary notation, with 0.0 dB represented by 0x8000. Implementation of this register is optional.

#### **45.2.1.87 Minimum margin channel C register (Register 1.139)**

The Minimum margin channel C register contains a latched copy of the lowest value observed in the SNR operating margin channel C register (1.135) since the last read. It is reported in units of 0.1 dB to an accuracy of 0.5 dB within the range of –12.7 dB to 12.7 dB. The number is in offset binary notation, with 0.0 dB represented by 0x8000. Implementation of this register is optional.

#### **45.2.1.88 Minimum margin channel D register (Register 1.140)**

The Minimum margin channel D register contains a latched copy of the lowest value observed in the SNR operating margin channel D register (1.136) since the last read. It is reported in units of 0.1 dB to an accuracy of 0.5 dB within the range of –12.7 dB to 12.7 dB. The number is in offset binary notation, with 0.0 dB represented by 0x8000. Implementation of this register is optional.

#### **45.2.1.89 RX signal power channel A register (Register 1.141)**

The RX signal power channel A register is read only and contains the receive signal power measured at the MDI during training as described in 55.4.3.1, 113.4.3.1, and 126.4.3.1. The RX signal power should reflect the power measured when the device transitions out of the state PMA\_Training\_Init\_M or PMA\_Training\_Init\_S (as appropriate, see 55.4.6.1, 113.4.6.1, and 126.4.6.1), when the link partner is transmitting with PBO\_tx = 4 (8 dB power backoff). It is reported in units of 0.1 dB to an accuracy of 0.5 dB within the range of –20 dBm to 5.5 dBm. The number is in offset binary notation, with 0.0 dB represented by 0x8000. Implementation of this register is optional.

#### 45.2.1.90 RX signal power channel B register (Register 1.142)

The RX signal power channel B register is read only and contains the receive signal power measured at the MDI during training as described in 55.4.3.1, 113.4.3.1, and 126.4.3.1. The RX signal power should reflect the power measured when the device transitions out of the state PMA\_Training\_Init\_M or PMA\_Training\_Init\_S (as appropriate, see 55.4.6.1, 113.4.6.1, and 126.4.6.1), when the link partner is transmitting with PBO\_tx = 4 (8 dB power backoff). It is reported in units of 0.1 dB to an accuracy of 0.5 dB within the range of –20 dBm to 5.5 dBm. The number is in offset binary notation, with 0.0 dB represented by 0x8000. Implementation of this register is optional.

#### 45.2.1.91 RX signal power channel C register (Register 1.143)

The RX signal power channel C register is read only and contains the receive signal power measured at the MDI during training as described in 55.4.3.1, 113.4.3.1, and 126.4.3.1. The RX signal power should reflect the power measured when the device transitions out of the state PMA\_Training\_Init\_M or PMA\_Training\_Init\_S (as appropriate, see 55.4.6.1, 113.4.6.1, and 126.4.6.1), when the link partner is transmitting with PBO\_tx = 4 (8 dB power backoff). It is reported in units of 0.1 dB to an accuracy of 0.5 dB within the range of –20 dBm to 5.5 dBm. The number is in offset binary notation, with 0.0 dB represented by 0x8000. Implementation of this register is optional.

#### 45.2.1.92 RX signal power channel D register (Register 1.144)

The RX signal power channel D register is read only and contains the receive signal power measured at the MDI during training as described in 55.4.3.1, 113.4.3.1, and 126.4.3.1. The RX signal power should reflect the power measured when the device transitions out of the state PMA\_Training\_Init\_M or PMA\_Training\_Init\_S (as appropriate, see 55.4.6.1, 113.4.6.1, and 126.4.6.1), when the link partner is transmitting with PBO\_tx = 4 (8 dB power backoff). It is reported in units of 0.1 dB to an accuracy of 0.5 dB within the range of –20 dBm to 5.5 dBm. The number is in offset binary notation, with 0.0 dB represented by 0x8000. Implementation of this register is optional.

#### 45.2.1.93 MultiGBASE-T skew delay register (Registers 1.145 and 1.146)

The skew delay register reports the current skew delay on each of the pair with respect to physical pair A (see Table 45–73). It is reported with resolution equal to one symbol period (see 55.1.3, 113.1.3, and 126.1.3) of the PHY (e.g., 1.25 ns for 10GBASE-T) to an accuracy of two symbol periods (e.g., 2.5 ns for 10GBASE-T). The number reported is in two’s complement notation with positive values representing delay and negative values representing advance with respect to physical pair A. If the delay exceeds the maximum amount that can be represented by the range (–64 symbols to +63 symbols), the field displays the maximum respective value. The value shall be updated at least once per second.

**Table 45–73—MultiGBASE-T skew delay register bit definitions**

| Bit(s)     | Name         | Description           | R/W <sup>a</sup> |
|------------|--------------|-----------------------|------------------|
| 1.145.15   | Reserved     | Value always 0        | RO               |
| 1.145.14:8 | Skew delay B | Skew delay for pair B | RO               |
| 1.145.7:0  | Reserved     | Value always 0        | RO               |
| 1.146.15   | Reserved     | Value always 0        | RO               |
| 1.146.14:8 | Skew delay D | Skew delay for pair D | RO               |

**Table 45–73—MultiGBASE-T skew delay register bit definitions (*continued*)**

| Bit(s)    | Name         | Description           | R/W <sup>a</sup> |
|-----------|--------------|-----------------------|------------------|
| 1.146.7   | Reserved     | Value always 0        | RO               |
| 1.146.6:0 | Skew delay C | Skew delay for pair C | RO               |

<sup>a</sup>RO = Read only

#### 45.2.1.94 MultiGBASE-T fast retrain status and control register (Register 1.147)

The assignment of bits in the MultiGBASE-T fast retrain status and control register is shown in Table 45–74.

**Table 45–74—MultiGBASE-T fast retrain status and control register bit definitions**

| Bit(s)      | Name                     | Description                                                                                                                                                              | R/W <sup>a</sup> |
|-------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.147.15:11 | LP fast retrain count    | Counts the number of fast retrains requested by the link partner                                                                                                         | RO/NR            |
| 1.147.10:6  | LD fast retrain count    | Counts the number of fast retrains requested by the local device                                                                                                         | RO/NR            |
| 1.147.5     | Reserved                 | Value always 0                                                                                                                                                           | RO               |
| 1.147.4     | Fast retrain ability     | 1 = Fast retrain capability is supported<br>0 = Fast retrain capability is not supported                                                                                 | RO               |
| 1.147.3     | Fast retrain negotiated  | 1 = Fast retrain capability was negotiated<br>0 = Fast retrain capability was not negotiated                                                                             | RO               |
| 1.147.2:1   | Fast retrain signal type | 11 = Reserved<br>10 = PHY signals Link Interruption during fast retrain<br>01 = PHY signals Local Fault during fast retrain<br>00 = PHY signals IDLE during fast retrain | R/W              |
| 1.147.0     | Fast retrain enable      | 1 = Fast retrain capability is enabled<br>0 = Fast retrain capability is disabled                                                                                        | R/W              |

<sup>a</sup> RO = Read only, R/W = Read/Write, NR = Non Roll-over

##### 45.2.1.94.1 LP fast retrain count (1.147.15:11)

These bits map to `fr_rx_counter` as defined in 55.4.5.4 for 10GBASE-T, 113.4.5.4 for 25GBASE-T and 40GBASE-T, and 126.4.5.4 for 2.5GBASE-T and 5GBASE-T. The counter is a 5-bit count of the number of fast retrains requested by the link partner. These bits shall be reset to all zeros when read or upon execution of the PMA reset. These bits shall be held at all ones in the case of overflow.

##### 45.2.1.94.2 LD fast retrain count (1.147.10:6)

These bits map to `fr_tx_counter` as defined in 55.4.5.4 for 10GBASE-T, 113.4.5.4 for 25GBASE-T and 40GBASE-T, and 126.4.5.4 for 2.5GBASE-T and 5GBASE-T. The counter is a 5-bit count of the number of fast retrains requested by the local device. These bits shall be reset to all zeros when read or upon execution of the PMA reset. These bits shall be held at all ones in the case of overflow.

#### 45.2.1.94.3 Fast retrain ability (1.147.4)

When read as a one, bit 1.147.4 indicates that the PHY supports fast retrain, as defined in 55.4.2.5.15. When read as a zero, bit 1.147.4 indicates that the PHY does not support fast retrain.

#### 45.2.1.94.4 Fast retrain negotiated (1.147.3)

When read as a one, bit 1.147.3 indicates that the PHY negotiated fast retrain, as defined in 55.4.2.5.15 during the most recent auto-negotiation. This is the condition where both the local device indicated fast retrain ability (bit 7.32.1 is one) and the link partner indicated fast retrain ability (bit 7.33.1 is one). When read as a zero, bit 1.147.3 indicates that the PHY did not negotiate fast retrain. See 45.2.7.10.14.

#### 45.2.1.94.5 Fast retrain signal type (1.147.2:1)

For PHYs that support fast retrain, these bits map to fr\_sigtype as defined in 55.3.6.2.2, 113.3.6.2.2, and 126.3.6.2.2. When Fast retrain signal type is set to 00, the PMA sends IDLE characters on the receive path during fast retrain. When Fast retrain signal type is set to 01, the PMA sends Local Fault on the receive path during fast retrain. When Fast retrain signal type is set to 10, the PMA sends Link Interruption on the receive path during fast retrain.

#### 45.2.1.94.6 Fast retrain enable (1.147.0)

For PHYs that support fast retrain, bit 1.147.0 controls fr\_enable as defined in 55.4.5.1, 113.4.5.1, and 126.4.5.1. When PMA reset is executed, this bit is set to one.

NOTE—Setting this bit to zero while a link is up will cause the PHY to stop supporting fast retrain, and the link will drop if the link partner initiates a fast retrain.

#### 45.2.1.95 BASE-R PMD control register (Register 1.150)

The BASE-R PMD control register is used for 10GBASE-KR and other PHY types using the PMDs described in Clause 72, Clause 84, Clause 85, Clause 92, Clause 93, Clause 94, Clause 110, Clause 111, Clause 136, or Clause 137.

The BASE-R PMD control register is also used by 5GBASE-KR described in Clause 130 to disable the transmitter equalizer for test purposes. 5GBASE-KR does not use the startup protocol.

The assignment of bits in the BASE-R PMD control register is shown in Table 45–75.

**Table 45–75—BASE-R PMD control register**

| Bit(s)     | Name                          | Description                                                                       | R/W <sup>a</sup> |
|------------|-------------------------------|-----------------------------------------------------------------------------------|------------------|
| 1.150.15:3 | Reserved                      | Value always 0                                                                    | RO               |
| 1.150.2    | Transmitter equalizer disable | 1 = Disable the 5GBASE-KR transmitter equalizer<br>0 = Normal operation           | R/W              |
| 1.150.1    | Training enable               | 1 = Enable the BASE-R startup protocol<br>0 = Disable the BASE-R startup protocol | R/W              |
| 1.150.0    | Restart training              | 1 = Restart BASE-R startup protocol<br>0 = Normal operation                       | R/W<br>SC        |

<sup>a</sup>R/W = Read/Write, SC = Self-clearing, RO = Read only

#### 45.2.1.95.1 Restart training (1.150.0)

This bit maps to the state variable `mr_restart_training` as defined in 72.6.10.3.1.

#### 45.2.1.95.2 Training enable (1.150.1)

This bit maps to the state variable `mr_training_enable` as defined in 72.6.10.3.1.

#### 45.2.1.95.3 Transmitter equalizer disable (1.150.2)

When bit 1.150.2 is set to one, 5GBASE-KR transmitter equalization is disabled. The default value of bit 1.150.2 is zero.

#### 45.2.1.96 BASE-R PMD status register (Register 1.151)

The BASE-R PMD status register is used for 10GBASE-KR and other PHY types using the PMDs described in Clause 72, Clause 84, Clause 85, Clause 92, Clause 93, Clause 94, Clause 110, Clause 111, Clause 136, or Clause 137. The assignment of bits in the BASE-R PMD status register is shown in Table 45–76.

**Table 45–76—BASE-R PMD status register**

| Bit(s)   | Name                      | Description                                                                                                  | R/W <sup>a</sup> |
|----------|---------------------------|--------------------------------------------------------------------------------------------------------------|------------------|
| 1.151.15 | Training failure 3        | 1 = Training failure has been detected for lane 3<br>0 = Training failure has not been detected for lane 3   | RO               |
| 1.151.14 | Startup protocol status 3 | 1 = Startup protocol in progress for lane 3<br>0 = Startup protocol complete for lane 3                      | RO               |
| 1.151.13 | Frame lock 3              | 1 = Training frame delineation detected for lane 3<br>0 = Training frame delineation not detected for lane 3 | RO               |
| 1.151.12 | Receiver status 3         | 1 = Receiver trained and ready to receive data for lane 3<br>0 = Receiver training for lane 3                | RO               |
| 1.151.11 | Training failure 2        | 1 = Training failure has been detected for lane 2<br>0 = Training failure has not been detected for lane 2   | RO               |
| 1.151.10 | Startup protocol status 2 | 1 = Startup protocol in progress for lane 2<br>0 = Startup protocol complete for lane 2                      | RO               |
| 1.151.9  | Frame lock 2              | 1 = Training frame delineation detected for lane 2<br>0 = Training frame delineation not detected for lane 2 | RO               |
| 1.151.8  | Receiver status 2         | 1 = Receiver trained and ready to receive data for lane 2<br>0 = Receiver training for lane 2                | RO               |
| 1.151.7  | Training failure 1        | 1 = Training failure has been detected for lane 1<br>0 = Training failure has not been detected for lane 1   | RO               |
| 1.151.6  | Startup protocol status 1 | 1 = Startup protocol in progress for lane 1<br>0 = Startup protocol complete for lane 1                      | RO               |
| 1.151.5  | Frame lock 1              | 1 = Training frame delineation detected for lane 1<br>0 = Training frame delineation not detected for lane 1 | RO               |
| 1.151.4  | Receiver status 1         | 1 = Receiver trained and ready to receive data for lane 1<br>0 = Receiver training for lane 1                | RO               |
| 1.151.3  | Training failure 0        | 1 = Training failure has been detected for lane 0<br>0 = Training failure has not been detected for lane 0   | RO               |

**Table 45–76—BASE-R PMD status register (*continued*)**

| Bit(s)  | Name                      | Description                                                                                                  | R/W <sup>a</sup> |
|---------|---------------------------|--------------------------------------------------------------------------------------------------------------|------------------|
| 1.151.2 | Startup protocol status 0 | 1 = Startup protocol in progress for lane 0<br>0 = Startup protocol complete for lane 0                      | RO               |
| 1.151.1 | Frame lock 0              | 1 = Training frame delineation detected for lane 0<br>0 = Training frame delineation not detected for lane 0 | RO               |
| 1.151.0 | Receiver status 0         | 1 = Receiver trained and ready to receive data for lane 0<br>0 = Receiver training for lane 0                | RO               |

<sup>a</sup>RO = Read only

#### **45.2.1.96.1 Receiver status 0 (1.151.0)**

This bit maps to the state variable rx\_trained as defined in 72.6.10.3.1 and local\_trained in 136.8.11.7.1.

#### **45.2.1.96.2 Frame lock 0 (1.151.1)**

This bit maps to the state variable frame\_lock as defined in 72.6.10.3.1 and local\_tf\_lock in 136.8.11.7.1.

#### **45.2.1.96.3 Startup protocol status 0 (1.151.2)**

This bit maps to the state variable training as defined in 72.6.10.3.1 and 136.8.11.7.1.

#### **45.2.1.96.4 Training failure 0 (1.151.3)**

This bit maps to the state variable training\_failure as defined in 72.6.10.3.1 and 136.8.11.7.1.

#### **45.2.1.96.5 Receiver status 1, 2, 3 (1.151.4, 1.151.8, 1.151.12)**

These bits are defined identically to 1.151.0 for lanes 1, 2, and 3 respectively.

#### **45.2.1.96.6 Frame lock 1, 2, 3 (1.151.5, 1.151.9, 1.151.13)**

These bits are defined identically to 1.151.1 for lanes 1, 2, and 3 respectively.

#### **45.2.1.96.7 Startup protocol status 1, 2, 3 (1.151.6, 1.151.10, 1.151.14)**

These bits are defined identically to 1.151.2 for lanes 1, 2, and 3 respectively.

#### **45.2.1.96.8 Training failure 1, 2, 3 (1.151.7, 1.151.11, 1.151.15)**

These bits are defined identically to 1.151.3 for lanes 1, 2, and 3 respectively.

#### **45.2.1.97 BASE-R LP coefficient update, lane 0 register (Register 1.152)**

The BASE-R LP coefficient update, lane 0 register is used for 10GBASE-KR and other PHY types using the PMDs described in Clause 72, Clause 84, Clause 85, Clause 92, Clause 93, Clause 94, Clause 110, or Clause 111. The BASE-R LP coefficient update, lane 0 register reflects the contents of the first 16-bit word of the training frame most recently received from the control channel for lane 0 or for a single-lane PHY.



The assignment of bits in the BASE-R LP coefficient update, lane 0 register is shown in Table 45–77. Normally the bits in this register are read only; however, when training is disabled by setting low bit 1 in the BASE-R PMD control register, the BASE-R LP coefficient update, lane 0 register becomes writeable.

A copy of this register may be implemented at address 1.1100 to assist PHY access for devices using post-read-increment-address access for a multi-lane PCS. If implemented, all accesses to the copy have identical behavior as accesses to the original register.

**Table 45–77—BASE-R LP coefficient update, lane 0 register bit definitions**

| Bit(s)      | Name                    | Description                                                               | R/W <sup>a</sup> |
|-------------|-------------------------|---------------------------------------------------------------------------|------------------|
| 1.152.15:14 | Reserved                | Value always 0                                                            | RO               |
| 1.152.13    | Preset                  | 1 = Preset coefficients<br>0 = Normal operation                           | R/W              |
| 1.152.12    | Initialize              | 1 = Initialize coefficients<br>0 = Normal operation                       | R/W              |
| 1.152.11:6  | Reserved                | Value always 0                                                            | RO               |
| 1.152.5:4   | Coefficient (+1) update | 5 4<br>1 1 = reserved<br>0 1 = increment<br>1 0 = decrement<br>0 0 = hold | R/W              |
| 1.152.3:2   | Coefficient (0) update  | 3 2<br>1 1 = reserved<br>0 1 = increment<br>1 0 = decrement<br>0 0 = hold | R/W              |
| 1.152.1:0   | Coefficient (–1) update | 1 0<br>1 1 = reserved<br>0 1 = increment<br>1 0 = decrement<br>0 0 = hold | R/W              |

<sup>a</sup>R/W = Read/Write, RO = Read only

#### 45.2.1.97.1 Preset (1.152.13)

The preset control bit requests that the coefficients be set to a state where equalization is turned off. The function and values of the preset bit is defined in 72.6.10.2.3.1.

#### 45.2.1.97.2 Initialize (1.152.12)

The initialize control is sent to request that the coefficients be set to configure the transmit equalizer to its INITIALIZE state. The function and values of the initialize bit is defined in 72.6.10.2.3.2.

#### 45.2.1.97.3 Coefficient (k) update (1.152.5:0)

Each coefficient, k, is assigned a 2-bit field describing a requested update. Three request encodings are defined: increment, decrement, and hold. The valid range for k is –1 to +1 where k = 0 denotes the main, or gain, tap. The function and values of the coefficient (k) update bits are defined in 72.6.10.2.3.3.

#### 45.2.1.98 BASE-R LP status report, lane 0 register (Register 1.153)

The BASE-R LP status report, lane 0 register is used for 10GBASE-KR and other PHY types using the PMDs described in Clause 72, Clause 84, Clause 85, Clause 92, Clause 93, Clause 94, Clause 110, or Clause 111. The BASE-R LP status report, lane 0 register reflects the contents of the second 16-bit word of the training frame most recently received from the control channel for lane 0 or for a single-lane PHY

The assignment of bits in the BASE-R LP status report, lane 0 register is shown in Table 45–78.

A copy of this register may be implemented at address 1.1200 to assist PHY access for devices using post-read-increment-address access for a multi-lane PCS. If implemented, all accesses to the copy have identical behavior as the original register.

**Table 45–78—BASE-R LP status report, lane 0 register bit definitions**

| Bit(s)     | Name                    | Description                                                                                                                                              | R/W <sup>a</sup> |
|------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.153.15   | Receiver ready          | 1 = The LP receiver has determined that training is complete and is prepared to receive data<br>0 = The LP receiver is requesting that training continue | RO               |
| 1.153.14:6 | Reserved                | Value always 0                                                                                                                                           | RO               |
| 1.153.5:4  | Coefficient (+1) status | 5 4<br>1 1 = maximum<br>1 0 = minimum<br>0 1 = updated<br>0 0 = not_updated                                                                              | RO               |
| 1.153.3:2  | Coefficient (0) status  | 3 2<br>1 1 = maximum<br>1 0 = minimum<br>0 1 = updated<br>0 0 = not_updated                                                                              | RO               |
| 1.153.1:0  | Coefficient (–1) status | 1 0<br>1 1 = maximum<br>1 0 = minimum<br>0 1 = updated<br>0 0 = not_updated                                                                              | RO               |

<sup>a</sup>RO = Read only

##### 45.2.1.98.1 Receiver ready (1.153.15)

The function and values for the receiver ready bit are defined in 72.6.10.2.4.4.

##### 45.2.1.98.2 Coefficient (k) status (1.153.5:0)

The function and values for the coefficient status bits are defined in 72.6.10.2.4.5.

#### 45.2.1.99 BASE-R LD coefficient update, lane 0 register (Register 1.154)

The BASE-R LD coefficient update, lane 0 register is used for 10GBASE-KR and other PHY types using the PMDs described in Clause 72, Clause 84, Clause 85, Clause 92, Clause 93, Clause 94, Clause 110, or Clause 111. The BASE-R LD coefficient update, lane 0 register reflects the contents of the first 16-bit word

of the outgoing training frame as defined by the LD receiver adaptation process in 72.6.10.2.5 for lane 0 or for a single-lane PHY.

The assignment of bits in the BASE-R LD coefficient update, lane 0 register is shown in Table 45–79.

A copy of this register may be implemented at address 1.1300 to assist PHY access for devices using post-read-increment-address access for a multi-lane PCS. If implemented, all accesses to the copy have identical behavior as the original register.

**Table 45–79—BASE-R LD coefficient update, lane 0 register bit definitions**

| Bit(s)      | Name                    | Description                                                               | R/W <sup>a</sup> |
|-------------|-------------------------|---------------------------------------------------------------------------|------------------|
| 1.154.15:14 | Reserved                | Value always 0                                                            | RO               |
| 1.154.13    | Preset                  | 1 = Preset coefficients<br>0 = Normal operation                           | RO               |
| 1.154.12    | Initialize              | 1 = Initialize coefficients<br>0 = Normal operation                       | RO               |
| 1.154.11:6  | Reserved                | Value always 0                                                            | RO               |
| 1.154.5:4   | Coefficient (+1) update | 5 4<br>1 1 = reserved<br>0 1 = increment<br>1 0 = decrement<br>0 0 = hold | RO               |
| 1.154.3:2   | Coefficient (0) update  | 3 2<br>1 1 = reserved<br>0 1 = increment<br>1 0 = decrement<br>0 0 = hold | RO               |
| 1.154.1:0   | Coefficient (–1) update | 1 0<br>1 1 = reserved<br>0 1 = increment<br>1 0 = decrement<br>0 0 = hold | RO               |

<sup>a</sup>RO = Read only

#### 45.2.1.99.1 Preset (1.154.13)

The function and values of the preset bit is defined in 72.6.10.2.3.1.

#### 45.2.1.99.2 Initialize (1.154.12)

The function and values of the initialize bit is defined in 72.6.10.2.3.2.

#### 45.2.1.99.3 Coefficient (k) update(1.154.5:0)

The function and values of the coefficient (k) update bits are defined in 72.6.10.2.3.3.

#### 45.2.1.100 BASE-R LD status report, lane 0 register (Register 1.155)

The BASE-R LD status report, lane 0 register is used for 10GBASE-KR and other PHY types using the PMDs described in Clause 72, Clause 84, Clause 85, Clause 92, Clause 93, Clause 94, Clause 110, or Clause 111. The BASE-R LD status report, lane 0 register reflects the contents of the second 16-bit word of the current outgoing training frame, as defined in the training state diagram in Figure 72–5 for lane 0 or for a single-lane PHY.

The assignment of bits in the BASE-R LD status report, lane 0 register is shown in Table 45–80.

A copy of this register may be implemented at address 1.1400 to assist PHY access for devices using post-read-increment-address access for a multi-lane PCS. If implemented, all accesses to the copy have identical behavior as the original register.

**Table 45–80—BASE-R LD status report, lane 0 register bit definitions**

| Bit(s)     | Name                    | Description                                                                                                                                              | R/W <sup>a</sup> |
|------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.155.15   | Receiver ready          | 1 = The LD receiver has determined that training is complete and is prepared to receive data<br>0 = The LD receiver is requesting that training continue | RO               |
| 1.155.14:6 | Reserved                | Value always 0                                                                                                                                           | RO               |
| 1.155.5:4  | Coefficient (+1) status | 5 4<br>1 1 = maximum<br>1 0 = minimum<br>0 1 = updated<br>0 0 = not_updated                                                                              | RO               |
| 1.155.3:2  | Coefficient (0) status  | 3 2<br>1 1 = maximum<br>1 0 = minimum<br>0 1 = updated<br>0 0 = not_updated                                                                              | RO               |
| 1.155.1:0  | Coefficient (–1) status | 1 0<br>1 1 = maximum<br>1 0 = minimum<br>0 1 = updated<br>0 0 = not_updated                                                                              | RO               |

<sup>a</sup>RO = Read only

##### 45.2.1.100.1 Receiver ready (1.155.15)

The function and values for the receiver ready bit are defined in 72.6.10.2.4.4.

##### 45.2.1.100.2 Coefficient (k) status (1.155.5:0)

The function and values for the coefficient status bits are defined in 72.6.10.2.4.5.

#### 45.2.1.101 BASE-R PMD status 2 register (Register 1.156)

The BASE-R PMD status 2 register is used for 100GBASE-CR10 and other PHY types using the PMDs described in Clause 72, Clause 84, or Clause 85 over more than 4 lanes. The assignment of bits in the BASE-R PMD status 2 register is shown in Table 45–81.

**Table 45–81—BASE-R PMD status 2 register**

| Bit(s)   | Name                      | Description                                                                                                  | R/W <sup>a</sup> |
|----------|---------------------------|--------------------------------------------------------------------------------------------------------------|------------------|
| 1.156.15 | Training failure 7        | 1 = Training failure has been detected for lane 7<br>0 = Training failure has not been detected for lane 7   | RO               |
| 1.156.14 | Startup protocol status 7 | 1 = Startup protocol in progress for lane 7<br>0 = Startup protocol complete for lane 7                      | RO               |
| 1.156.13 | Frame lock 7              | 1 = Training frame delineation detected for lane 7<br>0 = Training frame delineation not detected for lane 7 | RO               |
| 1.156.12 | Receiver status 7         | 1 = Receiver trained and ready to receive data for lane 7<br>0 = Receiver training for lane 7                | RO               |
| 1.156.11 | Training failure 6        | 1 = Training failure has been detected for lane 6<br>0 = Training failure has not been detected for lane 6   | RO               |
| 1.156.10 | Startup protocol status 6 | 1 = Startup protocol in progress for lane 6<br>0 = Startup protocol complete for lane 6                      | RO               |
| 1.156.9  | Frame lock 6              | 1 = Training frame delineation detected for lane 6<br>0 = Training frame delineation not detected for lane 6 | RO               |
| 1.156.8  | Receiver status 6         | 1 = Receiver trained and ready to receive data for lane 6<br>0 = Receiver training for lane 6                | RO               |
| 1.156.7  | Training failure 5        | 1 = Training failure has been detected for lane 5<br>0 = Training failure has not been detected for lane 5   | RO               |
| 1.156.6  | Startup protocol status 5 | 1 = Startup protocol in progress for lane 5<br>0 = Startup protocol complete for lane 5                      | RO               |
| 1.156.5  | Frame lock 5              | 1 = Training frame delineation detected for lane 5<br>0 = Training frame delineation not detected for lane 5 | RO               |
| 1.156.4  | Receiver status 5         | 1 = Receiver trained and ready to receive data for lane 5<br>0 = Receiver training for lane 5                | RO               |
| 1.156.3  | Training failure 4        | 1 = Training failure has been detected for lane 4<br>0 = Training failure has not been detected for lane 4   | RO               |
| 1.156.2  | Startup protocol status 4 | 1 = Startup protocol in progress for lane 4<br>0 = Startup protocol complete for lane 4                      | RO               |
| 1.156.1  | Frame lock 4              | 1 = Training frame delineation detected for lane 4<br>0 = Training frame delineation not detected for lane 4 | RO               |
| 1.156.0  | Receiver status 4         | 1 = Receiver trained and ready to receive data for lane 4<br>0 = Receiver training for lane 4                | RO               |

<sup>a</sup>RO = Read only

#### 45.2.1.101.1 Receiver status 4, 5, 6, 7 (1.156.0, 1.156.4, 1.156.8, 1.156.12)

These bits are defined identically to 1.151.0 for lanes 4, 5, 6, and 7 respectively.

#### 45.2.1.101.2 Frame lock 4, 5, 6, 7 (1.156.1, 1.156.5, 1.156.9, 1.156.13)

These bits are defined identically to 1.151.1 for lanes 4, 5, 6, and 7 respectively.

#### 45.2.1.101.3 Startup protocol status 4, 5, 6, 7 (1.156.2, 1.156.6, 1.156.10, 1.156.14)

These bits are defined identically to 1.151.2 for lanes 4, 5, 6, and 7 respectively.

#### 45.2.1.101.4 Training failure 4, 5, 6, 7 (1.156.3, 1.156.7, 1.156.11, 1.156.15)

These bits are defined identically to 1.151.3 for lanes 4, 5, 6, and 7 respectively.

#### 45.2.1.102 BASE-R PMD status 3 register (Register 1.157)

The BASE-R PMD status 3 register is used for 100GBASE-CR10 and other PHY types using the PMDs described in Clause 72, Clause 84, or Clause 85 over more than 8 lanes. The assignment of bits in the BASE-R PMD status 3 register is shown in Table 45–82.

**Table 45–82—BASE-R PMD status 3 register**

| Bit(s)     | Name                      | Description                                                                                                  | R/W <sup>a</sup> |
|------------|---------------------------|--------------------------------------------------------------------------------------------------------------|------------------|
| 1.157.15:8 | Reserved                  | Value always 0                                                                                               | RO               |
| 1.157.7    | Training failure 9        | 1 = Training failure has been detected for lane 9<br>0 = Training failure has not been detected for lane 9   | RO               |
| 1.157.6    | Startup protocol status 9 | 1 = Startup protocol in progress for lane 9<br>0 = Startup protocol complete for lane 9                      | RO               |
| 1.157.5    | Frame lock 9              | 1 = Training frame delineation detected for lane 9<br>0 = Training frame delineation not detected for lane 9 | RO               |
| 1.157.4    | Receiver status 9         | 1 = Receiver trained and ready to receive data for lane 9<br>0 = Receiver training for lane 9                | RO               |
| 1.157.3    | Training failure 8        | 1 = Training failure has been detected for lane 8<br>0 = Training failure has not been detected for lane 8   | RO               |
| 1.157.2    | Startup protocol status 8 | 1 = Startup protocol in progress for lane 8<br>0 = Startup protocol complete for lane 8                      | RO               |
| 1.157.1    | Frame lock 8              | 1 = Training frame delineation detected for lane 8<br>0 = Training frame delineation not detected for lane 8 | RO               |
| 1.157.0    | Receiver status 8         | 1 = Receiver trained and ready to receive data for lane 8<br>0 = Receiver training for lane 8                | RO               |

<sup>a</sup>RO = Read only

#### 45.2.1.102.1 Receiver status 8, 9 (1.157.0, 1.157.4)

These bits are defined identically to 1.151.0 for lanes 8 and 9 respectively.

#### 45.2.1.102.2 Frame lock 8, 9 (1.157.1, 1.157.5)

These bits are defined identically to 1.151.1 for lanes 8 and 9 respectively.

#### 45.2.1.102.3 Startup protocol status 8, 9 (1.157.2, 1.157.6)

These bits are defined identically to 1.151.2 for lanes 8 and 9 respectively.

#### 45.2.1.102.4 Training failure 8, 9 (1.157.3, 1.157.7)

These bits are defined identically to 1.151.3 for lanes 8 and 9 respectively.

#### 45.2.1.103 1000BASE-KX/2.5GBASE-KX control register (Register 1.160)

The assignment of bits in the 1000BASE-KX/2.5GBASE-KX control register is shown in Table 45–83.

**Table 45–83—1000BASE-KX/2.5GBASE-KX control register**

| Bit(s)     | Name                 | Description                                                     | R/W <sup>a</sup> |
|------------|----------------------|-----------------------------------------------------------------|------------------|
| 1.160.15:1 | Reserved             | Value always 0                                                  | RO               |
| 1.160.0    | PMD transmit disable | 1 = Disable transmitter output<br>0 = Enable transmitter output | R/W              |

<sup>a</sup>RO = Read only, R/W = Read/Write

##### 45.2.1.103.1 PMD transmit disable (1.160.0)

This bit disables the 1000BASE-KX transmitter as defined in 70.6.5 or the 2.5GBASE-KX transmitter as defined in 128.6.5.

#### 45.2.1.104 1000BASE-KX/2.5GBASE-KX status register (Register 1.161)

The assignment of bits in the 1000BASE-KX/2.5GBASE-KX status register is shown in Table 45–84.

**Table 45–84—1000BASE-KX/2.5GBASE-KX status register**

| Bit(s)      | Name                   | Description                                                                                                                                                             | R/W <sup>a</sup> |
|-------------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.161.15:14 | Reserved               | Value always 0                                                                                                                                                          | RO               |
| 1.161.13    | Transmit fault ability | 1 = PMA/PMD has the ability to detect a fault condition on the transmit path<br>0 = PMA/ PMD does not have the ability to detect a fault condition on the transmit path | RO               |
| 1.161.12    | Receive fault ability  | 1 = PMA/PMD has the ability to detect a fault condition on the receive path<br>0 = PMA/ PMD does not have the ability to detect a fault condition on the receive path   | RO               |
| 1.161.11    | Transmit fault         | 1 = Fault condition on transmit path<br>0 = No fault condition on transmit path                                                                                         | LH               |

**Table 45–84—1000BASE-KX/2.5GBASE-KX status register (*continued*)**

| Bit(s)    | Name                          | Description                                                                                                            | R/W <sup>a</sup> |
|-----------|-------------------------------|------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.161.10  | Receive fault                 | 1 = Fault condition on receive path<br>0 = No fault condition on receive path                                          | LH               |
| 1.161.9   | Reserved                      | Value always 0                                                                                                         | RO               |
| 1.161.8   | PMD transmit disable ability  | 1 = PMD has the ability to disable the transmit path<br>0 = PMD does not have the ability to disable the transmit path | RO               |
| 1.161.7:1 | Reserved                      | Value always 0                                                                                                         | RO               |
| 1.161.0   | Signal detect signal from PMD | 1 = PMD has asserted signal detect<br>0 = PMD has not asserted signal detect                                           | RO               |

<sup>a</sup>RO = Read only, LH = Latching high

#### **45.2.1.104.1 PMD transmit fault ability (1.161.13)**

When read as a one, bit 1.161.13 indicates that the PMA/PMD has the ability to detect a fault condition on the transmit path. When read as a zero, bit 1.161.13 indicates that the PMA/PMD does not have the ability to detect a fault condition on the transmit path.

#### **45.2.1.104.2 PMD receive fault ability (1.161.12)**

When read as a one, bit 1.161.12 indicates that the PMA/PMD has the ability to detect a fault condition on the receive path. When read as a zero, bit 1.161.12 indicates that the PMA/PMD does not have the ability to detect a fault condition on the receive path.

#### **45.2.1.104.3 PMD transmit fault (1.161.11)**

When read as a one, bit 1.161.11 indicates that the PMA/PMD has detected a fault condition on the transmit path. When read as a zero, bit 1.161.11 indicates that the PMA/PMD has not detected a fault condition on the transmit path. Detection of a fault condition on the transmit path is optional and the ability to detect such a condition is advertised by bit 1.161.13. A PMA/PMD that is unable to detect a fault condition on the transmit path shall return a value of zero for this bit. The transmit fault bit shall be implemented with latching high behavior.

The default value of bit 1.161.11 is zero.

#### **45.2.1.104.4 PMD receive fault (1.161.10)**

When read as a one, bit 1.161.10 indicates that the PMA/PMD has detected a fault condition on the receive path. When read as a zero, bit 1.161.10 indicates that the PMA/PMD has not detected a fault condition on the receive path. Detection of a fault condition on the receive path is optional and the ability to detect such a condition is advertised by bit 1.161.12. A PMA/PMD that is unable to detect a fault condition on the receive path shall return a value of zero for this bit. The receive fault bit shall be implemented with latching high behavior.

The default value of bit 1.161.10 is zero.



#### 45.2.1.104.5 PMD transmit disable ability (1.161.8)

When read as a one, bit 1.161.8 indicates that the PMD is able to perform the transmit disable function. When read as a zero, bit 1.161.8 indicates that the PMD is not able to perform the transmit disable function.

#### 45.2.1.104.6 1000BASE-KX/2.5GBASE-KX signal detect (1.161.0)

The PMD signal detect function for both 1000BASE-KX PMD (see 70.6.4) and 2.5GBASE-KX PMD (see 128.6.4) is mandatory if EEE is implemented and optional otherwise. The 1000BASE-X PCS and 2.5GBASE-X PCS require signal detect to be one before synchronization can occur. If the signal detect function is not implemented this bit is set to one.

#### 45.2.1.105 PMA overhead control 1, 2, and 3 registers (Register 1.162 through 1.164)

Assignment of bits in the PMA overhead control 1, 2, and 3 registers is shown in Table 45–85. These bits shall be reset to the default values indicated in Table 45–85 upon PHY reset. For the 100GBASE-KP4 PHY the use of these registers is specified in 94.2.2.3 and 94.2.3.1.

**Table 45–85—PMA overhead control 1, 2, and 3 register bit definitions**

| Bit(s)      | Name                             | Description                                                         | R/W <sup>a</sup> |
|-------------|----------------------------------|---------------------------------------------------------------------|------------------|
| 1.162.15:13 | Reserved                         | Value always 0                                                      | RO               |
| 1.162.12:8  | PMA transmit overhead sequence 0 | Sequence of overhead groups for lane 0<br>Default = 00110           | R/W              |
| 1.162.7:0   | PMA transmit overhead pattern    | Bit pattern for 8-bit transmit overhead group<br>Default = 01100110 | R/W              |
| 1.163.15    | Reserved                         | Value always 0                                                      | RO               |
| 1.163.14:10 | PMA transmit overhead sequence 3 | Sequence of overhead groups for lane 3<br>Default = 11001           | R/W              |
| 1.163.9:5   | PMA transmit overhead sequence 2 | Sequence of overhead groups for lane 2<br>Default = 10101           | R/W              |
| 1.163.4:0   | PMA transmit overhead sequence 1 | Sequence of overhead groups for lane 1<br>Default = 01010           | R/W              |
| 1.164.15:8  | Reserved                         | Value always 0                                                      | RO               |
| 1.164.7:0   | PMA receive overhead pattern     | Bit pattern for 8-bit receive overhead group<br>Default = 01100110  | R/W              |

<sup>a</sup>R/W = Read/Write, RO = Read only

#### 45.2.1.106 PMA overhead status 1 and 2 registers (Register 1.165, 1.166)

Assignment of bits in the PMA overhead status 1 and 2 registers is shown in Table 45–86. These bits shall be reset to all zeros upon PHY reset. For the 100GBASE-KP4 PHY the use of these registers is specified in 94.2.3.1.

**Table 45–86—PMA overhead status 1 and 2 register bit definitions**

| Bit(s)      | Name                            | Description                            | R/W <sup>a</sup> |
|-------------|---------------------------------|----------------------------------------|------------------|
| 1.165.15:12 | Reserved                        | Value always 0                         | RO               |
| 1.165.11:6  | PMA receive overhead sequence 1 | Sequence of overhead groups for lane 1 | RO               |
| 1.165.5:0   | PMA receive overhead sequence 0 | Sequence of overhead groups for lane 0 | RO               |
| 1.166.15:12 | Reserved                        | Value always 0                         | RO               |
| 1.166.11:6  | PMA receive overhead sequence 3 | Sequence of overhead groups for lane 3 | RO               |
| 1.166.5:0   | PMA receive overhead sequence 2 | Sequence of overhead groups for lane 2 | RO               |

<sup>a</sup>RO = Read only

#### 45.2.1.107 BASE-R FEC ability register (Register 1.170)

The assignment of bits in the BASE-R FEC ability register is shown in Table 45–87.

**Table 45–87—BASE-R FEC ability register bit definitions**

| Bit(s)     | Name                                | Description                                                                                                  | R/W <sup>a</sup> |
|------------|-------------------------------------|--------------------------------------------------------------------------------------------------------------|------------------|
| 1.170.15:2 | Reserved                            | Value always 0                                                                                               | RO               |
| 1.170.1    | BASE-R FEC error indication ability | A read of one in this bit indicates that the sublayer is able to report FEC decoding errors to the PCS layer | RO               |
| 1.170.0    | BASE-R FEC ability                  | A read of one in this bit indicates that the sublayer supports FEC                                           | RO               |

<sup>a</sup>RO Read only

##### 45.2.1.107.1 BASE-R FEC ability (1.170.0)

When read as a one, this bit indicates that the sublayer supports forward error correction (FEC). When read as a zero, the sublayer does not support forward error correction.

##### 45.2.1.107.2 BASE-R FEC error indication ability (1.170.1)

When read as a one, this bit indicates that the BASE-R FEC sublayer is able to indicate decoding errors to the PCS layer (see 74.8.3). When read as a zero, the BASE-R FEC sublayer is not able to indicate decoding errors to the PCS layer. BASE-R FEC error indication is controlled by the FEC enable error indication bit in the BASE-R FEC control register (see 45.2.1.108.2).

#### 45.2.1.108 BASE-R FEC control register (Register 1.171)

The assignment of bits in the BASE-R FEC control register is shown in Table 45–88.

**Table 45–88—BASE-R FEC control register bit definitions**

| Bit(s)     | Name                        | Description                                                                               | R/W <sup>a</sup> |
|------------|-----------------------------|-------------------------------------------------------------------------------------------|------------------|
| 1.171.15:2 | Reserved                    | Value always 0                                                                            | RO               |
| 1.171.1    | FEC enable error indication | A write of one to this bit configures the FEC decoder to indicate errors to the PCS layer | R/W              |
| 1.171.0    | FEC enable                  | A write of one to this bit enables FEC<br>A write of zero to this bit disables FEC        | R/W              |

<sup>a</sup>R/W = Read/Write, RO Read only

##### 45.2.1.108.1 FEC enable (1.171.0)

When written as a one, this bit enables FEC for the BASE-R PHY. When written as a zero, FEC is disabled in the BASE-R PHY. This bit shall be set to zero upon execution of PHY reset.

##### 45.2.1.108.2 FEC enable error indication (1.171.1)

This bit enables the BASE-R FEC decoder to indicate decoding errors to the upper layers (PCS) through the sync bits for the BASE-R PHY in the Local Device. When written as a one, this bit enables indication of decoding errors through the sync bits to the PCS layer. When written as zero the error indication function is disabled. Writes to this bit are ignored and reads return a zero if the BASE-R FEC does not have the ability to indicate decoding errors to the PCS layer (see 45.2.1.107.2 and 74.8.3).

#### 45.2.1.109 Single-lane PHY BASE-R FEC corrected blocks counter (Register 1.172, 1.173)

The assignment of bits in the single-lane PHY BASE-R FEC corrected blocks counter register is shown in Table 45–89. See 74.8.4.1 for a definition of this register. These bits shall be reset to all zeros when the register is read by the management function or upon PHY reset. These bits shall be held at all ones in the case of overflow. Registers 1.172, 1.173 are used to read the value of a 32-bit counter. When registers 1.172 and 1.173 are used to read the 32-bit counter value, the register 1.172 is read first, the value of the register 1.173 is latched when (and only when) register 1.172 is read and reads of register 1.173 returns the latched value rather than the current value of the counter.

**Table 45–89—Single-lane PHY BASE-R FEC corrected blocks counter register bit definitions**

| Bit(s)     | Name                       | Description                         | R/W <sup>a</sup> |
|------------|----------------------------|-------------------------------------|------------------|
| 1.172.15:0 | FEC corrected blocks lower | FEC_corrected_blocks_counter[15:0]  | RO,<br>NR        |
| 1.173.15:0 | FEC corrected blocks upper | FEC_corrected_blocks_counter[31:16] | RO,<br>NR        |

<sup>a</sup>RO = Read only, NR = Non Roll-over

For a multi-PCS lane PHY, this register may be a copy of register 1.300 BASE-R FEC corrected blocks counter, lane 0. If implemented, all accesses to the copy shall have identical behavior as the original register.

#### 45.2.1.110 Single-lane PHY BASE-R FEC uncorrected blocks counter (Register 1.174, 1.175)

The assignment of bits in the single-lane PHY BASE-R FEC uncorrected blocks counter register is shown in Table 45–90. See 74.8.4.2 for a definition of this register. These bits shall be reset to all zeros when the register is read by the management function or upon PHY reset. These bits shall be held at all ones in the case of overflow. Registers 1.174, 1.175 are used to read the value of a 32-bit counter. When registers 1.174 and 1.175 are used to read the 32-bit counter value, the register 1.174 is read first, the value of the register 1.175 is latched when (and only when) register 1.174 is read and reads of register 1.175 returns the latched value rather than the current value of the counter.

**Table 45–90—Single-lane PHY BASE-R FEC uncorrected blocks counter register bit definitions**

| Bit(s)     | Name                         | Description                           | R/W <sup>a</sup> |
|------------|------------------------------|---------------------------------------|------------------|
| 1.174.15:0 | FEC uncorrected blocks lower | FEC_uncorrected_blocks_counter[15:0]  | RO, NR           |
| 1.175.15:0 | FEC uncorrected blocks upper | FEC_uncorrected_blocks_counter[31:16] | RO, NR           |

<sup>a</sup>RO = Read only, NR = Non Roll-over

For a multi-PCS lane PHY, this register may be a copy of register 1.700 BASE-R FEC uncorrected blocks counter, lane 0. If implemented, all accesses to the copy shall have identical behavior as the original register.

#### 45.2.1.111 CAUI-4 C2M and 25GAUI C2M recommended CTLE register (Register 1.179)

The assignment of bits in the CAUI-4 C2M and 25GAUI C2M recommended CTLE register is shown in Table 45–91.

**Table 45–91—CAUI-4 C2M and 25GAUI C2M recommended CTLE register bit definitions**

| Bit(s)     | Name                     | Description                                                                                                                                                                                                                                 | R/W <sup>a</sup> |
|------------|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.179.15:5 | Reserved                 | Value always 0                                                                                                                                                                                                                              | RO               |
| 1.179.4:1  | Recommended CTLE peaking | 4 3 2 1<br>1 1 x x = Reserved<br>1 0 1 x = Reserved<br>1 0 0 1 = 9 dB<br>1 0 0 0 = 8 dB<br>0 1 1 1 = 7 dB<br>0 1 1 0 = 6 dB<br>0 1 0 1 = 5 dB<br>0 1 0 0 = 4 dB<br>0 0 1 1 = 3 dB<br>0 0 1 0 = 2 dB<br>0 0 0 1 = 1 dB<br>0 0 0 0 = Reserved | R/W              |
| 1.179.0    | Reserved                 | Value always 0                                                                                                                                                                                                                              | RO               |

<sup>a</sup>R/W = Read/Write, RO = Read only

#### 45.2.1.111.1 Recommended CTLE peaking (1.179.4:1)

The value of these bits sets the CTLE peaking value recommended by a host that implements the optional CAUI-4 C2M and 25GAUI C2M interface defined in Annex 83E and Annex 109B, respectively (see 83E.3.1.6). The module may optionally use this information to adjust its CTLE setting.

#### 45.2.1.112 25GAUI C2C and lane 0 CAUI-4 C2C transmitter equalization, receive direction register (Register 1.180)

The assignment of bits in the 25GAUI C2C and lane 0 CAUI-4 C2C transmitter equalization, receive direction register is shown in Table 45–92. The transmitter, receive direction, is the transmitter that sends data towards the PCS.

**Table 45–92—25GAUI C2C and lane 0 CAUI-4 C2C transmitter equalization, receive direction register bit definitions**

| Bit(s)      | Name                       | Description                                                                                                                                                                                                                                                                                                                                                                      | R/W <sup>a</sup> |
|-------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.180.15    | Request flag               | 1 = Change in equalization is requested<br>0 = No change in equalization is requested                                                                                                                                                                                                                                                                                            | RO               |
| 1.180.14:12 | Post-cursor request        | 14 13 12<br>1 1 1 Reserved<br>1 1 0 Reserved<br>1 0 1 <i>Requested_eq_cl</i> = 5 (c(1) ratio –0.25)<br>1 0 0 <i>Requested_eq_cl</i> = 4 (c(1) ratio –0.2)<br>0 1 1 <i>Requested_eq_cl</i> = 3 (c(1) ratio –0.15)<br>0 1 0 <i>Requested_eq_cl</i> = 2 (c(1) ratio –0.1)<br>0 0 1 <i>Requested_eq_cl</i> = 1 (c(1) ratio –0.05)<br>0 0 0 <i>Requested_eq_cl</i> = 0 (c(1) ratio 0) | RO               |
| 1.180.11:10 | Pre-cursor request         | 11 10<br>1 1 <i>Requested_eq_cm1</i> = 3 (c(–1) ratio –0.15)<br>1 0 <i>Requested_eq_cm1</i> = 2 (c(–1) ratio –0.1)<br>0 1 <i>Requested_eq_cm1</i> = 1 (c(–1) ratio –0.05)<br>0 0 <i>Requested_eq_cm1</i> = 0 (c(–1) ratio 0)                                                                                                                                                     | RO               |
| 1.180.9:7   | Post-cursor remote setting | 9 8 7<br>1 1 1 Reserved<br>1 1 0 Reserved<br>1 0 1 <i>Remote_eq_cl</i> = 5 (c(1) ratio –0.25)<br>1 0 0 <i>Remote_eq_cl</i> = 4 (c(1) ratio –0.2)<br>0 1 1 <i>Remote_eq_cl</i> = 3 (c(1) ratio –0.15)<br>0 1 0 <i>Remote_eq_cl</i> = 2 (c(1) ratio –0.1)<br>0 0 1 <i>Remote_eq_cl</i> = 1 (c(1) ratio –0.05)<br>0 0 0 <i>Remote_eq_cl</i> = 0 (c(1) ratio 0)                      | R/W              |
| 1.180.6:5   | Pre-cursor remote setting  | 6 5<br>1 1 <i>Remote_eq_cm1</i> = 3 (c(–1) ratio –0.15)<br>1 0 <i>Remote_eq_cm1</i> = 2 (c(–1) ratio –0.1)<br>0 1 <i>Remote_eq_cm1</i> = 1 (c(–1) ratio –0.05)<br>0 0 <i>Remote_eq_cm1</i> = 0 (c(–1) ratio 0)                                                                                                                                                                   | R/W              |

**Table 45–92—25GAUI C2C and lane 0 CAUI-4 C2C transmitter equalization, receive direction register bit definitions (*continued*)**

| Bit(s)    | Name                      | Description                                                                                                                                                                                                                                                                                                                                           | R/W <sup>a</sup> |
|-----------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.180.4:2 | Post-cursor local setting | 4 3 2<br>1 1 1 Reserved<br>1 1 0 Reserved<br>1 0 1 <i>Local_eq_cl</i> = 5 (c(1) ratio –0.25)<br>1 0 0 <i>Local_eq_cl</i> = 4 (c(1) ratio –0.2)<br>0 1 1 <i>Local_eq_cl</i> = 3 (c(1) ratio –0.15)<br>0 1 0 <i>Local_eq_cl</i> = 2 (c(1) ratio –0.1)<br>0 0 1 <i>Local_eq_cl</i> = 1 (c(1) ratio –0.05)<br>0 0 0 <i>Local_eq_cl</i> = 0 (c(1) ratio 0) | R/W              |
| 1.180.1:0 | Pre-cursor local setting  | 1 0<br>1 1 <i>Local_eq_cm1</i> = 3 (c(–1) ratio –0.15)<br>1 0 <i>Local_eq_cm1</i> = 2 (c(–1) ratio –0.1)<br>0 1 <i>Local_eq_cm1</i> = 1 (c(–1) ratio –0.05)<br>0 0 <i>Local_eq_cm1</i> = 0 (c(–1) ratio 0)                                                                                                                                            | R/W              |

<sup>a</sup>R/W = Read/Write, RO = Read only

#### 45.2.1.112.1 Request flag (1.180.15)

The value of this bit indicates the value of the variable *Request\_flag* in the 25GAUI or lane 0 CAUI-4 receiver in the receive direction (see 83D.3.3.2). This indicates whether the 25GAUI or CAUI-4 C2C device is issuing a request to change the remote transmitter equalization in the 25GAUI or lane 0 CAUI-4 C2C transmitter in the receive direction. If a 25GAUI or lane 0 CAUI-4 receiver in the receive direction is not present in the package, then the value returned for this bit should be zero.

#### 45.2.1.112.2 Post-cursor request (1.180.14:12)

The value of these bits indicates the value of the variable *Requested\_eq\_cl* in the 25GAUI or lane 0 CAUI-4 receiver in the receive direction (see 83D.3.3.2). When *Request\_flag* is equal to one, this value indicates the ratio of the post-cursor coefficient c(1), which is requested for the transmitter equalization in the 25GAUI or lane 0 CAUI-4 C2C transmitter in the receive direction.

#### 45.2.1.112.3 Pre-cursor request (1.180.11:10)

The value of these bits indicates the value of the variable *Requested\_eq\_cm1* in the 25GAUI or lane 0 CAUI-4 receiver in the receive direction (see 83D.3.3.2). When *Request\_flag* is equal to one, this value indicates the ratio of the pre-cursor coefficient c(–1), which is requested for the transmitter equalization in the 25GAUI or lane 0 CAUI-4 C2C transmitter in the receive direction.

#### 45.2.1.112.4 Post-cursor remote setting (1.180.9:7)

The value of these bits sets the variable *Remote\_eq\_cl* for the 25GAUI or lane 0 CAUI-4 receiver in the receive direction (see 83D.3.3.2). This is used by a 25GAUI or CAUI-4 receiver that implements the optional transmitter equalization feedback as an indication of the ratio of the post-cursor coefficient c(1) being used in the 25GAUI or lane 0 of the CAUI-4 transmitter in the receive direction (see 83D.3.1.1). It may be used to generate values for the request flag and the request bits. If a 25GAUI or lane 0 CAUI-4 receiver in the receive direction is not present in the package, then these bits have no effect.

#### 45.2.1.112.5 Pre-cursor remote setting (1.180.6:5)

The value of these bits sets the variable *Remote\_eq\_cm1* for the 25GAUI or lane 0 CAUI-4 receiver in the receive direction (see 83D.3.3.2). This is used by a 25GAUI or CAUI-4 receiver that implements the optional transmitter equalization feedback as an indication of the ratio of the pre-cursor coefficient  $c(-1)$  being used in the 25GAUI or lane 0 of the CAUI-4 transmitter in the receive direction (see 83D.3.1.1). It may be used to generate values for the request flag and the request bits. If a 25GAUI or lane 0 CAUI-4 receiver in the receive direction is not present in the package, then these bits have no effect.

#### 45.2.1.112.6 Post-cursor local setting (1.180.4:2)

The value of these bits sets the variable *Local\_eq\_c1* for the 25GAUI or lane 0 CAUI-4 transmitter in the receive direction (see 83D.3.1.1 and Table 83D–3), which controls the weight of the transmitter equalization post-cursor coefficient  $c(1)$ . If a 25GAUI or lane 0 CAUI-4 transmitter in the receive direction is not present in the package, then these bits have no effect.

#### 45.2.1.112.7 Pre-cursor local setting (1.180.1:0)

The value of these bits sets the variable *Local\_eq\_cm1* for the 25GAUI or lane 0 CAUI-4 transmitter in the receive direction (see 83D.3.1.1 and Table 83D–2), which controls the weight of the transmitter equalization pre-cursor coefficient  $c(-1)$ . If a 25GAUI or lane 0 CAUI-4 transmitter in the receive direction is not present in the package, then these bits have no effect.

#### 45.2.1.113 CAUI-4 chip-to-chip transmitter equalization, receive direction, lane 1 through lane 3 registers (Registers 1.181, 1.182, 1.183)

The CAUI-4 chip-to-chip transmitter equalization, receive direction, lane 1 through lane 3 registers are defined similarly to register 1.180 (which is used for lane 0, see 45.2.1.112) but for lanes 1 through 3, respectively. The transmitter, receive direction, is the transmitter that sends data towards the PCS.

#### 45.2.1.114 25GAUI C2C and lane 0 CAUI-4 C2C transmitter equalization, transmit direction register (Register 1.184)

The assignment of bits in the 25GAUI C2C and lane 0 CAUI-4 C2C transmitter equalization, transmit direction register is shown in Table 45–93. The transmitter, transmit direction, is the transmitter that sends data towards the PMD.

##### 45.2.1.114.1 Request flag (1.184.15)

The value of this bit indicates the value of the variable *Request\_flag* in the 25GAUI or lane 0 CAUI-4 receiver in the transmit direction (see 83D.3.3.2). This indicates whether the 25GAUI or CAUI-4 C2C device is issuing a request to change the remote transmitter equalization in the 25GAUI or lane 0 CAUI-4 C2C transmitter in the transmit direction. If a 25GAUI or lane 0 CAUI-4 receiver in the transmit direction is not present in the package, then the value returned for this bit should be zero.

##### 45.2.1.114.2 Post-cursor request (1.184.14:12)

The value of these bits indicates the value of the variable *Requested\_eq\_c1* in the 25GAUI or lane 0 CAUI-4 receiver in the transmit direction (see 83D.3.3.2). When *Request\_flag* is equal to one, this value indicates the ratio of the post-cursor coefficient  $c(1)$ , which is requested for the transmitter equalization in the 25GAUI or lane 0 CAUI-4 C2C transmitter in the transmit direction.

**Table 45–93—25GAUI C2C and lane 0 CAUI-4 C2C transmitter equalization, transmit direction register bit definitions**

| Bit(s)      | Name                       | Description                                                                                                                                                                                                                                                                                                                                                                      | R/W <sup>a</sup> |
|-------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.184.15    | Request flag               | 1 = Change in equalization is requested<br>0 = No change in equalization is requested                                                                                                                                                                                                                                                                                            | RO               |
| 1.184.14:12 | Post-cursor request        | 14 13 12<br>1 1 1 Reserved<br>1 1 0 Reserved<br>1 0 1 <i>Requested_eq_c1</i> = 5 (c(1) ratio –0.25)<br>1 0 0 <i>Requested_eq_c1</i> = 4 (c(1) ratio –0.2)<br>0 1 1 <i>Requested_eq_c1</i> = 3 (c(1) ratio –0.15)<br>0 1 0 <i>Requested_eq_c1</i> = 2 (c(1) ratio –0.1)<br>0 0 1 <i>Requested_eq_c1</i> = 1 (c(1) ratio –0.05)<br>0 0 0 <i>Requested_eq_c1</i> = 0 (c(1) ratio 0) | RO               |
| 1.184.11:10 | Pre-cursor request         | 11 10<br>1 1 <i>Requested_eq_cm1</i> = 3 (c(–1) ratio –0.15)<br>1 0 <i>Requested_eq_cm1</i> = 2 (c(–1) ratio –0.1)<br>0 1 <i>Requested_eq_cm1</i> = 1 (c(–1) ratio –0.05)<br>0 0 <i>Requested_eq_cm1</i> = 0 (c(–1) ratio 0)                                                                                                                                                     | RO               |
| 1.184.9:7   | Post-cursor remote setting | 9 8 7<br>1 1 1 Reserved<br>1 1 0 Reserved<br>1 0 1 <i>Remote_eq_c1</i> = 5 (c(1) ratio –0.25)<br>1 0 0 <i>Remote_eq_c1</i> = 4 (c(1) ratio –0.2)<br>0 1 1 <i>Remote_eq_c1</i> = 3 (c(1) ratio –0.15)<br>0 1 0 <i>Remote_eq_c1</i> = 2 (c(1) ratio –0.1)<br>0 0 1 <i>Remote_eq_c1</i> = 1 (c(1) ratio –0.05)<br>0 0 0 <i>Remote_eq_c1</i> = 0 (c(1) ratio 0)                      | R/W              |
| 1.184.6:5   | Pre-cursor remote setting  | 6 5<br>1 1 <i>Remote_eq_cm1</i> = 3 (c(–1) ratio –0.15)<br>1 0 <i>Remote_eq_cm1</i> = 2 (c(–1) ratio –0.1)<br>0 1 <i>Remote_eq_cm1</i> = 1 (c(–1) ratio –0.05)<br>0 0 <i>Remote_eq_cm1</i> = 0 (c(–1) ratio 0)                                                                                                                                                                   | R/W              |
| 1.184.4:2   | Post-cursor local setting  | 4 3 2<br>1 1 1 Reserved<br>1 1 0 Reserved<br>1 0 1 <i>Local_eq_c1</i> = 5 (c(1) ratio –0.25)<br>1 0 0 <i>Local_eq_c1</i> = 4 (c(1) ratio –0.2)<br>0 1 1 <i>Local_eq_c1</i> = 3 (c(1) ratio –0.15)<br>0 1 0 <i>Local_eq_c1</i> = 2 (c(1) ratio –0.1)<br>0 0 1 <i>Local_eq_c1</i> = 1 (c(1) ratio –0.05)<br>0 0 0 <i>Local_eq_c1</i> = 0 (c(1) ratio 0)                            | R/W              |
| 1.184.1:0   | Pre-cursor local setting   | 1 0<br>1 1 <i>Local_eq_cm1</i> = 3 (c(–1) ratio –0.15)<br>1 0 <i>Local_eq_cm1</i> = 2 (c(–1) ratio –0.1)<br>0 1 <i>Local_eq_cm1</i> = 1 (c(–1) ratio –0.05)<br>0 0 <i>Local_eq_cm1</i> = 0 (c(–1) ratio 0)                                                                                                                                                                       | R/W              |

<sup>a</sup>R/W = Read/Write, RO = Read only

#### 45.2.1.114.3 Pre-cursor request (1.184.11:10)

The value of these bits indicates the value of the variable *Requested\_eq\_cm1* in the 25GAUI or lane 0 CAUI-4 receiver in the transmit direction (see 83D.3.3.2). When *Request\_flag* is equal to one, this value



indicates the ratio of the pre-cursor coefficient  $c(-1)$ , which is requested for the transmitter equalization in the 25GAUI or lane 0 CAUI-4 C2C transmitter in the transmit direction.

#### **45.2.1.114.4 Post-cursor remote setting (1.184.9:7)**

The value of these bits sets the variable *Remote\_eq\_c1* for the 25GAUI or lane 0 CAUI-4 receiver in the transmit direction (see 83D.3.3.2). This is used by a 25GAUI or CAUI-4 receiver that implements the optional transmitter equalization feedback as an indication of the ratio of the post-cursor coefficient  $c(1)$  being used in the 25GAUI or lane 0 of the CAUI-4 transmitter in the transmit direction (see 83D.3.1.1). It may be used to generate values for the request flag and the request bits. If a 25GAUI or lane 0 CAUI-4 receiver in the transmit direction is not present in the package, then these bits have no effect.

#### **45.2.1.114.5 Pre-cursor remote setting (1.184.6:5)**

The value of these bits sets the variable *Remote\_eq\_cm1* for the 25GAUI or lane 0 CAUI-4 receiver in the transmit direction (see 83D.3.3.2). This is used by a 25GAUI or CAUI-4 receiver that implements the optional transmitter equalization feedback as an indication of the ratio of the pre-cursor coefficient  $c(-1)$  being used in the 25GAUI or lane 0 of the CAUI-4 transmitter in the transmit direction (see 83D.3.1.1). It may be used to generate values for the request flag and the request bits. If a 25GAUI or lane 0 CAUI-4 receiver in the transmit direction is not present in the package, then these bits have no effect.

#### **45.2.1.114.6 Post-cursor local setting (1.184.4:2)**

The value of these bits sets the variable *Local\_eq\_c1* for the 25GAUI or lane 0 CAUI-4 transmitter in the transmit direction (see 83D.3.1.1 and Table 83D–3), which controls the weight of the transmitter equalization post-cursor coefficient  $c(1)$ . If a 25GAUI or lane 0 CAUI-4 transmitter in the transmit direction is not present in the package, then these bits have no effect.

#### **45.2.1.114.7 Pre-cursor local setting (1.184.1:0)**

The value of these bits sets the variable *Local\_eq\_cm1* for the 25GAUI or lane 0 CAUI-4 transmitter in the transmit direction (see 83D.3.1.1 and Table 83D–2), which controls the weight of the transmitter equalization pre-cursor coefficient  $c(-1)$ . If a 25GAUI or lane 0 CAUI-4 transmitter in the transmit direction is not present in the package, then these bits have no effect.

#### **45.2.1.115 CAUI-4 chip-to-chip transmitter equalization, transmit direction, lane 1 through lane 3 registers (Registers 1.185, 1.186, 1.187)**

The CAUI-4 chip-to-chip transmitter equalization, transmit direction, lane 1 through lane 3 registers are defined similarly to register 1.184 (which is used for lane 0, see 45.2.1.114) but for lanes 1 through 3, respectively. The transmitter, transmit direction, is the transmitter that sends data towards the PMD.

#### 45.2.1.116 RS-FEC control register (Register 1.200)

The assignment of bits in the RS-FEC control register is shown in Table 45–94.

**Table 45–94—RS-FEC control register bit definitions**

| Bit(s)     | Name                         | Description                                                                                                                                                                                                             | R/W <sup>a</sup> |
|------------|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.200.15:5 | Reserved                     | Value always 0                                                                                                                                                                                                          | RO               |
| 1.200.4    | FEC degraded SER enable      | 1 = FEC decoder indicates degraded SER<br>0 = FEC decoder does not indicate degraded SER                                                                                                                                | R/W              |
| 1.200.3    | Four lane PMD indication     | 1 = FEC is not being used with a four-lane PMD (the four_lane_pmd variable defined in 91.6.1 is set to zero)<br>0 = FEC is being used with a four-lane PMD (the four_lane_pmd variable defined in 91.6.1 is set to one) | R/W              |
| 1.200.2    | RS-FEC Enable                | 1 = Clause 108 Reed-Solomon FEC is enabled<br>0 = Clause 108 Reed-Solomon FEC is disabled                                                                                                                               | R/W              |
| 1.200.1    | FEC bypass indication enable | 1 = FEC decoder does not indicate errors to the PCS<br>0 = FEC decoder indicates errors to the PCS layer                                                                                                                | R/W              |
| 1.200.0    | FEC bypass correction enable | 1 = FEC decoder performs error detection without error correction<br>0 = FEC decoder performs error detection and error correction                                                                                      | R/W              |

<sup>a</sup>R/W = Read/Write, RO = Read only

##### 45.2.1.116.1 FEC degraded SER enable (1.200.4)

This bit enables the RS-FEC decoder to indicate the presence of a degraded SER (see 91.5.3.3.1 and 134.5.3.3.2). When set to a one, this bit enables degraded SER detection. When set to a zero, degraded SER detection is disabled. Writes to this bit are ignored and reads return a zero if the FEC does not have the ability to signal the presence of a degraded SER.

##### 45.2.1.116.2 Four-lane PMD (1.200.3)

This bit enables the alignment marker mapping function in the RS-FEC to substitute the fixed bytes of the alignment markers corresponding to PCS lanes 17, 18, and 19 with the fixed bytes for the alignment marker corresponding to PCS lane 16 (see 91.5.2.6). When this bit is set to zero, the alignment markers corresponding to PCS lanes 17, 18, and 19 are passed through unmodified. The default value of this bit is zero.

##### 45.2.1.116.3 RS-FEC enable (1.200.2)

Bit 1.200.2 enables the Reed-Solomon FEC described in Clause 108. When set to a one, this bit enables the Reed-Solomon FEC. When set to a zero, bit 1.200.2 disables the Reed-Solomon FEC (see 108.6.3). Writes to bit 1.200.2 are ignored and reads return a one if the RS-FEC does not have the ability to disable the RS-FEC function.

#### 45.2.1.116.4 FEC bypass indication enable (1.200.1)

This bit enables the RS-FEC decoder to bypass error indication to the upper layers (PCS) through the sync bits for the BASE-R PHY in the Local Device. When set to a one, this bit enables bypass of the error indication. When set to a zero, errors are indicated to the PCS through the sync bits. Writes to bit 1.200.1 are ignored and reads return a zero if the RS-FEC does not have the ability to bypass indicating decoding errors to the PCS layer (see 91.5.3.3, 108.5.3.2, and 134.5.3.3.1).

#### 45.2.1.116.5 FEC bypass correction enable (1.200.0)

When bit 1.200.0 is set to one the Reed-Solomon decoder performs error detection without error correction (see 91.5.3.3 and 108.5.3.2). When this bit is set to zero, the decoder also performs error correction. Writes to this bit are ignored and reads return a zero if the RS-FEC does not have the ability to bypass correction.

#### 45.2.1.117 RS-FEC status register (Register 1.201)

The assignment of bits in the RS-FEC status register is shown in Table 45–95.

**Table 45–95—RS-FEC status register bit definitions**

| Bit(s)      | Name                          | Description                                                                                                                                                              | R/W <sup>a</sup> |
|-------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.201.15    | PCS align status              | 1 = FEC encoder has locked and aligned all PCS lanes<br>0 = FEC encoder has not locked and aligned all PCS lanes                                                         | RO               |
| 1.201.14    | RS-FEC align status           | 1 = RS-FEC receive lanes locked and aligned<br>0 = RS-FEC receive lanes not locked and aligned                                                                           | RO               |
| 1.201.13:12 | Reserved                      | Value always 0                                                                                                                                                           | RO               |
| 1.201.11    | FEC AM lock 3                 | 1 = RS-FEC receive lane 3 locked and aligned<br>0 = RS-FEC receive lane 3 not locked and aligned                                                                         | RO               |
| 1.201.10    | FEC AM lock 2                 | 1 = RS-FEC receive lane 2 locked and aligned<br>0 = RS-FEC receive lane 2 not locked and aligned                                                                         | RO               |
| 1.201.9     | FEC AM lock 1                 | 1 = RS-FEC receive lane 1 locked and aligned<br>0 = RS-FEC receive lane 1 not locked and aligned                                                                         | RO               |
| 1.201.8     | FEC AM lock 0                 | 1 = RS-FEC receive lane 0 locked and aligned<br>0 = RS-FEC receive lane 0 not locked and aligned                                                                         | RO               |
| 1.201.7     | FEC optional states supported | 1 = RS-FEC supports optional states in Figure 91–8<br>0 = RS-FEC does not support optional states in Figure 91–8                                                         | RO               |
| 1.201.6:5   | Reserved                      | Value always 0                                                                                                                                                           | RO               |
| 1.201.4     | FEC degraded SER              | 1 = SER is degraded<br>0 = SER is not degraded                                                                                                                           | RO               |
| 1.201.3     | FEC degraded SER ability      | 1 = RS-FEC decoder has the ability to indicate the presence of a degraded SER<br>0 = RS-FEC decoder does not have the ability to indicate the presence of a degraded SER | RO               |
| 1.201.2     | RS-FEC high SER               | 1 = FEC errors have exceeded threshold<br>0 = FEC errors have not exceeded threshold                                                                                     | RO/LH            |

**Table 45–95—RS-FEC status register bit definitions (*continued*)**

| Bit(s)  | Name                          | Description                                                                                                                        | R/W <sup>a</sup> |
|---------|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.201.1 | FEC bypass indication ability | 1 = FEC decoder has the ability to bypass error indication<br>0 = FEC decoder does not have the ability to bypass error indication | RO               |
| 1.201.0 | FEC bypass correction ability | 1 = FEC decoder has the ability to bypass error correction<br>0 = FEC decoder does not have the ability to bypass error correction | RO               |

<sup>a</sup>RO = Read only, LH = Latching high

#### **45.2.1.117.1 PCS align status (1.201.15)**

Bit 1.201.15 indicates the PCS alignment status of the RS-FEC. For the RS-FEC described in Clause 91 or Clause 134, PCS alignment is defined as block lock, alignment marker lock, and deskew of all transmit PCS lanes. For the RS-FEC described in Clause 108, PCS alignment is defined as block lock of the transmit PCS signal. When read as a zero, bit 1.201.15 indicates that the RS-FEC has not obtained PCS alignment. When read as a one, bit 1.201.15 indicates that the RS-FEC has obtained PCS alignment.

#### **45.2.1.117.2 RS-FEC align status (1.201.14)**

Bit 1.201.14 indicates the PMA alignment status of the RS-FEC. For the RS-FEC described in Clause 91 or Clause 134, PMA alignment is defined as alignment marker lock and deskew of all lanes on the PMA service interface. For the RS-FEC described in Clause 108, PMA alignment is defined as codeword marker lock on the PMA service interface. When read as a zero, bit 1.201.14 indicates that the RS-FEC has not obtained PMA alignment. When read as a one, bit 1.201.14 indicates that the RS-FEC has obtained PMA alignment.

#### **45.2.1.117.3 FEC AM lock 3 (1.201.11)**

When read as a one, bit 1.201.11 indicates that the RS-FEC described in Clause 91 has locked and aligned lane 3 of the PMA service interface. When read as a zero, bit 1.201.11 indicates that the RS-FEC has not locked and aligned lane 3 of the PMA service interface. This bit reflects the state of `amps_lock[3]` (see 91.5.3.1).

#### **45.2.1.117.4 FEC AM lock 2 (1.201.10)**

When read as a one, bit 1.201.10 indicates that the RS-FEC described in Clause 91 has locked and aligned lane 2 of the PMA service interface. When read as a zero, bit 1.201.10 indicates that the RS-FEC has not locked and aligned lane 2 of the PMA service interface. This bit reflects the state of `amps_lock[2]` (see 91.5.3.1).

#### **45.2.1.117.5 FEC AM lock 1 (1.201.9)**

When read as a one, bit 1.201.9 indicates that the RS-FEC described in Clause 91 or Clause 134 has locked and aligned lane 1 of the PMA service interface. When read as a zero, bit 1.201.9 indicates that the RS-FEC has not locked and aligned lane 1 of the PMA service interface. This bit reflects the state of `amps_lock[1]` (see 91.5.3.1).

#### **45.2.1.117.6 FEC AM lock 0 (1.201.8)**

When read as a one, bit 1.201.8 indicates that the RS-FEC described in Clause 91 or Clause 134 has locked and aligned lane 0 of the PMA service interface. When read as a zero, bit 1.201.8 indicates that the RS-FEC has not locked and aligned lane 0 of the PMA service interface. This bit reflects the state of `amps_lock[0]` (see 91.5.3.1).

#### **45.2.1.117.7 FEC optional states supported (1.201.7)**

When read as a one, bit 1.201.7 indicates that the RS-FEC described in Clause 91 implements the optional states in Figure 91–8. When read as a zero, bit 1.201.7 indicates that the optional states are not implemented.

#### **45.2.1.117.8 FEC degraded SER (1.201.4)**

This bit (1.201.4) reflects the state of the variable `FEC_degraded_SER` (see 91.6.12 and 134.6.9). When read as a one, bit 1.201.4 indicates the presence of a degraded received signal.

#### **45.2.1.117.9 FEC degraded SER ability (1.201.3)**

This bit is set to one to indicate that the decoder has the ability to indicate the presence of a degraded SER (see 91.5.3.3.1 and 134.5.3.3.2). This bit is set to zero if this ability is not supported.

#### **45.2.1.117.10 RS-FEC high SER (1.201.2)**

When `FEC_bypass_indication_enable` is set to one, bit 1.201.2 is set to one if the number of RS-FEC symbol errors in a window of 8192 codewords exceeds the threshold (see 91.5.3.3, 108.5.3.2, and 134.5.3.3.1) and is set to zero otherwise. The bit is set to zero if `FEC_bypass_indication_enable` is set to zero. This bit shall be implemented with latching high behavior.

#### **45.2.1.117.11 FEC bypass indication ability (1.201.1)**

The Reed-Solomon decoder may have the option to perform error detection without error indication (see 91.5.3.3, 108.5.3.2, and 134.5.3.3.1) to reduce the delay contributed by the RS-FEC sublayer. This bit is set to one to indicate that the decoder has this ability to bypass error indication. The bit is set to zero if this ability is not supported.

#### **45.2.1.117.12 FEC bypass correction ability (1.201.0)**

The Reed-Solomon decoder may have the option to perform error detection without error correction (see 91.5.3.3 and 108.5.3.2) to reduce the delay contributed by the RS-FEC sublayer. This bit is set to one to indicate that the decoder has this ability to bypass error correction. The bit is set to zero if this ability is not supported.

#### **45.2.1.118 RS-FEC corrected codewords counter (Register 1.202, 1.203)**

The assignment of bits in the RS-FEC corrected codewords counter register is shown in Table 45–96. See 91.6.16 and 108.6.7 for a definition of this register. These bits shall be reset to all zeros when the register is read by the management function or upon PHY reset. These bits shall be held at all ones in the case of overflow. Registers 1.202, 1.203 are used to read the value of a 32-bit counter. When registers 1.202 and 1.203 are used to read the 32-bit counter value, the register 1.202 is read first, the value of the register 1.203 is

latched when (and only when) register 1.202 is read, and reads of register 1.203 return the latched value rather than the current value of the counter.

**Table 45–96—RS-FEC corrected codewords counter register bit definitions**

| Bit(s)     | Name                          | Description                     | R/W <sup>a</sup> |
|------------|-------------------------------|---------------------------------|------------------|
| 1.202.15:0 | FEC corrected codewords lower | FEC_corrected_cw_counter[15:0]  | RO, NR           |
| 1.203.15:0 | FEC corrected codewords upper | FEC_corrected_cw_counter[31:16] | RO, NR           |

<sup>a</sup>RO = Read only, NR = Non Roll-over

#### 45.2.1.119 RS-FEC uncorrected codewords counter (Register 1.204, 1.205)

The assignment of bits in the RS-FEC uncorrected codewords counter register is shown in Table 45–97. See 91.6.17 and 108.6.8 for a definition of this register. These bits shall be reset to all zeros when the register is read by the management function or upon PHY reset. These bits shall be held at all ones in the case of overflow. Registers 1.204, 1.205 are used to read the value of a 32-bit counter. When registers 1.204 and 1.205 are used to read the 32-bit counter value, the register 1.204 is read first, the value of the register 1.205 is latched when (and only when) register 1.204 is read, and reads of register 1.205 return the latched value rather than the current value of the counter.

**Table 45–97—RS-FEC uncorrected codewords counter register bit definitions**

| Bit(s)     | Name                            | Description                       | R/W <sup>a</sup> |
|------------|---------------------------------|-----------------------------------|------------------|
| 1.204.15:0 | FEC uncorrected codewords lower | FEC_uncorrected_cw_counter[15:0]  | RO, NR           |
| 1.205.15:0 | FEC uncorrected codewords upper | FEC_uncorrected_cw_counter[31:16] | RO, NR           |

<sup>a</sup>RO = Read only, NR = Non Roll-over

#### 45.2.1.120 RS-FEC lane mapping register (Register 1.206)

The assignment of bits in the RS-FEC lane mapping register is shown in Table 45–98. When the RS-FEC detects and locks the RS-FEC for PMA service interface lane 0, the detected RS-FEC lane number is recorded bits 1:0 in this register. Similarly, the detected RS-FEC lane numbers for PMA service lanes 1, 2, and 3 are recorded in register bits 3:2, 5:4, and 7:6, respectively. The contents of the RS-FEC lane mapping register bits 7:0 are valid when RS-FEC align status (1.201.14) is set to one and are invalid otherwise.

**Table 45–98—RS-FEC lane mapping register**

| Bit(s)     | Name                  | Description                      | R/W <sup>a</sup> |
|------------|-----------------------|----------------------------------|------------------|
| 1.206.15:8 | Reserved              | Value always 0                   | RO               |
| 1.206.7:6  | RS-FEC lane 3 mapping | RS-FEC lane mapped to PMA lane 3 | RO               |
| 1.206.5:4  | RS-FEC lane 2 mapping | RS-FEC lane mapped to PMA lane 2 | RO               |

**Table 45–98—RS-FEC lane mapping register (*continued*)**

| Bit(s)    | Name                  | Description                      | R/W <sup>a</sup> |
|-----------|-----------------------|----------------------------------|------------------|
| 1.206.3:2 | RS-FEC lane 1 mapping | RS-FEC lane mapped to PMA lane 1 | RO               |
| 1.206.1:0 | RS-FEC lane 0 mapping | RS-FEC lane mapped to PMA lane 0 | RO               |

<sup>a</sup>RO = Read only

#### 45.2.1.121 RS-FEC symbol error counter lane 0 (Register 1.210, 1.211)

The assignment of bits in the RS-FEC symbol error counter lane 0 register is shown in Table 45–99. Symbol errors detected in FEC lane 0 are counted and shown in register 1.210.15:0 and 1.211.15:0. See 91.6.19 and 108.6.9 for a definition of this register. These bits shall be reset to all zeros when the register is read by the management function or upon PHY reset. These bits shall be held at all ones in the case of overflow. Registers 1.210, 1.211 are used to read the value of a 32-bit counter. When registers 1.210 and 1.211 are used to read the 32-bit counter value, the register 1.210 is read first, the value of the register 1.211 is latched when (and only when) register 1.210 is read, and reads of register 1.211 return the latched value rather than the current value of the counter.

**Table 45–99—RS-FEC symbol error counter register bit definitions**

| Bit(s)     | Name                            | Description                       | R/W <sup>a</sup> |
|------------|---------------------------------|-----------------------------------|------------------|
| 1.210.15:0 | FEC symbol errors, lane 0 lower | FEC_symbol_error_counter_0[15:0]  | RO, NR           |
| 1.211.15:0 | FEC symbol errors, lane 0 upper | FEC_symbol_error_counter_0[31:16] | RO, NR           |

<sup>a</sup>RO = Read only, NR = Non Roll-over

#### 45.2.1.122 RS-FEC symbol error counter lane 1 through 3 (Register 1.212, 1.213, 1.214, 1.215, 1.216, 1.217)

The behavior of the RS-FEC symbol error counters, lane 1 through 3 is identical to that described for FEC lane 0 in 45.2.1.121. Errors detected in each FEC lane are counted and shown in the corresponding register. FEC lane 1, lower 16 bits are shown in register 1.212; FEC lane 1, upper 16 bits are shown in register 1.213; FEC lane 2, lower 16 bits are shown in register 1.214; through register 1.217 for FEC lane 3, upper 16 bits.

#### 45.2.1.123 RS-FEC BIP error counter lane 0 (Register 1.230)

The assignment of bits in the RS-FEC BIP error counter lane 0 is shown in Table 45–100. The RS-FEC described in Clause 91 or Clause 134 calculates a BIP value for each PCS lane (see 91.5.2.4, 91.6.8). Errors detected in PCS lane 0 are counted and shown in register 1.230.15:0. The 16-bit counter shall be reset to all zeros when register 1.230 is read or upon PMA/PMD reset. The 16-bit counter shall be held at all ones in the case of overflow. A device that does not implement a separated RS-FEC shall return a zero for all bits in the RS-FEC BIP error counter, lane 0 register.

#### 45.2.1.124 RS-FEC BIP error counter, lane 1 through 19 (Registers 1.231 through 1.249)

The behavior of the RS-FEC BIP error counters, lane 1 through 19 is identical to that described for lane 0 in 45.2.1.123. Errors detected in each PCS lane are counted and shown in register bits 15:0 in the

**Table 45–100—RS-FEC BIP error counter lane 0 register bit definitions**

| Bit(s)     | Name                      | Description                          | R/W <sup>a</sup> |
|------------|---------------------------|--------------------------------------|------------------|
| 1.230.15:0 | BIP error counter, lane 0 | Errors detected by BIP in PCS lane 0 | RO, NR           |

<sup>a</sup>RO = Read only, NR = Non Roll-over

corresponding register. PCS lane 1 is shown in register 1.231; PCS lane 2 is shown in register 1.232; through register 1.249 for PCS lane 19.

#### 45.2.1.125 RS-FEC PCS lane 0 mapping register (Register 1.250)

The assignment of bits in the RS-FEC PCS lane 0 mapping register is shown in Table 45–101. When the RS-FEC instance of the multi-lane PCS described in Clause 82 detects and locks the alignment marker for service interface lane 0, the detected PCS lane number is recorded in this register. The contents of the Lane 0 mapping register is valid when the transmit PCS align status bit (register 1.201.15) is set to one and is invalid otherwise (see 45.2.1.117). A device that does not implement a separated RS-FEC shall return a zero for all bits in the RS-FEC PCS lane 0 mapping register.

**Table 45–101—RS-FEC PCS lane 0 mapping register bit definitions**

| Bit(s)     | Name           | Description                                   | R/W <sup>a</sup> |
|------------|----------------|-----------------------------------------------|------------------|
| 1.250.15:5 | Reserved       | Value always 0                                | RO               |
| 1.250.4:0  | Lane 0 mapping | PCS lane received in service interface lane 0 | RO               |

<sup>a</sup>RO = Read only

#### 45.2.1.126 RS-FEC PCS lanes 1 through 19 mapping registers (Registers 1.251 through 1.269)

The definition of Lanes 1 through 19 mapping registers is identical to that described for lane 0 in 45.2.1.125. The lane mapping for lane 1 is in register 1.251; lane 2 is in register 1.252; etc.

#### 45.2.1.127 RS-FEC PCS alignment status 1 register (Register 1.280)

The assignment of bits in the RS-FEC PCS alignment status 1 register is shown in Table 45–252. All the bits in the RS-FEC PCS alignment status 1 register are read only; a write to the RS-FEC PCS alignment status 1 register shall have no effect. A device that does not implement a separated RS-FEC shall return a zero for all



bits in the RS-FEC PCS alignment status 1 register. It is the responsibility of the STA management entity to ensure that a port type is supported by all MMDs before interrogating any of its status bits.

**Table 45–102—RS-FEC PCS alignment status 1 register bit definitions**

| Bit(s)     | Name         | Description                                      | R/W <sup>a</sup> |
|------------|--------------|--------------------------------------------------|------------------|
| 1.280.15:8 | Reserved     | Value always 0                                   | RO               |
| 1.280.7    | Block lock 7 | 1 = Lane 7 is locked<br>0 = Lane 7 is not locked | RO               |
| 1.280.6    | Block lock 6 | 1 = Lane 6 is locked<br>0 = Lane 6 is not locked | RO               |
| 1.280.5    | Block lock 5 | 1 = Lane 5 is locked<br>0 = Lane 5 is not locked | RO               |
| 1.280.4    | Block lock 4 | 1 = Lane 4 is locked<br>0 = Lane 4 is not locked | RO               |
| 1.280.3    | Block lock 3 | 1 = Lane 3 is locked<br>0 = Lane 3 is not locked | RO               |
| 1.280.2    | Block lock 2 | 1 = Lane 2 is locked<br>0 = Lane 2 is not locked | RO               |
| 1.280.1    | Block lock 1 | 1 = Lane 1 is locked<br>0 = Lane 1 is not locked | RO               |
| 1.280.0    | Block lock 0 | 1 = Lane 0 is locked<br>0 = Lane 0 is not locked | RO               |

<sup>a</sup>RO = Read only

#### 45.2.1.127.1 Block lock 7 (1.280.7)

When read as a one, bit 1.280.7 indicates that the RS-FEC transmit function has achieved block lock for service interface lane 7. When read as a zero, bit 1.280.7 indicates that the RS-FEC transmit function lane 7 has not achieved block lock. This bit reflects the state of block\_lock[7] (see 91.5.2.1).

#### 45.2.1.127.2 Block lock 6 (1.280.6)

When read as a one, bit 1.280.6 indicates that the RS-FEC transmit function has achieved block lock for service interface lane 6. When read as a zero, bit 1.280.6 indicates that the RS-FEC transmit function lane 6 has not achieved block lock. This bit reflects the state of block\_lock[6] (see 91.5.2.1).

#### 45.2.1.127.3 Block lock 5 (1.280.5)

When read as a one, bit 1.280.5 indicates that the RS-FEC transmit function has achieved block lock for service interface lane 5. When read as a zero, bit 1.280.5 indicates that the RS-FEC transmit function lane 5 has not achieved block lock. This bit reflects the state of block\_lock[5] (see 91.5.2.1).

#### 45.2.1.127.4 Block lock 4 (1.280.4)

When read as a one, bit 1.280.4 indicates that the RS-FEC transmit function has achieved block lock for service interface lane 4. When read as a zero, bit 1.280.4 indicates that the RS-FEC transmit function lane 4 has not achieved block lock. This bit reflects the state of block\_lock[4] (see 91.5.2.1).

#### 45.2.1.127.5 Block lock 3 (1.280.3)

When read as a one, bit 1.280.3 indicates that the RS-FEC transmit function has achieved block lock for service interface lane 3. When read as a zero, bit 1.280.3 indicates that the RS-FEC transmit function lane 3 has not achieved block lock. This bit reflects the state of block\_lock[3] (see 91.5.2.1).

#### 45.2.1.127.6 Block lock 2 (1.280.2)

When read as a one, bit 1.280.2 indicates that the RS-FEC transmit function has achieved block lock for service interface lane 2. When read as a zero, bit 1.280.2 indicates that the RS-FEC transmit function lane 2 has not achieved block lock. This bit reflects the state of block\_lock[2] (see 91.5.2.1).

#### 45.2.1.127.7 Block lock 1 (1.280.1)

When read as a one, bit 1.280.1 indicates that the RS-FEC transmit function has achieved block lock for service interface lane 1. When read as a zero, bit 1.280.1 indicates that the RS-FEC transmit function lane 1 has not achieved block lock. This bit reflects the state of block\_lock[1] (see 91.5.2.1).

#### 45.2.1.127.8 Block lock 0 (1.280.0)

When read as a one, bit 1.280.0 indicates that the RS-FEC transmit function has achieved block lock for service interface lane 0. When read as a zero, bit 1.280.0 indicates that the RS-FEC transmit function lane 0 has not achieved block lock. This bit reflects the state of block\_lock[0] (see 91.5.2.1).

#### 45.2.1.128 RS-FEC PCS alignment status 2 register (Register 1.281)

The assignment of bits in the RS-FEC PCS alignment status 2 register is shown in Table 45–253. All the bits in the RS-FEC PCS alignment status 2 register are read only; a write to the RS-FEC PCS alignment status 2 register shall have no effect. A device that does not implement a separated RS-FEC shall return a zero for all bits in the RS-FEC PCS alignment status 2 register. It is the responsibility of the STA management entity to ensure that a port type is supported by all MMDs before interrogating any of its status bits.

**Table 45–103—RS-FEC PCS alignment status 2 register bit definitions**

| Bit(s)      | Name          | Description                                        | R/W <sup>a</sup> |
|-------------|---------------|----------------------------------------------------|------------------|
| 1.281.15:12 | Reserved      | Value always 0                                     | RO               |
| 1.281.11    | Block lock 19 | 1 = Lane 19 is locked<br>0 = Lane 19 is not locked | RO               |
| 1.281.10    | Block lock 18 | 1 = Lane 18 is locked<br>0 = Lane 18 is not locked | RO               |
| 1.281.9     | Block lock 17 | 1 = Lane 17 is locked<br>0 = Lane 17 is not locked | RO               |

**Table 45–103—RS-FEC PCS alignment status 2 register bit definitions (*continued*)**

| Bit(s)  | Name          | Description                                        | R/W <sup>a</sup> |
|---------|---------------|----------------------------------------------------|------------------|
| 1.281.8 | Block lock 16 | 1 = Lane 16 is locked<br>0 = Lane 16 is not locked | RO               |
| 1.281.7 | Block lock 15 | 1 = Lane 15 is locked<br>0 = Lane 15 is not locked | RO               |
| 1.281.6 | Block lock 14 | 1 = Lane 14 is locked<br>0 = Lane 14 is not locked | RO               |
| 1.281.5 | Block lock 13 | 1 = Lane 13 is locked<br>0 = Lane 13 is not locked | RO               |
| 1.281.4 | Block lock 12 | 1 = Lane 12 is locked<br>0 = Lane 12 is not locked | RO               |
| 1.281.3 | Block lock 11 | 1 = Lane 11 is locked<br>0 = Lane 11 is not locked | RO               |
| 1.281.2 | Block lock 10 | 1 = Lane 10 is locked<br>0 = Lane 10 is not locked | RO               |
| 1.281.1 | Block lock 9  | 1 = Lane 9 is locked<br>0 = Lane 9 is not locked   | RO               |
| 1.281.0 | Block lock 8  | 1 = Lane 8 is locked<br>0 = Lane 8 is not locked   | RO               |

<sup>a</sup>RO = Read only

#### **45.2.1.128.1 Block lock 19 (1.281.11)**

When read as a one, bit 1.281.11 indicates that the RS-FEC transmit function has achieved block lock for service interface lane 19. When read as a zero, bit 1.281.11 indicates that the RS-FEC transmit function lane 19 has not achieved block lock. This bit reflects the state of block\_lock[19] (see 91.5.2.1).

#### **45.2.1.128.2 Block lock 18 (1.281.10)**

When read as a one, bit 1.281.10 indicates that the RS-FEC transmit function has achieved block lock for service interface lane 18. When read as a zero, bit 1.281.10 indicates that the RS-FEC transmit function lane 18 has not achieved block lock. This bit reflects the state of block\_lock[18] (see 91.5.2.1).

#### **45.2.1.128.3 Block lock 17 (1.281.9)**

When read as a one, bit 1.281.9 indicates that the RS-FEC transmit function has achieved block lock for service interface lane 17. When read as a zero, bit 1.281.9 indicates that the RS-FEC transmit function lane 17 has not achieved block lock. This bit reflects the state of block\_lock[17] (see 91.5.2.1).

#### **45.2.1.128.4 Block lock 16 (1.281.8)**

When read as a one, bit 1.281.8 indicates that the RS-FEC transmit function has achieved block lock for service interface lane 16. When read as a zero, bit 1.281.8 indicates that the RS-FEC transmit function lane 16 has not achieved block lock. This bit reflects the state of block\_lock[16] (see 91.5.2.1).

#### **45.2.1.128.5 Block lock 15 (1.281.7)**

When read as a one, bit 1.281.7 indicates that the RS-FEC transmit function has achieved block lock for service interface lane 15. When read as a zero, bit 1.281.7 indicates that the RS-FEC transmit function lane 15 has not achieved block lock. This bit reflects the state of `block_lock[15]` (see 91.5.2.1).

#### **45.2.1.128.6 Block lock 14 (1.281.6)**

When read as a one, bit 1.281.6 indicates that the RS-FEC transmit function has achieved block lock for service interface lane 14. When read as a zero, bit 1.281.6 indicates that the RS-FEC transmit function lane 14 has not achieved block lock. This bit reflects the state of `block_lock[14]` (see 91.5.2.1).

#### **45.2.1.128.7 Block lock 13 (1.281.5)**

When read as a one, bit 1.281.5 indicates that the RS-FEC transmit function has achieved block lock for service interface lane 13. When read as a zero, bit 1.281.5 indicates that the RS-FEC transmit function lane 13 has not achieved block lock. This bit reflects the state of `block_lock[13]` (see 91.5.2.1).

#### **45.2.1.128.8 Block lock 12 (1.281.4)**

When read as a one, bit 1.281.4 indicates that the RS-FEC transmit function has achieved block lock for service interface lane 12. When read as a zero, bit 1.281.4 indicates that the RS-FEC transmit function lane 12 has not achieved block lock. This bit reflects the state of `block_lock[12]` (see 91.5.2.1).

#### **45.2.1.128.9 Block lock 11 (1.281.3)**

When read as a one, bit 1.281.3 indicates that the RS-FEC transmit function has achieved block lock for service interface lane 11. When read as a zero, bit 1.281.3 indicates that the RS-FEC transmit function lane 11 has not achieved block lock. This bit reflects the state of `block_lock[11]` (see 91.5.2.1).

#### **45.2.1.128.10 Block lock 10 (1.281.2)**

When read as a one, bit 1.281.2 indicates that the RS-FEC transmit function has achieved block lock for service interface lane 10. When read as a zero, bit 1.281.2 indicates that the RS-FEC transmit function lane 10 has not achieved block lock. This bit reflects the state of `block_lock[10]` (see 91.5.2.1).

#### **45.2.1.128.11 Block lock 9 (1.281.1)**

When read as a one, bit 1.281.1 indicates that the RS-FEC transmit function has achieved block lock for service interface lane 9. When read as a zero, bit 1.281.1 indicates that the RS-FEC transmit function lane 9 has not achieved block lock. This bit reflects the state of `block_lock[9]` (see 91.5.2.1).

#### **45.2.1.128.12 Block lock 8 (1.281.0)**

When read as a one, bit 1.281.0 indicates that the RS-FEC transmit function has achieved block lock for service interface lane 8. When read as a zero, bit 1.281.0 indicates that the RS-FEC transmit function lane 8 has not achieved block lock. This bit reflects the state of `block_lock[8]` (see 91.5.2.1).

#### **45.2.1.129 RS-FEC PCS alignment status 3 register (Register 1.282)**

The assignment of bits in the RS-FEC PCS alignment status 3 register is shown in Table 45–104. All the bits in the RS-FEC PCS alignment status 3 register are read only; a write to the RS-FEC PCS alignment status 3 register shall have no effect. A device that does not implement a separated RS-FEC shall return a zero for all

bits in the RS-FEC PCS alignment status 3 register. It is the responsibility of the STA management entity to ensure that a port type is supported by all MMDs before interrogating any of its status bits.

**Table 45–104—RS-FEC PCS alignment status 3 register bit definitions**

| Bit(s)     | Name           | Description                                                                        | R/W <sup>a</sup> |
|------------|----------------|------------------------------------------------------------------------------------|------------------|
| 1.282.15:8 | Reserved       | Value always 0                                                                     | RO               |
| 1.282.7    | Lane 7 aligned | 1 = Lane 7 alignment marker is locked<br>0 = Lane 7 alignment marker is not locked | RO               |
| 1.282.6    | Lane 6 aligned | 1 = Lane 6 alignment marker is locked<br>0 = Lane 6 alignment marker is not locked | RO               |
| 1.282.5    | Lane 5 aligned | 1 = Lane 5 alignment marker is locked<br>0 = Lane 5 alignment marker is not locked | RO               |
| 1.282.4    | Lane 4 aligned | 1 = Lane 4 alignment marker is locked<br>0 = Lane 4 alignment marker is not locked | RO               |
| 1.282.3    | Lane 3 aligned | 1 = Lane 3 alignment marker is locked<br>0 = Lane 3 alignment marker is not locked | RO               |
| 1.282.2    | Lane 2 aligned | 1 = Lane 2 alignment marker is locked<br>0 = Lane 2 alignment marker is not locked | RO               |
| 1.282.1    | Lane 1 aligned | 1 = Lane 1 alignment marker is locked<br>0 = Lane 1 alignment marker is not locked | RO               |
| 1.282.0    | Lane 0 aligned | 1 = Lane 0 alignment marker is locked<br>0 = Lane 0 alignment marker is not locked | RO               |

<sup>a</sup>RO = Read only

#### **45.2.1.129.1 Lane 7 aligned (1.282.7)**

When read as a one, bit 1.282.7 indicates that the RS-FEC transmit function has achieved alignment marker lock for service interface lane 7. When read as a zero, bit 1.282.7 indicates that the RS-FEC transmit function lane 7 has not achieved alignment marker lock. This bit reflects the state of `am_lock[7]` (see 91.5.2.2).

#### **45.2.1.129.2 Lane 6 aligned (1.282.6)**

When read as a one, bit 1.282.6 indicates that the RS-FEC transmit function has achieved alignment marker lock for service interface lane 6. When read as a zero, bit 1.282.6 indicates that the RS-FEC transmit function lane 6 has not achieved alignment marker lock. This bit reflects the state of `am_lock[6]` (see 91.5.2.2).

#### **45.2.1.129.3 Lane 5 aligned (1.282.5)**

When read as a one, bit 1.282.5 indicates that the RS-FEC transmit function has achieved alignment marker lock for service interface lane 5. When read as a zero, bit 1.282.5 indicates that the RS-FEC transmit function lane 5 has not achieved alignment marker lock. This bit reflects the state of `am_lock[5]` (see 91.5.2.2).

#### 45.2.1.129.4 Lane 4 aligned (1.282.4)

When read as a one, bit 1.282.4 indicates that the RS-FEC transmit function has achieved alignment marker lock for service interface lane 4. When read as a zero, bit 1.282.4 indicates that the RS-FEC transmit function lane 4 has not achieved alignment marker lock. This bit reflects the state of am\_lock[4] (see 91.5.2.2).

#### 45.2.1.129.5 Lane 3 aligned (1.282.3)

When read as a one, bit 1.282.3 indicates that the RS-FEC transmit function has achieved alignment marker lock for service interface lane 3. When read as a zero, bit 1.282.3 indicates that the RS-FEC transmit function lane 3 has not achieved alignment marker lock. This bit reflects the state of am\_lock[3] (see 91.5.2.2).

#### 45.2.1.129.6 Lane 2 aligned (1.282.2)

When read as a one, bit 1.282.2 indicates that the RS-FEC transmit function has achieved alignment marker lock for service interface lane 2. When read as a zero, bit 1.282.2 indicates that the RS-FEC transmit function lane 2 has not achieved alignment marker lock. This bit reflects the state of am\_lock[2] (see 91.5.2.2).

#### 45.2.1.129.7 Lane 1 aligned (1.282.1)

When read as a one, bit 1.282.1 indicates that the RS-FEC transmit function has achieved alignment marker lock for service interface lane 1. When read as a zero, bit 1.282.1 indicates that the RS-FEC transmit function lane 1 has not achieved alignment marker lock. This bit reflects the state of am\_lock[1] (see 91.5.2.2).

#### 45.2.1.129.8 Lane 0 aligned (1.282.0)

When read as a one, bit 1.282.0 indicates that the RS-FEC transmit function has achieved alignment marker lock for service interface lane 0. When read as a zero, bit 1.282.0 indicates that the RS-FEC transmit function lane 0 has not achieved alignment marker lock. This bit reflects the state of am\_lock[0] (see 91.5.2.2).

#### 45.2.1.130 RS-FEC PCS alignment status 4 register (Register 1.283)

The assignment of bits in the RS-FEC PCS alignment status 4 register is shown in Table 45–105. All the bits in the RS-FEC PCS alignment status 4 register are read only; a write to the RS-FEC PCS alignment status 4 register shall have no effect. A device that does not implement a separated RS-FEC shall return a zero for all bits in the RS-FEC PCS alignment status 4 register. It is the responsibility of the STA management entity to ensure that a port type is supported by all MMDs before interrogating any of its status bits.

**Table 45–105—RS-FEC PCS alignment status 4 register bit definitions**

| Bit(s)      | Name            | Description                                                                          | R/W <sup>a</sup> |
|-------------|-----------------|--------------------------------------------------------------------------------------|------------------|
| 1.283.15:12 | Reserved        | Value always 0                                                                       | RO               |
| 1.283.11    | Lane 19 aligned | 1 = Lane 19 alignment marker is locked<br>0 = Lane 19 alignment marker is not locked | RO               |
| 1.283.10    | Lane 18 aligned | 1 = Lane 18 alignment marker is locked<br>0 = Lane 18 alignment marker is not locked | RO               |
| 1.283.9     | Lane 17 aligned | 1 = Lane 17 alignment marker is locked<br>0 = Lane 17 alignment marker is not locked | RO               |

**Table 45–105—RS-FEC PCS alignment status 4 register bit definitions (*continued*)**

| Bit(s)  | Name            | Description                                                                          | R/W <sup>a</sup> |
|---------|-----------------|--------------------------------------------------------------------------------------|------------------|
| 1.283.8 | Lane 16 aligned | 1 = Lane 16 alignment marker is locked<br>0 = Lane 16 alignment marker is not locked | RO               |
| 1.283.7 | Lane 15 aligned | 1 = Lane 15 alignment marker is locked<br>0 = Lane 15 alignment marker is not locked | RO               |
| 1.283.6 | Lane 14 aligned | 1 = Lane 14 alignment marker is locked<br>0 = Lane 14 alignment marker is not locked | RO               |
| 1.283.5 | Lane 13 aligned | 1 = Lane 13 alignment marker is locked<br>0 = Lane 13 alignment marker is not locked | RO               |
| 1.283.4 | Lane 12 aligned | 1 = Lane 12 alignment marker is locked<br>0 = Lane 12 alignment marker is not locked | RO               |
| 1.283.3 | Lane 11 aligned | 1 = Lane 11 alignment marker is locked<br>0 = Lane 11 alignment marker is not locked | RO               |
| 1.283.2 | Lane 10 aligned | 1 = Lane 10 alignment marker is locked<br>0 = Lane 10 alignment marker is not locked | RO               |
| 1.283.1 | Lane 9 aligned  | 1 = Lane 9 alignment marker is locked<br>0 = Lane 9 alignment marker is not locked   | RO               |
| 1.283.0 | Lane 8 aligned  | 1 = Lane 8 alignment marker is locked<br>0 = Lane 8 alignment marker is not locked   | RO               |

<sup>a</sup>RO = Read only

#### **45.2.1.130.1 Lane 19 aligned (1.283.11)**

When read as a one, bit 1.283.11 indicates that the RS-FEC transmit function has achieved alignment marker lock for service interface lane 19. When read as a zero, bit 1.283.11 indicates that the RS-FEC transmit function lane 19 has not achieved alignment marker lock. This bit reflects the state of am\_lock[19] (see 91.5.2.2).

#### **45.2.1.130.2 Lane 18 aligned (1.283.10)**

When read as a one, bit 1.283.10 indicates that the RS-FEC transmit function has achieved alignment marker lock for service interface lane 18. When read as a zero, bit 1.283.10 indicates that the RS-FEC transmit function lane 18 has not achieved alignment marker lock. This bit reflects the state of am\_lock[18] (see 91.5.2.2).

#### **45.2.1.130.3 Lane 17 aligned (1.283.9)**

When read as a one, bit 1.283.9 indicates that the RS-FEC transmit function has achieved alignment marker lock for service interface lane 17. When read as a zero, bit 1.283.9 indicates that the RS-FEC transmit function lane 17 has not achieved alignment marker lock. This bit reflects the state of am\_lock[17] (see 91.5.2.2).

#### **45.2.1.130.4 Lane 16 aligned (1.283.8)**

When read as a one, bit 1.283.8 indicates that the RS-FEC transmit function has achieved alignment marker lock for service interface lane 16. When read as a zero, bit 1.283.8 indicates that the RS-FEC transmit function lane 16 has not achieved alignment marker lock. This bit reflects the state of `am_lock[16]` (see 91.5.2.2).

#### **45.2.1.130.5 Lane 15 aligned (1.283.7)**

When read as a one, bit 1.283.7 indicates that the RS-FEC transmit function has achieved alignment marker lock for service interface lane 15. When read as a zero, bit 1.283.7 indicates that the RS-FEC transmit function lane 15 has not achieved alignment marker lock. This bit reflects the state of `am_lock[15]` (see 91.5.2.2).

#### **45.2.1.130.6 Lane 14 aligned (1.283.6)**

When read as a one, bit 1.283.6 indicates that the RS-FEC transmit function has achieved alignment marker lock for service interface lane 14. When read as a zero, bit 1.283.6 indicates that the RS-FEC transmit function lane 14 has not achieved alignment marker lock. This bit reflects the state of `am_lock[14]` (see 91.5.2.2).

#### **45.2.1.130.7 Lane 13 aligned (1.283.5)**

When read as a one, bit 1.283.5 indicates that the RS-FEC transmit function has achieved alignment marker lock for service interface lane 13. When read as a zero, bit 1.283.5 indicates that the RS-FEC transmit function lane 13 has not achieved alignment marker lock. This bit reflects the state of `am_lock[13]` (see 91.5.2.2).

#### **45.2.1.130.8 Lane 12 aligned (1.283.4)**

When read as a one, bit 1.283.4 indicates that the RS-FEC transmit function has achieved alignment marker lock for service interface lane 12. When read as a zero, bit 1.283.4 indicates that the RS-FEC transmit function lane 12 has not achieved alignment marker lock. This bit reflects the state of `am_lock[12]` (see 91.5.2.2).

#### **45.2.1.130.9 Lane 11 aligned (1.283.3)**

When read as a one, bit 1.283.3 indicates that the RS-FEC transmit function has achieved alignment marker lock for service interface lane 11. When read as a zero, bit 1.283.3 indicates that the RS-FEC transmit function lane 11 has not achieved alignment marker lock. This bit reflects the state of `am_lock[11]` (see 91.5.2.2).

#### **45.2.1.130.10 Lane 10 aligned (1.283.2)**

When read as a one, bit 1.283.2 indicates that the RS-FEC transmit function has achieved alignment marker lock for service interface lane 10. When read as a zero, bit 1.283.2 indicates that the RS-FEC transmit function lane 10 has not achieved alignment marker lock. This bit reflects the state of `am_lock[10]` (see 91.5.2.2).

#### **45.2.1.130.11 Lane 9 aligned (1.283.1)**

When read as a one, bit 1.283.1 indicates that the RS-FEC transmit function has achieved alignment marker lock for service interface lane 9. When read as a zero, bit 1.283.1 indicates that the RS-FEC transmit function lane 9 has not achieved alignment marker lock. This bit reflects the state of `am_lock[9]` (see 91.5.2.2).



#### 45.2.1.130.12 Lane 8 aligned (1.283.0)

When read as a one, bit 1.283.0 indicates that the RS-FEC transmit function has achieved alignment marker lock for service interface lane 8. When read as a zero, bit 1.283.0 indicates that the RS-FEC transmit function lane 8 has not achieved alignment marker lock. This bit reflects the state of `am_lock[8]` (see 91.5.2.2).

#### 45.2.1.131 BASE-R FEC corrected blocks counter, lanes 0 through 19

(Register 1.300, 1.301, 1.302, 1.303, 1.304, 1.305, 1.306, 1.307, 1.308, 1.309, 1.310, 1.311, 1.312, 1.313, 1.314, 1.315, 1.316, 1.317, 1.318, 1.319, 1.320, 1.321, 1.322, 1.323, 1.324, 1.325, 1.326, 1.327, 1.328, 1.329, 1.330, 1.331, 1.332, 1.333, 1.334, 1.335, 1.336, 1.337, 1.338, 1.339)

For multi-PCS lane BASE-R PHYs, the even-numbered registers in this set are defined similarly to register 1.172 (see 45.2.1.109) but for lanes 0 through 19 respectively of multi-PCS lane PHYs. The odd-numbered registers in this set are defined similarly to register 1.173 (see 45.2.1.109) but for lanes 0 through 19 respectively of multilane PHYs. Note that the lane numbers refer to the service interface lanes and do not necessarily correspond to the PCS lane numbers.

Registers corresponding to lanes that are not used for the selected PHY shall return all zeros.

#### 45.2.1.132 200GAUI-8 and 400GAUI-16 chip-to-module recommended CTLE, lane 0 register (Register 1.400)

The assignment of bits in the 200GAUI-8 and 400GAUI-16 chip-to-module recommended CTLE, lane 0 register is shown in Table 45–106.

**Table 45–106—200GAUI-8 and 400GAUI-16 chip-to-module recommended CTLE, lane 0 register bit definitions**

| Bit(s)     | Name                     | Description                                                                                                                                                                                                                                 | R/W <sup>a</sup> |
|------------|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.400.15:5 | Reserved                 | Value always 0                                                                                                                                                                                                                              | RO               |
| 1.400.4:1  | Recommended CTLE peaking | 4 3 2 1<br>1 1 x x = Reserved<br>1 0 1 x = Reserved<br>1 0 0 1 = 9 dB<br>1 0 0 0 = 8 dB<br>0 1 1 1 = 7 dB<br>0 1 1 0 = 6 dB<br>0 1 0 1 = 5 dB<br>0 1 0 0 = 4 dB<br>0 0 1 1 = 3 dB<br>0 0 1 0 = 2 dB<br>0 0 0 1 = 1 dB<br>0 0 0 0 = Reserved | R/W              |
| 1.400.0    | Reserved                 | Value always 0                                                                                                                                                                                                                              | RO               |

<sup>a</sup>R/W = Read/Write, RO = Read only

#### 45.2.1.132.1 Recommended CTLE peaking (1.400.4:1)

The value of these bits sets the CTLE peaking value recommended for lane 0 by a host that implements the optional 200GAUI-8 or 400GAUI-16 chip-to-module interface defined in Annex 120C (see 120C.3.1). The module may optionally use this information to adjust its CTLE setting.

#### 45.2.1.133 200GAUI-8 and 400GAUI-16 chip-to-module recommended CTLE, lane 1 through lane 7 registers (Registers 1.401 through 1.407)

The 200GAUI-8 and 400GAUI-16 chip-to-module recommended CTLE, lane 1 through lane 7 registers are defined similarly to register 1.400 (which is used for lane 0, see 45.2.1.111) but for lanes 1 through 7, respectively.

#### 45.2.1.134 400GAUI-16 chip-to-module recommended CTLE, lane 8 through lane 15 registers (Registers 1.408 through 1.415)

The 400GAUI-16 chip-to-module recommended CTLE, lane 8 through lane 15 registers are defined similarly to register 1.400 (which is used for lane 0, see 45.2.1.111) but for lanes 8 through 15, respectively.

#### 45.2.1.135 50GAUI-n, 100GAUI-2, 200GAUI-n, and 400GAUI-n chip-to-chip transmitter equalization, receive direction, lane 0 register (Register 1.500)

The assignment of bits in the 50GAUI-n, 100GAUI-2, 200GAUI-n, and 400GAUI-n chip-to-chip transmitter equalization, receive direction, lane 0 register is shown in Table 45–107. The transmitter, receive direction, is the transmitter that sends data towards the MAC.

**Table 45–107—50GAUI-n, 100GAUI-2, 200GAUI-n, and 400GAUI-n chip-to-chip transmitter equalization, receive direction, lane 0 register bit definitions**

| Bit(s)      | Name                | Description                                                                                                                                                                                                                                                                                                                                                                      | R/W <sup>a</sup> |
|-------------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.500.15    | Request flag        | 1 = Change in equalization is requested<br>0 = No change in equalization is requested                                                                                                                                                                                                                                                                                            | RO               |
| 1.500.14:12 | Post-cursor request | 14 13 12<br>1 1 1 Reserved<br>1 1 0 Reserved<br>1 0 1 <i>Requested_eq_c1</i> = 5 (c(1) ratio –0.25)<br>1 0 0 <i>Requested_eq_c1</i> = 4 (c(1) ratio –0.2)<br>0 1 1 <i>Requested_eq_c1</i> = 3 (c(1) ratio –0.15)<br>0 1 0 <i>Requested_eq_c1</i> = 2 (c(1) ratio –0.1)<br>0 0 1 <i>Requested_eq_c1</i> = 1 (c(1) ratio –0.05)<br>0 0 0 <i>Requested_eq_c1</i> = 0 (c(1) ratio 0) | RO               |
| 1.500.11:10 | Pre-cursor request  | 11 10<br>1 1 <i>Requested_eq_cm1</i> = 3 (c(–1) ratio –0.15)<br>1 0 <i>Requested_eq_cm1</i> = 2 (c(–1) ratio –0.1)<br>0 1 <i>Requested_eq_cm1</i> = 1 (c(–1) ratio –0.05)<br>0 0 <i>Requested_eq_cm1</i> = 0 (c(–1) ratio 0)                                                                                                                                                     | RO               |

**Table 45–107—50GAUI-n, 100GAUI-2, 200GAUI-n, and 400GAUI-n chip-to-chip transmitter equalization, receive direction, lane 0 register bit definitions (*continued*)**

| Bit(s)    | Name                       | Description                                                                                                                                                                                                                                                                                                                                                 | R/W <sup>a</sup> |
|-----------|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.500.9:7 | Post-cursor remote setting | 9 8 7<br>1 1 1 Reserved<br>1 1 0 Reserved<br>1 0 1 <i>Remote_eq_c1</i> = 5 (c(1) ratio –0.25)<br>1 0 0 <i>Remote_eq_c1</i> = 4 (c(1) ratio –0.2)<br>0 1 1 <i>Remote_eq_c1</i> = 3 (c(1) ratio –0.15)<br>0 1 0 <i>Remote_eq_c1</i> = 2 (c(1) ratio –0.1)<br>0 0 1 <i>Remote_eq_c1</i> = 1 (c(1) ratio –0.05)<br>0 0 0 <i>Remote_eq_c1</i> = 0 (c(1) ratio 0) | R/W              |
| 1.500.6:5 | Pre-cursor remote setting  | 6 5<br>1 1 <i>Remote_eq_cm1</i> = 3 (c(–1) ratio –0.15)<br>1 0 <i>Remote_eq_cm1</i> = 2 (c(–1) ratio –0.1)<br>0 1 <i>Remote_eq_cm1</i> = 1 (c(–1) ratio –0.05)<br>0 0 <i>Remote_eq_cm1</i> = 0 (c(–1) ratio 0)                                                                                                                                              | R/W              |
| 1.500.4:2 | Post-cursor local setting  | 4 3 2<br>1 1 1 Reserved<br>1 1 0 Reserved<br>1 0 1 <i>Local_eq_c1</i> = 5 (c(1) ratio –0.25)<br>1 0 0 <i>Local_eq_c1</i> = 4 (c(1) ratio –0.2)<br>0 1 1 <i>Local_eq_c1</i> = 3 (c(1) ratio –0.15)<br>0 1 0 <i>Local_eq_c1</i> = 2 (c(1) ratio –0.1)<br>0 0 1 <i>Local_eq_c1</i> = 1 (c(1) ratio –0.05)<br>0 0 0 <i>Local_eq_c1</i> = 0 (c(1) ratio 0)       | R/W              |
| 1.500.1:0 | Pre-cursor local setting   | 1 0<br>1 1 <i>Local_eq_cm1</i> = 3 (c(–1) ratio –0.15)<br>1 0 <i>Local_eq_cm1</i> = 2 (c(–1) ratio –0.1)<br>0 1 <i>Local_eq_cm1</i> = 1 (c(–1) ratio –0.05)<br>0 0 <i>Local_eq_cm1</i> = 0 (c(–1) ratio 0)                                                                                                                                                  | R/W              |

<sup>a</sup>R/W = Read/Write, RO = Read only

#### 45.2.1.135.1 Request flag (1.500.15)

The value of this bit indicates the value of the variable *Request\_flag* in the lane 0 50GAUI-n, 100GAUI-2, 200GAUI-n, or 400GAUI-n receiver in the receive direction (see 120B.3.2 and 120D.3.2.3). This indicates whether the 50GAUI-n, 100GAUI-2, 200GAUI-n, or 400GAUI-n chip-to-chip device is issuing a request to change the remote transmitter equalization in the 50GAUI-n, 100GAUI-2, 200GAUI-n, or 400GAUI-n chip-to-chip lane 0 transmitter in the receive direction. If a lane 0 50GAUI-n, 100GAUI-2, 200GAUI-n, or 400GAUI-n receiver in the receive direction is not present in the package, then the value returned for this bit should be zero.

#### 45.2.1.135.2 Post-cursor request (1.500.14:12)

The value of these bits indicates the value of the variable *Requested\_eq\_c1* in the lane 0 50GAUI-n, 100GAUI-2, 200GAUI-n, or 400GAUI-n receiver in the receive direction (see 120B.3.2 and 120D.3.2.3). When *Request\_flag* is equal to one, this value indicates the ratio of the post-cursor coefficient c(1), which is requested for the transmitter equalization in the 50GAUI-n, 100GAUI-2, 200GAUI-n, or 400GAUI-n chip-to-chip lane 0 transmitter in the receive direction.

#### 45.2.1.135.3 Pre-cursor request (1.500.11:10)

The value of these bits indicates the value of the variable *Requested\_eq\_cml* in the lane 0 50GAUI-n, 100GAUI-2, 200GAUI-n, or 400GAUI-n receiver in the receive direction (see 120B.3.2 and 120D.3.2.3). When *Request\_flag* is equal to one, this value indicates the ratio of the pre-cursor coefficient  $c(-1)$ , which is requested for the transmitter equalization in the 50GAUI-n, 100GAUI-2, 200GAUI-n, or 400GAUI-n chip-to-chip lane 0 transmitter in the receive direction.

#### 45.2.1.135.4 Post-cursor remote setting (1.500.9:7)

The value of these bits sets the variable *Remote\_eq\_c1* for the lane 0 50GAUI-n, 100GAUI-2, 200GAUI-n, or 400GAUI-n receiver in the receive direction (see 120B.3.2 and 120D.3.2.3). This is used by a 50GAUI-n, 100GAUI-2, 200GAUI-n, or 400GAUI-n receiver that implements the optional transmitter equalization feedback as an indication of the ratio of the post-cursor coefficient  $c(1)$  being used in lane 0 of the 50GAUI-n, 100GAUI-2, 200GAUI-n, or 400GAUI-n transmitter in the receive direction (see 120B.3.1 and 120D.3.1.5). It may be used to generate values for the request flag and the request bits. If a lane 0 50GAUI-n, 100GAUI-2, 200GAUI-n, or 400GAUI-n receiver in the receive direction is not present in the package, then these bits have no effect.

#### 45.2.1.135.5 Pre-cursor remote setting (1.500.6:5)

The value of these bits sets the variable *Remote\_eq\_cml* for the lane 0 50GAUI-n, 100GAUI-2, 200GAUI-n, or 400GAUI-n receiver in the receive direction (see 120B.3.2 and 120D.3.2.3). This is used by a 50GAUI-n, 100GAUI-2, 200GAUI-n, or 400GAUI-n receiver that implements the optional transmitter equalization feedback as an indication of the ratio of the pre-cursor coefficient  $c(-1)$  being used in lane 0 of the 50GAUI-n, 100GAUI-2, 200GAUI-n, or 400GAUI-n transmitter in the receive direction (see 120B.3.1 and 120D.3.1.5). It may be used to generate values for the request flag and the request bits. If a lane 0 50GAUI-n, 100GAUI-2, 200GAUI-n, or 400GAUI-n receiver in the receive direction is not present in the package, then these bits have no effect.

#### 45.2.1.135.6 Post-cursor local setting (1.500.4:2)

The value of these bits sets the variable *Local\_eq\_c1* for the lane 0 50GAUI-n, 100GAUI-2, 200GAUI-n, or 400GAUI-n transmitter in the receive direction (see 120B.3.1 and 120D.3.1.5), which controls the weight of the transmitter equalization post-cursor coefficient  $c(1)$ . If a lane 0 50GAUI-n, 100GAUI-2, 200GAUI-n, or 400GAUI-n transmitter in the receive direction is not present in the package, then these bits have no effect.

#### 45.2.1.135.7 Pre-cursor local setting (1.500.1:0)

The value of these bits sets the variable *Local\_eq\_cml* for the lane 0 50GAUI-n, 100GAUI-2, 200GAUI-n, or 400GAUI-n transmitter in the receive direction (see 120B.3.1 and 120D.3.1.5), which controls the weight of the transmitter equalization pre-cursor coefficient  $c(-1)$ . If a lane 0 50GAUI-n, 100GAUI-2, 200GAUI-n, or 400GAUI-n transmitter in the receive direction is not present in the package, then these bits have no effect.

#### 45.2.1.136 50GAUI-n, 100GAUI-2, 200GAUI-n, and 400GAUI-n chip-to-chip transmitter equalization, receive direction, lane 1 through lane 15 registers (Registers 1.501 through 1.515)

The 50GAUI-n, 100GAUI-2, 200GAUI-n, and 400GAUI-n chip-to-chip transmitter equalization, receive direction, lane 1 through lane 15 registers are defined similarly to register 1.500 (which is used for lane 0, see 45.2.1.135) but for lanes 1 through 15, respectively. The transmitter, receive direction, is the transmitter that sends data towards the MAC.

#### 45.2.1.137 50GAUI-n, 100GAUI-2, 200GAUI-n, and 400GAUI-n chip-to-chip transmitter equalization, transmit direction, lane 0 register (Register 1.516)

The assignment of bits in the 50GAUI-n, 100GAUI-2, 200GAUI-n, and 400GAUI-n chip-to-chip transmitter equalization, transmit direction, lane 0 register is shown in Table 45–108. The transmitter, transmit direction, is the transmitter that sends data towards the PMD.

**Table 45–108—50GAUI-n, 100GAUI-2, 200GAUI-n, and 400GAUI-n chip-to-chip transmitter equalization, transmit direction, lane 0 register bit definitions**

| Bit(s)      | Name                       | Description                                                                                                                                                                                                                                                                                                                                                                      | R/W <sup>a</sup> |
|-------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.516.15    | Request flag               | 1 = Change in equalization is requested<br>0 = No change in equalization is requested                                                                                                                                                                                                                                                                                            | RO               |
| 1.516.14:12 | Post-cursor request        | 14 13 12<br>1 1 1 Reserved<br>1 1 0 Reserved<br>1 0 1 <i>Requested_eq_c1</i> = 5 (c(1) ratio –0.25)<br>1 0 0 <i>Requested_eq_c1</i> = 4 (c(1) ratio –0.2)<br>0 1 1 <i>Requested_eq_c1</i> = 3 (c(1) ratio –0.15)<br>0 1 0 <i>Requested_eq_c1</i> = 2 (c(1) ratio –0.1)<br>0 0 1 <i>Requested_eq_c1</i> = 1 (c(1) ratio –0.05)<br>0 0 0 <i>Requested_eq_c1</i> = 0 (c(1) ratio 0) | RO               |
| 1.516.11:10 | Pre-cursor request         | 11 10<br>1 1 <i>Requested_eq_cm1</i> = 3 (c(–1) ratio –0.15)<br>1 0 <i>Requested_eq_cm1</i> = 2 (c(–1) ratio –0.1)<br>0 1 <i>Requested_eq_cm1</i> = 1 (c(–1) ratio –0.05)<br>0 0 <i>Requested_eq_cm1</i> = 0 (c(–1) ratio 0)                                                                                                                                                     | RO               |
| 1.516.9:7   | Post-cursor remote setting | 9 8 7<br>1 1 1 Reserved<br>1 1 0 Reserved<br>1 0 1 <i>Remote_eq_c1</i> = 5 (c(1) ratio –0.25)<br>1 0 0 <i>Remote_eq_c1</i> = 4 (c(1) ratio –0.2)<br>0 1 1 <i>Remote_eq_c1</i> = 3 (c(1) ratio –0.15)<br>0 1 0 <i>Remote_eq_c1</i> = 2 (c(1) ratio –0.1)<br>0 0 1 <i>Remote_eq_c1</i> = 1 (c(1) ratio –0.05)<br>0 0 0 <i>Remote_eq_c1</i> = 0 (c(1) ratio 0)                      | R/W              |
| 1.516.6:5   | Pre-cursor remote setting  | 6 5<br>1 1 <i>Remote_eq_cm1</i> = 3 (c(–1) ratio –0.15)<br>1 0 <i>Remote_eq_cm1</i> = 2 (c(–1) ratio –0.1)<br>0 1 <i>Remote_eq_cm1</i> = 1 (c(–1) ratio –0.05)<br>0 0 <i>Remote_eq_cm1</i> = 0 (c(–1) ratio 0)                                                                                                                                                                   | R/W              |
| 1.516.4:2   | Post-cursor local setting  | 4 3 2<br>1 1 1 Reserved<br>1 1 0 Reserved<br>1 0 1 <i>Local_eq_c1</i> = 5 (c(1) ratio –0.25)<br>1 0 0 <i>Local_eq_c1</i> = 4 (c(1) ratio –0.2)<br>0 1 1 <i>Local_eq_c1</i> = 3 (c(1) ratio –0.15)<br>0 1 0 <i>Local_eq_c1</i> = 2 (c(1) ratio –0.1)<br>0 0 1 <i>Local_eq_c1</i> = 1 (c(1) ratio –0.05)<br>0 0 0 <i>Local_eq_c1</i> = 0 (c(1) ratio 0)                            | R/W              |
| 1.516.1:0   | Pre-cursor local setting   | 1 0<br>1 1 <i>Local_eq_cm1</i> = 3 (c(–1) ratio –0.15)<br>1 0 <i>Local_eq_cm1</i> = 2 (c(–1) ratio –0.1)<br>0 1 <i>Local_eq_cm1</i> = 1 (c(–1) ratio –0.05)<br>0 0 <i>Local_eq_cm1</i> = 0 (c(–1) ratio 0)                                                                                                                                                                       | R/W              |

<sup>a</sup>R/W = Read/Write, RO = Read only

#### 45.2.1.137.1 Request flag (1.516.15)

The value of this bit indicates the value of the variable *Request\_flag* in the lane 0 50GAUI-n, 100GAUI-2, 200GAUI-n, or 400GAUI-n receiver in the transmit direction (see 120B.3.2 and 120D.3.2.3). This indicates whether the 50GAUI-n, 100GAUI-2, 200GAUI-n, or 400GAUI-n chip-to-chip device is issuing a request to change the remote transmitter equalization in the 50GAUI-n, 100GAUI-2, 200GAUI-n, or 400GAUI-n chip-to-chip lane 0 transmitter in the transmit direction. If a lane 0 50GAUI-n, 100GAUI-2, 200GAUI-n, or 400GAUI-n receiver in the transmit direction is not present in the package, then the value returned for this bit should be zero.

#### 45.2.1.137.2 Post-cursor request (1.516.14:12)

The value of these bits indicates the value of the variable *Requested\_eq\_c1* in the lane 0 50GAUI-n, 100GAUI-2, 200GAUI-n, or 400GAUI-n receiver in the transmit direction (see 120B.3.2 and 120D.3.2.3). When *Request\_flag* is equal to one, this value indicates the ratio of the post-cursor coefficient  $c(1)$ , which is requested for the transmitter equalization in the 50GAUI-n, 100GAUI-2, 200GAUI-n, or 400GAUI-n chip-to-chip lane 0 transmitter in the transmit direction.

#### 45.2.1.137.3 Pre-cursor request (1.516.11:10)

The value of these bits indicates the value of the variable *Requested\_eq\_cm1* in the lane 0 50GAUI-n, 100GAUI-2, 200GAUI-n, or 400GAUI-n receiver in the transmit direction (see 120B.3.2 and 120D.3.2.3). When *Request\_flag* is equal to one, this value indicates the ratio of the pre-cursor coefficient  $c(-1)$ , which is requested for the transmitter equalization in the 50GAUI-n, 100GAUI-2, 200GAUI-n, or 400GAUI-n chip-to-chip lane 0 transmitter in the transmit direction.

#### 45.2.1.137.4 Post-cursor remote setting (1.516.9:7)

The value of these bits sets the variable *Remote\_eq\_c1* for the lane 0 50GAUI-n, 100GAUI-2, 200GAUI-n, or 400GAUI-n receiver in the transmit direction (see 120B.3.2 and 120D.3.2.3). This is used by a 50GAUI-n, 100GAUI-2, 200GAUI-n, or 400GAUI-n receiver that implements the optional transmitter equalization feedback as an indication of the ratio of the post-cursor coefficient  $c(1)$  being used in lane 0 of the 50GAUI-n, 100GAUI-2, 200GAUI-n, or 400GAUI-n transmitter in the transmit direction (see 120B.3.1 and 120D.3.1.5). It may be used to generate values for the request flag and the request bits. If a lane 0 50GAUI-n, 100GAUI-2, 200GAUI-n, or 400GAUI-n receiver in the transmit direction is not present in the package, then these bits have no effect.

#### 45.2.1.137.5 Pre-cursor remote setting (1.516.6:5)

The value of these bits sets the variable *Remote\_eq\_cm1* for the lane 0 50GAUI-n, 100GAUI-2, 200GAUI-n, or 400GAUI-n receiver in the transmit direction (see 120B.3.2 and 120D.3.2.3). This is used by a 50GAUI-n, 100GAUI-2, 200GAUI-n, or 400GAUI-n receiver that implements the optional transmitter equalization feedback as an indication of the ratio of the pre-cursor coefficient  $c(-1)$  being used in lane 0 of the 50GAUI-n, 100GAUI-2, 200GAUI-n, or 400GAUI-n transmitter in the transmit direction (see 120B.3.1 and 120D.3.1.5). It may be used to generate values for the request flag and the request bits. If a lane 0 50GAUI-n, 100GAUI-2, 200GAUI-n, or 400GAUI-n receiver in the transmit direction is not present in the package, then these bits have no effect.

#### 45.2.1.137.6 Post-cursor local setting (1.516.4:2)

The value of these bits sets the variable *Local\_eq\_c1* for the lane 0 50GAUI-n, 100GAUI-2, 200GAUI-n, or 400GAUI-n transmitter in the transmit direction (see 120B.3.1 and 120D.3.1.5), which controls the weight of the transmitter equalization post-cursor coefficient  $c(1)$ . If a lane 0 50GAUI-n, 100GAUI-2, 200GAUI-n,

or 400GAUI-n transmitter in the transmit direction is not present in the package, then these bits have no effect.

#### 45.2.1.137.7 Pre-cursor local setting (1.516.1:0)

The value of these bits sets the variable *Local\_eq\_cml* for the lane 0 50GAUI-n, 100GAUI-2, 200GAUI-n, or 400GAUI-n transmitter in the transmit direction (see 120B.3.1 and 120D.3.1.5), which controls the weight of the transmitter equalization pre-cursor coefficient  $c(-1)$ . If a lane 0 50GAUI-n, 100GAUI-2, 200GAUI-n, or 400GAUI-n transmitter in the transmit direction is not present in the package, then these bits have no effect.

#### 45.2.1.138 50GAUI-n, 100GAUI-2, 200GAUI-n, and 400GAUI-n chip-to-chip transmitter equalization, transmit direction, lane 1 through lane 15 registers (Registers 1.517 through 1.531)

The 50GAUI-n, 100GAUI-2, 200GAUI-n, and 400GAUI-n chip-to-chip transmitter equalization, transmit direction, lane 1 through lane 15 registers are defined similarly to register 1.516 (which is used for lane 0, see 45.2.1.137) but for lanes 1 through 15, respectively. The transmitter, transmit direction, is the transmitter that sends data towards the PMD.

#### 45.2.1.139 PMA precoder control Tx output (Register 1.600)

The assignment of bits in the PMA precoder control Tx output register is shown in Table 45–109.

**Table 45–109—PMA precoder control Tx output register bit definitions**

| Bit(s)     | Name                             | Description                                                                     | R/W <sup>a</sup> |
|------------|----------------------------------|---------------------------------------------------------------------------------|------------------|
| 1.600.15:4 | Reserved                         | Value always 0                                                                  | RO               |
| 1.600.3    | Lane 3 Tx output precoder enable | 1 = Lane 3 Tx output precoder enabled<br>0 = Lane 3 Tx output precoder disabled | R/W              |
| 1.600.2    | Lane 2 Tx output precoder enable | 1 = Lane 2 Tx output precoder enabled<br>0 = Lane 2 Tx output precoder disabled | R/W              |
| 1.600.1    | Lane 1 Tx output precoder enable | 1 = Lane 1 Tx output precoder enabled<br>0 = Lane 1 Tx output precoder disabled | R/W              |
| 1.600.0    | Lane 0 Tx output precoder enable | 1 = Lane 0 Tx output precoder enabled<br>0 = Lane 0 Tx output precoder disabled | R/W              |

<sup>a</sup>R/W = Read/Write, RO = Read only

##### 45.2.1.139.1 Lane 3 Tx output precoder enable (1.600.3)

This bit enables the lane 3 Tx output precoder.

##### 45.2.1.139.2 Lane 2 Tx output precoder enable (1.600.2)

This bit enables the lane 2 Tx output precoder.

##### 45.2.1.139.3 Lane 1 Tx output precoder enable (1.600.1)

This bit enables the lane 1 Tx output precoder.

#### 45.2.1.139.4 Lane 0 Tx output precoder enable (1.600.0)

This bit enables the lane 0 Tx output precoder.

#### 45.2.1.140 PMA precoder control Rx input (Register 1.601)

The assignment of bits in the precoder control Rx input register is shown in Table 45–110.

**Table 45–110—PMA precoder control Rx input register bit definitions**

| Bit(s)     | Name                            | Description                                                                   | R/W <sup>a</sup> |
|------------|---------------------------------|-------------------------------------------------------------------------------|------------------|
| 1.601.15:4 | Reserved                        | Value always 0                                                                | RO               |
| 1.601.3    | Lane 3 Rx input precoder enable | 1 = Lane 3 Rx input precoder enabled<br>0 = Lane 3 Rx input precoder disabled | R/W              |
| 1.601.2    | Lane 2 Rx input precoder enable | 1 = Lane 2 Rx input precoder enabled<br>0 = Lane 2 Rx input precoder disabled | R/W              |
| 1.601.1    | Lane 1 Rx input precoder enable | 1 = Lane 1 Rx input precoder enabled<br>0 = Lane 1 Rx input precoder disabled | R/W              |
| 1.601.0    | Lane 0 Rx input precoder enable | 1 = Lane 0 Rx input precoder enabled<br>0 = Lane 0 Rx input precoder disabled | R/W              |

<sup>a</sup>R/W = Read/Write, RO = Read only

#### 45.2.1.140.1 Lane 3 Rx input precoder enable (1.601.3)

This bit enables the lane 3 Rx input precoder.

#### 45.2.1.140.2 Lane 2 Rx input precoder enable (1.601.2)

This bit enables the lane 2 Rx input precoder.

#### 45.2.1.140.3 Lane 1 Rx input precoder enable (1.601.1)

This bit enables the lane 1 Rx input precoder.

#### 45.2.1.140.4 Lane 0 Rx input precoder enable (1.601.0)

This bit enables the lane 0 Rx input precoder.



#### 45.2.1.141 PMA precoder control Rx output (Register 1.602)

The assignment of bits in the precoder control Rx output register is shown in Table 45–111.

**Table 45–111—PMA precoder control Rx output register bit definitions**

| Bit(s)     | Name                             | Description                                                                     | R/W <sup>a</sup> |
|------------|----------------------------------|---------------------------------------------------------------------------------|------------------|
| 1.602.15:2 | Reserved                         | Value always 0                                                                  | RO               |
| 1.602.1    | Lane 1 Rx output precoder enable | 1 = Lane 1 Rx output precoder enabled<br>0 = Lane 1 Rx output precoder disabled | R/W              |
| 1.602.0    | Lane 0 Rx output precoder enable | 1 = Lane 0 Rx output precoder enabled<br>0 = Lane 0 Rx output precoder disabled | R/W              |

<sup>a</sup>R/W = Read/Write, RO = Read only

##### 45.2.1.141.1 Lane 1 Rx output precoder enable (1.602.1)

This bit enables the lane 1 Rx output precoder.

##### 45.2.1.141.2 Lane 0 Rx output precoder enable (1.602.0)

This bit enables the lane 0 Rx output precoder.

#### 45.2.1.142 PMA precoder control Tx input (Register 1.603)

The assignment of bits in the precoder control Tx input register is shown in Table 45–112.

**Table 45–112—PMA precoder control Tx input register bit definitions**

| Bit(s)     | Name                            | Description                                                                   | R/W <sup>a</sup> |
|------------|---------------------------------|-------------------------------------------------------------------------------|------------------|
| 1.603.15:2 | Reserved                        | Value always 0                                                                | RO               |
| 1.603.1    | Lane 1 Tx input precoder enable | 1 = Lane 1 Tx input precoder enabled<br>0 = Lane 1 Tx input precoder disabled | R/W              |
| 1.603.0    | Lane 0 Tx input precoder enable | 1 = Lane 0 Tx input precoder enabled<br>0 = Lane 0 Tx input precoder disabled | R/W              |

<sup>a</sup>R/W = Read/Write, RO = Read only

##### 45.2.1.142.1 Lane 1 Tx input precoder enable (1.603.1)

This bit enables the lane 1 Tx input precoder.

##### 45.2.1.142.2 Lane 0 Tx input precoder enable (1.603.0)

This bit enables the lane 0 Tx input precoder.

#### 45.2.1.143 PMA precoder request flag (Register 1.604)

The assignment of bits in the precoder request flag register is shown in Table 45–113.

**Table 45–113—PMA precoder request flag register bit definitions**

| Bit(s)     | Name                           | Description                                                                           | R/W <sup>a</sup> |
|------------|--------------------------------|---------------------------------------------------------------------------------------|------------------|
| 1.604.15:2 | Reserved                       | Value always 0                                                                        | RO               |
| 1.604.1    | Tx input precoder request flag | 1 = Tx input precoding change requested<br>0 = Tx input precoding no change requested | RO               |
| 1.604.0    | Rx input precoder request flag | 1 = Rx input precoding change requested<br>0 = Rx input precoding no change requested | RO               |

<sup>a</sup>RO = Read only

##### 45.2.1.143.1 Tx input precoder request flag (1.604.1)

This bit indicates the Tx input precoder request flag.

##### 45.2.1.143.2 Rx input precoder request flag (1.604.0)

This bit indicates the Rx input precoder request flag.

#### 45.2.1.144 PMA precoder request Rx input status (Register 1.605)

The assignment of bits in the precoder request Rx input status register is shown in Table 45–114.

**Table 45–114—PMA precoder request Rx input status register bit definitions**

| Bit(s)     | Name                                    | Description                                                                          | R/W <sup>a</sup> |
|------------|-----------------------------------------|--------------------------------------------------------------------------------------|------------------|
| 1.605.15:2 | Reserved                                | Value always 0                                                                       | RO               |
| 1.605.1    | Lane 1 Rx input precoder request status | 1 = Lane 1 precoder Rx input requested<br>0 = Lane 1 precoder Rx input not requested | RO               |
| 1.605.0    | Lane 0 Rx input precoder request status | 1 = Lane 0 precoder Rx input requested<br>0 = Lane 0 precoder Rx input not requested | RO               |

<sup>a</sup>RO = Read only

##### 45.2.1.144.1 Lane 1 Rx input precoder request status (1.605.1)

This bit indicates the Rx input precoder request status for lane 1.

##### 45.2.1.144.2 Lane 0 Rx input precoder request status (1.605.0)

This bit indicates the Rx input precoder request status for lane 0.

#### 45.2.1.145 PMA precoder request Tx input status (Register 1.606)

The assignment of bits in the precoder request Tx input status register is shown in Table 45–115.

**Table 45–115—PMA precoder request Tx input status register bit definitions**

| Bit(s)     | Name                                    | Description                                                                          | R/W <sup>a</sup> |
|------------|-----------------------------------------|--------------------------------------------------------------------------------------|------------------|
| 1.606.15:2 | Reserved                                | Value always 0                                                                       | RO               |
| 1.606.1    | Lane 1 Tx input precoder request status | 1 = Lane 1 precoder Tx input requested<br>0 = Lane 1 precoder Tx input not requested | RO               |
| 1.606.0    | Lane 0 Tx input precoder request status | 1 = Lane 0 precoder Tx input requested<br>0 = Lane 0 precoder Tx input not requested | RO               |

<sup>a</sup>RO = Read only

##### 45.2.1.145.1 Lane 1 Tx input precoder request status(1.606.1)

This bit indicates the Tx input precoder request status for lane 1.

##### 45.2.1.145.2 Lane 0 Tx input precoder request status (1.606.0)

This bit indicates the Tx input precoder request status for lane 0.

#### 45.2.1.146 RS-FEC degraded SER activate threshold register (Register 1.650, 1.651)

The assignment of bits in the RS-FEC degraded SER activate threshold register is shown in Table 45–116. The value controls the threshold used to set the FEC degraded SER bit (1.201.4) as defined in 91.5.3.3.1 and 134.5.3.3.2.

**Table 45–116—RS-FEC degraded SER activate threshold register bit definitions**

| Bit(s)     | Name                                      | Description                                | R/W <sup>a</sup> |
|------------|-------------------------------------------|--------------------------------------------|------------------|
| 1.650.15:0 | FEC degraded SER activate threshold lower | FEC_degraded_SER_activate_threshold[15:0]  | R/W              |
| 1.651.15:0 | FEC degraded SER activate threshold upper | FEC_degraded_SER_activate_threshold[31:16] | R/W              |

<sup>a</sup>R/W = Read/Write

#### 45.2.1.147 RS-FEC degraded SER deactivate threshold register (Register 1.652, 1.653)

The assignment of bits in the RS-FEC degraded SER deactivate threshold register is shown in Table 45–117. The value controls the threshold used to clear the FEC degraded SER bit (1.201.4) as defined in 91.5.3.3.1 and 134.5.3.3.2.

**Table 45–117—RS-FEC degraded SER deactivate threshold register bit definitions**

| Bit(s)     | Name                                        | Description                                  | R/W <sup>a</sup> |
|------------|---------------------------------------------|----------------------------------------------|------------------|
| 1.652.15:0 | FEC degraded SER deactivate threshold lower | FEC_degraded_SER_deactivate_threshold[15:0]  | R/W              |
| 1.653.15:0 | FEC degraded SER deactivate threshold upper | FEC_degraded_SER_deactivate_threshold[31:16] | R/W              |

<sup>a</sup>R/W = Read/Write

#### 45.2.1.148 RS-FEC degraded SER interval register (Register 1.654, 1.655)

The assignment of bits in the RS-FEC degraded SER interval register is shown in Table 45–118. The value controls the interval used to set and clear the FEC degraded SER bit (1.201.4) as defined in 91.5.3.3.1 and 134.5.3.3.2.

**Table 45–118—RS-FEC degraded SER interval register bit definitions**

| Bit(s)     | Name                            | Description                      | R/W <sup>a</sup> |
|------------|---------------------------------|----------------------------------|------------------|
| 1.654.15:0 | FEC degraded SER interval lower | FEC_degraded_SER_interval[15:0]  | R/W              |
| 1.655.15:0 | FEC degraded SER interval upper | FEC_degraded_SER_interval[31:16] | R/W              |

<sup>a</sup>R/W = Read/Write

#### 45.2.1.149 BASE-R FEC uncorrected blocks counter, lanes 0 through 19

(Register 1.700, 1.701, 1.702, 1.703, 1.704, 1.705, 1.706, 1.707, 1.708, 1.709, 1.710, 1.711, 1.712, 1.713, 1.714, 1.715, 1.716, 1.717, 1.718, 1.719, 1.720, 1.721, 1.722, 1.723, 1.724, 1.725, 1.726, 1.727, 1.728, 1.729, 1.730, 1.731, 1.732, 1.733, 1.734, 1.735, 1.736, 1.737, 1.738, 1.739)

For multi-PCS lane BASE-R PHYs, the even-numbered registers in this set are defined similarly to register 1.174 (see 45.2.1.110) but for lanes 0 through 19 respectively of multi-PCS lane PHYs. The odd-numbered registers in this set are defined similarly to register 1.175 (see 45.2.1.110) but for lanes 0 through 19 respectively of multilane PHYs. Note that the lane numbers refer to the service interface lanes and do not necessarily correspond to the PCS lane numbers.

Registers corresponding to lanes that are not used for the selected PHY shall return all zeros.

#### 45.2.1.150 Tx optical channel control register (Register 1.800)

The assignment of bits in the Tx optical channel control register is shown in Table 45–119.

**Table 45–119—Tx optical channel control register bit definitions**

| Bit(s)     | Name                     | Description                                   | R/W <sup>a</sup> |
|------------|--------------------------|-----------------------------------------------|------------------|
| 1.800.15:6 | Reserved                 | Value always 0                                | RO               |
| 1.800.5:0  | Tx optical channel index | Integer value of the Tx optical channel index | R/W              |

<sup>a</sup>R/W = Read/Write, RO = Read only

##### 45.2.1.150.1 Tx optical channel index (1.800.5:0)

Bits 1.800.5:0 set the value of the Tx optical channel index (which directly relates to the optical channel and transmitter center frequency) with bit 1.800.0 being the LSB and bit 1.800.5 being the MSB. The channel index number indicates the corresponding optical frequency. For 100GBASE-ZR the specific optical frequency corresponding to each channel index number is listed in Table 154–5.

#### 45.2.1.151 Tx optical channel ability 1 register (Register 1.801)

The assignment of bits in the Tx optical channel ability 1 register is shown in Table 45–120.

**Table 45–120—Tx optical channel ability 1 register bit definitions**

| Bit(s)   | Name                | Description                                                                                                                        | R/W <sup>a</sup> |
|----------|---------------------|------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.801.15 | Tx index ability 15 | 1 = PMD is able to operate with Tx optical channel index = 15<br>0 = PMD is not able to operate with Tx optical channel index = 15 | RO               |
| 1.801.14 | Tx index ability 14 | 1 = PMD is able to operate with Tx optical channel index = 14<br>0 = PMD is not able to operate with Tx optical channel index = 14 | RO               |
| 1.801.13 | Tx index ability 13 | 1 = PMD is able to operate with Tx optical channel index = 13<br>0 = PMD is not able to operate with Tx optical channel index = 13 | RO               |
| 1.801.12 | Tx index ability 12 | 1 = PMD is able to operate with Tx optical channel index = 12<br>0 = PMD is not able to operate with Tx optical channel index = 12 | RO               |
| 1.801.11 | Tx index ability 11 | 1 = PMD is able to operate with Tx optical channel index = 11<br>0 = PMD is not able to operate with Tx optical channel index = 11 | RO               |
| 1.801.10 | Tx index ability 10 | 1 = PMD is able to operate with Tx optical channel index = 10<br>0 = PMD is not able to operate with Tx optical channel index = 10 | RO               |
| 1.801.9  | Tx index ability 9  | 1 = PMD is able to operate with Tx optical channel index = 9<br>0 = PMD is not able to operate with Tx optical channel index = 9   | RO               |
| 1.801.8  | Tx index ability 8  | 1 = PMD is able to operate with Tx optical channel index = 8<br>0 = PMD is not able to operate with Tx optical channel index = 8   | RO               |
| 1.801.7  | Tx index ability 7  | 1 = PMD is able to operate with Tx optical channel index = 7<br>0 = PMD is not able to operate with Tx optical channel index = 7   | RO               |
| 1.801.6  | Tx index ability 6  | 1 = PMD is able to operate with Tx optical channel index = 6<br>0 = PMD is not able to operate with Tx optical channel index = 6   | RO               |

**Table 45–120—Tx optical channel ability 1 register bit definitions (continued)**

| Bit(s)  | Name               | Description                                                                                                                      | R/W <sup>a</sup> |
|---------|--------------------|----------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.801.5 | Tx index ability 5 | 1 = PMD is able to operate with Tx optical channel index = 5<br>0 = PMD is not able to operate with Tx optical channel index = 5 | RO               |
| 1.801.4 | Tx index ability 4 | 1 = PMD is able to operate with Tx optical channel index = 4<br>0 = PMD is not able to operate with Tx optical channel index = 4 | RO               |
| 1.801.3 | Tx index ability 3 | 1 = PMD is able to operate with Tx optical channel index = 3<br>0 = PMD is not able to operate with Tx optical channel index = 3 | RO               |
| 1.801.2 | Tx index ability 2 | 1 = PMD is able to operate with Tx optical channel index = 2<br>0 = PMD is not able to operate with Tx optical channel index = 2 | RO               |
| 1.801.1 | Tx index ability 1 | 1 = PMD is able to operate with Tx optical channel index = 1<br>0 = PMD is not able to operate with Tx optical channel index = 1 | RO               |
| 1.801.0 | Tx index ability 0 | 1 = PMD is able to operate with Tx optical channel index = 0<br>0 = PMD is not able to operate with Tx optical channel index = 0 | RO               |

<sup>a</sup>RO = Read only

#### 45.2.1.151.1 Tx index ability 0 through 15 (1.801.0 through 1.801.15)

When read as a one, bit 1.801.0 indicates that the PMD selected by bits 1.7.6:0 is able to operate with a Tx optical channel index of 0. When read as a zero, bit 1.801.0 indicates that the PMD is not able to operate with a Tx optical channel index of 0. Bits 1.801.1 through 1.801.15 indicate the equivalent for index values 1 through 15, respectively. The optical frequencies that correspond to these index values are given in the appropriate PMD clause. For 100GBASE-ZR see Table 154–5.

#### 45.2.1.152 Tx optical channel ability 2 register (Register 1.802)

The assignment of bits in the Tx optical channel ability 2 register is shown in Table 45–121.

**Table 45–121—Tx optical channel ability 2 register bit definitions**

| Bit(s)   | Name                | Description                                                                                                                        | R/W <sup>a</sup> |
|----------|---------------------|------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.802.15 | Tx index ability 31 | 1 = PMD is able to operate with Tx optical channel index = 31<br>0 = PMD is not able to operate with Tx optical channel index = 31 | RO               |
| 1.802.14 | Tx index ability 30 | 1 = PMD is able to operate with Tx optical channel index = 30<br>0 = PMD is not able to operate with Tx optical channel index = 30 | RO               |
| 1.802.13 | Tx index ability 29 | 1 = PMD is able to operate with Tx optical channel index = 29<br>0 = PMD is not able to operate with Tx optical channel index = 29 | RO               |
| 1.802.12 | Tx index ability 28 | 1 = PMD is able to operate with Tx optical channel index = 28<br>0 = PMD is not able to operate with Tx optical channel index = 28 | RO               |
| 1.802.11 | Tx index ability 27 | 1 = PMD is able to operate with Tx optical channel index = 27<br>0 = PMD is not able to operate with Tx optical channel index = 27 | RO               |
| 1.802.10 | Tx index ability 26 | 1 = PMD is able to operate with Tx optical channel index = 26<br>0 = PMD is not able to operate with Tx optical channel index = 26 | RO               |
| 1.802.9  | Tx index ability 25 | 1 = PMD is able to operate with Tx optical channel index = 25<br>0 = PMD is not able to operate with Tx optical channel index = 25 | RO               |

**Table 45–121—Tx optical channel ability 2 register bit definitions (continued)**

| Bit(s)  | Name                | Description                                                                                                                        | R/W <sup>a</sup> |
|---------|---------------------|------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.802.8 | Tx index ability 24 | 1 = PMD is able to operate with Tx optical channel index = 24<br>0 = PMD is not able to operate with Tx optical channel index = 24 | RO               |
| 1.802.7 | Tx index ability 23 | 1 = PMD is able to operate with Tx optical channel index = 23<br>0 = PMD is not able to operate with Tx optical channel index = 23 | RO               |
| 1.802.6 | Tx index ability 22 | 1 = PMD is able to operate with Tx optical channel index = 22<br>0 = PMD is not able to operate with Tx optical channel index = 22 | RO               |
| 1.802.5 | Tx index ability 21 | 1 = PMD is able to operate with Tx optical channel index = 21<br>0 = PMD is not able to operate with Tx optical channel index = 21 | RO               |
| 1.802.4 | Tx index ability 20 | 1 = PMD is able to operate with Tx optical channel index = 20<br>0 = PMD is not able to operate with Tx optical channel index = 20 | RO               |
| 1.802.3 | Tx index ability 19 | 1 = PMD is able to operate with Tx optical channel index = 19<br>0 = PMD is not able to operate with Tx optical channel index = 19 | RO               |
| 1.802.2 | Tx index ability 18 | 1 = PMD is able to operate with Tx optical channel index = 18<br>0 = PMD is not able to operate with Tx optical channel index = 18 | RO               |
| 1.802.1 | Tx index ability 17 | 1 = PMD is able to operate with Tx optical channel index = 17<br>0 = PMD is not able to operate with Tx optical channel index = 17 | RO               |
| 1.802.0 | Tx index ability 16 | 1 = PMD is able to operate with Tx optical channel index = 16<br>0 = PMD is not able to operate with Tx optical channel index = 16 | RO               |

<sup>a</sup>RO = Read only

#### 45.2.1.152.1 Tx index ability 16 through 31 (1.802.0 through 1.802.15)

When read as a one, bit 1.802.0 indicates that the PMD selected by bits 1.7.6:0 is able to operate with a Tx optical channel index of 16. When read as a zero, bit 1.802.0 indicates that the PMD is not able to operate with a Tx optical channel index of 16. Bits 1.802.1 through 1.802.15 indicate the equivalent for index values 17 through 31, respectively. The optical frequencies that correspond to these index values are given in the appropriate PMD clause. For 100GBASE-ZR see Table 154–5.

#### 45.2.1.153 Tx optical channel ability 3 register (Register 1.803)

The assignment of bits in the Tx optical channel ability 3 register is shown in Table 45–122.

**Table 45–122—Tx optical channel ability 3 register bit definitions**

| Bit(s)   | Name                | Description                                                                                                                        | R/W <sup>a</sup> |
|----------|---------------------|------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.803.15 | Tx index ability 47 | 1 = PMD is able to operate with Tx optical channel index = 47<br>0 = PMD is not able to operate with Tx optical channel index = 47 | RO               |
| 1.803.14 | Tx index ability 46 | 1 = PMD is able to operate with Tx optical channel index = 46<br>0 = PMD is not able to operate with Tx optical channel index = 46 | RO               |
| 1.803.13 | Tx index ability 45 | 1 = PMD is able to operate with Tx optical channel index = 45<br>0 = PMD is not able to operate with Tx optical channel index = 45 | RO               |
| 1.803.12 | Tx index ability 44 | 1 = PMD is able to operate with Tx optical channel index = 44<br>0 = PMD is not able to operate with Tx optical channel index = 44 | RO               |

**Table 45–122—Tx optical channel ability 3 register bit definitions (*continued*)**

| Bit(s)   | Name                | Description                                                                                                                        | R/W <sup>a</sup> |
|----------|---------------------|------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.803.11 | Tx index ability 43 | 1 = PMD is able to operate with Tx optical channel index = 43<br>0 = PMD is not able to operate with Tx optical channel index = 43 | RO               |
| 1.803.10 | Tx index ability 42 | 1 = PMD is able to operate with Tx optical channel index = 42<br>0 = PMD is not able to operate with Tx optical channel index = 42 | RO               |
| 1.803.9  | Tx index ability 41 | 1 = PMD is able to operate with Tx optical channel index = 41<br>0 = PMD is not able to operate with Tx optical channel index = 41 | RO               |
| 1.803.8  | Tx index ability 40 | 1 = PMD is able to operate with Tx optical channel index = 40<br>0 = PMD is not able to operate with Tx optical channel index = 40 | RO               |
| 1.803.7  | Tx index ability 39 | 1 = PMD is able to operate with Tx optical channel index = 39<br>0 = PMD is not able to operate with Tx optical channel index = 39 | RO               |
| 1.803.6  | Tx index ability 38 | 1 = PMD is able to operate with Tx optical channel index = 38<br>0 = PMD is not able to operate with Tx optical channel index = 38 | RO               |
| 1.803.5  | Tx index ability 37 | 1 = PMD is able to operate with Tx optical channel index = 37<br>0 = PMD is not able to operate with Tx optical channel index = 37 | RO               |
| 1.803.4  | Tx index ability 36 | 1 = PMD is able to operate with Tx optical channel index = 36<br>0 = PMD is not able to operate with Tx optical channel index = 36 | RO               |
| 1.803.3  | Tx index ability 35 | 1 = PMD is able to operate with Tx optical channel index = 35<br>0 = PMD is not able to operate with Tx optical channel index = 35 | RO               |
| 1.803.2  | Tx index ability 34 | 1 = PMD is able to operate with Tx optical channel index = 34<br>0 = PMD is not able to operate with Tx optical channel index = 34 | RO               |
| 1.803.1  | Tx index ability 33 | 1 = PMD is able to operate with Tx optical channel index = 33<br>0 = PMD is not able to operate with Tx optical channel index = 33 | RO               |
| 1.803.0  | Tx index ability 32 | 1 = PMD is able to operate with Tx optical channel index = 32<br>0 = PMD is not able to operate with Tx optical channel index = 32 | RO               |

<sup>a</sup>RO = Read only

#### 45.2.1.153.1 Tx index ability 32 through 47 (1.803.0 through 1.803.15)

When read as a one, bit 1.803.0 indicates that the PMD selected by bits 1.7.6:0 is able to operate with a Tx optical channel index of 32. When read as a zero, bit 1.803.0 indicates that the PMD is not able to operate with a Tx optical channel index of 32. Bits 1.803.1 through 1.803.15 indicate the equivalent for index values 33 through 47, respectively. The optical frequencies that correspond to these index values are given in the appropriate PMD clause. For 100GBASE-ZR see Table 154–5.



#### 45.2.1.154 Rx optical channel control register (Register 1.820)

The assignment of bits in the Rx optical channel control register is shown in Table 45–123.

**Table 45–123—Rx optical channel control register bit definitions**

| Bit(s)     | Name                                    | Description                                                                                                                                      | R/W <sup>a</sup> |
|------------|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.820.15   | Tx Rx different optical channel ability | 1 = PMD is able to operate with different Tx and Rx optical channels<br>0 = PMD is not able to operate with different Tx and Rx optical channels | RO               |
| 1.820.14:6 | Reserved                                | Value always 0                                                                                                                                   | RO               |
| 1.820.5:0  | Rx optical channel index                | Integer value of the Rx optical channel index                                                                                                    | R/W              |

<sup>a</sup>R/W = Read/Write, RO = Read only

##### 45.2.1.154.1 Tx Rx different optical channel ability (1.820.15)

When read as a one, bit 1.820.15 indicates that the PMD is able to operate with an Rx optical channel index that is different from the Tx optical channel index. When read as a zero, bit 1.820.15 indicates that the PMD is not able to operate with an Rx optical channel index that is different from the Tx optical channel index.

##### 45.2.1.154.2 Rx optical channel index (1.820.5:0)

If the PMD is able to operate with an Rx optical channel index number that is different from the Tx optical channel index number (bit 1.820.15 is one), bits 1.820.5:0 set the value of the Rx optical channel index number (and hence the receiver optical frequency) with bit 1.820.0 being the LSB and bit 1.820.5 being the MSB. The channel index number indicates the corresponding optical frequency. For 100GBASE-ZR the specific optical frequency corresponding to each channel index number is listed in Table 154–5. If the PMD is not able to operate with an Rx optical channel index that is different from the Tx optical channel index (bit 1.820.15 is zero), bits 1.820.5:0 have no effect and the value of the Rx optical channel index is the same as the value of the Tx optical channel index as set by bits 1.800.5:0.

#### 45.2.1.155 Rx optical channel ability 1 register (Register 1.821)

The assignment of bits in the Rx optical channel ability 1 register is shown in Table 45–124.

**Table 45–124—Rx optical channel ability 1 register bit definitions**

| Bit(s)   | Name                | Description                                                                                                                        | R/W <sup>a</sup> |
|----------|---------------------|------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.821.15 | Rx index ability 15 | 1 = PMD is able to operate with Rx optical channel index = 15<br>0 = PMD is not able to operate with Rx optical channel index = 15 | RO               |
| 1.821.14 | Rx index ability 14 | 1 = PMD is able to operate with Rx optical channel index = 14<br>0 = PMD is not able to operate with Rx optical channel index = 14 | RO               |
| 1.821.13 | Rx index ability 13 | 1 = PMD is able to operate with Rx optical channel index = 13<br>0 = PMD is not able to operate with Rx optical channel index = 13 | RO               |
| 1.821.12 | Rx index ability 12 | 1 = PMD is able to operate with Rx optical channel index = 12<br>0 = PMD is not able to operate with Rx optical channel index = 12 | RO               |
| 1.821.11 | Rx index ability 11 | 1 = PMD is able to operate with Rx optical channel index = 11<br>0 = PMD is not able to operate with Rx optical channel index = 11 | RO               |
| 1.821.10 | Rx index ability 10 | 1 = PMD is able to operate with Rx optical channel index = 10<br>0 = PMD is not able to operate with Rx optical channel index = 10 | RO               |
| 1.821.9  | Rx index ability 9  | 1 = PMD is able to operate with Rx optical channel index = 9<br>0 = PMD is not able to operate with Rx optical channel index = 9   | RO               |
| 1.821.8  | Rx index ability 8  | 1 = PMD is able to operate with Rx optical channel index = 8<br>0 = PMD is not able to operate with Rx optical channel index = 8   | RO               |
| 1.821.7  | Rx index ability 7  | 1 = PMD is able to operate with Rx optical channel index = 7<br>0 = PMD is not able to operate with Rx optical channel index = 7   | RO               |
| 1.821.6  | Rx index ability 6  | 1 = PMD is able to operate with Rx optical channel index = 6<br>0 = PMD is not able to operate with Rx optical channel index = 6   | RO               |
| 1.821.5  | Rx index ability 5  | 1 = PMD is able to operate with Rx optical channel index = 5<br>0 = PMD is not able to operate with Rx optical channel index = 5   | RO               |
| 1.821.4  | Rx index ability 4  | 1 = PMD is able to operate with Rx optical channel index = 4<br>0 = PMD is not able to operate with Rx optical channel index = 4   | RO               |
| 1.821.3  | Rx index ability 3  | 1 = PMD is able to operate with Rx optical channel index = 3<br>0 = PMD is not able to operate with Rx optical channel index = 3   | RO               |
| 1.821.2  | Rx index ability 2  | 1 = PMD is able to operate with Rx optical channel index = 2<br>0 = PMD is not able to operate with Rx optical channel index = 2   | RO               |
| 1.821.1  | Rx index ability 1  | 1 = PMD is able to operate with Rx optical channel index = 1<br>0 = PMD is not able to operate with Rx optical channel index = 1   | RO               |
| 1.821.0  | Rx index ability 0  | 1 = PMD is able to operate with Rx optical channel index = 0<br>0 = PMD is not able to operate with Rx optical channel index = 0   | RO               |

<sup>a</sup>RO = Read only

##### 45.2.1.155.1 Rx index ability 0 through 15 (1.821.0 through 1.821.15)

When read as a one, bit 1.821.0 indicates that the PMD selected by bits 1.7.6:0 is able to operate with an Rx optical channel index of 0. When read as a zero, bit 1.821.0 indicates that the PMD is not able to operate with an Rx optical channel index of 0. Bits 1.821.1 through 1.821.15 indicate the equivalent for index values

1 through 15, respectively. The optical frequencies that correspond to these index values are given in the appropriate PMD clause. For 100GBASE-ZR see Table 154–5.

#### 45.2.1.156 Rx optical channel ability 2 register (Register 1.822)

The assignment of bits in the Rx optical channel ability 2 register is shown in Table 45–125.

**Table 45–125—Rx optical channel ability 2 register bit definitions**

| Bit(s)   | Name                | Description                                                                                                                        | R/W <sup>a</sup> |
|----------|---------------------|------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.822.15 | Rx index ability 31 | 1 = PMD is able to operate with Rx optical channel index = 31<br>0 = PMD is not able to operate with Rx optical channel index = 31 | RO               |
| 1.822.14 | Rx index ability 30 | 1 = PMD is able to operate with Rx optical channel index = 30<br>0 = PMD is not able to operate with Rx optical channel index = 30 | RO               |
| 1.822.13 | Rx index ability 29 | 1 = PMD is able to operate with Rx optical channel index = 29<br>0 = PMD is not able to operate with Rx optical channel index = 29 | RO               |
| 1.822.12 | Rx index ability 28 | 1 = PMD is able to operate with Rx optical channel index = 28<br>0 = PMD is not able to operate with Rx optical channel index = 28 | RO               |
| 1.822.11 | Rx index ability 27 | 1 = PMD is able to operate with Rx optical channel index = 27<br>0 = PMD is not able to operate with Rx optical channel index = 27 | RO               |
| 1.822.10 | Rx index ability 26 | 1 = PMD is able to operate with Rx optical channel index = 26<br>0 = PMD is not able to operate with Rx optical channel index = 26 | RO               |
| 1.822.9  | Rx index ability 25 | 1 = PMD is able to operate with Rx optical channel index = 25<br>0 = PMD is not able to operate with Rx optical channel index = 25 | RO               |
| 1.822.8  | Rx index ability 24 | 1 = PMD is able to operate with Rx optical channel index = 24<br>0 = PMD is not able to operate with Rx optical channel index = 24 | RO               |
| 1.822.7  | Rx index ability 23 | 1 = PMD is able to operate with Rx optical channel index = 23<br>0 = PMD is not able to operate with Rx optical channel index = 23 | RO               |
| 1.822.6  | Rx index ability 22 | 1 = PMD is able to operate with Rx optical channel index = 22<br>0 = PMD is not able to operate with Rx optical channel index = 22 | RO               |
| 1.822.5  | Rx index ability 21 | 1 = PMD is able to operate with Rx optical channel index = 21<br>0 = PMD is not able to operate with Rx optical channel index = 21 | RO               |
| 1.822.4  | Rx index ability 20 | 1 = PMD is able to operate with Rx optical channel index = 20<br>0 = PMD is not able to operate with Rx optical channel index = 20 | RO               |
| 1.822.3  | Rx index ability 19 | 1 = PMD is able to operate with Rx optical channel index = 19<br>0 = PMD is not able to operate with Rx optical channel index = 19 | RO               |
| 1.822.2  | Rx index ability 18 | 1 = PMD is able to operate with Rx optical channel index = 18<br>0 = PMD is not able to operate with Rx optical channel index = 18 | RO               |
| 1.822.1  | Rx index ability 17 | 1 = PMD is able to operate with Rx optical channel index = 17<br>0 = PMD is not able to operate with Rx optical channel index = 17 | RO               |
| 1.822.0  | Rx index ability 16 | 1 = PMD is able to operate with Rx optical channel index = 16<br>0 = PMD is not able to operate with Rx optical channel index = 16 | RO               |

<sup>a</sup>RO = Read only

#### 45.2.1.156.1 Rx index ability 16 through 31 (1.822.0 through 1.822.15)

When read as a one, bit 1.822.0 indicates that the PMD selected by bits 1.7.6:0 is able to operate with an Rx optical channel index of 16. When read as a zero, bit 1.822.0 indicates that the PMD is not able to operate with an Rx optical channel index of 16. Bits 1.822.1 through 1.822.15 indicate the equivalent for index values 17 through 31, respectively. The optical frequencies that correspond to these index values are given in the appropriate PMD clause. For 100GBASE-ZR see Table 154–5.

#### 45.2.1.157 Rx optical channel ability 3 register (Register 1.823)

The assignment of bits in the Rx optical channel ability 3 register is shown in Table 45–126.

**Table 45–126—Rx optical channel ability 3 register bit definitions**

| Bit(s)   | Name                | Description                                                                                                                        | R/W <sup>a</sup> |
|----------|---------------------|------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.823.15 | Rx index ability 47 | 1 = PMD is able to operate with Rx optical channel index = 47<br>0 = PMD is not able to operate with Rx optical channel index = 47 | RO               |
| 1.823.14 | Rx index ability 46 | 1 = PMD is able to operate with Rx optical channel index = 46<br>0 = PMD is not able to operate with Rx optical channel index = 46 | RO               |
| 1.823.13 | Rx index ability 45 | 1 = PMD is able to operate with Rx optical channel index = 45<br>0 = PMD is not able to operate with Rx optical channel index = 45 | RO               |
| 1.823.12 | Rx index ability 44 | 1 = PMD is able to operate with Rx optical channel index = 44<br>0 = PMD is not able to operate with Rx optical channel index = 44 | RO               |
| 1.823.11 | Rx index ability 43 | 1 = PMD is able to operate with Rx optical channel index = 43<br>0 = PMD is not able to operate with Rx optical channel index = 43 | RO               |
| 1.823.10 | Rx index ability 42 | 1 = PMD is able to operate with Rx optical channel index = 42<br>0 = PMD is not able to operate with Rx optical channel index = 42 | RO               |
| 1.823.9  | Rx index ability 41 | 1 = PMD is able to operate with Rx optical channel index = 41<br>0 = PMD is not able to operate with Rx optical channel index = 41 | RO               |
| 1.823.8  | Rx index ability 40 | 1 = PMD is able to operate with Rx optical channel index = 40<br>0 = PMD is not able to operate with Rx optical channel index = 40 | RO               |
| 1.823.7  | Rx index ability 39 | 1 = PMD is able to operate with Rx optical channel index = 39<br>0 = PMD is not able to operate with Rx optical channel index = 39 | RO               |
| 1.823.6  | Rx index ability 38 | 1 = PMD is able to operate with Rx optical channel index = 38<br>0 = PMD is not able to operate with Rx optical channel index = 38 | RO               |
| 1.823.5  | Rx index ability 37 | 1 = PMD is able to operate with Rx optical channel index = 37<br>0 = PMD is not able to operate with Rx optical channel index = 37 | RO               |
| 1.823.4  | Rx index ability 36 | 1 = PMD is able to operate with Rx optical channel index = 36<br>0 = PMD is not able to operate with Rx optical channel index = 36 | RO               |
| 1.823.3  | Rx index ability 35 | 1 = PMD is able to operate with Rx optical channel index = 35<br>0 = PMD is not able to operate with Rx optical channel index = 35 | RO               |
| 1.823.2  | Rx index ability 34 | 1 = PMD is able to operate with Rx optical channel index = 34<br>0 = PMD is not able to operate with Rx optical channel index = 34 | RO               |
| 1.823.1  | Rx index ability 33 | 1 = PMD is able to operate with Rx optical channel index = 33<br>0 = PMD is not able to operate with Rx optical channel index = 33 | RO               |
| 1.823.0  | Rx index ability 32 | 1 = PMD is able to operate with Rx optical channel index = 32<br>0 = PMD is not able to operate with Rx optical channel index = 32 | RO               |

<sup>a</sup>RO = Read only

#### 45.2.1.157.1 Rx index ability 32 through 47 (1.823.0 through 1.823.15)

When read as a one, bit 1.823.0 indicates that the PMD selected by bits 1.7.6:0 is able to operate with an Rx optical channel index of 32. When read as a zero, bit 1.823.0 indicates that the PMD is not able to operate with an Rx optical channel index of 32. Bits 1.823.1 through 1.823.15 indicate the equivalent for index values 33 through 47, respectively. The optical frequencies that correspond to these index values are given in the appropriate PMD clause. For 100GBASE-ZR see Table 154–5.

#### 45.2.1.158 BASE-H PMA/PMD control register (Register 1.900)

The assignment of bits in the BASE-H PMA/PMD control register is shown in Table 45–127.

**Table 45–127—BASE-H PMA/PMD control register bit definitions**

| Bit(s)     | Name           | Description                                                                                                                                             | R/W <sup>a</sup> |
|------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.900.15:4 | Reserved       | Value always 0                                                                                                                                          | RO               |
| 1.900.3:0  | Type selection | 3 2 1 0<br>1 x x x = Reserved<br>0 1 x x = Reserved<br>0 0 1 1 = Reserved<br>0 0 1 0 = 1000BASE-RHC<br>0 0 0 1 = 1000BASE-RHB<br>0 0 0 0 = 1000BASE-RHA | R/W              |

<sup>a</sup>RO = Read Only, R/W = Read Write

#### 45.2.1.158.1 Type selection (1.900.3:0)

Bits 1.900.3:0 are used to set the mode of operation. When these bits are set to 0000, the mode of operation is 1000BASE-RHA. When these bits are set to 0001, the mode of operation is 1000BASE-RHB. When these bits are set to 0010, the mode of operation is 1000BASE-RHC. Any attempt to configure a different value or a mode of operation not supported per register 1.22 shall have no effect.

#### 45.2.1.159 Nx25G-EPON PMA/PMD extended ability register (Registers 1.1000 through 1.1002)

The assignment of bits in the Nx25G-EPON PMA/PMD extended ability register is shown in Table 45–128.

**Table 45–128—Nx25G-EPON PMA/PMD extended ability register bit definitions**

| Bit(s)    | Name           | Description                                                                                        | R/W <sup>a</sup> |
|-----------|----------------|----------------------------------------------------------------------------------------------------|------------------|
| 1.1000.15 | 25GBASE-PQX-U3 | 1 = PMA/PMD is able to perform 25GBASE-PQX-U3<br>0 = PMA/PMD is not able to perform 25GBASE-PQX-U3 | RO               |
| 1.1000.14 | 25GBASE-PQX-U2 | 1 = PMA/PMD is able to perform 25GBASE-PQX-U2<br>0 = PMA/PMD is not able to perform 25GBASE-PQX-U2 | RO               |
| 1.1000.13 | 25GBASE-PQX-D3 | 1 = PMA/PMD is able to perform 25GBASE-PQX-D3<br>0 = PMA/PMD is not able to perform 25GBASE-PQX-D3 | RO               |
| 1.1000.12 | 25GBASE-PQX-D2 | 1 = PMA/PMD is able to perform 25GBASE-PQX-D2<br>0 = PMA/PMD is not able to perform 25GBASE-PQX-D2 | RO               |

**Table 45–128—Nx25G-EPON PMA/PMD extended ability register bit definitions (*continued*)**

| Bit(s)    | Name              | Description                                                                                              | R/W <sup>a</sup> |
|-----------|-------------------|----------------------------------------------------------------------------------------------------------|------------------|
| 1.1000.11 | 25GBASE-PQG-U3    | 1 = PMA/PMD is able to perform 25GBASE-PQG-U3<br>0 = PMA/PMD is not able to perform 25GBASE-PQG-U3       | RO               |
| 1.1000.10 | 25GBASE-PQG-U2    | 1 = PMA/PMD is able to perform 25GBASE-PQG-U2<br>0 = PMA/PMD is not able to perform 25GBASE-PQG-U2       | RO               |
| 1.1000.9  | 25GBASE-PQG-D3    | 1 = PMA/PMD is able to perform 25GBASE-PQG-D3<br>0 = PMA/PMD is not able to perform 25GBASE-PQG-D3       | RO               |
| 1.1000.8  | 25GBASE-PQG-D2    | 1 = PMA/PMD is able to perform 25GBASE-PQG-D2<br>0 = PMA/PMD is not able to perform 25GBASE-PQG-D2       | RO               |
| 1.1000.7  | 25/10GBASE-PQX-U3 | 1 = PMA/PMD is able to perform 25/10GBASE-PQX-U3<br>0 = PMA/PMD is not able to perform 25/10GBASE-PQX-U3 | RO               |
| 1.1000.6  | 25/10GBASE-PQX-U2 | 1 = PMA/PMD is able to perform 25/10GBASE-PQX-U2<br>0 = PMA/PMD is not able to perform 25/10GBASE-PQX-U2 | RO               |
| 1.1000.5  | 25/10GBASE-PQX-D3 | 1 = PMA/PMD is able to perform 25/10GBASE-PQX-D3<br>0 = PMA/PMD is not able to perform 25/10GBASE-PQX-D3 | RO               |
| 1.1000.4  | 25/10GBASE-PQX-D2 | 1 = PMA/PMD is able to perform 25/10GBASE-PQX-D2<br>0 = PMA/PMD is not able to perform 25/10GBASE-PQX-D2 | RO               |
| 1.1000.3  | 25/10GBASE-PQG-U3 | 1 = PMA/PMD is able to perform 25/10GBASE-PQG-U3<br>0 = PMA/PMD is not able to perform 25/10GBASE-PQG-U3 | RO               |
| 1.1000.2  | 25/10GBASE-PQG-U2 | 1 = PMA/PMD is able to perform 25/10GBASE-PQG-U2<br>0 = PMA/PMD is not able to perform 25/10GBASE-PQG-U2 | RO               |
| 1.1000.1  | 25/10GBASE-PQG-D3 | 1 = PMA/PMD is able to perform 25/10GBASE-PQG-D3<br>0 = PMA/PMD is not able to perform 25/10GBASE-PQG-D3 | RO               |
| 1.1000.0  | 25/10GBASE-PQG-D2 | 1 = PMA/PMD is able to perform 25/10GBASE-PQG-D2<br>0 = PMA/PMD is not able to perform 25/10GBASE-PQG-D2 | RO               |
| 1.1001.15 | 50/25GBASE-PQX-U3 | 1 = PMA/PMD is able to perform 50/25GBASE-PQX-U3<br>0 = PMA/PMD is not able to perform 50/25GBASE-PQX-U3 | RO               |
| 1.1001.14 | 50/25GBASE-PQX-U2 | 1 = PMA/PMD is able to perform 50/25GBASE-PQX-U2<br>0 = PMA/PMD is not able to perform 50/25GBASE-PQX-U2 | RO               |
| 1.1001.13 | 50/25GBASE-PQX-D3 | 1 = PMA/PMD is able to perform 50/25GBASE-PQX-D3<br>0 = PMA/PMD is not able to perform 50/25GBASE-PQX-D3 | RO               |
| 1.1001.12 | 50/25GBASE-PQX-D2 | 1 = PMA/PMD is able to perform 50/25GBASE-PQX-D2<br>0 = PMA/PMD is not able to perform 50/25GBASE-PQX-D2 | RO               |
| 1.1001.11 | 50/25GBASE-PQG-U3 | 1 = PMA/PMD is able to perform 50/25GBASE-PQG-U3<br>0 = PMA/PMD is not able to perform 50/25GBASE-PQG-U3 | RO               |
| 1.1001.10 | 50/25GBASE-PQG-U2 | 1 = PMA/PMD is able to perform 50/25GBASE-PQG-U2<br>0 = PMA/PMD is not able to perform 50/25GBASE-PQG-U2 | RO               |
| 1.1001.9  | 50/25GBASE-PQG-D3 | 1 = PMA/PMD is able to perform 50/25GBASE-PQG-D3<br>0 = PMA/PMD is not able to perform 50/25GBASE-PQG-D3 | RO               |
| 1.1001.8  | 50/25GBASE-PQG-D2 | 1 = PMA/PMD is able to perform 50/25GBASE-PQG-D2<br>0 = PMA/PMD is not able to perform 50/25GBASE-PQG-D2 | RO               |
| 1.1001.7  | 50/10GBASE-PQX-U3 | 1 = PMA/PMD is able to perform 50/10GBASE-PQX-U3<br>0 = PMA/PMD is not able to perform 50/10GBASE-PQX-U3 | RO               |

**Table 45–128—Nx25G-EPON PMA/PMD extended ability register bit definitions (*continued*)**

| Bit(s)      | Name              | Description                                                                                              | R/W <sup>a</sup> |
|-------------|-------------------|----------------------------------------------------------------------------------------------------------|------------------|
| 1.1001.6    | 50/10GBASE-PQX-U2 | 1 = PMA/PMD is able to perform 50/10GBASE-PQX-U2<br>0 = PMA/PMD is not able to perform 50/10GBASE-PQX-U2 | RO               |
| 1.1001.5    | 50/10GBASE-PQX-D3 | 1 = PMA/PMD is able to perform 50/10GBASE-PQX-D3<br>0 = PMA/PMD is not able to perform 50/10GBASE-PQX-D3 | RO               |
| 1.1001.4    | 50/10GBASE-PQX-D2 | 1 = PMA/PMD is able to perform 50/10GBASE-PQX-D2<br>0 = PMA/PMD is not able to perform 50/10GBASE-PQX-D2 | RO               |
| 1.1001.3    | 50/10GBASE-PQG-U3 | 1 = PMA/PMD is able to perform 50/10GBASE-PQG-U3<br>0 = PMA/PMD is not able to perform 50/10GBASE-PQG-U3 | RO               |
| 1.1001.2    | 50/10GBASE-PQG-U2 | 1 = PMA/PMD is able to perform 50/10GBASE-PQG-U2<br>0 = PMA/PMD is not able to perform 50/10GBASE-PQG-U2 | RO               |
| 1.1001.1    | 50/10GBASE-PQG-D3 | 1 = PMA/PMD is able to perform 50/10GBASE-PQG-D3<br>0 = PMA/PMD is not able to perform 50/10GBASE-PQG-D3 | RO               |
| 1.1001.0    | 50/10GBASE-PQG-D2 | 1 = PMA/PMD is able to perform 50/10GBASE-PQG-D2<br>0 = PMA/PMD is not able to perform 50/10GBASE-PQG-D2 | RO               |
| 1.1002.15:8 | Reserved          | Value always 0                                                                                           | RO               |
| 1.1002.7    | 50GBASE-PQX-U3    | 1 = PMA/PMD is able to perform 50GBASE-PQX-U3<br>0 = PMA/PMD is not able to perform 50GBASE-PQX-U3       | RO               |
| 1.1002.6    | 50GBASE-PQX-U2    | 1 = PMA/PMD is able to perform 50GBASE-PQX-U2<br>0 = PMA/PMD is not able to perform 50GBASE-PQX-U2       | RO               |
| 1.1002.5    | 50GBASE-PQX-D3    | 1 = PMA/PMD is able to perform 50GBASE-PQX-D3<br>0 = PMA/PMD is not able to perform 50GBASE-PQX-D3       | RO               |
| 1.1002.4    | 50GBASE-PQX-D2    | 1 = PMA/PMD is able to perform 50GBASE-PQX-D2<br>0 = PMA/PMD is not able to perform 50GBASE-PQX-D2       | RO               |
| 1.1002.3    | 50GBASE-PQG-U3    | 1 = PMA/PMD is able to perform 50GBASE-PQG-U3<br>0 = PMA/PMD is not able to perform 50GBASE-PQG-U3       | RO               |
| 1.1002.2    | 50GBASE-PQG-U2    | 1 = PMA/PMD is able to perform 50GBASE-PQG-U2<br>0 = PMA/PMD is not able to perform 50GBASE-PQG-U2       | RO               |
| 1.1002.1    | 50GBASE-PQG-D3    | 1 = PMA/PMD is able to perform 50GBASE-PQG-D3<br>0 = PMA/PMD is not able to perform 50GBASE-PQG-D3       | RO               |
| 1.1002.0    | 50GBASE-PQG-D2    | 1 = PMA/PMD is able to perform 50GBASE-PQG-D2<br>0 = PMA/PMD is not able to perform 50GBASE-PQG-D2       | RO               |

<sup>a</sup>RO = Read only

#### 45.2.1.159.1 25GBASE-PQX-U3 (1.1000.15)

When read as a one, bit 1.1000.15 indicates that the PMA/PMD is able to support a 25GBASE-PQX-U3 PMA/PMD type. When read as a zero, bit 1.1000.15 indicates that the PMA/PMD is not able to support a 25GBASE-PQX-U3 PMA/PMD type.

#### **45.2.1.159.2 25GBASE-PQX-U2 (1.1000.14)**

When read as a one, bit 1.1000.14 indicates that the PMA/PMD is able to support a 25GBASE-PQX-U2 PMA/PMD type. When read as a zero, bit 1.1000.14 indicates that the PMA/PMD is not able to support a 25GBASE-PQX-U2 PMA/PMD type.

#### **45.2.1.159.3 25GBASE-PQX-D3 (1.1000.13)**

When read as a one, bit 1.1000.13 indicates that the PMA/PMD is able to support a 25GBASE-PQX-D3 PMA/PMD type. When read as a zero, bit 1.1000.13 indicates that the PMA/PMD is not able to support a 25GBASE-PQX-D3 PMA/PMD type.

#### **45.2.1.159.4 25GBASE-PQX-D2 (1.1000.12)**

When read as a one, bit 1.1000.12 indicates that the PMA/PMD is able to support a 25GBASE-PQX-D2 PMA/PMD type. When read as a zero, bit 1.1000.12 indicates that the PMA/PMD is not able to support a 25GBASE-PQX-D2 PMA/PMD type.

#### **45.2.1.159.5 25GBASE-PQG-U3 (1.1000.11)**

When read as a one, bit 1.1000.11 indicates that the PMA/PMD is able to support a 25GBASE-PQG-U3 PMA/PMD type. When read as a zero, bit 1.1000.11 indicates that the PMA/PMD is not able to support a 25GBASE-PQG-U3 PMA/PMD type.

#### **45.2.1.159.6 25GBASE-PQG-U2 (1.1000.10)**

When read as a one, bit 1.1000.10 indicates that the PMA/PMD is able to support a 25GBASE-PQG-U2 PMA/PMD type. When read as a zero, bit 1.1000.10 indicates that the PMA/PMD is not able to support a 25GBASE-PQG-U2 PMA/PMD type.

#### **45.2.1.159.7 25GBASE-PQG-D3 (1.1000.9)**

When read as a one, bit 1.1000.9 indicates that the PMA/PMD is able to support a 25GBASE-PQG-D3 PMA/PMD type. When read as a zero, bit 1.1000.9 indicates that the PMA/PMD is not able to support a 25GBASE-PQG-D3 PMA/PMD type.

#### **45.2.1.159.8 25GBASE-PQG-D2 (1.1000.8)**

When read as a one, bit 1.1000.8 indicates that the PMA/PMD is able to support a 25GBASE-PQG-D2 PMA/PMD type. When read as a zero, bit 1.1000.8 indicates that the PMA/PMD is not able to support a 25GBASE-PQG-D2 PMA/PMD type.

#### **45.2.1.159.9 25/10GBASE-PQX-U3 (1.1000.7)**

When read as a one, bit 1.1000.7 indicates that the PMA/PMD is able to support a 25/10GBASE-PQX-U3 PMA/PMD type. When read as a zero, bit 1.1000.7 indicates that the PMA/PMD is not able to support a 25/10GBASE-PQX-U3 PMA/PMD type.

#### **45.2.1.159.10 25/10GBASE-PQX-U2 (1.1000.6)**

When read as a one, bit 1.1000.6 indicates that the PMA/PMD is able to support a 25/10GBASE-PQX-U2 PMA/PMD type. When read as a zero, bit 1.1000.6 indicates that the PMA/PMD is not able to support a 25/10GBASE-PQX-U2 PMA/PMD type.



**45.2.1.159.11 25/10GBASE-PQX-D3 (1.1000.5)**

When read as a one, bit 1.1000.5 indicates that the PMA/PMD is able to support a 25/10GBASE-PQX-D3 PMA/PMD type. When read as a zero, bit 1.1000.5 indicates that the PMA/PMD is not able to support a 25/10GBASE-PQX-D3 PMA/PMD type.

**45.2.1.159.12 25/10GBASE-PQX-D2 (1.1000.4)**

When read as a one, bit 1.1000.4 indicates that the PMA/PMD is able to support a 25/10GBASE-PQX-D2 PMA/PMD type. When read as a zero, bit 1.1000.4 indicates that the PMA/PMD is not able to support a 25/10GBASE-PQX-D2 PMA/PMD type.

**45.2.1.159.13 25/10GBASE-PQG-U3 (1.1000.3)**

When read as a one, bit 1.1000.3 indicates that the PMA/PMD is able to support a 25/10GBASE-PQG-U3 PMA/PMD type. When read as a zero, bit 1.1000.3 indicates that the PMA/PMD is not able to support a 25/10GBASE-PQG-U3 PMA/PMD type.

**45.2.1.159.14 25/10GBASE-PQG-U2 (1.1000.2)**

When read as a one, bit 1.1000.2 indicates that the PMA/PMD is able to support a 25/10GBASE-PQG-U2 PMA/PMD type. When read as a zero, bit 1.1000.2 indicates that the PMA/PMD is not able to support a 25/10GBASE-PQG-U2 PMA/PMD type.

**45.2.1.159.15 25/10GBASE-PQG-D3 (1.1000.1)**

When read as a one, bit 1.1000.1 indicates that the PMA/PMD is able to support a 25/10GBASE-PQG-D3 PMA/PMD type. When read as a zero, bit 1.1000.1 indicates that the PMA/PMD is not able to support a 25/10GBASE-PQG-D3 PMA/PMD type.

**45.2.1.159.16 25/10GBASE-PQG-D2 (1.1000.0)**

When read as a one, bit 1.1000.0 indicates that the PMA/PMD is able to support a 25/10GBASE-PQG-D2 PMA/PMD type. When read as a zero, bit 1.1000.0 indicates that the PMA/PMD is not able to support a 25/10GBASE-PQG-D2 PMA/PMD type.

**45.2.1.159.17 50/25GBASE-PQX-U3 (1.1001.15)**

When read as a one, bit 1.1001.15 indicates that the PMA/PMD is able to support a 50/25GBASE-PQX-U3 PMA/PMD type. When read as a zero, bit 1.1001.15 indicates that the PMA/PMD is not able to support a 50/25GBASE-PQX-U3 PMA/PMD type.

**45.2.1.159.18 50/25GBASE-PQX-U2 (1.1001.14)**

When read as a one, bit 1.1001.14 indicates that the PMA/PMD is able to support a 50/25GBASE-PQX-U2 PMA/PMD type. When read as a zero, bit 1.1001.14 indicates that the PMA/PMD is not able to support a 50/25GBASE-PQX-U2 PMA/PMD type.

**45.2.1.159.19 50/25GBASE-PQX-D3 (1.1001.13)**

When read as a one, bit 1.1001.13 indicates that the PMA/PMD is able to support a 50/25GBASE-PQX-D3 PMA/PMD type. When read as a zero, bit 1.1001.13 indicates that the PMA/PMD is not able to support a 50/25GBASE-PQX-D3 PMA/PMD type.

**45.2.1.159.20 50/25GBASE-PQX-D2 (1.1001.12)**

When read as a one, bit 1.1001.12 indicates that the PMA/PMD is able to support a 50/25GBASE-PQX-D2 PMA/PMD type. When read as a zero, bit 1.1001.12 indicates that the PMA/PMD is not able to support a 50/25GBASE-PQX-D2 PMA/PMD type.

**45.2.1.159.21 50/25GBASE-PQG-U3 (1.1001.11)**

When read as a one, bit 1.1001.11 indicates that the PMA/PMD is able to support a 50/25GBASE-PQG-U3 PMA/PMD type. When read as a zero, bit 1.1001.11 indicates that the PMA/PMD is not able to support a 50/25GBASE-PQG-U3 PMA/PMD type.

**45.2.1.159.22 50/25GBASE-PQG-U2 (1.1001.10)**

When read as a one, bit 1.1001.10 indicates that the PMA/PMD is able to support a 50/25GBASE-PQG-U2 PMA/PMD type. When read as a zero, bit 1.1001.10 indicates that the PMA/PMD is not able to support a 50/25GBASE-PQG-U2 PMA/PMD type.

**45.2.1.159.23 50/25GBASE-PQG-D3 (1.1001.9)**

When read as a one, bit 1.1001.9 indicates that the PMA/PMD is able to support a 50/25GBASE-PQG-D3 PMA/PMD type. When read as a zero, bit 1.1001.9 indicates that the PMA/PMD is not able to support a 50/25GBASE-PQG-D3 PMA/PMD type.

**45.2.1.159.24 50/25GBASE-PQG-D2 (1.1001.8)**

When read as a one, bit 1.1001.8 indicates that the PMA/PMD is able to support a 50/25GBASE-PQG-D2 PMA/PMD type. When read as a zero, bit 1.1001.8 indicates that the PMA/PMD is not able to support a 50/25GBASE-PQG-D2 PMA/PMD type.

**45.2.1.159.25 50/10GBASE-PQX-U3 (1.1001.7)**

When read as a one, bit 1.1001.7 indicates that the PMA/PMD is able to support a 50/10GBASE-PQX-U3 PMA/PMD type. When read as a zero, bit 1.1001.7 indicates that the PMA/PMD is not able to support a 50/10GBASE-PQX-U3 PMA/PMD type.

**45.2.1.159.26 50/10GBASE-PQX-U2 (1.1001.6)**

When read as a one, bit 1.1001.6 indicates that the PMA/PMD is able to support a 50/10GBASE-PQX-U2 PMA/PMD type. When read as a zero, bit 1.1001.6 indicates that the PMA/PMD is not able to support a 50/10GBASE-PQX-U2 PMA/PMD type.

**45.2.1.159.27 50/10GBASE-PQX-D3 (1.1001.5)**

When read as a one, bit 1.1001.5 indicates that the PMA/PMD is able to support a 50/10GBASE-PQX-D3 PMA/PMD type. When read as a zero, bit 1.1001.5 indicates that the PMA/PMD is not able to support a 50/10GBASE-PQX-D3 PMA/PMD type.

**45.2.1.159.28 50/10GBASE-PQX-D2 (1.1001.4)**

When read as a one, bit 1.1001.4 indicates that the PMA/PMD is able to support a 50/10GBASE-PQX-D2 PMA/PMD type. When read as a zero, bit 1.1001.4 indicates that the PMA/PMD is not able to support a 50/10GBASE-PQX-D2 PMA/PMD type.

#### **45.2.1.159.29 50/10GBASE-PQG-U3 (1.1001.3)**

When read as a one, bit 1.1001.3 indicates that the PMA/PMD is able to support a 50/10GBASE-PQG-U3 PMA/PMD type. When read as a zero, bit 1.1001.3 indicates that the PMA/PMD is not able to support a 50/10GBASE-PQG-U3 PMA/PMD type.

#### **45.2.1.159.30 50/10GBASE-PQG-U2 (1.1001.2)**

When read as a one, bit 1.1001.2 indicates that the PMA/PMD is able to support a 50/10GBASE-PQG-U2 PMA/PMD type. When read as a zero, bit 1.1001.2 indicates that the PMA/PMD is not able to support a 50/10GBASE-PQG-U2 PMA/PMD type.

#### **45.2.1.159.31 50/10GBASE-PQG-D3 (1.1001.1)**

When read as a one, bit 1.1001.1 indicates that the PMA/PMD is able to support a 50/10GBASE-PQG-D3 PMA/PMD type. When read as a zero, bit 1.1001.1 indicates that the PMA/PMD is not able to support a 50/10GBASE-PQG-D3 PMA/PMD type.

#### **45.2.1.159.32 50/10GBASE-PQG-D2 (1.1001.0)**

When read as a one, bit 1.1001.0 indicates that the PMA/PMD is able to support a 50/10GBASE-PQG-D2 PMA/PMD type. When read as a zero, bit 1.1001.0 indicates that the PMA/PMD is not able to support a 50/10GBASE-PQG-D2 PMA/PMD type.

#### **45.2.1.159.33 50GBASE-PQX-U3 (1.1002.7)**

When read as a one, bit 1.1002.7 indicates that the PMA/PMD is able to support a 50GBASE-PQX-U3 PMA/PMD type. When read as a zero, bit 1.1002.7 indicates that the PMA/PMD is not able to support a 50GBASE-PQX-U3 PMA/PMD type.

#### **45.2.1.159.34 50GBASE-PQX-U2 (1.1002.6)**

When read as a one, bit 1.1002.6 indicates that the PMA/PMD is able to support a 50GBASE-PQX-U2 PMA/PMD type. When read as a zero, bit 1.1002.6 indicates that the PMA/PMD is not able to support a 50GBASE-PQX-U2 PMA/PMD type.

#### **45.2.1.159.35 50GBASE-PQX-D3 (1.1002.5)**

When read as a one, bit 1.1002.5 indicates that the PMA/PMD is able to support a 50GBASE-PQX-D3 PMA/PMD type. When read as a zero, bit 1.1002.5 indicates that the PMA/PMD is not able to support a 50GBASE-PQX-D3 PMA/PMD type.

#### **45.2.1.159.36 50GBASE-PQX-D2 (1.1002.4)**

When read as a one, bit 1.1002.4 indicates that the PMA/PMD is able to support a 50GBASE-PQX-D2 PMA/PMD type. When read as a zero, bit 1.1002.4 indicates that the PMA/PMD is not able to support a 50GBASE-PQX-D2 PMA/PMD type.

#### **45.2.1.159.37 50GBASE-PQG-U3 (1.1002.3)**

When read as a one, bit 1.1002.3 indicates that the PMA/PMD is able to support a 50GBASE-PQG-U3 PMA/PMD type. When read as a zero, bit 1.1002.3 indicates that the PMA/PMD is not able to support a 50GBASE-PQG-U3 PMA/PMD type.

#### 45.2.1.159.38 50GBASE-PQG-U2 (1.1002.2)

When read as a one, bit 1.1002.2 indicates that the PMA/PMD is able to support a 50GBASE-PQG-U2 PMA/PMD type. When read as a zero, bit 1.1002.2 indicates that the PMA/PMD is not able to support a 50GBASE-PQG-U2 PMA/PMD type.

#### 45.2.1.159.39 50GBASE-PQG-D3 (1.1002.1)

When read as a one, bit 1.1002.1 indicates that the PMA/PMD is able to support a 50GBASE-PQG-D3 PMA/PMD type. When read as a zero, bit 1.1002.1 indicates that the PMA/PMD is not able to support a 50GBASE-PQG-D3 PMA/PMD type.

#### 45.2.1.159.40 50GBASE-PQG-D2 (1.1002.0)

When read as a one, bit 1.1002.0 indicates that the PMA/PMD is able to support a 50GBASE-PQG-D2 PMA/PMD type. When read as a zero, bit 1.1002.0 indicates that the PMA/PMD is not able to support a 50GBASE-PQG-D2 PMA/PMD type.

#### 45.2.1.160 BASE-R LP coefficient update register, lanes 1 through 9 (Register 1.1101, 1.1102, 1.1103, 1.1104, 1.1105, 1.1106, 1.1107, 1.1108, 1.1109)

For multi-PCS lane BASE-R PHYs, these registers are defined similarly to register 1.152 (which is used for lane 0, see 45.2.1.97) but for lanes 1 through 9 respectively of multi-PCS lane PHYs.

Registers corresponding to lanes that are not used for the selected PHY shall return all zeros.

#### 45.2.1.161 BASE-R PAM4 PMD training LP control, lane 0 through lane 3 registers (Register 1.1120 through 1.1123)

The BASE-R PAM4 PMD training LP control, lane 0 through lane 3 registers reflect the contents of the first 16-bit word of the training frame most recently received for each lane. Lane 0 maps to register 1.1120, lane 1 maps to register 1.1121, lane 2 maps to register 1.1122, and lane 3 maps to register 1.1123.

The assignment of bits in the BASE-R PAM4 PMD training LP control, lane 0 register is shown in Table 45–129. The assignment of bits in the registers for lane 1 through lane 3 is equivalent to the assignment for lane 0. When training is not disabled, the bits in registers 1.1120 through 1.1123 are read only; however, when training is disabled the R/W bits become writeable.

**Table 45–129—BASE-R PAM4 PMD training LP control, lane 0 register  
bit definitions**

| Bit(s)       | Name                      | Description                                                                                         | R/W <sup>a</sup> |
|--------------|---------------------------|-----------------------------------------------------------------------------------------------------|------------------|
| 1.1120.15:14 | Reserved                  | Value always 0                                                                                      | RO               |
| 1.1120.13:12 | Initial condition request | 13 12<br>1 1 = Preset 3<br>1 0 = Preset 2<br>0 1 = Preset 1<br>0 0 = Individual coefficient control | R/W              |
| 1.1120.11:10 | Reserved                  | Value always 0                                                                                      | RO               |

**Table 45–129—BASE-R PAM4 PMD training LP control, lane 0 register  
bit definitions (*continued*)**

| Bit(s)     | Name                             | Description                                                                      | R/W <sup>a</sup> |
|------------|----------------------------------|----------------------------------------------------------------------------------|------------------|
| 1.1120.9:8 | Modulation and precoding request | 9 8<br>1 1 = PAM4 with precoding<br>1 0 = PAM4<br>0 1 = Reserved<br>0 0 = PAM2   | R/W              |
| 1.1120.7:5 | Reserved                         | Value always 0                                                                   | RO               |
| 1.1120.4:2 | Coefficient select               | 4 3 2<br>1 1 0 = $c(-2)$<br>1 1 1 = $c(-1)$<br>0 0 0 = $c(0)$<br>0 0 1 = $c(1)$  | R/W              |
| 1.1120.1:0 | Coefficient request              | 1 0<br>1 1 = No equalization<br>1 0 = Decrement<br>0 1 = Increment<br>0 0 = Hold | R/W              |

<sup>a</sup>R/W = Read/Write, RO = Read only

#### **45.2.1.162 BASE-R LP status report register, lanes 1 through 9 (Register 1.1201, 1.1202, 1.1203, 1.1204, 1.1205, 1.1206, 1.1207, 1.1208, 1.1209)**

For multi-PCS lane BASE-R PHYs, these registers are defined similarly to register 1.153 (which is used for lane 0, see 45.2.1.98) but for lanes 1 through 9 respectively of multi-PCS lane PHYs.

Registers corresponding to lanes that are not used for the selected PHY shall return all zeros.

#### **45.2.1.163 BASE-R PAM4 PMD training LP status, lane 0 through lane 3 registers (Register 1.1220 through 1.1223)**

The BASE-R PAM4 PMD training LP status, lane 0 through lane 3 registers reflect the contents of the second 16-bit word of the training frame most recently received for each lane. Lane 0 maps to register 1.1220, lane 1 maps to register 1.1221, lane 2 maps to register 1.1222, and lane 3 maps to register 1.1223.

The assignment of bits in the BASE-R PAM4 PMD training LP status, lane 0 register is shown in Table 45–130. The assignment of bits in the registers for lane 1 through lane 3 is equivalent to the assignment for lane 0.

#### **45.2.1.164 BASE-R LD coefficient update register, lanes 1 through 9 (Register 1.1301, 1.1302, 1.1303, 1.1304, 1.1305, 1.1306, 1.1307, 1.1308, 1.1309)**

For multi-PCS lane BASE-R PHYs, these registers are defined similarly to register 1.154 (which is used for lane 0, see 45.2.1.99) but for lanes 1 through 9 respectively of multi-PCS lane PHYs.

Registers corresponding to lanes that are not used for the selected PHY shall return all zeros.

**Table 45–130—BASE-R PAM4 PMD training LP status, lane 0 register bit definitions**

| Bit(s)       | Name                            | Description                                                                                                                                                                                                                                           | R/W <sup>a</sup> |
|--------------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.1220.15    | Receiver ready                  | 1 = Training is complete and the receiver is ready for data<br>0 = Request for training to continue                                                                                                                                                   | RO               |
| 1.1220.14:12 | Reserved                        | Value always 0                                                                                                                                                                                                                                        | RO               |
| 1.1220.11:10 | Modulation and precoding status | 11 10<br>1 1 = PAM4 with precoding<br>1 0 = PAM4<br>0 1 = Reserved<br>0 0 = PAM2                                                                                                                                                                      | RO               |
| 1.1220.9     | Receiver frame lock             | 1 = Frame boundaries identified<br>0 = Frame boundaries not identified                                                                                                                                                                                | RO               |
| 1.1220.8     | Initial condition status        | 1 = Updated<br>0 = Not updated                                                                                                                                                                                                                        | RO               |
| 1.1220.7     | Parity                          | Even parity bit                                                                                                                                                                                                                                       | RO               |
| 1.1220.6     | Reserved                        | Value always 0                                                                                                                                                                                                                                        | RO               |
| 1.1220.5:3   | Coefficient select echo         | 5 4 3<br>1 1 0 = $c(-2)$<br>1 1 1 = $c(-1)$<br>0 0 0 = $c(0)$<br>0 0 1 = $c(1)$                                                                                                                                                                       | RO               |
| 1.1220.2:0   | Coefficient status              | 2 1 0<br>1 1 1 = Reserved<br>1 1 0 = Coefficient at limit and equalization limit<br><br>1 0 1 = Reserved<br>1 0 0 = Equalization limit<br>0 1 1 = Coefficient not supported<br>0 1 0 = Coefficient at limit<br>0 0 1 = Updated<br>0 0 0 = Not updated | RO               |

<sup>a</sup>RO = Read only

#### 45.2.1.165 BASE-R PAM4 PMD training LD control, lane 0 through lane 3 registers (Register 1.1320 through 1.1323)

The BASE-R PAM4 PMD training LD control, lane 0 through lane 3 registers reflect the contents of the control field of the outgoing training frame for each lane. Lane 0 maps to register 1.1320, lane 1 maps to register 1.1321, lane 2 maps to register 1.1322, and lane 3 maps to register 1.1323.

The assignment of bits in the BASE-R PAM4 PMD training LP control, lane 0 register is shown in Table 45–131. The assignment of bits in the registers for lane 1 through lane 3 is equivalent to the assignment for lane 0.

**Table 45–131—BASE-R PAM4 PMD training LD control, lane 0 register bit definitions**

| Bit(s)       | Name                             | Description                                                                                         | R/W <sup>a</sup> |
|--------------|----------------------------------|-----------------------------------------------------------------------------------------------------|------------------|
| 1.1320.15:14 | Reserved                         | Value always 0                                                                                      | RO               |
| 1.1320.13:12 | Initial condition request        | 13 12<br>1 1 = Preset 3<br>1 0 = Preset 2<br>0 1 = Preset 1<br>0 0 = Individual coefficient control | R/W              |
| 1.1320.11:10 | Reserved                         | Value always 0                                                                                      | RO               |
| 1.1320.9:8   | Modulation and precoding request | 9 8<br>1 1 = PAM4 with precoding<br>1 0 = PAM4<br>0 1 = Reserved<br>0 0 = PAM2                      | R/W              |
| 1.1320.7:5   | Reserved                         | Value always 0                                                                                      | RO               |
| 1.1320.4:2   | Coefficient select               | 4 3 2<br>1 1 0 = $c(-2)$<br>1 1 1 = $c(-1)$<br>0 0 0 = $c(0)$<br>0 0 1 = $c(1)$                     | R/W              |
| 1.1320.1:0   | Coefficient request              | 1 0<br>1 1 = No equalization<br>1 0 = Decrement<br>0 1 = Increment<br>0 0 = Hold                    | R/W              |

<sup>a</sup>R/W = Read/Write, RO = Read only

#### 45.2.1.166 BASE-R LD status report register, lanes 1 through 9 (Register 1.1401, 1.1402, 1.1403, 1.1404, 1.1405, 1.1406, 1.1407, 1.1408, 1.1409)

For multi-PCS lane BASE-R PHYs, these registers are defined similarly to register 1.155 (which is used for lane 0, see 45.2.1.100) but for lanes 1 through 9 respectively of multi-PCS lane PHYs.

Registers corresponding to lanes that are not used for the selected PHY shall return all zeros.

#### 45.2.1.167 BASE-R PAM4 PMD training LD status, lane 0 through lane 3 registers (Register 1.1420 through 1.1423)

The BASE-R PAM4 PMD training LD status, lane 0 through lane 3 registers reflect the contents of the status field of the outgoing training frame for each lane. Lane 0 maps to register 1.1420, lane 1 maps to register 1.1421, lane 2 maps to register 1.1422, and lane 3 maps to register 1.1423.

The assignment of bits in the BASE-R PAM4 PMD training LD status, lane 0 register is shown in Table 45–132. The assignment of bits in the registers for lane 1 through lane 3 is equivalent to the assignment for lane 0.

**Table 45–132—BASE-R PAM4 PMD training LD status, lane 0 register bit definitions**

| Bit(s)       | Name                            | Description                                                                                                                                                                                                                                       | R/W <sup>a</sup> |
|--------------|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.1420.15    | Receiver ready                  | 1 = Training is complete and the receiver is ready for data<br>0 = Request for training to continue                                                                                                                                               | RO               |
| 1.1420.14:12 | Reserved                        | Value always 0                                                                                                                                                                                                                                    | RO               |
| 1.1420.11:10 | Modulation and precoding status | 11 10<br>1 1 = PAM4 with precoding<br>1 0 = PAM4<br>0 1 = Reserved<br>0 0 = PAM2                                                                                                                                                                  | RO               |
| 1.1420.9     | Receiver frame lock             | 1 = Frame boundaries identified<br>0 = Frame boundaries not identified                                                                                                                                                                            | RO               |
| 1.1420.8     | Initial condition status        | 1 = Updated<br>0 = Not updated                                                                                                                                                                                                                    | RO               |
| 1.1420.7     | Parity                          | Even parity bit                                                                                                                                                                                                                                   | RO               |
| 1.1420.6     | Reserved                        | Value always 0                                                                                                                                                                                                                                    | RO               |
| 1.1420.5:3   | Coefficient select echo         | 5 4 3<br>1 1 0 = $c(-2)$<br>1 1 1 = $c(-1)$<br>0 0 0 = $c(0)$<br>0 0 1 = $c(1)$                                                                                                                                                                   | RO               |
| 1.1420.2:0   | Coefficient status              | 2 1 0<br>1 1 1 = Reserved<br>1 1 0 = Coefficient at limit and equalization limit<br>1 0 1 = Reserved<br>1 0 0 = Equalization limit<br>0 1 1 = Coefficient not supported<br>0 1 0 = Coefficient at limit<br>0 0 1 = Updated<br>0 0 0 = Not updated | RO               |

<sup>a</sup>RO = Read only

#### 45.2.1.168 PMD training pattern lanes 0 through 3 (Register 1.1450 through 1.1453)

The assignment of bits in the PMD training pattern lane 0 register is shown in Table 45–133. The assignment of bits in the PMD training pattern lanes 1 through 3 registers are defined similarly to lane 0. Register 1.1450



controls the PMD training pattern for PMD lane 0; register 1.1451 controls the PMD training pattern for PMD lane 1; etc.

**Table 45–133—PMD training pattern lane 0 bit definitions**

| Bit(s)       | Name                  | Description                                              | R/W <sup>a</sup> |
|--------------|-----------------------|----------------------------------------------------------|------------------|
| 1.1450.15:14 | Seed                  | Two most significant bits of PRBS13 seed                 | R/W              |
| 1.1450.13    | Reserved              | Value always 0                                           | RO               |
| 1.1450.12:11 | Polynomial identifier | Identifier (0, 1, 2, or 3) selecting polynomial for PRBS | R/W              |
| 1.1450.10:0  | Seed                  | 11 bit, binary seed for sequence                         | R/W              |

<sup>a</sup>R/W = Read/Write, RO = Read only

Register bits 12:11 contain a 2-bit identifier that selects the polynomial used for training in the particular PMD lane according to the definition in 92.7.12. The polynomial identifier for each lane should be unique; two lanes having the same identifier could impair operation of the PMD control function. The default identifiers are (binary): for lane 0, 00; for lane 1, 01; for lane 2, 10; for lane 3, 11.

For PHYs that use a PRBS11 training pattern, register bits 10:0 contain the 11-bit seed for the sequence, where register bit 0 gives seed bit S0; register bit 1 gives seed bit S1; etc., through register bit 10 gives seed bit S10. The default seeds are (binary, S0 is left-most bit): for lane 0, 1010111110; for lane 1, 11001000101; for lane 2, 11100101101; for lane 3, 11110110110. This produces the following initial output (hexadecimal representation where the hex symbols are transmitted from left to right and the most significant bit of each hex symbol is transmitted first): for lane 0, 0xFBFB1CB3E; for lane 1, 0xFBB1E665; for lane 2, 0xF3FDAE46; for lane 3, 0xF2FFA46B.

For PHYs that use a PRBS13 training pattern, the 13-bit seed value is composed from register bits 15:14 and register bits 10:0, where register bit 0 gives seed bit S0; register bit 1 gives seed bit S1; etc., through register bit 10 gives seed bit S10, register bit 14 gives seed bit S11 and register bit 15 gives seed bit S12. The default PRBS13 seed values are given in Table 136–8.

#### 45.2.1.169 Test-pattern ability (Register 1.1500)

The test-pattern ability register is used for PHY types that implement SSPRQ, square wave, and PRBS testing in the PMA. These functions are described in 127.3.4.1, 83.5.10, and 120.5.11. The assignment of bits in the test-pattern ability register is shown in Table 45–134.

**Table 45–134—Test-pattern ability register bit definitions**

| Bit(s)       | Name                       | Description                                                                                                              | R/W <sup>a</sup> |
|--------------|----------------------------|--------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.1500.15:14 | Reserved                   | Value always 0                                                                                                           | RO               |
| 1.1500.13    | SSPRQ Tx generator ability | 1 = SSPRQ transmit direction pattern generator supported<br>0 = SSPRQ transmit direction pattern generator not supported | RO               |
| 1.1500.12    | Square wave test ability   | 1 = Square wave testing supported<br>0 = Square wave testing not supported                                               | RO               |

**Table 45–134—Test-pattern ability register bit definitions (*continued*)**

| Bit(s)    | Name                         | Description                                                                                                                  | R/W <sup>a</sup> |
|-----------|------------------------------|------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.1500.11 | PRBS13Q Tx generator ability | 1 = PRBS13Q transmit direction pattern generator supported<br>0 = PRBS13Q transmit direction pattern generator not supported | RO               |
| 1.1500.10 | PRBS13Q Rx generator ability | 1 = PRBS13Q receive direction pattern generator supported<br>0 = PRBS13Q receive direction pattern generator not supported   | RO               |
| 1.1500.9  | PRBS31Q Tx generator ability | 1 = PRBS31Q transmit direction pattern generator supported<br>0 = PRBS31Q transmit direction pattern generator not supported | RO               |
| 1.1500.8  | PRBS31Q Tx checker ability   | 1 = PRBS31Q transmit direction pattern checker supported<br>0 = PRBS31Q transmit direction pattern checker not supported     | RO               |
| 1.1500.7  | PRBS31Q Rx generator ability | 1 = PRBS31Q receive direction pattern generator supported<br>0 = PRBS31Q receive direction pattern generator not supported   | RO               |
| 1.1500.6  | PRBS31Q Rx checker ability   | 1 = PRBS31Q receive direction pattern checker supported<br>0 = PRBS31Q receive direction pattern checker not supported       | RO               |
| 1.1500.5  | PRBS9 Tx generator ability   | 1 = PRBS9 transmit direction pattern generator supported<br>0 = PRBS9 transmit direction pattern generator not supported     | RO               |
| 1.1500.4  | PRBS9 Rx generator ability   | 1 = PRBS9 receive direction pattern generator supported<br>0 = PRBS9 receive direction pattern generator not supported       | RO               |
| 1.1500.3  | PRBS31 Tx generator ability  | 1 = PRBS31 transmit direction pattern generator supported<br>0 = PRBS31 transmit direction pattern generator not supported   | RO               |
| 1.1500.2  | PRBS31 Tx checker ability    | 1 = PRBS31 transmit direction pattern checker supported<br>0 = PRBS31 transmit direction pattern checker not supported       | RO               |
| 1.1500.1  | PRBS31 Rx generator ability  | 1 = PRBS31 receive direction pattern generator supported<br>0 = PRBS31 receive direction pattern generator not supported     | RO               |
| 1.1500.0  | PRBS31 Rx checker ability    | 1 = PRBS31 receive direction pattern checker supported<br>0 = PRBS31 receive direction pattern checker not supported         | RO               |

<sup>a</sup>RO = Read only

If SSPRQ pattern testing is supported and register 1.1500 is implemented then bit 1.1500.13 shall indicate the generation ability in the transmit direction. The SSPRQ pattern test is controlled by register 1.1501 (see 45.2.1.170).

If square wave testing is supported and register 1.1500 is implemented then bit 1.1500.12 shall be set to one. The square wave test is controlled by register 1.1510 (see 45.2.1.171).

If PRBS13Q pattern testing is supported and register 1.1500 is implemented then bit 1.1500.11 shall indicate the generation ability in the transmit direction and bit 1.1500.10 shall indicate the generation ability in the receive direction. The PRBS13Q pattern test is controlled by register 1.1512 (see 45.2.1.172).

If PRBS31Q pattern testing is supported and register 1.1500 is implemented then bit 1.1500.9 shall indicate the generation ability in the transmit direction and bit 1.1500.7 shall indicate the generation ability in the receive direction. Bit 1.1500.8 shall indicate the checker ability in the transmit direction and bit 1.1500.6

shall indicate the checker ability in the receive direction. The PRBS31Q pattern test is controlled by register 1.1501 (see 45.2.1.170).

If PRBS9 pattern testing is supported and register 1.1500 is implemented then bit 1.1500.5 shall indicate the generation ability in the transmit direction and bit 1.1500.4 shall indicate the generation ability in the receive direction. The PRBS pattern test is controlled by register 1.1501 (see 45.2.1.170).

If PRBS31 pattern testing is supported and register 1.1500 is implemented then bit 1.1500.3 shall indicate the generation ability in the transmit direction and bit 1.1500.1 shall indicate the generation ability in the receive direction. Bit 1.1500.2 shall indicate the checker ability in the transmit direction and bit 1.1500.0 shall indicate the checker ability in the receive direction. The PRBS pattern test is controlled by register 1.1501 (see 45.2.1.170).

#### 45.2.1.170 PRBS pattern testing control (Register 1.1501)

The PRBS pattern testing control register is used for PHY types that implement PRBS pattern testing in the PMA. This function is described in 83.5.10 and 120.5.11. The assignment of bits in the PRBS pattern testing control register is shown in Table 45–135.

**Table 45–135—PRBS pattern testing control register bit definitions**

| Bit(s)     | Name                                      | Description                                                                                                   | R/W <sup>a</sup> |
|------------|-------------------------------------------|---------------------------------------------------------------------------------------------------------------|------------------|
| 1.1501.15  | Reserved                                  | Value always 0                                                                                                | RO               |
| 1.1501.14  | SSPRQ pattern enable                      | 1 = Enable SSPRQ test-pattern<br>0 = Disable SSPRQ test-pattern                                               | R/W              |
| 1.1501.13  | PRBS31Q pattern enable                    | 1 = Enable PRBS31Q test-pattern<br>0 = Disable PRBS31Q test-pattern                                           | R/W              |
| 1.1501.12  | Reserved                                  | Value always 0                                                                                                | RO               |
| 1.1501.11  | Transmitter linearity test pattern enable | 1 = Enable transmitter linearity test-pattern<br>0 = Disable transmitter linearity test-pattern               | R/W              |
| 1.1501.10  | QPRBS13 pattern enable                    | 1 = Enable QPRBS13 test-pattern<br>0 = Disable QPRBS13 test-pattern                                           | R/W              |
| 1.1501.9   | JP03B pattern enable                      | 1 = Enable JP03B test-pattern<br>0 = Disable JP03B test-pattern                                               | R/W              |
| 1.1501.8   | JP03A pattern enable                      | 1 = Enable JP03A test-pattern<br>0 = Disable JP03A test-pattern                                               | R/W              |
| 1.1501.7   | PRBS31 pattern enable                     | 1 = Enable PRBS31 test-pattern<br>0 = Disable PRBS31 test-pattern                                             | R/W              |
| 1.1501.6   | PRBS9 pattern enable                      | 1 = Enable PRBS9 test-pattern<br>0 = Disable PRBS9 test-pattern                                               | R/W              |
| 1.1501.5:4 | Reserved                                  | Value always 0                                                                                                | RO               |
| 1.1501.3   | Tx generator enable                       | 1 = Enable transmit direction test-pattern generator<br>0 = Disable transmit direction test-pattern generator | R/W              |

**Table 45–135—PRBS pattern testing control register bit definitions (*continued*)**

| Bit(s)   | Name                | Description                                                                                                 | R/W <sup>a</sup> |
|----------|---------------------|-------------------------------------------------------------------------------------------------------------|------------------|
| 1.1501.2 | Tx checker enable   | 1 = Enable transmit direction test-pattern checker<br>0 = Disable transmit direction test-pattern checker   | R/W              |
| 1.1501.1 | Rx generator enable | 1 = Enable receive direction test-pattern generator<br>0 = Disable receive direction test-pattern generator | R/W              |
| 1.1501.0 | Rx checker enable   | 1 = Enable receive direction test-pattern checker<br>0 = Disable receive direction test-pattern checker     | R/W              |

<sup>a</sup>R/W = Read/Write, RO = Read only

Register 1.1501, bit 7 enables testing with the PRBS31 pattern defined in 83.5.10. Register 1.1501, bit 6 enables testing with the PRBS9 pattern defined in 83.5.10. The assertion of register 1.1501 bits 7 and 6 is mutually exclusive. If both bits are asserted the behavior is undefined. The assertion of register 1.1501, bits 7 and 6 works in conjunction with register 1.1501, bits 3:0. If none of the bits 3:0 are asserted then bits 7 and 6 have no effect.

Register 1.1501, bit 3 enables pattern generation in the transmit direction. Register 1.1501, bit 2 enables pattern checking in the transmit direction. Register 1.1501, bit 1 enables pattern generation in the receive direction. Register 1.1501, bit 0 enables pattern checking in the receive direction. If none of the bits 3:0 are asserted then bits 7 and 6 have no effect.

Register 1.1501 bit 8 enables testing with the JP03A pattern defined in 94.2.9.1 for 100GBASE-KP4 PMA/PMD. Register 1.1501 bit 9 enables testing with the JP03B pattern defined in 94.2.9.2 for 100GBASE-KP4 PMA/PMD. Register 1.1501 bit 10 enables testing with the QPRBS13 pattern defined in 94.2.9.3 for 100GBASE-KP4 PMA/PMD. Register 1.1501 bit 11 enables the transmitter linearity test pattern defined in 94.2.9.4 for 100GBASE-KP4 PMA/PMD. Register 1.1501 bit 13 enables the PRBS31Q test pattern defined in 120.5.11.2.2. Register 1.1501 bit 14 enables the SSPRQ test pattern defined in 120.5.11.2.3. The assertion of bits 1.1501.8, 1.1501.9, 1.1501.10, 1.1501.11, 1.1501.13, and 1.1501.14 are mutually exclusive. If more than one bit is asserted, the behavior is undefined. The assertion of 1.1501.8, 1.1501.9, 1.1501.10, and 1.1501.11 operates in conjunction with register 1.1501 bit 3 for 100GBASE-KP4 PMA/PMD. For other PMA/PMD types or if bit 1.1501.3 is not asserted, then 1.1501.8, 1.1501.9, 1.1501.10, and 1.1501.11 have no effect. For the 50GBASE-R, 100GBASE-R PAM4, 200GBASE-R, and 400GBASE-R PMAs, the assertion of register 1.1501 bit 13 operates in conjunction with register 1.1501 bits 3 and 1. If neither bit is asserted, then register 1.1501 bit 13 has no effect. Also for the 50GBASE-R, 100GBASE-R PAM4, 200GBASE-R, and 400GBASE-R PMAs, the assertion of register 1.1501 bit 14 operates in conjunction with register 1.1501 bit 3. If bit 1.1501.3 is not asserted, then register 1.1501 bit 14 has no effect. For other PMA/PMD types register 1.1501 bits 13 and 14 have no effect.

#### **45.2.1.171 Square wave testing control (Register 1.1510)**

The square wave testing control register is used for PHY types that implement transmit square wave testing in the PMA. This function is described in 83.5.10 and 120.5.11. The assignment of bits in the square wave testing control register is shown in Table 45–136.

Register 1.1510, bits 0 through 15 enable square wave output on PMA lanes 0 through 15 respectively. Lanes for which a square wave pattern is not enabled act as determined by other registers.

**Table 45–136—Square wave testing control register bit definitions**

| Bit(s)    | Name              | Description                                                             | R/W <sup>a</sup> |
|-----------|-------------------|-------------------------------------------------------------------------|------------------|
| 1.1510.15 | Lane 15 SW enable | 1 = Enable square wave on lane 15<br>0 = Disable square wave on lane 15 | R/W              |
| 1.1510.14 | Lane 14 SW enable | 1 = Enable square wave on lane 14<br>0 = Disable square wave on lane 14 | R/W              |
| 1.1510.13 | Lane 13 SW enable | 1 = Enable square wave on lane 13<br>0 = Disable square wave on lane 13 | R/W              |
| 1.1510.12 | Lane 12 SW enable | 1 = Enable square wave on lane 12<br>0 = Disable square wave on lane 12 | R/W              |
| 1.1510.11 | Lane 11 SW enable | 1 = Enable square wave on lane 11<br>0 = Disable square wave on lane 11 | R/W              |
| 1.1510.10 | Lane 10 SW enable | 1 = Enable square wave on lane 10<br>0 = Disable square wave on lane 10 | R/W              |
| 1.1510.9  | Lane 9 SW enable  | 1 = Enable square wave on lane 9<br>0 = Disable square wave on lane 9   | R/W              |
| 1.1510.8  | Lane 8 SW enable  | 1 = Enable square wave on lane 8<br>0 = Disable square wave on lane 8   | R/W              |
| 1.1510.7  | Lane 7 SW enable  | 1 = Enable square wave on lane 7<br>0 = Disable square wave on lane 7   | R/W              |
| 1.1510.6  | Lane 6 SW enable  | 1 = Enable square wave on lane 6<br>0 = Disable square wave on lane 6   | R/W              |
| 1.1510.5  | Lane 5 SW enable  | 1 = Enable square wave on lane 5<br>0 = Disable square wave on lane 5   | R/W              |
| 1.1510.4  | Lane 4 SW enable  | 1 = Enable square wave on lane 4<br>0 = Disable square wave on lane 4   | R/W              |
| 1.1510.3  | Lane 3 SW enable  | 1 = Enable square wave on lane 3<br>0 = Disable square wave on lane 3   | R/W              |
| 1.1510.2  | Lane 2 SW enable  | 1 = Enable square wave on lane 2<br>0 = Disable square wave on lane 2   | R/W              |
| 1.1510.1  | Lane 1 SW enable  | 1 = Enable square wave on lane 1<br>0 = Disable square wave on lane 1   | R/W              |
| 1.1510.0  | Lane 0 SW enable  | 1 = Enable square wave on lane 0<br>0 = Disable square wave on lane 0   | R/W              |

<sup>a</sup>R/W = Read/Write

#### 45.2.1.172 PRBS13Q testing control (Register 1.1512)

The PRBS13Q testing control register is used for PHY types that implement PRBS13Q testing in the PMA. This function is described in 120.5.11.2.1. The assignment of bits in the PRBS13Q testing control register is shown in Table 45–137.

Register 1.1512, bits 0 through 7 enable PRBS13Q output on PMA lanes 0 through 7 respectively. The assertion of register 1.1512 bits 0 through 7 operates in conjunction with register 1.1501 bits 3 and 1. If

**Table 45–137—PRBS13Q testing control register bit definitions**

| Bit(s)      | Name                  | Description                                                   | R/W <sup>a</sup> |
|-------------|-----------------------|---------------------------------------------------------------|------------------|
| 1.1512.15:8 | Reserved              | Value always 0                                                | RO               |
| 1.1512.7    | Lane 7 PRBS13Q enable | 1 = Enable PRBS13Q on lane 7<br>0 = Disable PRBS13Q on lane 7 | R/W              |
| 1.1512.6    | Lane 6 PRBS13Q enable | 1 = Enable PRBS13Q on lane 6<br>0 = Disable PRBS13Q on lane 6 | R/W              |
| 1.1512.5    | Lane 5 PRBS13Q enable | 1 = Enable PRBS13Q on lane 5<br>0 = Disable PRBS13Q on lane 5 | R/W              |
| 1.1512.4    | Lane 4 PRBS13Q enable | 1 = Enable PRBS13Q on lane 4<br>0 = Disable PRBS13Q on lane 4 | R/W              |
| 1.1512.3    | Lane 3 PRBS13Q enable | 1 = Enable PRBS13Q on lane 3<br>0 = Disable PRBS13Q on lane 3 | R/W              |
| 1.1512.2    | Lane 2 PRBS13Q enable | 1 = Enable PRBS13Q on lane 2<br>0 = Disable PRBS13Q on lane 2 | R/W              |
| 1.1512.1    | Lane 1 PRBS13Q enable | 1 = Enable PRBS13Q on lane 1<br>0 = Disable PRBS13Q on lane 1 | R/W              |
| 1.1512.0    | Lane 0 PRBS13Q enable | 1 = Enable PRBS13Q on lane 0<br>0 = Disable PRBS13Q on lane 0 | R/W              |

<sup>a</sup>R/W = Read/Write, RO = Read only

neither bit is asserted, then register 1.1512 bits 0 through 7 have no effect. Lanes for which a PRBS13Q pattern is not enabled act as determined by other registers.

#### 45.2.1.173 PRBS Tx pattern testing error counter (Register 1.1600 through 1.1615)

The PRBS Tx pattern testing error counter registers are used for PHY types that implement PRBS Tx pattern testing in the PMA. This function is described in 83.5.10 and 120.5.11. The assignment of bits in the PRBS Tx pattern testing error counter registers are shown in Table 45–138. Register 1.1600 contains the PRBS Tx pattern testing error counter for lane 0, register 1.1601 contains the PRBS Tx pattern testing error counter for lane 1, and registers 1.1602 through 1.1609 contain the PRBS Tx pattern testing error counters for lanes 2 through 15 respectively. Counters corresponding to lanes that are not implemented in a PMA shall read all zeros.

**Table 45–138—PRBS Tx pattern testing error counter**

| Bit(s)                    | Name          | Description | R/W <sup>a</sup> |
|---------------------------|---------------|-------------|------------------|
| 1.1600 <sup>b</sup> .15:0 | Error counter |             | RO, NR           |

<sup>a</sup>RO = Read only, NR = Non Roll-over.

<sup>b</sup>All instances of address 1.1600 also apply to addresses 1.1601 through 1.1615.

The PRBS Tx pattern testing error counter is a 16-bit counter as defined in 83.5.10 and 120.5.11. These bits shall be reset to all zeros when the register is read by the management function or upon execution of the PMA reset. These bits shall be held at all ones in the case of overflow.

#### 45.2.1.174 PRBS Rx pattern testing error counter (Register 1.1700 through 1.1715)

The PRBS Rx pattern testing error counter registers are used for PHY types that implement PRBS Rx pattern testing in the PMA. This function is described in 83.5.10 and 120.5.11. The assignment of bits in the PRBS Rx pattern testing error counter registers is identical to the PRBS Tx pattern testing error counter as shown in Table 45–138. Register 1.1700 contains the PRBS Rx pattern testing error counter for lane 0, register 1.1701 contains the PRBS Rx pattern testing error counter for lane 1, and registers 1.1702 through 1.1715 contain the PRBS Rx pattern testing error counters for lanes 2 through 15 respectively. Counters corresponding to lanes that are not implemented in a PMA shall read all zeros.

The PRBS Rx pattern testing error counter is a 16-bit counter as defined in 83.5.10 and 120.5.11. These bits shall be reset to all zeros when the register is read by the management function or upon execution of the PMA reset. These bits shall be held at all ones in the case of overflow.

#### 45.2.1.175 TimeSync PMA/PMD capability (Register 1.1800)

The TimeSync PMA/PMD capability register (see Table 45–139) indicates the capability of the PMA/PMD to report the transmit and receive data delay, stored in registers 1.1801 and 1.1805 through 1.1808 respectively.

**Table 45–139—TimeSync PMA/PMD capability**

| Bit(s)      | Name                              | Description                                                                                                                                                             | R/W <sup>a</sup> |
|-------------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.1800.15:2 | Reserved                          | Value always 0                                                                                                                                                          | RO               |
| 1.1800.1    | TimeSync transmit path data delay | 1 = PMA/PMD provides information on transmit path data delay in registers 1.1801 through 1.1804<br>0 = PMA/PMD does not provide information on transmit path data delay | RO               |
| 1.1800.0    | TimeSync receive path data delay  | 1 = PMA/PMD provides information on receive path data delay in registers 1.1805 through 1.1808<br>0 = PMA/PMD does not provide information on receive path data delay   | RO               |

<sup>a</sup>RO = Read only.

#### 45.2.1.176 TimeSync PMA/PMD transmit path data delay (Registers 1.1801, 1.1802, 1.1803, 1.1804)

The TimeSync PMA/PMD transmit path data delay register contains the maximum (registers 1.1801, 1.1802, see Table 45–140) and minimum (registers 1.1803, 1.1804, see Table 45–140) values of the transmit

path data delay. The transmit path data delay is expressed in units of ns. The values contained in these registers are valid when the link is established, as indicated by bit 2 in register 1.1 (see 45.2.1.2.4).

**Table 45–140—TimeSync PMA/PMD transmit path data delay register**

| Bit(s)      | Name                                            | Description                  | R/W <sup>a</sup> |
|-------------|-------------------------------------------------|------------------------------|------------------|
| 1.1801.15:0 | Maximum PMA/PMD transmit path data delay, lower | PMA/PMD_delay_TX_max [15:0]  | RO, MW           |
| 1.1802.15:0 | Maximum PMA/PMD transmit path data delay, upper | PMA/PMD_delay_TX_max [31:16] | RO, MW           |
| 1.1803.15:0 | Minimum PMA/PMD transmit path data delay, lower | PMA/PMD_delay_TX_min [15:0]  | RO, MW           |
| 1.1804.15:0 | Minimum PMA/PMD transmit path data delay, upper | PMA/PMD_delay_TX_min [31:16] | RO, MW           |

<sup>a</sup>RO = Read only, MW = Multi-word.

#### 45.2.1.177 TimeSync PMA/PMD receive path data delay (Registers 1.1805, 1.1806, 1.1807, 1.1808)

The TimeSync PMA/PMD receive path data delay register contains the maximum (registers 1.1805, 1.1806, see Table 45–141) and minimum (registers 1.1807, 1.1808, see Table 45–141) values of the receive path data delay. The receive path data delay is expressed in units of ns. The values contained in these registers are valid when the link is established, as indicated by bit 2 in register 1.1 (see 45.2.1.2.4).

**Table 45–141—TimeSync PMA/PMD receive path data delay register**

| Bit(s)      | Name                                           | Description                  | R/W <sup>a</sup> |
|-------------|------------------------------------------------|------------------------------|------------------|
| 1.1805.15:0 | Maximum PMA/PMD receive path data delay, lower | PMA/PMD_delay_RX_max [15:0]  | RO, MW           |
| 1.1806.15:0 | Maximum PMA/PMD receive path data delay, upper | PMA/PMD_delay_RX_max [31:16] | RO, MW           |
| 1.1807.15:0 | Minimum PMA/PMD receive path data delay, lower | PMA/PMD_delay_RX_min [15:0]  | RO, MW           |
| 1.1808.15:0 | Minimum PMA/PMD receive path data delay, upper | PMA/PMD_delay_RX_min [31:16] | RO, MW           |

<sup>a</sup>RO = Read only, MW = Multi-word.

#### 45.2.1.178 10GPASS-XR control and status register (Register 1.1900)

The assignment of bits in the 10GPASS-XR control and status register is shown in Table 45–142.



**Table 45–142—10GPASS-XR control and status register bit definitions**

| Bit(s)       | Name                            | Description                                                                                                                                                                        | R/W <sup>a</sup> |
|--------------|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.1900.15:14 | Reserved                        | Value always 0                                                                                                                                                                     | RO               |
| 1.1900.13    | Time synch capable              | 1 = The CNU supports time synchronization variables<br>0 = The CNU does not support time synchronization variables                                                                 | RO               |
| 1.1900.12    | US rate mismatch                | 1 = the upstream rate calculated at the CNU and the CLT matches within 10 b/s<br>0 = the upstream rate calculated at the CNU and the CLT is mismatched by greater than 10 b/s      | RO               |
| 1.1900.11    | DS rate mismatch                | 1 = the downstream rate calculated at the CNU and the CLT matches within 10 b/s<br>0 = the downstream rate calculated at the CNU and the CLT is mismatched by greater than 10 b/s  | RO               |
| 1.1900.10    | Link up ready                   | 1 = the CNU is ready to enter the Link-Up state<br>0 = the CNU is not ready to enter the Link-Up state                                                                             | R/W              |
| 1.1900.9:3   | Continuous pilot scaling factor | Number of continuous pilots in the downstream OFDM channels                                                                                                                        | R/W              |
| 1.1900.2     | CRC40 errored blocks            | 1 = 65-bit blocks with detected CRC40 errors are labeled as errored<br>0 = 65-bit blocks with detected CRC40 errors are not labeled as errored                                     | R/W              |
| 1.1900.1     | PHY Discovery complete          | 1 = The PMA/PMD has completed PHY Discovery on the coaxial cable distribution network<br>0 = The PMA/PMD has not completed PHY Discovery on the coaxial cable distribution network | R/W              |
| 1.1900.0     | PHY Discovery enable            | 1 = PMA/PMD is permitted to transmit on the media<br>0 = PMA/PMD is not permitted to transmit on the media                                                                         | R/W              |

<sup>a</sup>RO = Read only, R/W = Read/Write

#### 45.2.1.178.1 Time sync capable (1.1900.13)

When read as one, bit 1.1900.13 indicates that the 10G-PASS-XR PMA/PMD is capable of supporting the time synchronization variables PHY differential delay and PHY differential delay tolerance reflected in registers 1.1949 and 1.1950. This bit is a reflection of the *TimeSyncCapable* variable defined in 101.1.3.

#### 45.2.1.178.2 US rate mismatch (1.1900.12)

When read as one, bit 1.1900.12 indicates that the upstream rate calculated at the CNU and the CLT is mismatched by greater than 10 b/s. This bit is defined in 10GPASS-XR-U PMA/PMD only, in 10GPASS-XR-D always read as zero. This bit is a reflection of the *US\_RateMatchFail* variable defined in 100.3.2.3.

#### 45.2.1.178.3 DS rate mismatch (1.1900.11)

When read as one, bit 1.1900.11 indicates that the downstream rate calculated at the CNU and the CLT is mismatched by greater than 10 b/s. This bit is defined in 10GPASS-XR-U PMA/PMD only, in 10GPASS-XR-D always read as zero. This bit is a reflection of the *DS\_RateMatchFail* variable defined in 100.3.2.3.

#### 45.2.1.178.4 Link up ready (1.1900.10)

Bit 1.1900.10 indicates that the CNU is ready for the link-up state. This bit is a reflection of the *LinkUpRdy* variable defined in 102.4.1.8.

#### 45.2.1.178.5 Continuous pilot scaling factor (1.1900.9:3)

Bits 1.1900.9:3 form an unsigned integer that is used to determine the number of continuous pilots in the downstream OFDM channels. These bits are a reflection of the variable *CntPltSF* defined in 101.4.3.6.5.

#### 45.2.1.178.6 CRC40 errored blocks (1.1900.2)

Bit 1.1900.2 is used to control whether 65-bit blocks with detected CRC40 errors are labeled as errored before being passed higher layers as described in 101.3.3.1.4. This bit is a reflection of the variable *CRC40ErrCtrl* defined in 101.3.3.1.6.

#### 45.2.1.178.7 PHY Discovery complete (1.1900.1)

When read as one, bit 1.1900.1 indicates that the 10GPASS-XR PMA/PMD has completed PHY Discovery (see 102.4.1) on the coaxial cable distribution network. When read as zero, bit 1.1900.1 indicates that the PMA/PMD has not completed PHY Discovery on the coaxial cable distribution network. This bit is defined in 10GPASS-XR-U PMA/PMD only, in 10GPASS-XR-D always read as one. This bit is a reflection of the variable *PhyDiscCmplt* defined in 102.4.1.8.

The default value for bit 1.1900.1 is zero.

#### 45.2.1.178.8 PHY Discovery enable (1.1900.0)

When read as one, bit 1.1900.0 indicates that the 10GPASS-XR PMA/PMD is permitted to transmit on the media. When read as zero, bit 1.1900.0 indicates that the PMA/PMD is not permitted to transmit on the media. This bit is a reflection of the variable *PD\_Enable* defined in 102.2.7.3.

The default value for bit 1.1900.0 is zero.

#### 45.2.1.179 DS OFDM control register (Register 1.1901)

The assignment of bits in the DS OFDM control register is shown in Table 45–143.

**Table 45–143—DS OFDM control register bit definitions**

| Bit(s)       | Name                 | Description                                                                                       | R/W <sup>a</sup> |
|--------------|----------------------|---------------------------------------------------------------------------------------------------|------------------|
| 1.1901.15    | CLT tx mute          | 1 = CLT PHY muted state for test purposes<br>0 = CLT PHY not muted (normal operation)             | R/W              |
| 1.1901.14:12 | DS OFDM channels     | Indicates the number of OFDM channels the PMA/PMD is operating in the downstream direction        | R/W              |
| 1.1901.11:7  | DS time interleaving | Indicates the number of OFDM symbols the PMA/PMD is time interleaving in the downstream direction | R/W              |

**Table 45–143—DS OFDM control register bit definitions (*continued*)**

| Bit(s)     | Name             | Description                                                                             | R/W <sup>a</sup> |
|------------|------------------|-----------------------------------------------------------------------------------------|------------------|
| 1.1901.6:4 | DS windowing     | Indicates the size of the windowing control for the PMA/PMD in the downstream direction | R/W              |
| 1.1901.3:0 | DS cyclic prefix | Indicates the size of the cyclic prefix for the PMA/PMD in the downstream direction     | R/W              |

<sup>a</sup>R/W = Read/Write

#### **45.2.1.179.1 CLT tx mute (1.1901.15)**

When bit 1.1901.15 is set to one, the CLT PMD transmitter enters the test mode and it is muted. When bit 1.1901.15 is set to a zero, the CLT PMD enters the normal operating state. This bit has no effect in the CNU. This bit is a reflection of the variable *CLT\_TxMute* defined in 100.4.1.

#### **45.2.1.179.2 DS OFDM channels (1.1901.14:12)**

Bits 1.1901.14:12 indicate the integer number of downstream OFDM channels in use. The number is between 1 and 5; where bit 1.1901.12 is the LSB and bit 1.1901.14 is the MSB. These bits are a reflection of the counter *DS\_ChCnt* defined in 100.3.2.3.

#### **45.2.1.179.3 DS time interleaving (1.1901.11:7)**

Bits 1.1901.11:7 indicate the integer number of time interleaved OFDM symbols in the downstream direction. The number is between 1 and 32; where bit 1.1901.7 is the LSB and bit 1.1901.11 is the MSB. These bits are a reflection of the variable *DS\_TmIntrlv* defined in 101.4.3.9.5.

#### **45.2.1.179.4 DS windowing (1.1901.6:4)**

Bits 1.1901.6:4 indicate the size, in OFDM Clock periods (1/204.8 MHz), of the windowing control for the 10GPASS-XR PMA/PMD in the downstream direction. These bits are a reflection of the variable *DSNrp* defined in 101.4.3.12.1 with bits 1.1901.6:4 mapping to bits 2:0 of *DSNrp*, respectively.

#### **45.2.1.179.5 DS cyclic prefix (1.1901.3:0)**

Bits 1.1901.3:0 indicate the size, in OFDM Clock periods (1/204.8 MHz), of the cyclic prefix control for the 10GPASS-XR PMA/PMD in the downstream direction. These bits are a reflection of the variable *DSNcp* defined in 101.4.3.12.1 with bits 1.1901.3:0 mapping to bits 3:0 of *DSNcp*, respectively.

#### **45.2.1.180 DS OFDM channel frequency control register 1 through 5 (Register 1.1902 through 1.1906)**

The assignment of bits in the DS OFDM channel frequency control register 1 through 5 is shown in Table 45–144.

**Table 45–144—DS OFDM channel frequency control register 1 through 5 bit definitions**

| Bit(s)      | Name              | Description                                                                                       | R/W <sup>a</sup> |
|-------------|-------------------|---------------------------------------------------------------------------------------------------|------------------|
| 1.1902.15:0 | DS OFDM freq ch 1 | This register specifies the center frequency of subcarrier 0 of downstream OFDM channel number 1. | R/W              |
| 1.1903.15:0 | DS OFDM freq ch 2 | This register specifies the center frequency of subcarrier 0 of downstream OFDM channel number 2. | R/W              |
| 1.1904.15:0 | DS OFDM freq ch 3 | This register specifies the center frequency of subcarrier 0 of downstream OFDM channel number 3. | R/W              |
| 1.1905.15:0 | DS OFDM freq ch 4 | This register specifies the center frequency of subcarrier 0 of downstream OFDM channel number 4. | R/W              |
| 1.1906.15:0 | DS OFDM freq ch 5 | This register specifies the center frequency of subcarrier 0 of downstream OFDM channel number 5. | R/W              |

<sup>a</sup>R/W = Read/Write

#### **45.2.1.180.1 DS OFDM freq ch 1 (1.1902.15:0)**

Register 1.1902 specifies the center frequency for OFDM channel number 1 in units of 50 kHz. This register is a reflection of the variable *DS\_FreqCh(1)* defined in 100.3.2.3.

#### **45.2.1.180.2 DS OFDM freq ch 2 (1.1903.15:0)**

Register 1.1903 specifies the center frequency of OFDM channel number 2 in units of 50 kHz. This register is a reflection of the variable *DS\_FreqCh(2)* defined in 100.3.2.3.

#### **45.2.1.180.3 DS OFDM freq ch 3 (1.1904.15:0)**

Register 1.1904 specifies the center frequency of OFDM channel number 3 in units of 50 kHz. This register is a reflection of the variable *DS\_FreqCh(3)* defined in 100.3.2.3.

#### **45.2.1.180.4 DS OFDM freq ch 4 (1.1905.15:0)**

Register 1.1905 specifies the center frequency of OFDM channel number 4 in units of 50 kHz. This register is a reflection of the variable *DS\_FreqCh(4)* defined in 100.3.2.3.

#### **45.2.1.180.5 DS OFDM freq ch 5 (1.1906.15:0)**

Register 1.1906 specifies the center frequency of OFDM channel number 5 in units of 50 kHz. This register is a reflection of the variable *DS\_FreqCh(5)* defined in 100.3.2.3.

#### 45.2.1.181 US OFDM control register (Register 1.1907)

The assignment of bits in the US OFDM control register is shown in Table 45–145.

**Table 45–145—US OFDM control register bit definitions**

| Bit(s)      | Name                | Description                                                                                                           | R/W <sup>a</sup> |
|-------------|---------------------|-----------------------------------------------------------------------------------------------------------------------|------------------|
| 1.1907.15:8 | Random seed         | Random back-off seed for PHY Discovery                                                                                | R/W              |
| 1.1907.7    | Resource Block size | 1 = 16 OFDMA symbols in the upstream OFDMA Resource Block<br>0 = 8 OFDMA symbols in the upstream OFDMA Resource Block | R/W              |
| 1.1907.6:4  | US windowing        | Indicates the size of the windowing control for the PMA/PMD in the upstream direction                                 | R/W              |
| 1.1907.3:0  | US cyclic prefix    | Indicates the size of the cyclic prefix for the PMA/PMD in the upstream direction                                     | R/W              |

<sup>a</sup>R/W = Read/Write

##### 45.2.1.181.1 Random seed (1.1907.15:8)

Bits 1.1907.15:8 form an 8-bit integer that is used by the CNU for the seed of the PHY Discovery back-off algorithm. These bits are a reflection of the *Rnd* variable defined in 102.4.1.9.2.

##### 45.2.1.181.2 Resource Block size (1.1907.7)

Bit 1.1907.7 indicates the number of OFDM symbols in a Resource Block in the upstream direction. This bit is a reflection of the variable *RBsize* defined in 101.4.4.3.5.

##### 45.2.1.181.3 US windowing (1.1907.6:4)

Bits 1.1907.6:4 indicate the size, in units of OFDM Clock periods (1/204.8 MHz), of the windowing control for the 10GPASS-XR PMA/PMD in the upstream direction. These bits are a reflection of the variable *USNrp* defined in 101.4.4.10.1 with bits 1.1907.6:4 mapping to bits 2:0 of *USNrp*, respectively.

##### 45.2.1.181.4 US cyclic prefix (1.1907.3:0)

Bits 1.1907.3:0 indicate the size, in units of OFDM Clock periods (1/204.8 MHz), of the cyclic prefix control for the 10GPASS-XR PMA/PMD in the upstream direction. These bits are a reflection of the variable *USNcp* defined in 101.4.4.10.1 with bits 1.1907.3:0 mapping to bits 3:0 of *USNcp*, respectively.

#### 45.2.1.182 US OFDM channel frequency control register (Register 1.1908)

The assignment of bits in the US OFDM channel frequency control register is shown in Table 45–146.

Register 1.1908 indicates the center frequency, in units of 50 kHz, of subcarrier 0 for the upstream OFDM channel. This register is a reflection of the variable *US\_FreqCh1* defined in 100.3.2.3.

**Table 45–146—US OFDM channel frequency control register bit definitions**

| Bit(s)      | Name         | Description                                                                       | R/W <sup>a</sup> |
|-------------|--------------|-----------------------------------------------------------------------------------|------------------|
| 1.1908.15:0 | US OFDM freq | This specifies the center frequency of subcarrier 0 of the upstream OFDM channel. | R/W              |

<sup>a</sup>R/W = Read/Write

#### 45.2.1.183 US OFDMA pilot pattern register (Register 1.1909)

The assignment of bits in the US OFDMA pilot pattern register is shown in Table 45–147. For additional information on the use of the parameters in this register see 101.4.4.6.

**Table 45–147—US OFDMA pilot pattern register bit definitions**

| Bit(s)       | Name          | Description                                                                     | R/W <sup>a</sup> |
|--------------|---------------|---------------------------------------------------------------------------------|------------------|
| 1.1909.15    | Reserved      | Value always 0                                                                  | RO               |
| 1.1909.14:12 | Type 2 repeat | Indicates the number of subcarriers between Type 2 Pilots                       | R/W              |
| 1.1909.11:8  | Type 2 start  | Indicates the number of the subcarrier on which the Type 2 Pilot pattern starts | R/W              |
| 1.1909.7     | Reserved      | Value always 0                                                                  | RO               |
| 1.1909.6:4   | Type 1 repeat | Indicates the number of subcarriers between Type 1 Pilots                       | R/W              |
| 1.1909.3:0   | Type 1 start  | Indicates the number of the subcarrier on which the Type 1 Pilot pattern starts | R/W              |

<sup>a</sup>RO = Read only, R/W = Read/Write

##### 45.2.1.183.1 Type 2 repeat (1.1909.14:12)

Bits 1.1909.14:12 indicate the number of subcarriers between repeating Type 2 Pilots. Additional information on pilot patterns and encoding for these bits is located in 101.4.4.6 and Table 101–14. These bits are a reflection of the variable *Type2\_Repeat* defined in 101.4.4.6.1.

##### 45.2.1.183.2 Type 2 start (1.1909.11:8)

Bits 1.1909.11:8 indicate the number of the first subcarrier designated as a Type 2 Pilot. These bits are a reflection of the variable *Type2\_Start* defined in 101.4.4.6.1.

##### 45.2.1.183.3 Type 1 repeat (1.1909.6:4)

Bits 1.1909.6:4 indicate the number of subcarriers between repeating Type 1 Pilots. Additional information on pilot patterns and encoding for these bits is located in 101.4.4.6 and Table 101–14. These bits are a reflection of the variable *Type1\_Repeat* defined in 101.4.4.6.1.

##### 45.2.1.183.4 Type 1 start (1.1909.3:0)

Bits 1.1909.3:0 indicate the number of the first subcarrier designated as a Type 1 Pilot. These bits are a reflection of the variable *Type1\_Start* defined in 101.4.4.6.1.

#### 45.2.1.184 Profile control register (Register 1.1910)

The assignment of bits in the Profile control register is shown in Table 45–148. See 102.4.5 for additional information on the profile copy functionality.

**Table 45–148—Profile control register bit definitions**

| Bit(s)       | Name                | Description                                                                                                                                              | R/W <sup>a</sup> |
|--------------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.1910.15:12 | Reserved            | Value always 0                                                                                                                                           | RO               |
| 1.1910.11    | US copy in process  | 1 = the active upstream profile is being copied to the off-line profile<br>0 = the upstream off-line profile is not being modified by a profile copy     | RO               |
| 1.1910.10    | US profile copy     | 1 = initiates a copy of the active upstream profile to the off-line profile<br>0 = no copy initiated                                                     | R/W,<br>SC       |
| 1.1910.9:8   | US configuration ID | Controls switching the active upstream profile                                                                                                           | RO               |
| 1.1910.7     | Reserved            | Value always 0                                                                                                                                           | RO               |
| 1.1910.6:4   | DS copy channel ID  | Indicates which of the 5 downstream OFDM channel profiles is to be copied                                                                                |                  |
| 1.1910.3     | DS copy in process  | 1 = the active downstream profile is being copied to the off-line profile<br>0 = the downstream off-line profile is not being modified by a profile copy | RO               |
| 1.1910.2     | DS profile copy     | 1 = initiates a copy of the active downstream profile to the off-line profile<br>0 = normal                                                              | R/W,<br>SC       |
| 1.1910.1:0   | DS configuration ID | Controls switching the active downstream profile                                                                                                         | RO               |

<sup>a</sup>R/W = Read/Write, RO = Read only, SC = Self-clearing

##### 45.2.1.184.1 US copy in process (1.1910.11)

When read as one, bit 1.1910.11 indicates that a copy of the currently active upstream profile to the inactive profile is in process, writes to all upstream profile descriptors and their respective registers (see 45.2.8.3 and 102.4.5) are ignored, and switching between profiles (see 102.2.3.1.1) is prohibited. This bit is a reflection of the variable *US\_CpyInP* defined in 102.4.5.1.

##### 45.2.1.184.2 US profile copy (1.1910.10)

When bit 1.1910.10 is set to one, a copy of the currently active upstream profile to the inactive profile is initiated and will continue to completion. This bit is set to zero by the PMA/PMD on or before completion of the profile copy. This bit is a reflection of the variable *US\_ProfICpy* defined in 102.4.5.1.

##### 45.2.1.184.3 US configuration ID (1.1910.9:8)

Bits 1.1910.9:8 indicate the value of the most recently received upstream Configuration ID bits (see 102.2.3.1). These bits are a reflection of the variable *US\_CID* defined in 102.2.7.3.

#### 45.2.1.184.4 DS copy channel ID (1.1910.6:4)

Bits 1.1910.6:4 indicate which one of the five downstream OFDM channel profiles is to be copied. These bits are a reflection of the *DS\_CpyCh* variable defined in 102.4.5.1.

#### 45.2.1.184.5 DS copy in process (1.1910.3)

When read as one, bit 1.1910.3 indicates that a copy of the currently active downstream profile to the inactive profile is in process, writes to all downstream profile descriptors and their respective registers (see 45.2.8.2 and 102.4.5) are ignored, and switching between profiles (see 102.2.3.1.1) is prohibited. This bit is a reflection of the variable *DS\_CpyInP* defined in 102.4.5.1.

#### 45.2.1.184.6 DS profile copy (1.1910.2)

When bit 1.1910.2 is set to one, a copy of the currently active downstream profile to the inactive profile is initiated and will continue to completion. This bit is set to zero by the PMA/PMD on or before completion of the profile copy. This bit is a reflection of the variable *DS\_PrflCpy* defined in 102.4.5.1.

#### 45.2.1.184.7 DS configuration ID (1.1910.1:0)

Bits 1.1910.1:0 indicate the value of the most recently received downstream Configuration ID bits (see 102.2.3.1). These bits are a reflection of the variable *DS\_CID* defined in 102.2.7.3.

#### 45.2.1.185 DS PHY Link control register (Register 1.1911)

The assignment of bits in the DS PHY Link control register is shown in Table 45–149.

**Table 45–149—DS PHY Link control register bit definitions**

| Bit(s)       | Name              | Description                     | R/W <sup>a</sup> |
|--------------|-------------------|---------------------------------|------------------|
| 1.1911.15:12 | Reserved          | Value always 0                  | RO               |
| 1.1911.11:0  | DS PHY Link start | DS PHY Link starting subcarrier | R/W              |

<sup>a</sup>RO = Read only, R/W = Read/Write

#### 45.2.1.185.1 DS PHY Link start (1.1911.11:0)

Bits 1.1911.11:0 set the starting subcarrier number of the downstream 10GPASS-XR PHY Link. They indicate the lowest frequency subcarrier of the downstream PHY Link used to carry PHY Link information bits. See 102.2 for additional details on the downstream PHY Link. These bits are a reflection of the variable *DS\_PhyLinkStrt* defined in 102.2.6.

#### 45.2.1.186 US PHY Link control register (Register 1.1912)

The assignment of bits in the US PHY Link control register is shown in Table 45–150.

#### 45.2.1.186.1 US PHY Link modulation (1.1912.15:12)

Bits 1.1912.15:12 are used to set the modulation type of the upstream PHY Link. These bits are a reflection of the *US\_PhyLinkMod* variable defined in 102.3.5.3 with bits 1.1912.15:12 mapping to bits 3:0 of *US\_PhyLinkMod*, respectively.



**Table 45–150—US PHY Link control register bit definitions**

| Bit(s)       | Name                   | Description                     | R/W <sup>a</sup> |
|--------------|------------------------|---------------------------------|------------------|
| 1.1912.15:12 | US PHY Link modulation | US PHY Link modulation type     | R/W              |
| 1.1912.11:0  | US PHY Link start      | US PHY Link starting subcarrier | R/W              |

<sup>a</sup>R/W = Read/Write

#### 45.2.1.186.2 US PHY Link start (1.1912.11:0)

Bits 1.1912.11:0 set the starting subcarrier number of the upstream 10GPASS-XR PHY Link. They indicate the lowest frequency subcarrier of the upstream PHY Link used to carry PHY Link information bits. See 102.3 for additional details on the upstream PHY Link. These bits are a reflection of the variable *US\_PhyLinkStirt* defined in 102.3.5.3.

#### 45.2.1.187 PHY Discovery control registers (Registers 1.1913 and 1.1914)

The PHY Discovery process is used to bring up new CNUs on the EPoC coax cable distribution network. Registers 1.1913 and 1.1914 indicate when the next PHY Discovery window is opened relative to the downstream Timestamp with bit 1.1913.0 being the LSB and bit 1.1914.15 being the MSB. Setting the PHY Discovery start parameter to zero disables the PHY Discovery window. The PHY Discovery control registers direct this process, which is described in 102.4.1. The assignment of bits in the PHY Discovery control registers is shown in Table 45–151. These registers are a reflection of the *DiscStirt* variable defined in 102.2.3.2.4.

**Table 45–151—PHY Discovery control register bit definitions**

| Bit(s)      | Name                      | Description                                         | R/W <sup>a</sup> |
|-------------|---------------------------|-----------------------------------------------------|------------------|
| 1.1913.15:0 | PHY Discovery start lower | Time of next open PHY Discovery window bits [15:0]  | R/W, MW          |
| 1.1914.15:0 | PHY Discovery start upper | Time of next open PHY Discovery window bits [31:16] | R/W, MW          |

<sup>a</sup> R/W = Read/Write, MW = Multi-word

#### 45.2.1.188 New CNU control register (Register 1.1915)

The assignment of bits in the New CNU control register is shown in Table 45–152. Additional information on the use of the New CNU control register can be found in 102.4.1.6 and 102.4.3.

##### 45.2.1.188.1 CNU\_ID assigned flag 1 (1.1915.15)

Bit 1.1915.15 indicates if the associated CNU\_ID value has been assigned to a CNU by the PHY. When this bit is set to one, the associated CNU\_ID has been assigned to a CNU. When this bit is set to zero the associated CNU\_ID has not been assigned. See 102.4.1.6 and 102.4.3 for additional details on the use of this flag. This bit is a reflection of the variable *AssgndCNU\_ID* defined in 102.4.1.8.

**Table 45–152—New CNU control register bit definitions**

| Bit(s)      | Name                 | Description                                                                                                            | R/W <sup>a</sup> |
|-------------|----------------------|------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.1915.15   | CNU_ID assigned flag | 1 = the allowed CNU_ID value has been assigned to a CNU<br>0 = the allowed CNU_ID value has not been assigned to a CNU | R/W              |
| 1.1915.14:0 | Allowed CNU_ID       | A new CNU may be assigned this value for CNU_ID if the CNU_ID assigned flag is FALSE                                   | R/W              |

<sup>a</sup>R/W = Read/Write

#### 45.2.1.188.2 Allowed CNU\_ID (1.1915.14:0)

Bits 1.1915.14:0 indicate to the 10GPASS-XR PHY a valid CNU\_ID value. The value may be assigned to a new CNU when the CNU\_ID assigned flag (bit 1.1915.15) is set to zero, when the flag is set to one it is an indication that this value has already been assigned to a CNU and it should not be use for another CNU. These bits are a reflection of the *AllwdCNU\_ID* variable defined in 102.4.1.8.

#### 45.2.1.189 New CNU info registers 1 through 5 (Registers 1.1916 through 1.1920)

The assignment of bits in the New CNU info registers 1 through 5 is shown in Table 45–153. Additional information on the use of the New CNU info registers see 102.4.3.

**Table 45–153—New CNU info registers 1 through 5 bit definitions**

| Bit(s)      | Name          | Description                                                         | R/W <sup>a</sup> |
|-------------|---------------|---------------------------------------------------------------------|------------------|
| 1.1916.15:0 | New CNU range | The range of the CNU corresponding to Allowed CNU_ID                | RO               |
| 1.1917.15:0 | New CNU MAC 0 | MAC address bits [15:0] of the CNU corresponding to Allowed CNU_ID  | RO, MW           |
| 1.1918.15:0 | New CNU MAC 1 | MAC address bits [31:16] of the CNU corresponding to Allowed CNU_ID | RO, MW           |
| 1.1919.15:0 | New CNU MAC 2 | MAC address bits [47:32] of the CNU corresponding to Allowed CNU_ID | RO, MW           |
| 1.1920.15:0 | Reserved      | Value always 0                                                      | RO               |

<sup>a</sup>RO = Read only, MW = Multi-word

#### 45.2.1.189.1 New CNU range (1.1916.15:0)

Register 1.1916 forms an integer that indicates the range of the CNU corresponding to Allowed CNU\_ID (see 102.4.1.6) in units of OFDM Clock periods (1/204.8 MHz). This register is a reflection of the variable *NewCNU\_Rng* defined in 102.4.1.8.

#### 45.2.1.189.2 New CNU MAC 0 through 2 (1.1917.15:0 through 1.1919.15:0)

Registers 1.1917 through 1.1919 hold the MAC address of the CNU, as determined by the PHY Discovery process, corresponding to Allowed CNU\_ID (see 45.2.1.188) with bit 1.1917.0 being the LSB and 1.1919.15 being the MSB. These registers are a reflection of the variable *New\_MAC* defined in 102.4.1.9.2.

#### 45.2.1.190 DS PHY Link frame counter (Register 1.1921)

Register 1.1921 represents the DS PHY Link frame count. This counter is incremented at the beginning of the PHY Link frame and, on terminal count, rolls over to zero. The assignment of bits in the DS PHY Link frame counter is shown in Table 45–154.

**Table 45–154—DS PHY Link frame counter bit definitions**

| Bit(s)      | Name                   | Description                                                                                                                                                                | R/W <sup>a</sup> |
|-------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.1921.15:0 | PHY Link frame counter | Counter that indicates the PHY Link frame currently being processed by the PHY. This counter rolls over to zero and is incremented at the beginning of each PHY Link frame | RO               |

<sup>a</sup>RO = Read only

#### 45.2.1.191 PMA/PMD timing offset register (Registers 1.1922 and 1.1923)

The assignment of bits in the PMA/PMD timing offset registers is shown in Table 45–155. Registers 1.1923 and 1.1922 form an offset, in units of OFDM Clock periods (1/204.8 MHz), used to align the CNU to the upstream OFDM timing. For more information on the use of this register see 102.4.1.6. These registers are a reflection of the variable *PhyTimingOffset* defined in 102.4.1.8.

**Table 45–155—PMA/PMD timing offset register bit definitions**

| Bit(s)      | Name                        | Description                    | R/W <sup>a</sup> |
|-------------|-----------------------------|--------------------------------|------------------|
| 1.1922.15:0 | PMA/PMD timing offset lower | Transmit timing offset [15:0]  | R/W, MW          |
| 1.1923.15:0 | PMA/PMD timing offset upper | Transmit timing offset [31:16] | R/W, MW          |

<sup>a</sup>R/W = Read/Write, MW = Multi-word

#### 45.2.1.192 PMA/PMD power offset register (Register 1.1924)

The assignment of bits in the PMA/PMD power offset register is shown in Table 45–156.

**Table 45–156—PMA/PMD power offset register bit definitions**

| Bit(s)      | Name                 | Description     | R/W <sup>a</sup> |
|-------------|----------------------|-----------------|------------------|
| 1.1924.15:8 | Reserved             | Value always 0  | RO               |
| 1.1924.7:0  | PMA/PMD power offset | TX Power offset | R/W              |

<sup>a</sup> RO = Read Only, R/W = Read/Write

#### 45.2.1.192.1 PMA/PMD power offset (1.1924.7:0)

Bits 1.1924.7:0 represent a power offset, in units of 0.25 dB, the CNU is to make in order that transmissions arrive at the CLT at the desired power level. For more information on the use of these bits see 102.4.1.6. These bits are a reflection of the variable *PhyPowerOffset* defined in 102.4.1.8.

#### 45.2.1.193 PMA/PMD ranging offset registers (Registers 1.1925 and 1.1926)

Registers 1.1925 and 1.1926 represent the PMA/PMD ranging offset parameter, in units of OFDM Clock periods (1/204.8 MHz). The assignment of bits in the PMA/PMD ranging offset register is shown in Table 45–157. These registers are a reflection of the variable *PhyRngOffset* defined in 102.4.1.8.

**Table 45–157—PMA/PMD ranging offset registers bit definitions**

| Bit(s)      | Name                         | Description                                  | R/W <sup>a</sup> |
|-------------|------------------------------|----------------------------------------------|------------------|
| 1.1925.15:0 | PMA/PMD ranging offset lower | PMA/PMD ranging offset register bits [15:0]  | R/W, MW          |
| 1.1926.15:0 | PMA/PMD ranging offset upper | PMA/PMD ranging offset register bits [31:16] | R/W, MW          |

<sup>a</sup>R/W = Read/Write, MW = Multi-word

#### 45.2.1.194 DS PMA/PMD data rate registers (Registers 1.1927, 1.1928 and 1.1929)

Registers 1.1927, 1.1928, and 1.1929 represent the downstream data rate, in units of b/s. Bit 1.1929.4 is the MSB and bit 1.1927.0 is the LSB of the value. The bit assignments for the DS Data Rate registers is illustrated in Table 45–158. These registers are a reflection of the variable *DS\_DataRate* defined in 100.3.2.3.

**Table 45–158—DS PMA/PMD data rate register bit definitions**

| Bit(s)      | Name                            | Description                                                                | R/W <sup>a</sup> |
|-------------|---------------------------------|----------------------------------------------------------------------------|------------------|
| 1.1927.15:3 | DS PMA/PMD data rate lower      | The downstream PMA/PMD data rate bits [15:3] of a UQ34.3 formatted number  | RO, MW           |
| 1.1927.2:0  | DS PMA/PMD data rate fractional | The downstream PMA/PMD data rate bits [2:0] of a UQ34.3 formatted number   | RO, MW           |
| 1.1928.15:0 | DS PMA/PMD data rate mid        | The downstream PMA/PMD data rate bits [31:16] of a UQ34.3 formatted number | RO, MW           |
| 1.1929.15:5 | Reserved                        | Value always 0                                                             | RO               |
| 1.1929.4:0  | DS PMA/PMD data rate upper      | The downstream PMA/PMD data rate bits [36:32] of a UQ34.3 formatted number | RO, MW           |

<sup>a</sup>RO = read only, MW = Multi-word

#### 45.2.1.195 US PMA/PMD data rate registers (Registers 1.1930, 1.1931 and 1.1932)

Registers 1.1930, 1.1931, and 1.1932 represent the upstream data rate in units of b/s. Bit 1.1932.4 is the MSB and bit 1.1930.0 is the LSB of the value. The bit assignments for the US Data Rate registers is illustrated in Table 45–159. These registers are a reflection of the variable *US\_DataRate* defined in 100.3.2.3.

**Table 45–159—US PMA/PMD data rate register bit definitions**

| Bit(s)      | Name                            | Description                                                              | R/W <sup>a</sup> |
|-------------|---------------------------------|--------------------------------------------------------------------------|------------------|
| 1.1930.15:3 | US PMA/PMD data rate lower      | The upstream PMA/PMD data rate bits [15:3] of a UQ34.3 formatted number  | RO, MW           |
| 1.1930.2:0  | US PMA/PMD data rate fractional | The upstream PMA/PMD data rate bits [2:0] of a UQ34.3 formatted number   | RO, MW           |
| 1.1931.15:0 | US PMA/PMD data rate mid        | The upstream PMA/PMD data rate bits [31:16] of a UQ34.3 formatted number | RO, MW           |
| 1.1932.15:5 | Reserved                        | Value always 0                                                           | RO               |
| 1.1932.4:0  | US PMA/PMD data rate upper      | The upstream PMA/PMD data rate bits [36:32] of a UQ34.3 formatted number | RO, MW           |

<sup>a</sup>RO = read only, MW = Multi-word

#### 45.2.1.196 10GPASS-XR FEC codeword counter (Registers 1.1933, 1.1934)

The assignment of bits in the 10GPASS-XR FEC codeword counter is shown in Table 45–160. Registers 1.1933 and 1.1934 are used to read the value of a 32-bit counter. When registers 1.1933 and 1.1934 are used to read the 32-bit counter value, the register 1.1933 is read first, the value of the register 1.1934 is latched when (and only when) register 1.1933 is read and reads of register 1.1934 return the latched value rather than the current value of the counter. These registers are a reflection of the variable *FecCodeWordCount* defined in 101.3.3.1.6.

**Table 45–160—10GPASS-XR FEC codeword counter bit definitions**

| Bit(s)      | Name                       | Description                        | R/W <sup>a</sup> |
|-------------|----------------------------|------------------------------------|------------------|
| 1.1933.15:0 | FEC codeword counter lower | Total FEC codewords counter [15:0] | RO, MW           |
| 1.1934.15:0 | FEC codeword counter upper | Total FEC codewords counter [31:0] | RO, MW           |

<sup>a</sup>RO = Read only, MW = Multi-word

#### 45.2.1.197 10GPASS-XR FEC codeword success counter (Registers 1.1935 and 1.1936)

The assignment of bits in the 10GPASS-XR FEC codeword success counter is shown in Table 45–161. Registers 1.1935 and 1.1936 are used to read the value of a 32-bit counter. When registers 1.1935 and 1.1936 are used to read the 32-bit counter value, the register 1.1935 is read first, the value of the register 1.1936 is latched when (and only when) register 1.1935 is read and reads of register 1.1936 return the latched value

rather than the current value of the counter. These registers are a reflection of the variable *FecCodeWordSuccess* defined in 101.3.3.1.6.

**Table 45–161—10GPASS-XR FEC codeword success counter bit definitions**

| Bit(s)      | Name                               | Description                                                  | R/W <sup>a</sup> |
|-------------|------------------------------------|--------------------------------------------------------------|------------------|
| 1.1935.15:0 | FEC codeword success counter lower | Total FEC codewords successfully decoded counter bits [15:0] | RO, MW           |
| 1.1936.15:0 | FEC codeword success counter upper | Total FEC codewords successfully decoded counter bits [31:0] | RO, MW           |

<sup>a</sup>RO = Read only, MW = Multi-word

#### 45.2.1.198 10GPASS-XR FEC codeword fail counter (Registers 1.1937 and 1.1938)

The assignment of bits in the 10GPASS-XR FEC codeword fail counter is shown in Table 45–162. Registers 1.1937 and 1.1938 are used to read the value of a 32-bit counter. When registers 1.1937 and 1.1938 are used to read the 32-bit counter value, the register 1.1937 is read first, the value of the register 1.1938 is latched when (and only when) register 1.1937 is read and reads of register 1.1938 return the latched value rather than the current value of the counter. These registers are a reflection of the variable *FecCodeWordFail* defined in 101.3.3.1.6.

**Table 45–162—10GPASS-XR FEC codeword fail counter bit definitions**

| Bit(s)      | Name                            | Description                                                    | R/W <sup>a</sup> |
|-------------|---------------------------------|----------------------------------------------------------------|------------------|
| 1.1937.15:0 | FEC codeword fail counter lower | Total FEC codewords unsuccessfully decoded counter bits [15:0] | RO, MW           |
| 1.1938.15:0 | FEC codeword fail counter upper | Total FEC codewords unsuccessfully decoded counter bits [31:0] | RO, MW           |

<sup>a</sup>RO = Read only, MW = Multi-word

#### 45.2.1.199 PHY Link EPFH counter (Register 1.1939)

The assignment of bits in the PHY Link EPFH counter is shown in Table 45–163. This register is reset to all zeros when read by the management function or upon PMA/PMD reset. These bits are held at all ones in the case of overflow. This register is a reflection of the counter *EPFHcnt* defined in 102.2.7.2.

**Table 45–163—PHY Link EPFH counter bit definitions**

| Bit(s)      | Name                  | Description                                        | R/W <sup>a</sup> |
|-------------|-----------------------|----------------------------------------------------|------------------|
| 1.1939.15:0 | PHY Link EPFH counter | Total PHY Link EPFH message blocks received [15:0] | RO, NR           |

<sup>a</sup>RO = Read only, NR = Non Roll-over

#### 45.2.1.200 PHY Link EPFH error counter (Register 1.1940)

The assignment of bits in the PHY Link EPFH error counter is shown in Table 45–164. This register is reset to all zeros when read by the management function or upon PMA/PMD reset. These bits are held at all ones in the case of overflow. This register is a reflection of the counter *EPFHerr* defined in 102.2.7.2.

**Table 45–164—PHY Link EPFH error counter bit definitions**

| Bit(s)      | Name                        | Description                                                          | R/W <sup>a</sup> |
|-------------|-----------------------------|----------------------------------------------------------------------|------------------|
| 1.1940.15:0 | PHY Link EPFH error counter | Total PHY Link EPFH message blocks received with CRC32 errors [15:0] | RO, NR           |

<sup>a</sup>RO = Read only, NR = Non Roll-over

#### 45.2.1.201 PHY Link EPCH counter (Register 1.1941)

The assignment of bits in the PHY Link EPCH counter is shown in Table 45–165. This register is reset to all zeros when read by the management function or upon PMA/PMD reset. These bits are held at all ones in the case of overflow. This register is a reflection of the counter *EPCHcnt* defined in 102.2.7.2.

**Table 45–165—PHY Link EPCH counter bit definitions**

| Bit(s)      | Name                  | Description                                        | R/W <sup>a</sup> |
|-------------|-----------------------|----------------------------------------------------|------------------|
| 1.1941.15:0 | PHY Link EPCH counter | Total PHY Link EPCH message blocks received [15:0] | RO, NR           |

<sup>a</sup>RO = Read only, NR = Non Roll-over

#### 45.2.1.202 PHY Link EPCH error counter (Register 1.1942)

The assignment of bits in the PHY Link EPCH error counter is shown in Table 45–166. This register is reset to all zeros when read by the management function or upon PMA/PMD reset. These bits are held at all ones in the case of overflow. This register is a reflection of the counter *EPCHerr* defined in 102.2.7.2.

**Table 45–166—PHY Link EPCH error counter bit definitions**

| Bit(s)      | Name                        | Description                                                          | R/W <sup>a</sup> |
|-------------|-----------------------------|----------------------------------------------------------------------|------------------|
| 1.1942.15:0 | PHY Link EPCH error counter | Total PHY Link EPCH message blocks received with CRC32 errors [15:0] | RO, NR           |

<sup>a</sup>RO = Read only, NR = Non Roll-over

#### 45.2.1.203 PHY Link EMB counter (Register 1.1943)

The assignment of bits in the PHY Link EMB counter is shown in Table 45–167. This register is reset to all zeros when read by the management function or upon PMA/PMD reset. These bits are held at all ones in the case of overflow. This register is a reflection of the counter *EMBCnt* defined in 102.2.7.2.

**Table 45–167—PHY Link EMB counter bit definitions**

| Bit(s)      | Name                 | Description                                       | R/W <sup>a</sup> |
|-------------|----------------------|---------------------------------------------------|------------------|
| 1.1943.15:0 | PHY Link EMB counter | Total PHY Link EMB message blocks received [15:0] | RO, NR           |

<sup>a</sup>RO = Read only, NR = Non Roll-over

#### 45.2.1.204 PHY Link EMB error counter (Register 1.1944)

The assignment of bits in the PHY Link EMB error counter is shown in Table 45–168. This register is reset to all zeros when read by the management function or upon PMA/PMD reset. These bits are held at all ones in the case of overflow. This register is a reflection of the counter *EMBerr* defined in 102.2.7.2.

**Table 45–168—PHY Link EMB error counter bit definitions**

| Bit(s)      | Name                       | Description                                                         | R/W <sup>a</sup> |
|-------------|----------------------------|---------------------------------------------------------------------|------------------|
| 1.1944.15:0 | PHY Link EMB error counter | Total PHY Link EMB message blocks received with CRC32 errors [15:0] | RO, NR           |

<sup>a</sup>RO = Read only, NR = Non Roll-over

#### 45.2.1.205 PHY Link FPMB counter (Register 1.1945)

The assignment of bits in the PHY Link FPMB counter is shown in Table 45–169. This register is reset to all zeros when read by the management function or upon PMA/PMD reset. These bits are held at all ones in the case of overflow. This register is a reflection of the counter *FPMBcnt* defined in 102.1.8.

**Table 45–169—PHY Link FPMB counter bit definitions**

| Bit(s)      | Name                  | Description                                        | R/W <sup>a</sup> |
|-------------|-----------------------|----------------------------------------------------|------------------|
| 1.1945.15:0 | PHY Link FPMB counter | Total PHY Link FPMB message blocks received [15:0] | RO, NR           |

<sup>a</sup>RO = Read only, NR = Non Roll-over

#### 45.2.1.206 PHY Link FPMB error counter (Register 1.1946)

The assignment of bits in the PHY Link FPMB error counter is shown in Table 45–170. This register is reset to all zeros when read by the management function or upon PMA/PMD reset. These bits are held at all ones in the case of overflow. This register is a reflection of the counter *FPMBerr* defined in 102.1.8.

**Table 45–170—PHY Link FPMB error counter bit definitions**

| Bit(s)      | Name                        | Description                                                          | R/W <sup>a</sup> |
|-------------|-----------------------------|----------------------------------------------------------------------|------------------|
| 1.1946.15:0 | PHY Link FPMB error counter | Total PHY Link FPMB message blocks received with CRC32 errors [15:0] | RO, NR           |

<sup>a</sup>RO = Read only, NR = Non Roll-over



#### 45.2.1.207 US PHY Link response time register (Register 1.1947)

The assignment of bits in the US PHY Link response time register is shown in Table 45–171. These bits indicate the time, in units of OFDM Clock periods (1/204.8 MHz), required by a CNU to respond to an EPoC message block received on the PHY Link and are a reflection of the variable *PhyLinkRspTm* defined in 102.2.7.3.

**Table 45–171—US PHY Link response time register bit definitions**

| Bit(s)      | Name                      | Description                                                | R/W <sup>a</sup> |
|-------------|---------------------------|------------------------------------------------------------|------------------|
| 1.1947.15:0 | US PHY Link response time | Time required by a CNU to respond to an EPoC message block | RO               |

<sup>a</sup>RO = Read only

#### 45.2.1.208 10GPASS-XR modulation ability register (Register 1.1948)

The assignment of bits in the 10GPASS-XR modulation ability register is shown in Table 45–172.

**Table 45–172—10GPASS-XR modulation ability register bit definitions**

| Bit(s)       | Name                    | Description                                                                | R/W <sup>a</sup> |
|--------------|-------------------------|----------------------------------------------------------------------------|------------------|
| 1.1948.15:10 | Reserved                | Value always 0                                                             | RO               |
| 1.1948.9:8   | US modulation ability   | Indicates the PHYs ability to support optional upstream modulation types   | RO               |
| 1.1948.7:5   | DS OFDM channel ability | Indicates the number of downstream OFDM channels supported                 | RO               |
| 1.1948.4:0   | DS modulation ability   | Indicates the PHYs ability to support optional downstream modulation types | RO               |

<sup>a</sup>RO = Read only

##### 45.2.1.208.1 US modulation ability (1.1948.9:8)

Bits 1.1948.9:8 indicate the ability of the PHY to support optional upstream modulation formats 4096-QAM and 2048-QAM. These bits are a reflection of the variable *US\_ModAbility* defined in 101.4.4.4.4.

##### 45.2.1.208.2 DS OFDM channel ability (1.1948.7:5)

Bits 1.1948.7:5 indicate the number of OFDM channels the PMA/PMD is able to support in the downstream direction. The value of these bits is between 1 and 5 inclusive. These bits are a reflection of the variable *DS\_OFDM\_ChAbility* defined in 101.4.3.4.5.

##### 45.2.1.208.3 DS modulation ability (1.1948.4:0)

Bits 1.1948.4:0 indicate the ability of the PHY to support optional downstream modulation formats 16384-QAM, 8192-QAM, 32-QAM, 16-QAM, and 8-QAM. These bits are a reflection of variable *DS\_ModAbility* defined in 101.4.3.4.5.

#### 45.2.1.209 PHY Discovery Response power control register (Register 1.1949)

The assignment of bits in the PHY Discovery Response power control register is shown in Table 45–173.

**Table 45–173—PHY Discovery Response power control register bit definitions**

| Bit(s)      | Name                                | Description                                                                             | R/W <sup>a</sup> |
|-------------|-------------------------------------|-----------------------------------------------------------------------------------------|------------------|
| 1.1949.15:8 | PHY Discovery Response power step   | Indicate the power increase of the PHY Discovery Response if there is no acknowledgment | R/W              |
| 1.1949.7:0  | PHY Discover Response initial power | Initial power for CNU PHY Discovery Response                                            | R/W              |

<sup>a</sup>R/W = Read/Write

##### 45.2.1.209.1 PHY Discovery Response power step (1.1949.15:8)

Bits 1.1949.15:8 indicate to the CNU the amount of power to increase, in units of 0.25 dB, the PHY Discovery Response by if there is no acknowledgment from the CLT to a PHY Discovery Response from the CNU. These bits are a reflection of the *PdRespPwrStep* variable defined in 102.4.1.8.

##### 45.2.1.209.2 PHY Discover Response initial power (1.1949.7:0)

Bits 1.1949.7:0 indicate to the CNU the initial power to be used when transmitting the PHY Discovery Response in units of 0.25 dBmV/1.6 MHz. These bits are a reflection of the *PdRespInitPwr* variable defined in 102.4.1.8.

#### 45.2.1.210 US target receive power register (Register 1.1950)

The assignment of bits in the US target receive power register is shown in Table 45–174. Bits 1.1950.9:0 are used to set the target upstream receive power, in units of 0.1 dBmV/6.4 MHz, and are a reflection of the *CLT\_TargetReceivePower* variable defined in 100.3.5.1. These bits are valid only for 10GPASS-XR-D PMA/PMD and are reserved for 10GPASS-XR-U PMA/PMD and always read as zero.

**Table 45–174—US target receive power register bit definitions**

| Bit(s)       | Name                    | Description                            | R/W <sup>a</sup> |
|--------------|-------------------------|----------------------------------------|------------------|
| 1.1950.15:10 | Reserved                | Value always 0                         | RO               |
| 1.1950.9:0   | US target receive power | Sets the target upstream receive power | R/W              |

<sup>a</sup>R/W = Read/Write, RO = Read only

#### 45.2.1.211 DS transmit power registers (Registers 1.1951 through 1.1955)

The assignment of bits in the DS transmit power registers is shown in Table 45–175.

**Table 45–175—DS transmit power registers bit definitions**

| Bit(s)      | Name                   | Description                                | R/W <sup>a</sup> |
|-------------|------------------------|--------------------------------------------|------------------|
| 1.1951.15:9 | Reserved               | Value always 0                             | RO               |
| 1.1951.8:0  | DS transmit power Ch 1 | Sets the transmit power for OFDM channel 1 | R/W              |
| 1.1952.15:9 | Reserved               | Value always 0                             | RO               |
| 1.1952.8:0  | DS transmit power Ch 2 | Sets the transmit power for OFDM channel 2 | R/W              |
| 1.1953.15:9 | Reserved               | Value always 0                             | RO               |
| 1.1953.8:0  | DS transmit power Ch 3 | Sets the transmit power for OFDM channel 3 | R/W              |
| 1.1954.15:9 | Reserved               | Value always 0                             | RO               |
| 1.1954.8:0  | DS transmit power Ch 4 | Sets the transmit power for OFDM channel 4 | R/W              |
| 1.1955.15:9 | Reserved               | Value always 0                             | RO               |
| 1.1955.8:0  | DS transmit power Ch 5 | Sets the transmit power for OFDM channel 5 | R/W              |

<sup>a</sup>R/W = Read/Write, RO = Read only

#### 45.2.1.211.1 DS transmit power Ch1 (1.1951.8:0)

Bits 1.1951.8:0 are used to set the transmit level of the downstream OFDM channel 1, in units of 0.2 dBmV/6 MHz. These bits are a reflection of the *DS\_PowerCh(1)* variable defined in 100.3.3.2.1. These bits are valid only for 10GPASS-XR-D PMA/PMD and are reserved for 10GPASS-XR-U PMA/PMD and always read as zero.

#### 45.2.1.211.2 DS transmit power Ch2 (1.1952.8:0)

Bits 1.1952.8:0 are used to set the transmit level of the downstream OFDM channel 2, in units of 0.2 dBmV/6 MHz. These bits are a reflection of the *DS\_PowerCh(2)* variable defined in 100.3.3.2.1. These bits are valid only for 10GPASS-XR-D PMA/PMD and are reserved for 10GPASS-XR-U PMA/PMD and always read as zero.

#### 45.2.1.211.3 DS transmit power Ch3 (1.1953.8:0)

Bits 1.1953.8:0 are used to set the transmit level of the downstream OFDM channel 3, in units of 0.2 dBmV/6 MHz. These bits are a reflection of the *DS\_PowerCh(3)* variable defined in 100.3.3.2.1. These bits are valid only for 10GPASS-XR-D PMA/PMD and are reserved for 10GPASS-XR-U PMA/PMD and always read as zero.

#### 45.2.1.211.4 DS transmit power Ch4 (1.1954.8:0)

Bits 1.1954.8:0 are used to set the transmit level of the downstream OFDM channel 4, in units of 0.2 dBmV/6 MHz. These bits are a reflection of the *DS\_PowerCh(4)* variable defined in 100.3.3.2.1. These bits are valid only for 10GPASS-XR-D PMA/PMD and are reserved for 10GPASS-XR-U PMA/PMD and always read as zero.

#### 45.2.1.211.5 DS transmit power Ch5 (1.1955.8:0)

Bits 1.1955.8:0 are used to set the transmit level of the downstream OFDM channel 5, in units of 0.2 dBmV/6 MHz. These bits are a reflection of the *DS\_PowerCh(5)* variable defined in 100.3.3.2.1. These

bits are valid only for 10GPASS-XR-D PMA/PMD and are reserved for 10GPASS-XR-U PMA/PMD and always read as zero.

#### 45.2.1.212 US receive power measurement registers (1.1956 through 1.1957)

The assignment of bits in the US receive power measurement registers is shown in Table 45–176.

**Table 45–176—US receive power measurement registers bit definitions**

| Bit(s)      | Name                         | Description                                                                        | R/W <sup>a</sup> |
|-------------|------------------------------|------------------------------------------------------------------------------------|------------------|
| 1.1956.15   | US receive power valid       | Indicates the value in US receive power measurement is valid                       | RO               |
| 1.1956.14:9 | Reserved                     | Value always 0                                                                     | RO               |
| 1.1956.8:0  | US receive power measurement | Indicate the measured receive power for the CNU identified by US receive power CNU | RO               |
| 1.1957.15   | Reserved                     | Value always 0                                                                     | RO               |
| 1.1957.14:0 | US receive power CNU         | Indicates the CNU on which to measure the receive power                            | R/W              |

<sup>a</sup>R/W = Read/Write, RO = Read only

##### 45.2.1.212.1 US receive power valid (1.1956.15)

When read as one, bit 1.1956.15 indicates the value in the US receive power measurement is valid for the CNU identified by US receive power CNU. This bit is only valid for 10GPASS-XR-D PMA/PMD and is reserved for 10GPASS-XR-U PMA/PMD and always reads as zero. This bit is a reflection of the variable *RxPwrValid* defined in 100.4.3.1.

##### 45.2.1.212.2 US receive power measurement (1.1956.8:0)

Bits 1.1956.8:0 report the received power, in units of 0.1 dBmV, for the CNU identified by US receive power CNU. These bits are only valid for 10GPASS-XR-D PMA/PMD and are reserved for 10GPASS-XR-U PMA/PMD and always read as zero. These bits are a reflection of the variable *RxPwr* defined in 100.4.3.1.

##### 45.2.1.212.3 US receive power CNU (1.1957.14:0)

When set to a CNU\_ID bits 1.1957.14:0 indicate which CNU the CLT is to measure the received power on. These bits are only valid for 10GPASS-XR-D PMA/PMD and are reserved for 10GPASS-XR-U PMA/PMD and always read as zero. These bits are a reflection of the variable *RxPwr\_CNU\_ID* defined in 100.4.3.1.

#### 45.2.1.213 Reported power register (1.1958)

The assignment of bits in the Reported power register is shown in Table 45–177.

##### 45.2.1.213.1 Reported power (1.1958.8:0)

Bits 1.1958.8:0 indicate the reported power output of the PMA/PMD in units of 0.25 dBmV. These bits are a reflection of the variable *ReportedPwr* defined in 100.3.4.3.1.

**Table 45–177—Reported power register bit definitions**

| Bit(s)      | Name           | Description                               | R/W <sup>a</sup> |
|-------------|----------------|-------------------------------------------|------------------|
| 1.1958.15:9 | Reserved       | Value always 0                            | RO               |
| 1.1958.8:0  | Reported power | The reported output power for the PMA/PMD | RO               |

<sup>a</sup>RO = Read only

#### 45.2.1.214 BASE-T1 PMA/PMD control register (Register 1.2100)

The assignment of bits in the BASE-T1 PMA/PMD control register is shown in Table 45–178.

**Table 45–178—BASE-T1 PMA/PMD control register bit definitions**

| Bit(s)      | Name                      | Description                                                                                                                                                                                                                  | R/W <sup>a</sup> |
|-------------|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.2100.15   | Reserved                  | Value always 1                                                                                                                                                                                                               | RO               |
| 1.2100.14   | MASTER-SLAVE config value | 1 = Configure PHY as MASTER<br>0 = Configure PHY as SLAVE                                                                                                                                                                    | R/W              |
| 1.2100.13:4 | Reserved                  | Value always 0                                                                                                                                                                                                               | RO               |
| 1.2100.3:0  | Type Selection            | 3 2 1 0<br>1 x x x = Reserved<br>0 1 1 1 = Reserved<br>0 1 1 0 = 10GBASE-T1<br>0 1 0 1 = 5GBASE-T1<br>0 1 0 0 = 2.5GBASE-T1<br>0 0 1 1 = 10BASE-T1S<br>0 0 1 0 = 10BASE-T1L<br>0 0 0 1 = 1000BASE-T1<br>0 0 0 0 = 100BASE-T1 | R/W              |

<sup>a</sup>RO = Read only, R/W = Read/Write

NOTE—Annex K defines optional alternative terminology for “master” and “slave”.

##### 45.2.1.214.1 MASTER-SLAVE config value (1.2100.14)

Bit 1.2100.14 is used to select MASTER or SLAVE operation when Auto-Negotiation enable bit 7.512.12 is set to zero, or if Auto-Negotiation is not implemented. If bit 1.2100.14 is set to one the PHY shall operate as MASTER. If bit 1.2100.14 is set to zero the PHY shall operate as SLAVE. This bit shall be ignored when the Auto-Negotiation enable bit 7.512.12 is set to one.

##### 45.2.1.214.2 Type selection (1.2100.3:0)

Bits 1.2100.3:0 are used to set the mode of operation when Auto-Negotiation enable bit 7.512.12 is set to zero, or if Auto-Negotiation is not implemented. When these bits are set to 0000, the mode of operation is 100BASE-T1. When these bits are set to 0001, the mode of operation is 1000BASE-T1. When these bits are set to 0010, the mode of operation is 10BASE-T1L. When these bits are set to 0011, the mode of operation is 10BASE-T1S. When these bits are set to 0100, the mode of operation is 2.5GBASE-T1. When these bits are

set to 0101, the mode of operation is 5GBASE-T1. When these bits are set to 0110, the mode of operation is 10GBASE-T1. These bits shall be ignored when the Auto-Negotiation enable bit 7.512.12 is set to one.

#### 45.2.1.215 100BASE-T1 PMA/PMD test control register (Register 1.2102)

The assignment of bits in the 100BASE-T1 PMA/PMD test control register is shown in Table 45–179.

**Table 45–179—100BASE-T1 PMA/PMD test control register bit definitions**

| Bit(s)       | Name                         | Description                                                                                                                                                                                  | R/W <sup>a</sup> |
|--------------|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.2102.15:13 | 100BASE-T1 test mode control | 15 14 13<br>1 1 1 = Reserved<br>1 1 0 = Reserved<br>1 0 1 = Test mode 5<br>1 0 0 = Test mode 4<br>0 1 1 = Reserved<br>0 1 0 = Test mode 2<br>0 0 1 = Test mode 1<br>0 0 0 = Normal operation | R/W              |
| 1.2102.12:0  | Reserved                     | Value always 0                                                                                                                                                                               | RO               |

<sup>a</sup>RO = Read only, R/W = Read/Write

#### 45.2.1.215.1 100BASE-T1 test mode control (1.2102.15:13)

100BASE-T1 test mode control operations are selected using bits 1.2102.15:13. The default value for bits 1.2102.15:13 is 000.

#### 45.2.1.216 IFEC control register (Register 1.2200)

The assignment of bits in the IFEC control register is shown in Table 45–180.

**Table 45–180—IFEC control register bit definitions**

| Bit(s)      | Name                          | Description                                                                                                                          | R/W <sup>a</sup> |
|-------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.2200.15:2 | Reserved                      | Value always 0                                                                                                                       | RO               |
| 1.2200.1    | IFEC bypass indication enable | 1 = IFEC decoder does not indicate errors<br>0 = IFEC decoder indicates errors                                                       | R/W              |
| 1.2200.0    | IFEC bypass correction enable | 1 = IFEC decoder performs error detection without error correction<br>0 = IFEC decoder performs error detection and error correction | R/W              |

<sup>a</sup>R/W = Read/Write, RO = Read only

#### 45.2.1.216.1 IFEC bypass indication enable (1.2200.1)

This bit enables the Inverse RS-FEC decoder to bypass error indication to the far end (PCS) through the sync bits for the BASE-R PHY in the remote device. When set to one, this bit enables bypass of the error indication function. When set to zero, errors are indicated to the remote PCS through the sync bits. Writes to

bit 1.2200.1 are ignored and reads return a zero if the Inverse RS-FEC does not have the ability to bypass decoding error indications to the remote PCS layer (see 152.5.2.3).

#### 45.2.1.216.2 IFEC bypass correction enable (1.2200.0)

When bit 1.2200.0 is set to one the Inverse RS-FEC decoder performs error detection without error correction (see 152.5.2.3). When this bit is set to zero, the decoder also performs error correction. Writes to this bit are ignored and reads return a zero if the Inverse RS-FEC does not have the ability to bypass error correction.

#### 45.2.1.217 IFEC status register (Register 1.2201)

The assignment of bits in the IFEC status register is shown in Table 45–181.

**Table 45–181—IFEC status register bit definitions**

| Bit(s)       | Name                           | Description                                                                                                                          | R/W <sup>a</sup> |
|--------------|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.2201.15    | PCS align status               | 1 = IFEC encoder has locked and aligned all PCS lanes<br>0 = IFEC encoder has not locked and aligned all PCS lanes                   | RO               |
| 1.2201.14    | IFEC align status              | 1 = IFEC receive lanes locked and aligned<br>0 = IFEC receive lanes not locked and aligned                                           | RO               |
| 1.2201.13:12 | Reserved                       | Value always 0                                                                                                                       | RO               |
| 1.2201.11    | IFEC AM lock 3                 | 1 = IFEC lane 3 locked and aligned<br>0 = IFEC lane 3 not locked and aligned                                                         | RO               |
| 1.2201.10    | IFEC AM lock 2                 | 1 = IFEC lane 2 locked and aligned<br>0 = IFEC lane 2 not locked and aligned                                                         | RO               |
| 1.2201.9     | IFEC AM lock 1                 | 1 = IFEC lane 1 locked and aligned<br>0 = IFEC lane 1 not locked and aligned                                                         | RO               |
| 1.2201.8     | IFEC AM lock 0                 | 1 = IFEC lane 0 locked and aligned<br>0 = IFEC lane 0 not locked and aligned                                                         | RO               |
| 1.2201.7:3   | Reserved                       | Value always 0                                                                                                                       | RO               |
| 1.2201.2     | IFEC high SER                  | 1 = IFEC errors have exceeded threshold<br>0 = IFEC errors have not exceeded threshold                                               | RO/LH            |
| 1.2201.1     | IFEC bypass indication ability | 1 = IFEC decoder has the ability to bypass error indication<br>0 = IFEC decoder does not have the ability to bypass error indication | RO               |
| 1.2201.0     | IFEC bypass correction ability | 1 = IFEC decoder has the ability to bypass error correction<br>0 = IFEC decoder does not have the ability to bypass error correction | RO               |

<sup>a</sup>RO = Read only, LH = Latching high

##### 45.2.1.217.1 PCS align status (1.2201.15)

Bit 1.2201.15 indicates the PCS alignment status of the Inverse RS-FEC. For the Inverse RS-FEC described in Clause 152, PCS alignment is defined as block lock, alignment marker lock, and deskew of all 20 receive PCS lanes. When read as a zero, bit 1.2201.15 indicates that the Inverse RS-FEC has not obtained PCS

alignment. When read as a one, bit 1.2201.15 indicates that the Inverse RS-FEC has obtained PCS alignment.

#### **45.2.1.217.2 IFEC align status (1.2201.14)**

Bit 1.2201.14 indicates the alignment status of the Inverse RS-FEC. For the Inverse RS-FEC described in Clause 152, IFEC alignment is defined as alignment marker lock and deskew of all four lanes on the IFEC service interface. When read as a zero, bit 1.2201.14 indicates that the Inverse RS-FEC has not obtained IFEC alignment. When read as a one, bit 1.2201.14 indicates that the Inverse RS-FEC has obtained IFEC alignment.

#### **45.2.1.217.3 IFEC AM lock 3 (1.2201.11)**

When read as a one, bit 1.2201.11 indicates that the Inverse RS-FEC described in Clause 152 has locked and aligned lane 3 of the IFEC service interface. When read as a zero, bit 1.2201.11 indicates that the Inverse RS-FEC has not locked and aligned lane 3 of the IFEC service interface. This bit reflects the state of `amps_lock[3]` (see 152.5.2.1).

#### **45.2.1.217.4 IFEC AM lock 2 (1.2201.10)**

When read as a one, bit 1.2201.10 indicates that the Inverse RS-FEC described in Clause 152 has locked and aligned lane 2 of the IFEC service interface. When read as a zero, bit 1.2201.10 indicates that the Inverse RS-FEC has not locked and aligned lane 2 of the IFEC service interface. This bit reflects the state of `amps_lock[2]` (see 152.5.2.1).

#### **45.2.1.217.5 IFEC AM lock 1 (1.2201.9)**

When read as a one, bit 1.2201.9 indicates that the Inverse RS-FEC described in Clause 152 has locked and aligned lane 1 of the IFEC service interface. When read as a zero, bit 1.2201.9 indicates that the Inverse RS-FEC has not locked and aligned lane 1 of the IFEC service interface. This bit reflects the state of `amps_lock[1]` (see 152.5.2.1).

#### **45.2.1.217.6 IFEC AM lock 0 (1.2201.8)**

When read as a one, bit 1.2201.8 indicates that the Inverse RS-FEC described in Clause 152 has locked and aligned lane 0 of the IFEC service interface. When read as a zero, bit 1.2201.8 indicates that the Inverse RS-FEC has not locked and aligned lane 0 of the IFEC service interface. This bit reflects the state of `amps_lock[0]` (see 152.5.2.1).

#### **45.2.1.217.7 IFEC high SER (1.2201.2)**

When bit 1.2200.1 (IFEC bypass indication enable) is set to one, bit 1.2201.2 is set to one if the number of RS-FEC symbol errors in a window of 8192 codewords exceeds the threshold (see 152.5.2.3) and is set to zero otherwise. This bit is set to zero if bit 1.2200.1 (IFEC bypass indication enable) is set to zero. This bit shall be implemented with latching high behavior.

#### **45.2.1.217.8 IFEC bypass indication ability (1.2201.1)**

The Reed-Solomon decoder may have the option to perform error detection without error indication (see 152.5.2.3) to reduce the delay contributed by the Inverse RS-FEC sublayer. This bit is set to one to indicate that the decoder has this ability to bypass the error indication function. The bit is set to zero if this ability is not supported.



#### 45.2.1.217.9 IFEC bypass correction ability (1.2201.0)

The Reed-Solomon decoder may have the option to perform error detection without error correction (see 152.5.2.3) to reduce the delay contributed by the Inverse RS-FEC sublayer. This bit is set to one to indicate that the decoder has this ability to bypass error correction. The bit is set to zero if this ability is not supported.

#### 45.2.1.218 IFEC corrected codewords counter (Register 1.2202, 1.2203)

The assignment of bits in the IFEC corrected codewords counter is shown in Table 45–182. See 152.6.8 for a definition of this register. These bits shall be reset to all zeros when the register is read by the management function or upon PHY reset. These bits shall be held at all ones in the case of overflow. Registers 1.2202 and 1.2203 are used to read the value of a 32-bit counter. When registers 1.2202 and 1.2203 are used to read the 32-bit counter value, register 1.2202 is read first, the value of the register 1.2203 is latched when (and only when) register 1.2202 is read, and reads of register 1.2203 return the latched value rather than the current value of the counter.

**Table 45–182—IFEC corrected codewords counter bit definitions**

| Bit(s)      | Name                           | Description                      | R/W <sup>a</sup> |
|-------------|--------------------------------|----------------------------------|------------------|
| 1.2202.15:0 | IFEC corrected codewords lower | IFEC_corrected_cw_counter[15:0]  | RO, NR           |
| 1.2203.15:0 | IFEC corrected codewords upper | IFEC_corrected_cw_counter[31:16] | RO, NR           |

<sup>a</sup>RO = Read only, NR = Non Roll-over

#### 45.2.1.219 IFEC uncorrected codewords counter (Register 1.2204, 1.2205)

The assignment of bits in the IFEC uncorrected codewords counter is shown in Table 45–183. See 152.6.9 for a definition of this register. These bits shall be reset to all zeros when the register is read by the management function or upon PHY reset. These bits shall be held at all ones in the case of overflow. Registers 1.2204 and 1.2205 are used to read the value of a 32-bit counter. When registers 1.2204 and 1.2205 are used to read the 32-bit counter value, register 1.2204 is read first, the value of the register 1.2205 is latched when (and only when) register 1.2204 is read, and reads of register 1.2205 return the latched value rather than the current value of the counter.

**Table 45–183—IFEC uncorrected codewords counter bit definitions**

| Bit(s)      | Name                             | Description                        | R/W <sup>a</sup> |
|-------------|----------------------------------|------------------------------------|------------------|
| 1.2204.15:0 | IFEC uncorrected codewords lower | IFEC_uncorrected_cw_counter[15:0]  | RO, NR           |
| 1.2205.15:0 | IFEC uncorrected codewords upper | IFEC_uncorrected_cw_counter[31:16] | RO, NR           |

<sup>a</sup>RO = Read only, NR = Non Roll-over

#### 45.2.1.220 IFEC lane mapping register (Register 1.2206)

The assignment of bits in the IFEC lane mapping register is shown in Table 45–184. When the RS-FEC detects and locks the RS-FEC for IFEC service interface lane 0, the detected RS-FEC lane number is recorded in bits 1:0 in this register. Similarly, the detected RS-FEC lane numbers for IFEC service lanes 1, 2,

and 3 are recorded in bits 3:2, 5:4, and 7:6, respectively. The contents of the RS-FEC lane mapping register bits 7:0 are valid when Inverse RS-FEC align status (1.2201.14) is set to one and are invalid otherwise.

**Table 45–184—IFEC lane mapping register**

| Bit(s)      | Name                  | Description                       | R/W <sup>a</sup> |
|-------------|-----------------------|-----------------------------------|------------------|
| 1.2206.15:8 | Reserved              | Value always 0                    | RO               |
| 1.2206.7:6  | RS-FEC lane 3 mapping | RS-FEC lane mapped to IFEC lane 3 | RO               |
| 1.2206.5:4  | RS-FEC lane 2 mapping | RS-FEC lane mapped to IFEC lane 2 | RO               |
| 1.2206.3:2  | RS-FEC lane 1 mapping | RS-FEC lane mapped to IFEC lane 1 | RO               |
| 1.2206.1:0  | RS-FEC lane 0 mapping | RS-FEC lane mapped to IFEC lane 0 | RO               |

<sup>a</sup>RO = Read only

#### 45.2.1.221 IFEC symbol error counter, lane 0 (Register 1.2210, 1.2211)

The assignment of bits in the IFEC symbol error counter, lane 0 is shown in Table 45–185. Symbol errors detected in FEC lane 0 are counted and shown in bits 1.2210.15:0 and 1.2211.15:0. See 152.6.11 for a definition of this counter. These bits shall be reset to all zeros when the register is read by the management function or upon PHY reset. These bits shall be held at all ones in the case of overflow. Registers 1.2210 and 1.2211 are used to read the value of a 32-bit counter. When registers 1.2210 and 1.2211 are used to read the 32-bit counter value, register 1.2210 is read first, the value of the register 1.2211 is latched when (and only when) register 1.2210 is read, and reads of register 1.2211 return the latched value rather than the current value of the counter.

**Table 45–185—IFEC symbol error counter, lane 0 bit definitions**

| Bit(s)      | Name                             | Description                        | R/W <sup>a</sup> |
|-------------|----------------------------------|------------------------------------|------------------|
| 1.2210.15:0 | IFEC symbol errors, lane 0 lower | IFEC_symbol_error_counter_0[15:0]  | RO, NR           |
| 1.2211.15:0 | IFEC symbol errors, lane 0 upper | IFEC_symbol_error_counter_0[31:16] | RO, NR           |

<sup>a</sup>RO = Read only, NR = Non Roll-over

#### 45.2.1.222 IFEC symbol error counter, lane 1 through 3 (Register 1.2212, 1.2213, 1.2214, 1.2215, 1.2216, 1.2217)

The behavior of the IFEC symbol error counters, lane 1 through 3 is identical to that described for FEC lane 0 in 45.2.1.221. Errors detected in each FEC lane are counted and shown in the corresponding register. FEC lane 1, lower 16 bits are shown in register 1.2212; FEC lane 1, upper 16 bits are shown in register 1.2213; FEC lane 2, lower 16 bits are shown in register 1.2214; through register 1.2217 for FEC lane 3, upper 16 bits.

#### 45.2.1.223 SC-FEC alignment status 1 register (Register 1.2246)

The assignment of bits in the staircase FEC (SC-FEC) alignment status 1 register is shown in Table 45–186.

**Table 45–186—SC-FEC alignment status 1 register bit definitions**

| Bit(s)       | Name                | Description                                                                                    | R/W <sup>a</sup> |
|--------------|---------------------|------------------------------------------------------------------------------------------------|------------------|
| 1.2246.15:13 | Reserved            | Value always 0                                                                                 | RO               |
| 1.2246.12    | SC-FEC align status | 1 = SC-FEC receive lanes locked and aligned<br>0 = SC-FEC receive lanes not locked and aligned | RO               |
| 1.2246.11:8  | Reserved            | Value always 0                                                                                 | RO               |
| 1.2246.7     | SC-FEC FAS lock 7   | 1 = Lane 7 is locked<br>0 = Lane 7 is not locked                                               | RO               |
| 1.2246.6     | SC-FEC FAS lock 6   | 1 = Lane 6 is locked<br>0 = Lane 6 is not locked                                               | RO               |
| 1.2246.5     | SC-FEC FAS lock 5   | 1 = Lane 5 is locked<br>0 = Lane 5 is not locked                                               | RO               |
| 1.2246.4     | SC-FEC FAS lock 4   | 1 = Lane 4 is locked<br>0 = Lane 4 is not locked                                               | RO               |
| 1.2246.3     | SC-FEC FAS lock 3   | 1 = Lane 3 is locked<br>0 = Lane 3 is not locked                                               | RO               |
| 1.2246.2     | SC-FEC FAS lock 2   | 1 = Lane 2 is locked<br>0 = Lane 2 is not locked                                               | RO               |
| 1.2246.1     | SC-FEC FAS lock 1   | 1 = Lane 1 is locked<br>0 = Lane 1 is not locked                                               | RO               |
| 1.2246.0     | SC-FEC FAS lock 0   | 1 = Lane 0 is locked<br>0 = Lane 0 is not locked                                               | RO               |

<sup>a</sup>RO = Read only

#### **45.2.1.223.1 SC-FEC align status (1.2246.12)**

When read as a one, bit 1.2246.12 indicates that the SC-FEC has locked and aligned all receive lanes. When read as a zero, bit 1.2246.12 indicates that the SC-FEC has not locked and aligned all receive lanes.

#### **45.2.1.223.2 SC-FEC FAS lock 7 (1.2246.7)**

When read as a one, bit 1.2246.7 indicates that the SC-FEC receiver has achieved frame alignment signal (FAS) lock for lane 7 of the PMA service interface. When read as a zero, bit 1.2246.7 indicates that the SC-FEC receiver has not achieved FAS lock for lane 7 of the PMA service interface. This bit reflects the state of fas\_lock<7> (see 153.2.4.1.1).

#### **45.2.1.223.3 SC-FEC FAS lock 6 (1.2246.6)**

When read as a one, bit 1.2246.6 indicates that the SC-FEC receiver has achieved FAS lock for lane 6 of the PMA service interface. When read as a zero, bit 1.2246.6 indicates that the SC-FEC receiver has not achieved FAS lock for lane 6 of the PMA service interface. This bit reflects the state of fas\_lock<6> (see 153.2.4.1.1).

#### **45.2.1.223.4 SC-FEC FAS lock 5 (1.2246.5)**

When read as a one, bit 1.2246.5 indicates that the SC-FEC receiver has achieved FAS lock for lane 5 of the PMA service interface. When read as a zero, bit 1.2246.5 indicates that the SC-FEC receiver has not achieved FAS lock for lane 5 of the PMA service interface. This bit reflects the state of fas\_lock<5> (see 153.2.4.1.1).

#### **45.2.1.223.5 SC-FEC FAS lock 4 (1.2246.4)**

When read as a one, bit 1.2246.4 indicates that the SC-FEC receiver has achieved FAS lock for lane 4 of the PMA service interface. When read as a zero, bit 1.2246.4 indicates that the SC-FEC receiver has not achieved FAS lock for lane 4 of the PMA service interface. This bit reflects the state of fas\_lock<4> (see 153.2.4.1.1).

#### **45.2.1.223.6 SC-FEC FAS lock 3 (1.2246.3)**

When read as a one, bit 1.2246.3 indicates that the SC-FEC receiver has achieved FAS lock for lane 3 of the PMA service interface. When read as a zero, bit 1.2246.3 indicates that the SC-FEC receiver has not achieved FAS lock for lane 3 of the PMA service interface. This bit reflects the state of fas\_lock<3> (see 153.2.4.1.1).

#### **45.2.1.223.7 SC-FEC FAS lock 2 (1.2246.2)**

When read as a one, bit 1.2246.2 indicates that the SC-FEC receiver has achieved FAS lock for lane 2 of the PMA service interface. When read as a zero, bit 1.2246.2 indicates that the SC-FEC receiver has not achieved FAS lock for lane 2 of the PMA service interface. This bit reflects the state of fas\_lock<2> (see 153.2.4.1.1).

#### **45.2.1.223.8 SC-FEC FAS lock 1 (1.2246.1)**

When read as a one, bit 1.2246.1 indicates that the SC-FEC receiver has achieved FAS lock for lane 1 of the PMA service interface. When read as a zero, bit 1.2246.1 indicates that the SC-FEC receiver has not achieved FAS lock for lane 1 of the PMA service interface. This bit reflects the state of fas\_lock<1> (see 153.2.4.1.1).

#### **45.2.1.223.9 SC-FEC FAS lock 0 (1.2246.0)**

When read as a one, bit 1.2246.0 indicates that the SC-FEC receiver has achieved FAS lock for lane 0 of the PMA service interface. When read as a zero, bit 1.2246.0 indicates that the SC-FEC receiver has not achieved FAS lock for lane 0 of the PMA service interface. This bit reflects the state of fas\_lock<0> (see 153.2.4.1.1).

#### **45.2.1.224 SC-FEC alignment status 2 register (Register 1.2247)**

The assignment of bits in the SC-FEC alignment status 2 register is shown in Table 45–187.

#### **45.2.1.224.1 SC-FEC FAS lock 19 (1.2247.11)**

When read as a one, bit 1.2247.11 indicates that the SC-FEC receiver has achieved FAS lock for lane 19 of the PMA service interface. When read as a zero, bit 1.2247.11 indicates that the SC-FEC receiver has not achieved FAS lock for lane 19 of the PMA service interface. This bit reflects the state of fas\_lock<19> (see 153.2.4.1.1).

**Table 45–187—SC-FEC alignment status 2 register bit definitions**

| Bit(s)       | Name               | Description                                        | R/W <sup>a</sup> |
|--------------|--------------------|----------------------------------------------------|------------------|
| 1.2247.15:12 | Reserved           | Value always 0                                     | RO               |
| 1.2247.11    | SC-FEC FAS lock 19 | 1 = Lane 19 is locked<br>0 = Lane 19 is not locked | RO               |
| 1.2247.10    | SC-FEC FAS lock 18 | 1 = Lane 18 is locked<br>0 = Lane 18 is not locked | RO               |
| 1.2247.9     | SC-FEC FAS lock 17 | 1 = Lane 17 is locked<br>0 = Lane 17 is not locked | RO               |
| 1.2247.8     | SC-FEC FAS lock 16 | 1 = Lane 16 is locked<br>0 = Lane 16 is not locked | RO               |
| 1.2247.7     | SC-FEC FAS lock 15 | 1 = Lane 15 is locked<br>0 = Lane 15 is not locked | RO               |
| 1.2247.6     | SC-FEC FAS lock 14 | 1 = Lane 14 is locked<br>0 = Lane 14 is not locked | RO               |
| 1.2247.5     | SC-FEC FAS lock 13 | 1 = Lane 13 is locked<br>0 = Lane 13 is not locked | RO               |
| 1.2247.4     | SC-FEC FAS lock 12 | 1 = Lane 12 is locked<br>0 = Lane 12 is not locked | RO               |
| 1.2247.3     | SC-FEC FAS lock 11 | 1 = Lane 11 is locked<br>0 = Lane 11 is not locked | RO               |
| 1.2247.2     | SC-FEC FAS lock 10 | 1 = Lane 10 is locked<br>0 = Lane 10 is not locked | RO               |
| 1.2247.1     | SC-FEC FAS lock 9  | 1 = Lane 9 is locked<br>0 = Lane 9 is not locked   | RO               |
| 1.2247.0     | SC-FEC FAS lock 8  | 1 = Lane 8 is locked<br>0 = Lane 8 is not locked   | RO               |

<sup>a</sup>RO = Read only

#### **45.2.1.224.2 SC-FEC FAS lock 18 (1.2247.10)**

When read as a one, bit 1.2247.10 indicates that the SC-FEC receiver has achieved FAS lock for lane 18 of the PMA service interface. When read as a zero, bit 1.2247.10 indicates that the SC-FEC receiver has not achieved FAS lock for lane 18 of the PMA service interface This bit reflects the state of fas\_lock<18> (see 153.2.4.1.1).

#### **45.2.1.224.3 SC-FEC FAS lock 17 (1.2247.9)**

When read as a one, bit 1.2247.9 indicates that the SC-FEC receiver has achieved FAS lock for lane 17 of the PMA service interface. When read as a zero, bit 1.2247.9 indicates that the SC-FEC receiver has not achieved FAS lock for lane 17 of the PMA service interface This bit reflects the state of fas\_lock<17> (see 153.2.4.1.1).

#### **45.2.1.224.4 SC-FEC FAS lock 16 (1.2247.8)**

When read as a one, bit 1.2247.8 indicates that the SC-FEC receiver has achieved FAS lock for lane 16 of the PMA service interface. When read as a zero, bit 1.2247.8 indicates that the SC-FEC receiver has not achieved FAS lock for lane 16 of the PMA service interface This bit reflects the state of fas\_lock<16> (see 153.2.4.1.1).

#### **45.2.1.224.5 SC-FEC FAS lock 15 (1.2247.7)**

When read as a one, bit 1.2247.7 indicates that the SC-FEC receiver has achieved FAS lock for lane 15 of the PMA service interface. When read as a zero, bit 1.2247.7 indicates that the SC-FEC receiver has not achieved FAS lock for lane 15 of the PMA service interface This bit reflects the state of fas\_lock<15> (see 153.2.4.1.1).

#### **45.2.1.224.6 SC-FEC FAS lock 14 (1.2247.6)**

When read as a one, bit 1.2247.6 indicates that the SC-FEC receiver has achieved FAS lock for lane 14 of the PMA service interface. When read as a zero, bit 1.2247.6 indicates that the SC-FEC receiver has not achieved FAS lock for lane 14 of the PMA service interface This bit reflects the state of fas\_lock<14> (see 153.2.4.1.1).

#### **45.2.1.224.7 SC-FEC FAS lock 13 (1.2247.5)**

When read as a one, bit 1.2247.5 indicates that the SC-FEC receiver has achieved FAS lock for lane 13 of the PMA service interface. When read as a zero, bit 1.2247.5 indicates that the SC-FEC receiver has not achieved FAS lock for lane 13 of the PMA service interface This bit reflects the state of fas\_lock<13> (see 153.2.4.1.1).

#### **45.2.1.224.8 SC-FEC FAS lock 12 (1.2247.4)**

When read as a one, bit 1.2247.4 indicates that the SC-FEC receiver has achieved FAS lock for lane 12 of the PMA service interface. When read as a zero, bit 1.2247.4 indicates that the SC-FEC receiver has not achieved FAS lock for lane 12 of the PMA service interface This bit reflects the state of fas\_lock<12> (see 153.2.4.1.1).

#### **45.2.1.224.9 SC-FEC FAS lock 11 (1.2247.3)**

When read as a one, bit 1.2247.3 indicates that the SC-FEC receiver has achieved FAS lock for lane 11 of the PMA service interface. When read as a zero, bit 1.2247.3 indicates that the SC-FEC receiver has not achieved FAS lock for lane 11 of the PMA service interface This bit reflects the state of fas\_lock<11> (see 153.2.4.1.1).

#### **45.2.1.224.10 SC-FEC FAS lock 10 (1.2247.2)**

When read as a one, bit 1.2247.2 indicates that the SC-FEC receiver has achieved FAS lock for lane 10 of the PMA service interface. When read as a zero, bit 1.2247.2 indicates that the SC-FEC receiver has not achieved FAS lock for lane 10 of the PMA service interface This bit reflects the state of fas\_lock<10> (see 153.2.4.1.1).

#### **45.2.1.224.11 SC-FEC FAS lock 9 (1.2247.1)**

When read as a one, bit 1.2247.1 indicates that the SC-FEC receiver has achieved FAS lock for lane 9 of the PMA service interface. When read as a zero, bit 1.2247.1 indicates that the SC-FEC receiver has not

achieved FAS lock for lane 9 of the PMA service interface This bit reflects the state of fas\_lock<9> (see 153.2.4.1.1).

#### 45.2.1.224.12 SC-FEC FAS lock 8 (1.2247.0)

When read as a one, bit 1.2247.0 indicates that the SC-FEC receiver has achieved FAS lock for lane 8 of the PMA service interface. When read as a zero, bit 1.2247.0 indicates that the SC-FEC receiver has not achieved FAS lock for lane 8 of the PMA service interface This bit reflects the state of fas\_lock<8> (see 153.2.4.1.1).

#### 45.2.1.225 SC-FEC lane mapping, lane 0 register (Register 1.2250)

The assignment of bits in the SC-FEC lane mapping, lane 0 register is shown in Table 45–188. When the SC-FEC described in Clause 153 detects and locks the FAS on PMA service interface lane 0, the detected SC-FEC lane number is recorded in this register. The contents of the SC-FEC lane mapping, lane 0 register is valid when the SC-FEC FAS lock 0 bit (1.2246.0) is set to one and is invalid otherwise (see 45.2.1.223.9).

**Table 45–188—SC-FEC lane mapping, lane 0 register bit definitions**

| Bit(s)      | Name           | Description                                          | R/W <sup>a</sup> |
|-------------|----------------|------------------------------------------------------|------------------|
| 1.2250.15:5 | Reserved       | Value always 0                                       | RO               |
| 1.2250.4:0  | Lane 0 mapping | SC-FEC lane received on PMA service interface lane 0 | RO               |

<sup>a</sup>RO = Read only

#### 45.2.1.226 SC-FEC lane mapping, lane 1 through 19 registers (Registers 1.2251 through 1.2269)

The definition of the SC-FEC lane mapping, lane 1 through 19 registers is equivalent to that described for lane 0 in 45.2.1.225. The lane mapping for lane 1 is in register 1.2251; lane 2 is in register 1.2252; etc.

#### 45.2.1.227 SC-FEC corrected codewords counter (Register 1.2276, 1.2277)

The assignment of bits in the SC-FEC corrected codewords counter is shown in Table 45–189. See 153.2.5.1 for a definition of this counter. These bits shall be reset to all zeros when the register is read by the management function or upon PHY reset. These bits shall be held at all ones in the case of overflow. Registers 1.2276 and 1.2277 are used to read the value of a 32-bit counter. When registers 1.2276 and 1.2277 are used to read the 32-bit counter value, register 1.2276 is read first, the value of the register 1.2277 is latched when (and only when) register 1.2276 is read, and reads of register 1.2277 return the latched value rather than the current value of the counter.

**Table 45–189—SC-FEC corrected codewords counter bit definitions**

| Bit(s)      | Name                          | Description                     | R/W <sup>a</sup> |
|-------------|-------------------------------|---------------------------------|------------------|
| 1.2276.15:0 | FEC corrected codewords lower | FEC_corrected_cw_counter[15:0]  | RO, NR           |
| 1.2277.15:0 | FEC corrected codewords upper | FEC_corrected_cw_counter[31:16] | RO, NR           |

<sup>a</sup>RO = Read only, NR = Non Roll-over

#### 45.2.1.228 SC-FEC uncorrected codewords counter (Register 1.2278, 1.2279)

The assignment of bits in the SC-FEC uncorrected codewords counter is shown in Table 45–190. See 153.2.5.2 for a definition of this counter. These bits shall be reset to all zeros when the register is read by the management function or upon PHY reset. These bits shall be held at all ones in the case of overflow. Registers 1.2278 and 1.2279 are used to read the value of a 32-bit counter. When registers 1.2278 and 1.2279 are used to read the 32-bit counter value, register 1.2278 is read first, the value of the register 1.2279 is latched when (and only when) register 1.2278 is read, and reads of register 1.2279 return the latched value rather than the current value of the counter.

**Table 45–190—SC-FEC uncorrected codewords counter bit definitions**

| Bit(s)      | Name                            | Description                       | R/W <sup>a</sup> |
|-------------|---------------------------------|-----------------------------------|------------------|
| 1.2278.15:0 | FEC uncorrected codewords lower | FEC_uncorrected_cw_counter[15:0]  | RO, NR           |
| 1.2279.15:0 | FEC uncorrected codewords upper | FEC_uncorrected_cw_counter[31:16] | RO, NR           |

<sup>a</sup>RO = Read only, NR = Non Roll-over

#### 45.2.1.229 SC-FEC total bits register (Register 1.2280, 1.2281, 1.2282, 1.2283)

The assignment of bits in the SC-FEC total bits register is shown in Table 45–191. See 153.2.5.3 for a definition of this register. These bits shall be reset to all zeros when the register is read by the management function or upon PHY reset. These bits shall be held at all ones in the case of overflow. Registers 1.2280, 1.2281, 1.2282, and 1.2283 are used to read the value of a 64-bit counter. When registers 1.2280, 1.2281, 1.2282, and 1.2283 are used to read the 64-bit counter value, register 1.2280 is read first, the values of registers 1.2281, 1.2282, and 1.2283 are latched when (and only when) register 1.2280 is read, and reads of registers 1.2281, 1.2282, and 1.2283 return the latched value rather than the current value of the counter.

**Table 45–191—SC-FEC total bits register bit definitions**

| Bit(s)      | Name                   | Description           | R/W <sup>a</sup> |
|-------------|------------------------|-----------------------|------------------|
| 1.2280.15:0 | FEC total bits lowest  | FEC_total_bits[15:0]  | RO, NR           |
| 1.2281.15:0 | FEC total bits         | FEC_total_bits[31:16] | RO, NR           |
| 1.2282.15:0 | FEC total bits         | FEC_total_bits[47:32] | RO, NR           |
| 1.2283.15:0 | FEC total bits highest | FEC_total_bits[64:48] | RO, NR           |

<sup>a</sup>RO = Read only, NR = Non Roll-over

#### 45.2.1.230 SC-FEC corrected bits register (Register 1.2284, 1.2285, 1.2286, 1.2287)

The assignment of bits in the SC-FEC corrected bits register is shown in Table 45–192. See 153.2.5.4 for a definition of this register. These bits shall be reset to all zeros when the register is read by the management function or upon PHY reset. These bits shall be held at all ones in the case of overflow. Registers 1.2284, 1.2285, 1.2286, and 1.2287 are used to read the value of a 64-bit counter. When registers 1.2284, 1.2285, 1.2286, and 1.2287 are used to read the 64-bit counter value, register 1.2284 is read first, the values of



registers 1.2285, 1.2286, and 1.2287 are latched when (and only when) register 1.2284 is read, and reads of registers 1.2285, 1.2286, and 1.2287 return the latched value rather than the current value of the counter.

**Table 45–192—SC-FEC corrected bits register bit definitions**

| Bit(s)      | Name                       | Description               | R/W <sup>a</sup> |
|-------------|----------------------------|---------------------------|------------------|
| 1.2284.15:0 | FEC corrected bits lowest  | FEC_corrected_bits[15:0]  | RO, NR           |
| 1.2285.15:0 | FEC corrected bits         | FEC_corrected_bits[31:16] | RO, NR           |
| 1.2286.15:0 | FEC corrected bits         | FEC_corrected_bits[47:32] | RO, NR           |
| 1.2287.15:0 | FEC corrected bits highest | FEC_corrected_bits[64:48] | RO, NR           |

<sup>a</sup>RO = Read only, NR = Non Roll-over

#### 45.2.1.231 10BASE-T1L PMA control register (Register 1.2294)

The assignment of bits in the 10BASE-T1L PMA control register is shown in Table 45–193.

**Table 45–193—10BASE-T1L PMA control register bit definitions**

| Bit(s)     | Name                               | Description                                                                                    | R/W <sup>a</sup> |
|------------|------------------------------------|------------------------------------------------------------------------------------------------|------------------|
| 1.2294.15  | PMA reset                          | 1 = PMA reset<br>0 = Normal operation                                                          | R/W, SC          |
| 1.2294.14  | Transmit disable                   | 1 = Transmit disable<br>0 = Normal operation                                                   | R/W              |
| 1.2294.13  | Reserved                           | Value always zero                                                                              | RO               |
| 1.2294.12  | Transmit voltage amplitude control | 1 = Enable 2.4 V <sub>pp</sub> operating mode<br>0 = Enable 1.0 V <sub>pp</sub> operating mode | R/W              |
| 1.2294.11  | Low-power                          | 1 = Low-power mode<br>0 = Normal operation                                                     | R/W              |
| 1.2294.10  | EEE enable                         | 1 = Enable EEE mode<br>0 = Disable EEE mode                                                    | R/W              |
| 1.2294.9:1 | Reserved                           | Value always 0                                                                                 | RO               |
| 1.2294.0   | Loopback                           | 1 = Enable loopback mode<br>0 = Disable loopback mode                                          | R/W              |

<sup>a</sup>RO = Read only, R/W = Read/Write, SC = Self-clearing

##### 45.2.1.231.1 PMA reset (1.2294.15)

Resetting the 10BASE-T1L PMA is accomplished by setting bit 1.2294.15 to one. This action shall set all 10BASE-T1L PMA registers to their default states. As a consequence, this action may change the internal state of the 10BASE-T1L PMA and the state of the physical link. This action may also initiate a reset in any

other MMDs that are instantiated in the same package. This bit is self-clearing, and the 10BASE-T1L PMA shall return a value of one in bit 1.2294.15 when a reset is in progress; otherwise, it shall return a value of zero. The 10BASE-T1L PMA is not required to accept a write transaction to any of its registers until the reset process is completed. The control and management interface shall be restored to operation within 0.5 s from the setting of bit 1.2294.15.

During a reset, the 10BASE-T1L PMA shall respond to reads from bits 1.2294.15, 1.8.15:14, and 1.0.15. Reads for all other bits are indeterminate, and the values are invalid.

NOTE—This operation may interrupt data communication.

Bit 1.2294.15 is a copy of bit 1.0.15, and setting or clearing either bit shall set or clear the other bit. Setting either bit shall reset the 10BASE-T1L PMA.

#### **45.2.1.231.2 Transmit disable (1.2294.14)**

When bit 1.2294.14 is set to one, the PMA shall disable output on the transmit path. When bit 1.2294.14 is set to zero, the PMA shall enable output on the transmit path.

Bit 1.2294.14 is a copy of bit 1.9.0, and setting or clearing either bit shall set or clear the other bit. Setting either bit shall disable the transmitter.

#### **45.2.1.231.3 Transmit voltage amplitude control (1.2294.12)**

Bit 1.2294.12 is used to set the 2.4 Vpp operating mode when Auto-Negotiation enable bit 7.512.12 is set to zero or if Auto-Negotiation is not implemented. If bit 1.2294.12 is set to one, the PHY shall operate in 2.4 Vpp operating mode according to 146.5.4.1. If bit 1.2294.12 is set to zero, the PHY shall operate in 1.0 Vpp operating mode according to 146.5.4.1. The default value of bit 1.2294.12 is zero. This bit shall be ignored when the Auto-Negotiation enable bit 7.512.12 is set to one.

#### **45.2.1.231.4 Low-power (1.2294.11)**

When the low-power ability is supported, the 10BASE-T1L PMA may be placed into a low-power mode by setting bit 1.2294.11 to one. This action may also initiate a low-power mode in any other MMDs that are instantiated in the same package. The low-power mode is exited by resetting the 10BASE-T1L PMA. The behavior of the 10BASE-T1L PMA in transition to and from the low-power mode is implementation specific, and any interface signals should not be relied upon. While in the low-power mode, the device shall, as a minimum, respond to management transactions necessary to exit the low-power mode. The default value of bit 1.2294.11 is zero.

NOTE—This operation may interrupt data communication. The data path of the 10BASE-T1L PMA, depending on implementation, may take many seconds to run at optimum error ratio after exiting from reset or low-power mode.

Bit 1.2294.11 is a copy of bit 1.0.11, and setting or clearing either bit shall set or clear the other bit. Setting either bit shall put the 10BASE-T1L PMA in low-power mode.

#### **45.2.1.231.5 EEE enable (1.2294.10)**

Bit 1.2294.10 is used to enable EEE functionality when Auto-Negotiation enable bit 7.512.12 is set to zero or if Auto-Negotiation is not implemented. If bit 1.2294.10 is set to one, the PHY shall operate with EEE enabled. If bit 1.2294.10 is set to zero, the PHY shall operate with EEE disabled. This bit shall be ignored when the Auto-Negotiation enable bit 7.512.12 is set to one. The default value of bit 1.2294.10 is zero.

#### 45.2.1.231.6 Loopback (1.2294.0)

The 10BASE-T1L PMA shall be placed in near-end loopback mode of operation when bit 1.2294.0 is set to one. When in loopback mode, the 10BASE-T1L PMA shall accept data on the transmit path and return it on the receive path. The default value of bit 1.2294.0 is zero. Bit 1.2294.0 is a copy of 1.0.0, and setting or clearing either bit shall set or clear the other bit. Setting either bit shall enable loopback.

#### 45.2.1.232 10BASE-T1L PMA status register (Register 1.2295)

The assignment of bits in the 10BASE-T1L PMA status register is shown in Table 45–194.

**Table 45–194—10BASE-T1L PMA status register bit definitions**

| Bit(s)       | Name                           | Description                                                                                                                                                  | R/W <sup>a</sup> |
|--------------|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.2295.15:14 | Reserved                       | Value always 0                                                                                                                                               | RO               |
| 1.2295.13    | Loopback ability               | 1 = PHY has loopback ability<br>0 = PHY has no loopback ability                                                                                              | RO               |
| 1.2295.12    | 2.4 Vpp operating mode ability | 1 = PHY has 2.4 Vpp operating mode ability<br>0 = PHY does not have 2.4 Vpp operating mode ability                                                           | RO               |
| 1.2295.11    | Low-power ability              | 1 = PMA has low-power ability<br>0 = PMA does not have low-power ability                                                                                     | RO               |
| 1.2295.10    | EEE ability                    | 1 = PHY has EEE ability<br>0 = PHY does not have EEE ability                                                                                                 | RO               |
| 1.2295.9     | Receive fault ability          | 1 = PMA has the ability to detect a fault condition on the receive path<br>0 = PMA does not have the ability to detect a fault condition on the receive path | RO               |
| 1.2295.8:3   | Reserved                       | Value always 0                                                                                                                                               | RO               |
| 1.2295.2     | Receive polarity               | 1 = Receive polarity is reversed<br>0 = Receive polarity is not reversed                                                                                     | RO               |
| 1.2295.1     | Receive fault                  | 1 = Fault condition detected<br>0 = Fault condition not detected                                                                                             | RO/LH            |
| 1.2295.0     | Receive link status            | 1 = PMA receive link up<br>0 = PMA receive link down                                                                                                         | RO/LL            |

<sup>a</sup>RO = Read only, LL = Latching low, LH = Latching high

##### 45.2.1.232.1 Loopback ability (1.2295.13)

When read as a one, this bit indicates that the 10BASE-T1L PHY supports PMA loopback. When read as a zero, this bit indicates that the 10BASE-T1L PHY does not support PMA loopback.

#### **45.2.1.232.2 2.4 Vpp operating mode ability (1.2295.12)**

When read as a one, this bit indicates that the 10BASE-T1L PHY supports a transmit level of 2.4 Vpp. When read as a zero, this bit indicates that the 10BASE-T1L PHY does not support a transmit level of 2.4 Vpp.

#### **45.2.1.232.3 Low-power ability (1.2295.11)**

When read as a one, bit 1.2295.11 indicates that the 10BASE-T1L PMA supports the low-power ability. When read as a zero, bit 1.2295.11 indicates that the 10BASE-T1L PMA does not support the low-power ability. If the 10BASE-T1L PMA supports the low-power ability, then it is controlled using either bit 1.2294.11 or bit 1.0.11.

#### **45.2.1.232.4 EEE ability (1.2295.10)**

When read as a one, this bit indicates that the 10BASE-T1L PHY supports EEE. When read as a zero, this bit indicates that the 10BASE-T1L PHY does not support EEE.

#### **45.2.1.232.5 Receive fault ability (1.2295.9)**

When read as a one, bit 1.2295.9 indicates that the 10BASE-T1L PMA has the ability to detect a fault condition on the receive path. When read as a zero, bit 1.2295.9 indicates that the 10BASE-T1L PMA does not have the ability to detect a fault condition on the receive path.

#### **45.2.1.232.6 Receive polarity (1.2295.2)**

When read as a zero, bit 1.2295.2 indicates that the polarity of the receiver is not reversed. When read as a one, bit 1.2295.2 indicates that the polarity of the receiver is reversed.

#### **45.2.1.232.7 Receive fault (1.2295.1)**

When read as a one, bit 1.2295.1 indicates that the 10BASE-T1L PMA has detected a fault condition on the receive path. When read as a zero, bit 1.2295.1 indicates that the 10BASE-T1L PMA has not detected a fault condition on the receive path. Detection of a fault condition on the receive path is optional, and the ability to detect such a condition is advertised by bit 1.2295.9. The 10BASE-T1L PMA that is unable to detect a fault condition on the receive path shall return a value of zero for this bit. The receive fault bit shall be implemented with latching high behavior.

#### **45.2.1.232.8 Receive link status (1.2295.0)**

When read as a one, bit 1.2295.0 indicates that the 10BASE-T1L PMA receive link is up. When read as a zero, bit 1.2295.0 indicates that the 10BASE-T1L PMA receive link has been down one or more times since the register was last read. The receive link status bit shall be implemented with latching low behavior.

#### **45.2.1.233 10BASE-T1L test mode control register (Register 1.2296)**

The assignment of bits in the 10BASE-T1L test mode control register is shown in Table 45–195. The default values for each bit should be chosen so that the initial state of the device upon power up or reset is a normal operational state without management intervention.

**Table 45–195—10BASE-T1L test mode control register bit definitions**

| Bit(s)       | Name              | Description                                                                                                                              | R/W <sup>a</sup> |
|--------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.2296.15:13 | Test mode control | 15 14 13<br>1 x x = Reserved<br>0 1 1 = Test mode 3<br>0 1 0 = Test mode 2<br>0 0 1 = Test mode 1<br>0 0 0 = Normal (non-test) operation | R/W              |
| 1.2296.12:0  | Reserved          | Value always 0                                                                                                                           | RO               |

<sup>a</sup>RO = Read only, R/W = Read/Write

#### 45.2.1.233.1 Test mode control (1.2296.15:13)

Transmitter test mode operations defined by bits 1.2296.15:13 are described in 146.5.2. The default value for bits 1.2296.15:13 is zero.

#### 45.2.1.234 10BASE-T1S PMA control register (Register 1.2297)

The assignment of bits in the 10BASE-T1S PMA control register is shown in Table 45–196.

**Table 45–196—10BASE-T1S PMA control register bit definitions**

| Bit(s)       | Name             | Description                                                                                           | R/W <sup>a</sup> |
|--------------|------------------|-------------------------------------------------------------------------------------------------------|------------------|
| 1.2297.15    | PMA reset        | 1 = PMA reset<br>0 = Normal operation                                                                 | R/W, SC          |
| 1.2297.14    | Transmit disable | 1 = Transmit disable<br>0 = Normal operation                                                          | R/W              |
| 1.2297.13:12 | Reserved         | Value always 0                                                                                        | RO               |
| 1.2297.11    | Low-power        | 1 = Low-power mode<br>0 = Normal operation                                                            | R/W              |
| 1.2297.10    | Multidrop mode   | 1 = Enable operation over mixing segment network<br>0 = Disable operation over mixing segment network | R/W              |
| 1.2297.9:1   | Reserved         | Value always 0                                                                                        | RO               |
| 1.2297.0     | Loopback         | 1 = Enable loopback mode<br>0 = Disable loopback mode                                                 | R/W              |

<sup>a</sup>RO = Read only, R/W = Read/Write, SC = Self-clearing

#### 45.2.1.234.1 PMA reset (1.2297.15)

Resetting the 10BASE-T1S PMA is accomplished by setting bit 1.2297.15 to one. This action shall set all 10BASE-T1S PMA registers to their default states. As a consequence, this action may change the internal state of the 10BASE-T1S PMA and the state of the physical link. This action may also initiate a reset in any

other MMDs that are instantiated in the same package. This bit is self-clearing, and the 10BASE-T1S PMA shall return a value of one in bit 1.2297.15 when a reset is in progress; otherwise, it shall return a value of zero. The 10BASE-T1S PMA is not required to accept a write transaction to any of its registers until the reset process is completed. The control and management interface shall be restored to operation within 0.5 s from the setting of bit 1.2297.15.

During a reset, the 10BASE-T1S PMA shall respond to reads from bits 1.2297.15, 1.8.15:14, and 1.0.15. All other register bits should be ignored.

NOTE—This operation may interrupt communication.

Bit 1.2297.15 is a copy of 1.0.15, and setting or clearing either bit shall set or clear the other bit. Setting either bit shall reset the 10BASE-T1S PMA.

#### **45.2.1.234.2 Transmit disable (1.2297.14)**

When bit 1.2297.14 is set to one, the PMA shall disable output on the transmit path. When bit 1.2297.14 is set to zero, the PMA shall enable output on the transmit path.

Bit 1.2297.14 is a copy of bit 1.9.0, and setting or clearing either bit shall set or clear the other bit. Setting either bit shall disable the transmitter.

#### **45.2.1.234.3 Low-power (1.2297.11)**

When the low-power ability is supported, the 10BASE-T1S PMA may be placed into a low-power mode by setting bit 1.2297.11 to one. This action may also initiate a low-power mode in any other MMDs that are instantiated in the same package. The low-power mode is exited by resetting the 10BASE-T1S PMA. The behavior of the 10BASE-T1S PMA in transition to and from the low-power mode is implementation specific, and any interface signals should not be relied upon. While in the low-power mode, the device shall respond to management transactions necessary to exit the low-power mode. The default value of bit 1.2297.11 is zero.

NOTE—The time from low-power mode to full operation is implementation specific.

Bit 1.2297.11 is a copy of bit 1.0.11, and setting or clearing either bit shall set or clear the other bit. Setting either bit shall put the 10BASE-T1S PMA in low-power mode.

#### **45.2.1.234.4 Multidrop mode (1.2297.10)**

When Auto-Negotiation is implemented and enabled, writing to this bit shall have no effect on the PHY, and the PCS multidrop variable shall be set to FALSE. If multidrop mode is not supported according to bit 1.2298.10, then writing to bit 1.2297.10 shall have no effect, and the multidrop variable shall be set to FALSE. Otherwise, if bit 1.2297.10 is set to one, the 10BASE-T1S PMA shall operate in multidrop mode, and the multidrop variable is set to TRUE; and if bit 1.2297.10 is set to zero, the multidrop variable is set to FALSE. If multidrop mode is supported according to bit 1.2298.10, then the default value of bit 1.2297.10 should be one.

#### **45.2.1.234.5 Loopback (1.2297.0)**

The 10BASE-T1S PMA shall be placed in loopback mode of operation when loopback bit 1.2297.0 is set to one. When in loopback mode, the 10BASE-T1S PMA shall accept data on the transmit path and return it on the receive path. The default value of bit 1.2297.0 is zero. Bit 1.2297.0 is a copy of 1.0.0, and setting or clearing either bit shall set or clear the other bit. Setting either bit shall enable loopback.

#### 45.2.1.235 10BASE-T1S PMA status register (Register 1.2298)

The assignment of bits in the 10BASE-T1S PMA status register is shown in Table 45–197.

**Table 45–197—10BASE-T1S PMA status register bit definitions**

| Bit(s)       | Name                   | Description                                                                                                                                                  | R/W <sup>a</sup> |
|--------------|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.2298.15:14 | Reserved               | Value always 0                                                                                                                                               | RO               |
| 1.2298.13    | Loopback ability       | 1 = PHY has loopback ability<br>0 = PHY has no loopback ability                                                                                              | RO               |
| 1.2298.12    | Reserved               | Value always 0                                                                                                                                               | RO               |
| 1.2298.11    | Low-power ability      | 1 = PMA has low-power ability<br>0 = PMA does not have low-power ability                                                                                     | RO               |
| 1.2298.10    | Multidrop mode ability | 1 = PMA has the ability to operate over a mixing segment network<br>0 = PMA does not have the ability to operate over a mixing segment network               | RO               |
| 1.2298.9     | Receive fault ability  | 1 = PMA has the ability to detect a fault condition on the receive path<br>0 = PMA does not have the ability to detect a fault condition on the receive path | RO               |
| 1.2298.8:2   | Reserved               | Value always 0                                                                                                                                               | RO               |
| 1.2298.1     | Receive fault          | 1 = Fault condition detected<br>0 = Fault condition not detected                                                                                             | RO/LH            |
| 1.2298.0     | Reserved               | Value always 0                                                                                                                                               | RO               |

<sup>a</sup>RO = Read only, LH = Latching high

##### 45.2.1.235.1 10BASE-T1S loopback ability (1.2298.13)

When read as a one, this bit indicates that the 10BASE-T1S PHY supports PMA loopback. When read as a zero, this bit indicates that the 10BASE-T1S PHY does not support PMA loopback.

##### 45.2.1.235.2 Low-power ability (1.2298.11)

When read as a one, bit 1.2298.11 indicates that the 10BASE-T1S PMA supports the low-power ability. When read as a zero, bit 1.2298.11 indicates that the 10BASE-T1S PMA does not support the low-power feature. If the 10BASE-T1S PMA supports the low-power feature, then it is controlled using either bit 1.2297.11 or bit 1.0.11.

##### 45.2.1.235.3 Multidrop ability (1.2298.10)

When read as a one, bit 1.2298.10 indicates that the 10BASE-T1S PMA supports multidrop mode (see Clause 147). When read as a zero, bit 1.2298.10 indicates that the 10BASE-T1S PMA does not support multidrop mode. If the 10BASE-T1S PMA supports multidrop mode, then it is controlled using bit 1.2297.10; otherwise, bit 1.2297.10 has no effect.

#### 45.2.1.235.4 Receive fault ability (1.2298.9)

When read as a one, bit 1.2298.9 indicates that the 10BASE-T1S PMA has the ability to detect a fault condition on the receive path. When read as a zero, bit 1.2298.9 indicates that the 10BASE-T1S PMA does not have the ability to detect a fault condition on the receive path.

#### 45.2.1.235.5 Receive fault (1.2298.1)

When read as a one, bit 1.2298.1 indicates that the 10BASE-T1S PMA has detected a fault condition on the receive path. When read as a zero, bit 1.2298.1 indicates that the 10BASE-T1S PMA has not detected a fault condition on the receive path. Detection of a fault condition on the receive path is optional, and the ability to detect such a condition is advertised by bit 1.2298.9. The 10BASE-T1S PMA that is unable to detect a fault condition on the receive path shall return a value of zero for this bit. This bit shall be implemented with latching high behavior.

#### 45.2.1.236 10BASE-T1S test mode control register (Register 1.2299)

The assignment of bits in the 10BASE-T1S test mode control register is shown in Table 45–198. The default values for each bit should be chosen so that the initial state of the device upon power up or reset is a normal operational state without management intervention.

**Table 45–198—10BASE-T1S test mode control register bit definitions**

| Bit(s)       | Name              | Description                                                                                                                                                                         | R/W <sup>a</sup> |
|--------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.2299.15:13 | Test mode control | 15 14 13<br>1 1 x = Reserved<br>1 0 1 = Reserved<br>1 0 0 = Test mode 4<br>0 1 1 = Test mode 3<br>0 1 0 = Test mode 2<br>0 0 1 = Test mode 1<br>0 0 0 = Normal (non-test) operation | R/W              |
| 1.2299.12:0  | Reserved          | Value always 0                                                                                                                                                                      | RO               |

<sup>a</sup>RO = Read only, R/W = Read/Write

#### 45.2.1.236.1 Test mode control (1.2299.15:13)

Transmitter test mode operations defined by bits 1.2299.15:13 are described in 147.5.2. The default value for bits 1.2299.15:13 is zero.

#### 45.2.1.237 1000BASE-T1 PMA control register (Register 1.2304)

The assignment of bits in the 1000BASE-T1 PMA control register is shown in Table 45–199.



**Table 45–199—1000BASE-T1 PMA control register bit definitions**

| Bit(s)       | Name             | Description                                  | R/W <sup>a</sup> |
|--------------|------------------|----------------------------------------------|------------------|
| 1.2304.15    | PMA/PMD reset    | 1 = PMA/PMD reset<br>0 = Normal operation    | R/W, SC          |
| 1.2304.14    | Transmit disable | 1 = Transmit disable<br>0 = Normal operation | R/W              |
| 1.2304.13:12 | Reserved         | Value always 0                               | RO               |
| 1.2304.11    | Low-power        | 1 = Low-power mode<br>0 = Normal operation   | R/W              |
| 1.2304.10:0  | Reserved         | Value always 0                               | RO               |

<sup>a</sup>RO = Read only, R/W = Read/Write, SC = Self-clearing

#### **45.2.1.237.1 PMA/PMD reset (1.2304.15)**

Resetting the 1000BASE-T1 PMA/PMD is accomplished by setting bit 1.2304.15 to a one. This action shall set all 1000BASE-T1 PMA/PMD registers to their default states. As a consequence, this action may change the internal state of the 1000BASE-T1 PMA/PMD and the state of the physical link. This action may also initiate a reset in any other MMDs that are instantiated in the same package. This bit is self-clearing, and the 1000BASE-T1 PMA/PMD shall return a value of one in bit 1.2304.15 when a reset is in progress; otherwise, it shall return a value of zero. The 1000BASE-T1 PMA/PMD is not required to accept a write transaction to any of its registers until the reset process is completed. The control and management interface is restored to operation as defined in 97.4.2.1, starting when bit 1.2304.15 is set.

During a reset, the 1000BASE-T1 PMD/PMA shall respond to reads from register bits 1.2304.15, 1.8.15:14, and 1.0.15. All other register bits shall be ignored.

NOTE—This operation may interrupt data communication.

Bit 1.2304.15 is a copy of 1.0.15 and setting or clearing either bit shall set or clear the other bit. Setting either bit shall reset the 1000BASE-T1 PMA/PMD.

#### **45.2.1.237.2 Transmit disable (1.2304.14)**

When bit 1.2304.14 is set to a one, the PMA shall disable output on the transmit path. When bit 1.2304.14 is set to a zero, the PMA shall enable output on the transmit path.

Bit 1.2304.14 is a copy of 1.9.0 and setting or clearing either bit shall set or clear the other bit. Setting either bit shall disable the transmitter.

#### **45.2.1.237.3 Low power (1.2304.11)**

When the low-power ability is supported, the 1000BASE-T1 PMA/PMD may be placed into a low-power mode by setting bit 1.2304.11 to one. This action may also initiate a low-power mode in any other MMDs that are instantiated in the same package. The low-power mode is exited by resetting the 1000BASE-T1 PMA/PMD. The behavior of the 1000BASE-T1 PMA/PMD in transition to and from the low-power mode is implementation specific and any interface signals should not be relied upon. While in the low-power mode, the device shall, as a minimum, respond to management transactions necessary to exit the low-power mode. The default value of bit 1.2304.11 is zero.

This operation interrupts data communication. The data path of the 1000BASE-T1 PMD, depending on type and temperature, may take many seconds to run at optimum error ratio after exiting from reset or low-power mode.

Bit 1.2304.11 is a copy of 1.0.11 and setting or clearing either bit shall set or clear the other bit. Setting either bit shall put the 1000BASE-T1 PMA/PMD in low-power mode.

#### 45.2.1.238 1000BASE-T1 PMA status register (Register 1.2305)

The assignment of bits in the 1000BASE-T1 PMA status register is shown in Table 45–200.

**Table 45–200—1000BASE-T1 PMA status register bit definitions**

| Bit(s)       | Name                    | Description                                                                                                                                                          | R/W <sup>a</sup> |
|--------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.2305.15:12 | Reserved                | Value always 0                                                                                                                                                       | RO               |
| 1.2305.11    | 1000BASE-T1 OAM Ability | 1 = PHY has 1000BASE-T1 OAM ability<br>0 = PHY does not have 1000BASE-T1 OAM ability                                                                                 | RO               |
| 1.2305.10    | EEE Ability             | 1 = PHY has EEE ability<br>0 = PHY does not have EEE ability                                                                                                         | RO               |
| 1.2305.9     | Receive fault ability   | 1 = PMA/PMD has the ability to detect a fault condition on the receive path<br>0 = PMA/PMD does not have the ability to detect a fault condition on the receive path | RO               |
| 1.2305.8     | Low-power ability       | 1 = PMA/PMD has low-power ability<br>0 = PMA/PMD does not have low-power ability                                                                                     | RO               |
| 1.2305.7:3   | Reserved                | Value always 0                                                                                                                                                       | RO               |
| 1.2305.2     | Receive polarity        | 1 = Receive polarity is reversed<br>0 = Receive polarity is not reversed                                                                                             | RO               |
| 1.2305.1     | Receive fault           | 1 = Fault condition detected<br>0 = Fault condition not detected                                                                                                     | RO               |
| 1.2305.0     | Receive link status     | 1 = PMA/PMD receive link up<br>0 = PMA/PMD receive link down                                                                                                         | RO/LL            |

<sup>a</sup>RO = Read only, LL = Latching Low

##### 45.2.1.238.1 1000BASE-T1 OAM ability (1.2305.11)

When read as a one, this bit indicates that the 1000BASE-T1 PHY supports 1000BASE-T1 OAM (see 97.3.8). When read as a zero, this bit indicates that the 1000BASE-T1 PHY does not support 1000BASE-T1 OAM.

##### 45.2.1.238.2 EEE ability (1.2305.10)

When read as a one, this bit indicates that the 1000BASE-T1 PHY supports EEE. When read as a zero, this bit indicates that the 1000BASE-T1 PHY does not support EEE.

#### 45.2.1.238.3 Receive fault ability (1.2305.9)

When read as a one, bit 1.2305.9 indicates that the 1000BASE-T1 PMA/PMD has the ability to detect a fault condition on the receive path. When read as a zero, bit 1.2305.9 indicates that the 1000BASE-T1 PMA/PMD does not have the ability to detect a fault condition on the receive path.

#### 45.2.1.238.4 Low-power ability (1.2305.8)

When read as a one, bit 1.2305.8 indicates that the 1000BASE-T1 PMA/PMD supports the low-power feature. When read as a zero, bit 1.2305.8 indicates that the 1000BASE-T1 PMA/PMD does not support the low-power feature. If the 1000BASE-T1 PMA/PMD supports the low-power feature, then it is controlled using either bit 1.2304.11 or bit 1.0.11.

#### 45.2.1.238.5 Receive polarity (1.2305.2)

When read as zero, bit 1.2305.2 indicates that the polarity of the receiver is not reversed. When read as one, bit 1.2305.2 indicates that the polarity of receiver is reversed.

#### 45.2.1.238.6 Receive fault (1.2305.1)

When read as a one, bit 1.2305.1 indicates that the 1000BASE-T1 PMA/PMD has detected a fault condition on the receive path. When read as a zero, bit 1.2305.1 indicates that the 1000BASE-T1 PMA/PMD has not detected a fault condition on the receive path. Detection of a fault condition on the receive path is optional and the ability to detect such a condition is advertised by bit 1.2305.9. The 1000BASE-T1 PMA/PMD that is unable to detect a fault condition on the receive path shall return a value of zero for this bit.

#### 45.2.1.238.7 Receive link status (1.2305.0)

When read as a one, bit 1.2305.0 indicates that the 1000BASE-T1 PMA/PMD receive link is up. When read as a zero, bit 1.2305.0 indicates that the 1000BASE-T1 PMA/PMD receive link has been down one or more times since the register was last read. The receive link status bit shall be implemented with latching low behavior.

#### 45.2.1.239 1000BASE-T1 training register (Register 1.2306)

The assignment of bits in the 1000BASE-T1 training register is shown in Table 45–201.

**Table 45–201—1000BASE-T1 training register bit definitions**

| Bit(s)       | Name                          | Description                                                                                                          | R/W <sup>a</sup> |
|--------------|-------------------------------|----------------------------------------------------------------------------------------------------------------------|------------------|
| 1.2306.15:11 | Reserved                      | Value always 0                                                                                                       | RO               |
| 1.2306.10:4  | User field                    | 7-bit user defined field to send to the link partner                                                                 | R/W              |
| 1.2306.3:2   | Reserved                      | Value always 0                                                                                                       | RO               |
| 1.2306.1     | 1000BASE-T1 OAM advertisement | 1 = 1000BASE-T1 OAM ability advertised to link partner<br>0 = 1000BASE-T1 OAM ability not advertised to link partner | R/W              |
| 1.2306.0     | EEE advertisement             | 1 = EEE ability advertised to link partner<br>0 = EEE ability not advertised to link partner                         | R/W              |

<sup>a</sup>RO = Read only, R/W = Read/Write

#### 45.2.1.239.1 User field (1.2306.10:4)

This register is a user defined 7-bit field that is transmitted to the link partner during training.

#### 45.2.1.239.2 1000BASE-T1 OAM advertisement (1.2306.1)

When set as a one, this bit indicates to the link partner that the 1000BASE-T1 PHY is advertising 1000BASE-T1 OAM capability. When set as a zero, this bit indicates to the link partner that the 1000BASE-T1 PHY is not advertising 1000BASE-T1 OAM capability. This bit shall be set to zero if the 1000BASE-T1 PHY does not support 1000BASE-T1 OAM.

#### 45.2.1.239.3 EEE advertisement (1.2306.0)

When set as a one, this bit indicates to the link partner that the 1000BASE-T1 PHY is advertising EEE capability. When set as a zero, this bit indicates to the link partner that the 1000BASE-T1 PHY is not advertising EEE capability. This bit shall be set to zero if the 1000BASE-T1 PHY does not support EEE.

#### 45.2.1.240 1000BASE-T1 link partner training register (Register 1.2307)

The assignment of bits in the 1000BASE-T1 link partner training register is shown in Table 45–202. The values in this register are not valid until link is up.

**Table 45–202—1000BASE-T1 link partner training register bit definitions**

| Bit(s)       | Name                                       | Description                                                                                            | R/W <sup>a</sup> |
|--------------|--------------------------------------------|--------------------------------------------------------------------------------------------------------|------------------|
| 1.2307.15:11 | Reserved                                   | Value always 0                                                                                         | RO               |
| 1.2307.10:4  | Link partner user field                    | 7-bit user defined field received from the link partner                                                | RO               |
| 1.2307.3:2   | Reserved                                   | Value always 0                                                                                         | RO               |
| 1.2307.1     | Link partner 1000BASE-T1 OAM advertisement | 1 = Link partner has 1000BASE-T1 OAM ability<br>0 = Link partner does not have 1000BASE-T1 OAM ability | RO               |
| 1.2307.0     | Link partner EEE advertisement             | 1 = Link partner has EEE ability<br>0 = Link partner does not have EEE ability                         | RO               |

<sup>a</sup>RO = Read only

#### 45.2.1.240.1 Link partner user field (1.2307.10:4)

This register is a user defined 7-bit field that is received from the link partner during training.

#### 45.2.1.240.2 Link partner 1000BASE-T1 OAM advertisement (1.2307.1)

When read as a one, this bit indicates the link partner is advertising 1000BASE-T1 OAM capability. When read as a zero, this bit indicates the link partner is not advertising 1000BASE-T1 OAM capability. 1000BASE-T1 OAM capability shall be enabled only when both the local device and its link partner are advertising 1000BASE-T1 OAM capability.

#### 45.2.1.240.3 Link partner EEE advertisement (1.2307.0)

When read as a one, this bit indicates the link partner is advertising EEE capability. When read as a zero, this bit indicates the link partner is not advertising EEE capability. EEE capability shall be enabled only when both the local device and its link partner are advertising EEE capability.

#### 45.2.1.241 1000BASE-T1 test mode control register (Register 1.2308)

The assignment of bits in the 1000BASE-T1 test mode control register is shown in Table 45–203. The default values for each bit should be chosen so that the initial state of the device upon power up or reset is a normal operational state without management intervention.

**Table 45–203—1000BASE-T1 test mode control register bit definitions**

| Bit(s)       | Name              | Description                                                                                                                                                                                                   | R/W <sup>a</sup> |
|--------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.2308.15:13 | Test mode control | 15 14 13<br>1 1 1 = Test mode 7<br>1 1 0 = Test mode 6<br>1 0 1 = Test mode 5<br>1 0 0 = Test mode 4<br>0 1 1 = Reserved<br>0 1 0 = Test mode 2<br>0 0 1 = Test mode 1<br>0 0 0 = Normal (non-test) operation | R/W              |
| 1.2308.12:0  | Reserved          | Value always 0                                                                                                                                                                                                | RO               |

<sup>a</sup>RO = Read only

#### 45.2.1.241.1 Test mode control (1.2308.15:13)

Transmitter test mode operations defined by bits 1.2308.15:13, are described in 97.5.2 and Table 97–12. The default value for bits 1.2308.15:13 is zero.

#### 45.2.1.242 MultiGBASE-T1 PMA control register (Register 1.2309)

The assignment of bits in the MultiGBASE-T1 PMA control register is shown in Table 45–204.

**Table 45–204—MultiGBASE-T1 PMA control register bit definitions**

| Bit(s)       | Name             | Description                                  | R/W <sup>a</sup> |
|--------------|------------------|----------------------------------------------|------------------|
| 1.2309.15    | PMA/PMD reset    | 1 = PMA/PMD reset<br>0 = Normal operation    | R/W,<br>SC       |
| 1.2309.14    | Transmit disable | 1 = Transmit disable<br>0 = Normal operation | R/W              |
| 1.2309.13:12 | Reserved         | Value always 0                               | RO               |

**Table 45–204—MultiGBASE-T1 PMA control register bit definitions (*continued*)**

| Bit(s)      | Name      | Description                                | R/W <sup>a</sup> |
|-------------|-----------|--------------------------------------------|------------------|
| 1.2309.11   | Low-power | 1 = Low-power mode<br>0 = Normal operation | R/W              |
| 1.2309.10:0 | Reserved  | Value always 0                             | RO               |

<sup>a</sup>R/W = Read/Write, RO = Read only, SC = Self-clearing

#### 45.2.1.242.1 PMA/PMD reset (1.2309.15)

Resetting the MultiGBASE-T1 PMA/PMD is accomplished by setting bit 1.2309.15 to a one. This action shall set all PMA/PMD registers to their default states. As a consequence, this action may change the internal state of a MultiGBASE-T1 PMA/PMD and the state of the physical link. This action may also initiate a reset in any other MMDs that are instantiated in the same package. Bit 1.2309.15 is self-clearing, and the MultiGBASE-T1 PMA/PMD shall return a value of one in bit 1.2309.15 when a reset is in progress; otherwise, it shall return a value of zero. The MultiGBASE-T1 PMA/PMD is not required to accept a write transaction to any of its registers until the reset process is completed. The control and management interface is restored to operation as defined in 149.4.2.1, starting when bit 1.2309.15 is set.

During a reset, the MultiGBASE-T1 PMA/PMD shall respond to reads from register bits 1.2309.15, 1.8.15:14, and 1.0.15. All other register bits shall be ignored.

NOTE—This operation may interrupt data communication.

Bit 1.2309.15 is a copy of bit 1.0.15 and setting or clearing either bit shall set or clear the other bit. Setting either bit shall reset the MultiGBASE-T1 PMA/PMD.

#### 45.2.1.242.2 Transmit disable (1.2309.14)

When bit 1.2309.14 is set to a one, the PMA shall disable output on the transmit path. When bit 1.2309.14 is set to a zero, the PMA shall enable output on the transmit path.

Bit 1.2309.14 is a copy of bit 1.9.0 and setting or clearing either bit shall set or clear the other bit. Setting either bit shall disable the transmitter.

#### 45.2.1.242.3 Low power (1.2309.11)

When the low-power ability is supported, the MultiGBASE-T1 PMA/PMD may be placed into a low-power mode by setting bit 1.2309.11 to one. This action may also initiate a low-power mode in any other MMDs that are instantiated in the same package. The low-power mode is exited by resetting the MultiGBASE-T1 PMA/PMD. The behavior of the MultiGBASE-T1 PMA/PMD in transition to and from the low-power mode is implementation specific and any interface signals should not be relied upon. While in the low-power mode, the device shall, at a minimum, respond to management transactions necessary to exit the low-power mode.

The default value of bit 1.2309.11 is zero.

This operation interrupts data communication.

Bit 1.2309.11 is a copy of bit 1.0.11 and setting or clearing either bit shall set or clear the other bit. Setting either bit shall put the MultiGBASE-T1 PMA/PMD in low-power mode.

#### 45.2.1.243 MultiGBASE-T1 PMA status register (1.2310)

The assignment of bits in the MultiGBASE-T1 PMA status register is shown in Table 45–205.

**Table 45–205—MultiGBASE-T1 PMA status register bit definitions**

| Bit(s)       | Name                      | Description                                                                                                                                                          | R/W <sup>a</sup> |
|--------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.2310.15:12 | Reserved                  | Value always 0                                                                                                                                                       | RO               |
| 1.2310.11    | MultiGBASE-T1 OAM ability | 1 = PHY has MultiGBASE-T1 OAM ability<br>0 = PHY does not have MultiGBASE-T1 OAM ability                                                                             | RO               |
| 1.2310.10    | EEE ability               | 1 = PHY has EEE ability<br>0 = PHY does not have EEE ability                                                                                                         | RO               |
| 1.2310.9     | Receive fault ability     | 1 = PMA/PMD has the ability to detect a fault condition on the receive path<br>0 = PMA/PMD does not have the ability to detect a fault condition on the receive path | RO               |
| 1.2310.8     | Low-power ability         | 1 = PMA/PMD has low-power ability<br>0 = PMA/PMD does not have low-power ability                                                                                     | RO               |
| 1.2310.7:5   | Reserved                  | Value always 0                                                                                                                                                       | RO               |
| 1.2310.4:3   | PrecodeSel                | 4 3<br>0 0 = no precoder requested<br>0 1 = 1–D precoder requested<br>1 0 = 1+D precoder requested<br>1 1 = 1–D <sup>2</sup> precoder requested                      | RO               |
| 1.2310.2     | Receive polarity          | 1 = Receive polarity is reversed<br>0 = Receive polarity is not reversed                                                                                             | RO               |
| 1.2310.1     | Receive fault             | 1 = Fault condition detected<br>0 = Fault condition not detected                                                                                                     | RO/LH            |
| 1.2310.0     | Receive link status       | 1 = PMA/PMD receive link up<br>0 = PMA/PMD receive link down                                                                                                         | RO/LL            |

<sup>a</sup>RO = Read only, LH = Latching High, LL = Latching Low

#### 45.2.1.243.1 MultiGBASE-T1 OAM ability (1.2310.11)

When read as a one, bit 1.2310.11 indicates that the MultiGBASE-T1 PHY supports MultiGBASE-T1 OAM (see 149.3.9). When read as a zero, bit 1.2310.11 indicates that the MultiGBASE-T1 PHY does not support MultiGBASE-T1 OAM.

#### 45.2.1.243.2 EEE ability (1.2310.10)

When read as a one, bit 1.2310.10 indicates that the MultiGBASE-T1 PHY supports EEE. When read as a zero, bit 1.2310.10 indicates that the MultiGBASE-T1 PHY does not support EEE.

#### **45.2.1.243.3 Receive fault ability (1.2310.9)**

When read as a one, bit 1.2310.9 indicates that the MultiGBASE-T1 PMA/PMD has the ability to detect a fault condition on the receive path. When read as a zero, bit 1.2310.9 indicates that the MultiGBASE-T1 PMA/PMD does not have the ability to detect a fault condition on the receive path.

#### **45.2.1.243.4 Low-power ability (1.2310.8)**

When read as a one, bit 1.2310.8 indicates that the MultiGBASE-T1 PMA/PMD supports the low-power feature. When read as a zero, bit 1.2310.8 indicates that the MultiGBASE-T1 PMA/PMD does not support the low-power feature. If the MultiGBASE-T1 PMA/PMD supports the low-power feature, then it is controlled using either bit 1.2309.11 or bit 1.0.11.

#### **45.2.1.243.5 PrecodeSel (1.2310.4:3)**

Bits 1.2310.4:3 contain the requested precoder setting communicated by the PHY to the link partner via the PrecodeSel bits in the Infofield (see 149.4.2.4.5).

#### **45.2.1.243.6 Receive polarity (1.2310.2)**

When read as zero, bit 1.2310.2 indicates that the polarity of the receiver is not reversed. When read as one, bit 1.2310.2 indicates that the polarity of the receiver is reversed.

#### **45.2.1.243.7 Receive fault (1.2310.1)**

When read as a one, bit 1.2310.1 indicates that the MultiGBASE-T1 PMA/PMD has detected a fault condition on the receive path. When read as a zero, bit 1.2310.1 indicates that the MultiGBASE-T1 PMA/PMD has not detected a fault condition on the receive path. Detection of a fault condition on the receive path is optional and the ability to detect such a condition is advertised by bit 1.2310.9. A MultiGBASE-T1 PMA/PMD that is unable to detect a fault condition on the receive path shall return a value of zero for bit 1.2310.1. The receive fault bit shall be implemented with latching high behavior.

#### **45.2.1.243.8 Receive link status (1.2310.0)**

When read as a one, bit 1.2310.0 indicates that the MultiGBASE-T1 PMA/PMD receive link is up. When read as a zero, bit 1.2310.0 indicates that the MultiGBASE-T1 PMA/PMD receive link has been down one or more times since the register was last read. The receive link status bit shall be implemented with latching low behavior.

#### **45.2.1.244 MultiGBASE-T1 training register (1.2311)**

The assignment of bits in the MultiGBASE-T1 training register is shown in Table 45–206.

##### **45.2.1.244.1 Interleave request (1.2311.12:11)**

Bits 1.2311.12:11 control the Reed-Solomon interleave setting of this PHY. Reed-Solomon interleaving is described in 149.3.2.2.15. This is communicated to the link partner via Infofields as specified in 149.4.2.4.5.

The values of  $L = 2$  and  $L = 4$  are not defined for 2.5GBASE-T1 PHYs, and the value of  $L = 4$  is not defined for 5GBASE-T1 PHYs. If bits 1.2311.12:11 are set to these undefined values, the PHY will communicate these values to the link partner, but the requested interleaver depth is out of scope of this standard and may not be supported by the link partner.



**Table 45–206—MultiGBASE-T1 training register bit definitions**

| Bit(s)       | Name                            | Description                                                                                                                                                                                                                                                                              | R/W <sup>a</sup> |
|--------------|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.2311.15:13 | Reserved                        | Value always 0                                                                                                                                                                                                                                                                           | RO               |
| 1.2311.12:11 | Interleave request              | 12 11<br>0 0: L=1, no interleaving, default for 2.5GBASE-T1<br>0 1: L=2, RS-FEC interleaving factor two, default for 5GBASE-T1, undefined for 2.5GBASE-T1<br>1 0: L=4, RS-FEC interleaving factor four, default for 10GBASE-T1, undefined for 2.5GBASE-T1 and 5GBASE-T1<br>1 1: Reserved | R/W              |
| 1.2311.10:6  | Reserved                        | Value always 0                                                                                                                                                                                                                                                                           | RO               |
| 1.2311.5     | Precoder selection              | Controls requested precoder setting<br>0 = PHY<br>1 = User                                                                                                                                                                                                                               | R/W              |
| 1.2311.4     | Slow Wake request               | 1 = Alert window start is only immediate frame after refresh<br>0 = Alert window start is every 8th RS frame after refresh                                                                                                                                                               | R/W              |
| 1.2311.3:2   | User precoder selection         | 3 2<br>0 0 = no precoder requested<br>0 1 = 1-D precoder requested<br>1 0 = 1+D <sub>2</sub> precoder requested<br>1 1 = 1-D <sub>2</sub> <sup>2</sup> precoder requested                                                                                                                | R/W              |
| 1.2311.1     | MultiGBASE-T1 OAM advertisement | 1 = MultiGBASE-T1 OAM ability advertised to link partner<br>0 = MultiGBASE-T1 OAM ability not advertised to link partner                                                                                                                                                                 | R/W              |
| 1.2311.0     | EEE advertisement               | 1 = EEE ability advertised to link partner<br>0 = EEE ability not advertised to link partner                                                                                                                                                                                             | R/W              |

<sup>a</sup>RO = Read only, R/W = Read/Write

#### 45.2.1.244.2 Precoder selection (1.2311.5)

When bit 1.2311.5 is set to a one, the PHY shall use bits 1.2311.3:2 for the value of PrecoderSel, and when set to a zero the PHY controls the value of PrecoderSel. PrecoderSel is the desired precoder setting communicated to the link partner via the Infobits specified in 149.4.2.4.5.

#### 45.2.1.244.3 Slow Wake request (1.2311.4)

When set as a one, bit 1.2311.4 indicates to the link partner that the local MultiGBASE-T1 PHY may only transmit alert immediately following refresh. When set as a zero, bit 1.2311.4 indicates to the link partner that the local MultiGBASE-T1 PHY may transmit an alert at any eighth RS-FEC frame after a refresh.

#### 45.2.1.244.4 User precoder selection (1.2311.3:2)

When bit 1.2311.5 is a one, bits 1.2311.3:2 are the requested precoder setting communicated by the PHY to the link partner via the PrecoderSel bits in the Infobits (see 149.4.2.4.5).

#### 45.2.1.244.5 MultiGBASE-T1 OAM advertisement (1.2311.1)

Support for MultiGBASE-T1 OAM capability shall be advertised if bit 1.2311.1 is set to one. Support for MultiGBASE-T1 OAM capability shall not be advertised if bit 1.2311.1 is set to zero. Support for MultiGBASE-T1 OAM capability should only be advertised if it is supported by the PHY.

#### 45.2.1.244.6 EEE advertisement (1.2311.0)

Support for EEE capability shall be advertised if bit 1.2311.0 is set to one. Support for EEE capability shall not be advertised if bit 1.2311.0 is set to zero. Support for EEE operation should only be advertised if it is supported by the PHY.

#### 45.2.1.245 MultiGBASE-T1 link partner training register (1.2312)

The assignment of bits in the MultiGBASE-T1 link partner training register is shown in Table 45–207. The values in this register are not valid when the link is down.

**Table 45–207—MultiGBASE-T1 link partner training register bit definitions**

| Bit(s)       | Name                                         | Description                                                                                                                                                                                                                                                                              | R/W <sup>a</sup> |
|--------------|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.2312.15:13 | Reserved                                     | Value always 0                                                                                                                                                                                                                                                                           | RO               |
| 1.2312.12:11 | Link partner interleave request              | 12 11<br>0 0: L=1, no interleaving, default for 2.5GBASE-T1<br>0 1: L=2, RS-FEC interleaving factor two, default for 5GBASE-T1, undefined for 2.5GBASE-T1<br>1 0: L=4, RS-FEC interleaving factor four, default for 10GBASE-T1, undefined for 2.5GBASE-T1 and 5GBASE-T1<br>1 1: Reserved | RO               |
| 1.2312.10:5  | Reserved                                     | Value always 0                                                                                                                                                                                                                                                                           | RO               |
| 1.2312.4     | Link partner Slow Wake requested             | 1 = Alert window start is only immediate frame after refresh<br>0 = Alert window start is every 8th RS frame after refresh                                                                                                                                                               | RO               |
| 1.2312.3:2   | Link partner precoder requested              | 3 2<br>0 0 = no precoder requested<br>0 1 = 1–D precoder requested<br>1 0 = 1+D precoder requested<br>1 1 = 1–D <sup>2</sup> precoder requested                                                                                                                                          | RO               |
| 1.2312.1     | Link partner MultiGBASE-T1 OAM advertisement | 1 = Link partner has MultiGBASE-T1 OAM ability<br>0 = Link partner does not have MultiGBASE-T1 OAM ability                                                                                                                                                                               | RO               |
| 1.2312.0     | Link partner EEE advertisement               | 1 = Link partner has EEE ability<br>0 = Link partner does not have EEE ability                                                                                                                                                                                                           | RO               |

<sup>a</sup>RO = Read only

#### **45.2.1.245.1 Link partner interleave request (1.2312.12:11)**

Bits 1.2312.12:11 contain the Reed-Solomon interleave setting requested by the link partner. Reed-Solomon interleaving is described in 149.3.2.2.15. This is communicated by the link partner via Infofields as specified in 149.4.2.4.5.

The values of  $L = 2$  and  $L = 4$  are not defined for 2.5GBASE-T1 PHYs, and the value of  $L = 4$  is not defined for 5GBASE-T1 PHYs. Bits 1.2312.12:11 will indicate whatever value is received from the link partner, but if the undefined values are received, the requested interleaver depth is out of scope of this standard and may not be supported by the local PHY.

#### **45.2.1.245.2 Link partner Slow Wake requested (1.2312.4)**

Bit 1.2312.4 contains the Slow Wake setting received from the link partner. When set as a one, the link partner may only transmit alert immediately following refresh. When set as a zero, the link partner may only transmit an alert at any eighth RS-FEC frame after a refresh.

#### **45.2.1.245.3 Link partner precoder requested (1.2312.3:2)**

Bits 1.2312.3:2 contain the precoder setting requested by the link partner. The precoder requested is encoded as the value of `precoder_type` defined in 149.3.2.2.20.

#### **45.2.1.245.4 Link partner MultiGBASE-T1 OAM advertisement (1.2312.1)**

When read as a one, bit 1.2312.1 indicates the link partner is advertising MultiGBASE-T1 OAM capability. When read as a zero, bit 1.2312.1 indicates the link partner is not advertising MultiGBASE-T1 OAM capability. MultiGBASE-T1 OAM capability shall be enabled only when both the local PHY and its link partner are advertising MultiGBASE-T1 OAM capability.

#### **45.2.1.245.5 Link partner EEE advertisement (1.2312.0)**

When read as a one, bit 1.2312.0 indicates the link partner is advertising EEE capability. When read as a zero, bit 1.2312.0 indicates the link partner is not advertising EEE capability. EEE capability shall be enabled only when both the local PHY and its link partner are advertising EEE capability.

#### **45.2.1.246 MultiGBASE-T1 test mode control register (1.2313)**

The assignment of bits in the MultiGBASE-T1 test mode control register is shown in Table 45–208. The default values for each bit are chosen so that the initial state of the device upon power up or reset is a normal operational state without management intervention.

##### **45.2.1.246.1 Test mode control (1.2313.15:13)**

Transmitter test mode operations defined by bits 1.2313.15:13, are described in 149.5.1 and Table 149–17. The default value for bits 1.2313.15:13 is zero.

##### **45.2.1.246.2 Local transmitter precoder override (1.2313.11)**

When bit 1.2313.11 is set to one, the local transmitter's precoder shall be controlled by the value of bits 1.2313.10:9, and the precoder requested by the link partner in `PrecodeSel` shall be ignored. When bit 1.2313.11 is set to zero, the transmitter shall ignore bits 1.2313.10:9, and the precoder is set according to the value of `PrecodeSel` received from the link partner as specified in 149.3.2.2.20. The default value of 1.2313.11 is zero.

**Table 45–208—MultiGBASE-T1 test mode control register bit definitions**

| Bit(s)       | Name                                | Description                                                                                                                                                                                                      | R/W <sup>a</sup> |
|--------------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 1.2313.15:13 | Test mode control                   | 15 14 13<br>1 1 1 = Test mode 7<br>1 1 0 = Test mode 6<br>1 0 1 = Test mode 5<br>1 0 0 = Test mode 4<br>0 1 1 = Test mode 3<br>0 1 0 = Test mode 2<br>0 0 1 = Test mode 1<br>0 0 0 = Normal (non-test) operation | R/W              |
| 1.2313.12    | Reserved                            | Value always 0                                                                                                                                                                                                   | RO               |
| 1.2313.11    | Local transmitter precoder override | 0 = Normal operation<br>1 = User override                                                                                                                                                                        | R/W              |
| 1.2313.10:9  | Local transmit precoder setting     | 10 9<br>0 0 = transmit with no precoder<br>0 1 = transmit with 1–D precoder<br>1 0 = transmit with 1+D precoder<br>1 1 = transmit with 1–D <sup>2</sup> precoder                                                 | R/W              |
| 1.2313.8:2   | Reserved                            | Value always 0                                                                                                                                                                                                   | RO               |
| 1.2313.1:0   | Jitter test control                 | 1 0<br>0 0 = Square wave<br>0 1 = JP03A Pattern<br>1 0 = JP03B Pattern<br>1 1 = Reserved                                                                                                                         | R/W              |

<sup>a</sup>RO = Read only, R/W = Read/Write

#### 45.2.1.246.3 Local transmit precoder setting (1.2313.10:9)

When bit 1.2313.11 is set to one, bits 1.2313.10:9 control the precoder setting of the local transmitter, as defined in 149.3.2.2.20 in the variable precoder\_type.

#### 45.2.1.246.4 Jitter test control (1.2313.1:0)

When the transmitter is in test mode 2, bits 1.2313.1:0 control the pattern of the jitter test signal. When the transmitter is not in test mode 2, the setting of bits 1.2313.1:0 has no effect. A value of 00 transmits a square wave from the transmitter, a value of 01 transmits the JP03A pattern, and a value of 10 transmits the JP03B pattern. See 149.5.2.3.1 and 149.5.2.3.2 for more information.

#### 45.2.1.247 MultiGBASE-T1 SNR operating margin register (Register 1.2314)

Register 1.2314 contains the current SNR operating margin measured at the slicer input for the PMAs in the MultiGBASE-T1 set. It is reported with 0.1 dB of resolution to an accuracy of 0.5 dB within the range of –12.7 dB to 12.7 dB. The number is in offset binary notation, with 0.0 dB represented by 0x80, 12.7 dB represented by 0xFF, and –12.7 dB represented by 0x01. Implementation of this register is optional.

The assignment of bits in the MultiGBASE-T1 SNR operating margin register is shown in Table 45–209.

**Table 45–209—MultiGBASE-T1 SNR operating margin register bit definitions**

| Bit(s)      | Name                               | Description                                 | R/W <sup>a</sup> |
|-------------|------------------------------------|---------------------------------------------|------------------|
| 1.2314.15:8 | MultiGBASE-T1 SNR operating margin | Value of current SNR operating margin in dB | RO               |
| 1.2314.7:0  | Reserved                           | Value always 0                              | RO               |

<sup>a</sup>RO = Read only

#### 45.2.1.248 MultiGBASE-T1 minimum SNR margin register (Register 1.2315)

Register 1.2315 contains a latched copy of the lowest value observed in the MultiGBASE-T1 SNR operating margin register (1.2314) since the last read. It is reported with 0.1 dB of resolution to an accuracy of 0.5 dB within the range of –12.7 dB to 12.7 dB. The number is in offset binary notation, with 0.0 dB represented by 0x80, 12.7 dB represented by 0xFF, and –12.7 dB represented by 0x01. Implementation of this register is optional.

The assignment of bits in the MultiGBASE-T1 minimum SNR margin register is shown in Table 45–210.

**Table 45–210—MultiGBASE-T1 minimum SNR margin register bit definitions**

| Bit(s)      | Name                             | Description                                | R/W <sup>a</sup> |
|-------------|----------------------------------|--------------------------------------------|------------------|
| 1.2315.15:8 | MultiGBASE-T1 minimum SNR margin | Value of minimum observed SNR margin in dB | RO               |
| 1.2315.7:0  | Reserved                         | Value always 0                             | RO               |

<sup>a</sup>RO = Read only

#### 45.2.1.249 MultiGBASE-T1 user defined data (Register 1.2316)

The assignment of bits for the MultiGBASE-T1 user defined data register is shown in Table 45–211. The values of the bits in this register are outside the scope of this standard.

**Table 45–211—MultiGBASE-T1 user defined data register bit definitions**

| Bit(s)      | Name                            | Description                                                            | R/W <sup>a</sup> |
|-------------|---------------------------------|------------------------------------------------------------------------|------------------|
| 1.2316.15:0 | MultiGBASE-T1 user defined data | 16 bits of vendor specific data that the PHY sends to its link partner | R/W              |

<sup>a</sup>R/W = Read/Write

##### 45.2.1.249.1 MultiGBASE-T1 user defined data (1.2316.15:0)

Bits 1.2316.15:0 contain vendor specific data that the PHY may communicate to its link partner during training.

#### 45.2.1.250 MultiGBASE-T1 link partner user defined data register (Register 1.2317)

The assignment of bits for the MultiGBASE-T1 link partner user defined data register is shown in 45–212. The values of the bits in this register are outside the scope of this standard.

**Table 45–212—MultiGBASE-T1 link partner user defined data register bit definitions**

| Bit(s)      | Name                                         | Description                                                                    | R/W <sup>a</sup> |
|-------------|----------------------------------------------|--------------------------------------------------------------------------------|------------------|
| 1.2317.15:0 | MultiGBASE-T1 link partner user defined data | 16 bits of vendor specific data that the PHY may receive from its link partner | RO               |

<sup>a</sup>RO = Read only

##### 45.2.1.250.1 MultiGBASE-T1 link partner user defined data register (1.2317.15:0)

Bits 1.2317.15:0 contain vendor specific data that the PHY may receive from its link partner during training.

#### 45.2.2 WIS registers

The assignment of registers in the WIS is shown in Table 45–213. For the WIS octet fields, bit 8 of the corresponding field in the WIS frame maps to the lowest numbered bit of the field in the register.

**Table 45–213—WIS registers**

| Register address  | Register name                      | Subclause |
|-------------------|------------------------------------|-----------|
| 2.0               | WIS control 1                      | 45.2.2.1  |
| 2.1               | WIS status 1                       | 45.2.2.2  |
| 2.2, 2.3          | WIS device identifier              | 45.2.2.3  |
| 2.4               | WIS speed ability                  | 45.2.2.4  |
| 2.5, 2.6          | WIS devices in package             | 45.2.2.5  |
| 2.7               | 10G WIS control 2                  | 45.2.2.6  |
| 2.8               | 10G WIS status 2                   | 45.2.2.7  |
| 2.9               | 10G WIS test-pattern error counter | 45.2.2.8  |
| 2.10 through 2.13 | Reserved                           |           |
| 2.14, 2.15        | WIS package identifier             | 45.2.2.9  |
| 2.16 through 2.32 | Reserved                           |           |
| 2.33              | 10G WIS status 3                   | 45.2.2.10 |
| 2.34 through 2.36 | Reserved                           |           |

**Table 45–213—WIS registers (*continued*)**

| Register address          | Register name                          | Subclause |
|---------------------------|----------------------------------------|-----------|
| 2.37                      | 10G WIS far end path block error count | 45.2.2.11 |
| 2.38                      | Reserved                               |           |
| 2.39 through 2.46         | 10G WIS J1 transmit                    | 45.2.2.12 |
| 2.47 through 2.54         | 10G WIS J1 receive                     | 45.2.2.13 |
| 2.55, 2.56                | 10G WIS far end line BIP errors        | 45.2.2.14 |
| 2.57, 2.58                | 10G WIS line BIP errors                | 45.2.2.15 |
| 2.59                      | 10G WIS path block error count         | 45.2.2.16 |
| 2.60                      | 10G WIS section BIP error count        | 45.2.2.17 |
| 2.61 through 2.63         | Reserved                               |           |
| 2.64 through 2.71         | 10G WIS J0 transmit                    | 45.2.2.18 |
| 2.72 through 2.79         | 10G WIS J0 receive                     | 45.2.2.19 |
| 2.80 through 2.1799       | Reserved                               |           |
| 2.1800                    | TimeSync WIS capability                | 45.2.2.20 |
| 2.1801 through 2.1804     | TimeSync WIS transmit path data delay  | 45.2.2.21 |
| 2.1805 through 2.1808     | TimeSync WIS receive path data delay   | 45.2.2.22 |
| 2.1809 through 2.32767    | Reserved                               |           |
| 2.32 768 through 2.65 535 | Vendor specific                        |           |

#### 45.2.2.1 WIS control 1 register (Register 2.0)

The assignment of bits in the WIS control 1 register is shown in Table 45–214. The default value for each bit of the WIS control 1 register should be chosen so that the initial state of the device upon power up or reset is a normal operational state without management intervention.

**Table 45–214—WIS control 1 register bit definitions**

| Bit(s) | Name     | Description                                           | R/W <sup>a</sup> |
|--------|----------|-------------------------------------------------------|------------------|
| 2.0.15 | Reset    | 1 = WIS reset<br>0 = Normal operation                 | R/W<br>SC        |
| 2.0.14 | Loopback | 1 = Enable loopback mode<br>0 = Disable loopback mode | R/W              |

**Table 45–214—WIS control 1 register bit definitions (*continued*)**

| Bit(s)   | Name            | Description                                                                                                                                                                          | R/W <sup>a</sup> |
|----------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 2.0.13   | Speed selection | 1 = Operation at 10 Gb/s and above<br>0 = Unspecified                                                                                                                                | R/W              |
| 2.0.12   | Reserved        | Value always 0                                                                                                                                                                       | RO               |
| 2.0.11   | Low power       | 1 = Low-power mode<br>0 = Normal operation                                                                                                                                           | R/W              |
| 2.0.10:7 | Reserved        | Value always 0                                                                                                                                                                       | RO               |
| 2.0.6    | Speed selection | 1 = Operation at 10 Gb/s and above<br>0 = Unspecified                                                                                                                                | R/W              |
| 2.0.5:2  | Speed selection | <div> <div>5 4 3 2</div> <div>1 x x x = Reserved</div> <div>x 1 x x = Reserved</div> <div>x x 1 x = Reserved</div> <div>0 0 0 1 = Reserved</div> <div>0 0 0 0 = 10 Gb/s</div> </div> | R/W              |
| 2.0.1:0  | Reserved        | Value always 0                                                                                                                                                                       | RO               |

<sup>a</sup>R/W = Read/Write, SC = Self-clearing, RO = Read only

#### 45.2.2.1.1 Reset (2.0.15)

Resetting a WIS is accomplished by setting bit 2.0.15 to a one. This action shall set all to their default states. As a consequence, this action may change the internal state of the WIS and the state of the physical link. This action may also initiate a reset in any other MMDs that are instantiated in the same package. This bit is self-clearing, and a WIS shall return a value of one in bit 2.0.15 when a reset is in progress and a value of zero otherwise. A WIS is not required to accept a write transaction to any of its registers until the reset process is completed. The reset process shall be completed within 0.5 s from the setting of bit 2.0.15. During a reset, a WIS shall respond to reads from register bits 2.0.15 and 2.8.15:14. All other register bits should be ignored.

NOTE—This operation may interrupt data communication.

#### 45.2.2.1.2 Loopback (2.0.14)

The WIS shall be placed in a loopback mode of operation when bit 2.0.14 is set to a one. When bit 2.0.14 is set to a one, the WIS shall ignore all data presented to it by the PMA sublayer. When bit 2.0.14 is set to a one, the WIS shall accept data on the transmit path and return it on the receive path. For 10 Gb/s operation, the detailed behavior of the WIS during loopback is specified in 50.3.9.

The default value of bit 2.0.14 is zero.

NOTE—The signal path through the WIS that is exercised in the loopback mode of operation is implementation specific, but it is recommended that the signal path encompass as much of the WIS circuitry as is practical. The intention of providing this loopback mode of operation is to permit a diagnostic or self-test function to perform the transmission and reception of a PDU, thus testing the transmit and receive data paths. Other loopback signal paths may be enabled using loopback controls within other MMDs.



#### 45.2.2.1.3 Low power (2.0.11)

A WIS may be placed into a low-power mode by setting bit 2.0.11 to a one. This action may also initiate a low-power mode in any other MMDs that are instantiated in the same package. The low-power mode is exited by resetting the WIS. The behavior of the WIS in transition to and from the low-power mode is implementation specific and any interface signals should not be relied upon. While in the low-power mode, the device shall, as a minimum, respond to management transactions necessary to exit the low-power mode. The default value of bit 2.0.11 is zero.

#### 45.2.2.1.4 Speed selection (2.0.13, 2.0.6, and 2.0.5:2)

Speed selection bits 2.0.13 and 2.0.6 shall both be written as a one. Any attempt to change the bits to an invalid setting shall be ignored. These two bits are set to one in order to make them compatible with Clause 22.

The speed of the WIS may be selected using bits 5 through 2. The speed abilities of the WIS are advertised in the WIS speed ability register. A WIS may ignore writes to the WIS speed selection bits that select speeds it has not advertised in the WIS speed ability register. It is the responsibility of the STA entity to ensure that mutually acceptable speeds are applied consistently across all the MMDs on a particular PHY.

The WIS speed selection defaults to a supported ability.

#### 45.2.2.2 WIS status 1 register (Register 2.1)

The assignment of bits in the WIS status 1 register is shown in Table 45–215. All the bits in the WIS status 1 register are read only; a write to the WIS status 1 register shall have no effect.

**Table 45–215—WIS status 1 register bit definitions**

| Bit(s)   | Name              | Description                                                                | R/W <sup>a</sup> |
|----------|-------------------|----------------------------------------------------------------------------|------------------|
| 2.1.15:8 | Reserved          | Value always 0                                                             | RO               |
| 2.1.7    | Fault             | 1 = Fault condition<br>0 = No fault condition                              | RO/LH            |
| 2.1.6:3  | Reserved          | Value always 0                                                             | RO               |
| 2.1.2    | Link status       | 1 = WIS link up<br>0 = WIS link down                                       | RO/LL            |
| 2.1.1    | Low-power ability | 1 = WIS supports low-power mode<br>0 = WIS does not support low-power mode | RO               |
| 2.1.0    | Reserved          | Value always 0                                                             | RO               |

<sup>a</sup>RO = Read only, LH = Latching high, LL = Latching low

#### 45.2.2.2.1 Fault (2.1.7)

When read as a one, bit 2.1.7 indicates that the WIS has detected a fault condition. When read as a zero, bit 2.1.7 indicates that the WIS has not detected a fault condition. The fault bit shall be implemented with latching high behavior.

The default value of bit 2.1.7 is zero.

#### 45.2.2.2.2 Link status (2.1.2)

When read as a one, bit 2.1.2 indicates that the WIS receive link is up. When read as a zero, bit 2.1.2 indicates that the WIS receive link is down. The link status bit shall be implemented with latching low behavior.

#### 45.2.2.2.3 Low-power ability (2.1.1)

When read as a one, bit 2.1.1 indicates that the WIS supports the low-power feature. When read as a zero, bit 2.1.1 indicates that the WIS does not support the low-power feature. If a WIS supports the low-power feature, then it is controlled using the low-power bit in the WIS control register.

#### 45.2.2.3 WIS device identifier (Registers 2.2 and 2.3)

Registers 2.2 and 2.3 provide a 32-bit value, which may constitute a unique identifier for a particular type of WIS. The identifier shall be composed of the 3rd through 24th bits of the Organizationally Unique Identifier (OUI) assigned to the device manufacturer by the IEEE, plus a six-bit model number, plus a four-bit revision number. A WIS may return a value of zero in each of the 32 bits of the WIS device identifier.

The format of the WIS device identifier is specified in 22.2.4.3.1.

#### 45.2.2.4 WIS speed ability (Register 2.4)

The assignment of bits in the WIS speed ability register is shown in Table 45–216.

**Table 45–216—WIS speed ability register bit definitions**

| Bit(s)   | Name                       | Description                                                                                  | R/W <sup>a</sup> |
|----------|----------------------------|----------------------------------------------------------------------------------------------|------------------|
| 2.4.15:1 | Reserved for future speeds | Value always 0                                                                               | RO               |
| 2.4.0    | 10G capable                | 1 = WIS is capable of operating at 10 Gb/s<br>0 = WIS is not capable of operating at 10 Gb/s | RO               |

<sup>a</sup>RO = Read only

##### 45.2.2.4.1 10G capable (2.4.0)

When read as a one, bit 2.4.0 indicates that the WIS is able to operate at a data rate of 10 Gb/s (9.58 Gb/s payload rate). When read as a zero, bit 2.4.0 indicates that the WIS is not able to operate at a data rate of 10 Gb/s (9.58 Gb/s payload rate).

#### 45.2.2.5 WIS devices in package (Registers 2.5 and 2.6)

The WIS devices in package registers are defined in Table 45–2.

#### 45.2.2.6 10G WIS control 2 register (Register 2.7)

The assignment of bits in the 10G WIS control 2 register is shown in Table 45–217. The default value for each bit of the 10G WIS control 2 register should be chosen so that the initial state of the device upon power up or reset is a normal operational state without management intervention.

**Table 45–217—10G WIS control 2 register bit definitions**

| Bit(s)   | Name                                | Description                                                                                                           | R/W <sup>a</sup> |
|----------|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------|------------------|
| 2.7.15:6 | Reserved                            | Value always 0                                                                                                        | RO               |
| 2.7.5    | PRBS31 receive test-pattern enable  | 1 = Enable PRBS31 test-pattern mode on the receive path<br>0 = Disable PRBS31 test-pattern mode on the receive path   | R/W              |
| 2.7.4    | PRBS31 transmit test-pattern enable | 1 = Enable PRBS31 test-pattern mode on the transmit path<br>0 = Disable PRBS31 test-pattern mode on the transmit path | R/W              |
| 2.7.3    | Test-pattern selection              | 1 = Select square wave test pattern<br>0 = Select mixed-frequency test pattern                                        | R/W              |
| 2.7.2    | Receive test-pattern enable         | 1 = Enable test-pattern mode on the receive path<br>0 = Disable test-pattern mode on the receive path                 | R/W              |
| 2.7.1    | Transmit test-pattern enable        | 1 = Enable test-pattern mode on the transmit path<br>0 = Disable test-pattern mode on the transmit path               | R/W              |
| 2.7.0    | PCS type selection                  | 1 = Select 10GBASE-W PCS type<br>0 = Select 10GBASE-R PCS type                                                        | R/W              |

<sup>a</sup>RO = Read only, R/W = Read/Write

#### **45.2.2.6.1 PRBS31 receive test-pattern enable (2.7.5)**

If the WIS supports the optional PRBS31 (see 49.2.8) pattern testing advertised in bit 2.8.1 and the mandatory receive test-pattern enable bit (2.7.2) is not one, setting bit 2.7.5 to a one shall set the receive path of the WIS into the PRBS31 test-pattern mode. Setting bit 2.7.5 to a zero shall disable the PRBS31 test-pattern mode on the receive path of the WIS. The behavior of the WIS when in PRBS31 test-pattern mode is specified in 50.3.8.2.

#### **45.2.2.6.2 PRBS31 transmit test-pattern enable (2.7.4)**

If the WIS supports the optional PRBS31 pattern testing advertised in bit 2.8.1 and the mandatory transmit test-pattern enable bit (2.7.1) is not one, then setting bit 2.7.4 to a one shall set the transmit path of the WIS into the PRBS31 test-pattern mode. Setting bit 2.7.4 to a zero shall disable the PRBS31 test-pattern mode on the transmit path of the WIS. The behavior of the WIS when in PRBS31 test-pattern mode is specified in 50.3.8.2.

#### **45.2.2.6.3 Test-pattern selection (2.7.3)**

Bit 2.7.3 controls the type of pattern sent by the transmitter when in test-pattern mode. Setting bit 2.7.3 to a one shall select the square wave test pattern. Setting bit 2.7.3 to a zero shall select the mixed-frequency test pattern. The details of the test patterns are specified in Clause 50.

#### **45.2.2.6.4 Receive test-pattern enable (2.7.2)**

Setting bit 2.7.2 to a one shall set the receive path of the WIS into the test-pattern mode. Setting bit 2.7.2 to a zero shall disable the test-pattern mode on the receive path of the WIS. The behavior of the WIS when in test-pattern mode is specified in Clause 50.

#### 45.2.2.6.5 Transmit test-pattern enable (2.7.1)

Setting bit 2.7.1 to a one shall set the transmit path of the WIS into the test-pattern mode. Setting bit 2.7.1 to a zero shall disable the test-pattern mode on the transmit path of the WIS. The behavior of the WIS when in test-pattern mode is specified in Clause 50.

#### 45.2.2.6.6 PCS type selection (2.7.0)

Setting bit 2.7.0 to a one shall enable the 10GBASE-W logic and set the speed of the WIS-PMA interface to 9.95328 Gb/s. Setting bit 2.7.0 to a zero shall disable the 10GBASE-W logic, set the speed of the PCS-PMA interface to 10.3125 Gb/s and bypass the data around the 10GBASE-W logic. A WIS that is only capable of supporting 10GBASE-W operation and is unable to support 10GBASE-R operation shall ignore values written to this bit and shall return a value of one when read. It is the responsibility of the STA entity to ensure that mutually acceptable MMD types are applied consistently across all the MMDs on a particular PHY.

#### 45.2.2.7 10G WIS status 2 register (Register 2.8)

The assignment of bits in the 10G WIS status 2 register is shown in Table 45–218. All the bits in the 10G WIS status 2 register are read only; a write to the 10G WIS status 2 register shall have no effect.

**Table 45–218—10G WIS status 2 register bit definitions**

| Bit(s)    | Name                           | Description                                                                                                                                                                                | R/W <sup>a</sup> |
|-----------|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 2.8.15:14 | Device present                 | 15 14<br>1 0 = Device responding at this address<br>1 1 = No device responding at this address<br>0 1 = No device responding at this address<br>0 0 = No device responding at this address | RO               |
| 2.8.13:2  | Reserved                       | Value always 0                                                                                                                                                                             | RO               |
| 2.8.1     | PRBS31 pattern testing ability | 1 = WIS is able to support PRBS31 pattern testing<br>0 = WIS is not able to support PRBS31 pattern testing                                                                                 | RO               |
| 2.8.0     | 10GBASE-R ability              | 1 = WIS is able to support 10GBASE-R port types<br>0 = WIS is not able to support 10GBASE-R port types                                                                                     | RO               |

<sup>a</sup>RO = Read only

##### 45.2.2.7.1 Device present (2.8.15:14)

When read as <10>, bits 2.8.15:14 indicate that a device is present and responding at this register address. When read as anything other than <10>, bits 2.8.15:14 indicate that no device is present at this address or that the device is not functioning properly.

##### 45.2.2.7.2 PRBS31 pattern testing ability (2.8.1)

When read as a one, bit 2.8.1 indicates that the WIS is able to support PRBS31 pattern testing. When read as a zero, bit 2.8.1 indicates that the WIS is not able to support PRBS31 pattern testing. If the WIS is able to support PRBS31 pattern testing, then the pattern generation and checking is controlled using bits 2.7.5:4.

#### 45.2.2.7.3 10GBASE-R ability (2.8.0)

When read as a one, bit 2.8.0 indicates that the WIS is able to bypass the WIS logic and adjust the XSBI interface speed to support 10GBASE-R port types. When read as a zero, bit 2.8.0 indicates that the WIS is not able to bypass the WIS logic and cannot support 10GBASE-R port types.

#### 45.2.2.8 10G WIS test-pattern error counter register (Register 2.9)

The assignment of bits in the 10G WIS test-pattern error counter register is shown in Table 45–219. This register is only required when the PRBS31 pattern generation capability is supported.

**Table 45–219—10G WIS test-pattern error counter register bit definitions**

| Bit(s)   | Name                       | Description   | R/W <sup>a</sup> |
|----------|----------------------------|---------------|------------------|
| 2.9.15:0 | Test-pattern error counter | Error counter | RO               |

<sup>a</sup>RO = Read only

The test-pattern error counter is a sixteen bit counter that contains the number of errors received during a pattern test. These bits shall be reset to all zeros when the test-pattern error counter is read by the management function or upon execution of the WIS reset. These bits shall be held at all ones in the case of overflow. The test-pattern methodology is described in 49.2.8.

#### 45.2.2.9 WIS package identifier (Registers 2.14 and 2.15)

Registers 2.14 and 2.15 provide a 32-bit value, which may constitute a unique identifier for a particular type of package that the WIS is instantiated within. The identifier shall be composed of the 3rd through 24th bits of the Organizationally Unique Identifier (OUI) assigned to the package manufacturer by the IEEE, plus a six-bit model number, plus a four-bit revision number. A WIS may return a value of zero in each of the 32 bits of the WIS package identifier.

A non-zero package identifier may be returned by one or more MMDs in the same package. The package identifier may be the same as the device identifier.

The format of the WIS package identifier is specified in 22.2.4.3.1.

#### 45.2.2.10 10G WIS status 3 register (Register 2.33)

The assignment of bits in the 10G WIS status 3 register is shown in Table 45–220. All the bits in the 10G WIS status 3 register are read only; a write to the 10G WIS status 3 register shall have no effect.

##### 45.2.2.10.1 SEF (2.33.11)

When read as a one, bit 2.33.11 indicates that the SEF flag has been raised by the WIS. When read as a zero, bit 2.33.11 indicates that the SEF flag is lowered. The SEF bit shall be implemented with latching high behavior.

The SEF functionality implemented by the WIS is described in 50.3.2.5.

**Table 45–220—10G WIS status 3 register bit definitions**

| Bit(s)     | Name                | Description                                                                                                                             | R/W <sup>a</sup> |
|------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 2.33.15:12 | Reserved            | Value always 0                                                                                                                          | RO               |
| 2.33.11    | SEF                 | Severely errored frame                                                                                                                  | RO/LH            |
| 2.33.10    | Far end PLM-P/LCD-P | 1 = Far end path label mismatch / Loss of code-group delineation<br>0 = No far end path label mismatch / Loss of code-group delineation | RO/LH            |
| 2.33.9     | Far end AIS-P/LOP-P | 1 = Far end path alarm indication signal / Path loss of pointer<br>0 = No far end path alarm indication signal / Path loss of pointer   | RO/LH            |
| 2.33.8     | Reserved            | Value always 0                                                                                                                          | RO               |
| 2.33.7     | LOF                 | 1 = Loss of frame flag raised<br>0 = Loss of frame flag lowered                                                                         | RO/LH            |
| 2.33.6     | LOS                 | 1 = Loss of signal flag raised<br>0 = Loss of signal flag lowered                                                                       | RO/LH            |
| 2.33.5     | RDI-L               | 1 = Line remote defect flag raised<br>0 = Line remote defect flag lowered                                                               | RO/LH            |
| 2.33.4     | AIS-L               | 1 = Line alarm indication flag raised<br>0 = Line alarm indication flag lowered                                                         | RO/LH            |
| 2.33.3     | LCD-P               | 1 = Path loss of code-group delineation flag raised<br>0 = Path loss of code-group delineation flag lowered                             | RO/LH            |
| 2.33.2     | PLM-P               | 1 = Path label mismatch flag raised<br>0 = Path label mismatch flag lowered                                                             | RO/LH            |
| 2.33.1     | AIS-P               | 1 = Path alarm indication signal raised<br>0 = Path alarm indication signal lowered                                                     | RO/LH            |
| 2.33.0     | LOP-P               | 1 = Loss of pointer flag raised<br>0 = Loss of pointer flag lowered                                                                     | RO/LH            |

<sup>a</sup>RO = Read only, LH = Latching high

#### 45.2.2.10.2 Far end PLM-P/LCD-P (2.33.10)

When read as a one, bit 2.33.10 indicates that the far end path label mismatch/loss of code-group delineation flag has been raised. When read as a zero, bit 2.33.10 indicates that the far end path label mismatch/loss of code-group delineation flag is lowered. The far end PLM-P/LCD-P bit shall be implemented with latching high behavior.

The far end path label mismatch/loss of code-group delineation functionality implemented by the WIS is described in 50.3.2.5.

#### **45.2.2.10.3 Far end AIS-P/LOP-P (2.33.9)**

When read as a one, bit 2.33.9 indicates that the far end path alarm indication signal/path loss of pointer flag has been raised by the WIS. When read as a zero, bit 2.33.9 indicates that the far end path alarm indication signal/path loss of pointer flag is lowered. The far end AIS-P/LOP-P bit shall be implemented with latching high behavior.

The far end path alarm indication signal/path loss of pointer functionality implemented by the WIS is described in 50.3.2.5.

#### **45.2.2.10.4 LOF (2.33.7)**

When read as a one, bit 2.33.7 indicates that the loss of frame flag has been raised. When read as a zero, bit 2.33.7 indicates that the loss of frame flag is lowered. The LOF bit shall be implemented with latching high behavior.

The LOF functionality implemented by the WIS is described in 50.3.2.5.

#### **45.2.2.10.5 LOS (2.33.6)**

When read as a one, bit 2.33.6 indicates that the loss of signal flag has been raised. When read as a zero, bit 2.33.6 indicates that the loss of signal flag is lowered. The LOS bit shall be implemented with latching high behavior.

The LOS functionality implemented by the WIS is described in 50.3.2.5.

#### **45.2.2.10.6 RDI-L (2.33.5)**

When read as a one, bit 2.33.5 indicates that the line remote defect flag has been raised. When read as a zero, bit 2.33.5 indicates that the line remote defect flag is lowered. The RDI-L bit shall be implemented with latching high behavior.

The RDI-L functionality implemented by the WIS is described in 50.3.2.5.

#### **45.2.2.10.7 AIS-L (2.33.4)**

When read as a one, bit 2.33.4 indicates that the line alarm indication flag has been raised. When read as a zero, bit 2.33.4 indicates that the line alarm indication flag is lowered. The AIS-L bit shall be implemented with latching high behavior.

The AIS-L functionality implemented by the WIS is described in 50.3.2.5.

#### **45.2.2.10.8 LCD-P (2.33.3)**

When read as a one, bit 2.33.3 indicates that the loss of code-group delineation flag has been raised. When read as a zero, bit 2.33.3 indicates that the loss of code-group delineation flag is lowered. The LCD-P bit shall be implemented with latching high behavior.

The loss of code-group delineation functionality implemented by the WIS is described in 50.3.2.5.

#### 45.2.2.10.9 PLM-P (2.33.2)

When read as a one, bit 2.33.2 indicates that the path label mismatch flag has been raised. When read as a zero, bit 2.33.2 indicates that the path label mismatch flag is lowered. The PLM-P bit shall be implemented with latching high behavior.

The PLM-P functionality implemented by the WIS is described in 50.3.2.5.

#### 45.2.2.10.10 AIS-P (2.33.1)

When read as a one, bit 2.33.1 indicates that the path alarm indication signal has been raised. When read as a zero, bit 2.33.1 indicates that the path alarm indication signal is lowered. The AIS-P bit shall be implemented with latching high behavior.

The path alarm indication signal functionality implemented by the WIS is described in 50.3.2.5.

#### 45.2.2.10.11 LOP-P (2.33.0)

When read as a one, bit 2.33.0 indicates that the loss of pointer flag has been raised. When read as a zero, bit 2.33.0 indicates that the loss of pointer flag is lowered. The LOP-P bit shall be implemented with latching high behavior.

The LOP-P functionality implemented by the WIS is described in 50.3.2.5.

#### 45.2.2.11 10G WIS far end path block error count (Register 2.37)

The assignment of bits in the 10G WIS far end path block error count register is shown in Table 45–221.

**Table 45–221—10G WIS far end path block error count register bit definitions**

| Bit(s)    | Name                           | Description                    | R/W <sup>a</sup> |
|-----------|--------------------------------|--------------------------------|------------------|
| 2.37.15:0 | Far end path block error count | Far end path block error count | RO               |

<sup>a</sup>RO = Read only,

The 10G WIS far end path block error count is incremented by one whenever a far end path block error, defined in Annex 50A, is detected as described in 50.3.2.5. The counter wraps around to zero when it is incremented beyond its maximum value of 65 535. It is cleared to zero when the WIS is reset.

#### 45.2.2.12 10G WIS J1 transmit (Registers 2.39 through 2.46)

The assignment of octets in the 10G WIS J1 transmit registers is shown in Table 45–222.

The first transmitted path trace octet is J1 transmit 15, which contains the delineation octet. The default value for the J1 transmit 15 octet is 137 (hexadecimal 89). The last transmitted path trace octet is J1 transmit 0. The default value for the J1 transmit 0 through 14 octets is 0. The transmitted path trace is described in 50.3.2.1.



**Table 45–222—10G WIS J1 transmit 0–15 register bit definitions**

| Bit(s)    | Name           | Description                     | R/W <sup>a</sup> |
|-----------|----------------|---------------------------------|------------------|
| 2.46.15:8 | J1 transmit 15 | Transmitted path trace octet 15 | R/W              |
| 2.46.7:0  | J1 transmit 14 | Transmitted path trace octet 14 | R/W              |
| 2.45.15:8 | J1 transmit 13 | Transmitted path trace octet 13 | R/W              |
| 2.45.7:0  | J1 transmit 12 | Transmitted path trace octet 12 | R/W              |
| 2.44.15:8 | J1 transmit 11 | Transmitted path trace octet 11 | R/W              |
| 2.44.7:0  | J1 transmit 10 | Transmitted path trace octet 10 | R/W              |
| 2.43.15:8 | J1 transmit 9  | Transmitted path trace octet 9  | R/W              |
| 2.43.7:0  | J1 transmit 8  | Transmitted path trace octet 8  | R/W              |
| 2.42.15:8 | J1 transmit 7  | Transmitted path trace octet 7  | R/W              |
| 2.42.7:0  | J1 transmit 6  | Transmitted path trace octet 6  | R/W              |
| 2.41.15:8 | J1 transmit 5  | Transmitted path trace octet 5  | R/W              |
| 2.41.7:0  | J1 transmit 4  | Transmitted path trace octet 4  | R/W              |
| 2.40.15:8 | J1 transmit 3  | Transmitted path trace octet 3  | R/W              |
| 2.40.7:0  | J1 transmit 2  | Transmitted path trace octet 2  | R/W              |
| 2.39.15:8 | J1 transmit 1  | Transmitted path trace octet 1  | R/W              |
| 2.39.7:0  | J1 transmit 0  | Transmitted path trace octet 0  | R/W              |

<sup>a</sup>R/W = Read/Write

#### 45.2.2.13 10G WIS J1 receive (Registers 2.47 through 2.54)

The assignment of octets in the 10G WIS J1 receive registers is shown in Table 45–223.

**Table 45–223—10G WIS J1 receive 0–15 register bit definitions**

| Bit(s)    | Name          | Description                  | R/W <sup>a</sup> |
|-----------|---------------|------------------------------|------------------|
| 2.54.15:8 | J1 receive 15 | Received path trace octet 15 | RO               |
| 2.54.7:0  | J1 receive 14 | Received path trace octet 14 | RO               |
| 2.53.15:8 | J1 receive 13 | Received path trace octet 13 | RO               |
| 2.53.7:0  | J1 receive 12 | Received path trace octet 12 | RO               |
| 2.52.15:8 | J1 receive 11 | Received path trace octet 11 | RO               |

**Table 45–223—10G WIS J1 receive 0–15 register bit definitions (*continued*)**

| Bit(s)    | Name          | Description                  | R/W <sup>a</sup> |
|-----------|---------------|------------------------------|------------------|
| 2.52.7:0  | J1 receive 10 | Received path trace octet 10 | RO               |
| 2.51.15:8 | J1 receive 9  | Received path trace octet 9  | RO               |
| 2.51.7:0  | J1 receive 8  | Received path trace octet 8  | RO               |
| 2.50.15:8 | J1 receive 7  | Received path trace octet 7  | RO               |
| 2.50.7:0  | J1 receive 6  | Received path trace octet 6  | RO               |
| 2.49.15:8 | J1 receive 5  | Received path trace octet 5  | RO               |
| 2.49.7:0  | J1 receive 4  | Received path trace octet 4  | RO               |
| 2.48.15:8 | J1 receive 3  | Received path trace octet 3  | RO               |
| 2.48.7:0  | J1 receive 2  | Received path trace octet 2  | RO               |
| 2.47.15:8 | J1 receive 1  | Received path trace octet 1  | RO               |
| 2.47.7:0  | J1 receive 0  | Received path trace octet 0  | RO               |

<sup>a</sup>RO = Read only

The first received path trace octet is J1 receive 15. The last received path trace octet is J1 receive 0. The received path trace is described in 50.3.2.4.

#### 45.2.2.14 10G WIS far end line BIP errors (Registers 2.55 and 2.56)

The assignment of octets in the 10G WIS far end line BIP errors registers is shown in Table 45–224.

**Table 45–224—10G WIS far end line BIP errors 0–1 register bit definitions**

| Bit(s)    | Name                          | Description                                                       | R/W <sup>a</sup> |
|-----------|-------------------------------|-------------------------------------------------------------------|------------------|
| 2.56.15:0 | WIS far end line BIP errors 0 | Least significant word of the WIS far end line BIP errors counter | RO               |
| 2.55.15:0 | WIS far end line BIP errors 1 | Most significant word of the WIS far end line BIP errors counter  | RO               |

<sup>a</sup>RO = Read only

The 10G WIS far end line BIP Errors register pair reflects the contents of the far end line BIP errors counter (as described in 50.3.11.3) that is incremented on each WIS frame by the number of far end line BIP errors reported by the far end, as described in 50.3.2.5. Whenever the most significant 16 bit register of the counter (2.55) is read, the 32 bit counter value is latched into the register pair, with the most significant bits appearing in 2.55 and the least significant 16 bits appearing in 2.56, the value being latched before the contents of 2.55 (the most significant 16 bits) are driven on the MDIO interface. A subsequent read from register 2.56 will return the least significant 16 bits of the latched value, but will not change the register contents. Writes to these registers have no effect.

NOTE—These counters do not follow the behavior described in 45.2 for 32-bit counters.

#### 45.2.2.15 10G WIS line BIP errors (Registers 2.57 and 2.58)

The assignment of octets in the 10G WIS line BIP errors registers is shown in Table 45–225.

**Table 45–225—10G WIS line BIP errors 0–1 register bit definitions**

| Bit(s)    | Name                  | Description                                               | R/W <sup>a</sup> |
|-----------|-----------------------|-----------------------------------------------------------|------------------|
| 2.58.15:0 | WIS line BIP errors 0 | Least significant word of the WIS line BIP errors counter | RO               |
| 2.57.15:0 | WIS line BIP errors 1 | Most significant word of the WIS line BIP errors counter  | RO               |

<sup>a</sup>RO = Read only

The 10G WIS line BIP errors register pair reflects the contents of the line BIP errors counter (as described in 50.3.11.3) that is incremented on each WIS frame by the number of line BIP errors detected on the incoming data stream, as described in 50.3.2.5. Whenever the most significant 16 bit register of the counter (2.57) is read, the 32 bit counter value is latched into the register pair, with the most significant bits appearing in 2.57 and the least significant 16 bits appearing in 2.58, the value being latched before the contents of 2.57 (the most significant 16 bits) are driven on the MDIO interface. A subsequent read from register 2.58 will return the least significant 16 bits of the latched value, but will not change the register contents. Writes to these registers have no effect.

NOTE—These counters do not follow the behavior described in 45.2 for 32-bit counters.

#### 45.2.2.16 10G WIS path block error count (Register 2.59)

The assignment of bits in the 10G WIS path block error count register is shown in Table 45–226.

**Table 45–226—10G WIS path block error count register bit definitions**

| Bit(s)    | Name                   | Description              | R/W <sup>a</sup> |
|-----------|------------------------|--------------------------|------------------|
| 2.59.15:0 | Path block error count | Path block error counter | RO               |

<sup>a</sup>RO = Read only

##### 45.2.2.16.1 Path block error count (2.59.15:0)

The path block error count is incremented by one whenever a B3 parity error (defined in Annex 50A) is detected, as described in 50.3.2.5. The counter wraps around to zero when it is incremented beyond its maximum value of 65 535. It is cleared to zero when the WIS is reset.

#### 45.2.2.17 10G WIS section BIP error count (Register 2.60)

The assignment of bits in the 10G WIS section BIP error count register is shown in Table 45–227.

**Table 45–227—10G WIS section BIP error count register bit definitions**

| Bit(s)    | Name                    | Description             | R/W <sup>a</sup> |
|-----------|-------------------------|-------------------------|------------------|
| 2.60.15:0 | Section BIP error count | Section BIP error count | RO               |

<sup>a</sup>RO = Read only

#### 45.2.2.17.1 Section BIP error count (2.60.15:0)

The section BIP error count is incremented by the number of section BIP errors detected within each WIS frame, as described in 50.3.2.5. The counter wraps around to zero when it is incremented beyond its maximum value of 65 535. It is cleared to zero when the WIS is reset.

#### 45.2.2.18 10G WIS J0 transmit (Registers 2.64 through 2.71)

The assignment of octets in the 10G WIS J0 transmit registers is shown in Table 45–228.

**Table 45–228—10G WIS J0 transmit 0–15 register bit definitions**

| Bit(s)    | Name           | Description                        | R/W <sup>a</sup> |
|-----------|----------------|------------------------------------|------------------|
| 2.71.15:8 | J0 transmit 15 | Transmitted section trace octet 15 | R/W              |
| 2.71.7:0  | J0 transmit 14 | Transmitted section trace octet 14 | R/W              |
| 2.70.15:8 | J0 transmit 13 | Transmitted section trace octet 13 | R/W              |
| 2.70.7:0  | J0 transmit 12 | Transmitted section trace octet 12 | R/W              |
| 2.69.15:8 | J0 transmit 11 | Transmitted section trace octet 11 | R/W              |
| 2.69.7:0  | J0 transmit 10 | Transmitted section trace octet 10 | R/W              |
| 2.68.15:8 | J0 transmit 9  | Transmitted section trace octet 9  | R/W              |
| 2.68.7:0  | J0 transmit 8  | Transmitted section trace octet 8  | R/W              |
| 2.67.15:8 | J0 transmit 7  | Transmitted section trace octet 7  | R/W              |
| 2.67.7:0  | J0 transmit 6  | Transmitted section trace octet 6  | R/W              |
| 2.66.15:8 | J0 transmit 5  | Transmitted section trace octet 5  | R/W              |
| 2.66.7:0  | J0 transmit 4  | Transmitted section trace octet 4  | R/W              |
| 2.65.15:8 | J0 transmit 3  | Transmitted section trace octet 3  | R/W              |
| 2.65.7:0  | J0 transmit 2  | Transmitted section trace octet 2  | R/W              |
| 2.64.15:8 | J0 transmit 1  | Transmitted section trace octet 1  | R/W              |
| 2.64.7:0  | J0 transmit 0  | Transmitted section trace octet 0  | R/W              |

<sup>a</sup>R/W = Read/Write

The J0 transmit octets allow a receiver to verify its continued connection to the WIS transmitter. The first transmitted section trace octet is J0 transmit 15, which contains the delineation octet. The default value for the J0 transmit 15 octet is 137 (hexadecimal 89). The last transmitted section trace octet is J0 transmit 0. The default value for the J0 transmit 0 through 14 octets is 0. The transmitted section trace is described in 50.3.2.3.

#### 45.2.2.19 10G WIS J0 receive (Registers 2.72 through 2.79)

The assignment of octets in the 10G WIS J0 receive registers is shown in Table 45–229.

**Table 45–229— 10G WIS J0 receive 0–15 register bit definitions**

| Bit(s)    | Name          | Description                     | R/W <sup>a</sup> |
|-----------|---------------|---------------------------------|------------------|
| 2.79.15:8 | J0 receive 15 | Received section trace octet 15 | RO               |
| 2.79.7:0  | J0 receive 14 | Received section trace octet 14 | RO               |
| 2.78.15:8 | J0 receive 13 | Received section trace octet 13 | RO               |
| 2.78.7:0  | J0 receive 12 | Received section trace octet 12 | RO               |
| 2.77.15:8 | J0 receive 11 | Received section trace octet 11 | RO               |
| 2.77.7:0  | J0 receive 10 | Received section trace octet 10 | RO               |
| 2.76.15:8 | J0 receive 9  | Received section trace octet 9  | RO               |
| 2.76.7:0  | J0 receive 8  | Received section trace octet 8  | RO               |
| 2.75.15:8 | J0 receive 7  | Received section trace octet 7  | RO               |
| 2.75.7:0  | J0 receive 6  | Received section trace octet 6  | RO               |
| 2.74.15:8 | J0 receive 5  | Received section trace octet 5  | RO               |
| 2.74.7:0  | J0 receive 4  | Received section trace octet 4  | RO               |
| 2.73.15:8 | J0 receive 3  | Received section trace octet 3  | RO               |
| 2.73.7:0  | J0 receive 2  | Received section trace octet 2  | RO               |
| 2.72.15:8 | J0 receive 1  | Received section trace octet 1  | RO               |
| 2.72.7:0  | J0 receive 0  | Received section trace octet 0  | RO               |

<sup>a</sup>RO = Read only

The first received section trace octet is J0 receive 15. The last received section trace octet is J0 receive 0. The J0 receive octets allow a WIS receiver to verify its continued connection to the intended transmitter. The received section trace is described in 50.3.2.4.

#### 45.2.2.20 TimeSync WIS capability (Register 2.1800)

The TimeSync WIS capability register (see Table 45–230) indicates the capability of the WIS to report the transmit and receive data delay, stored in registers 2.1801 through 2.1804 and 2.1805 through 2.1808, respectively.

**Table 45–230—TimeSync WIS capability**

| Bit(s)      | Name                              | Description                                                                                                                                                     | R/W <sup>a</sup> |
|-------------|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 2.1800.15:2 | Reserved                          | Value always 0                                                                                                                                                  | RO               |
| 2.1800.1    | TimeSync transmit path data delay | 1 = WIS provides information on transmit path data delay in registers 2.1801 through 2.1804<br>0 = WIS does not provide information on transmit path data delay | RO               |
| 2.1800.0    | TimeSync receive path data delay  | 1 = WIS provides information on receive path data delay in registers 2.1805 through 2.1808<br>0 = WIS does not provide information on receive path data delay   | RO               |

<sup>a</sup>RO = Read only

#### 45.2.2.21 TimeSync WIS transmit path data delay (Registers 2.1801, 2.1802, 2.1803, 2.1804)

The TimeSync WIS transmit path data delay register contains the maximum (registers 2.1801, 2.1802, see Table 45–231) and minimum (registers 2.1803, 2.1804, see Table 45–231) values of the transmit path data delay. The transmit path data delay is expressed in units of ns. The values contained in these registers are valid when the link is established, as indicated by bit 2 in register 1.1 (see 45.2.1.2.4).

**Table 45–231—TimeSync WIS transmit path data delay register**

| Bit(s)      | Name                                        | Description              | R/W <sup>a</sup> |
|-------------|---------------------------------------------|--------------------------|------------------|
| 2.1801.15:0 | Maximum WIS transmit path data delay, lower | WIS_delay_TX_max [15:0]  | RO, MW           |
| 2.1802.15:0 | Maximum WIS transmit path data delay, upper | WIS_delay_TX_max [31:16] | RO, MW           |
| 2.1803.15:0 | Minimum WIS transmit path data delay, lower | WIS_delay_TX_min [15:0]  | RO, MW           |
| 2.1804.15:0 | Minimum WIS transmit path data delay, upper | WIS_delay_TX_min [31:16] | RO, MW           |

<sup>a</sup>RO = Read only, MW = Multi-word

#### 45.2.2.22 TimeSync WIS receive path data delay (Registers 2.1805, 2.1806, 2.1807, 2.1808)

The TimeSync WIS receive path data delay register contains the maximum (registers 2.1805, 2.1806, see Table 45–232) and minimum (registers 2.1807, 2.1808, see Table 45–232) values of the receive path data delay. The receive path data delay is expressed in units of ns. The values contained in these registers are valid when the link is established, as indicated by bit 2 in register 2.1 (see 45.2.2.2.2).

**Table 45–232—TimeSync WIS receive path data delay register**

| Bit(s)      | Name                                       | Description              | R/W <sup>a</sup> |
|-------------|--------------------------------------------|--------------------------|------------------|
| 2.1805.15:0 | Maximum WIS receive path data delay, lower | WIS_delay_RX_max [15:0]  | RO, MW           |
| 2.1806.15:0 | Maximum WIS receive path data delay, upper | WIS_delay_RX_max [31:16] | RO, MW           |
| 2.1807.15:0 | Minimum WIS receive path data delay, lower | WIS_delay_RX_min [15:0]  | RO, MW           |
| 2.1808.15:0 | Minimum WIS receive path data delay, upper | WIS_delay_RX_min [31:16] | RO, MW           |

<sup>a</sup>RO = Read only, MW = Multi-word

### 45.2.3 PCS registers

The assignment of registers in the PCS is shown in Table 45–233.

**Table 45–233—PCS registers**

| Register address  | Register name                          | Subclause |
|-------------------|----------------------------------------|-----------|
| 3.0               | PCS control 1                          | 45.2.3.1  |
| 3.1               | PCS status 1                           | 45.2.3.2  |
| 3.2, 3.3          | PCS device identifier                  | 45.2.3.3  |
| 3.4               | PCS speed ability                      | 45.2.3.4  |
| 3.5, 3.6          | PCS devices in package                 | 45.2.3.5  |
| 3.7               | PCS control 2                          | 45.2.3.6  |
| 3.8               | PCS status 2                           | 45.2.3.7  |
| 3.9               | PCS status 3                           | 45.2.3.8  |
| 3.10 through 3.13 | Reserved                               |           |
| 3.14, 3.15        | PCS package identifier                 | 45.2.3.9  |
| 3.16 through 3.19 | Reserved                               |           |
| 3.20              | EEE control and capability 1           | 45.2.3.10 |
| 3.21              | EEE control and capability 2           | 45.2.3.11 |
| 3.22              | EEE wake error counter                 | 45.2.3.12 |
| 3.23              | Reserved                               |           |
| 3.24              | 10GBASE-X PCS status                   | 45.2.3.13 |
| 3.25              | 10GBASE-X PCS test control             | 45.2.3.14 |
| 3.26 through 3.31 | Reserved                               |           |
| 3.32              | BASE-R and MultiGBASE-T PCS status 1   | 45.2.3.15 |
| 3.33              | BASE-R and MultiGBASE-T PCS status 2   | 45.2.3.16 |
| 3.34 through 3.37 | 5/10/25GBASE-R PCS test pattern seed A | 45.2.3.17 |
| 3.38 through 3.41 | 5/10/25GBASE-R PCS test pattern seed B | 45.2.3.18 |
| 3.42              | BASE-R PCS test-pattern control        | 45.2.3.19 |
| 3.43              | BASE-R PCS test-pattern error counter  | 45.2.3.20 |
| 3.44              | BER high order counter                 | 45.2.3.21 |
| 3.45              | Error blocks high order counter        | 45.2.3.22 |

**Table 45–233—PCS registers (*continued*)**

| Register address    | Register name                                                           | Subclause |
|---------------------|-------------------------------------------------------------------------|-----------|
| 3.46 through 3.49   | Reserved                                                                |           |
| 3.50 through 3.53   | Multi-lane BASE-R PCS alignment status 1 through 4                      | 45.2.3.23 |
| 3.54 through 3.59   | Reserved                                                                |           |
| 3.60                | 10P/2B capability                                                       | 45.2.3.27 |
| 3.61                | 10P/2B PCS control                                                      | 45.2.3.28 |
| 3.62, 3.63          | 10P/2B PME available                                                    | 45.2.3.29 |
| 3.64, 3.65          | 10P/2B PME aggregate                                                    | 45.2.3.30 |
| 3.66                | 10P/2B PAF RX error counter                                             | 45.2.3.31 |
| 3.67                | 10P/2B PAF small fragment counter                                       | 45.2.3.32 |
| 3.68                | 10P/2B PAF large fragments counter                                      | 45.2.3.33 |
| 3.69                | 10P/2B PAF overflow counter                                             | 45.2.3.34 |
| 3.70                | 10P/2B PAF bad fragments counter                                        | 45.2.3.35 |
| 3.71                | 10P/2B PAF lost fragments counter                                       | 45.2.3.36 |
| 3.72                | 10P/2B PAF lost starts of fragments counter                             | 45.2.3.37 |
| 3.73                | 10P/2B PAF lost ends of fragments counter                               | 45.2.3.38 |
| 3.74                | 10GBASE-PR and 10/1GBASE-PRX FEC ability                                | 45.2.3.39 |
| 3.75                | 10GBASE-PR and 10/1GBASE-PRX FEC control                                | 45.2.3.40 |
| 3.76, 3.77          | 10G-EPON and Nx25G-EPON corrected FEC codewords counter                 | 45.2.3.41 |
| 3.78, 3.79          | 10G-EPON and Nx25G-EPON uncorrected FEC codewords counter               | 45.2.3.42 |
| 3.80                | 10GBASE-PR, 10/1GBASE-PRX, and Nx25G-EPON BER monitor interval control  | 45.2.3.43 |
| 3.81                | 10GBASE-PR, 10/1GBASE-PRX, and Nx25G-EPON BER monitor status            | 45.2.3.44 |
| 3.82                | 10GBASE-PR, 10/1GBASE-PRX, and Nx25G-EPON BER monitor threshold control | 45.2.3.45 |
| 3.83 through 3.134  | Nx25G-EPON synchronization pattern                                      | 45.2.3.46 |
| 3.135 through 3.199 | Reserved                                                                |           |
| 3.200 through 3.219 | BIP error counters, lanes 0 through 19                                  | 45.2.3.47 |
| 3.220 through 3.399 | Reserved                                                                |           |
| 3.400 through 3.419 | PCS lane mapping registers, lanes 0 through 19                          | 45.2.3.49 |
| 3.420 through 3.499 | Reserved                                                                |           |
| 3.500               | 1000BASE-H OAM transmit control                                         | 45.2.3.51 |
| 3.501 through 3.508 | 1000BASE-H OAM transmit message                                         | 45.2.3.51 |
| 3.509               | 1000BASE-H OAM receive control                                          | 45.2.3.52 |
| 3.510 through 3.517 | 1000BASE-H OAM receive message                                          | 45.2.3.52 |
| 3.518               | 1000BASE-H PCS control                                                  | 45.2.3.53 |
| 3.519               | 1000BASE-H PCS status 1                                                 | 45.2.3.54 |
| 3.520               | 1000BASE-H PCS status 2                                                 | 45.2.3.55 |
| 3.521               | 1000BASE-H PCS status 3                                                 | 45.2.3.56 |
| 3.522               | 1000BASE-H PCS status 4                                                 | 45.2.3.57 |
| 3.523 through 3.599 | Reserved                                                                |           |



**Table 45–233—PCS registers (*continued*)**

| Register address          | Register name                                 | Subclause               |
|---------------------------|-----------------------------------------------|-------------------------|
| 3.600 through 3.631       | PCS FEC symbol error counter, lane 0 to 15    | 45.2.3.58,<br>45.2.3.59 |
| 3.632 through 3.799       | Reserved                                      |                         |
| 3.800                     | PCS FEC control                               | 45.2.3.60               |
| 3.801                     | PCS FEC status                                | 45.2.3.61               |
| 3.802, 3.803              | PCS FEC corrected codewords counter           | 45.2.3.62               |
| 3.804, 3.805              | PCS FEC uncorrected codewords counter         | 45.2.3.63               |
| 3.806, 3.807              | PCS FEC degraded SER activate threshold       | 45.2.3.64               |
| 3.808, 3.809              | PCS FEC degraded SER deactivate threshold     | 45.2.3.65               |
| 3.810, 3.811              | PCS FEC degraded SER interval                 | 45.2.3.66               |
| 3.812 through 3.1799      | Reserved                                      |                         |
| 3.1800                    | TimeSync PCS capability                       | 45.2.3.67               |
| 3.1801 through 3.1804     | TimeSync PCS transmit path data delay         | 45.2.3.68               |
| 3.1805 through 3.1808     | TimeSync PCS receive path data delay          | 45.2.3.69               |
| 3.1809 through 3.2277     | Reserved                                      |                         |
| 3.2278                    | 10BASE-T1L PCS control                        | 45.2.3.70               |
| 3.2279                    | 10BASE-T1L PCS status                         | 45.2.3.71               |
| 3.2280 through 3.2290     | Reserved                                      |                         |
| 3.2291                    | 10BASE-T1S PCS control                        | 45.2.3.72               |
| 3.2292                    | 10BASE-T1S PCS status                         | 45.2.3.73               |
| 3.2293                    | 10BASE-T1S PCS diagnostic 1                   | 45.2.3.74               |
| 3.2294                    | 10BASE-T1S PCS diagnostic 2                   | 45.2.3.75               |
| 3.2295 through 3.2303     | Reserved                                      |                         |
| 3.2304                    | 1000BASE-T1 PCS control                       | 45.2.3.76               |
| 3.2305                    | 1000BASE-T1 PCS status 1                      | 45.2.3.77               |
| 3.2306                    | 1000BASE-T1 PCS status 2                      | 45.2.3.78               |
| 3.2307                    | Reserved                                      |                         |
| 3.2308                    | BASE-T1 OAM transmit                          | 45.2.3.79               |
| 3.2309 through 3.2312     | BASE-T1 OAM message                           | 45.2.3.80               |
| 3.2313                    | BASE-T1 OAM receive                           | 45.2.3.81               |
| 3.2314 through 3.2317     | Link partner BASE-T1 OAM message              | 45.2.3.82               |
| 3.2318 through 3.2319     | MultiGBASE-T1 OAM status message              | 45.2.3.83               |
| 3.2320 through 3.2321     | Link partner MultiGBASE-T1 OAM status message | 45.2.3.84               |
| 3.2322                    | MultiGBASE-T1 PCS control                     | 45.2.3.85               |
| 3.2323                    | MultiGBASE-T1 PCS status 1                    | 45.2.3.86               |
| 3.2324                    | MultiGBASE-T1 PCS status 2                    | 45.2.3.87               |
| 3.2325 through 3.32767    | Reserved                                      |                         |
| 3.32 768 through 3.65 535 | Vendor specific                               |                         |

### 45.2.3.1 PCS control 1 register (Register 3.0)

The assignment of bits in the PCS control 1 register is shown in Table 45–234. The default value for each bit of the PCS control 1 register should be chosen so that the initial state of the device upon power up or reset is a normal operational state without management intervention.

**Table 45–234—PCS control 1 register bit definitions**

| Bit(s)  | Name              | Description                                                                                                                                                                                                                                                                                                  | R/W <sup>a</sup> |
|---------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 3.0.15  | Reset             | 1 = PCS reset<br>0 = Normal operation                                                                                                                                                                                                                                                                        | R/W<br>SC        |
| 3.0.14  | Loopback          | 1 = Enable loopback mode<br>0 = Disable loopback mode                                                                                                                                                                                                                                                        | R/W              |
| 3.0.13  | Speed selection   | 13 6<br>1 1 = bits 5:2 select speed<br>0 x = unspecified<br>x 0 = unspecified                                                                                                                                                                                                                                | R/W              |
| 3.0.12  | Reserved          | Value always 0                                                                                                                                                                                                                                                                                               | RO               |
| 3.0.11  | Low power         | 1 = Low-power mode<br>0 = Normal operation                                                                                                                                                                                                                                                                   | R/W              |
| 3.0.10  | Clock stop enable | 1 = The PHY may stop the clock during LPI<br>0 = Clock not stoppable                                                                                                                                                                                                                                         | R/W              |
| 3.0.9:7 | Reserved          | Value always 0                                                                                                                                                                                                                                                                                               | RO               |
| 3.0.6   | Speed selection   | 13 6<br>1 1 = bits 5:2 select speed<br>0 x = unspecified<br>x 0 = unspecified                                                                                                                                                                                                                                | R/W              |
| 3.0.5:2 | Speed selection   | 5 4 3 2<br>1 1 x x = Reserved<br>1 0 1 1 = 25/10 Gb/s<br>1 0 1 0 = 400 Gb/s<br>1 0 0 1 = 200 Gb/s<br>1 0 0 0 = 5 Gb/s<br>0 1 1 1 = 2.5 Gb/s<br>0 1 1 0 = 50 Gb/s<br>0 1 0 1 = 25 Gb/s<br>0 1 0 0 = 100 Gb/s<br>0 0 1 1 = 40 Gb/s<br>0 0 1 0 = 10/1 Gb/s<br>0 0 0 1 = 10PASS-TS/2BASE-TL<br>0 0 0 0 = 10 Gb/s | R/W              |
| 3.0.1:0 | Reserved          | Value always 0                                                                                                                                                                                                                                                                                               | RO               |

<sup>a</sup>RO = Read only, R/W = Read/Write, SC = Self-clearing

#### 45.2.3.1.1 Reset (3.0.15)

Resetting a PCS is accomplished by setting bit 3.0.15 to a one. This action shall set all PCS registers to their default states. As a consequence, this action may change the internal state of the PCS and the state of the physical link. This action may also initiate a reset in any other MMDs that are instantiated in the same package. This bit is self-clearing, and a PCS shall return a value of one in bit 3.0.15 when a reset is in progress and a value of zero otherwise. A PCS is not required to accept a write transaction to any of its registers until the reset process is completed. The reset process shall be completed within 0.5 s from the setting of bit 3.0.15. During a reset, a PCS shall respond to reads from register bits 3.0.15 and 3.8.15:14. All other register bits should be ignored.

NOTE—This operation may interrupt data communication.

#### 45.2.3.1.2 Loopback (3.0.14)

When the 100BASE-T1, any MultiGBASE-T, or the 5/10GBASE-R mode of operation is selected for the PCS using the PCS type selection field (3.7.3:0), the PCS shall be placed in a loopback mode of operation when bit 3.0.14 is set to a one. When bit 3.0.14 is set to a one, the 100BASE-T1, 5/10GBASE-R, or any PCS in the MultiGBASE-T set shall accept data on the transmit path and return it on the receive path. The speed of the loopback is selected by the PCS control 1 (register 3.0) defined in 45.2.3.1. The specific behavior of the 100BASE-T1 PCS during loopback is specified in 96.3.5. The specific behavior of the 5/10GBASE-R PCS during loopback is specified in 49.2. The specific behavior for the 10GBASE-T PCS during loopback is specified in 55.3.7.3. The specific behavior for the 25GBASE-T and 40GBASE-T PCS during loopback is specified in 113.3.7.3. The specific behavior for the 2.5GBASE-T or 5GBASE-T PCS during loopback is specified in 126.3.7.3. For all other port types, the PCS loopback functionality is not applicable and writes to this bit shall be ignored and reads from this bit shall return a value of zero.

The default value of bit 3.0.14 is zero.

NOTE—The signal path through the PCS that is exercised in the loopback mode of operation is implementation specific, but it is recommended that the signal path encompass as much of the PCS circuitry as is practical. The intention of providing this loopback mode of operation is to permit a diagnostic or self-test function to perform the transmission and reception of a PDU, thus testing the transmit and receive data paths. Other loopback signal paths may be enabled using loopback controls within other MMDs.

#### 45.2.3.1.3 Low power (3.0.11)

A PCS may be placed into a low-power mode by setting bit 3.0.11 to a one. This action may also initiate a low-power mode in any other MMDs that are instantiated in the same package. The low-power mode is exited by resetting the PCS. The behavior of the PCS in transition to and from the low-power mode is implementation specific and any interface signals should not be relied upon. While in the low-power mode, the device shall, as a minimum, respond to management transactions necessary to exit the low-power mode. The default value of bit 3.0.11 is zero.

#### 45.2.3.1.4 Clock stop enable (3.0.10)

If bit 3.0.10 is set to one then the PHY may stop the receive xMII clock while it is signaling LPI otherwise it shall keep the clock active. If the PHY does not support EEE capability or is not able to stop the receive clock then this bit has no effect (see 22.2.2.2, 35.2.2.10, and 46.3.2.4).

#### 45.2.3.1.5 Speed selection (3.0.13, 3.0.6, 3.0.5:2)

Speed selection bits 3.0.13 and 3.0.6 shall both be written as a one. Any attempt to change the bits to an invalid setting shall be ignored. These two bits are set to one in order to make them compatible with Clause 22.

The speed of the PCS may be selected using bits 5 through 2. The speed abilities of the PCS are advertised in the PCS speed ability register. A PCS may ignore writes to the PCS speed selection bits that select speeds it has not advertised in the PCS speed ability register. It is the responsibility of the STA entity to ensure that mutually acceptable speeds are applied consistently across all the MMDs on a particular PHY.

The PCS speed selection defaults to a supported ability.

The speed selection bits 3.0.5:2, when set to 0001, select the use of the 10PASS-TS and 2BASE-TL PCS.

#### 45.2.3.2 PCS status 1 register (Register 3.1)

The assignment of bits in the PCS status 1 register is shown in Table 45–235. All the bits in the PCS status 1 register are read only; a write to the PCS status 1 register shall have no effect.

**Table 45–235—PCS status 1 register bit definitions**

| Bit(s)    | Name                    | Description                                                                     | R/W <sup>a</sup> |
|-----------|-------------------------|---------------------------------------------------------------------------------|------------------|
| 3.1.15:12 | Reserved                | Value always 0                                                                  | RO               |
| 3.1.11    | Tx LPI received         | 1 = Tx PCS has received LPI<br>0 = LPI not received                             | RO/LH            |
| 3.1.10    | Rx LPI received         | 1 = Rx PCS has received LPI<br>0 = LPI not received                             | RO/LH            |
| 3.1.9     | Tx LPI indication       | 1 = Tx PCS is currently receiving LPI<br>0 = PCS is not currently receiving LPI | RO               |
| 3.1.8     | Rx LPI indication       | 1 = Rx PCS is currently receiving LPI<br>0 = PCS is not currently receiving LPI | RO               |
| 3.1.7     | Fault                   | 1 = Fault condition detected<br>0 = No fault condition detected                 | RO               |
| 3.1.6     | Clock stop capable      | 1 = The MAC may stop the clock during LPI<br>0 = Clock not stoppable            | RO               |
| 3.1.5:3   | Reserved                | Value always 0                                                                  | RO               |
| 3.1.2     | PCS receive link status | 1 = PCS receive link up<br>0 = PCS receive link down                            | RO/LL            |
| 3.1.1     | Low-power ability       | 1 = PCS supports low-power mode<br>0 = PCS does not support low-power mode      | RO               |
| 3.1.0     | Reserved                | Value always 0                                                                  | RO               |

<sup>a</sup>RO = Read only, LL = Latching low, LH = Latching high

##### 45.2.3.2.1 Transmit LPI received (3.1.11)

When read as a one, bit 3.1.11 indicates that the transmit PCS has received LPI signaling one or more times since the register was last read. When read as a zero, bit 3.1.11 indicates that the PCS has not received LPI signaling. This bit shall be implemented with latching high behavior.

#### **45.2.3.2.2 Receive LPI received (3.1.10)**

When read as a one, bit 3.1.10 indicates that the receive PCS has received LPI signaling one or more times since the register was last read. When read as a zero, bit 3.1.10 indicates that the PCS has not received LPI signaling. This bit shall be implemented with latching high behavior.

#### **45.2.3.2.3 Transmit LPI indication (3.1.9)**

When read as a one, bit 3.1.9 indicates that the transmit PCS is currently receiving LPI signals. When read as a zero, bit 3.1.9 indicates that the PCS is not currently receiving LPI signals. The behavior if read during a state transition is undefined.

#### **45.2.3.2.4 Receive LPI indication (3.1.8)**

When read as a one, bit 3.1.8 indicates that the receive PCS is currently receiving LPI signals. When read as a zero, bit 3.1.8 indicates that the PCS is not currently receiving LPI signals. The behavior if read during a state transition is undefined.

#### **45.2.3.2.5 Fault (3.1.7)**

When read as a one, bit 3.1.7 indicates that the PCS has detected a fault condition on either the transmit or receive paths. When read as a zero, bit 3.1.7 indicates that the PCS has not detected a fault condition. For 10 Gb/s operation, bit 3.1.7 is read as a one when either of the fault bits (3.8.11, 3.8.10) located in register 3.8 are read as a one. For 10BASE-TS or 2BASE-TL operation, this bit shall become a one when any 10P/2B PCS registers indicate a fault (see 45.2.3.31 through 45.2.3.38).

#### **45.2.3.2.6 Clock stop capable (3.1.6)**

If bit 3.1.6 is set to one then the RS may stop the transmit xMII clock while it is signaling LPI otherwise it shall keep the clock active. If the RS does not support EEE capability or is not able to stop the transmit clock then this bit has no effect (see 22.2.2.6, 35.2.2.6, and 46.3.2.4).

#### **45.2.3.2.7 PCS receive link status (3.1.2)**

When read as a one, bit 3.1.2 indicates that the PCS receive link is up. When read as a zero, bit 3.1.2 indicates that the PCS receive link is down. When a 10/25/40/50/100/200/400GBASE-R, 10GBASE-W, or any MultiGBASE-T mode of operation is selected for the PCS using the PCS type selection field (3.7.3:0), this bit is a latching low version of bit 3.32.12. When a 10GBASE-X mode of operation is selected for the PCS using the PCS type selection field (3.7.3:0), this bit is a latching low version of bit 3.24.12. The receive link status bit shall be implemented with latching low behavior.

#### **45.2.3.2.8 Low-power ability (3.1.1)**

When read as a one, bit 3.1.1 indicates that the PCS supports the low-power feature. When read as a zero, bit 3.1.1 indicates that the PCS does not support the low-power feature. If a PCS supports the low-power feature then it is controlled using the low-power bit 3.0.11.

#### **45.2.3.3 PCS device identifier (Registers 3.2 and 3.3)**

Registers 3.2 and 3.3 provide a 32-bit value, which may constitute a unique identifier for a particular type of PCS. The identifier shall be composed of the 3rd through 24th bits of the Organizationally Unique Identifier (OUI) assigned to the device manufacturer by the IEEE, plus a six-bit model number, plus a four-bit revision number. A PCS may return a value of zero in each of the 32 bits of the PCS device identifier.

The format of the PCS device identifier is specified in 22.2.4.3.1.

#### 45.2.3.4 PCS speed ability (Register 3.4)

The assignment of bits in the PCS speed ability register is shown in Table 45–236.

**Table 45–236—PCS speed ability register bit definitions**

| Bit(s)    | Name                       | Description                                                                                                | R/W <sup>a</sup> |
|-----------|----------------------------|------------------------------------------------------------------------------------------------------------|------------------|
| 3.4.15:10 | Reserved for future speeds | Value always 0                                                                                             | RO               |
| 3.4.9     | 400G capable               | 1 = PCS is capable of operating at 400 Gb/s<br>0 = PCS is not capable of operating at 400 Gb/s             | RO               |
| 3.4.8     | 200G capable               | 1 = PCS is capable of operating at 200 Gb/s<br>0 = PCS is not capable of operating at 200 Gb/s             | RO               |
| 3.4.7     | 5G capable                 | 1 = PCS is capable of operating at 5 Gb/s<br>0 = PCS is not capable of operating at 5 Gb/s                 | RO               |
| 3.4.6     | 2.5G capable               | 1 = PCS is capable of operating at 2.5 Gb/s<br>0 = PCS is not capable of operating at 2.5 Gb/s             | RO               |
| 3.4.5     | 50G capable                | 1 = PCS is capable of operating at 50 Gb/s<br>0 = PCS is not capable of operating at 50 Gb/s               | RO               |
| 3.4.4     | 25G capable                | 1 = PCS is capable of operating at 25 Gb/s<br>0 = PCS is not capable of operating at 25 Gb/s               | RO               |
| 3.4.3     | 100G capable               | 1 = PCS is capable of operating at 100 Gb/s<br>0 = PCS is not capable of operating at 100 Gb/s             | RO               |
| 3.4.2     | 40G capable                | 1 = PCS is capable of operating at 40 Gb/s<br>0 = PCS is not capable of operating at 40 Gb/s               | RO               |
| 3.4.1     | 10PASS-TS/2BASE-TL capable | 1 = PCS is capable of operating as the 10P/2B PCS<br>0 = PCS is not capable of operating as the 10P/2B PCS | RO               |
| 3.4.0     | 10G capable                | 1 = PCS is capable of operating at 10 Gb/s<br>0 = PCS is not capable of operating at 10 Gb/s               | RO               |

<sup>a</sup>RO = Read only

##### 45.2.3.4.1 10G capable (3.4.0)

When read as a one, bit 3.4.0 indicates that the PCS is able to operate at a data rate of 10 Gb/s. When read as a zero, bit 3.4.0 indicates that the PCS is not able to operate at a data rate of 10 Gb/s.

##### 45.2.3.4.2 10PASS-TS/2BASE-TL capable

When read as a one, this bit indicates that the PCS is able to operate as the 10PASS-TS/2BASE-TL PCS, as specified in Clause 61.

##### 45.2.3.4.3 40G capable (3.4.2)

When read as a one, bit 3.4.2 indicates that the PCS is able to operate at a data rate of 40 Gb/s. When read as a zero, bit 3.4.2 indicates that the PCS is not able to operate at a data rate of 40 Gb/s.

#### **45.2.3.4.4 100G capable (3.4.3)**

When read as a one, bit 3.4.3 indicates that the PCS is able to operate at a data rate of 100 Gb/s. When read as a zero, bit 3.4.3 indicates that the PCS is not able to operate at a data rate of 100 Gb/s.

#### **45.2.3.4.5 25G capable (3.4.4)**

When read as a one, bit 3.4.4 indicates that the PCS is able to operate at a data rate of 25 Gb/s. When read as a zero, bit 3.4.4 indicates that the PCS is not able to operate at a data rate of 25 Gb/s.

#### **45.2.3.4.6 50G capable (3.4.5)**

When read as a one, bit 3.4.5 indicates that the PCS is able to operate at a data rate of 50 Gb/s. When read as a zero, bit 3.4.5 indicates that the PCS is not able to operate at a data rate of 50 Gb/s.

#### **45.2.3.4.7 2.5G capable (3.4.6)**

When read as a one, bit 3.4.6 indicates that the PCS is able to operate at a data rate of 2.5 Gb/s. When read as a zero, bit 3.4.6 indicates that the PCS is not able to operate at a data rate of 2.5 Gb/s.

#### **45.2.3.4.8 5G Capable (3.4.7)**

When read as a one, bit 3.4.7 indicates that the PCS is able to operate at a data rate of 5 Gb/s. When read as a zero, bit 3.4.7 indicates that the PCS is not able to operate at a data rate of 5 Gb/s.

#### **45.2.3.4.9 200G capable (3.4.8)**

When read as a one, bit 3.4.8 indicates that the PCS is able to operate at a data rate of 200 Gb/s. When read as a zero, bit 3.4.8 indicates that the PCS is not able to operate at a data rate of 200 Gb/s.

#### **45.2.3.4.10 400G capable (3.4.9)**

When read as a one, bit 3.4.9 indicates that the PCS is able to operate at a data rate of 400 Gb/s. When read as a zero, bit 3.4.9 indicates that the PCS is not able to operate at a data rate of 400 Gb/s.

### **45.2.3.5 PCS devices in package (Registers 3.5 and 3.6)**

The PCS devices in package registers are defined in Table 45–2.

### **45.2.3.6 PCS control 2 register (Register 3.7)**

The assignment of bits in the PCS control 2 register is shown in Table 45–237. The default value for each bit of the PCS control 2 register should be chosen so that the initial state of the device upon power up or reset is a normal operational state without management intervention.

#### **45.2.3.6.1 PCS type selection (3.7.4:0)**

The PCS type shall be selected using bits 4 through 0. The PCS type abilities of the PCS are advertised in bits 3.8.9:0 and 3.9.7:0. A PCS shall ignore writes to the PCS type selection bits that select PCS types it has not advertised in the PCS status 2 register or the PCS status 3 register. It is the responsibility of the STA entity to ensure that mutually acceptable MMD types are applied consistently across all the MMDs on a particular PHY. The PCS type selection defaults to a supported ability.

**Table 45–237—PCS control 2 register bit definitions**

| Bit(s)   | Name               | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | R/W <sup>a</sup> |
|----------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 3.7.15:5 | Reserved           | Value always 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | RO               |
| 3.7.4:0  | PCS type selection | 4 3 2 1 0<br>1 1 x x x = reserved<br>1 0 1 x x = reserved<br>1 0 0 1 1 = Select 25GBASE-PQ PCS type<br>1 0 0 1 0 = Select 25/10GBASE-PQ PCS type<br>1 0 0 0 1 = Select 25GBASE-PQ PCS type, Tx only<br>1 0 0 0 0 = Select 25GBASE-PQ PCS type, Rx only<br>0 1 1 1 1 = Select 5GBASE-R PCS type<br>0 1 1 1 0 = Select 2.5GBASE-X PCS type<br>0 1 1 0 1 = Select 400GBASE-R PCS type<br>0 1 1 0 0 = Select 200GBASE-R PCS type<br>0 1 0 1 1 = Select 5GBASE-T PCS type<br>0 1 0 1 0 = Select 2.5GBASE-T PCS type<br>0 1 0 0 1 = Select 25GBASE-T PCS type<br>0 1 0 0 0 = Select 50GBASE-R PCS type<br>0 0 1 1 1 = Select 25GBASE-R PCS type<br>0 0 1 1 0 = Select 40GBASE-T PCS type<br>0 0 1 0 1 = Select 100GBASE-R PCS type<br>0 0 1 0 0 = Select 40GBASE-R PCS type<br>0 0 0 1 1 = Select 10GBASE-T PCS type<br>0 0 0 1 0 = Select 10GBASE-W PCS type<br>0 0 0 0 1 = Select 10GBASE-X PCS type<br>0 0 0 0 0 = Select 10GBASE-R PCS type | R/W              |

<sup>a</sup>RO = Read only, R/W = Read/Write

#### 45.2.3.7 PCS status 2 register (Register 3.8)

The assignment of bits in the PCS status 2 register is shown in Table 45–238. All the bits in the PCS status 2 register are read only; a write to the PCS status 2 register shall have no effect.

**Table 45–238—PCS status 2 register bit definitions**

| Bit(s)    | Name               | Description                                                                                                                                                                                | R/W <sup>a</sup> |
|-----------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 3.8.15:14 | Device present     | 15 14<br>1 0 = Device responding at this address<br>1 1 = No device responding at this address<br>0 1 = No device responding at this address<br>0 0 = No device responding at this address | RO               |
| 3.8.13    | 5GBASE-T capable   | 1 = PCS is able to support 5GBASE-T PCS type<br>0 = PCS is not able to support 5GBASE-T PCS type                                                                                           | RO               |
| 3.8.12    | 2.5GBASE-T capable | 1 = PCS is able to support 2.5GBASE-T PCS type<br>0 = PCS is not able to support 2.5GBASE-T PCS type                                                                                       | RO               |
| 3.8.11    | Transmit fault     | 1 = Fault condition on transmit path<br>0 = No fault condition on transmit path                                                                                                            | RO/LH            |
| 3.8.10    | Receive fault      | 1 = Fault condition on the receive path<br>0 = No fault condition on the receive path                                                                                                      | RO/LH            |



**Table 45–238—PCS status 2 register bit definitions (*continued*)**

| Bit(s) | Name               | Description                                                                                          | R/W <sup>a</sup> |
|--------|--------------------|------------------------------------------------------------------------------------------------------|------------------|
| 3.8.9  | 25GBASE-T capable  | 1 = PCS is able to support 25GBASE-T PCS type<br>0 = PCS is not able to support 25GBASE-T PCS type   | RO               |
| 3.8.8  | 50GBASE-R capable  | 1 = PCS is able to support 50GBASE-R PCS type<br>0 = PCS is not able to support 50GBASE-R PCS type   | RO               |
| 3.8.7  | 25GBASE-R capable  | 1 = PCS is able to support 25GBASE-R PCS type<br>0 = PCS is not able to support 25GBASE-R PCS type   | RO               |
| 3.8.6  | 40GBASE-T capable  | 1 = PCS is able to support 40GBASE-T PCS type<br>0 = PCS is not able to support 40GBASE-T PCS type   | RO               |
| 3.8.5  | 100GBASE-R capable | 1 = PCS is able to support 100GBASE-R PCS type<br>0 = PCS is not able to support 100GBASE-R PCS type | RO               |
| 3.8.4  | 40GBASE-R capable  | 1 = PCS is able to support 40GBASE-R PCS type<br>0 = PCS is not able to support 40GBASE-R PCS type   | RO               |
| 3.8.3  | 10GBASE-T capable  | 1 = PCS is able to support 10GBASE-T PCS type<br>0 = PCS is not able to support 10GBASE-T PCS type   | RO               |
| 3.8.2  | 10GBASE-W capable  | 1 = PCS is able to support 10GBASE-W PCS type<br>0 = PCS is not able to support 10GBASE-W PCS type   | RO               |
| 3.8.1  | 10GBASE-X capable  | 1 = PCS is able to support 10GBASE-X PCS type<br>0 = PCS is not able to support 10GBASE-X PCS type   | RO               |
| 3.8.0  | 10GBASE-R capable  | 1 = PCS is able to support 10GBASE-R PCS types<br>0 = PCS is not able to support 10GBASE-R PCS types | RO               |

<sup>a</sup>RO = Read only, LH = Latching high

#### 45.2.3.7.1 Device present (3.8.15:14)

When read as <10>, bits 3.8.15:14 indicate that a device is present and responding at this register address. When read as anything other than <10>, bits 3.8.15:14 indicate that no device is present at this address or that the device is not functioning properly.

#### 45.2.3.7.2 5GBASE-T capable (3.8.13)

When read as a one, bit 3.8.13 indicates that the PCS is able to support the 5GBASE-T PCS type. When read as a zero, bit 3.8.13 indicates that the PCS is not able to support the 5GBASE-T PCS type.

#### 45.2.3.7.3 2.5GBASE-T capable (3.8.12)

When read as a one, bit 3.8.12 indicates that the PCS is able to support the 2.5GBASE-T PCS type. When read as a zero, bit 3.8.12 indicates that the PCS is not able to support the 2.5GBASE-T PCS type.

#### 45.2.3.7.4 Transmit fault (3.8.11)

When read as a one, bit 3.8.11 indicates that the PCS has detected a fault condition on the transmit path. When read as a zero, bit 3.8.11 indicates that the PCS has not detected a fault condition on the transmit path. The transmit fault bit shall be implemented with latching high behavior.

The default value of bit 3.8.11 is zero.

#### **45.2.3.7.5 Receive fault (3.8.10)**

When read as a one, bit 3.8.10 indicates that the PCS has detected a fault condition on the receive path. When read as a zero, bit 3.8.10 indicates that the PCS has not detected a fault condition on the receive path. The receive fault bit shall be implemented with latching high behavior.

The default value of bit 3.8.10 is zero.

#### **45.2.3.7.6 25GBASE-T capable (3.8.9)**

When read as a one, bit 3.8.9 indicates that the PCS is able to support the 25GBASE-T PCS type. When read as a zero, bit 3.8.9 indicates that the PCS is not able to support the 25GBASE-T PCS type.

#### **45.2.3.7.7 50GBASE-R capable (3.8.8)**

When read as a one, bit 3.8.8 indicates that the PCS is able to support the 50GBASE-R PCS type. When read as a zero, bit 3.8.8 indicates that the PCS is not able to support the 50GBASE-R PCS type.

#### **45.2.3.7.8 25GBASE-R capable (3.8.7)**

When read as a one, bit 3.8.7 indicates that the PCS is able to support the 25GBASE-R PCS type. When read as a zero, bit 3.8.7 indicates that the PCS is not able to support the 25GBASE-R PCS type.

#### **45.2.3.7.9 40GBASE-T capable (3.8.6)**

When read as a one, bit 3.8.6 indicates that the PCS is able to support the 40GBASE-T PCS type. When read as a zero, bit 3.8.6 indicates that the PCS is not able to support the 40GBASE-T PCS type.

#### **45.2.3.7.10 100GBASE-R capable (3.8.5)**

When read as a one, bit 3.8.5 indicates that the PCS is able to support the 100GBASE-R PCS type. When read as a zero, bit 3.8.5 indicates that the PCS is not able to support the 100GBASE-R PCS type.

#### **45.2.3.7.11 40GBASE-R capable (3.8.4)**

When read as a one, bit 3.8.4 indicates that the PCS is able to support the 40GBASE-R PCS type. When read as a zero, bit 3.8.4 indicates that the PCS is not able to support the 40GBASE-R PCS type.

#### **45.2.3.7.12 10GBASE-T capable (3.8.3)**

When read as a one, bit 3.8.3 indicates that the PCS is able to support the 10GBASE-T PCS type. When read as a zero, bit 3.8.3 indicates that the PCS is not able to support the 10GBASE-T PCS type.

#### **45.2.3.7.13 10GBASE-W capable (3.8.2)**

When read as a one, bit 3.8.2 indicates that the 64B/66B PCS is able to support operation in a 10GBASE-W PHY (that is, supports operation with a WIS). When read as a zero, bit 3.8.2 indicates that the 64B/66B PCS is not able to support operation with a WIS in a 10GBASE-W PHY.

NOTE—This bit does not indicate that the PCS is performing the functionality contained in the WIS. This bit indicates whether the 64B/66B PCS would be able to support a WIS if it were to be attached.

#### 45.2.3.7.14 10GBASE-X capable (3.8.1)

When read as a one, bit 3.8.1 indicates that the PCS is able to support the 10GBASE-X PCS type. When read as a zero, bit 3.8.1 indicates that the PCS is not able to support the 10GBASE-X PCS type.

#### 45.2.3.7.15 10GBASE-R capable (3.8.0)

When read as a one, bit 3.8.0 indicates that the PCS is able to support operation in a 10GBASE-R PHY. When read as a zero, bit 3.8.0 indicates that the PCS is not able to support operation in a 10GBASE-R PHY.

#### 45.2.3.8 PCS status 3 register (Register 3.9)

The assignment of bits in the PCS status 3 register is shown in Table 45–239. All the bits in the PCS status 3 register are read only; a write to the PCS status 3 register shall have no effect.

**Table 45–239—PCS status 3 register bit definitions**

| Bit(s)   | Name                       | Description                                                                                                          | R/W <sup>a</sup> |
|----------|----------------------------|----------------------------------------------------------------------------------------------------------------------|------------------|
| 3.9.15:8 | Reserved                   | Value always 0                                                                                                       | RO               |
| 3.9.7    | 25GBASE-PQ capable         | 1 = PCS is able to support 25GBASE-PQ PCS type<br>0 = PCS is not able to support 25GBASE-PQ PCS type                 | RO               |
| 3.9.6    | 25/10GBASE-PQ capable      | 1 = PCS is able to support 25/10GBASE-PQ PCS type<br>0 = PCS is not able to support 25/10GBASE-PQ PCS type           | RO               |
| 3.9.5    | 25GBASE-PQ Rx only capable | 1 = PCS is able to support 25GBASE-PQ Rx only PCS type<br>0 = PCS is not able to support 25GBASE-PQ Rx only PCS type | RO               |
| 3.9.4    | 25GBASE-PQ Tx only capable | 1 = PCS is able to support 25GBASE-PQ Tx only PCS type<br>0 = PCS is not able to support 25GBASE-PQ Tx only PCS type | RO               |
| 3.9.3    | 5GBASE-R capable           | 1 = PCS is able to support 5GBASE-R PCS type<br>0 = PCS is not able to support 5GBASE-R PCS type                     | RO               |
| 3.9.2    | 2.5GBASE-X capable         | 1 = PCS is able to support 2.5GBASE-X PCS type<br>0 = PCS is not able to support 2.5GBASE-X PCS type                 | RO               |
| 3.9.1    | 400GBASE-R capable         | 1 = PCS is able to support 400GBASE-R PCS type<br>0 = PCS is not able to support 400GBASE-R PCS type                 | RO               |
| 3.9.0    | 200GBASE-R capable         | 1 = PCS is able to support 200GBASE-R PCS type<br>0 = PCS is not able to support 200GBASE-R PCS type                 | RO               |

<sup>a</sup>RO = Read only

#### **45.2.3.8.1 25GBASE-PQ capable (3.9.7)**

When read as a one, bit 3.9.7 indicates that the PCS is able to support the 25GBASE-PQ PCS type. When read as a zero, bit 3.9.7 indicates that the PCS is not able to support the 25GBASE-PQ PCS type.

#### **45.2.3.8.2 25/10GBASE-PQ capable (3.9.6)**

When read as a one, bit 3.9.6 indicates that the PCS is able to support the 25/10GBASE-PQ PCS type. When read as a zero, bit 3.9.6 indicates that the PCS is not able to support the 25/10GBASE-PQ PCS type.

#### **45.2.3.8.3 25GBASE-PQ Rx only capable (3.9.5)**

When read as a one, bit 3.9.5 indicates that the PCS is able to support the 25GBASE-PQ Rx only PCS type. When read as a zero, bit 3.9.5 indicates that the PCS is not able to support the 25GBASE-PQ Rx only PCS type.

#### **45.2.3.8.4 25GBASE-PQ Tx only capable (3.9.4)**

When read as a one, bit 3.9.4 indicates that the PCS is able to support the 25GBASE-PQ Tx only PCS type. When read as a zero, bit 3.9.4 indicates that the PCS is not able to support the 25GBASE-PQ Tx only PCS type.

#### **45.2.3.8.5 5GBASE-R capable (3.9.3)**

When read as a one, bit 3.9.3 indicates that the PCS is able to support the 5GBASE-R PCS type. When read as a zero, bit 3.9.3 indicates that the PCS is not able to support the 5GBASE-R PCS type.

#### **45.2.3.8.6 2.5GBASE-X capable (3.9.2)**

When read as a one, bit 3.9.2 indicates that the PCS is able to support operation in a 2.5GBASE-X PCS type. When read as a zero, bit 3.9.2 indicates that the PCS is not able to support operation in a 2.5GBASE-X PCS type.

#### **45.2.3.8.7 400GBASE-R capable (3.9.1)**

When read as a one, bit 3.9.1 indicates that the PCS is able to support the 400GBASE-R PCS type. When read as a zero, bit 3.9.1 indicates that the PCS is not able to support the 400GBASE-R PCS type.

#### **45.2.3.8.8 200GBASE-R capable (3.9.0)**

When read as a one, bit 3.9.0 indicates that the PCS is able to support the 200GBASE-R PCS type. When read as a zero, bit 3.9.0 indicates that the PCS is not able to support the 200GBASE-R PCS type.

#### **45.2.3.9 PCS package identifier (Registers 3.14 and 3.15)**

Registers 3.14 and 3.15 provide a 32-bit value, which may constitute a unique identifier for a particular type of package that the PCS is instantiated within. The identifier shall be composed of the 3rd through 24th bits of the Organizationally Unique Identifier (OUI) assigned to the package manufacturer by the IEEE, plus a six-bit model number, plus a four-bit revision number. A PCS may return a value of zero in each of the 32 bits of the PCS package identifier.

A non-zero package identifier may be returned by one or more MMDs in the same package. The package identifier may be the same as the device identifier.

The format of the package identifier is specified in 22.2.4.3.1.

#### 45.2.3.10 EEE control and capability 1 (Register 3.20)

This register is used to indicate the capability of the PCS to support EEE functions for each PHY type. The assignment of bits in the EEE capability 1 register is shown in Table 45–240.

**Table 45–240—EEE control and capability 1 register bit definitions**

| Bit(s)  | Name                  | Description                                                                                          | R/W <sup>a</sup> |
|---------|-----------------------|------------------------------------------------------------------------------------------------------|------------------|
| 3.20.15 | Reserved              | Value always 0                                                                                       | RO               |
| 3.20.14 | 50GBASE-R fast wake   | 1 = EEE fast wake is supported for 50GBASE-R<br>0 = EEE fast wake is not supported for 50GBASE-R     | RO               |
| 3.20.13 | 100GBASE-R deep sleep | 1 = EEE deep sleep is supported for 100GBASE-R<br>0 = EEE deep sleep is not supported for 100GBASE-R | RO               |
| 3.20.12 | 100GBASE-R fast wake  | 1 = EEE fast wake is supported for 100GBASE-R<br>0 = EEE fast wake is not supported for 100GBASE-R   | RO               |
| 3.20.11 | 25GBASE-R deep sleep  | 1 = EEE deep sleep is supported for 25GBASE-R<br>0 = EEE deep sleep is not supported for 25GBASE-R   | RO               |
| 3.20.10 | 25GBASE-R fast wake   | 1 = EEE fast wake is supported for 25GBASE-R<br>0 = EEE fast wake is not supported for 25GBASE-R     | RO               |
| 3.20.9  | 40GBASE-R deep sleep  | 1 = EEE deep sleep is supported for 40GBASE-R<br>0 = EEE deep sleep is not supported for 40GBASE-R   | RO               |
| 3.20.8  | 40GBASE-R fast wake   | 1 = EEE fast wake is supported for 40GBASE-R<br>0 = EEE fast wake is not supported for 40GBASE-R     | RO               |
| 3.20.7  | 40GBASE-T EEE         | 1 = EEE is supported for 40GBASE-T<br>0 = EEE is not supported for 40GBASE-T                         | RO               |
| 3.20.6  | 10GBASE-KR EEE        | 1 = EEE is supported for 10GBASE-KR<br>0 = EEE is not supported for 10GBASE-KR                       | RO               |
| 3.20.5  | 10GBASE-KX4 EEE       | 1 = EEE is supported for 10GBASE-KX4<br>0 = EEE is not supported for 10GBASE-KX4                     | RO               |
| 3.20.4  | 1000BASE-KX EEE       | 1 = EEE is supported for 1000BASE-KX<br>0 = EEE is not supported for 1000BASE-KX                     | RO               |
| 3.20.3  | 10GBASE-T EEE         | 1 = EEE is supported for 10GBASE-T<br>0 = EEE is not supported for 10GBASE-T                         | RO               |
| 3.20.2  | 1000BASE-T EEE        | 1 = EEE is supported for 1000BASE-T<br>0 = EEE is not supported for 1000BASE-T                       | RO               |
| 3.20.1  | 100BASE-TX EEE        | 1 = EEE is supported for 100BASE-TX<br>0 = EEE is not supported for 100BASE-TX                       | RO               |
| 3.20.0  | LPI_FW                | 1 = Fast wake mode is used for LPI function<br>0 = Deep sleep is used for LPI function               | R/W              |

<sup>a</sup>R/W = Read/Write, RO = Read only

#### **45.2.3.10.1 50GBASE-R EEE fast wake supported (3.20.14)**

If the PCS supports EEE fast wake operation for 50GBASE-R, this bit shall be set to a one; otherwise this bit shall be set to a zero.

#### **45.2.3.10.2 100GBASE-R EEE deep sleep supported (3.20.13)**

If the PCS supports EEE deep sleep operation for 100GBASE-R, this bit shall be set to a one; otherwise this bit shall be set to a zero.

#### **45.2.3.10.3 100GBASE-R EEE fast wake supported (3.20.12)**

If the PCS supports EEE fast wake operation for 100GBASE-R, this bit shall be set to a one; otherwise this bit shall be set to a zero.

#### **45.2.3.10.4 25GBASE-R deep sleep (3.20.11)**

If the device supports EEE deep sleep operation for 25GBASE-R bit 3.20.11 shall be set to a one; otherwise this bit is set to a zero.

#### **45.2.3.10.5 25GBASE-R fast wake (3.20.10)**

If the device supports EEE fast wake operation for 25GBASE-R bit 3.20.10 shall be set to a one; otherwise this bit is set to a zero.

#### **45.2.3.10.6 40GBASE-R EEE deep sleep supported (3.20.9)**

If the PCS supports EEE deep sleep operation for 40GBASE-R, as defined in 78.1, this bit shall be set to a one; otherwise this bit shall be set to a zero.

#### **45.2.3.10.7 40GBASE-R EEE fast wake supported (3.20.8)**

If the PCS supports EEE fast wake operation for 40GBASE-R, as defined in 78.1, this bit shall be set to a one; otherwise this bit shall be set to a zero.

#### **45.2.3.10.8 40GBASE-T EEE supported (3.20.7)**

If the device supports EEE operation for 40GBASE-T as defined in 113.1.3.3, this bit shall be set to one.

#### **45.2.3.10.9 10GBASE-KR EEE supported (3.20.6)**

If the device supports EEE operation for 10GBASE-KR as defined in 72.1, this bit shall be set to one.

#### **45.2.3.10.10 10GBASE-KX4 EEE supported (3.20.5)**

If the device supports EEE operation for 10GBASE-KX4 as defined in 71.2, this bit shall be set to one.

#### **45.2.3.10.11 1000BASE-KX EEE supported (3.20.4)**

If the device supports EEE operation for 1000BASE-KX as defined in 70.1, this bit shall be set to one.

#### **45.2.3.10.12 10GBASE-T EEE supported (3.20.3)**

If the device supports EEE operation for 10GBASE-T as defined in 55.1.3.3, this bit shall be set to one.

#### 45.2.3.10.13 1000BASE-T EEE supported (3.20.2)

If the device supports EEE operation for 1000BASE-T as defined in 40.1.3, this bit shall be set to one.

#### 45.2.3.10.14 100BASE-TX EEE supported (3.20.1)

If the device supports EEE operation for 100BASE-TX as defined in 24.1.1, this bit shall be set to one.

#### 45.2.3.10.15 LPI\_FW (3.20.0)

If the device supports fast wake as defined in 78.5, this bit selects fast wake or deep sleep operation. Setting 3.20.0 to one selects fast wake, setting to zero selects deep sleep. This bit is ignored by devices that do not support fast wake, and this bit defaults to one for devices that support fast wake.

#### 45.2.3.11 EEE control and capability 2 (Register 3.21)

Register 3.21 extends register 3.20 and is used to indicate the capability of the PCS to support EEE functions for each PHY type. The assignment of bits in the EEE control and capability 2 register is shown in Table 45–241.

**Table 45–241—EEE control and capability 2 register bit definitions**

| Bit(s)    | Name                 | Description                                                                                        | R/W <sup>a</sup> |
|-----------|----------------------|----------------------------------------------------------------------------------------------------|------------------|
| 3.21.15:9 | Reserved             | Value always 0                                                                                     | RO               |
| 3.21.8    | 5GBASE-KR EEE        | 1 = EEE is supported for 5GBASE-R<br>0 = EEE is not supported for 5GBASE-R                         | RO               |
| 3.21.7    | 2.5GBASE-KX EEE      | 1 = EEE is supported for 2.5GBASE-X<br>0 = EEE is not supported for 2.5GBASE-X                     | RO               |
| 3.21.6    | Reserved             | Value always 0                                                                                     | RO               |
| 3.21.5    | 400GBASE-R fast wake | 1 = EEE fast wake is supported for 400GBASE-R<br>0 = EEE fast wake is not supported for 400GBASE-R | RO               |
| 3.21.4    | Reserved             | Value always 0                                                                                     | RO               |
| 3.21.3    | 200GBASE-R fast wake | 1 = EEE fast wake is supported for 200GBASE-R<br>0 = EEE fast wake is not supported for 200GBASE-R | RO               |
| 3.21.2    | 25GBASE-T EEE        | 1 = EEE is supported for 25GBASE-T<br>0 = EEE is not supported for 25GBASE-T                       | RO               |
| 3.21.1    | 5GBASE-T EEE         | 1 = EEE is supported for 5GBASE-T<br>0 = EEE is not supported for 5GBASE-T                         | RO               |
| 3.21.0    | 2.5GBASE-T EEE       | 1 = EEE is supported for 2.5GBASE-T<br>0 = EEE is not supported for 2.5GBASE-T                     | RO               |

<sup>a</sup> RO = Read only

##### 45.2.3.11.1 5GBASE-KR EEE supported (3.21.8)

If the device supports EEE operation for 5GBASE-KR as defined in 130.1, this bit shall be set to one.

##### 45.2.3.11.2 2.5GBASE-KX EEE supported (3.21.7)

If the device supports EEE operation for 2.5GBASE-KX as defined in 128.1, this bit shall be set to one.

#### 45.2.3.11.3 40GBASE-R EEE fast wake supported (3.21.5)

If the PCS supports EEE fast wake operation for 40GBASE-R, this bit shall be set to a one; otherwise this bit shall be set to a zero.

#### 45.2.3.11.4 200GBASE-R EEE fast wake supported (3.21.3)

If the PCS supports EEE fast wake operation for 200GBASE-R, this bit shall be set to a one; otherwise this bit shall be set to a zero.

#### 45.2.3.11.5 25GBASE-T EEE supported (3.21.2)

If the device supports EEE operation for 25GBASE-T as defined in 113.1.3.3, this bit shall be set to one.

#### 45.2.3.11.6 5GBASE-T EEE supported (3.21.1)

If the device supports EEE operation for 5GBASE-T as defined in 126.1.3.3, bit 3.21.1 shall be set to one.

#### 45.2.3.11.7 2.5GBASE-T EEE supported (3.21.0)

If the device supports EEE operation for 2.5GBASE-T as defined in 126.1.3.3, bit 3.21.0 shall be set to one.

#### 45.2.3.12 EEE wake error counter (Register 3.22)

This register is used by PHY types that support EEE to count wake time faults where the PHY fails to complete its normal wake sequence within the time required for the specific PHY type. The definition of the fault event to be counted is defined for each PHY and may occur during a refresh or a wake-up as defined by the PHY. This 16-bit counter shall be reset to all zeros when the EEE wake error counter is read by the management function or upon execution of the PCS reset. This counter shall be held at all ones in the case of overflow.

#### 45.2.3.13 10GBASE-X PCS status register (Register 3.24)

The assignment of bits in the 10GBASE-X PCS status register is shown in Table 45–242. All the bits in the 10GBASE-X PCS status register are read only; a write to the 10GBASE-X PCS status register shall have no effect. A PCS device that does not implement 10GBASE-X shall return a zero for all bits in the 10GBASE-X PCS status register. It is the responsibility of the STA management entity to ensure that a port type is supported by all MMDs before interrogating any of its status bits.

**Table 45–242—10GBASE-X PCS status register bit definitions**

| Bit(s)     | Name                            | Description                                                                                                    | R/W <sup>a</sup> |
|------------|---------------------------------|----------------------------------------------------------------------------------------------------------------|------------------|
| 3.24.15:13 | Reserved                        | Value always 0                                                                                                 | RO               |
| 3.24.12    | 10GBASE-X lane alignment status | 1 = 10GBASE-X PCS receive lanes aligned<br>0 = 10GBASE-X PCS receive lanes not aligned                         | RO               |
| 3.24.11    | Pattern testing ability         | 1 = 10GBASE-X PCS is able to generate test patterns<br>0 = 10GBASE-X PCS is not able to generate test patterns | RO               |
| 3.24.10:4  | Reserved                        | Value always 0                                                                                                 | RO               |



**Table 45–242—10GBASE-X PCS status register bit definitions (*continued*)**

| Bit(s) | Name        | Description                                                  | R/W <sup>a</sup> |
|--------|-------------|--------------------------------------------------------------|------------------|
| 3.24.3 | Lane 3 sync | 1 = Lane 3 is synchronized<br>0 = Lane 3 is not synchronized | RO               |
| 3.24.2 | Lane 2 sync | 1 = Lane 2 is synchronized<br>0 = Lane 2 is not synchronized | RO               |
| 3.24.1 | Lane 1 sync | 1 = Lane 1 is synchronized<br>0 = Lane 1 is not synchronized | RO               |
| 3.24.0 | Lane 0 sync | 1 = Lane 0 is synchronized<br>0 = Lane 0 is not synchronized | RO               |

<sup>a</sup>RO = Read only

#### **45.2.3.13.1 10GBASE-X receive lane alignment status (3.24.12)**

When read as a one, bit 3.24.12 indicates that the 10GBASE-X PCS has synchronized and aligned all four receive lanes. When read as a zero, bit 3.24.12 indicates that the 10GBASE-X PCS has not synchronized and aligned all four receive lanes.

#### **45.2.3.13.2 Pattern testing ability (3.24.11)**

When read as a one, bit 3.24.11 indicates that the 10GBASE-X PCS is able to generate test patterns. When read as a zero, bit 3.24.11 indicates that the 10GBASE-X PCS is not able to generate test patterns. If the 10GBASE-X PCS is able to generate test patterns, then the functionality is controlled using the transmit test-pattern enable bit in register 3.25.

#### **45.2.3.13.3 Lane 3 sync (3.24.3)**

When read as a one, bit 3.24.3 indicates that the 10GBASE-X PCS receive lane 3 is synchronized. When read as a zero, bit 3.24.3 indicates that the 10GBASE-X PCS receive lane 3 is not synchronized.

#### **45.2.3.13.4 Lane 2 sync (3.24.2)**

When read as a one, bit 3.24.2 indicates that the 10GBASE-X PCS receive lane 2 is synchronized. When read as a zero, bit 3.24.2 indicates that the 10GBASE-X PCS receive lane 2 is not synchronized.

#### **45.2.3.13.5 Lane 1 sync (3.24.1)**

When read as a one, bit 3.24.1 indicates that the 10GBASE-X PCS receive lane 1 is synchronized. When read as a zero, bit 3.24.1 indicates that the 10GBASE-X PCS receive lane 1 is not synchronized.

#### **45.2.3.13.6 Lane 0 sync (3.24.0)**

When read as a one, bit 3.24.0 indicates that the 10GBASE-X PCS receive lane 0 is synchronized. When read as a zero, bit 3.24.0 indicates that the 10GBASE-X PCS receive lane 0 is not synchronized.

#### 45.2.3.14 10GBASE-X PCS test control register (Register 3.25)

The assignment of bits in the 10GBASE-X PCS test control register is shown in Table 45–243. The default value for each bit of the 10GBASE-X PCS test control register should be chosen so that the initial state of the device upon power up or reset is a normal operational state without management intervention.

**Table 45–243—10GBASE-X PCS test control register bit definitions**

| Bit(s)    | Name                         | Description                                                                                                                          | R/W <sup>a</sup> |
|-----------|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 3.25.15:3 | Reserved                     | Value always 0                                                                                                                       | RO               |
| 3.25.2    | Transmit test-pattern enable | 1 = Transmit test pattern enabled<br>0 = Transmit test pattern not enabled                                                           | R/W              |
| 3.25.1:0  | Test pattern select          | 1 0<br>1 1 = Reserved<br>1 0 = Mixed-frequency test pattern<br>0 1 = Low-frequency test pattern<br>0 0 = High-frequency test pattern | R/W              |

<sup>a</sup>RO = Read only, R/W = Read/Write

##### 45.2.3.14.1 Transmit test-pattern enable (3.25.2)

When bit 3.25.2 is set to a one, pattern testing is enabled on the transmit path. When bit 3.25.2 is set to a zero, pattern testing is disabled on the transmit path. Pattern testing is optional, and the ability of the 10GBASE-X PCS to generate test patterns is advertised by the pattern testing ability bit in register 3.24. A 10GBASE-X PCS that does not support the generation of test patterns shall ignore writes to this bit and always return a value of zero. The default of bit 3.25.2 is zero.

##### 45.2.3.14.2 Test pattern select (3.25.1:0)

The test pattern to be used when pattern testing is enabled using bit 3.25.2 is selected using bits 3.25.1:0. When bits 3.25.1:0 are set to <10>, the mixed-frequency test pattern shall be selected for pattern testing. When bits 3.25.1:0 are set to <01>, the low-frequency test pattern shall be selected for pattern testing. When bits 3.25.1:0 are set to <00>, the high-frequency test pattern shall be selected for pattern testing. The test patterns are defined in Annex 48A.

#### 45.2.3.15 BASE-R and MultiGBASE-T PCS status 1 register (Register 3.32)

The assignment of bits in the BASE-R and MultiGBASE-T PCS status 1 register is shown in Table 45–244. All the bits in the BASE-R and MultiGBASE-T PCS status 1 register are read only; a write to the BASE-R and MultiGBASE-T PCS status 1 register shall have no effect. A PCS device that implements neither BASE-R nor any member of the MultiGBASE-T set shall return a zero for all bits in the BASE-R and MultiGBASE-T PCS status 1 register. It is the responsibility of the STA management entity to ensure that a port type is supported by all MMDs before interrogating any of its status bits. The contents of register 3.32 are undefined when the BASE-R or MultiGBASE-T PCS is operating in seed test-pattern mode, PRBS31 test-pattern mode, or PRBS9 test-pattern mode.

##### 45.2.3.15.1 BASE-R and MultiGBASE-T receive link status (3.32.12)

When read as a one, bit 3.32.12 indicates that the PCS is in a fully operational state. When read as a zero, bit 3.32.12 indicates that the PCS is not fully operational. This bit is a reflection of the PCS\_status variable

**Table 45–244—BASE-R and MultiGBASE-T PCS status 1 register bit definitions**

| Bit(s)     | Name                                        | Description                                                                                                                      | R/W <sup>a</sup> |
|------------|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|------------------|
| 3.32.15:13 | Reserved                                    | Value always 0                                                                                                                   | RO               |
| 3.32.12    | BASE-R and MultiGBASE-T receive link status | 1 = BASE-R or any MultiGBASE-T PCS receive link up<br>0 = BASE-R or any MultiGBASE-T PCS receive link down                       | RO               |
| 3.32.11:4  | Reserved                                    | Value always 0                                                                                                                   | RO               |
| 3.32.3     | 10GBASE-R PRBS9 pattern testing ability     | 1 = PCS is able to support PRBS9 pattern testing<br>0 = PCS is not able to support PRBS9 pattern testing                         | RO               |
| 3.32.2     | 10GBASE-R PRBS31 pattern testing ability    | 1 = PCS is able to support PRBS31 pattern testing<br>0 = PCS is not able to support PRBS31 pattern testing                       | RO               |
| 3.32.1     | BASE-R and MultiGBASE-T PCS high BER        | 1 = BASE-R or any MultiGBASE-T PCS reporting a high BER<br>0 = BASE-R or any MultiGBASE-T PCS not reporting a high BER           | RO               |
| 3.32.0     | BASE-R and MultiGBASE-T PCS block lock      | 1 = BASE-R or any MultiGBASE-T PCS locked to received blocks<br>0 = BASE-R or any MultiGBASE-T PCS not locked to received blocks | RO               |

<sup>a</sup>RO = Read only

defined in 49.2.14.1 for 5/10/25GBASE-R, in 126.3.7.1 for 2.5GBASE-T and 5GBASE-T, in 55.3.7.1 for 10GBASE-T, in 113.3.7.1 for 25GBASE-T and 40GBASE-T, in 82.3.1 for 40/50/100GBASE-R, and in 119.3 for 200/400GBASE-R.

#### **45.2.3.15.2 PRBS9 pattern testing ability (3.32.3)**

When read as a one, bit 3.32.3 indicates that the PCS is able to support PRBS9 pattern testing of its transmitter. When read as a zero, bit 3.32.3 indicates that the PCS is not able to support PRBS9 pattern testing of its transmitter. If the PCS is able to support PRBS9 pattern testing of its transmitter then the pattern generation is controlled using bit 3.42.6.

#### **45.2.3.15.3 PRBS31 pattern testing ability (3.32.2)**

When read as a one, bit 3.32.2 indicates that the PCS is able to support PRBS31 pattern testing. When read as a zero, bit 3.32.2 indicates that the PCS is not able to support PRBS31 pattern testing. If the PCS is able to support PRBS31 pattern testing then the pattern generation and checking is controlled using bits 3.42.5:4.

#### **45.2.3.15.4 BASE-R and MultiGBASE-T PCS high BER (3.32.1)**

For BASE-R, when read as a one, bit 3.32.1 indicates that the 64B/66B receiver is detecting a BER of  $\geq 10^{-4}$ . When read as a zero, bit 3.32.1 indicates that the 64B/66B receiver is detecting a BER of  $< 10^{-4}$ . This bit is a direct reflection of the state of the `hi_ber` variable in the BER monitor state diagrams as defined in 49.2.13.2.2 for 5/10/25GBASE-R and in 82.2.19.2.2 for 40/50/100GBASE-R.

For any member of the MultiGBASE-T set, when read as a one, bit 3.32.1 indicates that the 64B/65B receiver is detecting a BER of  $\geq 10^{-4}$ . When read as a zero, bit 3.32.1 indicates that the 64B/65B receiver is

detecting a BER of  $< 10^{-4}$ . This bit is a direct reflection of the state of the hi\_lfer variable in the LFER monitor state diagrams as defined in 126.3.6.2.2 for 2.5GBASE-T and 5GBASE-T, in 55.3.6.2.2 for 10GBASE-T, and in 113.3.6.2.2 for 25GBASE-T and 40GBASE-T.

#### 45.2.3.15.5 BASE-R and MultiGBASE-T PCS block lock (3.32.0)

When read as a one, bit 3.32.0 indicates that the 64B/66B receiver for BASE-R or the 64B/65B receiver for a member of the MultiGBASE-T set has block lock. When read as a zero, bit 3.32.0 indicates that the 64B/66B receiver for BASE-R or the 64B/65B receiver for the member of the MultiGBASE-T set has not achieved block lock. For a 5/10/25GBASE-R PCS, this bit is a direct reflection of the state of the block\_lock variable defined in 49.2.13.2.2. For a 40/50/100GBASE-R PCS, this bit reflects the logical AND of the state of the block\_lock<x> variables defined in 82.2.19.2.2. For a MultiGBASE-R PCS, this bit is a direct reflection of the state of the block\_lock variable defined in 126.3.6.2.2 for 2.5GBASE-T and 5GBASE-T, in 55.3.2.3 for 10GBASE-T, and in 113.3.6.2.2 for 25GBASE-T and 40GBASE-T. For a multi-lane PCS, this bit indicates that the receiver has both block lock and alignment for all lanes and is identical to 3.50.12 (see 45.2.3.23.1).

#### 45.2.3.16 BASE-R and MultiGBASE-T PCS status 2 register (Register 3.33)

The assignment of bits in the BASE-R and MultiGBASE-T PCS status 2 register is shown in Table 45–245. All the bits in the BASE-R and MultiGBASE-T PCS status 2 register are read only; a write to the BASE-R and MultiGBASE-T PCS status 2 register shall have no effect. A PCS device that implements neither BASE-R nor any member of the MultiGBASE-T set shall return a zero for all bits in the BASE-R and MultiGBASE-T PCS status 2 register. It is the responsibility of the STA management entity to ensure that a port type is supported by all MMDs before interrogating any of its status bits. The contents of register 3.33 are undefined when the BASE-R or the MultiGBASE-T set PCS is operating seed test-pattern mode, PRBS31 test-pattern mode, or PRBS9 test-pattern mode.

**Table 45–245—BASE-R and MultiGBASE-T PCS status 2 register bit definitions**

| Bit(s)    | Name               | Description                                                                                                                  | R/W <sup>a</sup> |
|-----------|--------------------|------------------------------------------------------------------------------------------------------------------------------|------------------|
| 3.33.15   | Latched block lock | 1 = BASE-R or any MultiGBASE-T PCS has block lock<br>0 = BASE-R or any MultiGBASE-T PCS does not have block lock             | RO/LL            |
| 3.33.14   | Latched high BER   | 1 = BASE-R or any MultiGBASE-T PCS has reported a high BER<br>0 = BASE-R or any MultiGBASE-T PCS has not reported a high BER | RO/LH            |
| 3.33.13:8 | BER                | BER counter                                                                                                                  | RO/NR            |
| 3.33.7:0  | Errored blocks     | Errored blocks counter                                                                                                       | RO/NR            |

<sup>a</sup>RO = Read only, LL = Latching low, LH = Latching high, NR = Non Roll-over

##### 45.2.3.16.1 Latched block lock (3.33.15)

When read as a one, bit 3.33.15 indicates that the 5/10/25/40/50/100GBASE-R or a member of the MultiGBASE-T set PCS has achieved block lock. When read as a zero, bit 3.33.15 indicates that the 5/10/25/40/50/100GBASE-R or a member of the MultiGBASE-T set PCS has lost block lock.

The latched block lock bit shall be implemented with latching low behavior.

This bit is a latching low version of the 5/10/25/40/50/100GBASE-R and MultiGBASE-T PCS block lock status bit (3.32.0).

#### 45.2.3.16.2 Latched high BER (3.33.14)

When read as a one, bit 3.33.14 indicates that the 5/10/25/40/50/100GBASE-R or a member of the MultiGBASE-T set PCS has detected a high BER. When read as a zero, bit 3.33.14 indicates that the 5/10/25/40/50/100GBASE-R or a member of the MultiGBASE-T set PCS has not detected a high BER.

The latched high BER bit shall be implemented with latching high behavior.

This bit is a latching high version of the 5/10/25/40/50/100GBASE-R and MultiGBASE-T PCS high BER status bit (3.32.1).

#### 45.2.3.16.3 BER (3.33.13:8)

The BER counter is a six bit count as defined by the `ber_count` variable in 49.2.14.2 and 82.2.19.2.4 for 5/10/25/40/50/100GBASE-R and defined by counter `lfer_count` in 126.3.7.2 for 2.5GBASE-T and 5GBASE-T, in 55.3.7.2 for 10GBASE-T, and in 113.3.7.2 for 25GBASE-T and 40GBASE-T. These bits shall be reset to all zeros when the BASE-R and MultiGBASE-T PCS status 2 register is read by the management function or upon execution of the PCS reset. If the BER high order counter, 3.44 (see 45.2.3.21) is not implemented then these bits shall be held at all ones in the case of overflow.

#### 45.2.3.16.4 Errored blocks (3.33.7:0)

The errored blocks counter is an eight bit count defined by the counter `errored_block_count` specified in 49.2.14.2 for 5/10/25GBASE-R, in 82.3.1 for 40/50/100GBASE-R, in 126.3.7.2 for 2.5GBASE-T and 5GBASE-T, in 55.3.7.2 for 10GBASE-T, and in 113.3.7.2 for 25GBASE-T and 40GBASE-T. These bits shall be reset to all zeros when the errored blocks count is read by the management function or upon execution of the PCS reset. If the Errored blocks high order counter, 3.45 (see 45.2.3.22) is not implemented then these bits shall be held at all ones in the case of overflow.

#### 45.2.3.17 5/10/25GBASE-R PCS test pattern seed A (Registers 3.34 through 3.37)

The assignment of bits in the 5/10/25GBASE-R PCS test pattern seed A registers is shown in Table 45–246. This register is only required when the 5GBASE-R, 10GBASE-R, or 25GBASE-R capability is supported. If both 10GBASE-R and 10GBASE-W capability is supported, then this register may either ignore writes and return zeros for reads when in 10GBASE-W mode or may function as defined for 10GBASE-R. For each seed register, seed bits are assigned to register bits in order with the lowest numbered seed bit for that register being assigned to register bit 0.

**Table 45–246—5/10/25GBASE-R PCS test pattern seed A 0-3 register bit definitions**

| Bit(s)     | Name                  | Description                    | R/W <sup>a</sup> |
|------------|-----------------------|--------------------------------|------------------|
| 3.37.15:10 | Reserved              | Value always 0                 | RO               |
| 3.37.9:0   | Test pattern seed A 3 | Test pattern seed A bits 48-57 | R/W              |
| 3.36.15:0  | Test pattern seed A 2 | Test pattern seed A bits 32-47 | R/W              |

**Table 45–246—5/10/25GBASE-R PCS test pattern seed A 0-3 register bit definitions (*continued*)**

| Bit(s)    | Name                  | Description                    | R/W <sup>a</sup> |
|-----------|-----------------------|--------------------------------|------------------|
| 3.35.15:0 | Test pattern seed A 1 | Test pattern seed A bits 16-31 | R/W              |
| 3.34.15:0 | Test pattern seed A 0 | Test pattern seed A bits 0-15  | R/W              |

<sup>a</sup>RO = Read only, R/W = Read/Write

The A seed for the pseudo random test pattern is held in registers 3.34 through 3.37. The test-pattern methodology is described in 49.2.8.

#### 45.2.3.18 5/10/25GBASE-R PCS test pattern seed B (Registers 3.38 through 3.41)

The assignment of bits in the 5/10/25GBASE-R PCS test pattern seed B registers is shown in Table 45–247. This register is only required when the 5GBASE-R, 10GBASE-R, or 25GBASE-R capability is supported. If both 10GBASE-R and 10GBASE-W capability is supported, then this register may either ignore writes and return zeros for reads when in 10GBASE-W mode or may function as defined for 10GBASE-R. For each seed register, seed bits are assigned to register bits in order with the lowest numbered seed bit for that register being assigned to register bit 0.

**Table 45–247—5/10/25GBASE-R PCS test pattern seed B 0-3 register bit definitions**

| Bit(s)     | Name                  | Description                    | R/W <sup>a</sup> |
|------------|-----------------------|--------------------------------|------------------|
| 3.41.15:10 | Reserved              | Value always 0                 | RO               |
| 3.41.9:0   | Test pattern seed B 3 | Test pattern seed B bits 48-57 | R/W              |
| 3.40.15:0  | Test pattern seed B 2 | Test pattern seed B bits 32-47 | R/W              |
| 3.39.15:0  | Test pattern seed B 1 | Test pattern seed B bits 16-31 | R/W              |
| 3.38.15:0  | Test pattern seed B 0 | Test pattern seed B bits 0-15  | R/W              |

<sup>a</sup>RO = Read only, R/W = Read/Write

The B seed for the pseudo random test pattern is held in registers 3.38 through 3.41. The test-pattern methodology is described in 49.2.8.

#### 45.2.3.19 BASE-R PCS test-pattern control register (Register 3.42)

The assignment of bits in the BASE-R PCS test-pattern control register is shown in Table 45–248. This register is only required when the BASE-R capability is supported. If both BASE-R and 10GBASE-W capability is supported, then this register may either ignore writes and return zeros for reads when in 10GBASE-W mode or may function as defined for BASE-R. PRBS9, PRBS31, pseudo random, and square wave test patterns are defined for 10GBASE-R PCS only. Scrambled idle test patterns are defined for 25/40/50/100/200/400GBASE-R PCS only. The test-pattern methodology is described in 49.2.8 and 82.2.11.

**Table 45–248—BASE-R PCS test-pattern control register bit definitions**

| Bit(s)    | Name                                                       | Description                                                                                                           | R/W <sup>a</sup> |
|-----------|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|------------------|
| 3.42.15:8 | Reserved                                                   | Value always 0                                                                                                        | RO               |
| 3.42.7    | Scrambled idle test-pattern enable                         | 1 = Enable scrambled idle test-pattern mode<br>0 = Disable scrambled idle test-pattern mode                           | R/W              |
| 3.42.6    | Single Lane PHY BASE-R PRBS9 transmit test-pattern enable  | 1 = Enable PRBS9 test-pattern mode on the transmit path<br>0 = Disable PRBS9 test-pattern mode on the transmit path   | R/W              |
| 3.42.5    | Single Lane PHY BASE-R PRBS31 receive test-pattern enable  | 1 = Enable PRBS31 test-pattern mode on the receive path<br>0 = Disable PRBS31 test-pattern mode on the receive path   | R/W              |
| 3.42.4    | Single Lane PHY BASE-R PRBS31 transmit test-pattern enable | 1 = Enable PRBS31 test-pattern mode on the transmit path<br>0 = Disable PRBS31 test-pattern mode on the transmit path | R/W              |
| 3.42.3    | Transmit test-pattern enable                               | 1 = Enable transmit test pattern<br>0 = Disable transmit test pattern                                                 | R/W              |
| 3.42.2    | Receive test-pattern enable                                | 1 = Enable receive test-pattern testing<br>0 = Disable receive test-pattern testing                                   | R/W              |
| 3.42.1    | Test-pattern select                                        | 1 = Square wave test pattern<br>0 = Pseudo random test pattern                                                        | R/W              |
| 3.42.0    | Data pattern select                                        | 1 = Zeros data pattern<br>0 = LF data pattern                                                                         | R/W              |

<sup>a</sup>RO = Read only, R/W = Read/Write

#### **45.2.3.19.1 Scrambled idle test-pattern enable (3.42.7)**

When bit 3.42.7 is set to a one, scrambled idle pattern testing is enabled. When bit 3.42.7 is set to zero, scrambled idle pattern testing is disabled.

The default value for bit 3.42.3 is zero.

#### **45.2.3.19.2 Single Lane PHY BASE-R PRBS9 transmit test-pattern enable (3.42.6)**

If the PCS supports the optional PRBS9 pattern testing (indicated by bit 3.32.3), the mandatory transmit test-pattern enable bit (3.42.3) is not one, and the optional PRBS31 transmit test-pattern enable bit (3.42.4) is not one, then when bit 3.42.6 is set to one the PCS shall transmit PRBS9. When bit 3.42.6 is set to zero, the PCS shall not generate PRBS9. The PRBS9 test-pattern is specified in 68.6.1. The behavior of the PCS when in PRBS9 test-pattern mode is specified in Clause 49.

#### **45.2.3.19.3 Single Lane PHY BASE-R PRBS31 receive test-pattern enable (3.42.5)**

If the PCS supports the optional PRBS31 pattern testing advertised in bit 3.32.2 and the mandatory receive test-pattern enable bit (3.42.2) is not one, setting bit 3.42.5 to a one shall set the receive path of the PCS into the PRBS31 test-pattern mode. The number of errors received during a PRBS31 pattern test are recorded in register 3.43. Setting bit 3.42.5 to a zero shall disable the PRBS31 test-pattern mode on the receive path of the PCS. The behavior of the PCS when in PRBS31 test-pattern mode is specified in Clause 49.

#### 45.2.3.19.4 Single Lane PHY BASE-R PRBS31 transmit test-pattern enable (3.42.4)

If the PCS supports the optional PRBS31 pattern testing advertised in bit 3.32.2 and the mandatory transmit test-pattern enable bit (3.42.3) is not one, then setting bit 3.42.4 to a one shall set the transmit path of the PCS into the PRBS31 test-pattern mode. Setting bit 3.42.4 to a zero shall disable the PRBS31 test-pattern mode on the transmit path of the PCS. The behavior of the PCS when in PRBS31 test-pattern mode is specified in Clause 49.

#### 45.2.3.19.5 Transmit test-pattern enable (3.42.3)

When bit 3.42.3 is set to a one, pattern testing is enabled on the transmit path. When bit 3.42.3 is set to a zero, pattern testing is disabled on the transmit path.

The default value for bit 3.42.3 is zero.

#### 45.2.3.19.6 Receive test-pattern enable (3.42.2)

When bit 3.42.2 is set to a one, pattern testing is enabled on the receive path. When bit 3.42.2 is set to a zero, pattern testing is disabled on the receive path.

The default value for bit 3.42.2 is zero.

#### 45.2.3.19.7 Test-pattern select (3.42.1)

When bit 3.42.1 is set to a one, the square wave test pattern is used for pattern testing. When bit 3.42.1 is set to a zero, the pseudo random test pattern is used for pattern testing.

The default value for bit 3.42.1 is zero.

#### 45.2.3.19.8 Data pattern select (3.42.0)

When bit 3.42.0 is set to a one, the zeros data pattern is used for pattern testing. When bit 3.42.0 is set to a zero, the LF data pattern is used for pattern testing.

The default value for bit 3.42.0 is zero.

#### 45.2.3.20 BASE-R PCS test-pattern error counter register (Register 3.43)

The assignment of bits in the BASE-R PCS test-pattern error counter register is shown in Table 45–249. This register is only required when the BASE-R capability is supported. If both BASE-R and 10GBASE-W capability is supported, then this register may either ignore writes and return zeros for reads when in 10GBASE-W mode, or may function as defined for BASE-R.

**Table 45–249—BASE-R PCS test-pattern error counter register bit definitions**

| Bit(s)    | Name                       | Description   | R/W <sup>a</sup> |
|-----------|----------------------------|---------------|------------------|
| 3.43.15:0 | Test-pattern error counter | Error counter | RO               |

<sup>a</sup>RO = Read only

The test-pattern error counter is a sixteen bit counter that contains the number of errors received during a pattern test. These bits shall be reset to all zeros when the test-pattern error counter is read by the



management function or upon execution of the PCS reset. These bits shall be held at all ones in the case of overflow. The test-pattern methodology is described in 49.2.12 and 82.2.11. This counter will count either block errors or bit errors dependent on the test mode (see 49.2.12).

#### 45.2.3.21 BER high order counter (Register 3.44)

The assignment of bits in the BER high order counter register is shown in Table 45–250. This register is mandatory when the 40/100GBASE-R capability is supported and optional for other PHY types that implement register 3.33.

**Table 45–250—BER high order counter register bit definitions**

| Bit(s)    | Name           | Description              | R/W <sup>a</sup> |
|-----------|----------------|--------------------------|------------------|
| 3.44.15:0 | BER high order | Bits 21:6 of BER counter | RO               |

<sup>a</sup>RO = Read only

Bits 15:0 of this register are concatenated with bits 13:8 of register 3.33 (see 45.2.3.16) to read the value of a twenty-two bit BER counter. When registers 3.44 and 3.33 are used to read the 22-bit counter value, the register 3.33 is read first (indicating the lower 6 bits). The remaining 16 bits shall be latched when register 3.33 is read and reads to register 3.44 return the latched value. The counter continues to function as defined in 45.2.3.16.3 regardless of the state of the latched register. The 22-bit counter shall be reset to all zeros when register 3.33 is read or upon PCS reset. The 22-bit counter shall be held at all ones in the case of overflow.

#### 45.2.3.22 Errored blocks high order counter (Register 3.45)

The assignment of bits in the Errored blocks high order counter register is shown in Table 45–251. This register is mandatory when the 40/100GBASE-R capability is supported and optional for other PHY types that implement register 3.33. If this register is implemented then bit 3.45.15 shall be set to one.

**Table 45–251—Errored blocks high order counter register bit definitions**

| Bit(s)    | Name                       | Description                                         | R/W <sup>a</sup> |
|-----------|----------------------------|-----------------------------------------------------|------------------|
| 3.45.15   | High order counter present | Always reads as one if this register is implemented | RO               |
| 3.45.14   | Reserved                   | Value always 0                                      | RO               |
| 3.45.13:0 | Errored blocks high order  | Bits 21:8 of errored blocks counter                 | RO               |

<sup>a</sup>RO = Read only

Bits 13:0 of this register are concatenated with bits 7:0 of register 3.33 (see 45.2.3.16) to read the value of a twenty-two bit errored blocks counter. When registers 3.45 and 3.33 are used to read the 22-bit counter value, the register 3.33 is read first (indicating the lower 8 bits). The remaining 14 bits shall be latched when register 3.33 is read and reads to register 3.45 return the latched value. The counter continues to function as defined in 45.2.3.16.4 regardless of the state of the latched register. The 22-bit counter shall be reset to all zeros when register 3.33 is read or upon PCS reset. The 22-bit counter shall be held at all ones in the case of overflow.

### 45.2.3.23 Multi-lane BASE-R PCS alignment status 1 register (Register 3.50)

The assignment of bits in the multi-lane BASE-R PCS alignment status 1 register is shown in Table 45–252. All the bits in the multi-lane BASE-R PCS alignment status 1 register are read only; a write to the multi-lane BASE-R PCS alignment status 1 register shall have no effect. A PCS device that does not implement multi-lane BASE-R PCS shall return a zero for all bits in the multi-lane BASE-R PCS alignment status 1 register. A device that implements multi-lane BASE-R PCS shall return a zero for all bits in the multi-lane BASE-R PCS alignment status 1 register that are not required for the PCS configuration. It is the responsibility of the STA management entity to ensure that a port type is supported by all MMDs before interrogating any of its status bits.

**Table 45–252—Multi-lane BASE-R PCS alignment status 1 register bit definitions**

| Bit(s)     | Name                      | Description                                                                              | R/W <sup>a</sup> |
|------------|---------------------------|------------------------------------------------------------------------------------------|------------------|
| 3.50.15:13 | Reserved                  | Value always 0                                                                           | RO               |
| 3.50.12    | PCS lane alignment status | 1 = PCS receive lanes locked and aligned<br>0 = PCS receive lanes not locked and aligned | RO               |
| 3.50.11:8  | Reserved                  | Value always 0                                                                           | RO               |
| 3.50.7     | Block lock 7              | 1 = Lane 7 is locked<br>0 = Lane 7 is not locked                                         | RO               |
| 3.50.6     | Block lock 6              | 1 = Lane 6 is locked<br>0 = Lane 6 is not locked                                         | RO               |
| 3.50.5     | Block lock 5              | 1 = Lane 5 is locked<br>0 = Lane 5 is not locked                                         | RO               |
| 3.50.4     | Block lock 4              | 1 = Lane 4 is locked<br>0 = Lane 4 is not locked                                         | RO               |
| 3.50.3     | Block lock 3              | 1 = Lane 3 is locked<br>0 = Lane 3 is not locked                                         | RO               |
| 3.50.2     | Block lock 2              | 1 = Lane 2 is locked<br>0 = Lane 2 is not locked                                         | RO               |
| 3.50.1     | Block lock 1              | 1 = Lane 1 is locked<br>0 = Lane 1 is not locked                                         | RO               |
| 3.50.0     | Block lock 0              | 1 = Lane 0 is locked<br>0 = Lane 0 is not locked                                         | RO               |

<sup>a</sup>RO = Read only

#### 45.2.3.23.1 Multi-lane BASE-R PCS alignment status (3.50.12)

When read as a one, bit 3.50.12 indicates that the PCS has locked and aligned all receive lanes. When read as a zero, bit 3.50.12 indicates that the PCS has not locked and aligned all receive lanes.

#### **45.2.3.23.2 Block lock 7 (3.50.7)**

When read as a one, bit 3.50.7 indicates that the PCS receiver has achieved block lock for service interface lane 7. When read as a zero, bit 3.50.7 indicates that the PCS receiver lane 7 has not achieved block lock. This bit reflects the state of `block_lock[7]` (see 82.2.19.2.2).

#### **45.2.3.23.3 Block lock 6 (3.50.6)**

When read as a one, bit 3.50.6 indicates that the PCS receiver has achieved block lock for service interface lane 6. When read as a zero, bit 3.50.6 indicates that the PCS receiver lane 6 has not achieved block lock. This bit reflects the state of `block_lock[6]` (see 82.2.19.2.2).

#### **45.2.3.23.4 Block lock 5 (3.50.5)**

When read as a one, bit 3.50.5 indicates that the PCS receiver has achieved block lock for service interface lane 5. When read as a zero, bit 3.50.5 indicates that the PCS receiver lane 5 has not achieved block lock. This bit reflects the state of `block_lock[5]` (see 82.2.19.2.2).

#### **45.2.3.23.5 Block lock 4 (3.50.4)**

When read as a one, bit 3.50.4 indicates that the PCS receiver has achieved block lock for service interface lane 4. When read as a zero, bit 3.50.4 indicates that the PCS receiver lane 4 has not achieved block lock. This bit reflects the state of `block_lock[4]` (see 82.2.19.2.2).

#### **45.2.3.23.6 Block lock 3 (3.50.3)**

When read as a one, bit 3.50.3 indicates that the PCS receiver has achieved block lock for service interface lane 3. When read as a zero, bit 3.50.3 indicates that the PCS receiver lane 3 has not achieved block lock. This bit reflects the state of `block_lock[3]` (see 82.2.19.2.2).

#### **45.2.3.23.7 Block lock 2 (3.50.2)**

When read as a one, bit 3.50.2 indicates that the PCS receiver has achieved block lock for service interface lane 2. When read as a zero, bit 3.50.2 indicates that the PCS receiver lane 2 has not achieved block lock. This bit reflects the state of `block_lock[2]` (see 82.2.19.2.2).

#### **45.2.3.23.8 Block lock 1 (3.50.1)**

When read as a one, bit 3.50.1 indicates that the PCS receiver has achieved block lock for service interface lane 1. When read as a zero, bit 3.50.1 indicates that the PCS receiver lane 1 has not achieved block lock. This bit reflects the state of `block_lock[1]` (see 82.2.19.2.2).

#### **45.2.3.23.9 Block lock 0 (3.50.0)**

When read as a one, bit 3.50.0 indicates that the PCS receiver has achieved block lock for service interface lane 0. When read as a zero, bit 3.50.0 indicates that the PCS receiver lane 0 has not achieved block lock. This bit reflects the state of `block_lock[0]` (see 82.2.19.2.2).

#### **45.2.3.24 Multi-lane BASE-R PCS alignment status 2 register (Register 3.51)**

The assignment of bits in the multi-lane BASE-R PCS alignment status 2 register is shown in Table 45–253. All the bits in the multi-lane BASE-R PCS alignment status 2 register are read only; a write to the multi-lane BASE-R PCS alignment status 2 register shall have no effect. A PCS device that does not implement multi-lane BASE-R PCS shall return a zero for all bits in the multi-lane BASE-R PCS alignment status 2

register. A device that implements multi-lane BASE-R PCS shall return a zero for all bits in the multi-lane BASE-R PCS alignment status 2 register that are not required for the PCS configuration. It is the responsibility of the STA management entity to ensure that a port type is supported by all MMDs before interrogating any of its status bits.

**Table 45–253—Multi-lane BASE-R PCS alignment status 2 register bit definitions**

| Bit(s)     | Name          | Description                                        | R/W <sup>a</sup> |
|------------|---------------|----------------------------------------------------|------------------|
| 3.51.15:12 | Reserved      | Value always 0                                     | RO               |
| 3.51.11    | Block lock 19 | 1 = Lane 19 is locked<br>0 = Lane 19 is not locked | RO               |
| 3.51.10    | Block lock 18 | 1 = Lane 18 is locked<br>0 = Lane 18 is not locked | RO               |
| 3.51.9     | Block lock 17 | 1 = Lane 17 is locked<br>0 = Lane 17 is not locked | RO               |
| 3.51.8     | Block lock 16 | 1 = Lane 16 is locked<br>0 = Lane 16 is not locked | RO               |
| 3.51.7     | Block lock 15 | 1 = Lane 15 is locked<br>0 = Lane 15 is not locked | RO               |
| 3.51.6     | Block lock 14 | 1 = Lane 14 is locked<br>0 = Lane 14 is not locked | RO               |
| 3.51.5     | Block lock 13 | 1 = Lane 13 is locked<br>0 = Lane 13 is not locked | RO               |
| 3.51.4     | Block lock 12 | 1 = Lane 12 is locked<br>0 = Lane 12 is not locked | RO               |
| 3.51.3     | Block lock 11 | 1 = Lane 11 is locked<br>0 = Lane 11 is not locked | RO               |
| 3.51.2     | Block lock 10 | 1 = Lane 10 is locked<br>0 = Lane 10 is not locked | RO               |
| 3.51.1     | Block lock 9  | 1 = Lane 9 is locked<br>0 = Lane 9 is not locked   | RO               |
| 3.51.0     | Block lock 8  | 1 = Lane 8 is locked<br>0 = Lane 8 is not locked   | RO               |

<sup>a</sup>RO = Read only

#### 45.2.3.24.1 Block lock 19 (3.51.11)

When read as a one, bit 3.51.11 indicates that the PCS receiver has achieved block lock for service interface lane 19. When read as a zero, bit 3.51.11 indicates that the PCS receiver lane 19 has not achieved block lock. This bit reflects the state of block\_lock[19] (see 82.2.19.2.2).

#### **45.2.3.24.2 Block lock 18 (3.51.10)**

When read as a one, bit 3.51.10 indicates that the PCS receiver has achieved block lock for service interface lane 18. When read as a zero, bit 3.51.10 indicates that the PCS receiver lane 18 has not achieved block lock. This bit reflects the state of `block_lock[18]` (see 82.2.19.2.2).

#### **45.2.3.24.3 Block lock 17 (3.51.9)**

When read as a one, bit 3.51.9 indicates that the PCS receiver has achieved block lock for service interface lane 17. When read as a zero, bit 3.51.9 indicates that the PCS receiver lane 17 has not achieved block lock. This bit reflects the state of `block_lock[17]` (see 82.2.19.2.2).

#### **45.2.3.24.4 Block lock 16 (3.51.8)**

When read as a one, bit 3.51.8 indicates that the PCS receiver has achieved block lock for service interface lane 16. When read as a zero, bit 3.51.8 indicates that the PCS receiver lane 16 has not achieved block lock. This bit reflects the state of `block_lock[16]` (see 82.2.19.2.2).

#### **45.2.3.24.5 Block lock 15 (3.51.7)**

When read as a one, bit 3.51.7 indicates that the PCS receiver has achieved block lock for service interface lane 15. When read as a zero, bit 3.51.7 indicates that the PCS receiver lane 15 has not achieved block lock. This bit reflects the state of `block_lock[15]` (see 82.2.19.2.2).

#### **45.2.3.24.6 Block lock 14 (3.51.6)**

When read as a one, bit 3.51.6 indicates that the PCS receiver has achieved block lock for service interface lane 14. When read as a zero, bit 3.51.6 indicates that the PCS receiver lane 14 has not achieved block lock. This bit reflects the state of `block_lock[14]` (see 82.2.19.2.2).

#### **45.2.3.24.7 Block lock 13 (3.51.5)**

When read as a one, bit 3.51.5 indicates that the PCS receiver has achieved block lock for service interface lane 13. When read as a zero, bit 3.51.5 indicates that the PCS receiver lane 13 has not achieved block lock. This bit reflects the state of `block_lock[13]` (see 82.2.19.2.2).

#### **45.2.3.24.8 Block lock 12 (3.51.4)**

When read as a one, bit 3.51.4 indicates that the PCS receiver has achieved block lock for service interface lane 12. When read as a zero, bit 3.51.4 indicates that the PCS receiver lane 12 has not achieved block lock. This bit reflects the state of `block_lock[12]` (see 82.2.19.2.2).

#### **45.2.3.24.9 Block lock 11 (3.51.3)**

When read as a one, bit 3.51.3 indicates that the PCS receiver has achieved block lock for service interface lane 11. When read as a zero, bit 3.51.3 indicates that the PCS receiver lane 11 has not achieved block lock. This bit reflects the state of `block_lock[11]` (see 82.2.19.2.2).

#### **45.2.3.24.10 Block lock 10 (3.51.2)**

When read as a one, bit 3.51.2 indicates that the PCS receiver has achieved block lock for service interface lane 10. When read as a zero, bit 3.51.2 indicates that the PCS receiver lane 10 has not achieved block lock. This bit reflects the state of `block_lock[10]` (see 82.2.19.2.2).

#### 45.2.3.24.11 Block lock 9 (3.51.1)

When read as a one, bit 3.51.1 indicates that the PCS receiver has achieved block lock for service interface lane 9. When read as a zero, bit 3.51.1 indicates that the PCS receiver lane 9 has not achieved block lock. This bit reflects the state of block\_lock[9] (see 82.2.19.2.2).

#### 45.2.3.24.12 Block lock 8 (3.51.0)

When read as a one, bit 3.51.0 indicates that the PCS receiver has achieved block lock for service interface lane 8. When read as a zero, bit 3.51.0 indicates that the PCS receiver lane 8 has not achieved block lock. This bit reflects the state of block\_lock[8] (see 82.2.19.2.2).

#### 45.2.3.25 Multi-lane BASE-R PCS alignment status 3 register (Register 3.52)

The assignment of bits in the multi-lane BASE-R PCS alignment status 3 register is shown in Table 45–254. All the bits in the multi-lane BASE-R PCS alignment status 3 register are read only; a write to the multi-lane BASE-R PCS alignment status 3 register shall have no effect. A PCS device that does not implement multi-lane BASE-R PCS shall return a zero for all bits in the multi-lane BASE-R PCS alignment status 3 register. A device that implements multi-lane BASE-R PCS shall return a zero for all bits in the multi-lane BASE-R PCS alignment status 3 register that are not required for the PCS configuration. It is the responsibility of the STA management entity to ensure that a port type is supported by all MMDs before interrogating any of its status bits.

**Table 45–254—Multi-lane BASE-R PCS alignment status 3 register  
bit definitions**

| Bit(s)    | Name           | Description                                                                        | R/W <sup>a</sup> |
|-----------|----------------|------------------------------------------------------------------------------------|------------------|
| 3.52.15:8 | Reserved       | Value always 0                                                                     | RO               |
| 3.52.7    | Lane 7 aligned | 1 = Lane 7 alignment marker is locked<br>0 = Lane 7 alignment marker is not locked | RO               |
| 3.52.6    | Lane 6 aligned | 1 = Lane 6 alignment marker is locked<br>0 = Lane 6 alignment marker is not locked | RO               |
| 3.52.5    | Lane 5 aligned | 1 = Lane 5 alignment marker is locked<br>0 = Lane 5 alignment marker is not locked | RO               |
| 3.52.4    | Lane 4 aligned | 1 = Lane 4 alignment marker is locked<br>0 = Lane 4 alignment marker is not locked | RO               |
| 3.52.3    | Lane 3 aligned | 1 = Lane 3 alignment marker is locked<br>0 = Lane 3 alignment marker is not locked | RO               |
| 3.52.2    | Lane 2 aligned | 1 = Lane 2 alignment marker is locked<br>0 = Lane 2 alignment marker is not locked | RO               |
| 3.52.1    | Lane 1 aligned | 1 = Lane 1 alignment marker is locked<br>0 = Lane 1 alignment marker is not locked | RO               |
| 3.52.0    | Lane 0 aligned | 1 = Lane 0 alignment marker is locked<br>0 = Lane 0 alignment marker is not locked | RO               |

<sup>a</sup>RO = Read only

#### **45.2.3.25.1 Lane 7 aligned (3.52.7)**

When read as a one, bit 3.52.7 indicates that the PCS receiver has achieved alignment marker lock for service interface lane 7. When read as a zero, bit 3.52.7 indicates that the PCS receiver lane 7 has not achieved alignment marker lock. This bit reflects the state of `am_lock[7]` (see 82.2.19.2.2).

#### **45.2.3.25.2 Lane 6 aligned (3.52.6)**

When read as a one, bit 3.52.6 indicates that the PCS receiver has achieved alignment marker lock for service interface lane 6. When read as a zero, bit 3.52.6 indicates that the PCS receiver lane 6 has not achieved alignment marker lock. This bit reflects the state of `am_lock[6]` (see 82.2.19.2.2).

#### **45.2.3.25.3 Lane 5 aligned (3.52.5)**

When read as a one, bit 3.52.5 indicates that the PCS receiver has achieved alignment marker lock for service interface lane 5. When read as a zero, bit 3.52.5 indicates that the PCS receiver lane 5 has not achieved alignment marker lock. This bit reflects the state of `am_lock[5]` (see 82.2.19.2.2).

#### **45.2.3.25.4 Lane 4 aligned (3.52.4)**

When read as a one, bit 3.52.4 indicates that the PCS receiver has achieved alignment marker lock for service interface lane 4. When read as a zero, bit 3.52.4 indicates that the PCS receiver lane 4 has not achieved alignment marker lock. This bit reflects the state of `am_lock[4]` (see 82.2.19.2.2).

#### **45.2.3.25.5 Lane 3 aligned (3.52.3)**

When read as a one, bit 3.52.3 indicates that the PCS receiver has achieved alignment marker lock for service interface lane 3. When read as a zero, bit 3.52.3 indicates that the PCS receiver lane 3 has not achieved alignment marker lock. This bit reflects the state of `am_lock[3]` (see 82.2.19.2.2).

#### **45.2.3.25.6 Lane 2 aligned (3.52.2)**

When read as a one, bit 3.52.2 indicates that the PCS receiver has achieved alignment marker lock for service interface lane 2. When read as a zero, bit 3.52.2 indicates that the PCS receiver lane 2 has not achieved alignment marker lock. This bit reflects the state of `am_lock[2]` (see 82.2.19.2.2).

#### **45.2.3.25.7 Lane 1 aligned (3.52.1)**

When read as a one, bit 3.52.1 indicates that the PCS receiver has achieved alignment marker lock for service interface lane 1. When read as a zero, bit 3.52.1 indicates that the PCS receiver lane 1 has not achieved alignment marker lock. This bit reflects the state of `am_lock[1]` (see 82.2.19.2.2).

#### **45.2.3.25.8 Lane 0 aligned (3.52.0)**

When read as a one, bit 3.52.0 indicates that the PCS receiver has achieved alignment marker lock for service interface lane 0. When read as a zero, bit 3.52.0 indicates that the PCS receiver lane 0 has not achieved alignment marker lock. This bit reflects the state of `am_lock[0]` (see 82.2.19.2.2).

#### **45.2.3.26 Multi-lane BASE-R PCS alignment status 4 register (Register 3.53)**

The assignment of bits in the multi-lane BASE-R PCS alignment status 4 register is shown in Table 45–255. All the bits in the multi-lane BASE-R PCS alignment status 4 register are read only; a write to the multi-lane BASE-R PCS alignment status 4 register shall have no effect. A PCS device that does not implement multi-lane BASE-R PCS shall return a zero for all bits in the multi-lane BASE-R PCS alignment status 4

register. A device that implements multi-lane BASE-R PCS shall return a zero for all bits in the multi-lane BASE-R PCS alignment status 4 register that are not required for the PCS configuration. It is the responsibility of the STA management entity to ensure that a port type is supported by all MMDs before interrogating any of its status bits.

**Table 45–255—Multi-lane BASE-R PCS alignment status 4 register  
bit definitions**

| Bit(s)     | Name            | Description                                                                          | R/W <sup>a</sup> |
|------------|-----------------|--------------------------------------------------------------------------------------|------------------|
| 3.53.15:12 | Reserved        | Value always 0                                                                       | RO               |
| 3.53.11    | Lane 19 aligned | 1 = Lane 19 alignment marker is locked<br>0 = Lane 19 alignment marker is not locked | RO               |
| 3.53.10    | Lane 18 aligned | 1 = Lane 18 alignment marker is locked<br>0 = Lane 18 alignment marker is not locked | RO               |
| 3.53.9     | Lane 17 aligned | 1 = Lane 17 alignment marker is locked<br>0 = Lane 17 alignment marker is not locked | RO               |
| 3.53.8     | Lane 16 aligned | 1 = Lane 16 alignment marker is locked<br>0 = Lane 16 alignment marker is not locked | RO               |
| 3.53.7     | Lane 15 aligned | 1 = Lane 15 alignment marker is locked<br>0 = Lane 15 alignment marker is not locked | RO               |
| 3.53.6     | Lane 14 aligned | 1 = Lane 14 alignment marker is locked<br>0 = Lane 14 alignment marker is not locked | RO               |
| 3.53.5     | Lane 13 aligned | 1 = Lane 13 alignment marker is locked<br>0 = Lane 13 alignment marker is not locked | RO               |
| 3.53.4     | Lane 12 aligned | 1 = Lane 12 alignment marker is locked<br>0 = Lane 12 alignment marker is not locked | RO               |
| 3.53.3     | Lane 11 aligned | 1 = Lane 11 alignment marker is locked<br>0 = Lane 11 alignment marker is not locked | RO               |
| 3.53.2     | Lane 10 aligned | 1 = Lane 10 alignment marker is locked<br>0 = Lane 10 alignment marker is not locked | RO               |
| 3.53.1     | Lane 9 aligned  | 1 = Lane 9 alignment marker is locked<br>0 = Lane 9 alignment marker is not locked   | RO               |
| 3.53.0     | Lane 8 aligned  | 1 = Lane 8 alignment marker is locked<br>0 = Lane 8 alignment marker is not locked   | RO               |

<sup>a</sup>RO = Read only

#### 45.2.3.26.1 Lane 19 aligned (3.53.11)

When read as a one, bit 3.53.11 indicates that the PCS receiver has achieved alignment marker lock for service interface lane 19. When read as a zero, bit 3.53.11 indicates that the PCS receiver lane 19 has not achieved alignment marker lock. This bit reflects the state of `am_lock[19]` (see 82.2.19.2.2).



#### **45.2.3.26.2 Lane 18 aligned (3.53.10)**

When read as a one, bit 3.53.10 indicates that the PCS receiver has achieved alignment marker lock for service interface lane 18. When read as a zero, bit 3.53.10 indicates that the PCS receiver lane 18 has not achieved alignment marker lock. This bit reflects the state of `am_lock[18]` (see 82.2.19.2.2).

#### **45.2.3.26.3 Lane 17 aligned (3.53.9)**

When read as a one, bit 3.53.9 indicates that the PCS receiver has achieved alignment marker lock for service interface lane 17. When read as a zero, bit 3.53.9 indicates that the PCS receiver lane 17 has not achieved alignment marker lock. This bit reflects the state of `am_lock[17]` (see 82.2.19.2.2).

#### **45.2.3.26.4 Lane 16 aligned (3.53.8)**

When read as a one, bit 3.53.8 indicates that the PCS receiver has achieved alignment marker lock for service interface lane 16. When read as a zero, bit 3.53.8 indicates that the PCS receiver lane 16 has not achieved alignment marker lock. This bit reflects the state of `am_lock[16]` (see 82.2.19.2.2).

#### **45.2.3.26.5 Lane 15 aligned (3.53.7)**

When read as a one, bit 3.53.7 indicates that the PCS receiver has achieved alignment marker lock for service interface lane 15. When read as a zero, bit 3.53.7 indicates that the PCS receiver lane 15 has not achieved alignment marker lock. This bit reflects the state of `am_lock[15]` (see 82.2.19.2.2).

#### **45.2.3.26.6 Lane 14 aligned (3.53.6)**

When read as a one, bit 3.53.6 indicates that the PCS receiver has achieved alignment marker lock for service interface lane 14. When read as a zero, bit 3.53.6 indicates that the PCS receiver lane 14 has not achieved alignment marker lock. This bit reflects the state of `am_lock[14]` (see 82.2.19.2.2).

#### **45.2.3.26.7 Lane 13 aligned (3.53.5)**

When read as a one, bit 3.53.5 indicates that the PCS receiver has achieved alignment marker lock for service interface lane 13. When read as a zero, bit 3.53.5 indicates that the PCS receiver lane 13 has not achieved alignment marker lock. This bit reflects the state of `am_lock[13]` (see 82.2.19.2.2).

#### **45.2.3.26.8 Lane 12 aligned (3.53.4)**

When read as a one, bit 3.53.4 indicates that the PCS receiver has achieved alignment marker lock for service interface lane 12. When read as a zero, bit 3.53.4 indicates that the PCS receiver lane 12 has not achieved alignment marker lock. This bit reflects the state of `am_lock[12]` (see 82.2.19.2.2).

#### **45.2.3.26.9 Lane 11 aligned (3.53.3)**

When read as a one, bit 3.53.3 indicates that the PCS receiver has achieved alignment marker lock for service interface lane 11. When read as a zero, bit 3.53.3 indicates that the PCS receiver lane 11 has not achieved alignment marker lock. This bit reflects the state of `am_lock[11]` (see 82.2.19.2.2).

#### **45.2.3.26.10 Lane 10 aligned (3.53.2)**

When read as a one, bit 3.53.2 indicates that the PCS receiver has achieved alignment marker lock for service interface lane 10. When read as a zero, bit 3.53.2 indicates that the PCS receiver lane 10 has not achieved alignment marker lock. This bit reflects the state of `am_lock[10]` (see 82.2.19.2.2).

#### 45.2.3.26.11 Lane 9 aligned (3.53.1)

When read as a one, bit 3.53.1 indicates that the PCS receiver has achieved alignment marker lock for service interface lane 9. When read as a zero, bit 3.53.1 indicates that the PCS receiver lane 9 has not achieved alignment marker lock. This bit reflects the state of `am_lock[9]` (see 82.2.19.2.2).

#### 45.2.3.26.12 Lane 8 aligned (3.53.0)

When read as a one, bit 3.53.0 indicates that the PCS receiver has achieved alignment marker lock for service interface lane 8. When read as a zero, bit 3.53.0 indicates that the PCS receiver lane 8 has not achieved alignment marker lock. This bit reflects the state of `am_lock[8]` (see 82.2.19.2.2).

#### 45.2.3.27 10P/2B capability register (3.60)

The 10P/2B capability register reports which functions are supported by the PCS. This register is present at the PCS layer for each PHY. The bit definitions of the 10P/2B capability register are shown in Table 45–256.

**Table 45–256—10P/2B capability register bit definitions**

| Bit(s)     | Name                 | Description                                                            | R/W <sup>a</sup> |
|------------|----------------------|------------------------------------------------------------------------|------------------|
| 3.60.15:13 | Reserved             | Value always 0                                                         | RO               |
| 3.60.12    | PAF available        | 1 = PAF supported<br>0 = PAF not supported                             | RO               |
| 3.60.11    | Remote PAF supported | 1 = link partner supports PAF<br>0 = link partner does not support PAF | RO               |
| 3.60.10:0  | Reserved             | Value always 0                                                         | RO               |

<sup>a</sup>RO = Read only

##### 45.2.3.27.1 PAF available (3.60.12)

This bit indicates that the PHY supports the PME aggregation function. The PHY sets this bit to a one when the capability is supported and zero otherwise. This bit reflects the signal `PAF_available` in 61.2.3.

##### 45.2.3.27.2 Remote PAF supported (3.60.11)

This bit indicates that the remote, link-partner PHY supports the PME aggregation function. The PHY sets this bit to a one when the capability is supported and zero otherwise. This bit does not accurately report the capability of the remote PCS until a remote discovery operation has been completed by the -O PHY. In this case, this bit is set if the “Ethernet bonding” `NPar(2)` bit is set in the capabilities exchange message received from the other device. See 61.4.7, which discusses use of G.994.1 to access remote registers.

#### 45.2.3.28 10P/2B PCS control register (Register 3.61)

The assignment of bits in the 10P/2B PCS control register is shown in Table 45–257.

##### 45.2.3.28.1 MII receive during transmit (3.61.15)

This register bit is used to tell the PHY-MAC rate matching function if the MAC is capable of receiving frames from the PHY while the MAC is transmitting (i.e., sending frames to the PHY). The variable

**Table 45–257—10B/2B PCS control register bit definitions**

| Bit(s)    | Name                            | Description                                                                                           | R/W <sup>a</sup> |
|-----------|---------------------------------|-------------------------------------------------------------------------------------------------------|------------------|
| 3.61.15   | MII receive during transmit     | 1 = MII can TX/RX simultaneously<br>0 = MII cannot TX/RX simultaneously (default)                     | R/W              |
| 3.61.14   | TX_EN and CRS infer a collision | 1 = MII uses TX_EN and CRS to infer a collision<br>0 = MII uses COL to indicate a collision (default) | R/W              |
| 3.61.13:1 | Reserved                        | Value always 0                                                                                        | RO               |
| 3.61.0    | PAF enable                      | 1 = use PAF<br>0 = do not use PAF                                                                     | O: R/W<br>R: RO  |

<sup>a</sup>R/W = Read/Write, RO = Read only

tx\_rx\_simultaneously for the PHY-MAC Rate-Matching function takes on the value of this bit as defined in 61.2.3.

#### 45.2.3.28.2 TX\_EN and CRS infer a collision (3.61.14)

This bit is set by the STA to tell the MAC-PHY rate matching function that the MAC-PHY interface does not have a separate collision signal but instead infers a collision when TX\_EN and CRS are asserted simultaneously. The variable crs\_and\_tx\_en\_infer\_col in the PHY-MAC Rate-Matching function takes on the value of this bit as in 61.2.3. This bit will default to a supported mode, and writes to unsupported modes will be ignored.

#### 45.2.3.28.3 PAF enable (3.61.0)

Setting this bit to a one shall activate the PME aggregation function of the PCS when the link is established. Writes to this bit while link is up or initializing (see 45.2.1.28) or if the PAF is not supported shall be ignored. When link is established, handshake indicates the use of PAF to the -R PHY. This bit reflects the signal PAF\_enable in 61.2.3.

#### 45.2.3.29 10P/2B PME available (Registers 3.62 and 3.63)

The 10P/2B PME available registers are used to indicate which PMEs in the aggregation group are available to be attached to the queried PCS. A PME is marked as unavailable if the PME does not support PME aggregation or if the PME is currently marked to be aggregated with another PMD. For a device that does not support aggregation of multiple PMEs, a single bit of this register shall be set to one and all other bits cleared to zero.

These registers may be writeable for -R ports. For PMEs that may be accessed through more than one MII, the availability is limited such that no PME may be mapped to more than one MII prior to enabling the links. In this case, the reset state of the 10P/2B PME available registers shall reflect the capabilities of the device, the management entity should reset appropriate bits to meet the restriction described.

If the -R device is not capable of aggregating PMEs to multiple MIIs then these registers may be read only.

The 10P/2B PME available register shall be available per PCS. For example, a package implementing four PMEs and one MII would have only one set of 10P/2B PME available registers, addressed by a read or write to 3.62 and 3.63 on any of those PHYs.

For more information, see 61.2.2.8.3.

The assignment of bits in the 10P/2B PME available registers is shown in Table 45–258.

**Table 45–258—10P/2B PME available register bit definitions**

| Bit(s)    | Name                      | Description                                                                                           | R/W <sup>a</sup> |
|-----------|---------------------------|-------------------------------------------------------------------------------------------------------|------------------|
| 3.62.15:0 | PME [p = 31:16] available | For each bit in the sequence:<br>1 = PME[p] is available for aggregating<br>0 = PME[p] is unavailable | O: RO<br>R: R/W  |
| 3.63.15:0 | PME [p = 15:0] available  | For each bit in the sequence:<br>1 = PME[p] is available for aggregating<br>0 = PME[p] is unavailable | O: RO<br>R: R/W  |

<sup>a</sup>RO = Read only, R/W = Read/Write

#### 45.2.3.30 10P/2B PME aggregate registers (Registers 3.64 and 3.65)

The 10P/2B PME aggregate registers are used to select PMEs for aggregation. Attempts to activate aggregation with an unavailable PME (see 45.2.3.29) are ignored. The PCS shall use PME aggregation if one or more bits are set to a one and if PME aggregation is supported.

The 10P/2B PME aggregate register shall be available per PCS. For example, a package implementing four PMEs and one MII would have only one set of 10P/2B PME aggregate registers, accessed by a read or write to 3.64, 3.65 on any of those PHYs.

Upon MMD reset, these registers shall be reset to all zeros.

For more information, see 61.2.2.8.3.

The assignment of bits for the 10P/2B PME aggregate registers is shown in Table 45–259.

**Table 45–259—10P/2B PME aggregate register bit definitions**

| Bit(s)    | Name                           | Description                                                                                                     | R/W <sup>a</sup> |
|-----------|--------------------------------|-----------------------------------------------------------------------------------------------------------------|------------------|
| 3.64.15:0 | Aggregate with PME [p = 31:16] | For each bit in the sequence:<br>1 = activate aggregation with PME[p]<br>0 = deactivate aggregation with PME[p] | R/W              |
| 3.65.15:0 | Aggregate with PME [p = 15:0]  | For each bit in the sequence:<br>1 = activate aggregation with PME[p]<br>0 = deactivate aggregation with PME[p] | R/W              |

<sup>a</sup>R/W = Read/Write

#### 45.2.3.31 10P/2B PAF RX error register (Register 3.66)

The 10P/2B PAF RX error register is a 16 bit counter that contains the number of fragments that have been received across the gamma interface with RxErr asserted. The corresponding signal, TC\_PAF\_RxErrorReceived, is defined in 61.2.3. This counter is inactive when the PAF is unsupported or disabled. Upon disabling the PAF, the register retains its previous value. These bits shall be reset to all zeros

when the 10P/2B PAF RX error register is read by the management function or upon execution of the MMD reset. These bits shall be held at all ones in the case of overflow. The assignment of bits in the 10P/2B PAF RX error register is shown in Table 45–260.

**Table 45–260—10P/2B PAF RX error register bit definitions**

| Bit(s)    | Name                | Description              | R/W <sup>a</sup> |
|-----------|---------------------|--------------------------|------------------|
| 3.66.15:0 | PAF RX errors[15:0] | The bytes of the counter | RO, NR           |

<sup>a</sup>RO = Read only, NR = Non Roll-over

#### 45.2.3.32 10P/2B PAF small fragments register (Register 3.67)

The 10P/2B PAF small fragments register is a 16 bit counter that contains the number of small fragments that have been received across the gamma interface. The corresponding signal, TC\_PAF\_FragmentTooSmall, is defined in 61.2.3. These bits shall be reset to all zeros when the register is read by the management function or upon execution of the MMD reset. This counter is inactive when the PAF is unsupported or disabled. Upon disabling the PAF, the register retains its previous value. These bits shall be held at all ones in the case of overflow. The assignment of bits in the 10P/2B PAF small fragment register is shown in Table 45–261.

**Table 45–261—10P/2B PAF small fragments register bit definitions**

| Bit(s)    | Name                      | Description              | R/W <sup>a</sup> |
|-----------|---------------------------|--------------------------|------------------|
| 3.67.15:0 | PAF small fragments[15:0] | The bytes of the counter | RO, NR           |

<sup>a</sup>RO = Read only, NR = Non Roll-over

#### 45.2.3.33 10P/2B PAF large fragments register (Register 3.68)

The 10P/2B PAF large fragments register is a 16 bit counter that contains the number of large fragments that have been received across the gamma interface. The corresponding signal, TC\_PAF\_FragmentTooLarge, is defined in 61.2.3. These bits shall be reset to all zeros when the register is read by the management function or upon execution of the MMD reset. This counter is inactive when the PAF is unsupported or disabled. Upon disabling the PAF, the register retains its previous value. These bits shall be held at all ones in the case of overflow. The assignment of bits in the 10P/2B PAF large fragments register is shown in Table 45–262.

**Table 45–262—10P/2B PAF large fragments register bit definitions**

| Bit(s)    | Name                      | Description              | R/W <sup>a</sup> |
|-----------|---------------------------|--------------------------|------------------|
| 3.68.15:0 | PAF large fragments[15:0] | The bytes of the counter | RO, NR           |

<sup>a</sup>RO = Read only, NR = Non Roll-over

#### 45.2.3.34 10P/2B PAF overflow register (Register 3.69)

The 10P/2B PAF overflow register is a 16 bit counter that contains the number of fragments that have been received across the gamma interface which would have caused the receive buffer to overflow. The

corresponding signal, TC\_PAF\_Overflow, is defined in 61.2.3. These bits shall be reset to all zeros when the register is read by the management function or upon execution of the MMD reset. This counter is inactive when the PAF is unsupported or disabled. Upon disabling the PAF, the register retains its previous value. These bits shall be held at all ones in the case of overflow. The assignment of bits in the 10P/2B PAF overflow register is shown in Table 45–263.

**Table 45–263—10P/2B PAF overflow register bit definitions**

| Bit(s)    | Name                         | Description              | R/W <sup>a</sup> |
|-----------|------------------------------|--------------------------|------------------|
| 3.69.15:0 | PAF overflow fragments[15:0] | The bytes of the counter | RO, NR           |

<sup>a</sup>RO = Read only, NR = Non Roll-over

#### 45.2.3.35 10P/2B PAF bad fragments register (Register 3.70)

The 10P/2B PAF bad fragments register is a 16 bit counter that contains the number of bad fragments that have been received across the gamma interface. The corresponding signal, TC\_PAF\_BadFragmentReceived, is defined in 61.2.3. These bits shall be reset to all zeros when the register is read by the management function or upon execution of the MMD reset. This counter is inactive when the PAF is unsupported or disabled. Upon disabling the PAF, the register retains its previous value. These bits shall be held at all ones in the case of overflow. The assignment of bits in the 10P/2B PAF bad fragments register is shown in Table 45–264.

**Table 45–264—P10P/2B AF bad fragments register bit definitions**

| Bit(s)    | Name                    | Description              | R/W <sup>a</sup> |
|-----------|-------------------------|--------------------------|------------------|
| 3.70.15:0 | PAF bad fragments[15:0] | The bytes of the counter | RO, NR           |

<sup>a</sup>RO = Read only, NR = Non Roll-over

#### 45.2.3.36 10P/2B PAF lost fragments register (Register 3.71)

The 10P/2B PAF lost fragments register is a 16 bit counter that contains the number of gaps in the sequence of fragments that have been received across the gamma interface. The corresponding signal, TC\_PAF\_LostFragment, is defined in 61.2.3.

These bits shall be reset to all zeros when the register is read by the management function or upon execution of the MMD reset. This counter is inactive when the PAF is unsupported or disabled. Upon disabling the PAF, the register retains its previous value. These bits shall be held at all ones in the case of overflow. The assignment of bits in the 10P/2B PAF lost fragments register is shown in Table 45–265.

**Table 45–265—10P/2B PAF lost fragments register bit definitions**

| Bit(s)    | Name                     | Description              | R/W <sup>a</sup> |
|-----------|--------------------------|--------------------------|------------------|
| 3.71.15:0 | PAF lost fragments[15:0] | The bytes of the counter | RO, NR           |

<sup>a</sup>RO = Read only, NR = Non Roll-over

#### 45.2.3.37 10P/2B PAF lost starts of fragments register (Register 3.72)

The 10P/2B PAF lost starts of fragments register is a 16-bit counter that contains the number of missing start of fragment indicators expected by the frame assembly function. The corresponding signal, TC\_PAF\_LostStart, is defined in 61.2.3. These bits shall be reset to all zeros when the register is read by the management function or upon execution of the MMD reset. This counter is inactive when the PAF is unsupported or disabled. Upon disabling the PAF, the register retains its previous value. These bits shall be held at all ones in the case of overflow. The assignment of bits in the 10P/2B PAF lost starts of fragments register is shown in Table 45–266.

**Table 45–266—10P/2B PAF lost starts of fragments register bit definitions**

| Bit(s)    | Name                               | Description              | R/W <sup>a</sup> |
|-----------|------------------------------------|--------------------------|------------------|
| 3.72.15:0 | PAF lost starts of fragments[15:0] | The bytes of the counter | RO, NR           |

<sup>a</sup>RO = Read only, NR = Non Roll-over

#### 45.2.3.38 10P/2B PAF lost ends of fragments register (Register 3.73)

The 10P/2B PAF lost ends of fragments register is a 16 bit counter that contains the number of missing end of fragment indicators expected by the frame assembly function. The corresponding signal, TC\_PAF\_LostEnd, is defined in 61.2.3. These bits shall be reset to all zeros when the register is read by the management function or upon execution of the MMD reset. This counter is inactive when the PAF is unsupported or disabled. Upon disabling the PAF, the register retains its previous value. These bits shall be held at all ones in the case of overflow. The assignment of bits in the 10P/2B PAF lost ends of fragments register is shown in Table 45–267.

**Table 45–267—10P/2B PAF lost ends of fragments register bit definitions**

| Bit(s)    | Name                             | Description              | R/W <sup>a</sup> |
|-----------|----------------------------------|--------------------------|------------------|
| 3.73.15:0 | PAF lost ends of fragments[15:0] | The bytes of the counter | RO, NR           |

<sup>a</sup>RO = Read only, NR = Non Roll-over

#### 45.2.3.39 10GBASE-PR and 10/1GBASE-PRX FEC ability register (Register 3.74)

The assignment of bits in the 10GBASE-PR and 10/1GBASE-PRX FEC ability register is shown in Table 45–268.

**Table 45–268—10GBASE-PR and 10/1GBASE-PRX FEC ability register bit definitions**

| Bit(s)    | Name                | Description                                                                                                                                                  | R/W <sup>a</sup> |
|-----------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 3.74.15:1 | Reserved            | Value always 0                                                                                                                                               | RO               |
| 3.74.0    | 10 Gb/s FEC ability | A read of one in this bit indicates that the PCS supports the 10/1GBASE-PRX or 10GBASE-PR 10 Gb/s FEC (always reads as one for 10/1GBASE-PRX or 10GBASE-PR). | RO               |

<sup>a</sup>RO = Read only

#### 45.2.3.40 10GBASE-PR and 10/1GBASE-PRX FEC control register (Register 3.75)

The assignment of bits in the 10GBASE-PR FEC control register is shown in Table 45–269.

**Table 45–269—10GBASE-PR and 10/1GBASE-PRX FEC control register bit definitions**

| Bit(s)    | Name                        | Description                                                                                                                                                                 | R/W <sup>a</sup> |
|-----------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 3.75.15:2 | Reserved                    | Value always 0                                                                                                                                                              | RO               |
| 3.75.1    | Enable FEC error indication | A write of one to this bit configures the 10 Gb/s FEC decoder to indicate uncorrectable codeword errors to the higher layer. In a 10/1GBASE-PRX OLT, this bit is undefined. | R/W              |
| 3.75.0    | 10 Gb/s FEC enable          | Always reads as one for 10/1GBASE-PRX or 10GBASE-PR since 10 Gb/s FEC is always enabled.                                                                                    | RO               |

<sup>a</sup>RO = Read only, R/W = Read/Write

##### 45.2.3.40.1 FEC enable error indication (3.75.1)

This bit instructs the 10 Gb/s FEC decoder component of the 10GBASE-PR and 10/1GBASE-PRX PCS to indicate decoding errors to the upper layers (see 45.2.3.40 and 76.3.3.3).

When written as a one, the receiving PCS invalidates 66 bit blocks received in uncorrectable FEC codewords. As a consequence, the receiving MAC discards any packet which includes data that was received in an uncorrectable FEC codeword (even though the packet itself might or might not contain errors).

When written as a zero, the receiving PCS does not modify 66 bit blocks received in uncorrectable FEC codewords. As a consequence, the receiving MAC performs regular processing on a packet that includes data that was received in an uncorrectable FEC codeword (though the packet itself may contain errors which might or might not be detected by the MAC FCS)

##### 45.2.3.40.2 10 Gb/s FEC Enable (3.75.0)

This bit indicates whether 10 Gb/s FEC is enabled in the 10GBASE-PR and 10/1GBASE-PRX PCS and always reads as one.

The register for enabling and disabling forward error correction in the 10/1GBASE-PRX upstream is specified in 45.2.10.3.

#### 45.2.3.41 10G-EPON and Nx25G-EPON corrected FEC codewords counter (Register 3.76, 3.77)

The assignment of bits in the 10G-EPON and Nx25G-EPON corrected FEC codewords counter register is shown in Table 45–270. See 76.3.3.3.2 for a definition of the 10G-EPON counters and 142.3.5.2 for the definition of the Nx25-EPON counters. These bits shall be reset to all zeros when the register is read by the management function or upon PCS reset. These bits shall be held at all ones in the case of overflow.



**Table 45–270—10G-EPON and Nx25G-EPON corrected FEC codewords counter register bit definitions**

| Bit(s)    | Name                          | Description                            | R/W <sup>a</sup> |
|-----------|-------------------------------|----------------------------------------|------------------|
| 3.76.15:0 | corrected FEC codewords lower | corrected_FEC_codewords_counter[15:0]  | RO, MW, NR       |
| 3.77.15:0 | corrected FEC codewords upper | corrected_FEC_codewords_counter[31:16] | RO, MW, NR       |

<sup>a</sup>RO = Read only, MW = Multi-word, NR = Non Roll-over

#### 45.2.3.42 10G-EPON and Nx25G-EPON uncorrected FEC codewords counter (Register 3.78, 3.79)

The assignment of bits in the 10G-EPON and Nx25G-EPON uncorrected FEC codewords counter register is shown in Table 45–271. See 76.3.3.3.2 for a definition of the 10G-EPON counters and 142.3.5.2 for the definition of the Nx25-EPON counters. These bits shall be reset to all zeros when the register is read by the management function or upon PCS reset. These bits shall be held at all ones in the case of overflow.

**Table 45–271—10G-EPON and Nx25G-EPON uncorrected FEC codewords counter register bit definitions**

| Bit(s)    | Name                            | Description                              | R/W <sup>a</sup> |
|-----------|---------------------------------|------------------------------------------|------------------|
| 3.78.15:0 | uncorrected FEC codewords lower | uncorrected_FEC_codewords_counter[15:0]  | RO, MW, NR       |
| 3.79.15:0 | uncorrected FEC codewords lower | uncorrected_FEC_codewords_counter[32:16] | RO, MW, NR       |

<sup>a</sup>RO = Read only, MW = Multi-word, NR = Non Roll-over

#### 45.2.3.43 10GBASE-PR, 10/1GBASE-PRX, and Nx25G-EPON BER monitor interval control register (Register 3.80)

The assignment of bits in the 10GBASE-PR, 10/1GBASE-PRX, and Nx25G-EPON BER monitor interval control register is shown in Table 45–272. This register is defined only when 10GBASE-PR,

10/1GBASE-PRX, or Nx25G-EPON ONU capability is supported. The 10G-EPON BER monitor is described in 76.3.3.4. The Nx25G-EPON QC-LDPC BER Monitor is described in 142.3.5.6.

**Table 45–272—10GBASE-PR, 10/1GBASE-PRX, and Nx25G-EPON BER monitor interval control register bit definitions**

| Bit(s)    | Name                      | Description                                                                                                                                                                                                                                                                                                                                                                                                            | R/W <sup>a</sup> |
|-----------|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 3.80.15:8 | Reserved                  | Value always 0                                                                                                                                                                                                                                                                                                                                                                                                         | RO               |
| 3.80.7:0  | EPON BER monitor interval | For 10GBASE-PR and 10/1GBASE-PRX: Duration (in units of 5 microseconds) of the timer used by the BER monitor function. Default value is 25 (i.e., 125 microseconds). A value of zero indicates that the BER monitor function is disabled.<br>For Nx25G-EPON: QC-LDPC codeword count (in units of 16 codewords) of the monitoring interval used by the BER Monitor function. Default value is 12 (i.e., 192 codewords). | R/W              |

<sup>a</sup>RO = Read only, R/W = Read/Write

#### 45.2.3.44 10GBASE-PR, 10/1GBASE-PRX, and Nx25G-EPON BER monitor status (Register 3.81)

The assignment of bits in the 10GBASE-PR, 10/1GBASE-PRX, and Nx25G-EPON BER monitor status register is shown in Table 45–273. This register is defined only when 10GBASE-PR, 10/1GBASE-PRX, or Nx25G-EPON ONU capability is supported.

**Table 45–273—10GBASE-PR, 10/1GBASE-PRX, and Nx25G-EPON BER monitor status register bit definitions**

| Bit(s)    | Name             | Description                                                                                                 | R/W <sup>a</sup> |
|-----------|------------------|-------------------------------------------------------------------------------------------------------------|------------------|
| 3.81.15:2 | Reserved         | Value always 0                                                                                              | RO               |
| 3.81.1    | Latched high BER | 10GBASE-PR, 10/1GBASE-PRX, or Nx25G-EPON PCS:<br>1 = reported a high BER.<br>0 = did not report a high BER. | RO, LH           |
| 3.81.0    | high BER         | 10GBASE-PR, 10/1GBASE-PRX, or Nx25G-EPON PCS:<br>1 = reporting a high BER.<br>0 = not reporting a high BER. | RO               |

<sup>a</sup>RO Read only, LH = Latching high

##### 45.2.3.44.1 10GBASE-PR, 10/1GBASE-PRX, and Nx25G-EPON PCS high BER (3.81.0)

In the 10GBASE-PR, 10/1GBASE-PRX, and Nx25G-EPON PCS, when read as a one, bit 3.81.0 indicates that the receiver is detecting a BER greater than the configurable threshold (high BER state). When read as a zero, bit 3.81.0 indicates that the receiver is detecting a BER lower than the configurable threshold (low BER state). This bit mirrors the state of the hi\_ber variable, defined in 76.3.3.4 for 10GBASE-PR and 10/1GBASE-PRX, and the HiBer variable in 142.3.5.2 for Nx25G-EPON.

#### 45.2.3.44.2 10GBASE-PR, 10/1GBASE-PRX, and Nx25G-EPON PCS latched high BER (3.81.1)

In the 10GBASE-PR, 10/1GBASE-PRX, and Nx25G-EPON PCS, when read as a one, bit 3.81.1 indicates that the receiver detected a BER greater than the configurable threshold (high BER state). When read as a zero, bit 3.81.1 indicates that the receiver detected BER lower than the configurable threshold (low BER state).

This bit is a latching high version of the 10GBASE-PR, 10/1GBASE-PRX, and Nx25G-EPON high BER status bit (3.81.0).

#### 45.2.3.45 10GBASE-PR, 10/1GBASE-PRX, and Nx25G-EPON BER monitor threshold control (Register 3.82)

The assignment of bits in the 10GBASE-PR, 10/1GBASE-PRX, and Nx25G-EPON BER monitor threshold control register is shown in Table 45–274. This register is defined only when 10GBASE-PR, 10/1GBASE-PRX, or Nx25G-EPON ONU capability is supported. The 10G-EPON BER monitor is described in 76.3.3.4. The Nx25G-EPON QC-LDPC BER Monitor is described in 142.3.5.6.

**Table 45–274—10GBASE-PR, 10/1GBASE-PRX, and Nx25G-EPON BER monitor threshold control register bit definitions**

| Bit(s)    | Name                       | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | R/W <sup>a</sup> |
|-----------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 3.82.15:0 | EPON BER monitor threshold | For 10G-EPON: number of sync header errors within a timer interval that triggers a high BER condition for the BER monitor function. Default value is 1600. A value of zero indicates that the BER monitor function is disabled.<br>For Nx25G-EPON: number of invalid QC-LDPC codeword parity checks within a BER monitor interval that triggers a high BER condition for the BER Monitor function. Default value is 18. A value of zero indicates that the BER Monitor function is disabled. | R/W              |

<sup>a</sup>R/W = Read/Write

#### 45.2.3.46 Nx25G-EPON synchronization pattern registers (Registers 3.83 through 3.134)

The assignment of bits in registers 3.83 through 3.134 is shown in Table 45–275. The Nx25G-EPON synchronization pattern (see 142.1.3 and 144.3.6.7) is used in the upstream data transmissions to facilitate the OLT in locking to the incoming data burst.

**Table 45–275—Nx25G-EPON synchronization pattern registers bit definitions**

| Bit(s)    | Name         | Description                 | R/W <sup>a</sup> |
|-----------|--------------|-----------------------------|------------------|
| 3.83.15:6 | Reserved     | Value always 0              | RO               |
| 3.83.5    | SP3 bit 257  | The MSB of the 257-bit SP3. | R/W              |
| 3.83.4    | SP3 balanced | Balance setting for SP3.    | R/W              |
| 3.83.3    | SP2 bit 257  | The MSB of the 257-bit SP2. | R/W              |
| 3.83.2    | SP2 balanced | Balance setting for SP2.    | R/W              |

**Table 45–275—Nx25G-EPON synchronization pattern registers bit definitions (*continued*)**

| Bit(s)                   | Name         | Description                                                                               | R/W <sup>a</sup> |
|--------------------------|--------------|-------------------------------------------------------------------------------------------|------------------|
| 3.83.1                   | SP1 bit 257  | The MSB of the 257-bit SP1.                                                               | R/W              |
| 3.83.0                   | SP1 balanced | Balance setting for SP1.                                                                  | R/W              |
| 3.99.15 through 3.84.0   | SP1 pattern  | The lower 256 bits of SP1. Bit 0 is stored in 3.84.0, and bit 255 is stored in 3.99.15.   | R/W              |
| 3.100.15:0               | SP1 length   | The number of times SP1 is to be repeated.                                                | R/W              |
| 3.116.15 through 3.101.0 | SP2 pattern  | The lower 256 bits of SP2. Bit 0 is stored in 3.101.0, and bit 255 is stored in 3.116.15. | R/W              |
| 3.117.15:0               | SP2 length   | The number of times SP2 is to be repeated.                                                | R/W              |
| 3.133.15 through 3.118.0 | SP3 pattern  | The lower 256 bits of SP3. Bit 0 is stored in 3.118.0, and bit 255 is stored in 3.133.15. | R/W              |
| 3.134.15:0               | SP3 length   | The number of times SP3 is to be repeated.                                                | R/W              |

<sup>a</sup>RO = Read only, R/W = Read/Write

#### **45.2.3.46.1 SP3 bit 257 (3.83.5)**

In the Nx25G-EPON PCS, bit 3.83.5 indicates the value to be used for the 257th bit of SP3. See 142.1.3 and 144.3.6.7 for additional details.

#### **45.2.3.46.2 SP3 balanced (3.83.4)**

In the Nx25G-EPON PCS, bit 3.83.4 indicates that repeating SP3 synchronization patterns are to have a balanced number of one and zero bits transmitted. When this bit is set to a zero then SP3 is to remain unbalanced, i.e., SP3 is always transmitted using the values from 3.83.5 and 3.118.0 through 3.133.15. When this bit is set to a one SP3 is to be balanced, i.e., each 257-bit block of SP3 (starting with the second block) is an inversion of the preceding block. See 142.1.3 and 144.3.6.7 for additional details.

#### **45.2.3.46.3 SP2 bit 257 (3.83.3)**

In the Nx25G-EPON PCS, bit 3.83.3 indicates the value to be used for the 257th bit of SP2. See 142.1.3 and 144.3.6.7 for additional details.

#### **45.2.3.46.4 SP2 balanced (3.83.2)**

In the Nx25G-EPON PCS, bit 3.83.2 indicates that repeating SP2 synchronization patterns are to have a balanced number of one and zero bits transmitted. When this bit is set to a zero then SP2 is to remain unbalanced, i.e., SP2 is always transmitted using the values from 3.83.3 and 3.101.0 through 3.116.15. When this bit is set to a one SP2 is to be balanced, i.e., each 257-bit block of SP2 (starting with the second block) is an inversion of the preceding block. See 142.1.3 and 144.3.6.7 for additional details.

#### **45.2.3.46.5 SP1 bit 257 (3.83.1)**

In the Nx25G-EPON PCS, bit 3.83.1 indicates the value to be used for the 257th bit of SP1. See 142.1.3 and 144.3.6.7 for additional details.

#### **45.2.3.46.6 SP1 balanced (3.83.0)**

In the Nx25G-EPON PCS, bit 3.83.0 indicates that repeating SP1 synchronization patterns are to have a balanced number of one and zero bits transmitted. When this bit is set to a zero then SP1 is to remain unbalanced, i.e., SP1 is always transmitted using the values from 3.83.1 and 3.84.0 through 3.99.15. When this bit is set to a one SP1 is to be balanced, i.e., each 257-bit block of SP1 (starting with the second block) is an inversion of the preceding block. See 142.1.3 and 144.3.6.7 for additional details.

#### **45.2.3.46.7 SP1 pattern (3.84.0 through 3.99.15)**

In the Nx25G-EPON PCS, bits 3.84.0 through 3.99.15 indicate the value to be used for the lower 256 bits of the initial SP1 transmitted in a burst. If present, subsequent transmissions of the lower 256 bits of SP1 are determined by bit 3.83.0. See 142.1.3 and 144.3.6.7 for additional details.

#### **45.2.3.46.8 SP1 length (3.100.15:0)**

In the Nx25G-EPON PCS, bits 3.100.15:0 indicate the number of times the 257-bit SP1 is transmitted in a given burst. See 142.1.3 and 144.3.6.7 for additional details.

#### **45.2.3.46.9 SP2 pattern (3.101.0 through 3.116.15)**

In the Nx25G-EPON PCS, bits 3.101.0 through 3.116.15 indicate the value to be used for the lower 256 bits of the initial SP2 transmitted in a burst. If present, subsequent transmissions of the lower 256 bits of SP2 are determined by bit 3.100.15:0. See 142.1.3 and 144.3.6.7 for additional details.

#### **45.2.3.46.10 SP2 length (3.117.15:0)**

In the Nx25G-EPON PCS, bits 3.117.15:0 indicate the number of times the 257-bit SP2 is transmitted in a given burst. See 142.1.3 and 144.3.6.7 for additional details.

#### **45.2.3.46.11 SP3 pattern (3.118.0 through 3.133.15)**

In the Nx25G-EPON PCS, bits 3.118.0 through 3.133.15 indicate the value to be used for the lower 256 bits of the initial SP3 transmitted in a burst. If present, subsequent transmissions of the lower 256 bits of SP3 are determined by bit 3.117.15:0. See 142.1.3 and 144.3.6.7 for additional details.

#### **45.2.3.46.12 SP3 length (3.134.15:0)**

In the Nx25G-EPON PCS, bits 3.134.15:0 indicate the number of times the 257-bit SP3 is transmitted in a given burst. See 142.1.3 and 144.3.6.7 for additional details.

#### **45.2.3.47 BIP error counter lane 0 (Register 3.200)**

The assignment of bits in the BIP error counter lane 0 is shown in Table 45–276. The multi-lane PCS described in Clause 82 calculates a BIP value for each PCS lane (see 82.2.8, 82.2.15). Errors detected in PCS lane 0 are counted and shown in register 3.200.15:0. The 16-bit counter shall be reset to all zeros when register 3.200 is read or upon PCS reset. The 16-bit counter shall be held at all ones in the case of overflow.

#### **45.2.3.48 BIP error counter, lanes 1 through 19 (Registers 3.201 through 3.219)**

The behavior of the BIP error counters, lanes 1 through 19 is identical to that described for PCS lane 0 in 45.2.3.47. Errors detected in each PCS lane are counted and shown in register bits 15:0 in the corresponding register. PCS lane 1 is shown in register 3.201; PCS lane 2 is shown in register 3.202; through register 3.219 for PCS lane 19.

**Table 45–276—BIP error counter, lane 0 register bit definitions**

| Bit(s)     | Name                      | Description                          | R/W <sup>a</sup> |
|------------|---------------------------|--------------------------------------|------------------|
| 3.200.15:0 | BIP error counter, lane 0 | Errors detected by BIP in PCS lane 0 | RO               |

<sup>a</sup>RO = Read only

#### 45.2.3.49 Lane 0 mapping register (Register 3.400)

The assignment of bits in the Lane 0 mapping register is shown in Table 45–277. When the multi-lane PCS described in Clause 82 or Clause 119 detects and locks the alignment marker for service interface lane 0, the detected PCS lane number is recorded in this register. The contents of the Lane 0 mapping register is valid when Lane 0 aligned bit (3.52.0) is set to one and is invalid otherwise.

**Table 45–277—Lane 0 mapping register bit definitions**

| Bit(s)     | Name           | Description                                   | R/W <sup>a</sup> |
|------------|----------------|-----------------------------------------------|------------------|
| 3.400.15:5 | Reserved       | Value always 0                                | RO               |
| 3.400.4:0  | Lane 0 mapping | PCS lane received in service interface lane 0 | RO               |

<sup>a</sup>RO = Read only

#### 45.2.3.50 Lanes 1 through 19 mapping registers (Registers 3.401 through 3.419)

The definition of lanes 1 through 19 mapping registers is identical to that described for lane 0 in 45.2.3.49. The lane mapping for lane 1 is in register 3.401; lane 2 is in register 3.402; etc.

#### 45.2.3.51 1000BASE-H OAM transmit registers (Registers 3.500 through 3.508)

Registers 3.500 through 3.508 are used to transmit messages over the 1000BASE-H OAM channel established between two link partners (see 115.9.1). The 1000BASE-H OAM message is part of the physical header data (PHD) defined for 1000BASE-H (see Table 115–6).

The transmit registers are used to compose a 1000BASE-H OAM message and to control its transmission to the link partner. Register 3.500 contains control, status, identification, and the beginning of a 1000BASE-H OAM message. Eight additional registers (3.501 through 3.508) hold the remainder of the 1000BASE-H OAM message.

The assignment of bits in the 1000BASE-H OAM transmit registers is shown in Table 45–278.

##### 45.2.3.51.1 TXO\_REQ (3.500.15)

Bit 3.500.15 set to one requests the 1000BASE-H based PHY to transmit the 1000BASE-H OAM message written to registers 3.500 through 3.508. Bit 3.500.15 is set to zero by the 1000BASE-H based PHY to indicate that the 1000BASE-H OAM message has been accepted for transmission, and that the 1000BASE-H OAM transmit message registers are free to accept a new 1000BASE-H OAM message.

Bit 3.500.15, together with bits 3.500.14 (TXO\_PHYT), 3.500.13 (TXO\_MERT), and 3.500.12 (TXO\_MSGT), indicates the status of the 1000BASE-H OAM transmission channel (see 115.9.2).

**Table 45–278—1000BASE-H OAM transmit register bit definitions**

| Bit(s)     | Name      | Description                                                                                                                                                                       | R/W <sup>a</sup> |
|------------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 3.500.15   | TXO_REQ   | 1 = Transmission of a 1000BASE-H OAM message is pending; write as one to request transmission<br>0 = 1000BASE-H transmit registers are available for a new 1000BASE-H OAM message | R/W, SC          |
| 3.500.14   | TXO_PHYT  | The identifier of the last 1000BASE-H OAM message received by the remote PHY                                                                                                      | RO               |
| 3.500.13   | TXO_MERT  | The identifier of the last 1000BASE-H OAM message read by the STA attached to the remote PHY                                                                                      | RO               |
| 3.500.12   | TXO_MSGT  | 1000BASE-H OAM message identifier; its value change with each new transmitted message                                                                                             | RO               |
| 3.500.11:0 | TXO_DATA0 | Transmit 1000BASE-H OAM message first 12 bits                                                                                                                                     | R/W              |
| 3.501.15:0 | TXO_DATA1 | Transmit 1000BASE-H OAM message 16-bit data word 1                                                                                                                                | R/W              |
| 3.502.15:0 | TXO_DATA2 | Transmit 1000BASE-H OAM message 16-bit data word 2                                                                                                                                | R/W              |
| 3.503.15:0 | TXO_DATA3 | Transmit 1000BASE-H OAM message 16-bit data word 3                                                                                                                                | R/W              |
| 3.504.15:0 | TXO_DATA4 | Transmit 1000BASE-H OAM message 16-bit data word 4                                                                                                                                | R/W              |
| 3.505.15:0 | TXO_DATA5 | Transmit 1000BASE-H OAM message 16-bit data word 5                                                                                                                                | R/W              |
| 3.506.15:0 | TXO_DATA6 | Transmit 1000BASE-H OAM message 16-bit data word 6                                                                                                                                | R/W              |
| 3.507.15:0 | TXO_DATA7 | Transmit 1000BASE-H OAM message 16-bit data word 7                                                                                                                                | R/W              |
| 3.508.15:0 | TXO_DATA8 | Transmit 1000BASE-H OAM message 16-bit data word 8                                                                                                                                | R/W              |

<sup>a</sup>R/W = Read/Write, RO = Read only, SC = Self-clearing

#### 45.2.3.51.2 TXO\_PHYT (3.500.14)

Bit 3.500.14 reflects the value of the bit TXO\_MSGT in the last 1000BASE-H OAM message received by the remote 1000BASE-H based PHY.

#### 45.2.3.51.3 TXO\_MERT (3.500.13)

Bit 3.500.13 reflects the value of the bit TXO\_MSGT in the last 1000BASE-H OAM message read by the STA attached to the remote 1000BASE-H based PHY.

#### 45.2.3.51.4 TXO\_MSGT (3.500.12)

Bit 3.500.12 is used for 1000BASE-H OAM message identification. Bit 3.500.12 is changed by the 1000BASE-H based PHY when it accepts a new 1000BASE-H OAM message for transmission (simultaneously setting bit 3.500.15 to zero), acting as a 1-bit sequence number.

#### 45.2.3.51.5 TXO\_DATAx (Bits 3.500.11:0 and registers 3.501 through 3.508)

Bits 3.500.11:0 (TXO\_DATA0) contains the first 12 bits of a 1000BASE-H OAM message and registers 3.501 through 3.508 (TXO\_DATA1 through TXO\_DATA8) the remaining 128 bits of that 1000BASE-H OAM message.

TXO\_DATA0 and TXO\_DATA1 shall be a 28-bit 1000BASE-H OAM protocol identifier composed of a 24-bit Organizationally Unique Identifier (OUI) or Company ID (CID) followed by a 4-bit protocol number; where TXO\_DATA0[10:0] = OUI[23:12], TXO\_DATA1[15:3] = OUI[11:0], and TXO\_DATA1[3:0] = protocol number. The content of TXO\_DATA2 to TXO\_DATA8 is vendor specific to the assignee of the OUI or CID.

#### 45.2.3.52 1000BASE-H OAM receive registers (Registers 3.509 through 3.517)

Registers 3.509 through 3.517 store messages received over the 1000BASE-H OAM channel established between two link partners (see 115.9.3). Register 3.509 contains status information, identification, and the first 12 bits of a 1000BASE-H OAM message, and registers 3.510 through 3.517 the following 128 bits of the received 1000BASE-H OAM message.

The assignment of bits in the 1000BASE-H OAM receive registers is shown in Table 45–279.

**Table 45–279—1000BASE-H OAM receive register bit definitions**

| Bit(s)      | Name      | Description                                                                                                                               | R/W <sup>a</sup> |
|-------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 3.509.15    | RXO_VAL   | 1 = A new 1000BASE-H OAM message has arrived<br>0 = No new 1000BASE-H OAM message arrived since either last message was read or PMA reset | RO               |
| 3.509.14:13 | Reserved  | Value always 0                                                                                                                            | RO               |
| 3.509.12    | RXO_MSGT  | Identifier of the received 1000BASE-H OAM message                                                                                         | RO               |
| 3.509.11:0  | RXO_DATA0 | Receive 1000BASE-H OAM message first 12 bits                                                                                              | RO               |
| 3.510.15:0  | RXO_DATA1 | Receive 1000BASE-H OAM message 16-bit data word 1                                                                                         | RO               |
| 3.511.15:0  | RXO_DATA2 | Receive 1000BASE-H OAM message 16-bit data word 2                                                                                         | RO               |
| 3.512.15:0  | RXO_DATA3 | Receive 1000BASE-H OAM message 16-bit data word 3                                                                                         | RO               |
| 3.513.15:0  | RXO_DATA4 | Receive 1000BASE-H OAM message 16-bit data word 4                                                                                         | RO               |
| 3.514.15:0  | RXO_DATA5 | Receive 1000BASE-H OAM message 16-bit data word 5                                                                                         | RO               |
| 3.515.15:0  | RXO_DATA6 | Receive 1000BASE-H OAM message 16-bit data word 6                                                                                         | RO               |
| 3.516.15:0  | RXO_DATA7 | Receive 1000BASE-H OAM message 16-bit data word 7                                                                                         | RO               |
| 3.517.15:0  | RXO_DATA8 | Receive 1000BASE-H OAM message 16-bit data word 8                                                                                         | RO               |

<sup>a</sup>RO = Read only

##### 45.2.3.52.1 RXO\_VAL (3.509.15)

The 1000BASE-H based PHY sets bit 3.509.15 to one to indicate the reception of a new 1000BASE-H OAM message. The 1000BASE-H based PHY sets bit 3.509.15 to zero when the last register (3.517) containing the 1000BASE-H OAM message is read (see Figure 115–43). The 1000BASE-H based PHY does not update the receive message registers until the message is read by STA as indicated by a read of register 3.517.



#### 45.2.3.52.2 RXO\_MSGT (3.509.12)

Bit 3.509.12 is used for message identification. Bit 3.509.12 changes with every new received 1000BASE-H OAM message, acting as a 1-bit sequence number.

#### 45.2.3.52.3 RXO\_DATAx (Bits 3.509.11:0 and registers 3.510 through 3.517)

Bits 3.509.11:0 (RXO\_DATA0) contains the first 12 bits of a received 1000BASE-H OAM message, and registers 3.510 through 3.517 (RXO\_DATA1 through RXO\_DATA8) the remaining 128 bits of that received 1000BASE-H OAM message.

#### 45.2.3.53 1000BASE-H PCS control register (Register 3.518)

The assignment of bits in the 1000BASE-H PCS control register is shown in Table 45–280.

**Table 45–280—1000BASE-H PCS control register bit definitions**

| Bit(s)      | Name                  | Description                                                                                                                                                                                        | R/W <sup>a</sup> |
|-------------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 3.518.15:13 | Operation mode        | 15 14 13<br>0 0 0 = Normal operation<br>0 0 1 = Test mode 1<br>0 1 0 = Test mode 2<br>0 1 1 = Test mode 3<br>1 0 0 = Test mode 4<br>1 0 1 = Test mode 5<br>1 1 0 = Test mode 6<br>1 1 1 = Reserved | R/W              |
| 3.518.12:10 | Loopback mode         | 12 11 10<br>0 0 0 = No loopback<br>0 0 1 = GMII level loopback<br>0 1 0 = PMD interface level loopback<br>0 1 1 = Line loopback<br>1 x x = Reserved                                                | R/W              |
| 3.518.9:2   | Reserved              | Value always 0                                                                                                                                                                                     | RO               |
| 3.518.1     | 1000BASE-H OAM enable | 1 = Enable 1000BASE-H OAM functionality<br>0 = Disable 1000BASE-H OAM functionality                                                                                                                | R/W              |
| 3.518.0     | EEE enable            | 1 = Enable LPI mode<br>0 = Disable LPI mode                                                                                                                                                        | R/W              |

<sup>a</sup>R/W = Read/Write, RO = Read only

##### 45.2.3.53.1 Operation mode (3.518.15:13)

Bits 3.518.15:13 shall have a default value of binary 000, selecting normal 1000BASE-H operation. A value of binary 001 through binary 110 in bits 3.518.15:13 shall select the test mode as specified in Table 45–280 with behavior as defined in 115.5.

##### 45.2.3.53.2 Loopback mode (3.518.12:10)

Bits 3.518.12:10 shall select one of the loopback modes defined in 115.10. Bits 3.518.12:10 shall have a default value of binary 000 (no loopback operation). Loopback modes are only available when a 1000BASE-H based PHY is in the normal operation mode (no test mode is selected in 3.518.15:13).

#### 45.2.3.53.3 1000BASE-H OAM enable (3.518.1)

Bit 3.518.1 controls establishing the 1000BASE-H OAM channel with the link partner (115.9). If the 1000BASE-H based PHY does not have 1000BASE-H OAM ability (bit 3.519.1 = 0), setting bit 3.518.1 shall have no effect. Setting bit 3.518.1 to zero shall prevent establishment of a 1000BASE-H OAM channel with the link partner. Changes in a 1000BASE-H OAM enable value shall only take effect after a PMA reset. Bit 3.518.1 has no specified default value.

#### 45.2.3.53.4 EEE enable (3.518.0)

Setting bit 3.518.0 to one enables 1000BASE-H based PHY EEE capability (see 115.4). Setting bit 3.518.0 to zero shall prevent establishment of EEE operation with the link partner. If the 1000BASE-H based PHY does not have EEE ability (bit 3.519.0 = 0), setting bit 3.518.0 shall have no effect. Changes in EEE enable value shall only take effect after a PMA reset. Bit 3.518.0 has no specified default value.

#### 45.2.3.54 1000BASE-H PCS status 1 register (Register 3.519)

The assignment of bits in the 1000BASE-H PCS status 1 register is shown in Table 45–281.

**Table 45–281— 1000BASE-H PCS status 1 register bit definitions**

| Bit(s)   | Name                        | Description                                                                                                          | R/W <sup>a</sup> |
|----------|-----------------------------|----------------------------------------------------------------------------------------------------------------------|------------------|
| 3.519.15 | Local receiver status       | 1 = The local receive operation is reliable<br>0 = The local receive operation is unreliable                         | RO               |
| 3.519.14 | Remote receiver status      | 1 = The receive operation of the remote PHY is reliable<br>0 = The receive operation of the remote PHY is unreliable | RO               |
| 3.519.13 | Link status                 | 1 = A bidirectional reliable link is established<br>0 = A bidirectional reliable link is not established             | RO, LL           |
| 3.519.12 | Local PHD reception status  | 1 = PHD reception is reliable for the local PHY<br>0 = PHD reception is unreliable for the local PHY                 | RO               |
| 3.519.11 | Remote PHD reception status | 1 = PHD reception is reliable for the remote PHY<br>0 = PHD reception is unreliable for the remote PHY               | RO               |
| 3.519.10 | PHD lock status             | 1 = PHD transmission and reception are reliable<br>0 = PHD transmission or reception are unreliable                  | RO, LL           |
| 3.519.9  | THP lock status             | 1 = THP is initialized; payload data is received with THP processing<br>0 = THP is not initialized                   | RO               |
| 3.519.8  | Tx Assert LPI received      | 1 = Tx PCS has received LPI<br>0 = LPI not received by Tx PCS                                                        | RO, LH           |
| 3.519.7  | Rx Assert LPI generated     | 1 = Rx PCS has generated LPI<br>0 = LPI not generated by Rx PCS                                                      | RO, LH           |
| 3.519.6  | Tx LPI indication           | 1 = Tx PCS is currently transmitting LPI<br>0 = Tx PCS is not currently transmitting LPI                             | RO               |
| 3.519.5  | Rx LPI indication           | 1 = Rx PCS is currently receiving LPI<br>0 = Rx PCS is not currently receiving LPI                                   | RO               |
| 3.519.4  | Reserved                    | Value always 0                                                                                                       | RO               |

**Table 45–281— 1000BASE-H PCS status 1 register bit definitions (*continued*)**

| Bit(s)  | Name                          | Description                                                                                                                                  | R/W <sup>a</sup> |
|---------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 3.519.3 | Remote 1000BASE-H OAM ability | 1 = The remote PHY has 1000BASE-H OAM ability and it is enabled<br>0 = The remote PHY does not have 1000BASE-H OAM ability or it is disabled | RO               |
| 3.519.2 | Remote EEE ability            | 1 = The remote PHY has EEE ability and it is enabled<br>0 = The remote PHY does not have EEE ability or it is disabled                       | RO               |
| 3.519.1 | 1000BASE-H OAM ability        | 1 = The PHY has 1000BASE-H OAM ability<br>0 = The PHY does not have 1000BASE-H OAM ability                                                   | RO               |
| 3.519.0 | EEE ability                   | 1 = The PHY has EEE ability<br>0 = The PHY does not have EEE ability                                                                         | RO               |

<sup>a</sup>RO = Read only, LH = Latching high, LL = Latching low

#### **45.2.3.54.1 Local receiver status (3.519.15)**

Bit 3.519.15 reflects the value of the state variable `loc_rcvr_status` as determined by the PHY quality monitor state diagram (see 115.3.5.1).

#### **45.2.3.54.2 Remote receiver status (3.519.14)**

Bit 3.519.14 reflects the value of the state variable `rem_rcvr_status` as determined by the link monitor state diagram (see 115.3.5.1).

#### **45.2.3.54.3 Link status (3.519.13)**

Bit 3.519.13 reflects the value of the state variable `link_status` as determined by the link monitor state diagram (see 115.3.5.1). Bit 3.519.13 shall have latching low behavior. Bit 1.1.2, bit 3.1.2, and bit 3.519.13 are identical for 1000BASE-H; a read to any of these three bits shall release the latch for all the bits.

#### **45.2.3.54.4 Local PHD reception status (3.519.12)**

Bit 3.519.12 reflects the value of the state variable `loc_rcvr_hdr_lock` as determined by the local PHD reception monitor state diagram (see 115.3.5.1).

#### **45.2.3.54.5 Remote PHD reception status (3.519.11)**

Bit 3.519.11 reflects the value of the state variable `rem_rcvr_hdr_lock` as determined by the remote PHD reception monitor state diagram (see 115.3.5.1).

#### **45.2.3.54.6 PHD lock status (3.519.10)**

Bit 3.519.10 reflects the value of the state variable `rcvr_hdr_lock` as determined by the PHD monitor state diagram (see 115.3.5.1). Bit 3.519.10 shall have latching low behavior.

#### **45.2.3.54.7 THP lock status (3.519.9)**

Bit 3.519.9 reflects the value of the state variable `rcvr_thp_lock` as determined by the adaptive THP REQ state diagram (see 115.3.6.1).

#### **45.2.3.54.8 Tx Assert LPI received (3.519.8)**

When read as a one, bit 3.519.8 indicates that the transmit 1000BASE-H PCS has received Assert LPI signaling from the TX GMII one or more times since the register was last read. When read as a zero, bit 3.519.8 indicates that the 1000BASE-H PCS transmitter has not received Assert LPI. Bit 3.519.8 shall have latching high behavior. Bit 3.519.8 is a copy of bit 3.1.11; a read of either bit shall release the latch for all the bits.

#### **45.2.3.54.9 Rx Assert LPI generated (3.519.7)**

When read as a one, bit 3.519.7 indicates that the receive 1000BASE-H PCS has generated Assert LPI on the RX GMII one or more times since the register was last read. When read as a zero, bit 3.519.7 indicates that the 1000BASE-H PCS receiver has not generated Assert LPI on the RX GMII. Bit 3.519.7 shall have latching high behavior. Bit 3.519.7 is a copy of bit 3.1.10; a read of either bit shall release the latch for all the bits.

#### **45.2.3.54.10 Tx LPI indication (3.519.6)**

Bit 3.519.6 indicates the current LPI mode status of the local 1000BASE-H based PHY transmit path. When read as one, the PHY transmitter is currently in LPI mode (transmitting refresh and quiet). When read as zero, the PHY transmitter is not in LPI mode (see 115.4.1). Bit 3.519.6 is a copy of bit 3.1.9.

#### **45.2.3.54.11 Rx LPI indication (3.519.5)**

Bit 3.519.5 indicates the current LPI mode status of the local 1000BASE-H based PHY receive path. When read as a one, the PHY is receiving LPI mode signals (refresh and quiet). When read as a zero, the PHY is not receiving LPI mode signals (see 115.4.2). Bit 3.519.5 is a copy of bit 3.1.8.

#### **45.2.3.54.12 Remote 1000BASE-H OAM ability (3.519.3)**

Bit 3.519.3 indicates the 1000BASE-H OAM ability of the remote PHY received in the PHD field PHD.CAPOAM (see Table 115–6). When read as one, bit 3.519.3 indicates the remote PHY has both 1000BASE-H OAM ability and the 1000BASE-H OAM is enabled. When read as zero, bit 3.519.3 indicates that the remote PHY either does not have 1000BASE-H OAM ability or 1000BASE-H OAM is disabled.

#### **45.2.3.54.13 Remote EEE ability (3.519.2)**

Bit 3.519.2 indicates the EEE ability of the remote PHY received in the PHD field PHD.CAP.LPI (see Table 115–6). When read as one, bit 3.519.2 indicates the remote PHY has both EEE ability and EEE is enabled. When read as zero, bit 3.519.2 indicates that the remote PHY either does not have EEE ability or EEE is disabled.

#### **45.2.3.54.14 1000BASE-H OAM ability (3.519.1)**

Bit 3.519.1 indicates the 1000BASE-H OAM ability of the 1000BASE-H based PHY (see 115.9). When read as one, bit 3.519.1 indicates that the 1000BASE-H based PHY supports the 1000BASE-H OAM protocol. When read as zero, bit 3.519.1 indicates the 1000BASE-H based PHY does not support the 1000BASE-H OAM protocol.

A PHY indicating in bit 3.519.1 support of 1000BASE-H OAM shall provide the functions of registers 3.500 through 3.517 as specified in 115.9.

#### 45.2.3.54.15 EEE ability (3.519.0)

Bit 3.519.0 indicates the EEE ability of the 1000BASE-H based PHY. When read as one, bit 3.519.0 indicates that the 1000BASE-H based PHY implements EEE, and is able to enter LPI mode (see 115.4). When read as zero, bit 3.519.0 indicates the 1000BASE-H based PHY does not implement EEE operation.

#### 45.2.3.55 1000BASE-H PCS status 2 register (Register 3.520)

The assignment of bits in the 1000BASE-H PCS status 2 register is shown in Table 45–282.

**Table 45–282—1000BASE-H PCS status 2 register bit definitions**

| Bit(s)     | Name              | Description                   | R/W <sup>a</sup> |
|------------|-------------------|-------------------------------|------------------|
| 3.520.15:8 | Reserved          | Value always 0                | RO               |
| 3.520.7:0  | Local link margin | Reports the local link margin | RO               |

<sup>a</sup>RO = Read only

##### 45.2.3.55.1 Local link margin (3.520.7:0)

Bits 3.520.7:0 indicate the link margin of the receiver. Link margin is defined in 115.3.7.2 and is fixed-point formatted (8,3). See 115.3.8 for fixed-point format definition.

#### 45.2.3.56 1000BASE-H PCS status 3 register (Register 3.521)

The assignment of bits in the 1000BASE-H PCS status 3 register is shown in Table 45–283.

**Table 45–283—1000BASE-H PCS status 3 register bit definitions**

| Bit(s)     | Name               | Description                              | R/W <sup>a</sup> |
|------------|--------------------|------------------------------------------|------------------|
| 3.521.15:8 | Reserved           | Value always 0                           | RO               |
| 3.521.7:0  | Remote link margin | Link margin reported by the link partner | RO               |

<sup>a</sup>RO = Read only

##### 45.2.3.56.1 Remote link margin (3.521.7:0)

Bits 3.521.7:0 report the link margin of the remote PHY receiver as it is received in the PHD field PHD.RX.LINKMARGIN (see Table 115–6). Link margin is defined in 115.3.7.2 and is fixed-point formatted (8,3). See 115.3.8 for fixed-point format definition.

#### 45.2.3.57 1000BASE-H PCS status 4 register (Register 3.522)

The assignment of bits in the 1000BASE-H PCS status 4 register is shown in Table 45–284.

Bits 3.522.15:0 are a 16-bit counter that counts the number of bits received with value 1 at the output of the binary descrambler, when the 1000BASE-H based PHY receiver is operating in test mode 1. These bits shall be reset to all zeros when the counter is read. The counter shall be held at all ones in the case of overflow.

**Table 45–284—1000BASE-H PCS status 4 register bit definitions**

| Bit(s)     | Name                  | Description                                         | R/W <sup>a</sup> |
|------------|-----------------------|-----------------------------------------------------|------------------|
| 3.522.15:0 | BER test mode counter | A 16-bit counter used when operating in test mode 1 | RO,<br>NR        |

<sup>a</sup>RO = Read only, NR = Non Roll-over

#### 45.2.3.58 PCS FEC symbol error counter lane 0 (Register 3.600, 3.601)

The assignment of bits in the PCS FEC symbol error counter lane 0 register is shown in Table 45–285. See 119.3.4 for a definition of this counter. Symbol errors detected in PCS lane 0 are counted and shown in register 3.600.15:0 and 3.601.15:0. These bits shall be reset to all zeros when the register is read by the management function or upon PHY reset. These bits shall be held at all ones in the case of overflow. Registers 3.600, 3.601 are used to read the value of a 32-bit counter. When registers 3.600 and 3.601 are used to read the 32-bit counter value, the register 3.600 is read first, the value of the register 3.601 is latched when (and only when) register 3.600 is read, and reads of register 3.601 return the latched value rather than the current value of the counter.

**Table 45–285—PCS FEC symbol error counter lane 0 register bit definitions**

| Bit(s)     | Name                                | Description                       | R/W <sup>a</sup> |
|------------|-------------------------------------|-----------------------------------|------------------|
| 3.600.15:0 | PCS FEC symbol errors, lane 0 lower | FEC_symbol_error_counter_0[15:0]  | RO, NR           |
| 3.601.15:0 | PCS FEC symbol errors, lane 0 upper | FEC_symbol_error_counter_0[31:16] | RO, NR           |

<sup>a</sup>RO = Read only, NR = Non Roll-over

#### 45.2.3.59 PCS FEC symbol error counter lane 1 through 15 (Registers 3.602 through 3.631)

The behavior of the PCS FEC symbol error counters, lane 1 through 15 is identical to that described for PCS lane 0 in 45.2.3.58. Errors detected in each PCS lane are counted and shown in the corresponding register. PCS lane 1, lower 16 bits are shown in register 3.602; PCS lane 1, upper 16 bits are shown in register 3.603; PCS lane 2, lower 16 bits are shown in register 3.604; through register 3.631 for PCS lane 15, upper 16 bits.

#### 45.2.3.60 PCS FEC control register (Register 3.800)

The assignment of bits in the PCS FEC control register is shown in Table 45–286.

**Table 45–286—PCS FEC control register bit definitions**

| Bit(s)     | Name                             | Description                                                                          | R/W <sup>a</sup> |
|------------|----------------------------------|--------------------------------------------------------------------------------------|------------------|
| 3.800.15:3 | Reserved                         | Value always 0                                                                       | RO               |
| 3.800.2    | PCS FEC degraded SER enable      | 1 = FEC decoder signals degraded SER<br>0 = FEC decoder does not signal degraded SER | R/W              |
| 3.800.1    | PCS FEC bypass indication enable | 1 = FEC decoder does not indicate errors<br>0 = FEC decoder indicates errors         | R/W              |
| 3.800.0    | Reserved                         | Value always 0                                                                       | RO               |

<sup>a</sup>R/W = Read/Write, RO = Read only

##### 45.2.3.60.1 PCS FEC degraded SER enable (3.800.2)

This bit enables the PCS FEC decoder to signal the presence of a degraded SER. When set to a one, this bit enables degraded SER signaling. When set to a zero, degraded SER signaling is disabled. Writes to this bit are ignored and reads return a zero if the PCS FEC does not have the ability to signal the presence of a degraded SER (see 119.2.5.3).

##### 45.2.3.60.2 PCS FEC bypass indication enable (3.800.1)

This bit enables the PCS FEC decoder to bypass error indication to the upper layers through the sync bits. When set to a one, this bit enables bypass of the error indication. When set to a zero, errors are indicated to the upper layers through the sync bits. Writes to this bit are ignored and reads return a zero if the PCS FEC does not have the ability to bypass indicating decoding errors (see 119.2.5.3).

#### 45.2.3.61 PCS FEC status register (Register 3.801)

The assignment of bits in the PCS FEC status register is shown in Table 45–287.

**Table 45–287—PCS FEC status register bit definitions**

| Bit(s)     | Name                         | Description                                                             | R/W <sup>a</sup> |
|------------|------------------------------|-------------------------------------------------------------------------|------------------|
| 3.801.15:7 | Reserved                     | Value always 0                                                          | RO               |
| 3.801.6    | Local degraded SER received  | 1 = local degraded SER received<br>0 = no local degraded SER received   | RO               |
| 3.801.5    | Remote degraded SER received | 1 = remote degraded SER received<br>0 = no remote degraded SER received | RO               |
| 3.801.4    | PCS FEC degraded SER         | 1 = SER is degraded<br>0 = SER is not degraded                          | RO               |

**Table 45–287—PCS FEC status register bit definitions (*continued*)**

| Bit(s)  | Name                              | Description                                                                                                                                                    | R/W <sup>a</sup> |
|---------|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 3.801.3 | PCS FEC degraded SER ability      | 1 = FEC decoder has the ability to signal the presence of a degraded SER<br>0 = FEC decoder does not have the ability to signal the presence of a degraded SER | RO               |
| 3.801.2 | PCS FEC high SER                  | 1 = FEC errors have exceeded threshold<br>0 = FEC errors have not exceeded threshold                                                                           | RO/LH            |
| 3.801.1 | PCS FEC bypass indication ability | 1 = FEC decoder has the ability to bypass error indication<br>0 = FEC decoder does not have the ability to bypass error indication                             | RO               |
| 3.801.0 | Reserved                          | Value always 0                                                                                                                                                 | RO               |

<sup>a</sup>RO = Read only, LH = Latching high

#### **45.2.3.61.1 Local degraded SER received (3.801.6)**

When read as a one, bit 3.801.6 indicates that the local degraded SER signal has been received. This bit reflects the state of rx\_local\_degraded (see 119.2.6.2.2).

#### **45.2.3.61.2 Remote degraded SER received (3.801.5)**

When read as a one, bit 3.801.5 indicates that the remote degraded SER signal has been received. This bit reflects the state of rx\_rm\_degraded (see 119.2.6.2.2).

#### **45.2.3.61.3 PCS FEC degraded SER (3.801.4)**

When read as a one, bit 3.801.4 indicates that the local PCS has detected a degradation of the received signal. This bit reflects the state of the variable FEC\_degraded\_SER (see 119.2.5.3). The value of bit 3.801.4 is unspecified if the value of the PCS FEC degraded SER activate threshold (registers 3.806 and 3.807) is less than the value of the PCS FEC degraded SER deactivate threshold (registers 3.808 and 3.809).

#### **45.2.3.61.4 PCS FEC degraded SER ability (3.801.3)**

The PCS FEC decoder may have the option to signal the presence of a degraded SER (see 119.2.5.3). This bit is set to one to indicate that the decoder has the ability to signal the presence of a degraded SER. The bit is set to zero if this ability is not supported.

#### **45.2.3.61.5 PCS FEC high SER (3.801.2)**

When PCS FEC bypass indication enable (bit 3.800.1) is set to one, this bit is set to one if the number of FEC symbol errors in a window of 8192 codewords exceeds the threshold (see 119.2.5.3) and is set to zero otherwise. The bit is set to zero if PCS FEC bypass indication enable (bit 3.800.1) is set to zero. This bit shall be implemented with latching high behavior.

#### **45.2.3.61.6 PCS FEC bypass indication ability (3.801.1)**

The PCS FEC decoder may have the option to perform error detection without error indication (see 119.2.5.3) to reduce the delay contributed by the FEC decoder. This bit is set to one to indicate that the decoder has this ability to bypass error indication. The bit is set to zero if this ability is not supported.



#### 45.2.3.62 PCS FEC corrected codewords counter (Register 3.802, 3.803)

The assignment of bits in the PCS FEC corrected codewords counter register is shown in Table 45–288. See 119.3.2 for a definition of this counter. These bits shall be reset to all zeros when the register is read by the management function or upon PHY reset. These bits shall be held at all ones in the case of overflow. Registers 3.802, 3.803 are used to read the value of a 32-bit counter. When registers 3.802 and 3.803 are used to read the 32-bit counter value, the register 3.802 is read first, the value of the register 3.803 is latched when (and only when) register 3.802 is read, and reads of register 3.803 return the latched value rather than the current value of the counter.

**Table 45–288—PCS FEC corrected codewords counter register bit definitions**

| Bit(s)     | Name                          | Description                     | R/W <sup>a</sup> |
|------------|-------------------------------|---------------------------------|------------------|
| 3.802.15:0 | FEC corrected codewords lower | FEC_corrected_cw_counter[15:0]  | RO, NR           |
| 3.803.15:0 | FEC corrected codewords upper | FEC_corrected_cw_counter[31:16] | RO, NR           |

<sup>a</sup>RO = Read only, NR = Non Roll-over

#### 45.2.3.63 PCS FEC uncorrected codewords counter (Register 3.804, 3.805)

The assignment of bits in the PCS FEC uncorrected codewords counter register is shown in Table 45–289. See 119.3.3 for a definition of this counter. These bits shall be reset to all zeros when the register is read by the management function or upon PHY reset. These bits shall be held at all ones in the case of overflow. Registers 3.804, 3.805 are used to read the value of a 32-bit counter. When registers 3.804 and 3.805 are used to read the 32-bit counter value, the register 3.804 is read first, the value of the register 3.805 is latched when (and only when) register 3.804 is read, and reads of register 3.805 return the latched value rather than the current value of the counter.

**Table 45–289—PCS FEC uncorrected codewords counter register bit definitions**

| Bit(s)     | Name                            | Description                       | R/W <sup>a</sup> |
|------------|---------------------------------|-----------------------------------|------------------|
| 3.804.15:0 | FEC uncorrected codewords lower | FEC_uncorrected_cw_counter[15:0]  | RO, NR           |
| 3.805.15:0 | FEC uncorrected codewords upper | FEC_uncorrected_cw_counter[31:16] | RO, NR           |

<sup>a</sup>RO = Read only, NR = Non Roll-over

#### 45.2.3.64 PCS FEC degraded SER activate threshold register (Register 3.806, 3.807)

The assignment of bits in the PCS FEC degraded SER activate threshold register is shown in Table 45–290. The value controls the threshold used to set the PCS FEC degraded SER bit (3.801.4) as defined in 119.2.5.3.

**Table 45–290—PCS FEC degraded SER activate threshold register bit definitions**

| Bit(s)     | Name                                          | Description                                | R/W <sup>a</sup> |
|------------|-----------------------------------------------|--------------------------------------------|------------------|
| 3.806.15:0 | PCS FEC degraded SER activate threshold lower | FEC_degraded_SER_activate_threshold[15:0]  | R/W              |
| 3.807.15:0 | PCS FEC degraded SER activate threshold upper | FEC_degraded_SER_activate_threshold[31:16] | R/W              |

<sup>a</sup>R/W = Read/Write

#### 45.2.3.65 PCS FEC degraded SER deactivate threshold register (Register 3.808, 3.809)

The assignment of bits in the PCS FEC degraded SER deactivate threshold register is shown in Table 45–291. The value controls the threshold used to clear the PCS FEC degraded SER bit (3.801.4) as defined in 119.2.5.3.

**Table 45–291—PCS FEC degraded SER deactivate threshold register bit definitions**

| Bit(s)     | Name                                            | Description                                  | R/W <sup>a</sup> |
|------------|-------------------------------------------------|----------------------------------------------|------------------|
| 3.808.15:0 | PCS FEC degraded SER deactivate threshold lower | FEC_degraded_SER_deactivate_threshold[15:0]  | R/W              |
| 3.809.15:0 | PCS FEC degraded SER deactivate threshold upper | FEC_degraded_SER_deactivate_threshold[31:16] | R/W              |

<sup>a</sup>R/W = Read/Write

#### 45.2.3.66 PCS FEC degraded SER interval register (Register 3.810, 3.811)

The assignment of bits in the PCS FEC degraded SER interval register is shown in Table 45–292. The value controls the interval used to set and clear the PCS FEC degraded SER bit (3.801.4) as defined in 119.2.5.3.

**Table 45–292—PCS FEC degraded SER interval register bit definitions**

| Bit(s)     | Name                                | Description                      | R/W <sup>a</sup> |
|------------|-------------------------------------|----------------------------------|------------------|
| 3.810.15:0 | PCS FEC degraded SER interval lower | FEC_degraded_SER_interval[15:0]  | R/W              |
| 3.811.15:0 | PCS FEC degraded SER interval upper | FEC_degraded_SER_interval[31:16] | R/W              |

<sup>a</sup>R/W = Read/Write

#### 45.2.3.67 TimeSync PCS capability (Register 3.1800)

The TimeSync PCS capability register (see Table 45–293) indicates the capability of the PCS to report the transmit and receive data delay, stored in registers 3.1801 through 3.1804 and 3.1805 through 3.1808, respectively.

**Table 45–293—TimeSync PCS capability**

| Bit(s)      | Name                              | Description                                                                                                                                                     | R/W <sup>a</sup> |
|-------------|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 3.1800.15:2 | Reserved                          | Value always 0                                                                                                                                                  | RO               |
| 3.1800.1    | TimeSync transmit path data delay | 1 = PCS provides information on transmit path data delay in registers 3.1801 through 3.1804<br>0 = PCS does not provide information on transmit path data delay | RO               |
| 3.1800.0    | TimeSync receive path data delay  | 1 = PCS provides information on receive path data delay in registers 3.1805 through 3.1808<br>0 = PCS does not provide information on receive path data delay   | RO               |

<sup>a</sup>RO = Read only

#### 45.2.3.68 TimeSync PCS transmit path data delay (Registers 3.1801, 3.1802, 3.1803, 3.1804)

The TimeSync PCS transmit path data delay register contains the maximum (registers 3.1801, 3.1802, see Table 45–294) and minimum (registers 3.1803, 3.1804, see Table 45–294) values of the transmit path data delay. The transmit path data delay is expressed in units of ns. The values contained in these registers are valid when the link is established, as indicated by bit 2 in register 1.1 (see 45.2.1.2.4).

**Table 45–294—TimeSync PCS transmit path data delay register**

| Bit(s)      | Name                                        | Description              | R/W <sup>a</sup> |
|-------------|---------------------------------------------|--------------------------|------------------|
| 3.1801.15:0 | Maximum PCS transmit path data delay, lower | PCS_delay_TX_max [15:0]  | RO, MW           |
| 3.1802.15:0 | Maximum PCS transmit path data delay, upper | PCS_delay_TX_max [31:16] | RO, MW           |
| 3.1803.15:0 | Minimum PCS transmit path data delay, lower | PCS_delay_TX_min [15:0]  | RO, MW           |
| 3.1804.15:0 | Minimum PCS transmit path data delay, upper | PCS_delay_TX_min [31:16] | RO, MW           |

<sup>a</sup>RO = Read only, MW = Multi-word

#### 45.2.3.69 TimeSync PCS receive path data delay (Registers 3.1805, 3.1806, 3.1807, 3.1808)

The TimeSync PCS receive path data delay register contains the maximum (registers 3.1805, 3.1806, see Table 45–295) and minimum (registers 3.1807, 3.1808, see Table 45–295) values of the receive path data delay. The receive path data delay is expressed in units of ns. The values contained in these registers are valid when the link is established, as indicated by bit 2 in register 1.1 (see 45.2.1.2.4).

**Table 45–295—TimeSync PCS receive path data delay register**

| Bit(s)      | Name                                       | Description              | R/W <sup>a</sup> |
|-------------|--------------------------------------------|--------------------------|------------------|
| 3.1805.15:0 | Maximum PCS receive path data delay, lower | PCS_delay_RX_max [15:0]  | RO, MW           |
| 3.1806.15:0 | Maximum PCS receive path data delay, upper | PCS_delay_RX_max [31:16] | RO, MW           |
| 3.1807.15:0 | Minimum PCS receive path data delay, lower | PCS_delay_RX_min [15:0]  | RO, MW           |
| 3.1808.15:0 | Minimum PCS receive path data delay, upper | PCS_delay_RX_min [31:16] | RO, MW           |

<sup>a</sup>RO = Read only, MW = Multi-word

#### 45.2.3.70 10BASE-T1L PCS control register (Register 3.2278)

The assignment of bits in the 10BASE-T1L PCS control register is shown in Table 45–296. The default value for each bit of the 10BASE-T1L PCS control register should be chosen so that the initial state of the device upon power up or reset is a normal operational state without management intervention.

**Table 45–296—10BASE-T1L PCS control register bit definitions**

| Bit(s)      | Name      | Description                                           | R/W <sup>a</sup> |
|-------------|-----------|-------------------------------------------------------|------------------|
| 3.2278.15   | PCS reset | 1 = PCS reset<br>0 = Normal operation                 | R/W, SC          |
| 3.2278.14   | Loopback  | 1 = Enable loopback mode<br>0 = Disable loopback mode | R/W              |
| 3.2278.13:0 | Reserved  | Value always 0                                        | RO               |

<sup>a</sup>RO = Read only, R/W = Read/Write, SC = Self-clearing

##### 45.2.3.70.1 PCS reset (3.2278.15)

Resetting the 10BASE-T1L PCS is accomplished by setting bit 3.2278.15 to one. This action shall set all 10BASE-T1L PCS registers to their default states. As a consequence, this action may change the internal state of the 10BASE-T1L PCS and the state of the physical link. This action may also initiate a reset in any other MMDs that are instantiated in the same package. This bit is self-clearing, and the 10BASE-T1L PCS shall return a value of one in bit 3.2278.15 when a reset is in progress; otherwise, it shall return a value of zero. The 10BASE-T1L PCS is not required to accept a write transaction to any of its registers until the reset process is completed. The control and management interface shall be restored to operation within 0.5 s from the setting of bit 3.2278.15. During a reset, a PCS shall respond to reads from bits 3.0.15, 3.8.15:14, and 3.2278.15. Reads for all other bits shall be ignored.

NOTE—This operation may interrupt data communication.

Bit 3.2278.15 is a copy of 3.0.15, and setting or clearing either bit shall set or clear the other bit. Setting either bit shall reset the 10BASE-T1L PCS.

##### 45.2.3.70.2 Loopback (3.2278.14)

The 10BASE-T1L PCS shall be placed in a loopback mode of operation when bit 3.2278.14 is set to one. When in loopback mode, the 10BASE-T1L PCS shall accept data on the transmit path and return it on the receive path.

The default value of bit 3.2278.14 is zero.

Bit 3.2278.14 is a copy of 3.0.14, and setting or clearing either bit shall set or clear the other bit. Setting either bit shall enable loopback.

#### 45.2.3.71 10BASE-T1L PCS status register (Register 3.2279)

The assignment of bits in the 10BASE-T1L PCS status register is shown in Table 45–297. All the bits in the 10BASE-T1L PCS status register are read only; a write to the 10BASE-T1L PCS status register shall have no effect.

**Table 45–297—10BASE-T1L PCS status register bit definitions**

| Bit(s)       | Name                    | Description                                                                     | R/W <sup>a</sup> |
|--------------|-------------------------|---------------------------------------------------------------------------------|------------------|
| 3.2279.15:12 | Reserved                | Value always 0                                                                  | RO               |
| 3.2279.11    | Tx LPI received         | 1 = Tx PCS has received LPI<br>0 = LPI not received                             | RO/LH            |
| 3.2279.10    | Rx LPI received         | 1 = Rx PCS has received LPI<br>0 = LPI not received                             | RO/LH            |
| 3.2279.9     | Tx LPI indication       | 1 = Tx PCS is currently receiving LPI<br>0 = PCS is not currently receiving LPI | RO               |
| 3.2279.8     | Rx LPI indication       | 1 = Rx PCS is currently receiving LPI<br>0 = PCS is not currently receiving LPI | RO               |
| 3.2279.7     | Fault                   | 1 = Fault condition detected<br>0 = No fault condition detected                 | RO/LH            |
| 3.2279.6:3   | Reserved                | Value always 0                                                                  | RO               |
| 3.2279.2     | PCS receive link status | 1 = PCS receive link up<br>0 = PCS receive link down                            | RO/LL            |
| 3.2279.1:0   | Reserved                | Value always 0                                                                  | RO               |

<sup>a</sup>RO = Read only, LH = Latching high, LL = Latching low

#### **45.2.3.71.1 Tx LPI received (3.2279.11)**

When read as a one, bit 3.2279.11 indicates that the transmit 10BASE-T1L PCS has received LPI signaling one or more times since the register was last read. When read as a zero, bit 3.2279.11 indicates that the 10BASE-T1L PCS has not received LPI signaling. This bit shall be implemented with latching high behavior.

#### **45.2.3.71.2 Rx LPI received (3.2279.10)**

When read as a one, bit 3.2279.10 indicates that the receive 10BASE-T1L PCS has received LPI signaling one or more times since the register was last read. When read as a zero, bit 3.2279.10 indicates that the 10BASE-T1L PCS has not received LPI signaling. This bit shall be implemented with latching high behavior.

#### **45.2.3.71.3 Tx LPI indication (3.2279.9)**

When read as a one, bit 3.2279.9 indicates that the transmit 10BASE-T1L PCS is currently receiving LPI signals. When read as a zero, bit 3.2279.9 indicates that the 10BASE-T1L PCS is not currently receiving LPI signals. The behavior if read during a state transition is undefined.

#### **45.2.3.71.4 Rx LPI indication (3.2279.8)**

When read as a one, bit 3.2279.8 indicates that the receive 10BASE-T1L PCS is currently receiving LPI signals. When read as a zero, bit 3.2279.8 indicates that the 10BASE-T1L PCS is not currently receiving LPI signals. The behavior if read during a state transition is undefined.

#### 45.2.3.71.5 Fault (3.2279.7)

When read as a one, bit 3.2279.7 indicates that the 10BASE-T1L PCS has detected a fault condition on either the transmit or receive path. When read as a zero, bit 3.2279.7 indicates that the 10BASE-T1L PCS has not detected a fault condition. This bit shall be implemented with latching high behavior.

#### 45.2.3.71.6 PCS receive link status (3.2279.2)

When read as a one, bit 3.2279.2 indicates that the 10BASE-T1L PCS receive link is up. When read as a zero, bit 3.2279.2 indicates that the 10BASE-T1L PCS receive link was down since the last read from this bit. This bit shall be implemented with latching low behavior and is a reflection of the variable `scr_status`. If the bit is read while `scr_status` = OK, then this bit is set. If `scr_status` = NOT\_OK, then this bit is reset.

#### 45.2.3.72 10BASE-T1S PCS control register (Register 3.2291)

The assignment of bits in the 10BASE-T1S PCS control register is shown in Table 45–298. The default value for each bit of the 10BASE-T1S PCS control register should be chosen so that the initial state of the device upon power up or reset is a normal operational state without management intervention.

**Table 45–298—10BASE-T1S PCS control register bit definitions**

| Bit(s)      | Name        | Description                                           | R/W <sup>a</sup> |
|-------------|-------------|-------------------------------------------------------|------------------|
| 3.2291.15   | PCS reset   | 1 = PCS reset<br>0 = Normal operation                 | R/W, SC          |
| 3.2291.14   | Loopback    | 1 = Enable loopback mode<br>0 = Disable loopback mode | R/W              |
| 3.2291.13:9 | Reserved    | Value always 0                                        | RO               |
| 3.2291.8    | Duplex mode | 1 = Set to half duplex<br>0 = Set to full duplex      | R/W              |
| 3.2291.7:0  | Reserved    | Value always 0                                        | RO               |

<sup>a</sup>RO = Read only, R/W = Read/Write, SC = Self-clearing

##### 45.2.3.72.1 PCS reset (3.2291.15)

Resetting the 10BASE-T1S PCS is accomplished by setting bit 3.2291.15 to one. This action shall set all 10BASE-T1S PCS registers to their default states. As a consequence, this action may change the internal state of the 10BASE-T1S PCS and the state of the physical link. This action may also initiate a reset in any other MMDs that are instantiated in the same package. This bit is self-clearing, and the 10BASE-T1S PCS shall return a value of one in bit 3.2291.15 when a reset is in progress; otherwise, it shall return a value of zero. The 10BASE-T1S PCS is not required to accept a write transaction to any of its registers until the reset process is completed. The control and management interface shall be restored to operation within 0.5 s from the setting of bit 3.2291.15. During a reset, a PCS shall respond to reads from bits 3.0.15, 3.8.15:14, and 3.2291.15. Reads for all other bits shall be ignored.

NOTE—This operation may interrupt data communication.

Bit 3.2291.15 is a copy of 3.0.15, and setting or clearing either bit shall set or clear the other bit. Setting either bit shall reset the 10BASE-T1S PCS.

#### 45.2.3.72.2 Loopback (3.2291.14)

The 10BASE-T1S PCS shall be placed in a loopback mode of operation when bit 3.2291.14 is set to one. When in loopback mode, the 10BASE-T1S PCS shall accept data on the transmit path and return it on the receive path.

The default value of bit 3.2291.14 is zero.

Bit 3.2291.14 is a copy of 3.0.14, and setting or clearing either bit shall set or clear the other bit. Setting either bit shall enable loopback.

#### 45.2.3.72.3 Duplex mode (3.2291.8)

Bit 3.2291.8 is used to configure the PCS duplex\_mode variable when not operating in Multidrop mode and when Auto-Negotiation enable bit 7.512.12 is set to zero, or if Auto-Negotiation is not implemented. If bit 3.2291.8 is set to one, then duplex\_mode is set to DUPLEX\_HALF. If bit 3.2291.8 is set to zero, then duplex\_mode is set to DUPLEX\_FULL. This bit shall be ignored when the Auto-Negotiation enable bit 7.512.12 is set to one or when bit 3.2292.6 indicates the PHY is not capable of full-duplex operation. If the PHY reports via bit 3.2292.6 that it is not capable of operating in full-duplex mode, the value of bit 3.2291.8 is undefined.

Bit 3.2291.8 is an inverted copy of bit 0.8 (see Table 22–7). When the PHY reports via bit 3.2292.6 that it is capable of operating in full-duplex mode, setting bit 0.8 or bit 3.2291.8 shall clear the other bit, and clearing bit 0.8 or bit 3.2291.8 shall set the other bit.

#### 45.2.3.73 10BASE-T1S PCS status register (Register 3.2292)

The assignment of bits in the 10BASE-T1S PCS status register is shown in Table 45–299. All the bits in the 10BASE-T1S PCS status register are read only; a write to the 10BASE-T1S PCS status register shall have no effect.

**Table 45–299—10BASE-T1S PCS status register bit definitions**

| Bit(s)      | Name                   | Description                                                                              | R/W <sup>a</sup> |
|-------------|------------------------|------------------------------------------------------------------------------------------|------------------|
| 3.2292.15:8 | Reserved               | Value always 0                                                                           | RO               |
| 3.2292.7    | Fault                  | 1 = Fault condition detected<br>0 = No fault condition detected                          | RO/LH            |
| 3.2292.6    | Full-duplex capability | 1 = PHY capable of full-duplex operation<br>0 = PHY not capable of full-duplex operation | RO               |
| 3.2292.5:0  | Reserved               | Value always 0                                                                           | RO               |

<sup>a</sup>RO = Read only, LH = Latching high

#### 45.2.3.73.1 Fault (3.2292.7)

When read as a one, bit 3.2292.7 indicates that the 10BASE-T1S PCS has detected a fault condition on either the transmit or receive path. When read as a zero, bit 3.2292.7 indicates that the 10BASE-T1S PCS has not detected a fault condition. This bit shall be implemented with latching high behavior.

#### 45.2.3.73.2 Full-duplex capability (3.2292.6)

When read as a one, bit 3.2292.6 indicates that the 10BASE-T1S PHY is capable of full-duplex operation. When read as a zero, bit 3.2292.6 indicates that the 10BASE-T1S PHY is not capable of full-duplex operation.

#### 45.2.3.74 10BASE-T1S PCS diagnostic 1 (Register 3.2293)

The assignment of bits in the 10BASE-T1S PCS diagnostic 1 register is shown in Table 45–300. All the bits in the 10BASE-T1S PCS diagnostic 1 register are read only and self-clear on read; a write to the 10BASE-T1S PCS diagnostic 1 register shall have no effect.

**Table 45–300—10BASE-T1S PCS diagnostic 1 register bit definitions**

| Bit(s)      | Name                | Description                                                                                        | R/W <sup>a</sup> |
|-------------|---------------------|----------------------------------------------------------------------------------------------------|------------------|
| 3.2293.15:0 | Remote jabber count | 16-bit field counting the number of remote jabber errors received since last read of this register | RO, SC           |

<sup>a</sup>RO = Read only, SC = Self-clearing

##### 45.2.3.74.1 Remote jabber count (3.2293.15:0)

Bits 3.2293.15:0 report the number of received jabber events since the last time register 3.2293 was read. The remote jabber count shall not wrap. When the maximum allowed value (65 535) is reached, the count stops until this register is cleared by a read operation.

#### 45.2.3.75 10BASE-T1S PCS diagnostic 2 (Register 3.2294)

The assignment of bits in the 10BASE-T1S PCS diagnostic 2 register is shown in Table 45–301. All the bits in the 10BASE-T1S PCS diagnostic 2 register are read only and self-clear on read; a write to the 10BASE-T1S PCS diagnostic 2 register shall have no effect.

**Table 45–301—10BASE-T1S PCS diagnostic 2 register bit definitions**

| Bit(s)      | Name           | Description                                                                                                                                | R/W <sup>a</sup> |
|-------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 3.2294.15:0 | CorruptedTxCnt | 16-bit field counting each time a transmission initiated locally results in a corrupted signal at the MDI since last read of this register | RO, SC           |

<sup>a</sup>RO = Read only, SC = Self-clearing



#### 45.2.3.75.1 CorruptedTxCnt (3.2294.15:0)

Bits 3.2294.15:0 count up at each positive edge of the MII signal COL. When the maximum allowed value (65 535) is reached, the count stops until this register is cleared by a read operation.

#### 45.2.3.76 1000BASE-T1 PCS control register (Register 3.2304)

The assignment of bits in the 1000BASE-T1 PCS control register is shown in Table 45–302. The default value for each bit of the 1000BASE-T1 PCS control register should be chosen so that the initial state of the device upon power up or reset is a normal operational state without management intervention.

**Table 45–302—1000BASE-T1 PCS control register bit definitions**

| Bit(s)      | Name      | Description                                           | R/W <sup>a</sup> |
|-------------|-----------|-------------------------------------------------------|------------------|
| 3.2304.15   | PCS reset | 1 = PCS reset<br>0 = Normal operation                 | R/W, SC          |
| 3.2304.14   | Loopback  | 1 = Enable loopback mode<br>0 = Disable loopback mode | R/W              |
| 3.2304.13:0 | Reserved  | Value always 0                                        | RO               |

<sup>a</sup>RO = Read only, R/W = Read/Write, SC = Self-clearing

##### 45.2.3.76.1 PCS reset (3.2304.15)

Resetting the 1000BASE-T1 PCS is accomplished by setting bit 3.2304.15 to a one. This action shall set all 1000BASE-T1 PCS registers to their default states. As a consequence, this action may change the internal state of the 1000BASE-T1 PCS and the state of the physical link. This action may also initiate a reset in any other MMDs that are instantiated in the same package. This bit is self-clearing, and the 1000BASE-T1 PCS shall return a value of one in bit 3.2304.15 when a reset is in progress; otherwise, it shall return a value of zero. The 1000BASE-T1 PCS is not required to accept a write transaction to any of its registers until the reset process is completed. The control and management interface is restored to operation as defined in 97.3.2.1, starting when bit 3.2304.15 is set. During a reset, a PCS shall respond to reads from register bits 3.0.15, 3.8.15:14, and 3.2304.15. All other register bits shall be ignored.

NOTE—This operation may interrupt data communication.

Bit 3.2304.15 is a copy of 3.0.15 and setting or clearing either bit shall set or clear the other bit. Setting either bit shall reset the 1000BASE-T1 PCS.

##### 45.2.3.76.2 Loopback (3.2304.14)

The 1000BASE-T1 PCS shall be placed in a loopback mode of operation when bit 3.2304.14 is set to a one. When bit 3.2304.14 is set to a one, the 1000BASE-T1 PCS shall accept data on the transmit path and return it on the receive path.

The default value of bit 3.2304.14 is zero.

Bit 3.2304.14 is a copy of 3.0.14 and setting or clearing either bit shall set or clear the other bit. Setting either bit shall enable loopback.

#### 45.2.3.77 1000BASE-T1 PCS status 1 register (Register 3.2305)

The assignment of bits in the 1000BASE-T1 PCS status 1 register is shown in Table 45–303. All the bits in the 1000BASE-T1 PCS status 1 register are read only; a write to the 1000BASE-T1 PCS status 1 register shall have no effect.

**Table 45–303—1000BASE-T1 PCS status 1 register bit definitions**

| Bit(s)       | Name                    | Description                                                                     | R/W <sup>a</sup> |
|--------------|-------------------------|---------------------------------------------------------------------------------|------------------|
| 3.2305.15:12 | Reserved                | Value always 0                                                                  | RO               |
| 3.2305.11    | Tx LPI received         | 1 = Tx PCS has received LPI<br>0 = LPI not received                             | RO/LH            |
| 3.2305.10    | Rx LPI received         | 1 = Rx PCS has received LPI<br>0 = LPI not received                             | RO/LH            |
| 3.2305.9     | Tx LPI indication       | 1 = Tx PCS is currently receiving LPI<br>0 = PCS is not currently receiving LPI | RO               |
| 3.2305.8     | Rx LPI indication       | 1 = Rx PCS is currently receiving LPI<br>0 = PCS is not currently receiving LPI | RO               |
| 3.2305.7     | Fault                   | 1 = Fault condition detected<br>0 = No fault condition detected                 | RO               |
| 3.2305.6:3   | Reserved                | Value always 0                                                                  | RO               |
| 3.2305.2     | PCS receive link status | 1 = PCS receive link up<br>0 = PCS receive link down                            | RO/LL            |
| 3.2305.1:0   | Reserved                | Value always 0                                                                  | RO               |

<sup>a</sup>RO = Read only, LH = Latching high, LL = Latching low

##### 45.2.3.77.1 Tx LPI received (3.2305.11)

When read as a one, bit 3.2305.11 indicates that the transmit 1000BASE-T1 PCS has received LPI signaling one or more times since the register was last read. When read as a zero, bit 3.2305.11 indicates that the 1000BASE-T1 PCS has not received LPI signaling. This bit shall be implemented with latching high behavior.

##### 45.2.3.77.2 Rx LPI received (3.2305.10)

When read as a one, bit 3.2305.10 indicates that the receive 1000BASE-T1 PCS has received LPI signaling one or more times since the register was last read. When read as a zero, bit 3.2305.10 indicates that the 1000BASE-T1 PCS has not received LPI signaling. This bit shall be implemented with latching high behavior.

##### 45.2.3.77.3 Tx LPI indication (3.2305.9)

When read as a one, bit 3.2305.9 indicates that the transmit 1000BASE-T1 PCS is currently receiving LPI signals. When read as a zero, bit 3.2305.9 indicates that the 1000BASE-T1 PCS is not currently receiving LPI signals. The behavior if read during a state transition is undefined.

#### 45.2.3.77.4 Rx LPI indication (3.2305.8)

When read as a one, bit 3.2305.8 indicates that the receive 1000BASE-T1 PCS is currently receiving LPI signals. When read as a zero, bit 3.2305.8 indicates that the 1000BASE-T1 PCS is not currently receiving LPI signals. The behavior if read during a state transition is undefined.

#### 45.2.3.77.5 Fault (3.2305.7)

When read as a one, bit 3.2305.7 indicates that the 1000BASE-T1 PCS has detected a fault condition on either the transmit or receive paths. When read as a zero, bit 3.2305.7 indicates that the 1000BASE-T1 PCS has not detected a fault condition.

#### 45.2.3.77.6 PCS receive link status (3.2305.2)

When read as a one, bit 3.2305.2 indicates that the 1000BASE-T1 PCS receive link is up. When read as a zero, bit 3.2305.2 indicates that the 1000BASE-T1 PCS receive link was down since the last read from this register. This bit is a latching low version of bit 3.2306.10. The PCS receive link status bit shall be implemented with latching low behavior.

#### 45.2.3.78 1000BASE-T1 PCS status 2 register (Register 3.2306)

The assignment of bits in the 1000BASE-T1 PCS status 2 register is shown in Table 45–304. All the bits in the 1000BASE-T1 PCS status 2 register are read only; a write to the 1000BASE-T1 PCS status 2 register shall have no effect.

**Table 45–304—1000BASE-T1 PCS status 2 register bit definitions**

| Bit(s)       | Name                | Description                                                                | R/W <sup>a</sup> |
|--------------|---------------------|----------------------------------------------------------------------------|------------------|
| 3.2306.15:11 | Reserved            | Value always 0                                                             | RO               |
| 3.2306.10    | Receive link status | 1 = PCS receive link up<br>0 = PCS receive link down                       | RO               |
| 3.2306.9     | PCS high BER        | 1 = PCS reporting a high BER<br>0 = PCS not reporting a high BER           | RO               |
| 3.2306.8     | PCS block lock      | 1 = PCS locked to received blocks<br>0 = PCS not locked to received blocks | RO               |
| 3.2306.7     | Latched high BER    | 1 = PCS has reported a high BER<br>0 = PCS has not reported a high BER     | RO/LH            |
| 3.2306.6     | Latched block lock  | 1 = PCS has block lock<br>0 = PCS does not have block lock                 | RO/LL            |
| 3.2306.5:0   | BER count           | BER counter                                                                | RO/NR            |

<sup>a</sup>RO = Read only, LH = Latching High, LL = Latching Low, NR = Non Roll-over

#### 45.2.3.78.1 Receive link status (3.2306.10)

When read as a one, bit 3.2306.10 indicates that the 1000BASE-T1 PCS is in a fully operational state. When read as a zero, bit 3.2306.10 indicates that the 1000BASE-T1 PCS is not fully operational. This bit is a reflection of the PCS\_status variable defined in 97.3.7.1.

#### 45.2.3.78.2 PCS high BER (3.2306.9)

When read as a one, bit 3.2306.9 indicates that the 1000BASE-T1 PCS receiver is detecting a BER of  $> 4 \times 10^{-4}$ . When read as a zero, bit 3.2306.9 indicates that the 1000BASE-T1 PCS is not detecting a BER of  $> 4 \times 10^{-4}$ . This bit is a reflection of the state of the hi\_rfer variable defined in 97.3.7.1.

#### 45.2.3.78.3 PCS block lock (3.2306.8)

When read as a one, bit 3.2306.8 indicates that the 1000BASE-T1 PCS receiver has block lock. When read as a zero, bit 3.2306.8 indicates that the 1000BASE-T1 PCS receiver has not achieved block lock. This bit is a reflection of the state of the block\_lock variable defined in 97.3.7.1.

#### 45.2.3.78.4 Latched high BER (3.2306.7)

When read as a one, bit 3.2306.7 indicates that the 1000BASE-T1 PCS has detected a high BER one or more times since the register was last read. When read as a zero, bit 3.2306.7 indicates that the 1000BASE-T1 PCS has not detected a high BER. The latched high BER bit shall be implemented with latching high behavior. This bit is a latching high version of the 1000BASE-T1 PCS high BER status bit (3.2306.9).

#### 45.2.3.78.5 Latched block lock (3.2306.6)

When read as a one, bit 3.2306.6 indicates that the 1000BASE-T1 PCS has achieved block lock. When read as a zero, bit 3.2306.6 indicates that the 1000BASE-T1 PCS has lost block lock one or more times since the register was last read. The latched block lock bit shall be implemented with latching low behavior.

This bit is a latching low version of the 1000BASE-T1 PCS block lock status bit (3.2306.8).

#### 45.2.3.78.6 BER count (3.2306.5:0)

The BER counter formed by bits 3.2306.5:0 is a six bit count as defined by RFER\_count in 97.3.7.2. These bits shall be reset to all zeros when the 1000BASE-T1 PCS status 2 register is read by the management function or upon execution of the 1000BASE-T1 PCS reset. These bits shall be held at all ones in the case of overflow.

#### 45.2.3.79 BASE-T1 OAM transmit register (Register 3.2308)

The assignment of bits in the BASE-T1 OAM transmit register is shown in Table 45–305.

**Table 45–305—BASE-T1 OAM transmit register bit definitions**

| Bit(s)    | Name                      | Description                                                                                                                                                                                                                                                                                                          | R/W <sup>a</sup> |
|-----------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 3.2308.15 | BASE-T1 OAM message valid | This bit is used to indicate message data in registers 3.2308.11:8, 3.2309, 3.2310, 3.2311, and 3.2312 are valid and ready to be loaded. This bit self-clears when registers are loaded by the OAM transmit state diagram.<br>1 = Message data in registers are valid<br>0 = Message data in registers are not valid | R/W, SC          |
| 3.2308.14 | Toggle value              | Toggle value to be transmitted with message. This bit is set by the state machine and cannot be overridden by the user.                                                                                                                                                                                              | RO               |

**Table 45–305—BASE-T1 OAM transmit register bit definitions (*continued*)**

| Bit(s)      | Name                          | Description                                                                                                                                                                                                                                                                                                                | R/W <sup>a</sup> |
|-------------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 3.2308.13   | BASE-T1 OAM message received  | This bit shall self clear on read.<br>1 = BASE-T1 OAM message received by link partner<br>0 = BASE-T1 OAM message not received by link partner                                                                                                                                                                             | RO, LH           |
| 3.2308.12   | Received message toggle value | Toggle value of message that was received by link partner as indicated in 3.2308.13.                                                                                                                                                                                                                                       | RO               |
| 3.2308.11:8 | Message number                | User-defined message number to send                                                                                                                                                                                                                                                                                        | R/W              |
| 3.2308.7:4  | Reserved                      | Value always 0                                                                                                                                                                                                                                                                                                             | RO               |
| 3.2308.3    | Ping received                 | Received PingTx value from latest good BASE-T1 OAM frame received                                                                                                                                                                                                                                                          | RO               |
| 3.2308.2    | Ping transmit                 | Ping value to send to link partner                                                                                                                                                                                                                                                                                         | R/W              |
| 3.2308.1:0  | Local SNR                     | 00 = PHY link is failing and will drop link and relink within 2 ms to 4 ms after the end of the current BASE-T1 OAM frame.<br>01 = LPI refresh is insufficient to maintain PHY SNR. Request link partner to exit LPI and send idles (used only when EEE is enabled).<br>10 = PHY SNR is marginal.<br>11 = PHY SNR is good. | RO               |

<sup>a</sup>RO = Read only, R/W = Read/Write, LH = Latching High, SC = Self-clearing

#### 45.2.3.79.1 BASE-T1 OAM message valid (3.2308.15)

Bit 3.2308.15 shall be set to one when the BASE-T1 OAM message to be transmitted in registers 3.2309, 3.2310, 3.2311, and 3.2312 and the message number in 3.2308.11:8 are properly configured to be transmitted. Bit 3.2308.15 self-clears to indicate whether the next BASE-T1 OAM message can be written into the registers.

#### 45.2.3.79.2 Toggle value (3.2308.14)

The state machine shall assign a value alternating between zero and one to associate with the 8-octet BASE-T1 OAM message transmitted by the PHY. Bit 3.2308.14 should be read and recorded prior to setting 3.2308.15 to one. The recorded value can be correlated with 3.2308.12 as a confirmation that the BASE-T1 OAM message is received by the link partner.

#### 45.2.3.79.3 BASE-T1 OAM message received (3.2308.13)

Bit 3.2308.13 shall indicate whether the most recently transmitted BASE-T1 OAM message with a toggle bit value in 3.2308.12 was received, read, and acknowledged by the link partner. This variable shall clear on read.

#### 45.2.3.79.4 Received message toggle value (3.2308.12)

Bit 3.2308.12 indicates the toggle bit value of the BASE-T1 OAM message that was received, read, and most recently acknowledged by the link partner. This bit is valid only if 3.2308.13 is one.

#### 45.2.3.79.5 Message number (3.2308.11:8)

Bits 3.2308.11:8 contain the BASE-T1 OAM message number to be transmitted. This field is user defined but it is recommended that it be used to indicate the meaning of the 8-octet BASE-T1 OAM message. If used this way, up to 16 different 8-octet messages can be exchanged. The message number is user defined and its definition is outside the scope of this standard.

#### 45.2.3.79.6 Ping received (3.2308.3)

Bit 3.2308.3 represents the value of the most recent Ping RX received from the link partner (see 97.3.8.2.3).

#### 45.2.3.79.7 Ping transmit (3.2308.2)

Bit 3.2308.2 represents the value to be sent to the link partner via the Ping TX function (see 97.3.8.2.4).

#### 45.2.3.79.8 Local SNR (3.2308.1:0)

Bits 3.2308.1:0 are set by the BASE-T1 PHY to indicate the status of the receiver. The definitions of good, marginal, when to request idles, and when to request retrain are implementation dependent.

#### 45.2.3.80 BASE-T1 OAM message register (Registers 3.2309 to 3.2312)

The BASE-T1 OAM message register contains the 8-octet BASE-T1 OAM message data to be transmitted. The 8-octet message data is user defined and its definition is outside the scope of this standard. See Table 45–306.

**Table 45–306—BASE-T1 OAM message register bit definitions**

| Bit(s)      | Name                  | Description                             | R/W <sup>a</sup> |
|-------------|-----------------------|-----------------------------------------|------------------|
| 3.2309.15:8 | BASE-T1 OAM message 1 | Message octet 1. LSB transmitted first. | R/W              |
| 3.2309.7:0  | BASE-T1 OAM message 0 | Message octet 0. LSB transmitted first. | R/W              |
| 3.2310.15:8 | BASE-T1 OAM message 3 | Message octet 3. LSB transmitted first. | R/W              |
| 3.2310.7:0  | BASE-T1 OAM message 2 | Message octet 2. LSB transmitted first. | R/W              |
| 3.2311.15:8 | BASE-T1 OAM message 5 | Message octet 5. LSB transmitted first. | R/W              |
| 3.2311.7:0  | BASE-T1 OAM message 4 | Message octet 4. LSB transmitted first. | R/W              |
| 3.2312.15:8 | BASE-T1 OAM message 7 | Message octet 7. LSB transmitted first. | R/W              |
| 3.2312.7:0  | BASE-T1 OAM message 6 | Message octet 6. LSB transmitted first. | R/W              |

<sup>a</sup>R/W = Read/Write

#### 45.2.3.81 BASE-T1 OAM receive register (Register 3.2313)

The assignment of bits in the BASE-T1 OAM receive register is shown in Table 45–307.

**Table 45–307—BASE-T1 OAM receive register bit definitions**

| Bit(s)       | Name                                   | Description                                                                                                                                                                                                                                                                                                                                                               | R/W <sup>a</sup> |
|--------------|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 3.2313.15    | Link partner BASE-T1 OAM message valid | This bit is used to indicate message data in registers 3.2313.11:8, 3.2314, 3.2315, 3.2316, and 3.2317 are stored and ready to be read. See 45.2.3.81.1 for self-clearing behavior.<br>1 = Message data in registers are valid<br>0 = Message data in registers are not valid                                                                                             | RO, SC           |
| 3.2313.14    | Link partner toggle value              | Toggle value received with message.                                                                                                                                                                                                                                                                                                                                       | RO               |
| 3.2313.13:12 | Reserved                               | Value always 0                                                                                                                                                                                                                                                                                                                                                            | RO               |
| 3.2313.11:8  | Link partner message number            | Message number from link partner                                                                                                                                                                                                                                                                                                                                          | RO               |
| 3.2313.7:2   | Reserved                               | Value always 0                                                                                                                                                                                                                                                                                                                                                            | RO               |
| 3.2313.1:0   | Link partner SNR                       | 00 = Link partner link is failing and will drop link and relink within 2 ms to 4 ms after the end of the current BASE-T1 OAM frame.<br>01 = LPI refresh is insufficient to maintain link partner SNR. Link partner requests local PHY to exit LPI and send idles (used only when EEE is enabled).<br>10 = Link partner SNR is marginal.<br>11 = Link partner SNR is good. | RO               |

<sup>a</sup>RO = Read only, SC = Self-clearing

##### 45.2.3.81.1 Link partner BASE-T1 OAM message valid (3.2313.15)

Bit 3.2313.15 shall be set to one when the BASE-T1 OAM message from the link partner is stored into registers 3.2314, 3.2315, 3.2316, and 3.2317 and the message number in 3.2313.11:8. Bit 3.2313.15 shall self-clear when register 3.2317 is read.

##### 45.2.3.81.2 Link partner toggle value (3.2313.14)

Bit 3.2313.14 indicates the toggle value associated with the 8-octet BASE-T1 OAM message from the link partner.

##### 45.2.3.81.3 Link partner message number (3.2313.11:8)

Bits 3.2313.11:8 contain the BASE-T1 OAM message number from the link partner.

##### 45.2.3.81.4 Link partner SNR (3.2313.1:0)

Bits 3.2313.1:0 indicate the status of the link partner receiver. The definitions of good, marginal, when to request idles, and when to request retrain are implementation dependent.

#### 45.2.3.82 Link partner BASE-T1 OAM message register (Registers 3.2314 to 3.2317)

The link partner BASE-T1 OAM message register contains the 8-octet BASE-T1 OAM message data from the link partner. See Table 45–308.

**Table 45–308—Link partner BASE-T1 OAM message register bit definitions**

| Bit(s)      | Name                               | Description                          | R/W <sup>a</sup> |
|-------------|------------------------------------|--------------------------------------|------------------|
| 3.2314.15:8 | Link partner BASE-T1 OAM message 1 | Message octet 1. LSB received first. | RO               |
| 3.2314.7:0  | Link partner BASE-T1 OAM message 0 | Message octet 0. LSB received first. | RO               |
| 3.2315.15:8 | Link partner BASE-T1 OAM message 3 | Message octet 3. LSB received first. | RO               |
| 3.2315.7:0  | Link partner BASE-T1 OAM message 2 | Message octet 2. LSB received first. | RO               |
| 3.2316.15:8 | Link partner BASE-T1 OAM message 5 | Message octet 5. LSB received first. | RO               |
| 3.2316.7:0  | Link partner BASE-T1 OAM message 4 | Message octet 4. LSB received first. | RO               |
| 3.2317.15:8 | Link partner BASE-T1 OAM message 7 | Message octet 7. LSB received first. | RO               |
| 3.2317.7:0  | Link partner BASE-T1 OAM message 6 | Message octet 6. LSB received first. | RO               |

<sup>a</sup>RO = Read only

#### 45.2.3.83 MultiGBASE-T1 OAM status message register (Register 3.2318 and 3.2319)

The MultiGBASE-T1 OAM status message register contains octets 8 through 11 of the MultiGBASE-T1 OAM message data to be transmitted. See 149.3.9.2.12 for details on the OAM status message definition. See Table 45–309.

**Table 45–309—MultiGBASE-T1 OAM status message register bit definitions**

| Bit(s)      | Name                                | Description                              | R/W <sup>a</sup> |
|-------------|-------------------------------------|------------------------------------------|------------------|
| 3.2318.15:8 | MultiGBASE-T1 OAM status message 9  | Message octet 9. LSB transmitted first.  | R/W              |
| 3.2318.7:0  | MultiGBASE-T1 OAM status message 8  | Message octet 8. LSB transmitted first.  | R/W              |
| 3.2319.15:8 | MultiGBASE-T1 OAM status message 11 | Message octet 11. LSB transmitted first. | R/W              |
| 3.2319.7:0  | MultiGBASE-T1 OAM status message 10 | Message octet 10. LSB transmitted first. | R/W              |

<sup>a</sup>R/W = Read/Write



#### 45.2.3.84 Link partner MultiGBASE-T1 OAM status message register (Register 3.2320 and 3.2321)

The link partner MultiGBASE-T1 OAM status message register contains octets 8 through 11 of the MultiGBASE-T1 OAM message data from the link partner. See 149.3.9.2.12 for details on the OAM status message definition. See Table 45–310.

**Table 45–310—Link partner MultiGBASE-T1 OAM status message register bit definitions**

| Bit(s)      | Name                                             | Description                           | R/W <sup>a</sup> |
|-------------|--------------------------------------------------|---------------------------------------|------------------|
| 3.2320.15:8 | Link partner MultiGBASE-T1 OAM status message 9  | Message octet 9. LSB received first.  | RO               |
| 3.2320.7:0  | Link partner MultiGBASE-T1 OAM status message 8  | Message octet 8. LSB received first.  | RO               |
| 3.2321.15:8 | Link partner MultiGBASE-T1 OAM status message 11 | Message octet 11. LSB received first. | RO               |
| 3.2321.7:0  | Link partner MultiGBASE-T1 OAM status message 10 | Message octet 10. LSB received first. | RO               |

<sup>a</sup>RO = Read only

#### 45.2.3.85 MultiGBASE-T1 PCS control register (Register 3.2322)

The assignment of bits in the MultiGBASE-T1 PCS control register is shown in Table 45–311. The default value for each bit of the MultiGBASE-T1 PCS control register is chosen so that the initial state of the device upon power up or reset is a normal operational state without management intervention.

**Table 45–311—MultiGBASE-T1 PCS control register bit definitions**

| Bit(s)      | Name      | Description                                           | R/W <sup>a</sup> |
|-------------|-----------|-------------------------------------------------------|------------------|
| 3.2322.15   | PCS reset | 1 = PCS reset<br>0 = Normal operation                 | R/W, SC          |
| 3.2322.14   | Loopback  | 1 = Enable loopback mode<br>0 = Disable loopback mode | R/W              |
| 3.2322.13:0 | Reserved  | Value always 0                                        | RO               |

<sup>a</sup>RO = Read only, R/W = Read/Write, SC = Self-clearing

##### 45.2.3.85.1 PCS reset (3.2322.15)

Resetting the MultiGBASE-T1 PCS is accomplished by setting bit 3.2322.15 to a one. This action shall set all MultiGBASE-T1 PCS registers to their default states. As a consequence, this action may change the internal state of the MultiGBASE-T1 PCS and the state of the physical link. This action may also initiate a reset in any other MMDs that are instantiated in the same package. Bit 3.2322.15 is self-clearing, and the MultiGBASE-T1 PCS shall return a value of one in bit 3.2322.15 when a reset is in progress; otherwise, it shall return a value of zero. The MultiGBASE-T1 PCS is not required to accept a write transaction to any of its registers until the reset process is completed. The control and management interface is restored to

operation as defined in 149.3.2.1 starting when bit 3.2322.15 is set. During a reset, a PCS shall respond to reads from register bits 3.0.15, 3.8.15:14, and 3.2322.15. All other register bits shall be ignored.

NOTE—This operation may interrupt data communication.

Bit 3.2322.15 is a copy of 3.0.15 and setting or clearing either bit shall set or clear the other bit. Setting either bit shall reset the MultiGBASE-T1 PCS.

#### 45.2.3.85.2 Loopback (3.2322.14)

The MultiGBASE-T1 PCS shall be placed in a loopback mode of operation when bit 3.2322.14 is set to a one. When bit 3.2322.14 is set to a one, the MultiGBASE-T1 PCS shall accept data on the transmit path and return it on the receive path.

The default value of bit 3.2322.14 is zero.

Bit 3.2322.14 is a copy of 3.0.14 and setting or clearing either bit shall set or clear the other bit. Setting either bit shall enable loopback.

#### 45.2.3.86 MultiGBASE-T1 PCS status 1 register (Register 3.2323)

The assignment of bits in the MultiGBASE-T1 PCS status 1 register is shown in Table 45–312. All the bits in the MultiGBASE-T1 PCS status 1 register are read only; a write to the MultiGBASE-T1 PCS status 1 register shall have no effect.

**Table 45–312—MultiGBASE-T1 PCS status 1 register bit definitions**

| Bit(s)       | Name                    | Description                                                                     | R/W <sup>a</sup> |
|--------------|-------------------------|---------------------------------------------------------------------------------|------------------|
| 3.2323.15:12 | Reserved                | Value always 0                                                                  | RO               |
| 3.2323.11    | Tx LPI received         | 1 = Tx PCS has received LPI<br>0 = LPI not received                             | RO/LH            |
| 3.2323.10    | Rx LPI received         | 1 = Rx PCS has received LPI<br>0 = LPI not received                             | RO/LH            |
| 3.2323.9     | Tx LPI indication       | 1 = Tx PCS is currently receiving LPI<br>0 = PCS is not currently receiving LPI | RO               |
| 3.2323.8     | Rx LPI indication       | 1 = Rx PCS is currently receiving LPI<br>0 = PCS is not currently receiving LPI | RO               |
| 3.2323.7     | Fault                   | 1 = Fault condition detected<br>0 = No fault condition detected                 | RO               |
| 3.2323.6:3   | Reserved                | Value always 0                                                                  | RO               |
| 3.2323.2     | PCS receive link status | 1 = PCS receive link up<br>0 = PCS receive link down                            | RO/LL            |
| 3.2323.1:0   | Reserved                | Value always 0                                                                  | RO               |

<sup>a</sup>RO = Read only, LH = Latching high, LL = Latching low

#### **45.2.3.86.1 Tx LPI received (3.2323.11)**

When read as a one, bit 3.2323.11 indicates that the transmit MultiGBASE-T1 PCS has received LPI signaling one or more times since the register was last read. When read as a zero, bit 3.2323.11 indicates that the transmit MultiGBASE-T1 PCS has not received LPI signaling. Bit 3.2323.11 shall be implemented with latching high behavior.

#### **45.2.3.86.2 Rx LPI received (3.2323.10)**

When read as a one, bit 3.2323.10 indicates that the receive MultiGBASE-T1 PCS has received LPI signaling one or more times since the register was last read. When read as a zero, bit 3.2323.10 indicates that the receive MultiGBASE-T1 PCS has not received LPI signaling. Bit 3.2323.10 shall be implemented with latching high behavior.

#### **45.2.3.86.3 Tx LPI indication (3.2323.9)**

When read as a one, bit 3.2323.9 indicates that the transmit MultiGBASE-T1 PCS is currently receiving LPI signals. When read as a zero, bit 3.2323.9 indicates that the transmit MultiGBASE-T1 PCS is not currently receiving LPI signals. The behavior if read during a state transition is undefined.

#### **45.2.3.86.4 Rx LPI indication (3.2323.8)**

When read as a one, bit 3.2323.8 indicates that the receive MultiGBASE-T1 PCS is currently receiving LPI signals. When read as a zero, bit 3.2323.8 indicates that the receive MultiGBASE-T1 PCS is not currently receiving LPI signals. The behavior if read during a state transition is undefined.

#### **45.2.3.86.5 Fault (3.2323.7)**

When read as a one, bit 3.2323.7 indicates that the MultiGBASE-T1 PCS has detected a fault condition on either the transmit or receive paths. When read as a zero, bit 3.2323.7 indicates that the MultiGBASE-T1 PCS has not detected a fault condition.

#### **45.2.3.86.6 PCS receive link status (3.2323.2)**

When read as a one, bit 3.2323.2 indicates that the MultiGBASE-T1 PCS receive link is up. When read as a zero, bit 3.2323.2 indicates that the MultiGBASE-T1 PCS receive link was down since the last read from this register. Bit 3.2323.2 is a latching low version of bit 3.2324.10. The PCS receive link status bit shall be implemented with latching low behavior.

#### **45.2.3.87 MultiGBASE-T1 PCS status 2 register (Register 3.2324)**

The assignment of bits in the MultiGBASE-T1 PCS status 2 register is shown in Table 45–313. All the bits in the MultiGBASE-T1 PCS status 2 register are read only; a write to the MultiGBASE-T1 PCS status 2 register shall have no effect.

##### **45.2.3.87.1 Receive link status (3.2324.10)**

When read as a one, bit 3.2324.10 indicates that the MultiGBASE-T1 PCS is in a fully operational state. When read as a zero, bit 3.2324.10 indicates that the MultiGBASE-T1 PCS is not fully operational. Bit 3.2324.10 is a reflection of the `pcs_status` variable defined in 149.3.8.1.

**Table 45–313—MultiGBASE-T1 PCS status 2 register bit definitions**

| Bit(s)       | Name                | Description                                                                | R/W <sup>a</sup> |
|--------------|---------------------|----------------------------------------------------------------------------|------------------|
| 3.2324.15:11 | Reserved            | Value always 0                                                             | RO               |
| 3.2324.10    | Receive link status | 1 = PCS receive link up<br>0 = PCS receive link down                       | RO               |
| 3.2324.9     | PCS high RFER       | 1 = PCS reporting a high RFER<br>0 = PCS not reporting a high RFER         | RO               |
| 3.2324.8     | PCS block lock      | 1 = PCS locked to received blocks<br>0 = PCS not locked to received blocks | RO               |
| 3.2324.7     | Latched high BER    | 1 = PCS has reported a high BER<br>0 = PCS has not reported a high BER     | RO/LH            |
| 3.2324.6     | Latched block lock  | 1 = PCS has block lock<br>0 = PCS does not have block lock                 | RO/LL            |
| 3.2324.5:0   | BER count           | Count of RS-FEC frame errors since last read                               | RO/NR            |

<sup>a</sup>RO = Read only, LH = Latching High, LL = Latching Low, NR = Non Roll-over

#### 45.2.3.87.2 PCS high RFER (3.2324.9)

When read as a one, bit 3.2324.9 indicates that the MultiGBASE-T1 PCS receiver is detecting 16 or more RS-FEC errored blocks in 312 500 bit times (one `rfer_timer` interval). When read as a zero, bit 3.2324.9 indicates that the MultiGBASE-T1 PCS is detecting fewer than 16 RS-FEC errored blocks in 312 500 bit times. Bit 3.2324.9 is a reflection of the state of the `hi_rfer` variable defined in 149.3.8.1.

#### 45.2.3.87.3 PCS block lock (3.2324.8)

When read as a one, bit 3.2324.8 indicates that the MultiGBASE-T1 PCS receiver has block lock. When read as a zero, bit 3.2324.8 indicates that the MultiGBASE-T1 PCS receiver has not achieved block lock. Bit 3.2324.8 is a reflection of the state of the `block_lock` variable defined in 149.3.8.1.

#### 45.2.3.87.4 Latched high BER (3.2324.7)

When read as a one, bit 3.2324.7 indicates that the MultiGBASE-T1 PCS has detected a high BER one or more times since the register was last read. When read as a zero, bit 3.2324.7 indicates that the MultiGBASE-T1 PCS has not detected a high BER. The latched high BER bit shall be implemented with latching high behavior. Bit 3.2324.7 is a latching high version of the MultiGBASE-T1 PCS high RFER status bit (3.2324.9).

#### 45.2.3.87.5 Latched block lock (3.2324.6)

When read as a one, bit 3.2324.6 indicates that the MultiGBASE-T1 PCS has achieved block lock. When read as a zero, bit 3.2324.6 indicates that the MultiGBASE-T1 PCS has lost block lock one or more times since the register was last read. The latched block lock bit shall be implemented with latching low behavior.

Bit 3.2324.6 is a latching low version of the MultiGBASE-T1 PCS block lock status bit (3.2324.8).

#### 45.2.3.87.6 BER count (3.2324.5:0)

The BER counter formed by bits 3.2324.5:0 is a six bit counter as defined by RFER\_count in 149.3.8.2. These bits shall be reset to all zeros when the MultiGBASE-T1 PCS status 2 register is read by the management function or upon execution of the MultiGBASE-T1 PCS reset. These bits shall be held at all ones in the case of overflow.

#### 45.2.4 PHY XS registers

The assignment of registers in the PHY XS is shown in Table 45–314.

**Table 45–314—PHY XS registers**

| Register address  | Register name                               | Subclause |
|-------------------|---------------------------------------------|-----------|
| 4.0               | PHY XS control 1                            | 45.2.4.1  |
| 4.1               | PHY XS status 1                             | 45.2.4.2  |
| 4.2, 4.3          | PHY XS device identifier                    | 45.2.4.3  |
| 4.4               | PHY XS speed ability                        | 45.2.4.4  |
| 4.5, 4.6          | PHY XS devices in package                   | 45.2.4.5  |
| 4.7               | Reserved                                    |           |
| 4.8               | PHY XS status 2                             | 45.2.4.6  |
| 4.9 through 4.13  | Reserved                                    |           |
| 4.14, 4.15        | PHY XS package identifier                   | 45.2.4.7  |
| 4.16 through 4.19 | Reserved                                    |           |
| 4.20              | EEE capability                              | 45.2.4.8  |
| 4.21              | Reserved                                    |           |
| 4.22              | EEE wake error counter                      | 45.2.4.9  |
| 4.23              | Reserved                                    |           |
| 4.24              | 10G PHY XGXS lane status                    | 45.2.4.10 |
| 4.25              | 10G PHY XGXS test control                   | 45.2.4.11 |
| 4.26 through 4.31 | Reserved                                    |           |
| 4.32              | BASE-R PHY XS status 1                      | 45.2.4.12 |
| 4.33 through 4.41 | Reserved                                    |           |
| 4.42              | BASE-R PHY XS test pattern control          | 45.2.4.13 |
| 4.43 through 4.49 | Reserved                                    |           |
| 4.50              | Multi-lane BASE-R PHY XS alignment status 1 | 45.2.4.14 |
| 4.51              | Reserved                                    |           |
| 4.52              | Multi-lane BASE-R PHY XS alignment status 3 | 45.2.4.15 |
| 4.53              | Multi-lane BASE-R PHY XS alignment status 4 | 45.2.4.16 |
| 4.54              | Reserved                                    |           |

**Table 45–314—PHY XS registers (*continued*)**

| Register address          | Register name                                 | Subclause               |
|---------------------------|-----------------------------------------------|-------------------------|
| 4.400 through 4.415       | PHY XS lane mapping, lane 0 through 15        | 45.2.4.17,<br>45.2.4.18 |
| 4.415 through 4.599       | Reserved                                      |                         |
| 4.600 through 4.631       | PHY XS FEC symbol error counter, lane 0 to 15 | 45.2.4.19,<br>45.2.4.20 |
| 4.632 through 4.799       | Reserved                                      |                         |
| 4.800                     | PHY XS FEC control                            | 45.2.4.21               |
| 4.801                     | PHY XS FEC status                             | 45.2.4.22               |
| 4.802, 4.803              | PHY XS FEC corrected codewords counter        | 45.2.4.23               |
| 4.804, 4.805              | PHY XS FEC uncorrected codewords counter      | 45.2.4.24               |
| 4.806, 4.807              | PHY XS FEC degraded SER activate threshold    | 45.2.4.25               |
| 4.808, 4.809              | PHY XS FEC degraded SER deactivate threshold  | 45.2.4.26               |
| 4.810, 4.811              | PHY XS FEC degraded SER interval              | 45.2.4.27               |
| 4.812 through 4.1799      | Reserved                                      |                         |
| 4.1800                    | TimeSync PHY XS capability                    | 45.2.4.28               |
| 4.1801 through 4.1804     | TimeSync PHY XS transmit path data delay      | 45.2.4.29               |
| 4.1805 through 4.1808     | TimeSync PHY XS receive path data delay       | 45.2.4.30               |
| 4.1809 through 4.32767    | Reserved                                      |                         |
| 4.32 768 through 4.65 535 | Vendor specific                               |                         |

#### 45.2.4.1 PHY XS control 1 register (Register 4.0)

The assignment of bits in the PHY XS control 1 register is shown in Table 45–315. The default value for each bit of the PHY XS control 1 register should be chosen so that the initial state of the device upon power up or reset is a normal operational state without management intervention.

**Table 45–315—PHY XS control 1 register bit definitions**

| Bit(s) | Name            | Description                                           | R/W <sup>a</sup> |
|--------|-----------------|-------------------------------------------------------|------------------|
| 4.0.15 | Reset           | 1 = PHY XS reset<br>0 = Normal operation              | R/W<br>SC        |
| 4.0.14 | Loopback        | 1 = Enable loopback mode<br>0 = Disable loopback mode | R/W              |
| 4.0.13 | Speed selection | 1 = Operation at 10 Gb/s and above<br>0 = Unspecified | R/W              |
| 4.0.12 | Reserved        | Value always 0                                        | RO               |

**Table 45–315—PHY XS control 1 register bit definitions (*continued*)**

| Bit(s)  | Name              | Description                                                                                                                                                                                                                                                                                                  | R/W <sup>a</sup> |
|---------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 4.0.11  | Low power         | 1 = Low-power mode<br>0 = Normal operation                                                                                                                                                                                                                                                                   | R/W              |
| 4.0.10  | Clock stop enable | 1 = The PHY XS may stop the clock during LPI<br>0 = Clock not stoppable                                                                                                                                                                                                                                      | R/W              |
| 4.0.9   | XAUI stop enable  | 1 = The PHY XS may stop XAUI signals during LPI<br>0 = XAUI not stoppable                                                                                                                                                                                                                                    | R/W              |
| 4.0.8:7 | Reserved          | Value always 0                                                                                                                                                                                                                                                                                               | RO               |
| 4.0.6   | Speed selection   | 1 = Operation at 10 Gb/s and above<br>0 = Unspecified                                                                                                                                                                                                                                                        | R/W              |
| 4.0.5:2 | Speed selection   | <div> <div>5 4 3 2</div> <div>1 1 x x = Reserved</div> <div>1 0 1 1 = Reserved</div> <div>1 0 1 0 = 400 Gb/s</div> <div>1 0 0 1 = 200 Gb/s</div> <div>1 0 0 0 = Reserved</div> <div>0 1 x x = Reserved</div> <div>0 0 1 x = Reserved</div> <div>0 0 0 1 = Reserved</div> <div>0 0 0 0 = 10 Gb/s</div> </div> | R/W              |
| 4.0.1:0 | Reserved          | Value always 0                                                                                                                                                                                                                                                                                               | RO               |

<sup>a</sup>R/W = Read/Write, SC = Self-clearing, RO = Read only

#### 45.2.4.1.1 Reset (4.0.15)

Resetting a PHY XS is accomplished by setting bit 4.0.15 to a one. This action shall set all PHY XS registers to their default states. As a consequence, this action may change the internal state of the PHY XS and the state of the physical link. This action may also initiate a reset in any other MMDs that are instantiated in the same package. This bit is self-clearing, and a PHY XS shall return a value of one in bit 4.0.15 when a reset is in progress and a value of zero otherwise. A PHY XS is not required to accept a write transaction to any of its registers until the reset process is completed. The reset process shall be completed within 0.5 s from the setting of bit 4.0.15. During a reset, a PHY XS shall respond to reads from register bits 4.0.15 and 4.8.15:14. All other register bits should be ignored.

NOTE—This operation may interrupt data communication.

#### 45.2.4.1.2 Loopback (4.0.14)

The PHY XS shall be placed in a loopback mode of operation when bit 4.0.14 is set to a one. When bit 4.0.14 is set to a one, the PHY XS shall accept data on the receive path and return it on the transmit path. The direction of the loopback path for the PHY XS is opposite to all other MMD loopbacks.

The loopback function is optional. A device's ability to perform the loopback function is advertised in the loopback ability bit of the related speed-dependent status register. A PHY XS that is unable to perform the loopback function shall ignore writes to this bit and return a value of zero when read. For 10 Gb/s operation,

the loopback functionality is detailed in 48.3.3 and the loopback ability bit is specified in the 10G PHY XGXS Lane status register.

The default value of bit 4.0.14 is zero.

NOTE—The signal path through the PHY XS that is exercised in the loopback mode of operation is implementation specific, but it is recommended that the signal path encompass as much of the PHY XS circuitry as is practical. The intention of providing this loopback mode of operation is to permit a diagnostic or self-test function to perform the transmission and reception of a PDU, thus testing the transmit and receive data paths. Other loopback signal paths may be enabled using loopback controls within other MMDs.

#### **45.2.4.1.3 Low power (4.0.11)**

A PHY XS may be placed into a low-power mode by setting bit 4.0.11 to a one. This action may also initiate a low-power mode in any other MMDs that are instantiated in the same package. The low-power mode is exited by resetting the PHY XS. The behavior of the PHY XS in transition to and from the low-power mode is implementation specific and any interface signals should not be relied upon. While in the low-power mode, the device shall, as a minimum, respond to management transactions necessary to exit the low-power mode. The default value of bit 4.0.11 is zero.

#### **45.2.4.1.4 Clock stop enable (4.0.10)**

If bit 4.0.10 is set to one, then the PHY XS may stop the transmit direction xMII clock while it is signaling LPI, otherwise it shall keep the clock active. If the PHY XS does not support EEE capability or is not able to stop the transmit clock, then this bit has no effect (see 46.3.2.4).

#### **45.2.4.1.5 XAUI stop enable (4.0.9)**

If bit 4.0.9 is set to one, then the PHY XS may stop signaling on the XAUI in the receive direction during LPI, otherwise the PHY XS shall keep the XAUI signals active. If the PHY XS does not support EEE capability or is not able to stop the receive path XAUI signals, then this bit has no effect.

#### **45.2.4.1.6 Speed selection (4.0.13, 4.0.6, 4.0.5:2)**

Speed selection bits 4.0.13 and 4.0.6 shall both be written as a one. Any attempt to change the bits to an invalid setting shall be ignored. These two bits are set to one in order to make them compatible with Clause 22.

The speed of the PHY XS may be selected using bits 5 through 2. The speed abilities of the PHY XS are advertised in the PHY XS speed ability register. A PHY XS may ignore writes to the PHY XS speed selection bits that select speeds it has not advertised in the PHY XS speed ability register. It is the responsibility of the STA entity to ensure that mutually acceptable speeds are applied consistently across all the MMDs on a particular PHY.

The PHY XS speed selection defaults to a supported ability.

#### **45.2.4.2 PHY XS status 1 register (Register 4.1)**

The assignment of bits in the PHY XS status 1 register is shown in Table 45–316. All the bits in the PHY XS status 1 register are read only; a write to the PHY XS status 1 register shall have no effect.



**Table 45–316—PHY XS status 1 register bit definitions**

| Bit(s)    | Name                        | Description                                                                        | R/W <sup>a</sup> |
|-----------|-----------------------------|------------------------------------------------------------------------------------|------------------|
| 4.1.15:12 | Reserved                    | Value always 0                                                                     | RO               |
| 4.1.11    | Tx LPI received             | 1 = Tx PHY XS has received LPI<br>0 = LPI not received                             | RO/LH            |
| 4.1.10    | Rx LPI received             | 1 = Rx PHY XS has received LPI<br>0 = LPI not received                             | RO/LH            |
| 4.1.9     | Tx LPI indication           | 1 = Tx PHY XS is currently receiving LPI<br>0 = PCS is not currently receiving LPI | RO               |
| 4.1.8     | Rx LPI indication           | 1 = Rx PHY XS is currently receiving LPI<br>0 = PCS is not currently receiving LPI | RO               |
| 4.1.7     | Fault                       | 1 = Fault condition detected<br>0 = No fault condition detected                    | RO               |
| 4.1.6     | Clock stop capable          | 1 = The attached PHY may stop the clock during LPI<br>0 = Clock not stoppable      | RO               |
| 4.1.5:3   | Reserved                    | Value always 0                                                                     | RO               |
| 4.1.2     | PHY XS transmit link status | 1 = The PHY XS transmit link is up<br>0 = The PHY XS transmit link is down         | RO/LL            |
| 4.1.1     | Low-power ability           | 1 = PHY XS supports low-power mode<br>0 = PHY XS does not support low-power mode   | RO               |
| 4.1.0     | Reserved                    | Value always 0                                                                     | RO               |

<sup>a</sup>RO = Read only, LL = Latching low, LH = Latching high

#### 45.2.4.2.1 Transmit LPI received (4.1.11)

When read as a one, bit 4.1.11 indicates that the transmit PHY XS has received LPI signaling one or more times since the register was last read. When read as a zero, bit 4.1.11 indicates that the PHY XS has not received LPI signaling. This bit shall be implemented with latching high behavior.

#### 45.2.4.2.2 Receive LPI received (4.1.10)

When read as a one, bit 4.1.10 indicates that the receive PHY XS has received LPI signaling one or more times since the register was last read. When read as a zero, bit 4.1.10 indicates that the PHY XS has not received LPI signaling. This bit shall be implemented with latching high behavior.

#### 45.2.4.2.3 Transmit LPI indication (4.1.9)

When read as a one, bit 4.1.9 indicates that the transmit PHY XS is currently receiving LPI signals. When read as a zero, bit 4.1.9 indicates that the PHY XS is not currently receiving LPI signals. The behavior if read during a state transition is undefined.

#### **45.2.4.2.4 Receive LPI indication (4.1.8)**

When read as a one, bit 4.1.8 indicates that the receive PHY XS is currently receiving LPI signals. When read as a zero, bit 4.1.8 indicates that the PHY XS is not currently receiving LPI signals. The behavior if read during a state transition is undefined.

#### **45.2.4.2.5 Fault (4.1.7)**

When read as a one, bit 4.1.7 indicates that the PHY XS has detected a fault condition on either the transmit or receive paths. When read as a zero, bit 4.1.7 indicates that the PHY XS has not detected a fault condition. Bit 4.1.7 is set to a one when either of the fault bits (4.8.11, 4.8.10) located in register 4.8 are set to a one.

#### **45.2.4.2.6 Clock stop capable (4.1.6)**

If bit 4.1.6 is set to one then the PHY XS is indicating that the attached PHY is permitted to stop the receive direction xMII clock while it is signaling LPI. If the bit is set to zero then the PHY XS is indicating that the attached PHY is not permitted to stop the receive xMII clock while it is signaling LPI. If the attached PHY does not support EEE capability or is not able to stop the receive direction xMII clock then this bit has no effect (see 46.3.2.4).

#### **45.2.4.2.7 PHY XS transmit link status (4.1.2)**

When read as a one, bit 4.1.2 indicates that the PHY XS transmit link is aligned. When read as a zero, bit 4.1.2 indicates that the PHY XS transmit link is not aligned. The transmit link status bit shall be implemented with latching low behavior.

For 10 Gb/s operation, bit 4.1.2 is a latching low version of bit 4.24.12.

#### **45.2.4.2.8 Low-power ability (4.1.1)**

When read as a one, bit 4.1.1 indicates that the PHY XS supports the low-power feature. When read as a zero, bit 4.1.1 indicates that the PHY XS does not support the low-power feature. If a PHY XS supports the low-power feature then it is controlled using the low-power bit in the PHY XS control register.

#### **45.2.4.3 PHY XS device identifier (Registers 4.2 and 4.3)**

Registers 4.2 and 4.3 provide a 32-bit value, which may constitute a unique identifier for a PHY XS. The identifier shall be composed of the 3rd through 24th bits of the Organizationally Unique Identifier (OUI) assigned to the device manufacturer by the IEEE, plus a six-bit model number, plus a four-bit revision number. A PHY XS may return a value of zero in each of the 32 bits of the PHY XS device identifier.

The format of the PHY XS device identifier is specified in 22.2.4.3.1.

#### **45.2.4.4 PHY XS speed ability (Register 4.4)**

The assignment of bits in the PHY XS speed ability register is shown in Table 45–317.

#### **45.2.4.4.1 400G capable (4.4.9)**

When read as a one, bit 4.4.9 indicates that the PHY XS is able to operate at a data rate of 400 Gb/s. When read as a zero, bit 4.4.9 indicates that the PHY-XS is not able to operate at a data rate of 400 Gb/s.

**Table 45–317—PHY XS speed ability register bit definitions**

| Bit(s)    | Name         | Description                                                                                          | R/W <sup>a</sup> |
|-----------|--------------|------------------------------------------------------------------------------------------------------|------------------|
| 4.4.15:10 | Reserved     | Value always 0                                                                                       | RO               |
| 4.4.9     | 400G capable | 1 = PHY XS is capable of operating at 400 Gb/s<br>0 = PHY XS is not capable of operating at 400 Gb/s | RO               |
| 4.4.8     | 200G capable | 1 = PHY XS is capable of operating at 200 Gb/s<br>0 = PHY XS is not capable of operating at 200 Gb/s | RO               |
| 4.4.7:1   | Reserved     | Value always 0                                                                                       | RO               |
| 4.4.0     | 10G capable  | 1 = PHY XS is capable of operating at 10 Gb/s<br>0 = PHY XS is not capable of operating at 10 Gb/s   | RO               |

<sup>a</sup>RO = Read only

#### 45.2.4.4.2 200G capable (4.4.8)

When read as a one, bit 4.4.8 indicates that the PHY XS is able to operate at a data rate of 200 Gb/s. When read as a zero, bit 4.4.8 indicates that the PHY-XS is not able to operate at a data rate of 200 Gb/s.

#### 45.2.4.4.3 10G capable (4.4.0)

When read as a one, bit 4.4.0 indicates that the PHY XS is able to operate at a data rate of 10 Gb/s. When read as a zero, bit 4.4.0 indicates that the PHY-XS is not able to operate at a data rate of 10 Gb/s.

#### 45.2.4.5 PHY XS devices in package (Registers 4.5 and 4.6)

The PHY XS devices in package registers are defined in Table 45–2.

#### 45.2.4.6 PHY XS status 2 register (Register 4.8)

The assignment of bits in the PHY XS status 2 register is shown in Table 45–318. All the bits in the PHY XS status 2 register are read only; a write to the PHY XS status 2 register shall have no effect.

**Table 45–318—PHY XS status 2 register bit definitions**

| Bit(s)    | Name           | Description                                                                                                                                                                                | R/W <sup>a</sup> |
|-----------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 4.8.15:14 | Device present | 15 14<br>1 0 = Device responding at this address<br>1 1 = No device responding at this address<br>0 1 = No device responding at this address<br>0 0 = No device responding at this address | RO               |
| 4.8.13:12 | Reserved       | Value always 0                                                                                                                                                                             | RO               |
| 4.8.11    | Transmit fault | 1 = Fault condition on transmit path<br>0 = No fault condition on transmit path                                                                                                            | RO/LH            |

**Table 45–318—PHY XS status 2 register bit definitions (*continued*)**

| Bit(s)  | Name          | Description                                                                   | R/W <sup>a</sup> |
|---------|---------------|-------------------------------------------------------------------------------|------------------|
| 4.8.10  | Receive fault | 1 = Fault condition on receive path<br>0 = No fault condition on receive path | RO/LH            |
| 4.8.9:0 | Reserved      | Value always 0                                                                | RO               |

<sup>a</sup>RO = Read only, LH = Latching high

#### 45.2.4.6.1 Device present (4.8.15:14)

When read as <10>, bits 4.8.15:14 indicate that a device is present and responding at this register address. When read as anything other than <10>, bits 4.8.15:14 indicate that no device is present at this register address or that the device is not functioning properly.

#### 45.2.4.6.2 Transmit fault (4.8.11)

When read as a one, bit 4.8.11 indicates that the PHY XS has detected a fault condition on the transmit path. When read as a zero, bit 4.8.11 indicates that the PHY XS has not detected a fault condition on the transmit path. The transmit fault bit shall be implemented with latching high behavior.

The default value for bit 4.8.11 is zero.

#### 45.2.4.6.3 Receive fault (4.8.10)

When read as a one, bit 4.8.10 indicates that the PHY XS has detected a fault condition on the receive path. When read as a zero, bit 4.8.10 indicates that the PHY XS has not detected a fault condition on the receive path. The receive fault bit shall be implemented with latching high behavior.

The default value of bit 4.8.10 is zero.

#### 45.2.4.7 PHY XS package identifier (Registers 4.14 and 4.15)

Registers 4.14 and 4.15 provide a 32-bit value, which may constitute a unique identifier for a particular type of package that the PHY XS is instantiated within. The identifier shall be composed of the 3rd through 24th bits of the Organizationally Unique Identifier (OUI) assigned to the package manufacturer by the IEEE, plus a six-bit model number, plus a four-bit revision number. A PHY XS may return a value of zero in each of the 32 bits of the PHY XS package identifier.

A non-zero package identifier may be returned by one or more MMDs in the same package. The package identifier may be the same as the device identifier.

The format of the PHY XS package identifier is specified in 22.2.4.3.1.

#### 45.2.4.8 EEE capability (Register 4.20)

This register is used to indicate the capability of the PHY XS to support EEE functions. The assignment of bits in the EEE capability register is shown in Table 45–319.

**Table 45–319—EEE capability register bit definitions**

| Bit(s)    | Name              | Description                                                                       | R/W <sup>a</sup> |
|-----------|-------------------|-----------------------------------------------------------------------------------|------------------|
| 4.20.15:5 | Reserved          | Value always 0                                                                    | RO               |
| 4.20.4    | PHY XS EEE        | 1 = EEE is supported for PHY XS<br>0 = EEE is not supported for PHY XS            | RO               |
| 4.20.3:1  | Reserved          | Value always 0                                                                    | RO               |
| 4.20.0    | XAUI stop capable | 1 = The DTE XS may stop XAUI signals during LPI<br>0 = XAUI signals not stoppable | RO               |

<sup>a</sup> RO = Read only

##### 45.2.4.8.1 PHY XS EEE supported (4.20.4)

If the device supports EEE operation for PHY XS as defined in 48.2, this bit shall be set to one.

##### 45.2.4.8.2 XAUI stop capable (4.20.0)

If bit 4.20.0 is set to one, then the PHY XS is indicating that the attached DTE XS is permitted to stop transmitting XAUI signals during LPI. If the bit is set to zero then the PHY XS is indicating that the attached DTE XS is not permitted to stop transmitting XAUI signals during LPI. If the DTE XS does not support EEE capability or is not able to stop the transmit direction XAUI, then this bit has no effect.

#### 45.2.4.9 EEE wake error counter (Register 4.22)

This register is used by PHY XS that support EEE to count wake time faults where the PHY XS fails to complete its normal wake sequence after a period of quiescence for XAUI transmit signals. The fault event to be counted may occur during a refresh or a wake-up. This 16-bit counter shall be reset to all zeros when the EEE wake error counter is read by the management function or upon execution of the PHY XS reset. This counter shall be held at all ones in the case of overflow.

#### 45.2.4.10 10G PHY XGXS lane status register (Register 4.24)

The assignment of bits in the 10G PHY XGXS lane status register is shown in Table 45–320. All the bits in the 10G PHY XGXS lane status register are read only; a write to the 10G PHY XGXS lane status register shall have no effect.

##### 45.2.4.10.1 PHY XGXS transmit lane alignment status (4.24.12)

When read as a one, bit 4.24.12 indicates that the PHY XGXS has synchronized and aligned all four transmit lanes. When read as a zero, bit 4.24.12 indicates that the PHY XGXS has not synchronized and aligned all four transmit lanes.

**Table 45–320—10G PHY XGXS lane status register bit definitions**

| Bit(s)     | Name                           | Description                                                                                                                          | R/W <sup>a</sup> |
|------------|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 4.24.15:13 | Reserved                       | Value always 0                                                                                                                       | RO               |
| 4.24.12    | PHY XGXS lane alignment status | 1 = PHY XGXS transmit lanes aligned<br>0 = PHY XGXS transmit lanes not aligned                                                       | RO               |
| 4.24.11    | Pattern testing ability        | 1 = PHY XGXS is able to generate test patterns<br>0 = PHY XGXS is not able to generate test patterns                                 | RO               |
| 4.24.10    | PHY XGXS loopback ability      | 1 = PHY XGXS has the ability to perform a loopback function<br>0 = PHY XGXS does not have the ability to perform a loopback function | RO               |
| 4.24.9:4   | Reserved                       | Value always 0                                                                                                                       | RO               |
| 4.24.3     | Lane 3 sync                    | 1 = Lane 3 is synchronized<br>0 = Lane 3 is not synchronized                                                                         | RO               |
| 4.24.2     | Lane 2 sync                    | 1 = Lane 2 is synchronized<br>0 = Lane 2 is not synchronized                                                                         | RO               |
| 4.24.1     | Lane 1 sync                    | 1 = Lane 1 is synchronized<br>0 = Lane 1 is not synchronized                                                                         | RO               |
| 4.24.0     | Lane 0 sync                    | 1 = Lane 0 is synchronized<br>0 = Lane 0 is not synchronized                                                                         | RO               |

<sup>a</sup>RO = Read only

#### 45.2.4.10.2 Pattern testing ability (4.24.11)

When read as a one, bit 4.24.11 indicates that the 10G PHY XGXS is able to generate test patterns. When read as a zero, bit 4.24.11 indicates that the 10G PHY XGXS is not able to generate test patterns. If the 10G PHY XGXS is able to generate test patterns, then the functionality is controlled using the transmit test-pattern enable bit in register 4.25.

#### 45.2.4.10.3 PHY XS loopback ability (4.24.10)

When read as a one, bit 4.24.10 indicates that the PHY XGXS is able to perform the loopback function. When read as a zero, bit 4.24.10 indicates that the PHY XGXS is not able to perform the loopback function. If a 10G PHY XGXS is able to perform the loopback function, then it is controlled using the PHY XGXS loopback bit 4.0.14.

#### 45.2.4.10.4 Lane 3 sync (4.24.3)

When read as a one, bit 4.24.3 indicates that the 10G PHY XGXS transmit lane 3 is synchronized. When read as a zero, bit 4.24.3 indicates that the 10G PHY XGXS transmit lane 3 is not synchronized.

#### 45.2.4.10.5 Lane 2 sync (4.24.2)

When read as a one, bit 4.24.2 indicates that the 10G PHY XGXS transmit lane 2 is synchronized. When read as a zero, bit 4.24.2 indicates that the 10G PHY XGXS transmit lane 2 is not synchronized.

#### 45.2.4.10.6 Lane 1 sync (4.24.1)

When read as a one, bit 4.24.1 indicates that the 10G PHY XGXS transmit lane 1 is synchronized. When read as a zero, bit 4.24.1 indicates that the 10G PHY XGXS transmit lane 1 is not synchronized.

#### 45.2.4.10.7 Lane 0 sync (4.24.0)

When read as a one, bit 4.24.0 indicates that the 10G PHY XGXS transmit lane 0 is synchronized. When read as a zero, bit 4.24.0 indicates that the 10G PHY XGXS transmit lane 0 is not synchronized.

#### 45.2.4.11 10G PHY XGXS test control register (Register 4.25)

The assignment of bits in the 10G PHY XGXS test control register is shown in Table 45–321. The default value for each bit of the 10G PHY XGXS test control register should be chosen so that the initial state of the device upon power up or reset is a normal operational state without management intervention.

**Table 45–321—10G PHY XGXS test control register bit definitions**

| Bit(s)    | Name                        | Description                                                                                                                          | R/W <sup>a</sup> |
|-----------|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 4.25.15:3 | Reserved                    | Value always 0                                                                                                                       | RO               |
| 4.25.2    | Receive test-pattern enable | 1 = Receive test pattern enabled<br>0 = Receive test pattern not enabled                                                             | R/W              |
| 4.25.1:0  | Test-pattern select         | 1 0<br>1 1 = Reserved<br>1 0 = Mixed-frequency test pattern<br>0 1 = Low-frequency test pattern<br>0 0 = High-frequency test pattern | R/W              |

<sup>a</sup>RO = Read only, R/W = Read/Write

##### 45.2.4.11.1 10G PHY XGXS test-pattern enable (4.25.2)

When bit 4.25.2 is set to a one, pattern testing is enabled on the receive path. When bit 4.25.2 is set to a zero, pattern testing is disabled on the receive path. Pattern testing is optional, and the ability of the 10G PHY XGXS to generate test patterns is advertised by the pattern testing ability bit in register 4.24. A 10G PHY XGXS that does not support the generation of test patterns shall ignore writes to this bit and always return a value of zero. The default of bit 4.25.2 is zero.

##### 45.2.4.11.2 10G PHY XGXS test-pattern select (4.25.1:0)

The test pattern to be used when pattern testing is enabled using bit 4.25.2 is selected using bits 4.25.1:0. When bits 4.25.1:0 are set to <10>, the mixed-frequency test pattern shall be selected for pattern testing. When bits 4.25.1:0 are set to <01>, the low-frequency test pattern shall be selected for pattern testing. When bits 4.25.1:0 are set to <00>, the high-frequency test pattern shall be selected for pattern testing. The test patterns are defined in Annex 48A.

#### 45.2.4.12 BASE-R PHY XS status 1 register (Register 4.32)

The assignment of bits in the BASE-R PHY XS status 1 register is shown in Table 45–322. A PHY XS device that does not implement BASE-R shall return a zero for all bits in the BASE-R PHY XS status 1 register. It is the responsibility of the STA management entity to ensure that a port type is supported by all MMDs before interrogating any of its status bits.

**Table 45–322—BASE-R PHY XS status 1 register bit definitions**

| Bit(s)     | Name                              | Description                                                              | R/W <sup>a</sup> |
|------------|-----------------------------------|--------------------------------------------------------------------------|------------------|
| 4.32.15:13 | Reserved                          | Value always 0                                                           | RO               |
| 4.32.12    | BASE-R PHY XS receive link status | 1 = BASE-R PHY XS receive link up<br>0 = BASE-R PHY XS receive link down | RO               |
| 4.32.11:0  | Reserved                          | Value always 0                                                           | RO               |

<sup>a</sup>RO = Read only

#### 45.2.4.12.1 BASE-R PHY XS receive link status (4.32.12)

When read as a one, bit 4.32.12 indicates that the PHY XS is in a fully operational state. When read as a zero, bit 4.32.12 indicates that the PHY XS is not fully operational. This bit is a reflection of the PHY XS equivalent of the PCS\_status variable defined in 119.3 for 200/400GBASE-R.

#### 45.2.4.13 BASE-R PHY XS test-pattern control register (Register 4.42)

The assignment of bits in the BASE-R PHY XS test-pattern control register is shown in Table 45–323. This register is only required when the 200/400GBASE-R capability is supported. The test-pattern methodology is described in 119.2.4.9.

**Table 45–323—BASE-R PHY XS test-pattern control register bit definitions**

| Bit(s)    | Name                         | Description                                                           | R/W <sup>a</sup> |
|-----------|------------------------------|-----------------------------------------------------------------------|------------------|
| 4.42.15:4 | Reserved                     | Value always 0                                                        | RO               |
| 4.42.3    | Transmit test-pattern enable | 1 = Enable transmit test pattern<br>0 = Disable transmit test pattern | R/W              |
| 4.42.2:0  | Reserved                     | Value always 0                                                        | RO               |

<sup>a</sup>RO = Read only, R/W = Read/Write

#### 45.2.4.13.1 Transmit test-pattern enable (4.42.3)

When bit 4.42.3 is set to a one, pattern testing is enabled on the transmit path. When bit 4.42.3 is set to a zero, pattern testing is disabled on the transmit path.

The default value for bit 4.42.3 is zero.

#### 45.2.4.14 Multi-lane BASE-R PHY XS alignment status 1 register (Register 4.50)

The assignment of bits in the multi-lane BASE-R PHY XS alignment status 1 register is shown in Table 45–324. A PHY XS device that does not implement multi-lane BASE-R PHY XS shall return a zero for all bits in the multi-lane BASE-R PHY XS alignment status 1 register. It is the responsibility of the STA management entity to ensure that a port type is supported by all MMDs before interrogating any of its status bits.



**Table 45–324—Multi-lane BASE-R PHY XS alignment status 1 register bit definitions**

| Bit(s)     | Name                         | Description                                                                                    | R/W <sup>a</sup> |
|------------|------------------------------|------------------------------------------------------------------------------------------------|------------------|
| 4.50.15:13 | Reserved                     | Value always 0                                                                                 | RO               |
| 4.50.12    | PHY XS lane alignment status | 1 = PHY XS receive lanes locked and aligned<br>0 = PHY XS receive lanes not locked and aligned | RO               |
| 4.50.11:0  | Reserved                     | Value always 0                                                                                 | RO               |

<sup>a</sup>RO = Read only

#### 45.2.4.14.1 PHY XS lane alignment status (4.50.12)

When read as a one, bit 4.50.12 indicates that the PHY XS has locked and aligned all receive lanes. When read as a zero, bit 4.50.12 indicates that the PHY XS has not locked and aligned all receive lanes.

#### 45.2.4.15 Multi-lane BASE-R PHY XS alignment status 3 register (Register 4.52)

The assignment of bits in the multi-lane BASE-R PHY XS alignment status 3 register is shown in Table 45–325. A PHY XS device that does not implement multi-lane BASE-R PHY XS shall return a zero for all bits in the multi-lane BASE-R PHY XS alignment status 3 register. A device that implements multi-lane BASE-R PHY XS shall return a zero for all bits in the multi-lane BASE-R PHY XS alignment status 3 register that are not required for the PHY XS configuration. It is the responsibility of the STA management entity to ensure that a port type is supported by all MMDs before interrogating any of its status bits.

**Table 45–325—Multi-lane BASE-R PHY XS alignment status 3 register bit definitions**

| Bit(s)    | Name           | Description                                                                        | R/W <sup>a</sup> |
|-----------|----------------|------------------------------------------------------------------------------------|------------------|
| 4.52.15:8 | Reserved       | Value always 0                                                                     | RO               |
| 4.52.7    | Lane 7 aligned | 1 = Lane 7 alignment marker is locked<br>0 = Lane 7 alignment marker is not locked | RO               |
| 4.52.6    | Lane 6 aligned | 1 = Lane 6 alignment marker is locked<br>0 = Lane 6 alignment marker is not locked | RO               |
| 4.52.5    | Lane 5 aligned | 1 = Lane 5 alignment marker is locked<br>0 = Lane 5 alignment marker is not locked | RO               |
| 4.52.4    | Lane 4 aligned | 1 = Lane 4 alignment marker is locked<br>0 = Lane 4 alignment marker is not locked | RO               |
| 4.52.3    | Lane 3 aligned | 1 = Lane 3 alignment marker is locked<br>0 = Lane 3 alignment marker is not locked | RO               |
| 4.52.2    | Lane 2 aligned | 1 = Lane 2 alignment marker is locked<br>0 = Lane 2 alignment marker is not locked | RO               |

**Table 45–325—Multi-lane BASE-R PHY XS alignment status 3 register  
bit definitions (*continued*)**

| Bit(s) | Name           | Description                                                                        | R/W <sup>a</sup> |
|--------|----------------|------------------------------------------------------------------------------------|------------------|
| 4.52.1 | Lane 1 aligned | 1 = Lane 1 alignment marker is locked<br>0 = Lane 1 alignment marker is not locked | RO               |
| 4.52.0 | Lane 0 aligned | 1 = Lane 0 alignment marker is locked<br>0 = Lane 0 alignment marker is not locked | RO               |

<sup>a</sup>RO = Read only

#### **45.2.4.15.1 Lane 7 aligned (4.52.7)**

When read as a one, bit 4.52.7 indicates that the PHY XS receiver has achieved alignment marker lock for service interface lane 7. When read as a zero, bit 4.52.7 indicates that the PHY XS receiver lane 7 has not achieved alignment marker lock. This bit reflects the state of `amps_lock[7]` (see 119.2.6.2.2).

#### **45.2.4.15.2 Lane 6 aligned (4.52.6)**

When read as a one, bit 4.52.6 indicates that the PHY XS receiver has achieved alignment marker lock for service interface lane 6. When read as a zero, bit 4.52.6 indicates that the PHY XS receiver lane 6 has not achieved alignment marker lock. This bit reflects the state of `amps_lock[6]` (see 119.2.6.2.2).

#### **45.2.4.15.3 Lane 5 aligned (4.52.5)**

When read as a one, bit 4.52.5 indicates that the PHY XS receiver has achieved alignment marker lock for service interface lane 5. When read as a zero, bit 4.52.5 indicates that the PHY XS receiver lane 5 has not achieved alignment marker lock. This bit reflects the state of `amps_lock[5]` (see 119.2.6.2.2).

#### **45.2.4.15.4 Lane 4 aligned (4.52.4)**

When read as a one, bit 4.52.4 indicates that the PHY XS receiver has achieved alignment marker lock for service interface lane 4. When read as a zero, bit 4.52.4 indicates that the PHY XS receiver lane 4 has not achieved alignment marker lock. This bit reflects the state of `amps_lock[4]` (see 119.2.6.2.2).

#### **45.2.4.15.5 Lane 3 aligned (4.52.3)**

When read as a one, bit 4.52.3 indicates that the PHY XS receiver has achieved alignment marker lock for service interface lane 3. When read as a zero, bit 4.52.3 indicates that the PHY XS receiver lane 3 has not achieved alignment marker lock. This bit reflects the state of `amps_lock[3]` (see 119.2.6.2.2).

#### **45.2.4.15.6 Lane 2 aligned (4.52.2)**

When read as a one, bit 4.52.2 indicates that the PHY XS receiver has achieved alignment marker lock for service interface lane 2. When read as a zero, bit 4.52.2 indicates that the PHY XS receiver lane 2 has not achieved alignment marker lock. This bit reflects the state of `amps_lock[2]` (see 119.2.6.2.2).

#### 45.2.4.15.7 Lane 1 aligned (4.52.1)

When read as a one, bit 4.52.1 indicates that the PHY XS receiver has achieved alignment marker lock for service interface lane 1. When read as a zero, bit 4.52.1 indicates that the PHY XS receiver lane 1 has not achieved alignment marker lock. This bit reflects the state of `amps_lock[1]` (see 119.2.6.2.2).

#### 45.2.4.15.8 Lane 0 aligned (4.52.0)

When read as a one, bit 4.52.0 indicates that the PHY XS receiver has achieved alignment marker lock for service interface lane 0. When read as a zero, bit 4.52.0 indicates that the PHY XS receiver lane 0 has not achieved alignment marker lock. This bit reflects the state of `amps_lock[0]` (see 119.2.6.2.2).

#### 45.2.4.16 Multi-lane BASE-R PHY XS alignment status 4 register (Register 4.53)

The assignment of bits in the multi-lane BASE-R PHY XS alignment status 4 register is shown in Table 45–326. A PHY XS device that does not implement multi-lane BASE-R PHY XS shall return a zero for all bits in the multi-lane BASE-R PHY XS alignment status 4 register. A device that implements multi-lane BASE-R PHY XS shall return a zero for all bits in the multi-lane BASE-R PHY XS alignment status 4 register that are not required for the PHY XS configuration. It is the responsibility of the STA management entity to ensure that a port type is supported by all MMDs before interrogating any of its status bits.

**Table 45–326—Multi-lane BASE-R PHY XS alignment status 4 register  
bit definitions**

| Bit(s)    | Name            | Description                                                                          | R/W <sup>a</sup> |
|-----------|-----------------|--------------------------------------------------------------------------------------|------------------|
| 4.53.15:8 | Reserved        | Value always 0                                                                       | RO               |
| 4.53.7    | Lane 15 aligned | 1 = Lane 15 alignment marker is locked<br>0 = Lane 15 alignment marker is not locked | RO               |
| 4.53.6    | Lane 14 aligned | 1 = Lane 14 alignment marker is locked<br>0 = Lane 14 alignment marker is not locked | RO               |
| 4.53.5    | Lane 13 aligned | 1 = Lane 13 alignment marker is locked<br>0 = Lane 13 alignment marker is not locked | RO               |
| 4.53.4    | Lane 12 aligned | 1 = Lane 12 alignment marker is locked<br>0 = Lane 12 alignment marker is not locked | RO               |
| 4.53.3    | Lane 11 aligned | 1 = Lane 11 alignment marker is locked<br>0 = Lane 11 alignment marker is not locked | RO               |
| 4.53.2    | Lane 10 aligned | 1 = Lane 10 alignment marker is locked<br>0 = Lane 10 alignment marker is not locked | RO               |
| 4.53.1    | Lane 9 aligned  | 1 = Lane 9 alignment marker is locked<br>0 = Lane 9 alignment marker is not locked   | RO               |
| 4.53.0    | Lane 8 aligned  | 1 = Lane 8 alignment marker is locked<br>0 = Lane 8 alignment marker is not locked   | RO               |

<sup>a</sup>RO = Read only

#### **45.2.4.16.1 Lane 15 aligned (4.53.7)**

When read as a one, bit 4.53.7 indicates that the PHY XS receiver has achieved alignment marker lock for service interface lane 15. When read as a zero, bit 4.53.7 indicates that the PHY XS receiver lane 15 has not achieved alignment marker lock. This bit reflects the state of `amps_lock[15]` (see 119.2.6.2.2).

#### **45.2.4.16.2 Lane 14 aligned (4.53.6)**

When read as a one, bit 4.53.6 indicates that the PHY XS receiver has achieved alignment marker lock for service interface lane 14. When read as a zero, bit 4.53.6 indicates that the PHY XS receiver lane 14 has not achieved alignment marker lock. This bit reflects the state of `amps_lock[14]` (see 119.2.6.2.2).

#### **45.2.4.16.3 Lane 13 aligned (4.53.5)**

When read as a one, bit 4.53.5 indicates that the PHY XS receiver has achieved alignment marker lock for service interface lane 13. When read as a zero, bit 4.53.5 indicates that the PHY XS receiver lane 13 has not achieved alignment marker lock. This bit reflects the state of `amps_lock[13]` (see 119.2.6.2.2).

#### **45.2.4.16.4 Lane 12 aligned (4.53.4)**

When read as a one, bit 4.53.4 indicates that the PHY XS receiver has achieved alignment marker lock for service interface lane 12. When read as a zero, bit 4.53.4 indicates that the PHY XS receiver lane 12 has not achieved alignment marker lock. This bit reflects the state of `amps_lock[12]` (see 119.2.6.2.2).

#### **45.2.4.16.5 Lane 11 aligned (4.53.3)**

When read as a one, bit 4.53.3 indicates that the PHY XS receiver has achieved alignment marker lock for service interface lane 11. When read as a zero, bit 4.53.3 indicates that the PHY XS receiver lane 11 has not achieved alignment marker lock. This bit reflects the state of `amps_lock[11]` (see 119.2.6.2.2).

#### **45.2.4.16.6 Lane 10 aligned (4.53.2)**

When read as a one, bit 4.53.2 indicates that the PHY XS receiver has achieved alignment marker lock for service interface lane 10. When read as a zero, bit 4.53.2 indicates that the PHY XS receiver lane 10 has not achieved alignment marker lock. This bit reflects the state of `amps_lock[10]` (see 119.2.6.2.2).

#### **45.2.4.16.7 Lane 9 aligned (4.53.1)**

When read as a one, bit 4.53.1 indicates that the PHY XS receiver has achieved alignment marker lock for service interface lane 9. When read as a zero, bit 4.53.1 indicates that the PHY XS receiver lane 9 has not achieved alignment marker lock. This bit reflects the state of `amps_lock[9]` (see 119.2.6.2.2).

#### **45.2.4.16.8 Lane 8 aligned (4.53.0)**

When read as a one, bit 4.53.0 indicates that the PHY XS receiver has achieved alignment marker lock for service interface lane 8. When read as a zero, bit 4.53.0 indicates that the PHY XS receiver lane 8 has not achieved alignment marker lock. This bit reflects the state of `amps_lock[8]` (see 119.2.6.2.2).

#### **45.2.4.17 PHY XS lane mapping, lane 0 register (Register 4.400)**

The assignment of bits in the PHY XS lane mapping, lane 0 register is shown in Table 45–327. When the multi-lane PHY XS described in Clause 118 detects and locks the alignment marker for service interface lane 0, the detected PCS lane number is recorded in this register. The contents of the PHY XS lane mapping, lane 0 register is valid when Lane 0 aligned bit (4.52.0) is set to one and is invalid otherwise.

**Table 45–327—PHY XS lane mapping, lane 0 register bit definitions**

| Bit(s)     | Name           | Description                                      | R/W <sup>a</sup> |
|------------|----------------|--------------------------------------------------|------------------|
| 4.400.15:5 | Reserved       | Value always 0                                   | RO               |
| 4.400.4:0  | Lane 0 mapping | PHY XS lane received in service interface lane 0 | RO               |

<sup>a</sup>RO = Read only

#### 45.2.4.18 PHY XS lane mapping, lane 1 through lane 15 registers (Registers 4.401 through 4.415)

The definition of the PHY XS lane mapping, lane 1 through 15 registers is identical to that described for lane 0 in 45.2.4.17. The lane mapping for lane 1 is in register 4.401; lane 2 is in register 4.402; etc.

#### 45.2.4.19 PHY XS FEC symbol error counter lane 0 (Register 4.600, 4.601)

The assignment of bits in the PHY XS FEC symbol error counter lane 0 register is shown in Table 45–328. See 119.3.4 for a definition of this counter. Symbol errors detected in PHY XS FEC lane 0 are counted and shown in register 4.600.15:0 and 4.601.15:0. These bits shall be reset to all zeros when the register is read by the management function or upon PHY reset. These bits shall be held at all ones in the case of overflow. Registers 4.600, 4.601 are used to read the value of a 32-bit counter. When registers 4.600 and 4.601 are used to read the 32-bit counter value, the register 4.600 is read first, the value of the register 4.601 is latched when (and only when) register 4.600 is read, and reads of register 4.601 return the latched value rather than the current value of the counter.

**Table 45–328—PHY XS FEC symbol error counter lane 0 register bit definitions**

| Bit(s)     | Name                                   | Description                       | R/W <sup>a</sup> |
|------------|----------------------------------------|-----------------------------------|------------------|
| 4.600.15:0 | PHY XS FEC symbol errors, lane 0 lower | FEC_symbol_error_counter_0[15:0]  | RO, NR           |
| 4.601.15:0 | PHY XS FEC symbol errors, lane 0 upper | FEC_symbol_error_counter_0[31:16] | RO, NR           |

<sup>a</sup>RO = Read only, NR = Non Roll-over

#### 45.2.4.20 PHY XS FEC symbol error counter lane 1 through 15 (Registers 4.602 through 4.631)

The behavior of the PHY XS FEC symbol error counters, lane 1 through 15 is identical to that described for PHY XS FEC lane 0 in 45.2.4.19. Errors detected in each PHY XS FEC lane are counted and shown in the corresponding register. PHY XS FEC lane 1, lower 16 bits are shown in register 4.602; PHY XS FEC lane 1, upper 16 bits are shown in register 4.603; PHY XS FEC lane 2, lower 16 bits are shown in register 4.604; through register 4.631 for PHY XS FEC lane 15, upper 16 bits.

#### 45.2.4.21 PHY XS FEC control register (Register 4.800)

The assignment of bits in the PHY XS FEC control register is shown in Table 45–329.

**Table 45–329—PHY XS FEC control register bit definitions**

| Bit(s)     | Name                                | Description                                                                          | R/W <sup>a</sup> |
|------------|-------------------------------------|--------------------------------------------------------------------------------------|------------------|
| 4.800.15:3 | Reserved                            | Value always 0                                                                       | RO               |
| 4.800.2    | PHY XS FEC degraded SER enable      | 1 = FEC decoder signals degraded SER<br>0 = FEC decoder does not signal degraded SER | R/W              |
| 4.800.1    | PHY XS FEC bypass indication enable | 1 = FEC decoder does not indicate errors<br>0 = FEC decoder indicates errors         | R/W              |
| 4.800.0    | Reserved                            | Value always 0                                                                       | RO               |

<sup>a</sup>R/W = Read/Write, RO = Read only

#### 45.2.4.21.1 PHY XS FEC degraded SER enable (4.800.2)

This bit enables the PHY XS FEC decoder to signal the presence of a degraded SER. When set to a one, this bit enables degraded SER signaling. When set to a zero, degraded SER signaling is disabled. Writes to this bit are ignored and reads return a zero if the PHY XS FEC does not have the ability to signal the presence of a degraded SER (see 119.2.5.3 for equivalent PCS behavior).

#### 45.2.4.21.2 PHY XS FEC bypass indication enable (4.800.1)

This bit enables the PHY XS FEC decoder to bypass error indication to the upper layers through the sync bits. When set to a one, this bit enables bypass of the error indication. When set to a zero, errors are indicated to the upper layers through the sync bits. Writes to this bit are ignored and reads return a zero if the PHY XS FEC does not have the ability to bypass indicating decoding errors (see 119.2.5.3 for equivalent PCS behavior).

#### 45.2.4.22 PHY XS FEC status register (Register 4.801)

The assignment of bits in the PHY XS FEC status register is shown in Table 45–330.

**Table 45–330—PHY XS FEC status register bit definitions**

| Bit(s)     | Name                            | Description                                                                                                                                                    | R/W <sup>a</sup> |
|------------|---------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 4.801.15:6 | Reserved                        | Value always 0                                                                                                                                                 | RO               |
| 4.801.5    | Remote degraded SER received    | 1 = remote degraded SER received<br>0 = no remote degraded SER received                                                                                        | RO               |
| 4.801.4    | PHY XS FEC degraded SER         | 1 = SER is degraded<br>0 = SER is not degraded                                                                                                                 | RO               |
| 4.801.3    | PHY XS FEC degraded SER ability | 1 = FEC decoder has the ability to signal the presence of a degraded SER<br>0 = FEC decoder does not have the ability to signal the presence of a degraded SER | RO               |

**Table 45–330—PHY XS FEC status register bit definitions (*continued*)**

| Bit(s)  | Name                                 | Description                                                                                                                        | R/W <sup>a</sup> |
|---------|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 4.801.2 | PHY XS FEC high SER                  | 1 = FEC errors have exceeded threshold<br>0 = FEC errors have not exceeded threshold                                               | RO/LH            |
| 4.801.1 | PHY XS FEC bypass indication ability | 1 = FEC decoder has the ability to bypass error indication<br>0 = FEC decoder does not have the ability to bypass error indication | RO               |
| 4.801.0 | Reserved                             | Value always 0                                                                                                                     | RO               |

<sup>a</sup>RO = Read only, LH = Latching high

#### 45.2.4.22.1 Remote degraded SER received (4.801.5)

When read as a one, bit 4.801.5 indicates that the remote degraded SER signal has been received. This bit reflects the state of rx\_rm\_degraded for the PHY XS (see 119.2.6.2.2 for equivalent PCS behavior).

#### 45.2.4.22.2 PHY XS FEC degraded SER (4.801.4)

When PHY XS FEC degraded SER enable (bit 4.800.2) is set to one, bit 4.801.4 is set to one if the number of FEC symbol errors in a window of PHY XS FEC degraded SER interval (registers 4.810 and 4.811) codewords exceeds the PHY XS FEC degraded SER activate threshold (registers 4.806 and 4.807) and is cleared if the number of FEC symbol errors in the same window is below the PHY XS FEC degraded SER deactivate threshold (registers 4.808 and 4.809). If the number of FEC symbol errors in the window is between the two thresholds, then bit 4.801.4 remains in its previous state. The bit is set to zero if PHY XS FEC degraded SER enable (bit 4.800.2) is set to zero (see 119.2.5.3 for equivalent PCS behavior). The value of bit 4.801.4 is undefined if the value of the PHY XS FEC degraded SER activate threshold is less than the value of the PHY XS FEC degraded SER deactivate threshold.

#### 45.2.4.22.3 PHY XS FEC degraded SER ability (4.801.3)

The PHY XS FEC decoder may have the option to signal the presence of a degraded SER (see 119.2.5.3 for equivalent PCS behavior). This bit is set to one to indicate that the decoder has the ability to signal the presence of a degraded SER. The bit is set to zero if this ability is not supported.

#### 45.2.4.22.4 PHY XS FEC high SER (4.801.2)

When PHY XS FEC bypass indication enable (bit 4.800.1) is set to one, this bit is set to one if the number of FEC symbol errors in a window of 8192 codewords exceeds the threshold (see 119.2.5.3 for equivalent PCS behavior) and is set to zero otherwise. The bit is set to zero if PHY XS FEC bypass indication enable (bit 4.800.1) is set to zero. This bit shall be implemented with latching high behavior.

#### 45.2.4.22.5 PHY XS FEC bypass indication ability (4.801.1)

The PHY XS FEC decoder may have the option to perform error detection without error indication (see 119.2.5.3 for equivalent PCS behavior) to reduce the delay contributed by the FEC decoder. This bit is set to one to indicate that the decoder has this ability to bypass error indication. The bit is set to zero if this ability is not supported.

#### 45.2.4.23 PHY XS FEC corrected codewords counter (Register 4.802, 4.803)

The assignment of bits in the PHY XS FEC corrected codewords counter register is shown in Table 45–331. See 119.3.2 for a definition of this counter. These bits shall be reset to all zeros when the register is read by the management function or upon PHY reset. These bits shall be held at all ones in the case of overflow. Registers 4.802, 4.803 are used to read the value of a 32-bit counter. When registers 4.802 and 4.803 are used to read the 32-bit counter value, the register 4.802 is read first, the value of the register 4.803 is latched when (and only when) register 4.802 is read, and reads of register 4.803 return the latched value rather than the current value of the counter.

**Table 45–331—PHY XS FEC corrected codewords counter register bit definitions**

| Bit(s)     | Name                          | Description                     | R/W <sup>a</sup> |
|------------|-------------------------------|---------------------------------|------------------|
| 4.802.15:0 | FEC corrected codewords lower | FEC_corrected_cw_counter[15:0]  | RO, NR           |
| 4.803.15:0 | FEC corrected codewords upper | FEC_corrected_cw_counter[31:16] | RO, NR           |

<sup>a</sup>RO = Read only, NR = Non Roll-over

#### 45.2.4.24 PHY XS FEC uncorrected codewords counter (Register 4.804, 4.805)

The assignment of bits in the PHY XS FEC uncorrected codewords counter register is shown in Table 45–332. See 119.3.3 for a definition of this counter. These bits shall be reset to all zeros when the register is read by the management function or upon PHY reset. These bits shall be held at all ones in the case of overflow. Registers 4.804, 4.805 are used to read the value of a 32-bit counter. When registers 4.804 and 4.805 are used to read the 32-bit counter value, the register 4.804 is read first, the value of the register 4.805 is latched when (and only when) register 4.804 is read, and reads of register 4.805 return the latched value rather than the current value of the counter.

**Table 45–332—PHY XS FEC uncorrected codewords counter register bit definitions**

| Bit(s)     | Name                            | Description                       | R/W <sup>a</sup> |
|------------|---------------------------------|-----------------------------------|------------------|
| 4.804.15:0 | FEC uncorrected codewords lower | FEC_uncorrected_cw_counter[15:0]  | RO, NR           |
| 4.805.15:0 | FEC uncorrected codewords upper | FEC_uncorrected_cw_counter[31:16] | RO, NR           |

<sup>a</sup>RO = Read only, NR = Non Roll-over



#### 45.2.4.25 PHY XS FEC degraded SER activate threshold register (Register 4.806, 4.807)

The assignment of bits in the PHY XS FEC degraded SER activate threshold register is shown in Table 45–333. The value controls the threshold used to set the PHY XS FEC degraded SER bit (4.801.4) in an equivalent manner to that defined for the FEC\_degraded\_SER bit in 119.2.5.3.

**Table 45–333—PHY XS FEC degraded SER activate threshold register bit definitions**

| Bit(s)     | Name                                             | Description                                | R/W <sup>a</sup> |
|------------|--------------------------------------------------|--------------------------------------------|------------------|
| 4.806.15:0 | PHY XS FEC degraded SER activate threshold lower | FEC_degraded_SER_activate_threshold[15:0]  | R/W              |
| 4.807.15:0 | PHY XS FEC degraded SER activate threshold upper | FEC_degraded_SER_activate_threshold[31:16] | R/W              |

<sup>a</sup>R/W = Read/Write

#### 45.2.4.26 PHY XS FEC degraded SER deactivate threshold register (Register 4.808, 4.809)

The assignment of bits in the PHY XS FEC degraded SER deactivate threshold register is shown in Table 45–334. The value controls the threshold used to clear the PHY XS FEC degraded SER bit (4.801.4) in an equivalent manner to that defined for the FEC\_degraded\_SER bit in 119.2.5.3.

**Table 45–334—PHY XS FEC degraded SER deactivate threshold register bit definitions**

| Bit(s)     | Name                                               | Description                                  | R/W <sup>a</sup> |
|------------|----------------------------------------------------|----------------------------------------------|------------------|
| 4.808.15:0 | PHY XS FEC degraded SER deactivate threshold lower | FEC_degraded_SER_deactivate_threshold[15:0]  | R/W              |
| 4.809.15:0 | PHY XS FEC degraded SER deactivate threshold upper | FEC_degraded_SER_deactivate_threshold[31:16] | R/W              |

<sup>a</sup>R/W = Read/Write

#### 45.2.4.27 PHY XS FEC degraded SER interval register (Register 4.810, 4.811)

The assignment of bits in the PHY XS FEC degraded SER interval register is shown in Table 45–335. The value controls the interval used to set and clear the PHY XS FEC degraded SER bit (4.801.4) in an equivalent manner to that defined for the FEC\_degraded\_SER bit in 119.2.5.3.

**Table 45–335—PHY XS FEC degraded SER interval register bit definitions**

| Bit(s)     | Name                                   | Description                      | R/W <sup>a</sup> |
|------------|----------------------------------------|----------------------------------|------------------|
| 4.810.15:0 | PHY XS FEC degraded SER interval lower | FEC_degraded_SER_interval[15:0]  | R/W              |
| 4.811.15:0 | PHY XS FEC degraded SER interval upper | FEC_degraded_SER_interval[31:16] | R/W              |

<sup>a</sup>R/W = Read/Write

#### 45.2.4.28 TimeSync PHY XS capability (Register 4.1800)

The TimeSync PHY XS capability register (see Table 45–336) indicates the capability of the PHY XS to report the transmit and receive data delay, stored in registers 4.1801 through 4.1804 and 4.1805 through 4.1808, respectively.

**Table 45–336—TimeSync PHY XS capability**

| Bit(s)      | Name                              | Description                                                                                                                                                           | R/W <sup>a</sup> |
|-------------|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 4.1800.15:2 | Reserved                          | Value always 0                                                                                                                                                        | RO               |
| 4.1800.1    | TimeSync transmit path data delay | 1 = PHY XS provides information on transmit path data delay in registers 4.1801 through 4.1804<br>0 = PHY XS does not provide information on transmit path data delay | RO               |
| 4.1800.0    | TimeSync receive path data delay  | 1 = PHY XS provides information on receive path data delay in registers 4.1805 through 4.1808<br>0 = PHY XS does not provide information on receive path data delay   | RO               |

<sup>a</sup>RO = Read only

#### 45.2.4.29 TimeSync PHY XS transmit path data delay (Registers 4.1801, 4.1802, 4.1803, 4.1804)

The TimeSync PHY XS transmit path data delay register contains the maximum (registers 4.1801, 4.1802, see Table 45–337) and minimum (registers 4.1803, 4.1804, see Table 45–337) values of the transmit path data delay. The transmit path data delay is expressed in units of ns. The values contained in these registers are valid when the link is established, as indicated by bit 2 in register 1.1 (see 45.2.1.2.4).

**Table 45–337—TimeSync PHY XS transmit path data delay register**

| Bit(s)      | Name                                           | Description                 | R/W <sup>a</sup> |
|-------------|------------------------------------------------|-----------------------------|------------------|
| 4.1801.15:0 | Maximum PHY XS transmit path data delay, lower | PHY_XS_delay_TX_max [15:0]  | RO, MW           |
| 4.1802.15:0 | Maximum PHY XS transmit path data delay, upper | PHY_XS_delay_TX_max [31:16] | RO, MW           |
| 4.1803.15:0 | Minimum PHY XS transmit path data delay, lower | PHY_XS_delay_TX_min [15:0]  | RO, MW           |
| 4.1804.15:0 | Minimum PHY XS transmit path data delay, upper | PHY_XS_delay_TX_min [31:16] | RO, MW           |

<sup>a</sup>RO = Read only, MW = Multi-word

#### 45.2.4.30 TimeSync PHY XS receive path data delay (Registers 4.1805, 4.1806, 4.1807, 4.1808)

The TimeSync PHY XS receive path data delay register contains the maximum (registers 4.1805, 4.1806, see Table 45–338) and minimum (registers 4.1807, 4.1808, see Table 45–338) values of the receive path data delay. The receive path data delay is expressed in units of ns. The values contained in these registers are valid when the link is established, as indicated by bit 2 in register 4.1 (see 45.2.4.2.7).

**Table 45–338—TimeSync PHY XS receive path data delay register**

| Bit(s)      | Name                                          | Description                 | R/W <sup>a</sup> |
|-------------|-----------------------------------------------|-----------------------------|------------------|
| 4.1805.15:0 | Maximum PHY XS receive path data delay, lower | PHY_XS_delay_RX_max [15:0]  | RO, MW           |
| 4.1806.15:0 | Maximum PHY XS receive path data delay, upper | PHY_XS_delay_RX_max [31:16] | RO, MW           |
| 4.1807.15:0 | Minimum PHY XS receive path data delay, lower | PHY_XS_delay_RX_min [15:0]  | RO, MW           |
| 4.1808.15:0 | Minimum PHY XS receive path data delay, upper | PHY_XS_delay_RX_min [31:16] | RO, MW           |

<sup>a</sup>RO = Read only, MW = Multi-word

#### 45.2.5 DTE XS registers

The assignment of registers in the DTE XS is shown in Table 45–339.

**Table 45–339—DTE XS registers**

| Register address  | Register name             | Subclause |
|-------------------|---------------------------|-----------|
| 5.0               | DTE XS control 1          | 45.2.5.1  |
| 5.1               | DTE XS status 1           | 45.2.5.2  |
| 5.2, 5.3          | DTE XS device identifier  | 45.2.5.3  |
| 5.4               | DTE XS speed ability      | 45.2.5.4  |
| 5.5, 5.6          | DTE XS devices in package | 45.2.5.5  |
| 5.7               | Reserved                  |           |
| 5.8               | DTE XS status 2           | 45.2.5.6  |
| 5.9 through 5.13  | Reserved                  |           |
| 5.14, 5.15        | DTE XS package identifier | 45.2.5.7  |
| 5.16 through 5.19 | Reserved                  |           |
| 5.20              | EEE capability            | 45.2.5.8  |
| 5.21              | Reserved                  |           |
| 5.22              | EEE wake error counter    | 45.2.5.9  |
| 5.23              | Reserved                  |           |
| 5.24              | 10G DTE XGXS lane status  | 45.2.5.10 |
| 5.25              | 10G DTE XGXS test control | 45.2.5.11 |
| 5.26 through 5.31 | Reserved                  |           |

**Table 45–339—DTE XS registers (*continued*)**

| Register address        | Register name                                 | Subclause               |
|-------------------------|-----------------------------------------------|-------------------------|
| 5.32                    | BASE-R DTE XS status 1                        | 45.2.5.12               |
| 5.33 through 5.41       | Reserved                                      |                         |
| 5.42                    | BASE-R DTE XS test pattern control            | 45.2.5.13               |
| 5.43 through 5.49       | Reserved                                      |                         |
| 5.50                    | Multi-lane BASE-R DTE XS alignment status 1   | 45.2.5.14               |
| 5.51                    | Reserved                                      |                         |
| 5.52                    | Multi-lane BASE-R DTE XS alignment status 3   | 45.2.5.15               |
| 5.53                    | Multi-lane BASE-R DTE XS alignment status 4   | 45.2.5.16               |
| 5.54                    | Reserved                                      |                         |
| 5.400 through 5.415     | DTE XS lane mapping, lane 0 through 15        | 45.2.5.17,<br>45.2.5.18 |
| 5.415 through 5.599     | Reserved                                      |                         |
| 5.600 through 5.631     | DTE XS FEC symbol error counter, lane 0 to 15 | 45.2.5.19,<br>45.2.5.20 |
| 5.632 through 5.799     | Reserved                                      |                         |
| 5.800                   | DTE XS FEC control                            | 45.2.5.21               |
| 5.801                   | DTE XS FEC status                             | 45.2.5.22               |
| 5.802, 5.803            | DTE XS FEC corrected codewords counter        | 45.2.5.23               |
| 5.804, 5.805            | DTE XS FEC uncorrected codewords counter      | 45.2.5.24               |
| 5.806, 5.807            | DTE XS FEC degraded SER activate threshold    | 45.2.5.25               |
| 5.808, 5.809            | DTE XS FEC degraded SER deactivate threshold  | 45.2.5.26               |
| 5.810, 5.811            | DTE XS FEC degraded SER interval              | 45.2.5.27               |
| 5.812 through 5.1799    | Reserved                                      |                         |
| 5.1800                  | TimeSync DTE XS capability                    | 45.2.5.28               |
| 5.1801 through 5.1804   | TimeSync DTE XS transmit path data delay      | 45.2.5.29               |
| 5.1805 through 5.1808   | TimeSync DTE XS receive path data delay       | 45.2.5.30               |
| 5.1809 through 5.32767  | Reserved                                      |                         |
| 5.32768 through 5.65535 | Vendor specific                               |                         |

#### 45.2.5.1 DTE XS control 1 register (Register 5.0)

The assignment of bits in the DTE XS control 1 register is shown in Table 45–340. The default value for each bit of the DTE XS control 1 register should be chosen so that the initial state of the device upon power up or reset is a normal operational state without management intervention.

**Table 45–340—DTE XS control 1 register bit definitions**

| Bit(s)  | Name              | Description                                                                                                                                                                                                  | R/W <sup>a</sup> |
|---------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 5.0.15  | Reset             | 1 = DTE XS reset<br>0 = Normal operation                                                                                                                                                                     | R/W<br>SC        |
| 5.0.14  | Loopback          | 1 = Enable loopback mode<br>0 = Disable loopback mode                                                                                                                                                        | R/W              |
| 5.0.13  | Speed selection   | 1 = Operation at 10 Gbp/s and above<br>0 = Unspecified                                                                                                                                                       | R/W              |
| 5.0.12  | Reserved          | Value always 0                                                                                                                                                                                               | RO               |
| 5.0.11  | Low power         | 1 = Low-power mode<br>0 = Normal operation                                                                                                                                                                   | R/W              |
| 5.0.10  | Clock stop enable | 1 = The DTE XS may stop the clock during LPI<br>0 = Clock not stoppable                                                                                                                                      | R/W              |
| 5.0.9   | XAUI stop enable  | 1 = The DTE XS may stop XAUI signals during LPI<br>0 = XAUI not stoppable                                                                                                                                    | R/W              |
| 5.0.8:7 | Reserved          | Value always 0                                                                                                                                                                                               | RO               |
| 5.0.6   | Speed selection   | 1 = Operation at 10 Gb/s and above<br>0 = Unspecified                                                                                                                                                        | R/W              |
| 5.0.5:2 | Speed selection   | 5 4 3 2<br>1 1 x x = Reserved<br>1 0 1 1 = Reserved<br>1 0 1 0 = 400 Gb/s<br>1 0 0 1 = 200 Gb/s<br>1 0 0 0 = Reserved<br>0 1 x x = Reserved<br>0 0 1 x = Reserved<br>0 0 0 1 = Reserved<br>0 0 0 0 = 10 Gb/s | R/W              |
| 5.0.1:0 | Reserved          | Value always 0                                                                                                                                                                                               | RO               |

<sup>a</sup>R/W = Read/Write, SC = Self-clearing, RO = Read only

##### 45.2.5.1.1 Reset (5.0.15)

Resetting a DTE XS is accomplished by setting bit 5.0.15 to a one. This action shall set all DTE XS registers to their default states. As a consequence, this action may change the internal state of the DTE XS and the state of the physical link. This action may also initiate a reset in any other MMDs that are instantiated in the same package. This bit is self-clearing, and a DTE XS shall return a value of one in bit 5.0.15 when a reset is

in progress and a value of zero otherwise. A DTE XS is not required to accept a write transaction to any of its registers until the reset process is completed. The reset process shall be completed within 0.5 s from the setting of bit 5.0.15. During a reset, a DTE XS shall respond to reads to register bits 5.0.15 and 5.8.15:14. All other register bits should be ignored.

NOTE—This operation may interrupt data communication.

#### **45.2.5.1.2 Loopback (5.0.14)**

The DTE XS shall be placed in a loopback mode of operation when bit 5.0.14 is set to a one. When bit 5.0.14 is set to a one, the DTE XS shall accept data on the transmit path and return it on the receive path. For 10 Gb/s operation, the specific behavior of a DTE XS during loopback is specified in 48.3.3.

The default value of bit 5.0.14 is zero.

NOTE—The signal path through the DTE XS that is exercised in the loopback mode of operation is implementation specific, but it is recommended that the signal path encompass as much of the DTE XS circuitry as is practical. The intention of providing this loopback mode of operation is to permit a diagnostic or self-test function to perform the transmission and reception of a PDU, thus testing the transmit and receive data paths. Other loopback signal paths may be enabled using loopback controls within other MMDs.

#### **45.2.5.1.3 Low power (5.0.11)**

A DTE XS may be placed into a low-power mode by setting bit 5.0.11 to a one. This action may also initiate a low-power mode in any other MMDs that are instantiated in the same package. The low-power mode is exited by resetting the DTE XS. The behavior of the DTE XS in transition to and from the low-power mode is implementation specific and any interface signals should not be relied upon. While in the low-power mode, the device shall, as a minimum, respond to management transactions necessary to exit the low-power mode. The default value of bit 5.0.11 is zero.

#### **45.2.5.1.4 Clock stop enable (5.0.10)**

If bit 5.0.10 is set to one, then the DTE XS may stop the receive xMII clock while it is signaling LPI, otherwise it shall keep the clock active. If the DTE XS does not support EEE capability or is not able to stop the receive clock, then this bit has no effect (see 46.3.2.4).

#### **45.2.5.1.5 XAUI stop enable (5.0.9)**

If bit 5.0.9 is set to one, then the DTE XS may stop signaling on the XAUI in the transmit direction during LPI, otherwise the DTE XS shall keep the XAUI signals active. If the DTE XS does not support EEE capability or is not able to stop the transmit path XAUI signals, then this bit has no effect.

#### **45.2.5.1.6 Speed selection (5.0.13, 5.0.6, 5.0.5:2)**

Speed selection bits 5.0.13 and 5.0.6 shall both be written as a one. Any attempt to change the bits to an invalid setting shall be ignored. These two bits are set to one in order to make them compatible with Clause 22.

The speed of the DTE XS may be selected using bits 5 through 2. The speed abilities of the DTE XS are advertised in the DTE XS speed ability register. A DTE XS may ignore writes to the DTE XS speed selection bits that select speeds it has not advertised in the DTE XS speed ability register. It is the responsibility of the STA entity to ensure that mutually acceptable speeds are applied consistently across all the MMDs on a particular PHY.

The DTE XS speed selection defaults to a supported ability.

#### 45.2.5.2 DTE XS status 1 register (Register 5.1)

The assignment of bits in the DTE XS status 1 register is shown in Table 45–341. All the bits in the DTE XS status 1 register are read only; a write to the DTE XS status 1 register shall have no effect.

**Table 45–341—DTE XS status 1 register bit definitions**

| Bit(s)    | Name                       | Description                                                                      | R/W <sup>a</sup> |
|-----------|----------------------------|----------------------------------------------------------------------------------|------------------|
| 5.1.15:12 | Reserved                   | Value always 0                                                                   | RO               |
| 5.1.11    | Tx LPI received            | 1 = Tx PCS has received LPI<br>0 = LPI not received                              | RO/LH            |
| 5.1.10    | Rx LPI received            | 1 = Rx PCS has received LPI<br>0 = LPI not received                              | RO/LH            |
| 5.1.9     | Tx LPI indication          | 1 = Tx PCS is currently receiving LPI<br>0 = PCS is not currently receiving LPI  | RO               |
| 5.1.8     | Rx LPI indication          | 1 = Rx PCS is currently receiving LPI<br>0 = PCS is not currently receiving LPI  | RO               |
| 5.1.7     | Fault                      | 1 = Fault condition detected<br>0 = No fault condition detected                  | RO               |
| 5.1.6     | Clock stop capable         | 1 = The MAC may stop the clock during LPI<br>0 = Clock not stoppable             | RO               |
| 5.1.5:3   | Reserved                   | Value always 0                                                                   | RO               |
| 5.1.2     | DTE XS receive link status | 1 = The DTE XS receive link is up<br>0 = The DTE XS receive link is down         | RO/LL            |
| 5.1.1     | Low-power ability          | 1 = DTE XS supports low-power mode<br>0 = DTE XS does not support low-power mode | RO               |
| 5.1.0     | Reserved                   | Value always 0                                                                   | RO               |

<sup>a</sup>RO = Read only, LL = Latching low, LH = Latching high

##### 45.2.5.2.1 Transmit LPI received (5.1.11)

When read as a one, bit 5.1.11 indicates that the transmit DTE XS has received LPI signaling one or more times since the register was last read. When read as a zero, bit 5.1.11 indicates that the DTE XS has not received LPI signaling. This bit shall be implemented with latching high behavior.

##### 45.2.5.2.2 Receive LPI received (5.1.10)

When read as a one, bit 5.1.10 indicates that the receive DTE XS has received LPI signaling one or more times since the register was last read. When read as a zero, bit 5.1.10 indicates that the DTE XS has not received LPI signaling. This bit shall be implemented with latching high behavior.

#### **45.2.5.2.3 Transmit LPI indication (5.1.9)**

When read as a one, bit 5.1.9 indicates that the transmit DTE XS is currently receiving LPI signals. When read as a zero, bit 5.1.9 indicates that the DTE XS is not currently receiving LPI signals. The behavior if read during a state transition is undefined.

#### **45.2.5.2.4 Receive LPI indication (5.1.8)**

When read as a one, bit 5.1.8 indicates that the receive DTE XS is currently receiving LPI signals. When read as a zero, bit 5.1.8 indicates that the DTE XS is not currently receiving LPI signals. The behavior if read during a state transition is undefined.

#### **45.2.5.2.5 Fault (5.1.7)**

When read as a one, bit 5.1.7 indicates that the DTE XS has detected a fault condition on either the transmit or receive paths. When read as a zero, bit 5.1.7 indicates that the DTE XS has not detected a fault condition. Bit 5.1.7 is set to a one when either of the fault bits (5.8.11, 5.8.10) located in register 5.8 are set to a one.

#### **45.2.5.2.6 Clock stop capable (5.1.6)**

If bit 5.1.6 is set to one, then the DTE XS is indicating that the attached RS is permitted to stop the transmit xMII clock while it is signaling LPI. If the bit is set to zero, then the DTE XS is indicating that the attached RS is not permitted to stop the transmit xMII clock while it is signaling LPI. If the RS does not support EEE capability or is not able to stop the transmit direction xMII clock, then this bit has no effect (see 46.3.2.4).

#### **45.2.5.2.7 DTE XS receive link status (5.1.2)**

When read as a one, bit 5.1.2 indicates that the DTE XS receive link is aligned. When read as a zero, bit 5.1.2 indicates that the DTE XS receive link is not aligned. The receive link status bit shall be implemented with latching low behavior.

For 10 Gb/s operation, this bit is a latching low version of bit 5.24.12.

#### **45.2.5.2.8 Low-power ability (5.1.1)**

When read as a one, bit 5.1.1 indicates that the DTE XS supports the low-power feature. When read as a zero, bit 5.1.1 indicates that the DTE XS does not support the low-power feature. If a DTE XS supports the low-power feature then it is controlled using the low-power bit in the DTE XS control register.

#### **45.2.5.3 DTE XS device identifier (Registers 5.2 and 5.3)**

Registers 5.2 and 5.3 provide a 32-bit value, which may constitute a unique identifier for a DTE XS. The identifier shall be composed of the 3rd through 24th bits of the Organizationally Unique Identifier (OUI) assigned to the device manufacturer by the IEEE, plus a six-bit model number, plus a four-bit revision number. A DTE XS may return a value of zero in each of the 32 bits of the DTE XS device identifier.

The format of the DTE XS device identifier is specified in 22.2.4.3.1.

#### **45.2.5.4 DTE XS speed ability (Register 5.4)**

The assignment of bits in the DTE XS speed ability register is shown in Table 45–342.



**Table 45–342— DTE XS speed ability register bit definitions**

| Bit(s)    | Name         | Description                                                                                          | R/W <sup>a</sup> |
|-----------|--------------|------------------------------------------------------------------------------------------------------|------------------|
| 5.4.15:10 | Reserved     | Value always 0                                                                                       | RO               |
| 5.4.9     | 400G capable | 1 = DTE XS is capable of operating at 400 Gb/s<br>0 = DTE XS is not capable of operating at 400 Gb/s | RO               |
| 5.4.8     | 200G capable | 1 = DTE XS is capable of operating at 200 Gb/s<br>0 = DTE XS is not capable of operating at 200 Gb/s | RO               |
| 5.4.7:1   | Reserved     | Value always 0                                                                                       | RO               |
| 5.4.0     | 10G capable  | 1 = DTE XS is capable of operating at 10 Gb/s<br>0 = DTE XS is not capable of operating at 10 Gb/s   | RO               |

<sup>a</sup>RO = Read only

#### 45.2.5.4.1 400G capable (5.4.9)

When read as a one, bit 5.4.9 indicates that the DTE XS is able to operate at a data rate of 400 Gb/s. When read as a zero, bit 5.4.9 indicates that the DTE XS is not able to operate at a data rate of 400 Gb/s.

#### 45.2.5.4.2 200G capable (5.4.8)

When read as a one, bit 5.4.8 indicates that the DTE XS is able to operate at a data rate of 200 Gb/s. When read as a zero, bit 5.4.8 indicates that the DTE XS is not able to operate at a data rate of 200 Gb/s.

#### 45.2.5.4.3 10G capable (5.4.0)

When read as a one, bit 5.4.0 indicates that the DTE XS is able to operate at a data rate of 10 Gb/s. When read as a zero, bit 5.4.0 indicates that the DTE XS is not able to operate at a data rate of 10 Gb/s.

#### 45.2.5.5 DTE XS devices in package (Registers 5.5 and 5.6)

The DTE XS devices in package registers are defined in Table 45–2.

#### 45.2.5.6 DTE XS status 2 register (Register 5.8)

The assignment of bits in the DTE XS status 2 register is shown in Table 45–343. All the bits in the DTE XS status 2 register are read only; a write to the DTE XS status 2 register shall have no effect.

**Table 45–343—DTE XS status 2 register bit definitions**

| Bit(s)    | Name           | Description                                                                                                                                                                                                                           | R/W <sup>a</sup> |
|-----------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 5.8.15:14 | Device present | <div>15 14</div> <div>1 0 = Device responding at this address</div> <div>1 1 = No device responding at this address</div> <div>0 1 = No device responding at this address</div> <div>0 0 = No device responding at this address</div> | RO               |
| 5.8.13:12 | Reserved       | Value always 0                                                                                                                                                                                                                        | RO               |

**Table 45–343—DTE XS status 2 register bit definitions (*continued*)**

| Bit(s)  | Name           | Description                                                                     | R/W <sup>a</sup> |
|---------|----------------|---------------------------------------------------------------------------------|------------------|
| 5.8.11  | Transmit fault | 1 = Fault condition on transmit path<br>0 = No fault condition on transmit path | RO/LH            |
| 5.8.10  | Receive fault  | 1 = Fault condition on receive path<br>0 = No fault condition on receive path   | RO/LH            |
| 5.8.9:0 | Reserved       | Value always 0                                                                  | RO               |

<sup>a</sup>RO = Read only, LH = Latching high

#### 45.2.5.6.1 Device present (5.8.15:14)

When read as <10>, bits 5.8.15:14 indicate that a device is present and responding at this register address. When read as anything other than <10>, bits 5.8.15:14 indicate that no device is present at this register address or that the device is not functioning properly.

#### 45.2.5.6.2 Transmit fault (5.8.11)

When read as a one, bit 5.8.11 indicates that the DTE XS has detected a fault condition on the transmit path. When read as a zero, bit 5.8.11 indicates that the DTE XS has not detected a fault condition on the transmit path. The transmit fault bit shall be implemented with latching high behavior.

The default value of bit 5.8.11 is zero.

#### 45.2.5.6.3 Receive fault (5.8.10)

When read as a one, bit 5.8.10 indicates that the DTE XS has detected a fault condition on the receive path. When read as a zero, bit 5.8.10 indicates that the DTE XS has not detected a fault condition on the receive path. The receive fault bit shall be implemented with latching high behavior.

The default value of bit 5.8.10 is zero.

#### 45.2.5.7 DTE XS package identifier (Registers 5.14 and 5.15)

Registers 5.14 and 5.15 provide a 32-bit value, which may constitute a unique identifier for a particular type of package that the DTE XS is instantiated within. The identifier shall be composed of the 3rd through 24th bits of the Organizationally Unique Identifier (OUI) assigned to the package manufacturer by the IEEE, plus a six-bit model number, plus a four-bit revision number. A DTE XS may return a value of zero in each of the 32 bits of the DTE XS package identifier.

A non-zero package identifier may be returned by one or more MMDs in the same package. The package identifier may be the same as the device identifier.

The format of the DTE XS package identifier is specified in 22.2.4.3.1.

#### 45.2.5.8 EEE capability (Register 5.20)

This register is used to indicate the capability of the DTE XS to support EEE functions. The assignment of bits in the EEE capability register is shown in Table 45–344.

**Table 45–344—EEE capability register bit definitions**

| Bit(s)    | Name              | Description                                                                       | R/W <sup>a</sup> |
|-----------|-------------------|-----------------------------------------------------------------------------------|------------------|
| 5.20.15:5 | Reserved          | Value always 0                                                                    | RO               |
| 5.20.4    | DTE XS EEE        | 1 = EEE is supported for DTE XS<br>0 = EEE is not supported for DTE XS            | RO               |
| 5.20.3:1  | Reserved          | Value always 0                                                                    | RO               |
| 5.20.0    | XAUI stop capable | 1 = The PHY XS may stop XAUI signals during LPI<br>0 = XAUI signals not stoppable | RO               |

<sup>a</sup> RO = Read only

##### 45.2.5.8.1 PHY XS EEE supported (5.20.4)

If the device supports EEE operation for DTE XS as defined in 48.2, this bit shall be set to one.

##### 45.2.5.8.2 XAUI stop capable (5.20.0)

If bit 5.20.0 is set to one, then the DTE XS is indicating that the attached PHY XS is permitted to stop signaling the XAUI in the receive direction during LPI. If the bit is set to zero, then the DTE XS is indicating that the attached PHY XS is not permitted to stop signaling the XAUI in the receive direction during LPI. If the PHY XS does not support EEE capability or is not able to stop the receive direction XAUI, then this bit has no effect.

#### 45.2.5.9 EEE wake error counter (Register 5.22)

This register is used by DTE XS that support EEE to count wake time faults where the DTE XS fails to complete its normal wake sequence after a period of quiescence for XAUI receive signals. The fault event to be counted may occur during a refresh or a wake-up. This 16-bit counter shall be reset to all zeros when the EEE wake error counter is read by the management function or upon execution of the DTE XS reset. This counter shall be held at all ones in the case of overflow.

#### 45.2.5.10 10G DTE XGXS lane status register (Register 5.24)

The assignment of bits in the 10G DTE XGXS lane status register is shown in Table 45–345. All the bits in the 10G DTE XGXS lane status register are read only; a write to the 10G DTE XGXS lane status register shall have no effect.

##### 45.2.5.10.1 DTE XGXS receive lane alignment status (5.24.12)

When read as a one, bit 5.24.12 indicates that the DTE XGXS has synchronized and aligned all four receive lanes. When read as a zero, bit 5.24.12 indicates that the DTE XGXS has not synchronized and aligned all four receive lanes.

**Table 45–345—10G DTE XGXS lane status register bit definitions**

| Bit(s)     | Name                           | Description                                                                                          | R/W <sup>a</sup> |
|------------|--------------------------------|------------------------------------------------------------------------------------------------------|------------------|
| 5.24.15:13 | Reserved                       | Value always 0                                                                                       | RO               |
| 5.24.12    | DTE XGXS lane alignment status | 1 = DTE XGXS receive lanes aligned<br>0 = DTE XGXS receive lanes not aligned                         | RO               |
| 5.24.11    | Pattern testing ability        | 1 = DTE XGXS is able to generate test patterns<br>0 = DTE XGXS is not able to generate test patterns | RO               |
| 5.24.10    | Ignored                        | Value 0 or 1, writes ignored                                                                         | RO               |
| 5.24.9:4   | Reserved                       | Value always 0                                                                                       | RO               |
| 5.24.3     | Lane 3 sync                    | 1 = Lane 3 is synchronized<br>0 = Lane 3 is not synchronized                                         | RO               |
| 5.24.2     | Lane 2 sync                    | 1 = Lane 2 is synchronized<br>0 = Lane 2 is not synchronized                                         | RO               |
| 5.24.1     | Lane 1 sync                    | 1 = Lane 1 is synchronized<br>0 = Lane 1 is not synchronized                                         | RO               |
| 5.24.0     | Lane 0 sync                    | 1 = Lane 0 is synchronized<br>0 = Lane 0 is not synchronized                                         | RO               |

<sup>a</sup>RO = Read only

#### 45.2.5.10.2 Pattern testing ability (5.24.11)

When read as a one, bit 5.24.11 indicates that the 10G DTE XGXS is able to generate test patterns. When read as a zero, bit 5.24.11 indicates that the 10G DTE XGXS is not able to generate test patterns. If the 10G DTE XGXS is able to generate test patterns then the functionality is controlled using the transmit test-pattern enable bit in register 5.25.

#### 45.2.5.10.3 Ignored (5.24.10)

So that a single device can implement either register 4.24 or register 5.24, bit 5.24.10 can return either a one or a zero and should be ignored.

#### 45.2.5.10.4 Lane 3 sync (5.24.3)

When read as a one, bit 5.24.3 indicates that the XGXS receive lane 3 is synchronized. When read as a zero, bit 5.24.3 indicates that the XGXS receive lane 3 is not synchronized.

#### 45.2.5.10.5 Lane 2 sync (5.24.2)

When read as a one, bit 5.24.2 indicates that the XGXS receive lane 2 is synchronized. When read as a zero, bit 5.24.2 indicates that the XGXS receive lane 2 is not synchronized.

#### 45.2.5.10.6 Lane 1 sync (5.24.1)

When read as a one, bit 5.24.1 indicates that the XGXS receive lane 1 is synchronized. When read as a zero, bit 5.24.1 indicates that the XGXS receive lane 1 is not synchronized.

#### 45.2.5.10.7 Lane 0 sync (5.24.0)

When read as a one, bit 5.24.0 indicates that the XGXS receive lane 0 is synchronized. When read as a zero, bit 5.24.0 indicates that the XGXS receive lane 0 is not synchronized.

#### 45.2.5.11 10G DTE XGXS test control register (Register 5.25)

The assignment of bits in the 10G DTE XGXS test control register is shown in Table 45–346. The default value for each bit of the 10G DTE XGXS test control register should be chosen so that the initial state of the device upon power up or reset is a normal operational state without management intervention.

**Table 45–346—10G DTE XGXS test control register bit definitions**

| Bit(s)    | Name                         | Description                                                                                                                          | R/W <sup>a</sup> |
|-----------|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 5.25.15:3 | Reserved                     | Value always 0                                                                                                                       | RO               |
| 5.25.2    | Transmit test-pattern enable | 1 = Transmit test pattern enabled<br>0 = Transmit test pattern not enabled                                                           | R/W              |
| 5.25.1:0  | Test-pattern select          | 1 0<br>1 1 = Reserved<br>1 0 = Mixed-frequency test pattern<br>0 1 = Low-frequency test pattern<br>0 0 = High-frequency test pattern | R/W              |

<sup>a</sup>RO = Read only, R/W = Read/Write

##### 45.2.5.11.1 10G DTE XGXS test-pattern enable (5.25.2)

When bit 5.25.2 is set to a one, pattern testing is enabled on the transmit path. When bit 5.25.2 is set to a zero, pattern testing is disabled on the transmit path. Pattern testing is optional, and the ability of the 10G DTE XGXS to generate test patterns is advertised by the pattern testing ability bit in register 5.24. A 10G DTE XGXS that does not support the generation of test patterns shall ignore writes to this bit and always return a value of zero. The default of bit 5.25.2 is zero.

##### 45.2.5.11.2 10G DTE XGXS test-pattern select (5.25.1:0)

The test pattern to be used when pattern testing is enabled using bit 5.25.2 is selected using bits 5.25.1:0. When bits 5.25.1:0 are set to <10>, the mixed-frequency test pattern shall be selected for pattern testing. When bits 5.25.1:0 are set to <01>, the low-frequency test pattern shall be selected for pattern testing. When bits 5.25.1:0 are set to <00>, the high-frequency test pattern shall be selected for pattern testing. The test patterns are defined in Annex 48A.

#### 45.2.5.12 BASE-R DTE XS status 1 register (Register 5.32)

The assignment of bits in the BASE-R DTE XS status 1 register is shown in Table 45–347. A DTE XS device that does not implement BASE-R shall return a zero for all bits in the BASE-R DTE XS status 1 register. It is the responsibility of the STA management entity to ensure that a port type is supported by all MMDs before interrogating any of its status bits.

**Table 45–347—BASE-R DTE XS status 1 register bit definitions**

| Bit(s)     | Name                              | Description                                                              | R/W <sup>a</sup> |
|------------|-----------------------------------|--------------------------------------------------------------------------|------------------|
| 5.32.15:13 | Reserved                          | Value always 0                                                           | RO               |
| 5.32.12    | BASE-R DTE XS receive link status | 1 = BASE-R DTE XS receive link up<br>0 = BASE-R DTE XS receive link down | RO               |
| 5.32.11:0  | Reserved                          | Value always 0                                                           | RO               |

<sup>a</sup>RO = Read only

#### 45.2.5.12.1 BASE-R DTE XS receive link status (5.32.12)

When read as a one, bit 5.32.12 indicates that the DTE XS is in a fully operational state. When read as a zero, bit 5.32.12 indicates that the DTE XS is not fully operational. This bit is a reflection of the DTE XS equivalent of the PCS\_status variable defined in 119.3 for 200/400GBASE-R.

#### 45.2.5.13 BASE-R DTE XS test-pattern control register (Register 5.42)

The assignment of bits in the BASE-R DTE XS test-pattern control register is shown in Table 45–348. This register is only required when the 200/400GBASE-R capability is supported. The test-pattern methodology is described in 119.2.4.9.

**Table 45–348—BASE-R DTE XS test-pattern control register bit definitions**

| Bit(s)    | Name                         | Description                                                           | R/W <sup>a</sup> |
|-----------|------------------------------|-----------------------------------------------------------------------|------------------|
| 5.42.15:4 | Reserved                     | Value always 0                                                        | RO               |
| 5.42.3    | Transmit test-pattern enable | 1 = Enable transmit test pattern<br>0 = Disable transmit test pattern | R/W              |
| 5.42.2:0  | Reserved                     | Value always 0                                                        | RO               |

<sup>a</sup>RO = Read only, R/W = Read/Write

#### 45.2.5.13.1 Transmit test-pattern enable (5.42.3)

When bit 5.42.3 is set to a one, pattern testing is enabled on the transmit path. When bit 5.42.3 is set to a zero, pattern testing is disabled on the transmit path.

The default value for bit 5.42.3 is zero.

#### 45.2.5.14 Multi-lane BASE-R DTE XS alignment status 1 register (Register 5.50)

The assignment of bits in the multi-lane BASE-R DTE XS alignment status 1 register is shown in Table 45–349. A DTE XS device that does not implement multi-lane BASE-R DTE XS shall return a zero for all bits in the multi-lane BASE-R DTE XS alignment status 1 register. It is the responsibility of the STA management entity to ensure that a port type is supported by all MMDs before interrogating any of its status bits.

**Table 45–349—Multi-lane BASE-R DTE XS alignment status 1 register bit definitions**

| Bit(s)     | Name                         | Description                                                                                    | R/W <sup>a</sup> |
|------------|------------------------------|------------------------------------------------------------------------------------------------|------------------|
| 5.50.15:13 | Reserved                     | Value always 0                                                                                 | RO               |
| 5.50.12    | DTE XS lane alignment status | 1 = DTE XS receive lanes locked and aligned<br>0 = DTE XS receive lanes not locked and aligned | RO               |
| 5.50.11:0  | Reserved                     | Value always 0                                                                                 | RO               |

<sup>a</sup>RO = Read only

#### 45.2.5.14.1 DTE XS lane alignment status (5.50.12)

When read as a one, bit 5.50.12 indicates that the DTE XS has locked and aligned all receive lanes. When read as a zero, bit 5.50.12 indicates that the DTE XS has not locked and aligned all receive lanes.

#### 45.2.5.15 Multi-lane BASE-R DTE XS alignment status 3 register (Register 5.52)

The assignment of bits in the multi-lane BASE-R DTE XS alignment status 3 register is shown in Table 45–350. A DTE XS device that does not implement multi-lane BASE-R DTE XS shall return a zero for all bits in the multi-lane BASE-R DTE XS alignment status 3 register. A device that implements multi-lane BASE-R DTE XS shall return a zero for all bits in the multi-lane BASE-R DTE XS alignment status 3 register that are not required for the DTE XS configuration. It is the responsibility of the STA management entity to ensure that a port type is supported by all MMDs before interrogating any of its status bits.

**Table 45–350—Multi-lane BASE-R DTE XS alignment status 3 register bit definitions**

| Bit(s)    | Name           | Description                                                                        | R/W <sup>a</sup> |
|-----------|----------------|------------------------------------------------------------------------------------|------------------|
| 5.52.15:8 | Reserved       | Value always 0                                                                     | RO               |
| 5.52.7    | Lane 7 aligned | 1 = Lane 7 alignment marker is locked<br>0 = Lane 7 alignment marker is not locked | RO               |
| 5.52.6    | Lane 6 aligned | 1 = Lane 6 alignment marker is locked<br>0 = Lane 6 alignment marker is not locked | RO               |
| 5.52.5    | Lane 5 aligned | 1 = Lane 5 alignment marker is locked<br>0 = Lane 5 alignment marker is not locked | RO               |
| 5.52.4    | Lane 4 aligned | 1 = Lane 4 alignment marker is locked<br>0 = Lane 4 alignment marker is not locked | RO               |
| 5.52.3    | Lane 3 aligned | 1 = Lane 3 alignment marker is locked<br>0 = Lane 3 alignment marker is not locked | RO               |
| 5.52.2    | Lane 2 aligned | 1 = Lane 2 alignment marker is locked<br>0 = Lane 2 alignment marker is not locked | RO               |

**Table 45–350—Multi-lane BASE-R DTE XS alignment status 3 register bit definitions**

| Bit(s) | Name           | Description                                                                        | R/W <sup>a</sup> |
|--------|----------------|------------------------------------------------------------------------------------|------------------|
| 5.52.1 | Lane 1 aligned | 1 = Lane 1 alignment marker is locked<br>0 = Lane 1 alignment marker is not locked | RO               |
| 5.52.0 | Lane 0 aligned | 1 = Lane 0 alignment marker is locked<br>0 = Lane 0 alignment marker is not locked | RO               |

<sup>a</sup>RO = Read only

#### **45.2.5.15.1 Lane 7 aligned (5.52.7)**

When read as a one, bit 5.52.7 indicates that the DTE XS receiver has achieved alignment marker lock for service interface lane 7. When read as a zero, bit 5.52.7 indicates that the DTE XS receiver lane 7 has not achieved alignment marker lock. This bit reflects the state of `amps_lock[7]` (see 119.2.6.2.2).

#### **45.2.5.15.2 Lane 6 aligned (5.52.6)**

When read as a one, bit 5.52.6 indicates that the DTE XS receiver has achieved alignment marker lock for service interface lane 6. When read as a zero, bit 5.52.6 indicates that the DTE XS receiver lane 6 has not achieved alignment marker lock. This bit reflects the state of `amps_lock[6]` (see 119.2.6.2.2).

#### **45.2.5.15.3 Lane 5 aligned (5.52.5)**

When read as a one, bit 5.52.5 indicates that the DTE XS receiver has achieved alignment marker lock for service interface lane 5. When read as a zero, bit 5.52.5 indicates that the DTE XS receiver lane 5 has not achieved alignment marker lock. This bit reflects the state of `amps_lock[5]` (see 119.2.6.2.2).

#### **45.2.5.15.4 Lane 4 aligned (5.52.4)**

When read as a one, bit 5.52.4 indicates that the DTE XS receiver has achieved alignment marker lock for service interface lane 4. When read as a zero, bit 5.52.4 indicates that the DTE XS receiver lane 4 has not achieved alignment marker lock. This bit reflects the state of `amps_lock[4]` (see 119.2.6.2.2).

#### **45.2.5.15.5 Lane 3 aligned (5.52.3)**

When read as a one, bit 5.52.3 indicates that the DTE XS receiver has achieved alignment marker lock for service interface lane 3. When read as a zero, bit 5.52.3 indicates that the DTE XS receiver lane 3 has not achieved alignment marker lock. This bit reflects the state of `amps_lock[3]` (see 119.2.6.2.2).

#### **45.2.5.15.6 Lane 2 aligned (5.52.2)**

When read as a one, bit 5.52.2 indicates that the DTE XS receiver has achieved alignment marker lock for service interface lane 2. When read as a zero, bit 5.52.2 indicates that the DTE XS receiver lane 2 has not achieved alignment marker lock. This bit reflects the state of `amps_lock[2]` (see 119.2.6.2.2).

#### **45.2.5.15.7 Lane 1 aligned (5.52.1)**

When read as a one, bit 5.52.1 indicates that the DTE XS receiver has achieved alignment marker lock for service interface lane 1. When read as a zero, bit 5.52.1 indicates that the DTE XS receiver lane 1 has not achieved alignment marker lock. This bit reflects the state of `amps_lock[1]` (see 119.2.6.2.2).



#### 45.2.5.15.8 Lane 0 aligned (5.52.0)

When read as a one, bit 5.52.0 indicates that the DTE XS receiver has achieved alignment marker lock for service interface lane 0. When read as a zero, bit 5.52.0 indicates that the DTE XS receiver lane 0 has not achieved alignment marker lock. This bit reflects the state of `amps_lock[0]` (see 119.2.6.2.2).

#### 45.2.5.16 Multi-lane BASE-R DTE XS alignment status 4 register (Register 5.53)

The assignment of bits in the multi-lane BASE-R DTE XS alignment status 4 register is shown in Table 45–351. A DTE XS device that does not implement multi-lane BASE-R DTE XS shall return a zero for all bits in the multi-lane BASE-R DTE XS alignment status 4 register. A device that implements multi-lane BASE-R DTE XS shall return a zero for all bits in the multi-lane BASE-R DTE XS alignment status 4 register that are not required for the DTE XS configuration. It is the responsibility of the STA management entity to ensure that a port type is supported by all MMDs before interrogating any of its status bits.

**Table 45–351—Multi-lane BASE-R DTE XS alignment status 4 register bit definitions**

| Bit(s)    | Name            | Description                                                                          | R/W <sup>a</sup> |
|-----------|-----------------|--------------------------------------------------------------------------------------|------------------|
| 5.53.15:8 | Reserved        | Value always 0                                                                       | RO               |
| 5.53.7    | Lane 15 aligned | 1 = Lane 15 alignment marker is locked<br>0 = Lane 15 alignment marker is not locked | RO               |
| 5.53.6    | Lane 14 aligned | 1 = Lane 14 alignment marker is locked<br>0 = Lane 14 alignment marker is not locked | RO               |
| 5.53.5    | Lane 13 aligned | 1 = Lane 13 alignment marker is locked<br>0 = Lane 13 alignment marker is not locked | RO               |
| 5.53.4    | Lane 12 aligned | 1 = Lane 12 alignment marker is locked<br>0 = Lane 12 alignment marker is not locked | RO               |
| 5.53.3    | Lane 11 aligned | 1 = Lane 11 alignment marker is locked<br>0 = Lane 11 alignment marker is not locked | RO               |
| 5.53.2    | Lane 10 aligned | 1 = Lane 10 alignment marker is locked<br>0 = Lane 10 alignment marker is not locked | RO               |
| 5.53.1    | Lane 9 aligned  | 1 = Lane 9 alignment marker is locked<br>0 = Lane 9 alignment marker is not locked   | RO               |
| 5.53.0    | Lane 8 aligned  | 1 = Lane 8 alignment marker is locked<br>0 = Lane 8 alignment marker is not locked   | RO               |

<sup>a</sup>RO = Read only

#### 45.2.5.16.1 Lane 15 aligned (5.53.7)

When read as a one, bit 5.53.7 indicates that the DTE XS receiver has achieved alignment marker lock for service interface lane 15. When read as a zero, bit 5.53.7 indicates that the DTE XS receiver lane 15 has not achieved alignment marker lock. This bit reflects the state of `amps_lock[15]` (see 119.2.6.2.2).

#### **45.2.5.16.2 Lane 14 aligned (5.53.6)**

When read as a one, bit 5.53.6 indicates that the DTE XS receiver has achieved alignment marker lock for service interface lane 14. When read as a zero, bit 5.53.6 indicates that the DTE XS receiver lane 14 has not achieved alignment marker lock. This bit reflects the state of `amps_lock[14]` (see 119.2.6.2.2).

#### **45.2.5.16.3 Lane 13 aligned (5.53.5)**

When read as a one, bit 5.53.5 indicates that the DTE XS receiver has achieved alignment marker lock for service interface lane 13. When read as a zero, bit 5.53.5 indicates that the DTE XS receiver lane 13 has not achieved alignment marker lock. This bit reflects the state of `amps_lock[13]` (see 119.2.6.2.2).

#### **45.2.5.16.4 Lane 12 aligned (5.53.4)**

When read as a one, bit 5.53.4 indicates that the DTE XS receiver has achieved alignment marker lock for service interface lane 12. When read as a zero, bit 5.53.4 indicates that the DTE XS receiver lane 12 has not achieved alignment marker lock. This bit reflects the state of `amps_lock[12]` (see 119.2.6.2.2).

#### **45.2.5.16.5 Lane 11 aligned (5.53.3)**

When read as a one, bit 5.53.3 indicates that the DTE XS receiver has achieved alignment marker lock for service interface lane 11. When read as a zero, bit 5.53.3 indicates that the DTE XS receiver lane 11 has not achieved alignment marker lock. This bit reflects the state of `amps_lock[11]` (see 119.2.6.2.2).

#### **45.2.5.16.6 Lane 10 aligned (5.53.2)**

When read as a one, bit 5.53.2 indicates that the DTE XS receiver has achieved alignment marker lock for service interface lane 10. When read as a zero, bit 5.53.2 indicates that the DTE XS receiver lane 10 has not achieved alignment marker lock. This bit reflects the state of `amps_lock[10]` (see 119.2.6.2.2).

#### **45.2.5.16.7 Lane 9 aligned (5.53.1)**

When read as a one, bit 5.53.1 indicates that the DTE XS receiver has achieved alignment marker lock for service interface lane 9. When read as a zero, bit 5.53.1 indicates that the DTE XS receiver lane 9 has not achieved alignment marker lock. This bit reflects the state of `amps_lock[9]` (see 119.2.6.2.2).

#### **45.2.5.16.8 Lane 8 aligned (5.53.0)**

When read as a one, bit 5.53.0 indicates that the DTE XS receiver has achieved alignment marker lock for service interface lane 8. When read as a zero, bit 5.53.0 indicates that the DTE XS receiver lane 8 has not achieved alignment marker lock. This bit reflects the state of `amps_lock[8]` (see 119.2.6.2.2).

#### **45.2.5.17 DTE XS lane mapping, lane 0 register (Register 5.400)**

The assignment of bits in the DTE XS lane mapping, lane 0 register is shown in Table 45–352. When the multi-lane DTE XS described in Clause 118 detects and locks the alignment marker for service interface lane 0, the detected PCS lane number is recorded in this register. The contents of the DTE XS lane mapping, lane 0 register is valid when Lane 0 aligned bit (5.52.0) is set to one and is invalid otherwise.

#### **45.2.5.18 DTE XS lane mapping, lane 1 through lane 15 registers (Registers 5.401 through 5.415)**

The definition of the DTE XS lane mapping, lane 1 through 15 registers is identical to that described for lane 0 in 45.2.5.17. The lane mapping for lane 1 is in register 5.401; lane 2 is in register 5.402; etc.

**Table 45–352—DTE XS lane mapping, lane 0 register bit definitions**

| Bit(s)     | Name           | Description                                      | R/W <sup>a</sup> |
|------------|----------------|--------------------------------------------------|------------------|
| 5.400.15:5 | Reserved       | Value always 0                                   | RO               |
| 5.400.4:0  | Lane 0 mapping | DTE XS lane received in service interface lane 0 | RO               |

<sup>a</sup>RO = Read only

#### 45.2.5.19 DTE XS FEC symbol error counter lane 0 (Register 5.600, 5.601)

The assignment of bits in the DTE XS FEC symbol error counter lane 0 register is shown in Table 45–353. See 119.3.4 for a definition of this counter. Symbol errors detected in DTE XS FEC lane 0 are counted and shown in register 5.600.15:0 and 5.601.15:0. These bits shall be reset to all zeros when the register is read by the management function or upon PHY reset. These bits shall be held at all ones in the case of overflow. Registers 5.600, 5.601 are used to read the value of a 32-bit counter. When registers 5.600 and 5.601 are used to read the 32-bit counter value, the register 5.600 is read first, the value of the register 5.601 is latched when (and only when) register 5.600 is read, and reads of register 5.601 return the latched value rather than the current value of the counter.

**Table 45–353—DTE XS FEC symbol error counter lane 0 register bit definitions**

| Bit(s)     | Name                                   | Description                       | R/W <sup>a</sup> |
|------------|----------------------------------------|-----------------------------------|------------------|
| 5.600.15:0 | DTE XS FEC symbol errors, lane 0 lower | FEC_symbol_error_counter_0[15:0]  | RO, NR           |
| 5.601.15:0 | DTE XS FEC symbol errors, lane 0 upper | FEC_symbol_error_counter_0[31:16] | RO, NR           |

<sup>a</sup>RO = Read only, NR = Non Roll-over

#### 45.2.5.20 DTE XS FEC symbol error counter lane 1 through 15 (Registers 5.602 through 5.631)

The behavior of the DTE XS FEC symbol error counters, lane 1 through 15 is identical to that described for DTE XS FEC lane 0 in 45.2.5.19. Errors detected in each DTE XS FEC lane are counted and shown in the corresponding register. DTE XS FEC lane 1, lower 16 bits are shown in register 5.602; DTE XS FEC lane 1, upper 16 bits are shown in register 5.603; DTE XS FEC lane 2, lower 16 bits are shown in register 5.604; through register 5.631 for DTE XS FEC lane 15, upper 16 bits.

#### 45.2.5.21 DTE XS FEC control register (Register 5.800)

The assignment of bits in the DTE XS FEC control register is shown in Table 45–354.

**Table 45–354—DTE XS FEC control register bit definitions**

| Bit(s)     | Name                                | Description                                                                          | R/W <sup>a</sup> |
|------------|-------------------------------------|--------------------------------------------------------------------------------------|------------------|
| 5.800.15:2 | Reserved                            | Value always 0                                                                       | RO               |
| 5.800.2    | DTE XS FEC degraded SER enable      | 1 = FEC decoder signals degraded SER<br>0 = FEC decoder does not signal degraded SER | R/W              |
| 5.800.1    | DTE XS FEC bypass indication enable | 1 = FEC decoder does not indicate errors<br>0 = FEC decoder indicates errors         | R/W              |
| 5.800.0    | Reserved                            | Value always 0                                                                       | RO               |

<sup>a</sup>R/W = Read/Write, RO = Read only

##### 45.2.5.21.1 DTE XS FEC degraded SER enable (5.800.2)

This bit enables the DTE XS FEC decoder to signal the presence of a degraded SER. When set to a one, this bit enables degraded SER signaling. When set to a zero, degraded SER signaling is disabled. Writes to this bit are ignored and reads return a zero if the DTE XS FEC does not have the ability to signal the presence of a degraded SER (see 119.2.5.3 for equivalent PCS behavior).

##### 45.2.5.21.2 DTE XS FEC bypass indication enable (5.800.1)

This bit enables the DTE XS FEC decoder to bypass error indication to the upper layers through the sync bits. When set to a one, this bit enables bypass of the error indication. When set to a zero, errors are indicated to the upper layers through the sync bits. Writes to this bit are ignored and reads return a zero if the DTE XS FEC does not have the ability to bypass indicating decoding errors (see 119.2.5.3 for equivalent PCS behavior).

#### 45.2.5.22 DTE XS FEC status register (Register 5.801)

The assignment of bits in the DTE XS FEC status register is shown in Table 45–355.

**Table 45–355—DTE XS FEC status register bit definitions**

| Bit(s)     | Name                         | Description                                                             | R/W <sup>a</sup> |
|------------|------------------------------|-------------------------------------------------------------------------|------------------|
| 5.801.15:7 | Reserved                     | Value always 0                                                          | RO               |
| 5.801.6    | Local degraded SER received  | 1 = local degraded SER received<br>0 = no local degraded SER received   | RO               |
| 5.801.5    | Remote degraded SER received | 1 = remote degraded SER received<br>0 = no remote degraded SER received | RO               |
| 5.801.4    | DTE XS FEC degraded SER      | 1 = SER is degraded<br>0 = SER is not degraded                          | RO               |

**Table 45–355—DTE XS FEC status register bit definitions (*continued*)**

| Bit(s)  | Name                                 | Description                                                                                                                                                    | R/W <sup>a</sup> |
|---------|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 5.801.3 | DTE XS FEC degraded SER ability      | 1 = FEC decoder has the ability to signal the presence of a degraded SER<br>0 = FEC decoder does not have the ability to signal the presence of a degraded SER | RO               |
| 5.801.2 | DTE XS FEC high SER                  | 1 = FEC errors have exceeded threshold<br>0 = FEC errors have not exceeded threshold                                                                           | RO/LH            |
| 5.801.1 | DTE XS FEC bypass indication ability | 1 = FEC decoder has the ability to bypass error indication<br>0 = FEC decoder does not have the ability to bypass error indication                             | RO               |
| 5.801.0 | Reserved                             | Value always 0                                                                                                                                                 | RO               |

<sup>a</sup>RO = Read only, LH = Latching high

#### 45.2.5.22.1 Local degraded SER received (5.801.6)

When read as a one, bit 5.801.6 indicates that the local degraded SER signal has been received. This bit reflects the state of rx\_local\_degraded (see 118.2.1) for the DTE XS.

#### 45.2.5.22.2 Remote degraded SER received (5.801.5)

When read as a one, bit 5.801.5 indicates that the remote degraded SER signal has been received. This bit reflects the state of rx\_rm\_degraded for the DTE XS (see 119.2.6.2.2 for equivalent PCS behavior).

#### 45.2.5.22.3 DTE XS FEC degraded SER (5.801.4)

When DTE XS FEC degraded SER enable (bit 5.800.2) is set to one, bit 5.801.4 is set to one if the number of FEC symbol errors in a window of DTE XS FEC degraded SER interval (registers 5.810 and 5.811) codewords exceeds the DTE XS FEC degraded SER activate threshold (registers 5.806 and 5.807) and is cleared if the number of FEC symbol errors in the same window is below the DTE XS FEC degraded SER deactivate threshold (registers 5.808 and 5.809). If the number of FEC symbol errors in the window is between the two thresholds, then bit 5.801.4 remains in its previous state. The bit is set to zero if DTE XS FEC degraded SER enable (bit 5.800.2) is set to zero (see 119.2.5.3 for equivalent PCS behavior). The value of bit 5.801.4 is undefined if the value of the DTE XS FEC degraded SER activate threshold is less than the value of the DTE XS FEC degraded SER deactivate threshold.

#### 45.2.5.22.4 DTE XS FEC degraded SER ability (5.801.3)

The DTE XS FEC decoder may have the option to signal the presence of a degraded SER (see 119.2.5.3 for equivalent PCS behavior). This bit is set to one to indicate that the decoder has the ability to signal the presence of a degraded SER. The bit is set to zero if this ability is not supported.

#### 45.2.5.22.5 DTE XS FEC high SER (5.801.2)

When DTE XS FEC bypass indication enable (bit 5.800.1) is set to one, this bit is set to one if the number of FEC symbol errors in a window of 8192 codewords exceeds the threshold (see 119.2.5.3) and is set to zero otherwise. The bit is set to zero if DTE XS FEC bypass indication enable (bit 5.800.1) is set to zero. This bit shall be implemented with latching high behavior.

#### 45.2.5.22.6 DTE XS FEC bypass indication ability (5.801.1)

The DTE XS FEC decoder may have the option to perform error detection without error indication (see 119.2.5.3) to reduce the delay contributed by the FEC decoder. This bit is set to one to indicate that the decoder has this ability to bypass error indication. The bit is set to zero if this ability is not supported.

#### 45.2.5.23 DTE XS FEC corrected codewords counter (Register 5.802, 5.803)

The assignment of bits in the DTE XS FEC corrected codewords counter register is shown in Table 45–356. See 119.3.2 for a definition of this counter. These bits shall be reset to all zeros when the register is read by the management function or upon PHY reset. These bits shall be held at all ones in the case of overflow. Registers 5.802, 5.803 are used to read the value of a 32-bit counter. When registers 5.802 and 5.803 are used to read the 32-bit counter value, the register 5.802 is read first, the value of the register 5.803 is latched when (and only when) register 5.802 is read, and reads of register 5.803 return the latched value rather than the current value of the counter.

**Table 45–356—DTE XS FEC corrected codewords counter register bit definitions**

| Bit(s)     | Name                          | Description                     | R/W <sup>a</sup> |
|------------|-------------------------------|---------------------------------|------------------|
| 5.802.15:0 | FEC corrected codewords lower | FEC_corrected_cw_counter[15:0]  | RO, NR           |
| 5.803.15:0 | FEC corrected codewords upper | FEC_corrected_cw_counter[31:16] | RO, NR           |

<sup>a</sup>RO = Read only, NR = Non Roll-over

#### 45.2.5.24 DTE XS FEC uncorrected codewords counter (Register 5.804, 5.805)

The assignment of bits in the DTE XS FEC uncorrected codewords counter register is shown in Table 45–357. See 119.3.3 for a definition of this counter. These bits shall be reset to all zeros when the register is read by the management function or upon PHY reset. These bits shall be held at all ones in the case of overflow. Registers 5.804, 5.805 are used to read the value of a 32-bit counter. When registers 5.804 and 5.805 are used to read the 32-bit counter value, the register 5.804 is read first, the value of the register 5.805 is latched when (and only when) register 5.804 is read, and reads of register 5.805 return the latched value rather than the current value of the counter.

**Table 45–357—DTE XS FEC uncorrected codewords counter register bit definitions**

| Bit(s)     | Name                            | Description                       | R/W <sup>a</sup> |
|------------|---------------------------------|-----------------------------------|------------------|
| 5.804.15:0 | FEC uncorrected codewords lower | FEC_uncorrected_cw_counter[15:0]  | RO, NR           |
| 5.805.15:0 | FEC uncorrected codewords upper | FEC_uncorrected_cw_counter[31:16] | RO, NR           |

<sup>a</sup>RO = Read only, NR = Non Roll-over

#### 45.2.5.25 DTE XS FEC degraded SER activate threshold register (Register 5.806, 5.807)

The assignment of bits in the DTE XS FEC degraded SER activate threshold register is shown in Table 45–358. The value controls the threshold used to set the DTE XS FEC degraded SER bit (5.801.4) in an equivalent manner to that defined for the FEC\_degraded\_SER bit in 119.2.5.3.

**Table 45–358—DTE XS FEC degraded SER activate threshold register bit definitions**

| Bit(s)     | Name                                             | Description                                | R/W <sup>a</sup> |
|------------|--------------------------------------------------|--------------------------------------------|------------------|
| 5.806.15:0 | DTE XS FEC degraded SER activate threshold lower | FEC_degraded_SER_activate_threshold[15:0]  | R/W              |
| 5.807.15:0 | DTE XS FEC degraded SER activate threshold upper | FEC_degraded_SER_activate_threshold[31:16] | R/W              |

<sup>a</sup>R/W = Read/Write

#### 45.2.5.26 DTE XS FEC degraded SER deactivate threshold register (Register 5.808, 5.809)

The assignment of bits in the DTE XS FEC degraded SER deactivate threshold register is shown in Table 45–359. The value controls the threshold used to clear the DTE XS FEC degraded SER bit (5.801.4) in an equivalent manner to that defined for the FEC\_degraded\_SER bit in 119.2.5.3.

**Table 45–359—DTE XS FEC degraded SER deactivate threshold register bit definitions**

| Bit(s)     | Name                                               | Description                                  | R/W <sup>a</sup> |
|------------|----------------------------------------------------|----------------------------------------------|------------------|
| 5.808.15:0 | DTE XS FEC degraded SER deactivate threshold lower | FEC_degraded_SER_deactivate_threshold[15:0]  | R/W              |
| 5.809.15:0 | DTE XS FEC degraded SER deactivate threshold upper | FEC_degraded_SER_deactivate_threshold[31:16] | R/W              |

<sup>a</sup>R/W = Read/Write

#### 45.2.5.27 DTE XS FEC degraded SER interval register (Register 5.810, 5.811)

The assignment of bits in the DTE XS FEC degraded SER interval register is shown in Table 45–360. The value controls the interval used to set and clear the DTE XS FEC degraded SER bit (5.801.4) in an equivalent manner to that defined for the FEC\_degraded\_SER bit in 119.2.5.3.

**Table 45–360—DTE XS FEC degraded SER interval register bit definitions**

| Bit(s)     | Name                                   | Description                      | R/W <sup>a</sup> |
|------------|----------------------------------------|----------------------------------|------------------|
| 5.810.15:0 | DTE XS FEC degraded SER interval lower | FEC_degraded_SER_interval[15:0]  | R/W              |
| 5.811.15:0 | DTE XS FEC degraded SER interval upper | FEC_degraded_SER_interval[31:16] | R/W              |

<sup>a</sup>R/W = Read/Write

#### 45.2.5.28 TimeSync DTE XS capability (Register 5.1800)

The TimeSync DTE XS capability register (see Table 45–361) indicates the capability of the DTE XS to report the transmit and receive data delay, stored in registers 5.1801 through 5.1804 and 5.1805 through 5.1808, respectively.

**Table 45–361—TimeSync DTE XS capability**

| Bit(s)      | Name                              | Description                                                                                                                                                           | R/W <sup>a</sup> |
|-------------|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 5.1800.15:2 | Reserved                          | Value always 0                                                                                                                                                        | RO               |
| 5.1800.1    | TimeSync transmit path data delay | 1 = DTE XS provides information on transmit path data delay in registers 5.1801 through 5.1804<br>0 = DTE XS does not provide information on transmit path data delay | RO               |
| 5.1800.0    | TimeSync receive path data delay  | 1 = DTE XS provides information on receive path data delay in registers 5.1805 through 5.1808<br>0 = DTE XS does not provide information on receive path data delay   | RO               |

<sup>a</sup>RO = Read only

#### 45.2.5.29 TimeSync DTE XS transmit path data delay (Registers 5.1801, 5.1802, 5.1803, 5.1804)

The TimeSync DTE XS transmit path data delay register contains the maximum (registers 5.1801, 5.1802, see Table 45–362) and minimum (registers 5.1803, 5.1804, see Table 45–362) values of the transmit path data delay. The transmit path data delay is expressed in units of ns. The values contained in these registers are valid when the link is established, as indicated by bit 2 in register 1.1 (see 45.2.1.2.4).

**Table 45–362—TimeSync DTE XS transmit path data delay register**

| Bit(s)      | Name                                           | Description                 | R/W <sup>a</sup> |
|-------------|------------------------------------------------|-----------------------------|------------------|
| 5.1801.15:0 | Maximum DTE XS transmit path data delay, lower | DTE_XS_delay_TX_max [15:0]  | RO, MW           |
| 5.1802.15:0 | Maximum DTE XS transmit path data delay, upper | DTE_XS_delay_TX_max [31:16] | RO, MW           |
| 5.1803.15:0 | Minimum DTE XS transmit path data delay, lower | DTE_XS_delay_TX_min [15:0]  | RO, MW           |
| 5.1804.15:0 | Minimum DTE XS transmit path data delay, upper | DTE_XS_delay_TX_min [31:16] | RO, MW           |

<sup>a</sup>RO = Read only, MW = Multi-word

#### 45.2.5.30 TimeSync DTE XS receive path data delay (Registers 5.1805, 5.1806, 5.1807, 5.1808)

The TimeSync DTE XS receive path data delay register contains the maximum (registers 5.1805, 5.1806, see Table 45–363) and minimum (registers 5.1807, 5.1808, see Table 45–363) values of the receive path



data delay. The receive path data delay is expressed in units of ns. The values contained in these registers are valid when the link is established, as indicated by bit 2 in register 5.1 (see 45.2.5.2.7).

**Table 45–363—TimeSync DTE XS receive path data delay register**

| Bit(s)      | Name                                          | Description                 | R/W <sup>a</sup> |
|-------------|-----------------------------------------------|-----------------------------|------------------|
| 5.1805.15:0 | Maximum DTE XS receive path data delay, lower | DTE_XS_delay_RX_max [15:0]  | RO, MW           |
| 5.1806.15:0 | Maximum DTE XS receive path data delay, upper | DTE_XS_delay_RX_max [31:16] | RO, MW           |
| 5.1807.15:0 | Minimum DTE XS receive path data delay, lower | DTE_XS_delay_RX_min [15:0]  | RO, MW           |
| 5.1808.15:0 | Minimum DTE XS receive path data delay, upper | DTE_XS_delay_RX_min [31:16] | RO, MW           |

<sup>a</sup>RO = Read only, MW = Multi-word

## 45.2.6 TC registers

The assignment of registers in the TC MMD is shown in Table 45–364.

**Table 45–364—TC registers**

| Register address | Register name                                          | Subclause |
|------------------|--------------------------------------------------------|-----------|
| 6.0              | TC control                                             | 45.2.6.1  |
| 6.1              | Reserved                                               |           |
| 6.2, 6.3         | TC device identifier                                   | 45.2.6.2  |
| 6.4              | TC speed ability                                       | 45.2.6.3  |
| 6.5, 6.6         | TC devices in package                                  | 45.2.6.4  |
| 6.7 through 6.13 | Reserved                                               |           |
| 6.14, 6.15       | TC package identifier                                  | 45.2.6.5  |
| 6.16             | 10P/2B aggregation discovery control <sup>a</sup>      | 45.2.6.6  |
| 6.17             | 10P/2B aggregation and discovery status <sup>a</sup>   | 45.2.6.7  |
| 6.18, 6.19, 6.20 | 10P/2B aggregation discovery code <sup>a</sup>         | 45.2.6.8  |
| 6.21             | 10P/2B link partner PME aggregate control <sup>a</sup> | 45.2.6.9  |
| 6.22, 6.23       | 10P/2B link partner PME aggregate data <sup>a</sup>    | 45.2.6.10 |
| 6.24             | 10P/2B TC CRC error counter                            | 45.2.6.11 |
| 6.25, 6.26       | 10P/2B TPS-TC coding violations counter                | 45.2.6.12 |
| 6.27             | 10P/2B TC indications                                  | 45.2.6.13 |

**Table 45–364—TC registers (*continued*)**

| Register address          | Register name                        | Subclause |
|---------------------------|--------------------------------------|-----------|
| 6.28 through 6.1799       | Reserved                             |           |
| 6.1800                    | TimeSync TC capability               | 45.2.6.14 |
| 6.1801 through 6.1804     | TimeSync TC transmit path data delay | 45.2.6.15 |
| 6.1805 through 6.1808     | TimeSync TC receive path data delay  | 45.2.6.16 |
| 6.1809 through 6.32767    | Reserved                             |           |
| 6.32 768 through 6.65 535 | Vendor specific                      |           |

<sup>a</sup>Register is defined only for -O port types and is reserved for -R ports

#### 45.2.6.1 TC control register (Register 6.0)

The assignment of bits in the TC control register is shown in Table 45–365. The default value for each bit of the TC control register has been chosen so that the initial state of the device upon power up or completion of reset is a normal operational state without management intervention.

**Table 45–365—TC control register bit definitions**

| Bit(s)   | Name            | Description                                                                                                                                                                                     | R/W <sup>a</sup> |
|----------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 6.0.15   | Reset           | 1 = TC reset<br>0 = Normal operation                                                                                                                                                            | R/W<br>SC        |
| 6.0.14   | Reserved        | Value always 0                                                                                                                                                                                  | RO               |
| 6.0.13   | Speed selection | <div> <div>13 6</div> <div>1 1 = bits 5:2 select speed</div> <div>0 x = unspecified</div> <div>x 0 = unspecified</div> </div>                                                                   | R/W              |
| 6.0.12:7 | Reserved        | Value always 0                                                                                                                                                                                  | RO               |
| 6.0.6    | Speed selection | <div> <div>13 6</div> <div>1 1 = bits 5:2 select speed</div> <div>0 x = unspecified</div> <div>x 0 = unspecified</div> </div>                                                                   | R/W              |
| 6.0.5:2  | Speed selection | <div> <div>5 4 3 2</div> <div>1 x x x = Reserved</div> <div>x 1 x x = Reserved</div> <div>x x 1 x = Reserved</div> <div>0 0 0 1 = 10PASS-TS/2BASE-TL</div> <div>0 0 0 0 = Reserved</div> </div> | R/W              |
| 6.0.1:0  | Reserved        | Value always 0                                                                                                                                                                                  | RO               |

<sup>a</sup>R/W = Read/Write, SC = Self-clearing, RO = Read only

#### 45.2.6.1.1 Reset (6.0.15)

Resetting a TC is accomplished by setting bit 6.0.15 to a one. This action shall set all TC registers to their default states. As a consequence, this action may change the internal state of the TC and the state of the physical link. This action may also initiate a reset in any other MMDs that are instantiated in the same package. This bit is self-clearing, and a TC shall return a value of one in bit 6.0.15 when a reset is in progress; otherwise, it shall return a value of zero. A TC is not required to accept a write transaction to any of its registers until the reset process is completed. The control and management interface shall be restored to operation within 0.5 s from the setting of bit 6.0.15. During a reset, a TC shall respond to reads from register bit 6.0.15.

NOTE—This operation may interrupt data communication. The data path of a TC, depending on type and temperature, may take many seconds to run at optimum error ratio after exiting from reset.

#### 45.2.6.1.2 Speed selection (6.0.13, 6.0.6, 6.0.5:2)

Speed selection bits 6.0.13 and 6.0.6 shall both be written as a one. Any attempt to change the bits to an invalid setting shall be ignored. These two bits are set to one in order to make them compatible with Clause 22.

The operating mode of the TC may be selected using bits 5 through 2. The abilities of the TC are advertised in the TC speed ability register. A TC may ignore writes to the TC speed selection bits that select speeds it has not advertised in the TC speed ability register. It is the responsibility of the STA entity to ensure that mutually acceptable speeds are applied consistently across all the MMDs on a particular PHY.

The speed selection bits 6.0.5:2, when set to 0001, select the use of the 10PASS-TS and 2BASE-TL TC.

The TC speed selection defaults to a supported ability.

#### 45.2.6.2 TC device identifier (Registers 6.2 and 6.3)

Registers 6.2 and 6.3 provide a 32-bit value, which may constitute a unique identifier for a particular type of TC. The identifier shall be composed of the 3rd through 24th bits of the Organizationally Unique Identifier (OUI) assigned to the device manufacturer by the IEEE, plus a six-bit model number, plus a four-bit revision number. A TC may return a value of zero in each of the 32 bits of the TC device identifier.

The format of the TC device identifier is specified in 22.2.4.3.1.

#### 45.2.6.3 TC speed ability (Register 6.4)

The assignment of bits in the TC speed ability register is shown in Table 45–366.

**Table 45–366—TC speed ability register bit definitions**

| Bit(s)   | Name                       | Description                                                                                            | R/W <sup>a</sup> |
|----------|----------------------------|--------------------------------------------------------------------------------------------------------|------------------|
| 6.4.15:1 | Reserved for future speeds | Value always 0                                                                                         | RO               |
| 6.4.1    | 10PASS-TS/2BASE-TL capable | 1 = TC is capable of operating as the 10P/2B TC<br>0 = TC is not capable of operating as the 10P/2B TC | RO               |
| 6.4.0    | Reserved                   | Value always 0                                                                                         | RO               |

<sup>a</sup>RO = Read only

#### 45.2.6.3.1 10PASS-TS/2BASE-TL capable (6.4.1)

When read as a one, this bit indicates that the TC is able to operate as the 10PASS-TS/2BASE-TL TC, as specified in Clause 61.

#### 45.2.6.4 TC devices in package registers (Registers 6.5, 6.6)

The TC devices in package registers are defined in Table 45–2.

#### 45.2.6.5 TC package identifier registers (Registers 6.14, 6.15)

Registers 6.14 and 6.15 provide a 32-bit value, which may constitute a unique identifier for a particular type of package that the TC MMD is instantiated within. The identifier shall be composed of the 3rd through 24th bits of the Organizationally Unique Identifier (OUI) assigned to the package manufacturer by the IEEE, plus a six-bit model number, plus a four-bit revision number. A TC may return a value of zero in each of the 32 bits of the package identifier.

A non-zero package identifier may be returned by one or more MMDs in the same package. The package identifier may be the same as the device identifier.

The format of the package identifier is specified in 22.2.4.3.1.

#### 45.2.6.6 10P/2B aggregation discovery control register (Register 6.16)

The 10P/2B aggregation discovery control register allows the STA of an -O port to determine the aggregation capabilities of an -R link-partner.

The 10P/2B aggregation discovery control register shall be implemented as a unique register for each TC MMD in a package. For example, a package implementing four PHYs would have four independent instances of the 10P/2B aggregation discovery control register, accessed by a read or write to each PHY.

For information on the encoding of this function on the physical link, see 61.4.7.

This register is defined for -O port subtypes only. The register bit definitions for the 10P/2B aggregation discovery control register are shown in Table 45–367.

**Table 45–367—10P/2B aggregation discovery control register bit definitions**

| Bit(s)    | Name                | Description                                                                 | R/W <sup>a</sup> |
|-----------|---------------------|-----------------------------------------------------------------------------|------------------|
| 6.16.15:2 | Reserved            | Value always 0                                                              | RO               |
| 6.16.1:0  | Discovery operation | 01 = Ready (default)<br>00 = Set if clear<br>11 = Clear if same<br>10 = Get | R/W              |

<sup>a</sup>RO = Read only, R/W = Read/Write

##### 45.2.6.6.1 Discovery operation (6.16.1:0)

The Discovery operation bits are used to query and manipulate the remote discovery register. The remote discover register is not a Clause 45 object, but a variable of the PME aggregation PCS function on -R ports. The Discovery operation makes use of G.994.1 handshaking messages, therefore valid only when the link

status is down (i.e., neither Initializing nor Up). Attempts to perform an operation while the link is Initializing or Up shall be ignored.

The default state of these bits is “Ready”. The bits shall indicate “Ready” any time the PME aggregation function is capable of performing an operation on the remote discovery register. If PAF is not supported, the discovery operation bits shall indicate “Ready” and ignore writes. These bits shall return to the “Ready” state upon MMD Reset.

If the STA sets the bits to “Get,” the PME aggregation function queries the remote discovery register and returns its contents to the aggregation discovery code register.

If the STA sets the bits to “Set if clear,” the PME aggregation function passes a message to the -R PCS instructing it to set the remote discovery register to the contents of the aggregation discovery code register, but only if the remote discovery register is all zeros.

If the STA sets the bits to “Clear if same,” the PME aggregation function passes a message to the -R PCS instructing it to clear the remote discovery register, but only if the contents of the remote discovery register currently match the contents of the aggregation discovery code register.

While the requested operation is in progress, the PHY maintains the operation value in the bits. After the operation is complete, the PHY shall set the bits to indicate “Ready”. If the operation does not complete within a 255 second time-out, the discovery operation result bit (6.17.0) will be set to “1” (operation unsuccessful), and the discovery operation bits will be set to “Ready”.

#### 45.2.6.7 10P/2B aggregation and discovery status register (Register 6.17)

The 10P/2B aggregation and discovery status register is defined for -O port subtypes only.

The assignment of bits in the 10P/2B aggregation and discovery status register is shown in Table 45–368.

**Table 45–368—10P/2B aggregation and discovery status register bit definitions**

| Bit(s)    | Name                                    | Description                                                                            | R/W <sup>a</sup> |
|-----------|-----------------------------------------|----------------------------------------------------------------------------------------|------------------|
| 6.17.15:2 | Reserved                                | Value always 0                                                                         | RO               |
| 6.17.1    | Link partner aggregate operation result | 1 = operation unsuccessful<br>0 = operation completed successfully (default)           | RO, LH           |
| 6.17.0    | Discovery operation result              | 1 = operation unsuccessful<br>0 = discovery operation completed successfully (default) | RO, LH           |

<sup>a</sup>RO = Read only, LH = Latching high

##### 45.2.6.7.1 Link partner aggregate operation result (6.17.1)

When a link partner aggregate operation is complete, the PHY sets this bit to indicate the result of the operation. A “1” indicates that the operation could not be completed. This may be for a variety of reasons:

- PMA/PMD link status is initializing or up.
- The link partner is not present or not responding.

If PAF is not supported, this bit shall remain set to zero.

#### 45.2.6.7.2 Discovery operation result (6.17.0)

When a discovery operation is complete, the PHY sets this bit to indicate the result of the operation. A “1” indicates that the operation could not be completed. This may be for a variety of reasons:

- a) PMA/PMD link status is initializing or up.
- b) A “Set if clear” operation was requested but the remote discovery register was not clear.
- c) A “Clear if same” operation was requested but the remote discovery register did not match the aggregation discovery code register.
- d) The link partner is not present or not responding.

If PAF is not supported, this bit shall read as zero.

#### 45.2.6.8 10P/2B aggregation discovery code (Registers 6.18, 6.19, 6.20)

The 10P/2B aggregation discovery code registers store the value of the remote\_discovery\_register exchanged with the -R link partner.

This register is defined for -O port subtypes only.

These registers shall be implemented as unique registers for each TC MMD in a package. For example, a package implementing four TCs would have four independent instances of the 10P/2B aggregation discovery code registers, accessed by a read or write to each PHY.

For information on the encoding of this function on the physical link, please see 61.4.7.

The assignment of bits for the 10P/2B aggregation discovery code registers are shown in Table 45–369.

**Table 45–369—10P/2B aggregation discovery code bit definitions**

| Bit(s)    | Name         | Description                                                        | R/W <sup>a</sup> |
|-----------|--------------|--------------------------------------------------------------------|------------------|
| 6.18.15:0 | Code [47:32] | The two most significant octets of the aggregation discovery code  | R/W              |
| 6.19.15:0 | Code [31:16] | The two middle octets of the aggregation discovery code            | R/W              |
| 6.20.15:0 | Code [15:0]  | The two least significant octets of the aggregation discovery code | R/W              |

<sup>a</sup>R/W = Read/Write

#### 45.2.6.9 10P/2B link partner PME aggregate control register (Register 6.21)

The 10P/2B link partner PME aggregate control register allows the STA of an -O port to read and write the remote PME\_Aggregate\_register (see 61.2.2.8.3).

The 10P/2B link partner PME aggregate control register shall be implemented as a unique register for each TC MMD in a package. For example, a package implementing four TCs would have four independent instances of the 10P/2B link partner PME aggregate control register, accessed by a read or write to each PHY.

This register is defined for -O port subtypes only.

The register bit definitions for the 10P/2B link partner PME aggregate control register are shown in Table 45–370.

**Table 45–370—10P/2B link partner PME aggregate control register bit definitions**

| Bit(s)    | Name                             | Description                                                  | R/W <sup>a</sup> |
|-----------|----------------------------------|--------------------------------------------------------------|------------------|
| 6.21.15:2 | Reserved                         | Value always 0                                               | RO               |
| 6.21.1:0  | Link partner aggregate operation | 01 = Ready (default)<br>00 = Set<br>11 = invalid<br>10 = Get | R/W              |

<sup>a</sup>RO = Read only, R/W = Read/Write

#### 45.2.6.9.1 Link partner aggregate operation (6.21.1:0)

The Link partner aggregate operation bits are used to query and manipulate the remote PME Aggregate register. This operation makes use of G.994.1 handshaking messages and therefore is performed only when the link status is down (i.e., neither Initializing nor Up). Attempts to perform an operation while the link is Initializing or Up shall be ignored.

The default state of these bits is “Ready.” The bits shall indicate “Ready” any time the PME aggregation function is capable of performing an operation on the remote PME\_Aggregate\_register. If PAF is not supported, the link partner aggregate operation bits shall indicate “Ready” ignore writes. These bits shall return to the “Ready” state upon MMD Reset.

If the STA sets the bits to “Get,” the PME aggregation function queries the remote PME\_Aggregate\_register and returns its contents to the 10P/2B link partner PME aggregate data register (see 45.2.6.10).

If the STA sets the bits to “Set,” the PME aggregation function passes a message to the -R PCS instructing it to set the bit location in the remote PME\_Aggregate\_register corresponding to the TC on which the message was received to the contents of bit 0 of the 10P/2B link partner PME aggregate data register.

While the requested operation is in progress, the PHY maintains the operation value in the bits. After the operation is complete, the PHY shall set the bits to indicate “Ready”.

#### 45.2.6.10 10P/2B link partner PME aggregate data (Registers 6.22, 6.23)

The 10P/2B link partner PME aggregate data registers store the data for the link partner aggregate operation. This register either contains the result of a “Get” operation, the data sent in a “Set” operation, or all zeros following an MMD reset.

These registers are defined for -O port subtypes only.

These registers shall be implemented as unique registers for each TC MMD in a package. For example, a package implementing four TCs would have four independent instances of the registers, accessed by a read or write to each PHY.

The assignment of bits for the 10P/2B link partner PME aggregate data registers are shown in Table 45–371.

**Table 45–371—10P/2B link partner PME aggregate data registers bit definitions**

| Bit(s)    | Name        | Description                                                             | R/W <sup>a</sup> |
|-----------|-------------|-------------------------------------------------------------------------|------------------|
| 6.22.15:0 | Data[31:16] | The two most significant octets of the link partner PME aggregate data  | R/W              |
| 6.23.15:0 | Data[15:0]  | The two least significant octets of the link partner PME aggregate data | R/W              |

<sup>a</sup>R/W = Read/Write

#### 45.2.6.11 10P/2B TC CRC error register (Register 6.24)

The 10P/2B TC CRC error register is a 16 bit counter that contains the number of TC frames received with the TC\_CRC\_error primitive asserted, defined in 61.2.3. These bits shall be reset to all zeros when the register is read by the management function or upon execution of the MMD reset. These bits shall be held at all ones in the case of overflow. The assignment of bits in the 10P/2B TC CRC error register are shown in Table 45–372.

**Table 45–372—10P/2B TC CRC error register bit definitions**

| Bit(s)    | Name             | Description              | R/W <sup>a</sup> |
|-----------|------------------|--------------------------|------------------|
| 6.24.15:0 | CRC errors[15:0] | The bytes of the counter | RO, NR           |

<sup>a</sup>RO = Read only, NR = Non Roll-over

#### 45.2.6.12 10P/2B TPS-TC coding violations counter (Registers 6.25, 6.26)

The 10P/2B TPS-TC coding violations counter is a 32-bit counter that contains the number of 64/64-octet encapsulation errors, defined in 61.3.3.1. This counter increments for each 64/65-octet received with the TC\_coding\_error signal asserted. These bits shall be reset to all zeros when the register is read by the management function or upon execution of the MMD reset. These bits shall be held at all ones in the case of overflow. The assignment of bits in the 10P/2B TPS-TC coding violations counter are shown in Table 45–373.

**Table 45–373—10P/2B TPS-TC coding violations counter bit definitions**

| Bit(s)    | Name                     | Description                         | R/W <sup>a</sup> |
|-----------|--------------------------|-------------------------------------|------------------|
| 6.25.15:0 | Coding violations[31:16] | The high order bytes of the counter | RO, MW           |
| 6.26.15:0 | Coding violations[15:0]  | The low order bytes of the counter  | RO, MW           |

<sup>a</sup>RO = Read only, MW = Multi-word

#### 45.2.6.13 10P/2B TC indications register (Register 6.27)

The 10P/2B TC indications register reflects the state of the TC sync detect state diagram and the state of the link partner TC sync detect state diagram (if present) (see 61.3.3.5). The assignment of bits in the 10P/2B TC indications register is shown in Table 45–374.

**Table 45–374—10P/2B TC indications register bit definitions**

| Bit(s)    | Name                   | Description                                                             | R/W <sup>a</sup> |
|-----------|------------------------|-------------------------------------------------------------------------|------------------|
| 6.27.15:9 | Reserved               | Value always 0                                                          | RO               |
| 6.27.8    | Local TC synchronized  | 1 = TC_synchronized is TRUE<br>0 = TC_synchronized is FALSE             | RO               |
| 6.27.7:1  | Reserved               | Value always 0                                                          | RO               |
| 6.27.0    | Remote TC synchronized | 1 = remote_TC_out_of_sync is FALSE<br>0 = remote_TC_out_of_sync is TRUE | RO               |

<sup>a</sup>RO = Read only



#### 45.2.6.13.1 Local TC synchronized (6.27.8)

This bit is read as a one when the TC\_synchronized variable in the TC sync detect state diagram is TRUE (see 61.3.3.8). In all other cases, this bit is read as zero.

#### 45.2.6.13.2 Remote TC synchronized (6.27.0)

This bit is read as a one when the remote\_TC\_out\_of\_sync variable in the link partner TC sync detect state diagram is FALSE (see 61.3.3.8). In all other cases, this bit is read as zero.

#### 45.2.6.14 TimeSync TC capability (Register 6.1800)

The TimeSync TC capability register (see Table 45–375) indicates the capability of the TC to report the transmit and receive data delay, stored in registers 6.1801 through 6.1804 and 6.1805 through 6.1808, respectively.

**Table 45–375—TimeSync TC capability**

| Bit(s)      | Name                              | Description                                                                                                                                                   | R/W <sup>a</sup> |
|-------------|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 6.1800.15:2 | Reserved                          | Value always 0                                                                                                                                                | RO               |
| 6.1800.1    | TimeSync transmit path data delay | 1 = TC provides information on transmit path data delay in registers 6.1801 through 6.1804<br>0 = TC does not provide information on transmit path data delay | RO               |
| 6.1800.0    | TimeSync receive path data delay  | 1 = TC provides information on receive path data delay in registers 6.1805 through 6.1808<br>0 = TC does not provide information on receive path data delay   | RO               |

<sup>a</sup>RO = Read only

#### 45.2.6.15 TimeSync TC transmit path data delay (Registers 6.1801, 6.1802, 6.1803, 6.1804)

The TimeSync TC transmit path data delay register contains the maximum (registers 6.1801, 6.1802, see Table 45–376) and minimum (registers 6.1803, 6.1804, see Table 45–376) values of the transmit path data delay. The transmit path data delay is expressed in units of ns. The values contained in these registers are valid when the link is established, as indicated by bit 2 in register 1.1 (see 45.2.1.2.4).

**Table 45–376—TimeSync TC transmit path data delay register**

| Bit(s)      | Name                                       | Description             | R/W <sup>a</sup> |
|-------------|--------------------------------------------|-------------------------|------------------|
| 6.1801.15:0 | Maximum TC transmit path data delay, lower | TC_delay_TX_max [15:0]  | RO, MW           |
| 6.1802.15:0 | Maximum TC transmit path data delay, upper | TC_delay_TX_max [31:16] | RO, MW           |
| 6.1803.15:0 | Minimum TC transmit path data delay, lower | TC_delay_TX_min [15:0]  | RO, MW           |
| 6.1804.15:0 | Minimum TC transmit path data delay, upper | TC_delay_TX_min [31:16] | RO, MW           |

<sup>a</sup>RO = Read only, MW = Multi-word

#### 45.2.6.16 TimeSync TC receive path data delay (Registers 6.1805, 6.1806, 6.1807, 6.1808)

The TimeSync TC receive path data delay register contains the maximum (registers 6.1805, 6.1806, see Table 45–377) and minimum (registers 6.1807, 6.1808, see Table 45–377) values of the receive path data delay. The receive path data delay is expressed in units of ns. The values contained in these registers are valid when the link is established, as indicated by bit 2 in register 1.1 (see 45.2.1.2.4).

**Table 45–377—TimeSync TC receive path data delay register**

| Bit(s)      | Name                                      | Description             | R/W <sup>a</sup> |
|-------------|-------------------------------------------|-------------------------|------------------|
| 6.1805.15:0 | Maximum TC receive path data delay, lower | TC_delay_RX_max [15:0]  | RO, MW           |
| 6.1806.15:0 | Maximum TC receive path data delay, upper | TC_delay_RX_max [31:16] | RO, MW           |
| 6.1807.15:0 | Minimum TC receive path data delay, lower | TC_delay_RX_min [15:0]  | RO, MW           |
| 6.1808.15:0 | Minimum TC receive path data delay, upper | TC_delay_RX_min [31:16] | RO, MW           |

<sup>a</sup>RO = Read only, MW = Multi-word

#### 45.2.7 Auto-Negotiation registers

The assignment of registers in the Auto-Negotiation (AN) MMD is shown in Table 45–378.

**Table 45–378—Auto-Negotiation MMD registers**

| Register address  | Register name                              | Subclause |
|-------------------|--------------------------------------------|-----------|
| 7.0               | AN control                                 | 45.2.7.1  |
| 7.1               | AN status                                  | 45.2.7.2  |
| 7.2, 7.3          | AN device identifier                       | 45.2.7.3  |
| 7.4               | Reserved                                   |           |
| 7.5, 7.6          | AN devices in package                      | 45.2.7.4  |
| 7.7 through 7.13  | Reserved                                   |           |
| 7.14, 7.15        | AN package identifier                      | 45.2.7.5  |
| 7.16 through 7.18 | AN advertisement                           | 45.2.7.6  |
| 7.19 through 7.21 | AN LP Base Page ability                    | 45.2.7.7  |
| 7.22 through 7.24 | AN XNP transmit                            | 45.2.7.8  |
| 7.25 through 7.27 | AN LP XNP ability                          | 45.2.7.9  |
| 7.28 through 7.31 | Reserved                                   |           |
| 7.32              | MultiGBASE-T AN control 1                  | 45.2.7.10 |
| 7.33              | MultiGBASE-T AN status 1                   | 45.2.7.11 |
| 7.34 through 7.47 | Reserved                                   |           |
| 7.48              | Backplane Ethernet, BASE-R copper status   | 45.2.7.12 |
| 7.49              | Backplane Ethernet, BASE-R copper status 2 | 45.2.7.13 |
| 7.50 through 7.59 | Reserved                                   |           |

**Table 45–378—Auto-Negotiation MMD registers (*continued*)**

| Register address        | Register name                   | Subclause |
|-------------------------|---------------------------------|-----------|
| 7.60                    | EEE advertisement 1             | 45.2.7.14 |
| 7.61                    | EEE link partner ability 1      | 45.2.7.15 |
| 7.62                    | EEE advertisement 2             | 45.2.7.16 |
| 7.63                    | EEE link partner ability 2      | 45.2.7.17 |
| 7.64                    | MultiGBASE-T AN control 2       | 45.2.7.18 |
| 7.65                    | MultiGBASE-T AN status 2        | 45.2.7.19 |
| 7.66 through 7.511      | Reserved                        |           |
| 7.512                   | BASE-T1 AN control              | 45.2.7.20 |
| 7.513                   | BASE-T1 AN status               | 45.2.7.21 |
| 7.514 through 7.516     | BASE-T1 AN advertisement        | 45.2.7.22 |
| 7.517 through 7.519     | BASE-T1 AN LP Base Page ability | 45.2.7.23 |
| 7.520 through 7.522     | BASE-T1 AN Next Page transmit   | 45.2.7.24 |
| 7.523 through 7.525     | BASE-T1 AN LP Next Page ability | 45.2.7.25 |
| 7.526                   | 10BASE-T1 AN control            | 45.2.7.26 |
| 7.527                   | 10BASE-T1 AN status             | 45.2.7.27 |
| 7.528 through 7.32767   | Reserved                        |           |
| 7.32768 through 7.65535 | Vendor specific                 |           |

#### 45.2.7.1 AN control register (Register 7.0)

The assignment of bits in the AN control register is shown in Table 45–379. The default value for each bit of the AN control register has been chosen so that the initial state of the device upon power up or completion of reset is a normal operational state without management intervention.

**Table 45–379—AN control register bit definitions**

| Bit(s)    | Name                       | Description                                                                 | R/W <sup>a</sup> |
|-----------|----------------------------|-----------------------------------------------------------------------------|------------------|
| 7.0.15    | AN reset                   | 1 = AN reset<br>0 = AN normal operation                                     | R/W<br>SC        |
| 7.0.14    | Reserved                   | Value always 0                                                              | RO               |
| 7.0.13    | Extended Next Page control | 1 = Extended Next Pages are enabled<br>0 = Extended Next Pages are disabled | R/W              |
| 7.0.12    | Auto-Negotiation enable    | 1 = enable Auto-Negotiation process<br>0 = disable Auto-Negotiation process | R/W              |
| 7.0.11:10 | Reserved                   | Value always 0                                                              | RO               |

**Table 45–379—AN control register bit definitions (*continued*)**

| Bit(s)  | Name                     | Description                                                                                         | R/W <sup>a</sup> |
|---------|--------------------------|-----------------------------------------------------------------------------------------------------|------------------|
| 7.0.9   | Restart Auto-Negotiation | 1 = Restart Auto-Negotiation process<br>0 = Auto-Negotiation in process, disabled, or not supported | R/W<br>SC        |
| 7.0.8:0 | Reserved                 | Value always 0                                                                                      | RO               |

<sup>a</sup>R/W = Read/Write, SC = Self-clearing, RO = Read only

A device that supports multiple port types may implement both Clause 22 control register operation and Clause 45 control register operation. Some control functions have been duplicated in both definitions. The register bits to control these functions are simply echoed in both locations; any reads or writes to these bits behave identically whether made through the Clause 22 location or the Clause 45 location.

#### 45.2.7.1.1 AN reset (7.0.15)

Resetting AN is accomplished by setting bit 7.0.15 to a one. This action shall set all AN registers to their default states. As a consequence, this action may change the internal state of AN and the state of the physical link. This action may also initiate a reset in any other MMDs that are instantiated in the same package. This bit is self-clearing, and AN shall return a value of one in bit 7.0.15 when a reset is in progress and a value of zero otherwise. AN is not required to accept a write transaction to any of its registers until the reset process is complete. The reset process shall be completed within 0.5 s from the setting of bit 7.0.15. During an AN reset, AN shall respond to reads from register bit 7.0.15. All other register bits should be ignored.

The default value for bit 7.0.15 is zero. See 22.2.4.1.1.

NOTE—This operation may interrupt data communication.

#### 45.2.7.1.2 Extended Next Page control (7.0.13)

When bit 7.0.13 is set to one, Extended Next Page(s) are exchanged if the device(s) is/are capable. If a device intends to enable the exchange of Extended Next Page, it shall set bit 7.0.13 to one. A device may choose not to exchange Extended Next Page by setting bit 7.0.13 to zero. The Extended Next Page function is defined in 28.2.3.4. Setting of this bit shall have no effect if Extended Next Page ability bit 7.16.12 is set to zero.

The default value for bit 7.0.13 is one if bit 7.16.12 is one. Otherwise, bit 7.0.13 defaults to zero.

#### 45.2.7.1.3 Auto-Negotiation enable (7.0.12)

The Auto-Negotiation function shall be enabled by setting bit 7.0.12 to a one. If bit 7.0.12 is set to one, then speed selection bits 1.0.13, 1.0.6, and bits 1.0.5:2 in PHY control register 1 and PHY type selection bits 1.7.3:0 in PHY control register 2 shall have no effect on the link configuration, and the Auto-Negotiation process determines the link configuration. If bit 7.0.12 is cleared to zero, then bits 1.0.13, 1.0.6, and bits 1.0.5:2 in control register 1 and type selection bit 1.7.3:0 in control register 2 determines the link configuration regardless of the prior state of the link configuration and the Auto-Negotiation process. Bit 7.0.12 is a copy of bit 0.12 in register 0 if present (see 22.2.4.1.4).

The default value of bit 7.0.12 is one, unless the PHY reports via bit 7.1.3 or 1.3 if present (see 22.2.4.2.12) that it lacks the ability to perform Auto-Negotiation, in which case the default value of bit 7.0.12 is zero (see 22.2.4.1.4).

#### 45.2.7.1.4 Restart Auto-Negotiation (7.0.9)

If the PMA/PMD reports (via bit 7.1.3) that it lacks the ability to perform Auto-Negotiation, or if Auto-Negotiation is disabled, the PMA/PMD shall return a value of zero in bit 7.0.9 and any attempt to write a one to bit 7.0.9 shall be ignored.

Otherwise, the Auto-Negotiation process shall be restarted by setting bit 7.0.9 to one. This bit is self-clearing, and a PMA/PMD shall return a value of one in bit 7.0.9 until the Auto-Negotiation process has been initiated. If Auto-Negotiation was completed prior to this bit being set, the process shall be reinitiated. The Auto-Negotiation process shall not be affected by clearing this bit to zero. Bit 7.0.9 is a copy of 0.9 in register 0, if present (see 22.2.4.1.7).

The default value for 7.0.9 is zero (see 22.2.4.1.7).

#### 45.2.7.2 AN status (Register 7.1)

The assignment of bits in the AN status register is shown in Table 45–380. All the bits in the AN status register are read only; therefore, a write to the AN status register shall have no effect.

**Table 45–380—AN status register**

| Bit(s)    | Name                                  | Description                                                                                                                                  | R/W <sup>a</sup> |
|-----------|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 7.1.15:10 | Reserved                              | Value always 0                                                                                                                               | RO               |
| 7.1.9     | Parallel detection fault              | 1 = A fault has been detected via the parallel detection function.<br>0 = A fault has not been detected via the parallel detection function. | RO<br>LH         |
| 7.1.8     | Reserved                              | Value always 0                                                                                                                               | RO               |
| 7.1.7     | Extended Next Page status             | 1 = Extended Next Page format is used<br>0 = Extended Next Page is not allowed                                                               | RO               |
| 7.1.6     | Page received                         | 1 = A page has been received<br>0 = A page has not been received                                                                             | RO<br>LH         |
| 7.1.5     | Auto-Negotiation complete             | 1 = Auto-Negotiation process completed<br>0 = Auto-Negotiation process not completed                                                         | RO               |
| 7.1.4     | Remote fault                          | 1 = remote fault condition detected<br>0 = no remote fault condition detected                                                                | RO<br>LH         |
| 7.1.3     | Auto-Negotiation ability              | 1 = PHY is able to perform Auto-Negotiation<br>0 = PHY is not able to perform Auto-Negotiation                                               | RO               |
| 7.1.2     | Link status                           | 1 = Link is up<br>0 = Link is down                                                                                                           | RO<br>LL         |
| 7.1.1     | Reserved                              | Value always 0                                                                                                                               | RO               |
| 7.1.0     | Link partner Auto-Negotiation ability | 1 = LP is able to perform Auto-Negotiation<br>0 = LP is not able to perform Auto-Negotiation                                                 | RO               |

<sup>a</sup>RO = Read only, LH = Latching high, LL = Latching low

##### 45.2.7.2.1 Parallel detection fault (7.1.9)

The parallel detection Fault bit (7.1.9) shall be set to one to indicate that more than one of 1000BASE-KX, 2.5GBASE-KX, or 10GBASE-KX4 PMAs have indicated link\_status=OK when the autoneg\_wait\_timer expires. The parallel detection fault bit shall be reset to zero on a read of the AN status register (register 7.1).

#### **45.2.7.2.2 Extended Next Page status (7.1.7)**

When set to one, bit 7.1.7 indicates that both the local device and the link partner have indicated support for Extended Next Page. When set to zero, bit 7.1.7 indicates that Extended Next Page shall not be used. If bit 7.0.13 or bit 7.16.12 are set to zero then this bit shall not be set to one.

#### **45.2.7.2.3 Page received (7.1.6)**

The Page received bit (7.1.6) shall be set to one to indicate that a new link codeword has been received and stored in the AN LP Base Page ability registers 7.19 to 7.21 or AN LP XNP ability registers 7.25 to 7.27. The contents of the AN LP Base Page ability registers 7.19 to 7.21 are valid when bit 7.1.6 is set the first time during the Auto-Negotiation. The Page received bit shall be reset to zero on a read of the AN status register (register 7.1) or if present, the Auto-Negotiation expansion register 6 (see 28.2.4.1.5). This bit is a copy of bit 6.1 in register 6, if present (see 28.2.4.1.5).

#### **45.2.7.2.4 Auto-Negotiation complete (7.1.5)**

When read as a one, bit 7.1.5 indicates that the Auto-Negotiation process has been completed, and that the contents of the Auto-Negotiation register 7.16 and 7.19 are valid. When read as a zero, bit 7.1.5 indicates that the Auto-Negotiation process has not been completed, and that the contents of 7.19, 7.22 through 7.27, and 7.33 registers are as defined by the current state of the Auto-Negotiation protocol, or as written for manual configuration. A PMA/PMD shall return a value of zero in bit 7.1.5 if Auto-Negotiation is disabled by clearing bit 7.0.12. A PMA/PMD shall also return a value of zero in bit 7.1.5 if it lacks the ability to perform Auto-Negotiation. Bit 7.1.5 is a copy of bit 1.5 in register 1, if present (see 22.2.4).

#### **45.2.7.2.5 Remote fault (7.1.4)**

When read as one, bit 7.1.4 indicates that a remote fault condition has been detected. The type of fault as well as the criteria and method of fault detection is AN specific. The remote fault bit shall be implemented with a latching function, such that the occurrence of a remote fault causes the bit 7.1.4 to become set and remain set until it is cleared. Bit 7.1.4 shall be cleared each time register 7.1 is read via the management interface, and shall also be cleared by a AN reset. Bit 7.1.4 is a copy of bit 1.4 in register 1, if present (see 22.2.4).

#### **45.2.7.2.6 Auto-Negotiation ability (7.1.3)**

When read as a one, bit 7.1.3 indicates that the PMA/PMD has the ability to perform Auto-Negotiation. When read as a zero, bit 7.1.3 indicates that the PMA/PMD lacks the ability to perform Auto-Negotiation. Bit 7.1.3 is a copy of bit 1.3 in register 1, if present (see 22.2.4).

#### **45.2.7.2.7 Link status (7.1.2)**

When read as a one, bit 7.1.2 indicates that the PMA/PMD has determined that a valid link has been established. When read as a zero, bit 7.1.2 indicates that the link has been invalid after this bit was last read. Bit 7.1.2 is set to one when the variable link\_status equals OK and is cleared to zero when the variable link\_status equals FAIL. The link status bit shall be implemented with a latching function, such that the occurrence of a link\_status equals FAIL condition causes the link status bit to become cleared and remain cleared until it is read via the management interface. Bit 7.1.2 shall be cleared upon AN reset. This status indication is intended to support the management attribute defined in 30.5.1.1.4, aMediaAvailable.

#### **45.2.7.2.8 Link partner Auto-Negotiation ability (7.1.0)**

The link partner Auto-Negotiation ability bit shall be set to one to indicate that the link partner is able to participate in the Auto-Negotiation function. This bit shall be reset to zero if the link partner is not Auto-Negotiation able.

#### **45.2.7.3 Auto-Negotiation device identifier (Registers 7.2 and 7.3)**

Registers 7.2 and 7.3 provide a 32-bit value, which may constitute a unique identifier for an Auto-Negotiation. The identifier shall be composed of the 3rd through 24th bits of the Organizationally Unique Identifier (OUI) assigned to the device manufacturer by the IEEE, plus a six-bit model number, plus a 4-bit revision number. An Auto-Negotiation may return a value of zero in each of the 32 bits of the Auto-Negotiation device identifier.

The format of the Auto-Negotiation device identifier is specified in 22.2.4.3.1.

#### **45.2.7.4 AN devices in package (Registers 7.5 and 7.6)**

The AN devices in package registers are defined in Table 45–2.

#### **45.2.7.5 AN package identifier (Registers 7.14 and 7.15)**

Registers 7.14 and 7.15 provide a 32-bit value, which may constitute a unique identifier for a particular type of package that the Auto-Negotiation is instantiated within. The identifier shall be composed of the 3rd through 24th bits of the OUI assigned to the package manufacturer by the IEEE, plus a six-bit model number, plus a four-bit revision number. An Auto-Negotiation MMD may return a value of zero in each of the 32 bits of the Auto-Negotiation package identifier.

A non-zero package identifier may be returned by one or more MMDs in the same package. The package identifier may be the same as the device identifier.

The format of the Auto-Negotiation package identifier is specified in 22.2.4.3.1.

#### **45.2.7.6 AN advertisement register (7.16, 7.17, and 7.18)**

If the BP AN ability bit (7.48.0) in the BP Ethernet status register is set to zero then only sixteen bits of the AN advertisement register are used and they are defined in 28.2.1.2.

If the BP AN ability bit (7.48.0) in the BP Ethernet status register is set to one then all forty eight bits are used and they are defined in 73.6.

The Selector field (7.16.4:0) is set to the IEEE 802.3 code as specified in Annex 28A. The Acknowledge bit (7.16.14) is set to zero.

The technology ability field, as defined in 28.2.1.2 and 73.6, represents the technologies supported by the local device. Only bits representing supported technologies may be set. Management may clear bits in the technology ability field and restart Auto-Negotiation to negotiate an alternate common mode.

The management entity initiates renegotiation with the link partner using alternate abilities by setting the Restart Auto-Negotiation bit (7.0.9) in the AN control register to one.

Any writes to this register prior to completion of Auto-Negotiation, as indicated by bit 7.1.5, should be followed by a renegotiation for the new values to take effect. Once Auto-Negotiation has completed, software may examine this register along with the LP Base Page ability register to determine the highest common denominator technology.

If the Auto-Negotiation advertisement register (register 4) is present, (see 28.2.4.1.3), then this register is a copy of the Auto-Negotiation advertisement register (register 4). In this case, reads to the AN advertisement register (7.16) report the value of the Auto-Negotiation advertisement register (register 4); writes to the AN advertisement register (7.16) cause a write to occur to the Auto-Negotiation advertisement register.

The Base Page value is transferred to *mr\_adv\_ability* when register 7.16 is written. Therefore, if used, registers 7.17 and 7.18 should be written before 7.16.

**Table 45–381—AN advertisement register bit definitions**

| Bit(s)    | Name           | Description                    | R/W <sup>a</sup> |
|-----------|----------------|--------------------------------|------------------|
| 7.16.15   | Next Page      | See 28.2.1.2 and 73.6.9        | R/W              |
| 7.16.14   | Acknowledge    | Value always 0, writes ignored | RO               |
| 7.16.13   | Remote fault   | See 28.2.1.2 and 73.6.7        | R/W              |
| 7.16.12:5 | D12:D5         | See 28.2.1.2 and 73.6          | R/W              |
| 7.16.4:0  | Selector field | See Annex 28A                  | R/W              |
| 7.17.15:0 | D31:D16        | See 73.6                       | R/W              |
| 7.18.15:0 | D47:D32        | See 73.6                       | R/W              |

<sup>a</sup>R/W = Read/Write, RO = Read only

NOTE 1—Auto-Negotiation that supports a 48-bit Base Page uses registers 7.16, 7.17, and 7.18 for storing Base Page information; Auto-Negotiation that supports a 16-bit Base Page only uses register 7.16 for storing Base Page information.

NOTE 2—Clause 37 1000BASE-X Auto-Negotiation is controlled through Clause 22 registers.

#### 45.2.7.7 AN LP Base Page ability register (7.19, 7.20, and 7.21)

All of the bits in the AN LP Base Page ability register are read only. A write to the AN LP Base Page ability register shall have no effect.

Register 7.19 is a copy of register 5, if present (see 28.2.4.1).

When registers 7.20 and 7.21 are used, the value of the registers 7.20 and 7.21 is latched when register 7.19 is read and reads of registers 7.20 and 7.21 return the latched value rather than the current value.

Bit 7.19.12 is reserved for Backplane Ethernet port types as they do not use Extended Next Pages.

**Table 45–382—AN LP Base Page ability register bit definitions**

| Bit(s)    | Name    | Description           | R/W <sup>a</sup> |
|-----------|---------|-----------------------|------------------|
| 7.19.15:0 | D15:D0  | See 28.2.1.2 and 73.6 | RO               |
| 7.20.15:0 | D31:D16 | See 73.6              | RO               |
| 7.21.15:0 | D47:D32 | See 73.6              | RO               |

<sup>a</sup>RO = Read only



NOTE—Auto-Negotiation that supports a 48-bit Base Page uses registers 7.19, 7.20, and 7.21 for storing LP Base Page information; Auto-Negotiation that supports a 16-bit Base Page only use register 7.19 for storing LP Base Page information.

#### 45.2.7.8 AN XNP transmit register (7.22, 7.23, and 7.24)

If the BP AN ability bit (7.48.0) in the BP Ethernet status register is set to zero then Extended Next Pages are enabled by setting bit 7.0.13 to one and the AN XNP transmit register contains the Next Page link codeword to be transmitted as defined in 28.2.3.4.

If the BP AN ability bit (7.48.0) is set to one then Next Page transmission is always enabled and the register contains the AN LD Next Page link codeword of the BP Ethernet PHY as defined in 73.7.7.1. Even though Backplane Ethernet does not use Extended Next Pages XNP is still used in the register name.

On power-up or AN reset, this register shall contain the default value, which represents a Message Page with the message code set to Null Message. This value may be replaced by any valid Extended Next Page message code that the device intends to transmit.

A write to register 7.23 or 7.24 does not set `mr_next_page_loaded`. Only a write to register 7.22 sets `mr_next_page_loaded` true as described in 28.2.4.1.8. Therefore registers 7.23 and 7.24 register should be written before register 7.22.

**Table 45–383—AN XNP transmit register bit definitions**

| Bit(s)    | Name                           | Description               | R/W <sup>a</sup> |
|-----------|--------------------------------|---------------------------|------------------|
| 7.22.15   | Next Page                      | See 28.2.3.4 and 73.7.7.1 | R/W              |
| 7.22.14   | Reserved                       | Value always 0            | RO               |
| 7.22.13   | Message Page                   | See 28.2.3.4 and 73.7.7.1 | R/W              |
| 7.22.12   | Acknowledge 2                  | See 28.2.3.4 and 73.7.7.1 | R/W              |
| 7.22.11   | Toggle                         | See 28.2.3.4 and 73.7.7.1 | RO               |
| 7.22.10:0 | Message/Unformatted Code Field | See 28.2.3.4 and 73.7.7.1 | R/W              |
| 7.23.15:0 | Unformatted Code Field 1       | See 28.2.3.4 and 73.7.7.1 | R/W              |
| 7.24.15:0 | Unformatted Code Field 2       | See 28.2.3.4 and 73.7.7.1 | R/W              |

<sup>a</sup>R/W = Read/Write, RO = Read only

#### 45.2.7.9 AN LP XNP ability register (7.25, 7.26, and 7.27)

AN LP XNP ability register (registers 7.25, 7.26, and 7.27) store link partner Extended Next Pages as shown in Table 45–384. All of the bits in the AN LP XNP ability register are read only. A write to the AN LP XNP ability register shall have no effect.

The value of registers 7.26 and 7.27 is latched when register 7.25 is read and reads of registers 7.26 and 7.27 return the latched value rather than the current value.

NOTE—If this register is used to store multiple link partner Extended Next Pages, the previous value of this register is assumed to be stored by a management entity that needs the information overwritten by subsequent link partner Extended Next Pages.

**Table 45–384—AN LP XNP ability register bit definitions**

| Bit(s)    | Name                           | Description               | R/W <sup>a</sup> |
|-----------|--------------------------------|---------------------------|------------------|
| 7.25.15   | Next Page                      | See 28.2.3.4 and 73.7.7.1 | RO               |
| 7.25.14   | Acknowledge                    | See 28.2.3.4 and 73.7.7.1 | RO               |
| 7.25.13   | Message Page                   | See 28.2.3.4 and 73.7.7.1 | RO               |
| 7.25.12   | Acknowledge 2                  | See 28.2.3.4 and 73.7.7.1 | RO               |
| 7.25.11   | Toggle                         | See 28.2.3.4 and 73.7.7.1 | RO               |
| 7.25.10:0 | Message/Unformatted Code Field | See 28.2.3.4 and 73.7.7.1 | RO               |
| 7.26.15:0 | Unformatted Code Field 1       | See 28.2.3.4 and 73.7.7.1 | RO               |
| 7.27.15:0 | Unformatted Code Field 2       | See 28.2.3.4 and 73.7.7.1 | RO               |

<sup>a</sup>RO = Read only

#### 45.2.7.10 MultiGBASE-T AN control 1 register (Register 7.32)

The assignment of bits in the MultiGBASE-T AN control 1 register is shown in Table 45–385. The default values for each bit of the MultiGBASE-T AN control 1 register should be chosen so that the initial state of the device upon power up or reset is a normal operational state without management intervention.

**Table 45–385—MultiGBASE-T AN control 1 register**

| Bit(s)  | Name                              | Description                                                                                                       | R/W <sup>a</sup> |
|---------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------|------------------|
| 7.32.15 | MASTER-SLAVE manual config enable | 1=Enable MASTER-SLAVE manual configuration<br>0=Disable MASTER-SLAVE manual configuration<br>Default value is 0   | R/W              |
| 7.32.14 | MASTER-SLAVE config value         | 1=Configure PHY as MASTER<br>0=Configure PHY as SLAVE<br>Default value is 0                                       | R/W              |
| 7.32.13 | Port type                         | 1=Multiport device<br>0=single-port device                                                                        | R/W              |
| 7.32.12 | 10GBASE-T ability                 | 1 = Advertise PHY as 10GBASE-T capable.<br>0 = Do not advertise the PHY as 10GBASE-T capable                      | R/W              |
| 7.32.11 | 40GBASE-T ability                 | 1 = Advertise PHY as 40GBASE-T capable<br>0 = Do not advertise the PHY as 40GBASE-T capable                       | R/W              |
| 7.32.10 | 25GBASE-T ability                 | 1 = Advertise PHY as 25GBASE-T capable<br>0 = Do not advertise the PHY as 25GBASE-T capable                       | R/W              |
| 7.32.9  | 25GBASE-T Fast retrain ability    | 1 = Advertise PHY as 25GBASE-T fast retrain capable<br>0 = Do not advertise PHY as 25GBASE-T fast retrain capable | R/W              |
| 7.32.8  | 5GBASE-T ability                  | 1 = Advertise PHY as 5GBASE-T capable<br>0 = Do not advertise PHY as 5GBASE-T capable                             | R/W              |
| 7.32.7  | 2.5GBASE-T ability                | 1 = Advertise PHY as 2.5GBASE-T capable<br>0 = Do not advertise PHY as 2.5GBASE-T capable                         | R/W              |

**Table 45–385—MultiGBASE-T AN control 1 register (*continued*)**

| Bit(s) | Name                                    | Description                                                                                                                                                                                                                                                                                                                                          | R/W <sup>a</sup> |
|--------|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 7.32.6 | 5GBASE-T Fast retrain ability           | 1 = Advertise PHY as 5GBASE-T fast retrain capable<br>0 = Do not advertise PHY as 5GBASE-T fast retrain capable                                                                                                                                                                                                                                      | R/W              |
| 7.32.5 | 2.5GBASE-T Fast retrain ability         | 1 = Advertise PHY as 2.5GBASE-T fast retrain capable<br>0 = Do not advertise PHY as 2.5GBASE-T fast retrain capable                                                                                                                                                                                                                                  | R/W              |
| 7.32.4 | Reserved                                | Value always 0                                                                                                                                                                                                                                                                                                                                       | RO               |
| 7.32.3 | 40GBASE-T Fast retrain ability          | 1 = Advertise PHY as 40GBASE-T fast retrain capable<br>0 = Do not advertise PHY as 40GBASE-T fast retrain capable                                                                                                                                                                                                                                    | R/W              |
| 7.32.2 | 10GBASE-T LD PMA training reset request | 1 = Local device requests that link partner reset PMA training PRBS every frame<br>0 = Local device requests that link partner run PMA training PRBS continuously<br>NOTE—the periodic training sequence request functionality is deprecated. Link partners may ignore a value of one in this bit. It is recommended to always set this bit to zero. | R/W              |
| 7.32.1 | 10GBASE-T Fast retrain ability          | 1 = Advertise PHY as 10GBASE-T fast retrain capable<br>0 = Do not advertise PHY as 10GBASE-T fast retrain capable                                                                                                                                                                                                                                    | R/W              |
| 7.32.0 | 10GBASE-T LD loop timing ability        | 1 = Advertise PHY as capable of 10GBASE-T loop timing<br>0 = Do not advertise PHY as capable of 10GBASE-T loop timing                                                                                                                                                                                                                                | R/W              |

<sup>a</sup>R/W = Read/Write, RO = Read only

NOTE—Annex K defines optional alternative terminology for “master” and “slave”.

#### 45.2.7.10.1 MASTER-SLAVE manual config enable (7.32.15)

If bit 7.32.15 is set to one then MASTER-SLAVE config value bit 7.32.14 is used to determine if the PHY operates as MASTER or SLAVE. If bit 7.32.15 is set to zero the PHY shall use Auto-Negotiation to determine the selection of MASTER or SLAVE. This bit is a copy of 9.12, if present (see 40.5.1.1), and any write to either 7.32.15 or 9.12 is reflected in both. Additional information regarding the resolution and selection of MASTER-SLAVE and bit 7.32.15 is contained in 55.6.2. The default value for bit 7.32.15 is zero.

#### 45.2.7.10.2 MASTER-SLAVE config value (7.32.14)

Bit 7.32.14 is used to select MASTER or SLAVE operation if MASTER-SLAVE manual config bit 7.32.15 is set to one. If bit 7.32.14 is set to one the PHY shall operate as MASTER. If bit 7.32.14 is set to zero the PHY shall operate as SLAVE. This bit is a copy of 9.11 (see 40.5.1.1), if present, and any write to either 7.32.14 or 9.11 is reflected in both. Additional information regarding the resolution and selection of MASTER-SLAVE and bit 7.32.14 is contained in 55.6.2. The default value for bit 7.32.14 is zero.

#### 45.2.7.10.3 Port type (7.32.13)

Bit 7.32.13 is to be used to indicate the preference during Auto-Negotiation to operate as MASTER (multiport device) or as SLAVE (single-port device) if the MASTER-SLAVE manual configuration enable bit 7.32.15 is zero. If bit 7.32.13 is set to one a preference to operate as MASTER is indicated. If bit 7.32.13 is set to zero a preference to operate as SLAVE is indicated. This bit is a copy of 9.10 (see 40.5.1.1), if present, and any write to either 7.32.13 or 9.10 is reflected in both. Subclause 55.6.2 contains additional information regarding the resolution and selection of MASTER-SLAVE and this bit.

#### **45.2.7.10.4 10GBASE-T capability (7.32.12)**

Bit 7.32.12 is to be used to select whether or not Auto-Negotiation advertises the ability to operate as a 10GBASE-T PHY. If bit 7.32.12 is set to one the PHY shall advertise 10GBASE-T PHY capability. If bit 7.32.12 is set to zero the PHY shall not advertise 10GBASE-T PHY capability.

#### **45.2.7.10.5 40GBASE-T capability (7.32.11)**

Bit 7.32.11 is used to select whether or not Auto-Negotiation advertises the ability to operate as a 40GBASE-T PHY. If bit 7.32.11 is set to one the PHY shall advertise 40GBASE-T PHY capability. If bit 7.32.11 is set to zero the PHY shall not advertise 40GBASE-T PHY capability.

#### **45.2.7.10.6 25GBASE-T capability (7.32.10)**

Bit 7.32.10 is used to select whether or not Auto-Negotiation advertises the ability to operate as a 25GBASE-T PHY. If bit 7.32.10 is set to one the PHY shall advertise 25GBASE-T PHY capability. If bit 7.32.10 is set to zero the PHY shall not advertise 25GBASE-T PHY capability.

#### **45.2.7.10.7 25GBASE-T Fast retrain ability (7.32.9)**

Bit 7.32.9 is used to select whether or not the 25GBASE-T PHY advertises the ability to support 25GBASE-T fast retrain. Fast retrain ability is exchanged during link training, see 113.4.2.5.10. If bit 7.32.9 is set to one, the PHY shall advertise fast retrain ability. If bit 7.32.9 is set to zero, the PHY shall not advertise fast retrain ability.

#### **45.2.7.10.8 5GBASE-T capability (7.32.8)**

Bit 7.32.8 is used to select whether or not Auto-Negotiation advertises the ability to operate as a 5GBASE-T PHY. If bit 7.32.8 is set to one the PHY shall advertise 5GBASE-T PHY capability. If bit 7.32.8 is set to zero the PHY shall not advertise 5GBASE-T PHY capability.

#### **45.2.7.10.9 2.5GBASE-T capability (7.32.7)**

Bit 7.32.7 is used to select whether or not Auto-Negotiation advertises the ability to operate as a 2.5GBASE-T PHY. If bit 7.32.7 is set to one the PHY shall advertise 2.5GBASE-T PHY capability. If bit 7.32.7 is set to zero the PHY shall not advertise 2.5GBASE-T PHY capability.

#### **45.2.7.10.10 5GBASE-T Fast retrain ability (7.32.6)**

Bit 7.32.6 is used to select whether or not the 5GBASE-T PHY advertises the ability to support 5GBASE-T fast retrain. Fast retrain ability is exchanged during link training, see 126.4.2.5.10. If bit 7.32.6 is set to one, the PHY shall advertise fast retrain ability. If bit 7.32.6 is set to zero, the PHY shall not advertise fast retrain ability.

#### **45.2.7.10.11 2.5GBASE-T Fast retrain ability (7.32.5)**

Bit 7.32.5 is used to select whether or not the 2.5GBASE-T PHY advertises the ability to support 2.5GBASE-T fast retrain. Fast retrain ability is exchanged during link training, see 126.4.2.5.10. If bit 7.32.5 is set to one, the PHY shall advertise fast retrain ability. If bit 7.32.5 is set to zero, the PHY shall not advertise fast retrain ability.

#### **45.2.7.10.12 40GBASE-T Fast retrain ability (7.32.3)**

Bit 7.32.3 is used to select whether or not the 40GBASE-T PHY advertises the ability to support 40GBASE-T fast retrain. Fast retrain ability is exchanged during link training, see 113.4.2.5.10. If bit 7.32.3 is set to one, the PHY shall advertise fast retrain ability. If bit 7.32.3 is set to zero, the PHY shall not advertise fast retrain ability.

#### **45.2.7.10.13 10GBASE-T LD PMA training reset request (7.32.2)**

For 10GBASE-T, if bit 7.32.2 is set to one then the local device expects the link partner to reset the PMA training PRBS for every PMA training frame. If bit 7.32.2 is zero then the local device expects link partner to run PMA training PRBS continuously through every PMA training frame.

The periodic training sequence request functionality is deprecated and may be unsupported by some implementations. The link partner may ignore a request caused by setting this bit to one. It is recommended to always set this bit to zero.

#### **45.2.7.10.14 10GBASE-T Fast retrain ability (7.32.1)**

Bit 7.32.1 is used to select whether or not Auto-Negotiation advertises the ability to support 10GBASE-T fast retrain. If bit 7.32.1 is set to one, the PHY shall advertise fast retrain ability. If bit 7.32.1 is set to zero, the PHY shall not advertise fast retrain ability.

#### **45.2.7.10.15 10GBASE-T LD loop timing ability (7.32.0)**

Bit 7.32.0 is to be used to select whether or not Auto-Negotiation advertises the ability to perform loop timing. If bit 7.32.0 is set to one the PHY shall advertise loop timing capability. If bit 7.32.0 is set to zero the PHY shall not advertise loop timing capability.

#### **45.2.7.11 MultiGBASE-T AN status 1 register (Register 7.33)**

The assignment of bits in the MultiGBASE-T AN status 1 register is shown in Table 45–386. All the bits in the MultiGBASE-T AN status 1 register are read only; a write shall have no effect.

##### **45.2.7.11.1 MASTER-SLAVE configuration fault (7.33.15)**

MASTER-SLAVE configuration fault bit 7.33.15 shall be set in the event that determination of the MASTER-SLAVE cannot be successfully concluded. MASTER-SLAVE configuration fault, as well as the criteria and method of fault detection, is PHY specific. Additional information regarding the resolution and selection of MASTER-SLAVE and bit 7.33.15 for 10GBASE-T is contained in 55.6.2. Additional information regarding the resolution and selection of MASTER-SLAVE and bit 7.33.15 for 25GBASE-T and 40GBASE-T is contained in 113.6.2. Additional information regarding the resolution and selection of MASTER-SLAVE and bit 7.33.15 for 2.5GBASE-T and 5GBASE-T is contained in 126.6.2. The MASTER-SLAVE configuration fault bit 7.33.15 shall be cleared each time the MultiGBASE-T status register 7.33 is read via the management interface and shall be cleared by a PMA reset of a PHY in the MultiGBASE-T set. Bit 7.33.15 shall self clear upon Auto-Negotiation enable.

##### **45.2.7.11.2 MASTER-SLAVE configuration resolution (7.33.14)**

Bit 7.33.14 is determined by the MASTER-SLAVE configuration resolution function. These are described in 126.6.2, 55.6.2, and 113.6.2, for 2.5G/5GBASE-T, 10GBASE-T, and 25G/40GBASE-T, respectively. If the MASTER-SLAVE configuration resolution bit 7.33.14 is set to one and the Auto-Negotiation complete bit 7.1.5 is set and MASTER-SLAVE configuration fault bit 7.33.15 in the MultiGBASE-T status register is zero, then MASTER mode of operation has been selected. If the MASTER-SLAVE configuration resolution

**Table 45–386—MultiGBASE-T AN status 1 register**

| Bit(s)                                                                                                                                                         | Name                                    | Description                                                                                                                                                                                                 | R/W <sup>a</sup> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 7.33.15                                                                                                                                                        | MASTER-SLAVE configuration fault        | 1 = MASTER-SLAVE configuration fault detected<br>0 = No MASTER-SLAVE configuration fault detected.                                                                                                          | RO<br>LH<br>SC   |
| 7.33.14                                                                                                                                                        | MASTER-SLAVE configuration resolution   | 1 = Local PHY configuration resolved to MASTER<br>0 = Local PHY configuration resolved to SLAVE                                                                                                             | RO               |
| 7.33.13                                                                                                                                                        | Local receiver status                   | 1 = Local receiver OK<br>0 = Local receiver not OK                                                                                                                                                          | RO               |
| 7.33.12                                                                                                                                                        | Remote receiver status                  | 1 = Remote receiver OK<br>0 = Remote receiver not OK                                                                                                                                                        | RO               |
| 7.33.11                                                                                                                                                        | Link partner 10GBASE-T capability       | 1 = Link partner is able to operate as 10GBASE-T<br>0 = Link partner is not able to operate as 10GBASE-T<br>This bit is guaranteed to be valid only when the Page received bit (7.1.6) has been set to one. | RO               |
| 7.33.10                                                                                                                                                        | LP loop timing ability                  | 1 = Link partner is capable of loop timing<br>0 = Link partner is not capable of loop timing                                                                                                                | RO               |
| 7.33.9                                                                                                                                                         | 10GBASE-T LP PMA training reset request | 1 = Link partner requests that local device reset PMA training PRBS every frame<br>0 = Link partner requests that local device run PMA training PRBS continuously<br>See NOTE                               | RO               |
| 7.33.8                                                                                                                                                         | Link partner 40GBASE-T capability       | 1 = Link partner is able to operate as 40GBASE-T<br>0 = Link partner is not able to operate as 40GBASE-T                                                                                                    | RO               |
| 7.33.7                                                                                                                                                         | Link partner 25GBASE-T capability       | 1 = Link partner is able to operate as 25GBASE-T<br>0 = Link partner is not able to operate as 25GBASE-T                                                                                                    | RO               |
| 7.33.6                                                                                                                                                         | Link partner 5GBASE-T capability        | 1 = Link partner is able to operate as 5GBASE-T<br>0 = Link partner is not able to operate as 5GBASE-T                                                                                                      | RO               |
| 7.33.5                                                                                                                                                         | Link partner 2.5GBASE-T capability      | 1 = Link partner is able to operate as 2.5GBASE-T<br>0 = Link partner is not able to operate as 2.5GBASE-T                                                                                                  | RO               |
| 7.33.4                                                                                                                                                         | 5GBASE-T Fast retrain ability           | 1 = Link partner is capable of 5GBASE-T fast retrain<br>0 = Link partner is not capable of 5GBASE-T fast retrain                                                                                            | RO               |
| 7.33.3                                                                                                                                                         | 2.5GBASE-T Fast retrain ability         | 1 = Link partner is capable of 2.5GBASE-T fast retrain<br>0 = Link partner is not capable of 2.5GBASE-T fast retrain                                                                                        | RO               |
| 7.33.2                                                                                                                                                         | 25GBASE-T Fast retrain ability          | 1 = Link partner is capable of 25GBASE-T fast retrain<br>0 = Link partner is not capable of 25GBASE-T fast retrain                                                                                          | RO               |
| 7.33.1                                                                                                                                                         | 10GBASE-T Fast retrain ability          | 1 = Link partner is capable of 10GBASE-T fast retrain<br>0 = Link partner is not capable of 10GBASE-T fast retrain                                                                                          | RO               |
| 7.33.0                                                                                                                                                         | 40GBASE-T Fast retrain ability          | 1 = Link partner is capable of 40GBASE-T fast retrain<br>0 = Link partner is not capable of 40GBASE-T fast retrain                                                                                          | RO               |
| NOTE—The periodic training sequence request functionality is deprecated. Implementations may ignore a value of one in this bit or have it always read as zero. |                                         |                                                                                                                                                                                                             |                  |

<sup>a</sup>RO = Read only, SC = Self-clearing, LH = Latching high

bit 7.33.14 is set to zero and the Auto-Negotiation complete bit 7.1.5 is set and MASTER-SLAVE configuration fault bit 7.33.15 in the MultiGBASE-T status register is zero, then SLAVE mode of operation has been selected. In all other cases, neither SLAVE mode nor MASTER mode has been selected.

#### **45.2.7.11.3 Local receiver status (7.33.13)**

Local receiver status bit 7.33.13 shall be set if the local receiver status is OK as defined in 55.2.2.7. If the local receiver status bit 7.33.13 is zero, the local receiver status is not OK as defined in 55.2.2.7.

#### **45.2.7.11.4 Remote receiver status (7.33.12)**

Remote receiver status bit 7.33.12 shall be set if the remote receiver status is OK as defined in 55.2.2.8. If the remote receiver status bit 7.33.12 is zero, the remote receiver status is not OK as defined in 55.2.2.8.

#### **45.2.7.11.5 Link partner 10GBASE-T capability (7.33.11)**

The bit is only valid when page receive bit 7.1.6 is set to one. When read as a one, bit 7.33.11 indicates that the link partner is able to operate as 10GBASE-T. When read as a zero, bit 7.33.11 indicates that the link partner is not able to operate as 10GBASE-T.

#### **45.2.7.11.6 Link partner loop timing ability (7.33.10)**

When read as a one, bit 7.33.10 indicates that the link partner has the ability to support loop timing as specified in 55.1.3. When read as a zero, bit 7.33.10 indicates that the link partner lacks the ability to support loop timing.

#### **45.2.7.11.7 10GBASE-T Link partner PMA training reset request (7.33.9)**

If bit 7.33.9 is set to one then the link partner is expecting the local device to reset the PMA training PRBS for every PMA training frame. If bit 7.33.9 is zero then the link partner expects the local device to run PMA training PRBS continuously through every PMA training frame. The periodic training sequence request functionality is deprecated. Implementations may ignore a value of one in this bit or have it always read as zero.

#### **45.2.7.11.8 Link partner 40GBASE-T capability (7.33.8)**

Bit 7.33.8 is only valid when page received bit 7.1.6 is set to one. When read as a one, bit 7.33.8 indicates that the link partner is able to operate as 40GBASE-T. When read as a zero, bit 7.33.8 indicates that the link partner is not able to operate as 40GBASE-T.

#### **45.2.7.11.9 Link partner 25GBASE-T capability (7.33.7)**

Bit 7.33.7 is only valid when page received bit 7.1.6 is set to one. When read as a one, bit 7.33.7 indicates that the link partner is able to operate as 25GBASE-T. When read as a zero, bit 7.33.7 indicates that the link partner is not able to operate as 25GBASE-T.

#### **45.2.7.11.10 Link partner 5GBASE-T capability (7.33.6)**

Bit 7.33.6 is only valid when page received bit 7.1.6 is set to one. When read as a one, bit 7.33.6 indicates that the link partner is able to operate as 5GBASE-T. When read as a zero, bit 7.33.6 indicates that the link partner is not able to operate as 5GBASE-T.

#### 45.2.7.11.11 Link partner 2.5GBASE-T capability (7.33.5)

Bit 7.33.5 is only valid when page received bit 7.1.5 is set to one. When read as a one, bit 7.33.5 indicates that the link partner is able to operate as 2.5GBASE-T. When read as a zero, bit 7.33.5 indicates that the link partner is not able to operate as 2.5GBASE-T.

#### 45.2.7.11.12 5GBASE-T Fast retrain ability (7.33.4)

When read as a one, bit 7.33.4 indicates that the link partner has the ability to support the 5GBASE-T fast retrain capability as specified in 126.4.2.5.16. When read as a zero, bit 7.33.4 indicates that the PHY lacks the ability to support the 5GBASE-T fast retrain capability. This bit is only valid after link is established.

#### 45.2.7.11.13 2.5GBASE-T Fast retrain ability (7.33.3)

When read as a one, bit 7.33.3 indicates that the link partner has the ability to support the 2.5GBASE-T fast retrain capability as specified in 126.4.2.5.16. When read as a zero, bit 7.33.3 indicates that the PHY lacks the ability to support the 2.5GBASE-T fast retrain capability. This bit is only valid after link is established.

#### 45.2.7.11.14 25GBASE-T Fast retrain ability (7.33.2)

When read as a one, bit 7.33.2 indicates that the link partner has the ability to support the 25GBASE-T fast retrain capability as specified in 113.4.2.5.16. When read as a zero, bit 7.33.2 indicates that the PHY lacks the ability to support the 25GBASE-T fast retrain capability. This bit is valid only after link is established.

#### 45.2.7.11.15 10GBASE-T Fast retrain ability (7.33.1)

When read as a one, bit 7.33.1 indicates that the link partner has the ability to support the 10GBASE-T fast retrain capability as specified in 55.4.2.5.15. When read as a zero, bit 7.33.1 indicates that the PHY lacks the ability to support the 10GBASE-T fast retrain capability.

#### 45.2.7.11.9 40GBASE-T Fast retrain ability (7.33.0)

When read as a one, bit 7.33.0 indicates that the link partner has the ability to support the 40GBASE-T fast retrain capability as specified in 113.4.2.5.16. When read as a zero, bit 7.33.0 indicates that the PHY lacks the ability to support the 40GBASE-T fast retrain capability. This bit is valid only after link is established.

#### 45.2.7.12 Backplane Ethernet, BASE-R copper status (Register 7.48)

The assignment of bits in the Backplane Ethernet, BASE-R copper status register is shown in Table 45–387.

**Table 45–387—Backplane Ethernet, BASE-R copper status register (register 7.48)  
bit definitions**

| Bit(s)  | Name        | Description                                                                                                | R/W <sup>a</sup> |
|---------|-------------|------------------------------------------------------------------------------------------------------------|------------------|
| 7.48.15 | 5GBASE-KR   | 1 = PMA/PMD is negotiated to perform 5GBASE-KR<br>0 = PMA/PMD is not negotiated to perform 5GBASE-KR       | RO               |
| 7.48.14 | 2.5GBASE-KX | 1 = PMA/PMD is negotiated to perform 2.5GGBASE-KX<br>0 = PMA/PMD is not negotiated to perform 2.5GGBASE-KX | RO               |



**Table 45–387—Backplane Ethernet, BASE-R copper status register (register 7.48)  
bit definitions (*continued*)**

| Bit(s)  | Name                         | Description                                                                                                                                | R/W <sup>a</sup> |
|---------|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 7.48.13 | 25GBASE-KR or 25GBASE-CR     | 1 = PMA/PMD is negotiated to perform 25GBASE-KR or 25GBASE-CR<br>0 = PMA/PMD is not negotiated to perform 25GBASE-KR or 25GBASE-CR         | RO               |
| 7.48.12 | 25GBASE-KR-S or 25GBASE-CR-S | 1 = PMA/PMD is negotiated to perform 25GBASE-KR-S or 25GBASE-CR-S<br>0 = PMA/PMD is not negotiated to perform 25GBASE-KR-S or 25GBASE-CR-S | RO               |
| 7.48.11 | 100GBASE-CR4                 | 1 = PMA/PMD is negotiated to perform 100GBASE-CR4<br>0 = PMA/PMD is not negotiated to perform 100GBASE-CR4                                 | RO               |
| 7.48.10 | 100GBASE-KR4                 | 1 = PMA/PMD is negotiated to perform 100GBASE-KR4<br>0 = PMA/PMD is not negotiated to perform 100GBASE-KR4                                 | RO               |
| 7.48.9  | 100GBASE-KP4                 | 1 = PMA/PMD is negotiated to perform 100GBASE-KP4<br>0 = PMA/PMD is not negotiated to perform 100GBASE-KP4                                 | RO               |
| 7.48.8  | 100GBASE-CR10                | 1 = PMA/PMD is negotiated to perform 100GBASE-CR10<br>0 = PMA/PMD is not negotiated to perform 100GBASE-CR10                               | RO               |
| 7.48.7  | RS-FEC negotiated            | 1 = PMA/PMD is negotiated to perform RS-FEC<br>0 = PMA/PMD is not negotiated to perform RS-FEC                                             | RO               |
| 7.48.6  | 40GBASE-CR4                  | 1 = PMA/PMD is negotiated to perform 40GBASE-CR4<br>0 = PMA/PMD is not negotiated to perform 40GBASE-CR4                                   | RO               |
| 7.48.5  | 40GBASE-KR4                  | 1 = PMA/PMD is negotiated to perform 40GBASE-KR4<br>0 = PMA/PMD is not negotiated to perform 40GBASE-KR4                                   | RO               |
| 7.48.4  | BASE-R FEC negotiated        | 1 = PMA/PMD is negotiated to perform BASE-R FEC<br>0 = PMA/PMD is not negotiated to perform BASE-R FEC                                     | RO               |
| 7.48.3  | 10GBASE-KR                   | 1 = PMA/PMD is negotiated to perform 10GBASE-KR<br>0 = PMA/PMD is not negotiated to perform 10GBASE-KR                                     | RO               |
| 7.48.2  | 10GBASE-KX4                  | 1 = PMA/PMD is negotiated to perform 10GBASE-KX4 or CX4<br>0 = PMA/PMD is not negotiated to perform 10GBASE-KX4/CX4                        | RO               |
| 7.48.1  | 1000BASE-KX                  | 1 = PMA/PMD is negotiated to perform 1000BASE-KX<br>0 = PMA/PMD is not negotiated to perform 1000BASE-KX                                   | RO               |
| 7.48.0  | BP AN ability                | If a Backplane, BASE-R copper PHY type is implemented, this bit is set to one                                                              | RO               |

<sup>a</sup>RO = Read only

#### 45.2.7.12.1 BASE-R FEC negotiated (7.48.4)

When the Auto-Negotiation process has completed as indicated by the AN complete bit (7.1.5), bit 7.48.4 indicates that BASE-R FEC operation has been negotiated. This bit is set only if a BASE-R PHY supporting FEC operation has also been negotiated.

#### 45.2.7.12.2 RS-FEC negotiated (7.48.7)

When the Auto-Negotiation process has completed as indicated by the AN complete bit (7.1.5), bit 7.48.7 indicates that RS-FEC operation has been negotiated. This bit is set only if RS-FEC operation has been negotiated for a BASE-R PHY supporting negotiation of RS-FEC operation.

#### 45.2.7.12.3 Negotiated Port Type (7.48.1, 7.48.2, 7.48.3, 7.48.5, 7.48.6, 7.48.8, 7.48.9, 7.48.10, 7.48.11, 7.48.12, 7.48.13, 7.48.14, 7.48.15, 7.49.0, 7.49.1, 7.49.2)

When the AN process has been completed as indicated by the AN complete bit, these bits in register 7.48 and register 7.49 indicate the negotiated port type. Only one of these bits is set depending on the priority resolution function. System developers need to distinguish between parallel detection of 10GBASE-KX4 and 10GBASE-CX4 based on the MDI and media type present.

#### 45.2.7.12.4 Backplane Ethernet, BASE-R copper AN ability (7.48.0)

If a Backplane, BASE-R copper PHY type is implemented, this bit shall be set to one.

When read as a one, bit 7.48.0 indicates that the PMA/PMD has the ability to perform Backplane Ethernet, BASE-R copper AN. When read as a zero, bit 7.48.0 indicates that the PMA/PMD lacks the ability to perform Backplane Ethernet BASE-R copper AN.

#### 45.2.7.13 Backplane Ethernet, BASE-R copper status 2 (Register 7.49)

The assignment of bits in the Backplane Ethernet, BASE-R copper status 2 register is shown in Table 45–388.

**Table 45–388—Backplane Ethernet, BASE-R copper status 2 register (register 7.49)  
bit definitions**

| Bit(s)    | Name                         | Description                                                                                                                                | RO <sup>a</sup> |
|-----------|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| 7.49.15:3 | Reserved                     | Value always 0                                                                                                                             | RO              |
| 7.49.2    | 200GBASE-KR4 or 200GBASE-CR4 | 1 = PMA/PMD is negotiated to perform 200GBASE-KR4 or 200GBASE-CR4<br>0 = PMA/PMD is not negotiated to perform 200GBASE-KR4 or 200GBASE-CR4 | RO              |
| 7.49.1    | 100GBASE-KR2 or 100GBASE-CR2 | 1 = PMA/PMD is negotiated to perform 100GBASE-KR2 or 100GBASE-CR2<br>0 = PMA/PMD is not negotiated to perform 100GBASE-KR2 or 100GBASE-CR2 | RO              |
| 7.49.0    | 50GBASE-KR or 50GBASE-CR     | 1 = PMA/PMD is negotiated to perform 50GBASE-KR or 50GBASE-CR<br>0 = PMA/PMD is not negotiated to perform 50GBASE-KR or 50GBASE-CR         | RO              |

<sup>a</sup>RO = Read only

#### 45.2.7.13.1 Negotiated Port Type

See 45.2.7.12.3.

#### 45.2.7.14 EEE advertisement 1 (Register 7.60)

This register defines the EEE advertisement for several device types. Devices that use Clause 28 Auto-Negotiation send EEE advertisement in the Unformatted Next Page following a EEE technology message code as defined in 28C.12 or as part of the 10GBASE-T and 1000BASE-T technology message code as defined in 28C.11. Devices that use Clause 73 Auto-Negotiation send EEE advertisement in the unformatted code field of Message Next Page with EEE technology message code as defined in 73A.4. 25GBASE-T and 40GBASE-T EEE advertisement is exchanged in the Infofield during training as defined in 113.4.2.5.10. For 2.5GBASE-T and 5GBASE-T, the EEE advertisement is exchanged in the Infofield during training as defined in 126.4.2.5.10. The assignment of bits in the EEE advertisement 1 register and the correspondence with the bits in the Next Page messages or in the training Infofield are shown in Table 45–389.

Bits 10:0 of register 7.60 map to bits U10 through U0 respectively of the Unformatted Next Page following a EEE technology message code as defined in 28C.12. Bits 15:0 of register 7.60 map to bits U15 through U0 respectively of the unformatted code field of Message Next Page with EEE technology message code as defined in 73A.4. Bits 3:1 of register 7.60 also map to bits U24 through U22 respectively of the 10GBASE-T and 1000BASE-T technology message code as defined in 28C.11. Devices using Clause 28 auto-negotiation may ignore bits defined for Clause 73 auto-negotiation, and devices using Clause 73 auto-negotiation may ignore bits defined for Clause 28 auto-negotiation.

**Table 45–389—EEE advertisement register 1 (register 7.60) bit definitions**

| Bit(s)  | Name              | Description                                                                                                                                     | Clause reference;<br>Next Page bit number | R/W <sup>a</sup> |
|---------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|------------------|
| 7.60.15 | Reserved          | Value always 0                                                                                                                                  |                                           | RO               |
| 7.60.14 | 25GBASE-R EEE     | 1 = Advertise that the 25GBASE-R PHY has EEE deep sleep capability<br>0 = Do not advertise that the 25GBASE-R PHY has EEE deep sleep capability | 73.7.7.1; U14                             | R/W              |
| 7.60.13 | 100GBASE-CR4 EEE  | 1 = Advertise that the 100GBASE-CR4 has EEE deep sleep capability<br>0 = Do not advertise that the 100GBASE-CR4 has EEE deep sleep capability   | 73.7.7.1; U13                             | R/W              |
| 7.60.12 | 100GBASE-KR4 EEE  | 1 = Advertise that the 100GBASE-KR4 has EEE deep sleep capability<br>0 = Do not advertise that the 100GBASE-KR4 has EEE deep sleep capability   | 73.7.7.1; U12                             | R/W              |
| 7.60.11 | 100GBASE-KP4 EEE  | 1 = Advertise that the 100GBASE-KP4 has EEE deep sleep capability<br>0 = Do not advertise that the 100GBASE-KP4 has EEE deep sleep capability   | 73.7.7.1; U11                             | R/W              |
| 7.60.10 | 100GBASE-CR10 EEE | 1 = Advertise that the 100GBASE-CR10 has EEE deep sleep capability<br>0 = Do not advertise that the 100GBASE-CR10 has EEE deep sleep capability | 73.7.7.1; U10                             | R/W              |

**Table 45–389—EEE advertisement register 1 (register 7.60) bit definitions (*continued*)**

| Bit(s) | Name            | Description                                                                                                                                 | Clause reference;<br>Next Page bit number | R/W <sup>a</sup> |
|--------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|------------------|
| 7.60.9 | 40GBASE-T EEE   | 1 = Advertise that the 40GBASE-T has EEE capability<br>0 = Do not advertise that the 40GBASE-T has EEE capability                           | 113.4.2.5.10; Infofield Octet 12 bit 7    | R/W              |
| 7.60.8 | 40GBASE-CR4 EEE | 1 = Advertise that the 40GBASE-CR4 has EEE deep sleep capability<br>0 = Do not advertise that the 40GBASE-CR4 has EEE deep sleep capability | 73.7.7.1; U8                              | R/W              |
| 7.60.7 | 40GBASE-KR4 EEE | 1 = Advertise that the 40GBASE-KR4 has EEE deep sleep capability<br>0 = Do not advertise that the 40GBASE-KR4 has EEE deep sleep capability | 73.7.7.1; U7                              | R/W              |
| 7.60.6 | 10GBASE-KR EEE  | 1 = Advertise that the 10GBASE-KR has EEE capability<br>0 = Do not advertise that the 10GBASE-KR has EEE capability                         | 73.7.7.1; U6                              | R/W              |
| 7.60.5 | 10GBASE-KX4 EEE | 1 = Advertise that the 10GBASE-KX4 has EEE capability<br>0 = Do not advertise that the 10GBASE-KX4 has EEE capability                       | 73.7.7.1; U5                              | R/W              |
| 7.60.4 | 1000BASE-KX EEE | 1 = Advertise that the 1000BASE-KX has EEE capability<br>0 = Do not advertise that the 1000BASE-KX has EEE capability                       | 73.7.7.1; U4                              | R/W              |
| 7.60.3 | 10GBASE-T EEE   | 1 = Advertise that the 10GBASE-T has EEE capability<br>0 = Do not advertise that the 10GBASE-T has EEE capability                           | 28.2.3.4.1; U3 /<br>28.2.3.4.2; U24       | R/W              |
| 7.60.2 | 1000BASE-T EEE  | 1 = Advertise that the 1000BASE-T has EEE capability<br>0 = Do not advertise that the 1000BASE-T has EEE capability                         | 28.2.3.4.1; U2 /<br>28.2.3.4.2; U23       | R/W              |
| 7.60.1 | 100BASE-TX EEE  | 1 = Advertise that the 100BASE-TX has EEE capability<br>0 = Do not advertise that the 100BASE-TX has EEE capability                         | 28.2.3.4.1; U1 /<br>28.2.3.4.2; U22       | R/W              |
| 7.60.0 | 25GBASE-T EEE   | 1 = Advertise that the 25GBASE-T has EEE capability<br>0 = Do not advertise that the 25GBASE-T has EEE capability                           | 113.4.2.5.10; Infofield Octet 12 bit 7    | R/W              |

<sup>a</sup>R/W = Read/Write, RO = Read only

#### **45.2.7.14.1 25GBASE-R EEE supported (7.60.14)**

If set to a one, bit 7.60.14 indicates that the technologies advertised in bits A9 and A10 of Table 73–4 are capable of supporting EEE deep sleep operation. If set to a zero, EEE deep sleep operation is not supported for 25GBASE-R.

#### **45.2.7.14.2 100GBASE-CR4 EEE supported (7.60.13)**

Support for EEE deep sleep operation for 100GBASE-CR4, as defined in 92.1, shall be advertised if this bit is set to one. Support for EEE deep sleep operation should only be advertised if it is supported on all sublayers of the PHY as well as physical instantiations of the PMA service interface as appropriate.

#### **45.2.7.14.3 100GBASE-KR4 EEE supported (7.60.12)**

Support for EEE deep sleep operation for 100GBASE-KR4, as defined in 93.1, shall be advertised if this bit is set to one. Support for EEE deep sleep operation should only be advertised if it is supported on all sublayers of the PHY as well as physical instantiations of the PMA service interface as appropriate.

#### **45.2.7.14.4 100GBASE-KP4 EEE supported (7.60.11)**

Support for EEE deep sleep operation for 100GBASE-KP4, as defined in 94.1, shall be advertised if this bit is set to one. Support for EEE deep sleep operation should only be advertised if it is supported on all sublayers of the PHY as well as physical instantiations of the PMA service interface as appropriate.

#### **45.2.7.14.5 100GBASE-CR10 EEE supported (7.60.10)**

Support for EEE deep sleep operation for 100GBASE-CR10, as defined in 85.1, shall be advertised if this bit is set to one. Support for EEE deep sleep operation should only be advertised if it is supported on all sublayers of the PHY as well as physical instantiations of the PMA service interface as appropriate.

#### **45.2.7.14.6 40GBASE-T EEE supported (7.60.9)**

If the device supports EEE operation for 40GBASE-T as defined in 113.1.3.3, and EEE operation is desired, this bit shall be set to one.

#### **45.2.7.14.7 40GBASE-CR4 EEE supported (7.60.8)**

Support for EEE deep sleep operation for 40GBASE-CR4, as defined in 85.1, shall be advertised if this bit is set to one. Support for EEE deep sleep operation should only be advertised if it is supported on all sublayers of the PHY as well as physical instantiations of the PMA service interface as appropriate.

#### **45.2.7.14.8 40GBASE-KR4 EEE supported (7.60.7)**

Support for EEE deep sleep operation for 40GBASE-KR4, as defined in 84.1, shall be advertised if this bit is set to one. Support for EEE deep sleep operation should only be advertised if it is supported on all sublayers of the PHY as well as physical instantiations of the PMA service interface as appropriate.

#### **45.2.7.14.9 10GBASE-KR EEE supported (7.60.6)**

If the device supports EEE operation for 10GBASE-KR as defined in 72.1, and EEE operation is desired, this bit shall be set to one.

#### 45.2.7.14.10 10GBASE-KX4 EEE supported (7.60.5)

If the device supports EEE operation for 10GBASE-KX4 as defined in 71.2, and EEE operation is desired, this bit shall be set to one.

#### 45.2.7.14.11 1000BASE-KX EEE supported (7.60.4)

If the device supports EEE operation for 1000BASE-KX as defined in 70.1, and EEE operation is desired, this bit shall be set to one.

#### 45.2.7.14.12 10GBASE-T EEE supported (7.60.3)

If the device supports EEE operation for 10GBASE-T as defined in 55.1.3.3, and EEE operation is desired, this bit shall be set to one.

#### 45.2.7.14.13 1000BASE-T EEE supported (7.60.2)

If the device supports EEE operation for 1000BASE-T as defined in 40.2.11, and EEE operation is desired, this bit shall be set to one.

#### 45.2.7.14.14 100BASE-TX EEE supported (7.60.1)

If the device supports EEE operation for 100BASE-TX as defined in 25.5, and EEE operation is desired, this bit shall be set to one.

#### 45.2.7.14.15 25GBASE-T EEE supported (7.60.0)

If the device supports EEE operation for 25GBASE-T as defined in 113.1.3.3, and EEE operation is desired, this bit shall be set to one.

#### 45.2.7.15 EEE link partner ability 1 (Register 7.61)

All of the bits in the EEE LP ability 1 register are read-only. A write to the EEE LP ability 1 register shall have no effect. Except for 10GBASE-T, members of the MultiGBASE-T PHY set exchange the EEE ability in the Infocfield during link training. For these PHYs, the EEE LP ability register is updated after link is established. For all other PHYs, when the AN process has been completed, this register shall reflect the contents of the link partner's EEE advertisement 1 register. The assignment of bits in the EEE link partner ability 1 register and the correspondence with the bits in the Next Page messages are shown in Table 45–390.

**Table 45–390—EEE link partner ability 1 (register 7.61) bit definitions**

| Bit(s)  | Name          | Description                                                                                                                                            | Clause reference;<br>Next Page bit number | R/W <sup>a</sup> |
|---------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|------------------|
| 7.61.15 | Reserved      | Value always 0                                                                                                                                         |                                           | RO               |
| 7.61:14 | 25GBASE-R EEE | 1 = Link partner is advertising EEE deep sleep capability for 25GBASE-R<br>0 = Link partner is not advertising EEE deep sleep capability for 25GBASE-R | 73.7.7.1;U14                              | RO               |

**Table 45–390—EEE link partner ability 1 (register 7.61) bit definitions (*continued*)**

| Bit(s)  | Name                 | Description                                                                                                                                                    | Clause reference;<br>Next Page bit number | R/W <sup>a</sup> |
|---------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|------------------|
| 7.61.13 | 100GBASE-CR4<br>EEE  | 1 = Link partner is advertising EEE deep sleep capability for 100GBASE-CR4<br>0 = Link partner is not advertising EEE deep sleep capability for 100GBASE-CR4   | 73.7.7.1; U13                             | RO               |
| 7.61.12 | 100GBASE-KR4<br>EEE  | 1 = Link partner is advertising EEE deep sleep capability for 100GBASE-KR4<br>0 = Link partner is not advertising EEE deep sleep capability for 100GBASE-KR4   | 73.7.7.1; U12                             | RO               |
| 7.61.11 | 100GBASE-KP4<br>EEE  | 1 = Link partner is advertising EEE deep sleep capability for 100GBASE-KP4<br>0 = Link partner is not advertising EEE deep sleep capability for 100GBASE-KP4   | 73.7.7.1; U11                             | RO               |
| 7.61.10 | 100GBASE-CR10<br>EEE | 1 = Link partner is advertising EEE deep sleep capability for 100GBASE-CR10<br>0 = Link partner is not advertising EEE deep sleep capability for 100GBASE-CR10 | 73.7.7.1; U10                             | RO               |
| 7.61.9  | 40GBASE-T EEE        | 1 = Link partner is advertising EEE capability for 40GBASE-T<br>0 = Link partner is not advertising EEE capability for 40GBASE-T                               | 113.4.2.5.10; Infofield Octet 12 bit 7    | RO               |
| 7.61.8  | 40GBASE-CR4<br>EEE   | 1 = Link partner is advertising EEE deep sleep capability for 40GBASE-CR4<br>0 = Link partner is not advertising EEE deep sleep capability for 40GBASE-CR4     | 73.7.7.1; U8                              | RO               |
| 7.61.7  | 40GBASE-KR4<br>EEE   | 1 = Link partner is advertising EEE deep sleep capability for 40GBASE-KR4<br>0 = Link partner is not advertising EEE deep sleep capability for 40GBASE-KR4     | 73.7.7.1; U7                              | RO               |
| 7.61.6  | 10GBASE-KR<br>EEE    | 1 = Link partner is advertising EEE capability for 10GBASE-KR<br>0 = Link partner is not advertising EEE capability for 10GBASE-KR                             | 73.7.7.1; U6                              | RO               |
| 7.61.5  | 10GBASE-KX4<br>EEE   | 1 = Link partner is advertising EEE capability for 10GBASE-KX4<br>0 = Link partner is not advertising EEE capability for 10GBASE-KX4                           | 73.7.7.1; U5                              | RO               |
| 7.61.4  | 1000BASE-KX<br>EEE   | 1 = Link partner is advertising EEE capability for 1000BASE-KX<br>0 = Link partner is not advertising EEE capability for 1000BASE-KX                           | 73.7.7.1; U4                              | RO               |
| 7.61.3  | 10GBASE-T EEE        | 1 = Link partner is advertising EEE capability for 10GBASE-T<br>0 = Link partner is not advertising EEE capability for 10GBASE-T                               | 28.2.3.4.1; U3 / 55.6.1; U24              | RO               |

**Table 45–390—EEE link partner ability 1 (register 7.61) bit definitions (continued)**

| Bit(s) | Name           | Description                                                                                                                        | Clause reference;<br>Next Page bit number | R/W <sup>a</sup> |
|--------|----------------|------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|------------------|
| 7.61.2 | 1000BASE-T EEE | 1 = Link partner is advertising EEE capability for 1000BASE-T<br>0 = Link partner is not advertising EEE capability for 1000BASE-T | 28.2.3.4.1; U2 / 55.6.1; U23              | RO               |
| 7.61.1 | 100BASE-TX EEE | 1 = Link partner is advertising EEE capability for 100BASE-TX<br>0 = Link partner is not advertising EEE capability for 100BASE-TX | 28.2.3.4.1; U1 / 55.6.1; U22              | RO               |
| 7.61.0 | 25GBASE-T EEE  | 1 = Link partner is advertising EEE capability for 25GBASE-T<br>0 = Link partner is not advertising EEE capability for 25GBASE-T   | 113.4.2.5.10; Infofield Octet 12 bit 7    | RO               |

<sup>a</sup>RO = Read only

The definitions for the contents of the EEE LP ability register are given by the definitions for the contents on the link partner's EEE advertisement 1 register, 7.60 (see 45.2.7.14).

#### 45.2.7.16 EEE advertisement 2 (Register 7.62)

EEE advertisement 2 register is a continuation of EEE advertisement 1 register. The assignment of bits in the EEE advertisement 2 register are shown in Table 45–391.

**Table 45–391—EEE advertisement 2 register (register 7.62) bit definitions**

| Bit(s)    | Name            | Description                                                                                                           | Clause reference;<br>Next Page bit number | R/W <sup>a</sup> |
|-----------|-----------------|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------|------------------|
| 7.62.15:4 | Reserved        | Value always 0                                                                                                        |                                           | RO               |
| 7.62.3    | 5GBASE-KR EEE   | 1 = Advertise that the 5GBASE-KR has EEE capability<br>0 = Do not advertise that the 5GBASE-KR has EEE capability     | 73.7.7.1; U17                             | R/W              |
| 7.62.2    | 2.5GBASE-KX EEE | 1 = Advertise that the 2.5GBASE-KX has EEE capability<br>0 = Do not advertise that the 2.5GBASE-KX has EEE capability | 73.7.7.1; U16                             | R/W              |
| 7.62.1    | 5GBASE-T EEE    | 1 = Advertise that the 5GBASE-T has EEE capability<br>0 = Do not advertise that the 5GBASE-T has EEE capability       | 126.4.2.5.10                              | R/W              |
| 7.62.0    | 2.5GBASE-T EEE  | 1 = Advertise that the 2.5GBASE-T has EEE capability<br>0 = Do not advertise that the 2.5GBASE-T has EEE capability   | 126.4.2.5.10                              | R/W              |

<sup>a</sup>R/W = Read/Write, RO = Read only



#### 45.2.7.16.1 5GBASE-KR EEE (7.62.3)

If the device supports EEE operation for 5GBASE-KR as defined in 130.1 and EEE operation is desired, this bit shall be set to one.

#### 45.2.7.16.2 2.5GBASE-KX EEE (7.62.2)

If the device supports EEE operation for 2.5GBASE-KX as defined in 128.1 and EEE operation is desired, this bit shall be set to one.

#### 45.2.7.16.3 5GBASE-T EEE (7.62.1)

If the device supports EEE operation for 5GBASE-T as defined in 126.6.1, and EEE operation is desired, bit 7.62.1 shall be set to one.

#### 45.2.7.16.4 2.5GBASE-T EEE (7.62.0)

If the device supports EEE operation for 2.5GBASE-T as defined in 126.6.1, and EEE operation is desired, bit 7.62.0 shall be set to one.

#### 45.2.7.17 EEE link partner ability 2 (Register 7.63)

All of the bits in the EEE link partner ability 2 register are read-only. A write to the EEE link partner ability 2 register shall have no effect. When the AN and training processes have been completed, this register shall reflect the contents of the link partner's EEE advertisement 2 register. The assignment of bits in the EEE link partner ability 2 register and the correspondence with the bits in the Next Page messages (for PHYs that exchange EEE abilities during Auto-Negotiation) are shown in Table 45–392.

The definitions for the contents of the EEE link partner ability 2 register are given by the definitions for the contents on the link partner's EEE advertisement 2 register, 7.62 (see 45.2.7.16).

**Table 45–392—EEE link partner ability 2 (register 7.63) bit definitions**

| Bit(s)    | Name            | Description                                                                                                                                            | Clause reference;<br>Next Page bit number | R/W <sup>a</sup> |
|-----------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|------------------|
| 7.63.15:4 | Reserved        | Value always 0                                                                                                                                         |                                           | RO               |
| 7.63.3    | 5GBASE-KR EEE   | 1 = Link partner is advertising that the 5GBASE-KR has EEE capability<br>0 = Link partner is not advertising that the 5GBASE-KR has EEE capability     | 73.7.7.1; U17                             | RO               |
| 7.63.2    | 2.5GBASE-KX EEE | 1 = Link partner is advertising that the 2.5GBASE-KX has EEE capability<br>0 = Link partner is not advertising that the 2.5GBASE-KX has EEE capability | 73.7.7.1; U16                             | RO               |

**Table 45–392—EEE link partner ability 2 (register 7.63) bit definitions (*continued*)**

| Bit(s) | Name           | Description                                                                                                                                          | Clause reference;<br>Next Page bit number | R/W <sup>a</sup> |
|--------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|------------------|
| 7.63.1 | 5GBASE-T EEE   | 1 = Link partner is advertising that the 5GBASE-T has EEE capability<br>0 = Link partner is not advertising that the 5GBASE-T has EEE capability     | 126.4.2.5.10                              | RO               |
| 7.63.0 | 2.5GBASE-T EEE | 1 = Link partner is advertising that the 2.5GBASE-T has EEE capability<br>0 = Link partner is not advertising that the 2.5GBASE-T has EEE capability | 126.4.2.5.10                              | RO               |

<sup>a</sup>RO = Read only

#### 45.2.7.18 MultiGBASE-T AN control 2 (Register 7.64)

Register 7.64 is a continuation of register 7.32. The assignment of bits in the MultiGBASE-T AN control 2 register is shown in Table 45–393. The default values of each bit of the MultiGBASE-T AN control 2 register should be chosen so that the initial state of the devices upon power up or reset is a normal operational state without management intervention.

**Table 45–393—MultiGBASE-T AN control 2 (register 7.64) bit definitions**

| Bit(s)    | Name                          | Description                                                                                                                                                                | R/W <sup>a</sup> |
|-----------|-------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 7.64.15:4 | Reserved                      | Value always 0                                                                                                                                                             | RO               |
| 7.64.3    | 2.5GBASE-T THP Bypass Request | 1 = Local device requests link partner to initially reset THP during fast retrain<br>0 = Local device requests link partner not to initially reset THP during fast retrain | R/W              |
| 7.64.2    | 5GBASE-T THP Bypass Request   | 1 = Local device requests link partner to initially reset THP during fast retrain<br>0 = Local device requests link partner not to initially reset THP during fast retrain | R/W              |
| 7.64.1    | 25GBASE-T THP Bypass Request  | 1 = Local device requests link partner to initially reset THP during fast retrain<br>0 = Local device requests link partner not to initially reset THP during fast retrain | R/W              |
| 7.64.0    | 40GBASE-T THP Bypass Request  | 1 = Local device requests link partner to initially reset THP during fast retrain<br>0 = Local device requests link partner not to initially reset THP during fast retrain | R/W              |

<sup>a</sup>R/W = Read/Write, RO = Read only

##### 45.2.7.18.1 2.5GBASE-T THP Bypass Request

Bit 7.64.3 is valid only if 7.32.5 is set to one advertising fast retrain ability, and is used to request the link partner whether to initially reset the THP during fast retrain. THP Bypass Request is exchanged during link training, see 126.4.2.5.10. If bit 7.64.3 is set to zero the local device requests link partner not to reset THP

during fast retrain. If bit 7.64.3 is set to one the local device requests link partner to initially reset THP during fast retrain.

#### 45.2.7.18.2 5GBASE-T THP Bypass Request

Bit 7.64.2 is valid only if 7.32.6 is set to one advertising fast retrain ability, and is used to request the link partner whether to initially reset the THP during fast retrain. THP Bypass Request is exchanged during link training, see 126.4.2.5.10. If bit 7.64.2 is set to zero the local device requests link partner not to reset THP during fast retrain. If bit 7.64.2 is set to one the local device requests link partner to initially reset THP during fast retrain.

#### 45.2.7.18.3 25GBASE-T THP Bypass Request

Bit 7.64.1 is valid only if 7.32.9 is set to one advertising fast retrain ability, and is used to request the link partner whether to initially reset the THP during fast retrain. THP Bypass Request is exchanged during link training, see 113.4.2.5.10. If bit 7.64.1 is set to zero the local device requests link partner not to reset THP during fast retrain. If bit 7.64.1 is set to one the local device requests link partner to initially reset THP during fast retrain.

#### 45.2.7.18.4 40GBASE-T THP Bypass Request

Bit 7.64.0 is valid only if 7.32.3 is set to one advertising fast retrain ability, and is used to request the link partner whether to initially reset the THP during fast retrain. THP Bypass Request is exchanged during link training, see 113.4.2.5.10. If bit 7.64.0 is set to zero the local device requests link partner not to reset THP during fast retrain. If bit 7.64.0 is set to one the local device requests link partner to initially reset THP during fast retrain.

#### 45.2.7.19 MultiGBASE-T AN status 2 (Register 7.65)

Register 7.65 is a continuation of register 7.33. The assignment of bits in the MultiGBASE-T AN status 2 register is shown in Table 45–394. All the bits in the MultiGBASE-T AN status 2 register are read only; a write shall have no effect.

**Table 45–394—MultiGBASE-T AN status 2 (register 7.65) bit definitions**

| Bit(s)    | Name                                       | Description                                                                                                                                                                | R/W <sup>a</sup> |
|-----------|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 7.65.15:4 | Reserved                                   | Value always 0                                                                                                                                                             | RO               |
| 7.65.3    | 2.5GBASE-T Link Partner THP Bypass Request | 1 = Link partner requests local device to initially reset THP during fast retrain<br>0 = Link partner requests local device not to initially reset THP during fast retrain | RO               |
| 7.65.2    | 5GBASE-T Link Partner THP Bypass Request   | 1 = Link partner requests local device to initially reset THP during fast retrain<br>0 = Link partner requests local device not to initially reset THP during fast retrain | RO               |

**Table 45–394—MultiGBASE-T AN status 2 (register 7.65) bit definitions (*continued*)**

| Bit(s) | Name                                      | Description                                                                                                                                                                | R/W <sup>a</sup> |
|--------|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 7.65.1 | 25GBASE-T Link Partner THP Bypass Request | 1 = Link partner requests local device to initially reset THP during fast retrain<br>0 = Link partner requests local device not to initially reset THP during fast retrain | RO               |
| 7.65.0 | 40GBASE-T Link Partner THP Bypass Request | 1 = Link partner requests local device to initially reset THP during fast retrain<br>0 = Link partner requests local device not to initially reset THP during fast retrain | RO               |

<sup>a</sup>RO = Read only

#### 45.2.7.19.1 2.5GBASE-T Link Partner THP Bypass Request

Bit 7.65.3 is valid only if 7.33.3 is set to one indicating that the link partner has fast retrain ability. THP Bypass Request is exchanged during link training, see 126.4.2.5.10. Bit 7.65.3 is updated after link is established. When read as a zero, the link partner requests local device not to reset THP during fast retrain. When read as a one, the link partner requests local device to initially reset THP during fast retrain.

#### 45.2.7.19.2 5GBASE-T Link Partner THP Bypass Request

Bit 7.65.2 is valid only if 7.33.0 is set to one indicating that the link partner has fast retrain ability. THP Bypass Request is exchanged during link training, see 126.4.2.5.10. Bit 7.65.2 is updated after link is established. When read as a zero, the link partner requests local device not to reset THP during fast retrain. When read as a one, the link partner requests local device to initially reset THP during fast retrain.

#### 45.2.7.19.3 25GBASE-T Link Partner THP Bypass Request

Bit 7.65.1 is valid only if 7.33.2 is set to one indicating that the link partner has fast retrain ability. THP Bypass Request is exchanged during link training, see 113.4.2.5.10. Bit 7.65.1 is updated after link is established. When read as a zero, the link partner requests local device not to reset THP during fast retrain. When read as a one, the link partner requests local device to initially reset THP during fast retrain.

#### 45.2.7.19.4 40GBASE-T Link Partner THP Bypass Request

Bit 7.65.0 is valid only if 7.33.0 is set to one indicating that the link partner has fast retrain ability. THP Bypass Request is exchanged during link training, see 113.4.2.5.10. Bit 7.65.0 is updated after link is established. When read as a zero, the link partner requests local device not to reset THP during fast retrain. When read as a one, the link partner requests local device to initially reset THP during fast retrain.

#### 45.2.7.20 BASE-T1 AN control register (Register 7.512)

The assignment of bits in the BASE-T1 AN control register is shown in Table 45–395. The default value for each bit of the BASE-T1 AN control register has been chosen so that the initial state of the device upon power up or completion of reset is a normal operational state without management intervention.

##### 45.2.7.20.1 AN reset (7.512.15)

Resetting AN is accomplished by setting bit 7.512.15 to a one. This action shall set all BASE-T1 AN registers to their default states. As a consequence, this action may change the internal state of AN and the state of the physical link. This action may also initiate a reset in any other MMDs that are instantiated in the

**Table 45–395—BASE-T1 AN control register bit definitions**

| Bit(s)      | Name                     | Description                                                                                         | R/W <sup>a</sup> |
|-------------|--------------------------|-----------------------------------------------------------------------------------------------------|------------------|
| 7.512.15    | AN reset                 | 1 = AN reset<br>0 = AN normal operation                                                             | R/W, SC          |
| 7.512.14:13 | Reserved                 | Value always 0                                                                                      | RO               |
| 7.512.12    | Auto-Negotiation enable  | 1 = enable Auto-Negotiation process<br>0 = disable Auto-Negotiation process                         | R/W              |
| 7.512.11:10 | Reserved                 | Value always 0                                                                                      | RO               |
| 7.512.9     | Restart Auto-Negotiation | 1 = Restart Auto-Negotiation process<br>0 = Auto-Negotiation in process, disabled, or not supported | R/W, SC          |
| 7.512.8:0   | Reserved                 | Value always 0                                                                                      | RO               |

<sup>a</sup>RO = Read only, R/W = Read/Write, SC = Self-clearing

same package. This bit is self-clearing, and AN shall return a value of one in bit 7.512.15 when a reset is in progress and a value of zero otherwise. AN is not required to accept a write transaction to any of its registers until the reset process is complete. The reset process shall be completed within 0.5 s from the setting of bit 7.512.15. During an AN reset, AN shall respond to reads from register bit 7.512.15. All other register bits shall be ignored.

The default value for bit 7.512.15 is zero.

NOTE—This operation may interrupt data communication.

#### **45.2.7.20.2 Auto-Negotiation enable (7.512.12)**

The Auto-Negotiation function shall be enabled by setting bit 7.512.12 to a one. If bit 7.512.12 is set to one, then type selection bits 1.2100.3:0 and MASTER-SLAVE bits 1.2100.14 shall have no effect on the link configuration, and the Auto-Negotiation process determines the link configuration. If bit 7.512.12 is cleared to zero, then bits 1.2100.3:0 and bit 1.2100.14 determine the link configuration regardless of the prior state of the link configuration and the Auto-Negotiation process.

If the BASE-T1 PHY reports via bit 7.513.3 that it lacks the ability to perform Auto-Negotiation, then the value of bit 7.512.12 shall be zero.

#### **45.2.7.20.3 Restart Auto-Negotiation (7.512.9)**

If the BASE-T1 PMA/PMD reports (via bit 7.513.3) that it lacks the ability to perform Auto-Negotiation, or if Auto-Negotiation is disabled, the BASE-T1 PMA/PMD shall return a value of zero in bit 7.512.9 and any attempt to write a one to bit 7.512.9 shall be ignored.

Otherwise, the Auto-Negotiation process shall be restarted by setting bit 7.512.9 to one. This bit is self-clearing, and the BASE-T1 PMA/PMD shall return a value of one in bit 7.512.9 until the Auto-Negotiation process has been initiated. If Auto-Negotiation was completed prior to this bit being set, the process shall be reinitialized. The Auto-Negotiation process shall not be affected by clearing this bit to zero.

The default value for 7.512.9 is zero.

#### 45.2.7.21 BASE-T1 AN status (Register 7.513)

The assignment of bits in the BASE-T1 AN status register is shown in Table 45–396. All the bits in the BASE-T1 AN status register are read only; therefore, a write to the BASE-T1 AN status register shall have no effect.

**Table 45–396—BASE-T1 AN status register bit definitions**

| Bit(s)     | Name                      | Description                                                                                    | R/W <sup>a</sup> |
|------------|---------------------------|------------------------------------------------------------------------------------------------|------------------|
| 7.513.15:7 | Reserved                  | Value always 0                                                                                 | RO               |
| 7.513.6    | Page received             | 1 = A page has been received<br>0 = A page has not been received                               | RO, LH           |
| 7.513.5    | Auto-Negotiation complete | 1 = Auto-Negotiation process completed<br>0 = Auto-Negotiation process not completed           | RO               |
| 7.513.4    | Remote fault              | 1 = remote fault condition detected<br>0 = no remote fault condition detected                  | RO, LH           |
| 7.513.3    | Auto-Negotiation ability  | 1 = PHY is able to perform Auto-Negotiation<br>0 = PHY is not able to perform Auto-Negotiation | RO               |
| 7.513.2    | Link status               | 1 = Link is up<br>0 = Link is down                                                             | RO, LL           |
| 7.513.1:0  | Reserved                  | Value always 0                                                                                 | RO               |

<sup>a</sup>RO = Read only, LH = Latching High, LL = Latching Low

##### 45.2.7.21.1 Page received (7.513.6)

The Page received bit (7.513.6) shall be set to one to indicate that a new link codeword has been received and stored in the BASE-T1 AN LP Base Page ability registers 7.517 to 7.519 or the BASE-T1 AN LP Next Page ability registers 7.523 to 7.525. The contents of the BASE-T1 AN LP Base Page ability registers 7.517 to 7.519 are valid when bit 7.513.6 is set the first time during the Auto-Negotiation. The Page received bit shall be reset to zero on a read of the BASE-T1 AN status register (register 7.513).

##### 45.2.7.21.2 Auto-Negotiation complete (7.513.5)

When read as a one, bit 7.513.5 indicates that the Auto-Negotiation process has been completed, and that the contents of the Auto-Negotiation registers 7.514 to 7.516 and 7.517 to 7.519 are valid. When read as a zero, bit 7.513.5 indicates that the Auto-Negotiation process has not been completed, and that the contents of 7.517 through 7.525 registers are as defined by the current state of the Auto-Negotiation protocol, or as written for manual configuration. The BASE-T1 PMA/PMD shall return a value of zero in bit 7.513.5 if Auto-Negotiation is disabled by clearing bit 7.512.12. The BASE-T1 PMA/PMD shall also return a value of zero in bit 7.513.5 if it lacks the ability to perform Auto-Negotiation.

##### 45.2.7.21.3 Remote fault (7.513.4)

When read as one, bit 7.513.4 indicates that a remote fault condition has been detected. The type of fault as well as the criteria and method of fault detection is AN specific. The remote fault bit shall be implemented with a latching function, such that the occurrence of a remote fault causes the bit 7.513.4 to become set and

remain set until it is cleared. Bit 7.513.4 shall be cleared each time register 7.513 is read via the management interface, and shall also be cleared by a AN reset.

#### 45.2.7.21.4 Auto-Negotiation ability (7.513.3)

When read as a one, bit 7.513.3 indicates that the BASE-T1 PMA/PMD has the ability to perform BASE-T1 Auto-Negotiation. When read as a zero, bit 7.513.3 indicates that the BASE-T1 PMA/PMD lacks the ability to perform BASE-T1 Auto-Negotiation.

#### 45.2.7.21.5 Link status (7.513.2)

When read as a one, bit 7.513.2 indicates that the BASE-T1 PMA/PMD has determined that a valid link has been established. When read as a zero, bit 7.513.2 indicates that the link has been invalid after this bit was last read. Bit 7.513.2 is set to one when the variable link\_status equals OK and is cleared to zero when the variable link\_status equals FAIL. The link status bit shall be implemented with a latching function, such that the occurrence of a link\_status equals FAIL condition causes the link status bit to become cleared and remain cleared until it is read via the management interface. Bit 7.513.2 shall be cleared upon AN reset. This status indication is intended to support the management attribute defined in 30.5.1.1.4, aMediaAvailable.

#### 45.2.7.22 BASE-T1 AN advertisement register (Registers 7.514, 7.515, and 7.516)

The assignment of bits in the BASE-T1 AN advertisement register is shown in Table 45–397.

The Selector field (7.514.4:0) shall be set to the IEEE 802.3 code as specified in Annex 98A. The Acknowledge bit (7.514.14) is set to zero.

The technology ability field, as defined in 98.2.1.2, represents the technologies supported by the local device. Only bits representing supported technologies may be set. Management may clear bits in the technology ability field and restart Auto-Negotiation to negotiate an alternate common mode.

The management entity initiates renegotiation with the link partner using alternate abilities by setting the Restart Auto-Negotiation bit (7.512.9) in the AN control register to one.

Any writes to this register prior to completion of Auto-Negotiation, as indicated by bit 7.513.5, should be followed by a renegotiation for the new values to take effect. Once Auto-Negotiation has completed, software may examine this register along with the LP Base Page ability register to determine the highest common denominator technology.

The Base Page value is transferred to mr\_adv\_ability when register 7.514 is written. Therefore, registers 7.515 and 7.516 should be written before 7.514.

**Table 45–397—BASE-T1 AN advertisement register bit definitions**

| Bit(s)     | Name         | Description    | R/W <sup>a</sup> |
|------------|--------------|----------------|------------------|
| 7.514.15   | Next Page    | See 98.2.1.2.9 | R/W              |
| 7.514.14   | Acknowledge  | Value always 0 | RO               |
| 7.514.13   | Remote fault | See 98.2.1.2.7 | R/W              |
| 7.514.12:5 | D12:D5       | See 98.2.1.2   | R/W              |

**Table 45–397—BASE-T1 AN advertisement register bit definitions (*continued*)**

| Bit(s)     | Name           | Description   | R/W <sup>a</sup> |
|------------|----------------|---------------|------------------|
| 7.514.4:0  | Selector field | See Annex 98A | R/W              |
| 7.515.15:0 | D31:D16        | See 98.2.1.2  | R/W              |
| 7.516.15:0 | D47:D32        | See 98.2.1.2  | R/W              |

<sup>a</sup>RO = Read only, R/W = Read/Write

#### 45.2.7.23 BASE-T1 AN LP Base Page ability register (Registers 7.517, 7.518, and 7.519)

The assignment of bits in the BASE-T1 AN LP Base Page ability register is shown in Table 45–398.

All of the bits in the BASE-T1 AN LP Base Page ability register are read only. A write to the BASE-T1 AN LP Base Page ability register shall have no effect.

The value of the registers 7.518 and 7.519 is latched when register 7.517 is read and reads of registers 7.518 and 7.519 return the latched value rather than the current value.

**Table 45–398—BASE-T1 AN LP Base Page ability register bit definitions**

| Bit(s)     | Name    | Description  | R/W <sup>a</sup> |
|------------|---------|--------------|------------------|
| 7.517.15:0 | D15:D0  | See 98.2.1.2 | RO               |
| 7.518.15:0 | D31:D16 | See 98.2.1.2 | RO               |
| 7.519.15:0 | D47:D32 | See 98.2.1.2 | RO               |

<sup>a</sup>RO = Read only

#### 45.2.7.24 BASE-T1 AN Next Page transmit register (Registers 7.520, 7.521, and 7.522)

The assignment of bits in the BASE-T1 AN Next Page transmit register is shown in Table 45–399.

The register contains the BASE-T1 AN Next Page link codeword as defined in 98.2.4.3.

On power-up or AN reset, this register shall contain the default value, which represents a Message Page with the message code set to Null Message. This value may be replaced by any valid Next Page Message Code that the device intends to transmit. The BASE-T1 AN shall use Next Page Message Codes as defined in Annex 98C.

A write to register 7.521 or 7.522 does not set `mr_next_page_loaded`. Only a write to register 7.520 sets `mr_next_page_loaded` true as described in 98.5.1. Therefore registers 7.521 and 7.522 should be written before register 7.520.



**Table 45–399—BASE-T1 AN Next Page transmit register bit definitions**

| Bit(s)     | Name                           | Description    | R/W <sup>a</sup> |
|------------|--------------------------------|----------------|------------------|
| 7.520.15   | Next page                      | See 98.2.4.3   | R/W              |
| 7.520.14   | Reserved                       | Value always 0 | RO               |
| 7.520.13   | Message page                   | See 98.2.4.3   | R/W              |
| 7.520.12   | Acknowledge 2                  | See 98.2.4.3   | R/W              |
| 7.520.11   | Toggle                         | See 98.2.4.3   | RO               |
| 7.520.10:0 | Message/Unformatted code field | See 98.2.4.3   | R/W              |
| 7.521.15:0 | Unformatted code field 1       | See 98.2.4.3   | R/W              |
| 7.522.15:0 | Unformatted code field 2       | See 98.2.4.3   | R/W              |

<sup>a</sup>RO = Read only, R/W = Read/Write

#### 45.2.7.25 BASE-T1 AN LP Next Page ability register (Registers 7.523, 7.524, and 7.525)

BASE-T1 AN LP Next Page ability register (registers 7.523, 7.524, and 7.525) store BASE-T1 link partner Next Pages as shown in Table 45–400. All of the bits in the BASE-T1 AN LP Next Page ability register are read only. A write to the BASE-T1 AN LP Next Page ability register shall have no effect.

**Table 45–400—BASE-T1 AN LP Next Page ability register bit definitions**

| Bit(s)     | Name                           | Description  | R/W <sup>a</sup> |
|------------|--------------------------------|--------------|------------------|
| 7.523.15   | Next Page                      | See 98.2.4.3 | RO               |
| 7.523.14   | Acknowledge                    | See 98.2.4.3 | RO               |
| 7.523.13   | Message page                   | See 98.2.4.3 | RO               |
| 7.523.12   | Acknowledge 2                  | See 98.2.4.3 | RO               |
| 7.523.11   | Toggle                         | See 98.2.4.3 | RO               |
| 7.523.10:0 | Message/Unformatted code field | See 98.2.4.3 | RO               |
| 7.524.15:0 | Unformatted code field 1       | See 98.2.4.3 | RO               |
| 7.525.15:0 | Unformatted code field 2       | See 98.2.4.3 | RO               |

<sup>a</sup>RO = Read only

The value of registers 7.524 and 7.525 is latched when register 7.523 is read and reads of registers 7.524 and 7.525 return the latched value rather than the current value.

NOTE—If this register is used to store multiple link partner Next Pages, the previous value of this register is assumed to be stored by a management entity that needs the information overwritten by subsequent link partner Next Pages.

#### 45.2.7.26 10BASE-T1 AN control register (Register 7.526)

The assignment of bits in the 10BASE-T1 AN control register is shown in Table 45–401. The default value for each bit of the 10BASE-T1 AN control register has been chosen so that the initial state of the device upon power up or completion of reset is a normal operational state without management intervention.

**Table 45–401—10BASE-T1 AN control register bit definitions**

| Bit(s)     | Name                                                              | Description                                                                                                                                                                               | R/W <sup>a</sup> |
|------------|-------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 7.526.15   | 10BASE-T1L capability advertisement                               | 1 = Advertise PHY as 10BASE-T1L capable<br>0 = Do not advertise PHY as 10BASE-T1L capable                                                                                                 | R/W              |
| 7.526.14   | 10BASE-T1L EEE ability advertisement                              | 1 = Advertise that the 10BASE-T1L PHY has EEE ability<br>0 = Do not advertise that the 10BASE-T1L PHY has EEE ability (default)                                                           | R/W              |
| 7.526.13   | 10BASE-T1L increased transmit/receive level ability advertisement | 1 = Advertise that the 10BASE-T1L PHY has increased transmit/receive level ability<br>0 = Do not advertise that the 10BASE-T1L PHY has increased transmit/receive level ability (default) | R/W              |
| 7.526.12   | 10BASE-T1L increased transmit level request                       | 1 = Request 10BASE-T1L increased transmit level<br>0 = Do not request 10BASE-T1L increased transmit level (default)                                                                       | R/W              |
| 7.526.11:8 | Reserved                                                          | Value always 0                                                                                                                                                                            | RO               |
| 7.526.7    | 10BASE-T1S full duplex ability advertisement                      | 1 = Advertise that the 10BASE-T1S PHY has full duplex ability<br>0 = Do not advertise that the 10BASE-T1S PHY has full duplex ability                                                     | R/W              |
| 7.526.6    | 10BASE-T1S half duplex capability advertisement                   | 1 = Advertise PHY as 10BASE-T1S half duplex capable<br>0 = Do not advertise PHY as 10BASE-T1S half duplex capable                                                                         | R/W              |
| 7.526.5:0  | Reserved                                                          | Value always 0                                                                                                                                                                            | RO               |

<sup>a</sup>RO = Read only, R/W = Read/Write

##### 45.2.7.26.1 10BASE-T1L capability advertisement (7.526.15)

Bit 7.526.15 is used to select whether Auto-Negotiation advertises the capability to operate as a 10BASE-T1L PHY. If bit 7.526.15 is set to one, the PHY shall advertise 10BASE-T1L capability. If bit 7.526.15 is set to zero, the PHY shall not advertise the capability to operate as a 10BASE-T1L PHY.

##### 45.2.7.26.2 10BASE-T1L EEE ability advertisement (7.526.14)

If the device supports EEE ability for 10BASE-T1L, as defined in 146.1.2.3, and EEE operation is desired, bit 7.526.14 shall be set to one.

#### 45.2.7.26.3 10BASE-T1L increased transmit/receive level ability advertisement (7.526.13)

If the device supports the 2.4 Vpp operating mode for 10BASE-T1L, as defined in 146.5.4.1, bit 7.526.13 shall be set to one.

#### 45.2.7.26.4 10BASE-T1L increased transmit level request (7.526.12)

If the device supports the 2.4 Vpp operating mode for 10BASE-T1L, as defined in 146.5.4.1, and the 2.4 Vpp transmit voltage operation is desired, bit 7.526.12 is set to one. Bit 7.526.12 is used to select whether Auto-Negotiation advertises a request to operate the 10BASE-T1L PHY in increased transmit level mode. If bit 7.526.12 is set to one, the PHY shall advertise a request to operate the 10BASE-T1L PHY in increased transmit level mode. If bit 7.526.12 is set to zero, the PHY shall not advertise a request to operate the 10BASE-T1L PHY in increased transmit level mode.

#### 45.2.7.26.5 10BASE-T1S full duplex ability advertisement (7.526.7)

Bit 7.526.7 is used to select whether Auto-Negotiation advertises the ability to operate the 10BASE-T1S PHY in full duplex mode. If bit 7.526.7 is set to one, the PHY shall advertise 10BASE-T1S full duplex capability. If bit 7.526.7 is set to zero, the PHY shall not advertise the ability to operate in 10BASE-T1S full duplex mode.

#### 45.2.7.26.6 10BASE-T1S half duplex capability advertisement (7.526.6)

Bit 7.526.6 is used to select whether Auto-Negotiation advertises the capability to operate the 10BASE-T1S PHY in half duplex mode. If bit 7.526.6 is set to one, the PHY shall advertise 10BASE-T1S half duplex capability. If bit 7.526.6 is set to zero, the PHY shall not advertise the capability to operate in 10BASE-T1S half duplex mode.

#### 45.2.7.27 10BASE-T1 AN status register (Register 7.527)

The assignment of bits in the 10BASE-T1 AN status register is shown in Table 45–402. All the bits in the 10BASE-T1 AN status register are read only; therefore, a write to the 10BASE-T1 AN status register shall have no effect.

When the AN process has been completed, this register shall reflect the contents of the link partner's 10BASE-T1 AN control register. The definitions for the contents of the 10BASE-T1 AN status register are given by the definitions for the contents on the link partner's 10BASE-T1 control register, 7.526 (see 45.2.7.26).

**Table 45–402—10BASE-T1 AN status register bit definitions**

| Bit(s)   | Name                                              | Description                                                                                                                                            | R/W <sup>a</sup> |
|----------|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 7.527.15 | 10BASE-T1L link partner capability advertisement  | 1 = Link partner is advertising PHY as 10BASE-T1L capable<br>0 = Link partner is not advertising PHY as 10BASE-T1L capable                             | RO               |
| 7.527.14 | 10BASE-T1L link partner EEE ability advertisement | 1 = Link partner is advertising that the 10BASE-T1L PHY has EEE ability<br>0 = Link partner is not advertising that the 10BASE-T1L PHY has EEE ability | RO               |

**Table 45–402—10BASE-T1 AN status register bit definitions (*continued*)**

| Bit(s)     | Name                                                                           | Description                                                                                                                                                                                                      | R/W <sup>a</sup> |
|------------|--------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 7.527.13   | 10BASE-T1L link partner increased transmit/receive level ability advertisement | 1 = Link partner is advertising that the 10BASE-T1L PHY has increased transmit/receive level ability<br>0 = Link partner is not advertising that the 10BASE-T1L PHY has increased transmit/receive level ability | RO               |
| 7.527.12   | 10BASE-T1L link partner increased transmit level request                       | 1 = Link partner is requesting 10BASE-T1L link partner increased transmit level<br>0 = Link partner is not requesting 10BASE-T1L link partner increased transmit level                                           | RO               |
| 7.527.11:8 | Reserved                                                                       | Value always 0                                                                                                                                                                                                   | RO               |
| 7.527.7    | 10BASE-T1S link partner full duplex ability advertisement                      | 1 = Link partner is advertising that the 10BASE-T1S PHY has full duplex ability<br>0 = Link partner is not advertising that the 10BASE-T1S PHY has full duplex ability                                           | RO               |
| 7.527.6    | 10BASE-T1S link partner half duplex capability advertisement                   | 1 = Link partner is advertising PHY as 10BASE-T1S half duplex capable<br>0 = Link partner is not advertising PHY as 10BASE-T1S half duplex capable                                                               | RO               |
| 7.527.5:0  | Reserved                                                                       | Value always 0                                                                                                                                                                                                   | RO               |

<sup>a</sup>RO = Read only

#### 45.2.8 OFDM PMA/PMD registers

The assignment of registers in the OFDM PMA/PMD MMD is shown in Table 45–403.

**Table 45–403—OFDM PMA/PMD registers**

| Register address          | Register name                            | Subclause |
|---------------------------|------------------------------------------|-----------|
| 12.0                      | 10GPASS-XR DS OFDM channel ID            | 45.2.8.1  |
| 12.1 through 12.1023      | 10GPASS-XR DS profile descriptor control | 45.2.8.2  |
| 12.1024 through 12.2047   | 10GPASS-XR US profile descriptor control | 45.2.8.3  |
| 12.2048 through 12.10239  | 10GPASS-XR US pre-equalizer coefficients | 45.2.8.4  |
| 12.10240 through 12.10241 | 10GPASS-XR receive MER control           | 45.2.8.5  |
| 12.10242 through 12.12287 | 10GPASS-XR receive MER measurement       | 45.2.8.6  |
| 12.12288 through 12.32767 | Reserved                                 |           |

#### 45.2.8.1 10GPASS-XR DS OFDM channel ID register (Register 12.0)

The assignment of bits in the DS OFDM channel ID register is shown in Table 45–404.

**Table 45–404—10GPASS-XR DS OFDM channel ID register bit definitions**

| Bit(s)    | Name               | Description                                                                              | R/W <sup>a</sup> |
|-----------|--------------------|------------------------------------------------------------------------------------------|------------------|
| 12.0.15:3 | Reserved           | Value always 0                                                                           | RO               |
| 12.0.2:0  | DS OFDM channel ID | Identifies to which OFDM channel (1 to 5) registers 12.1 through 12.1023 currently apply | R/W              |

<sup>a</sup>RO = Read only, R/W = Read/Write

##### 45.2.8.1.1 DS OFDM channel ID (12.0.2:0)

Bits 12.0.2:0 form a pointer to one of the five possible OFDM channels in the downstream EPoC network. Thus when this register is set to a value of one registers 12.1 through 12.1023 reflect the OFDM profile descriptor for OFDM channel one. When this register is set to a value of two registers 12.1 through 12.1023 reflect the OFDM profile descriptor for OFDM channel two, etc. These bits are a reflection of the *DS\_OFDM\_ID* variable defined in 101.4.3.4.5.

#### 45.2.8.2 10GPASS-XR DS profile descriptor control 1 through 1023 (Registers 12.1 through 12.1023)

The 10GPASS-XR DS profile descriptor control registers determine the modulation parameters for each downstream OFDM subcarrier. Each register in the group controls 4 of the 4096 subcarriers that are transmitted over the OFDM channel. Register 12.1 describes modulation parameters for downstream OFDM subcarriers number 4 through 7. Register 12.2 describes modulation parameters for downstream OFDM subcarriers number 8 through 11. Finally, register 12.1023 describes modulation parameters for downstream OFDM subcarriers number 4092 through 4095. Note that the first four subcarriers (i.e., subcarriers number 0 through 3) are not specified and are always excluded.

The assignment of bits in register 12.1 is shown in Table 45–405. The remaining registers 12.2 through 12.1023 have the same bit structure as that of register 12.1. Changing registers 12.1 through 12.1023 does not affect the active profile, only the inactive profile (see 102.2.3.1.1 for a description of the Configuration ID bits in the PHY Link frame for information on active profile control). Registers 12.1 through 12.1023 are a reflection of variables *DS\_ModTypeSC(n)* defined in 101.4.3.4.5.

**Table 45–405—10GPASS-XR DS profile descriptor control 1 register bit definitions**

| Bit(s)     | Name                   | Description                         | R/W <sup>a</sup> |
|------------|------------------------|-------------------------------------|------------------|
| 12.1.15:12 | DS modulation type SC7 | Modulation profile for subcarrier 7 | R/W              |
| 12.1.11:8  | DS modulation type SC6 | Modulation profile for subcarrier 6 | R/W              |
| 12.1.7:4   | DS modulation type SC5 | Modulation profile for subcarrier 5 | R/W              |
| 12.1.3:0   | DS modulation type SC4 | Modulation profile for subcarrier 4 | R/W              |

<sup>a</sup>R/W = Read/Write

#### 45.2.8.2.1 DS modulation type SC7 (12.1.15:12)

Bits 12.1.15:12 indicate the modulation profile for downstream OFDM subcarrier number 7. These bits are a reflection of variable *DS\_ModTypeSC(7)* defined in 101.4.3.4.5. See the variable definition for interpretation of individual bits.

#### 45.2.8.2.2 DS modulation type SC6 (12.1.11:8)

Bits 12.1.11:8 indicate the modulation profile for downstream OFDM subcarrier number 6. These bits are a reflection of variable *DS\_ModTypeSC(6)* defined in 101.4.3.4.5. See the variable definition for interpretation of individual bits.

#### 45.2.8.2.3 DS modulation type SC5 (12.1.7:4)

Bits 12.1.7:4 indicate the modulation profile for downstream OFDM subcarrier number 5. These bits are a reflection of variable *DS\_ModTypeSC(5)* defined in 101.4.3.4.5. See the variable definition for interpretation of individual bits.

#### 45.2.8.2.4 DS modulation type SC4 (12.1.3:0)

Bits 12.1.3:0 indicate the modulation profile for downstream OFDM subcarrier number 4. These bits are a reflection of variable *DS\_ModTypeSC(4)* defined in 101.4.3.4.5. See the variable definition for interpretation of individual bits.

### 45.2.8.3 10GPASS-XR US profile descriptor control 0 through 1023 registers (Registers 12.1024 through 12.2047)

The 10GPASS-XR US profile descriptor control 0 through 1023 registers determine the inactive modulation settings for the upstream OFDMA spectrum. Each register in the group controls 4 subcarriers of the 4096 subcarriers that are transmitted over the OFDMA channel, with the first register in the group (12.1024) controlling subcarriers 0 through 3 and the second register controlling subcarriers 4 through 7, etc. The assignment of bits in the first 10GPASS-XR US profile descriptor control register (register 12.1024) is shown in Table 45–406. Bit assignment per subcarrier for the remaining registers is identical to that of register 12.1024. See 101.4.5 for details on each modulation type. Registers 12.1024 through 12.2047 are a reflection of variables *US\_ModTypeSC(n)* defined in 101.4.4.4.4.

**Table 45–406—10GPASS-XR US profile descriptor control 0 register bit definitions**

| Bit(s)        | Name                   | Description                         | R/W <sup>a</sup> |
|---------------|------------------------|-------------------------------------|------------------|
| 12.1024.15:12 | US modulation type SC3 | Modulation profile for subcarrier 3 | R/W              |
| 12.1024.11:8  | US modulation type SC2 | Modulation profile for subcarrier 2 | R/W              |
| 12.1024.7:4   | US modulation type SC1 | Modulation profile for subcarrier 1 | R/W              |
| 12.1024.3:0   | US modulation type SC0 | Modulation profile for subcarrier 0 | R/W              |

<sup>a</sup>R/W = Read/Write

#### 45.2.8.3.1 US modulation type SC3 (12.1024.15:12)

Bits 12.1024.15:12 indicate the modulation profile for upstream OFDM subcarrier number 3. These bits are a reflection of variable *US\_ModTypeSC(3)* defined in 101.4.4.4.4. See the variable definition for interpretation of individual bits.

#### 45.2.8.3.2 US modulation type SC2 (12.1024.11:8)

Bits 12.1024.11:8 indicate the modulation profile for upstream OFDM subcarrier number 2. These bits are a reflection of variable *US\_ModTypeSC(2)* defined in 101.4.4.4.4. See the variable definition for interpretation of individual bits.

#### 45.2.8.3.3 US modulation type SC1 (12.1024.7:4)

Bits 12.1024.7:4 indicate the modulation profile for upstream OFDM subcarrier number 1. These bits are a reflection of variable *US\_ModTypeSC(1)* defined in 101.4.4.4.4. See the variable definition for interpretation of individual bits.

#### 45.2.8.3.4 US modulation type SC0 (12.1024.3:0)

Bits 12.1024.3:0 indicate the modulation profile for upstream OFDM subcarrier number 0. These bits are a reflection of variable *US\_ModTypeSC(0)* defined in 101.4.4.4.4. See the variable definition for interpretation of individual bits.

#### 45.2.8.4 10GPASS-XR US pre-equalizer coefficients 0 through 4095 (Registers 12.2048 through 12.10239)

The 10GPASS-XR US pre-equalizer coefficients 0 through 4095 registers determine the real and imaginary parts of the pre-equalizer settings for the upstream transmitter. Each register pair in the group controls one subcarrier of the 4096 subcarriers that are transmitted over the OFDMA channel, with the first register in the group (12.2048) controlling the real number setting for subcarrier 0 and the second register (12.2049) controlling the imaginary number setting for subcarrier 0. The second register pair (12.2050 and 12.2051) respectively controls the real and imaginary settings for subcarrier 1. Thus the last register pair (12.10238 and 12.10239) controls the real and imaginary settings for subcarrier 4095. The value in each register is in a Q2.14 format. The assignment of bits in the first and second 10GPASS-XR US pre-equalizer coefficients registers (registers 12.2048 and 12.2049) is shown in Table 45–407. Bit assignment per subcarrier for the remaining even and odd registers is identical to that of registers 12.2048 and 12.2049, respectively. See 101.4.4.9 for details on use of these registers. These registers are a reflection of the variables *EQ\_CoeffR(n)* and *EQ\_CoeffI(n)* defined in 101.4.4.9.2.

**Table 45–407—10GPASS-XR US pre-equalizer coefficients registers bit definitions**

| Bit(s)       | Name                                    | Description                                                      | R/W <sup>a</sup> |
|--------------|-----------------------------------------|------------------------------------------------------------------|------------------|
| 12.2048.15:0 | Real pre-equalizer coefficient SC0      | Real part of the pre-equalizer coefficient for subcarrier 0      | R/W              |
| 12.2049.15:0 | Imaginary pre-equalizer coefficient SC0 | Imaginary part of the pre-equalizer coefficient for subcarrier 0 | R/W              |

<sup>a</sup>R/W = Read/Write

#### 45.2.8.4.1 Real pre-equalizer coefficient SC0 (12.2048.15:0)

Bits 12.2048.15:0 indicate the real part of the pre-equalizer coefficient for subcarrier 0 for the upstream OFDMA channel. These bits are a reflection of the variable  $EQ\_CoefR(0)$  defined in 101.4.4.9.2.

#### 45.2.8.4.2 Imaginary pre-equalizer coefficient SC0 (12.2049.15:0)

Bits 12.2049.15:0 indicate the imaginary part of the pre-equalizer coefficient for subcarrier 0 for the upstream OFDMA channel. These bits are a reflection of the variable  $EQ\_CoefI(0)$  defined in 101.4.4.9.2.

#### 45.2.8.5 10GPASS-XR receive MER control registers (Registers 12.10240 and 12.10241)

The assignment of bits in the 10GPASS-XR receive MER control registers is shown in Table 45–408.

**Table 45–408—10GPASS-XR receive MER control register bit definitions**

| Bit(s)        | Name                   | Description                                                                                                                                                     | R/W <sup>a</sup> |
|---------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 12.10240.15:4 | Reserved               | Value always 0                                                                                                                                                  | RO               |
| 12.10240.3    | MER measurement valid  | 1 = all values in the 10GPASS-XR receive MER measurement registers are valid<br>0 = some values in the 10GPASS-XR receive MER measurement registers are invalid | RO               |
| 12.10240.2:0  | Receive MER channel ID | Identifies to which OFDM channel (1 to 5) registers 12.10242 through 12.12287 currently apply                                                                   | R/W <sup>b</sup> |
| 12.10241.15   | Reserved               | Value always 0                                                                                                                                                  | RO               |
| 12.10241.14:0 | Receive MER CNU ID     | Identifies to which CNU registers 12.10242 through 12.12287 currently apply                                                                                     |                  |

<sup>a</sup>RO = Read only, R/W = Read/Write

<sup>b</sup>These bits are valid only in the CNU; in the CLT these bits are reserved and always 0.

##### 45.2.8.5.1 MER measurement valid (12.10240.3)

When read as one, bit 12.10240.3 indicates the 10GPASS-XR receive MER measurement registers are valid. When read as zero, this bit indicates that the 10GPASS-XR receive MER measurement registers are not valid. This bit is a reflection of the variable  $RxMER\_Valid$  defined in 100.3.5.3.1.

##### 45.2.8.5.2 Receive MER Channel ID (12.10240.2:0)

Bits 12.10240.2:0 form a pointer to one of the five possible OFDM channels in the EPoC network. These bits are only valid for 10GPASS-XR-U PMA/PMD and are reserved for 10GPASS-XR-D PMA/PMD and always read as zero. These bits are a reflection of the variable  $RxMER\_ChID$  defined in 100.3.5.3.1.

##### 45.2.8.5.3 Receive MER CNU ID (12.10241.14:0)

Bits 12.10241.14:0 indicate the CNU\_ID of the CNU on which to measure the MER and report in registers 12.10242 through 12.12287. These bits are only valid for 10GPASS-XR-D PMA/PMD and are reserved for 10GPASS-XR-U PMA/PMD and always read as zero. These bits are a reflection of the variable  $RxMER\_CNU\_ID$  defined in 100.3.5.3.1.



#### 45.2.8.6 10GPASS-XR receive MER measurement registers (Registers 12.10242 through 12.12287)

The 10GPASS-XR receive MER measurement registers reflect the MER measured on each OFDM subcarrier for the OFDM channel indicated by the Receive MER Channel ID except subcarriers zero through three. Each register in the group reflects two of the 4096 subcarriers that are transmitted over the OFDM channel. Register 12.10242 reflects the receive MER measured on OFDM subcarriers number 4 and 5. Register 12.10243 reflects the receive MER measured on OFDM subcarriers number 6 and 7. Finally, register 12.12287 reflects the receive MER measured on OFDM subcarriers number 4094 and 4095. Note that the first four subcarriers (i.e., subcarriers number 0 through 3) are not reflected in register group 12.10242 through 12.12287 (10GPASS-XR receive MER measurement registers).

The assignment of individual bits in register 12.10242 is shown in Table 45–409. The remaining registers 12.10243 through 12.12287 have the same bit structure as that of register 12.10242. These registers are a reflection of the variable  $RxMER\_SC(n)$  defined in 100.3.5.3.1.

**Table 45–409—10GPASS-XR receive MER measurement register bit definitions**

| Bit(s)        | Name            | Description                                                                                          | R/W <sup>a</sup> |
|---------------|-----------------|------------------------------------------------------------------------------------------------------|------------------|
| 12.10242.15:8 | Receive MER SC5 | Receive MER measurement for subcarrier 5 on the OFDM channel indicated by the Receive MER Channel ID | RO               |
| 12.10242.7:0  | Receive MER SC4 | Receive MER measurement for subcarrier 4 on the OFDM channel indicated by the Receive MER Channel ID | RO               |

<sup>a</sup>RO = Read only

##### 45.2.8.6.1 Receive MER SC5 (12.10242.15:8)

Bits 12.10242.15:8 reflect the MER measured on OFDM subcarrier 5 for the OFDM channel indicated by the Receive MER channel ID.

##### 45.2.8.6.2 Receive MER SC4 (12.10242.7:0)

Bits 12.10242.7:0 reflect the MER measured on OFDM subcarrier 4 for the OFDM channel indicated by the Receive MER channel ID.

## 45.2.9 Power Unit registers

The assignment of registers in the Power Unit MMD is shown in Table 45–410.

**Table 45–410—Power Unit MMD registers**

| Register address | Register name     | Subclause |
|------------------|-------------------|-----------|
| 13.0             | PoDL PSE Control  | 45.2.9.1  |
| 13.1             | PoDL PSE Status 1 | 45.2.9.2  |
| 13.2             | PoDL PSE Status 2 | 45.2.9.3  |
| 13.3             | PoDL PSE Status 3 | 45.2.9.4  |
| 13.4             | PoDL PSE Status 4 | 45.2.9.5  |

### 45.2.9.1 PoDL PSE Control register (Register 13.0)

The assignment of bits in the PoDL PSE Control register is shown in Table 45–411. The default value for each bit of the PoDL PSE Control register should be chosen so that the initial state of the PSE upon power up or reset is a normal operational state without management intervention.

**Table 45–411—PoDL PSE Control register bit definitions**

| Bit(s)    | Name                                | Description                                                                           | R/W <sup>a</sup> |
|-----------|-------------------------------------|---------------------------------------------------------------------------------------|------------------|
| 13.0.15:3 | Reserved                            | Value always 0                                                                        | RO               |
| 13.0.2    | Enable cable resistance measurement | 1 = Cable resistance measurement enabled<br>0 = Cable resistance measurement disabled | R/W              |
| 13.0.1    | Enable power classification         | 1 = Power classification enabled<br>0 = Power classification disabled                 | R/W              |
| 13.0.0    | PSE Enable                          | 1 = PSE Enabled<br>0 = PSE Disabled                                                   | R/W              |

<sup>a</sup>R/W = Read/Write, RO = Read Only

#### 45.2.9.1.1 Enable power classification (13.0.1)

The power classification function is enabled by setting bit 13.0.1 to one and disabled by setting bit 13.0.1 to zero. This bit maps to the `mr_sccp_enabled` variable (see 104.4.4.3).

#### 45.2.9.1.2 PSE Enable (13.0.0)

When bit 13.0.0 is set to zero, the PSE function shall be disabled. When bit 13.0.0 is set to one, the PSE function shall be enabled. This register bit maps to the `mr_pse_enable` variable (see 104.4.4.3).

### 45.2.9.2 PoDL PSE Status 1 register (Register 13.1)

The assignment of bits in the PoDL PSE Status 1 register is shown in Table 45–412.

### Table 45-412—PoDL PSE Status 1 register bit definitions

| Bit(s)   | Name              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | R/W <sup>a</sup> |
|----------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 13.1.15  | Power Denied      | 1 = Power has been denied due to power not available<br>0 = Power has not been denied due to power not available                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | RO/<br>LH        |
| 13.1.14  | Valid Signature   | 1 = Valid PD signature detected<br>0 = No valid PD signature detected                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | RO/<br>LH        |
| 13.1.13  | Invalid Signature | 1 = Invalid PD signature detected<br>0 = No invalid PD signature detected                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | RO/<br>LH        |
| 13.1.12  | Class Timeout     | 1 = Classification timeout condition detected<br>0 = No Classification timeout condition detected                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | RO/<br>LH        |
| 13.1.11  | Overload          | 1 = Overload condition detected<br>0 = No overload condition detected                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | RO/<br>LH        |
| 13.1.10  | MFVS Absent       | 1 = MFVS absent condition detected<br>0 = No MFVS absent condition detected                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | RO/<br>LH        |
| 13.1.9:7 | PSE Type          | 9      8      7<br>1    1    x = Reserved<br>1    0    1 = Type F PSE<br>1    0    0 = Type E PSE<br>0    1    1 = Type D PSE<br>0    1    0 = Type C PSE<br>0    0    1 = Type B PSE<br>0    0    0 = Type A PSE                                                                                                                                                                                                                                                                                                                                                                                                                 | RO               |
| 13.1.6:3 | PD Class          | 6    5    4    3<br>1    1    1    1 = Extend to PoDL PSE status 2 register<br><br>1    1    1    0 = Class code 14<br>1    1    0    1 = Class code 13<br>1    1    0    0 = Class code 12<br>1    0    1    1 = Class code 11<br>1    0    1    0 = Class code 10<br>1    0    0    1 = Class code 9<br>1    0    0    0 = Class code 8<br>0    1    1    1 = Class code 7<br>0    1    1    0 = Class code 6<br>0    1    0    1 = Class code 5<br>0    1    0    0 = Class code 4<br>0    0    1    1 = Class code 3<br>0    0    1    0 = Class code 2<br>0    0    0    1 = Class code 1<br>0    0    0    0 = Class code 0 | RO               |
| 13.1.2:0 | PSE Status        | 2    1    0<br>1    1    1 = Unknown<br>1    1    0 = Reserved<br>1    0    1 = Idle<br>1    0    0 = Error<br>0    1    1 = Searching<br>0    1    0 = Delivering power<br>0    0    1 = Sleep<br>0    0    0 = Disabled                                                                                                                                                                                                                                                                                                                                                                                                         | RO               |

<sup>a</sup>RO = Read Only, LH = Latching High

#### **45.2.9.2.1 Power Denied (13.1.15)**

When read as a one, bit 13.1.15 indicates that application of full operating voltage at the PI has been denied. This bit shall be set to one when power\_available transitions from true to false (see 104.4.4.3). The Power Denied bit shall be implemented with latching high behavior as defined in 45.2.

#### **45.2.9.2.2 Valid Signature (13.1.14)**

When read as a one, bit 13.1.14 indicates that a valid signature has been detected. This bit shall be set to one when mr\_valid\_signature transitions from false to true (see 104.4.4.3). The Valid Signature bit shall be implemented with latching high behavior as defined in 45.2.

#### **45.2.9.2.3 Invalid Signature (13.1.13)**

When read as a one, bit 13.1.13 indicates that an invalid signature has been detected. This bit maps to the PSE state diagram variable mr\_invalid\_signature, and latches high when mr\_invalid\_signature transitions from false to true (see 104.4.4.3). The Invalid Signature bit shall be implemented with latching high behavior as defined in 45.2.

#### **45.2.9.2.4 Class Timeout (13.1.12)**

When read as a one, bit 13.1.12 indicates that a Classification timeout condition has been detected. The Class Timeout bit shall be set to one when tclass\_timer\_done transitions from false to true (see 104.4.4.3). The Class Timeout bit shall be implemented with latching high behavior as defined in 45.2.

#### **45.2.9.2.5 Overload (13.1.11)**

When read as a one, bit 13.1.11 indicates that an overload condition has been detected. This bit shall be set to one when the PSE state diagram variable overload\_held transitions from false to true (see 104.4.4.3). The Overload bit shall be implemented with latching high behavior as defined in 45.2.

#### **45.2.9.2.6 MFVS Absent (13.1.10)**

When read as a one, bit 13.1.10 indicates that an MFVS Absent condition has been detected. The MFVS Absent bit shall be set to one when mfvs\_timeout transitions from false to true (see 104.4.4.3). The MFVS Absent bit shall be implemented with latching high behavior as defined in 45.2.

#### **45.2.9.2.7 PSE Type (13.1.9:7)**

Bits 13.1.9:7 report the PSE Type of the PSE as specified in 104.4.1. When read as 000, bits 13.1.9:7 indicate a Type A PSE; when read as 001, a Type B PSE is indicated; when read as 010, a Type C PSE is indicated; when read as 011, a Type D PSE is indicated; when read as 100, a Type E PSE is indicated; and when read as 101, a Type F PSE is indicated. Values of 11x are reserved.

#### **45.2.9.2.8 PD Class (13.1.6:3)**

Bits 13.1.6:3 report the PD Class of a detected PD as specified in 104.5.3. The value in this register is valid while a PD is connected, i.e., while the PSE Status (13.1.2:0) bits are reporting “delivering power”. When read as 0000, a Class 0 PD is indicated; when read as 0001, a Class 1 PD is indicated; when read as 0010, a Class 2 PD is indicated; when read as 0011, a Class 3 PD is indicated; when read as 0100, a Class 4 PD is indicated; when read as 0101, a Class 5 PD is indicated; when read as 0110, a Class 6 PD is indicated; when read as 0111, a Class 7 PD is indicated; when read as 1000, a Class 8 PD is indicated; when read as 1001, a Class 9 PD is indicated; when read as 1010, a Class 10 PD is indicated; when read as 1011, a Class 11 PD is indicated; when read as 1100, a Class 12 PD is indicated; when read as 1101, a Class 13 PD is indicated;

when read as 1110, a Class 14 PD is indicated; and when read as 1111, the Class is indicated by the PD Extended Class (13.2.10:9) bits.

#### 45.2.9.2.9 PSE Status (13.1.2:0)

Bits 13.1.2:0 report the current status of the PSE. When read as 000, bits 13.1.2:0 indicate that `mr_pse_enable` is asserted false (see 104.4.4.3). When read as 001, bits 13.1.2:0 indicate that `pi_sleeping` is asserted true. When read as 010, bits 13.1.2:0 indicate that `pi_powered` is asserted true. When read as 011, bits 13.1.2:0 indicate that either `pi_detecting` or `pi_classifying` is asserted true. When read as 100, bits 13.1.2:0 indicate that `overload_held` is asserted true. When read as 101, bits 13.1.2:0 indicate that the logical combination `pi_prebiased*!pi_sleeping` is true. When read as 111, bits 13.1.2:0 indicate that the PSE status is unknown. Bit combination 110 is reserved.

#### 45.2.9.3 PoDL PSE Status 2 register (Register 13.2)

The PoDL PSE Status 2 register is an extension of the PoDL PSE Status 1 register. Assignment of bits in the PoDL PSE Status 2 register is shown in Table 45–413.

**Table 45–413—PoDL PSE Status 2 register bit definitions**

| Bit(s)     | Name              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | R/W <sup>a</sup> |   |   |   |   |            |   |           |            |   |   |            |   |   |                 |             |   |   |   |             |   |   |   |             |   |   |   |             |   |   |   |             |   |   |   |             |    |
|------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|---|---|---|---|------------|---|-----------|------------|---|---|------------|---|---|-----------------|-------------|---|---|---|-------------|---|---|---|-------------|---|---|---|-------------|---|---|---|-------------|---|---|---|-------------|----|
| 13.2.15    | Invalid Class     | 1 = Invalid PD class detected<br>0 = No invalid PD class detected                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | RO/LH            |   |   |   |   |            |   |           |            |   |   |            |   |   |                 |             |   |   |   |             |   |   |   |             |   |   |   |             |   |   |   |             |   |   |   |             |    |
| 13.2.14:11 | Reserved          | Value always 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | RO               |   |   |   |   |            |   |           |            |   |   |            |   |   |                 |             |   |   |   |             |   |   |   |             |   |   |   |             |   |   |   |             |   |   |   |             |    |
| 13.2.10:9  | PD Extended Class | <table><tr><td>10</td><td>9</td><td></td></tr><tr><td>1</td><td>1</td><td>= Reserved</td></tr><tr><td>1</td><td>0</td><td>= Reserved</td></tr><tr><td>0</td><td>1</td><td>= Reserved</td></tr><tr><td>0</td><td>0</td><td>= Class code 15</td></tr></table>                                                                                                                                                                                                                                                                                          | 10               | 9 |   | 1 | 1 | = Reserved | 1 | 0         | = Reserved | 0 | 1 | = Reserved | 0 | 0 | = Class code 15 | RO          |   |   |   |             |   |   |   |             |   |   |   |             |   |   |   |             |   |   |   |             |    |
| 10         | 9                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                  |   |   |   |   |            |   |           |            |   |   |            |   |   |                 |             |   |   |   |             |   |   |   |             |   |   |   |             |   |   |   |             |   |   |   |             |    |
| 1          | 1                 | = Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                  |   |   |   |   |            |   |           |            |   |   |            |   |   |                 |             |   |   |   |             |   |   |   |             |   |   |   |             |   |   |   |             |   |   |   |             |    |
| 1          | 0                 | = Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                  |   |   |   |   |            |   |           |            |   |   |            |   |   |                 |             |   |   |   |             |   |   |   |             |   |   |   |             |   |   |   |             |   |   |   |             |    |
| 0          | 1                 | = Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                  |   |   |   |   |            |   |           |            |   |   |            |   |   |                 |             |   |   |   |             |   |   |   |             |   |   |   |             |   |   |   |             |   |   |   |             |    |
| 0          | 0                 | = Class code 15                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                  |   |   |   |   |            |   |           |            |   |   |            |   |   |                 |             |   |   |   |             |   |   |   |             |   |   |   |             |   |   |   |             |   |   |   |             |    |
| 13.2.8:3   | Reserved          | Value always 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | RO               |   |   |   |   |            |   |           |            |   |   |            |   |   |                 |             |   |   |   |             |   |   |   |             |   |   |   |             |   |   |   |             |   |   |   |             |    |
| 13.2.2:0   | PD Type           | <table><tr><td>2</td><td>1</td><td>0</td><td></td></tr><tr><td>1</td><td>1</td><td>1</td><td>= Unknown</td></tr><tr><td>1</td><td>1</td><td>0</td><td>= Reserved</td></tr><tr><td>1</td><td>0</td><td>1</td><td>= Type F PD</td></tr><tr><td>1</td><td>0</td><td>0</td><td>= Type E PD</td></tr><tr><td>0</td><td>1</td><td>1</td><td>= Type D PD</td></tr><tr><td>0</td><td>1</td><td>0</td><td>= Type C PD</td></tr><tr><td>0</td><td>0</td><td>1</td><td>= Type B PD</td></tr><tr><td>0</td><td>0</td><td>0</td><td>= Type A PD</td></tr></table> | 2                | 1 | 0 |   | 1 | 1          | 1 | = Unknown | 1          | 1 | 0 | = Reserved | 1 | 0 | 1               | = Type F PD | 1 | 0 | 0 | = Type E PD | 0 | 1 | 1 | = Type D PD | 0 | 1 | 0 | = Type C PD | 0 | 0 | 1 | = Type B PD | 0 | 0 | 0 | = Type A PD | RO |
| 2          | 1                 | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                  |   |   |   |   |            |   |           |            |   |   |            |   |   |                 |             |   |   |   |             |   |   |   |             |   |   |   |             |   |   |   |             |   |   |   |             |    |
| 1          | 1                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | = Unknown        |   |   |   |   |            |   |           |            |   |   |            |   |   |                 |             |   |   |   |             |   |   |   |             |   |   |   |             |   |   |   |             |   |   |   |             |    |
| 1          | 1                 | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | = Reserved       |   |   |   |   |            |   |           |            |   |   |            |   |   |                 |             |   |   |   |             |   |   |   |             |   |   |   |             |   |   |   |             |   |   |   |             |    |
| 1          | 0                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | = Type F PD      |   |   |   |   |            |   |           |            |   |   |            |   |   |                 |             |   |   |   |             |   |   |   |             |   |   |   |             |   |   |   |             |   |   |   |             |    |
| 1          | 0                 | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | = Type E PD      |   |   |   |   |            |   |           |            |   |   |            |   |   |                 |             |   |   |   |             |   |   |   |             |   |   |   |             |   |   |   |             |   |   |   |             |    |
| 0          | 1                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | = Type D PD      |   |   |   |   |            |   |           |            |   |   |            |   |   |                 |             |   |   |   |             |   |   |   |             |   |   |   |             |   |   |   |             |   |   |   |             |    |
| 0          | 1                 | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | = Type C PD      |   |   |   |   |            |   |           |            |   |   |            |   |   |                 |             |   |   |   |             |   |   |   |             |   |   |   |             |   |   |   |             |   |   |   |             |    |
| 0          | 0                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | = Type B PD      |   |   |   |   |            |   |           |            |   |   |            |   |   |                 |             |   |   |   |             |   |   |   |             |   |   |   |             |   |   |   |             |   |   |   |             |    |
| 0          | 0                 | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | = Type A PD      |   |   |   |   |            |   |           |            |   |   |            |   |   |                 |             |   |   |   |             |   |   |   |             |   |   |   |             |   |   |   |             |   |   |   |             |    |

<sup>a</sup>RO = Read Only, LH = Latching High

##### 45.2.9.3.1 Invalid Class (13.2.15)

When read as a one, bit 13.2.15 indicates that an invalid class has been detected. This bit maps to the PSE state diagram variable `valid_class`, and latches high when `do_classification_done` is true and `valid_class` is false (see 104.4.4.3). The Invalid Class bit shall be implemented with latching high behavior as defined in 45.2.

#### 45.2.9.3.2 PD Extended Class (13.2.10:9)

When bits 13.2.10:9 are read as 00, a Class 15 PD is indicated. Values of 01 and 1x are reserved.

#### 45.2.9.3.3 PD Type (13.2.2:0)

Bits 13.2.2:0 report a value of 111 until a valid classification has taken place, or if no PD is present. A value of 111 indicates that the PSE has not performed classification and therefore cannot indicate the proper value for the PD Type. Once a valid classification has occurred, the value of these bits reflect the PD Type of an attached PD as specified in 104.5.1. When read as 000, bits 13.2.2:0 indicate a Type A PD; when read as 001, a Type B PD is indicated; when read as 010, a Type C PD is indicated; when read as 011, a Type D PD is indicated; when read as 100, a Type E PD is indicated; and when read as 101, a Type F PD is indicated. A value of 110 is reserved.

#### 45.2.9.4 PoDL PSE Status 3 register (Register 13.3)

The PoDL PSE Status 3 register is defined if cable resistance measurement is supported.

**Table 45–414—PoDL PSE Status 3 register bit definitions**

| Bit(s)     | Name              | Description                        | R/W <sup>a</sup> |
|------------|-------------------|------------------------------------|------------------|
| 13.3.15:12 | Reserved          | Value always 0                     | RO               |
| 13.3.11:0  | PD Assigned Power | PD Assigned Power, 0.025 W per LSB | RO               |

<sup>a</sup>RO = Read only

##### 45.2.9.4.1 PD Assigned Power (13.3.11:0)

The PD Assigned Power is the maximum average available power at the PD PI.

#### 45.2.9.5 PoDL PSE Status 4 register (Register 13.4)

The PoDL PSE Status 4 register is defined if cable resistance measurement is supported.

**Table 45–415—PoDL PSE Status 4 register bit definitions**

| Bit(s)     | Name               | Description                         | R/W <sup>a</sup> |
|------------|--------------------|-------------------------------------|------------------|
| 13.4.15:12 | Reserved           | Value always 0                      | RO               |
| 13.4.11:0  | PD Requested Power | PD Requested Power, 0.025 W per LSB | RO               |

<sup>a</sup>RO = Read only

##### 45.2.9.5.1 PD Requested Power (13.4.11:0)

The PD Requested Power is the requested average available power at the PD PI.

#### 45.2.10 Clause 22 extension registers

As new management features are added to 10, 100, and 1000 Mb/s PHYs, more register space is required beyond that defined in Clause 22. The Clause 22 extension MMD provides this space. This MMD is defined only for 10, 100, and 1000 Mb/s PHYs. Since these PHYs do not segment their management by their sublayers, all management extensions to these PHYs will appear in the Clause 22 extension MMD.

The assignment of registers in the Clause 22 extension MMD is shown in Table 45–416.

**Table 45–416—Clause 22 extension registers**

| Register address        | Register name                            | Subclause |
|-------------------------|------------------------------------------|-----------|
| 29.0 through 29.4       | Reserved                                 |           |
| 29.5, 29.6              | Clause 22 extension devices in package   | 45.2.10.1 |
| 29.7                    | FEC capability                           | 45.2.10.2 |
| 29.8                    | FEC control                              | 45.2.10.3 |
| 29.9                    | FEC buffer head coding violation counter | 45.2.10.4 |
| 29.10                   | FEC corrected blocks counter             | 45.2.10.5 |
| 29.11                   | FEC uncorrected blocks counter           | 45.2.10.6 |
| 29.12 through 29.32 767 | Reserved                                 |           |

##### 45.2.10.1 Clause 22 extension devices in package registers (Registers 29.5, 29.6)

The Clause 22 extension devices in package registers are defined in Table 45–2.

##### 45.2.10.2 FEC capability register (Register 29.7)

The assignment of bits in the FEC capability register is shown in Table 45–417.

**Table 45–417—FEC capability register bit definitions**

| Bit(s)    | Name        | Description                              | R/W <sup>a</sup> |
|-----------|-------------|------------------------------------------|------------------|
| 29.7.15:1 | Reserved    | Value always 0                           | RO               |
| 29.7.0    | FEC capable | 1 = FEC supported<br>0 = FEC unsupported | RO               |

<sup>a</sup>RO = Read only

##### 45.2.10.2.1 FEC capable (29.7.0)

When read as a one, this bit indicates that the PHY supports forward error correction. When read as a zero, the PHY does not support forward error correction.

### 45.2.10.3 FEC control register (Register 29.8)

The assignment of bits in the FEC control register is shown in Table 45–418.

**Table 45–418—FEC control register bit definitions**

| Bit(s)    | Name       | Description                         | R/W <sup>a</sup> |
|-----------|------------|-------------------------------------|------------------|
| 29.8.15:1 | Reserved   | Value always 0                      | RO               |
| 29.8.0    | FEC enable | 1 = FEC enabled<br>0 = FEC disabled | R/W              |

<sup>a</sup>RO = Read only, R/W = Read/Write

#### 45.2.10.3.1 FEC enable (29.8.0)

When written as a one, this bit enables the PHY's forward error correction. When written as a zero, FEC is disabled. This bit shall be set to zero upon execution of a PHY reset.

### 45.2.10.4 FEC buffer head coding violation counter (Register 29.9)

The assignment of bits in the FEC buffer head coding violation counter register is shown in Table 45–419. See 65.2.3.6.1 for a definition of this register. These bits shall be reset to all zeros when the register is read by the management function or upon PHY reset. These bits shall be held at all ones in the case of overflow.

**Table 45–419—FEC buffer head coding violation counter register bit definitions**

| Bit(s)    | Name                                     | Description   | R/W <sup>a</sup> |
|-----------|------------------------------------------|---------------|------------------|
| 29.9.15:0 | FEC buffer head coding violation counter | Error counter | RO, NR           |

<sup>a</sup>RO = Read only, NR = Non Roll-over

### 45.2.10.5 FEC corrected blocks counter (Register 29.10)

The assignment of bits in the FEC corrected blocks counter register is shown in Table 45–420. See 65.2.3.6.2 for a definition of this register. These bits shall be reset to all zeros when the register is read by the management function or upon PHY reset. These bits shall be held at all ones in the case of overflow.

**Table 45–420—FEC corrected blocks counter register bit definitions**

| Bit(s)     | Name                         | Description   | R/W <sup>a</sup> |
|------------|------------------------------|---------------|------------------|
| 29.10.15:0 | FEC corrected blocks counter | Error counter | RO, NR           |

<sup>a</sup>RO = Read only, NR = Non Roll-over



#### 45.2.10.6 FEC uncorrected blocks counter (Register 29.11)

The assignment of bits in the FEC uncorrected blocks counter register is shown in Table 45–421. See 65.2.3.6.3 for a definition of this register. These bits shall be reset to all zeros when the register is read by the management function or upon PHY reset. These bits shall be held at all ones in the case of overflow.

**Table 45–421—FEC uncorrected blocks counter register bit definitions**

| Bit(s)     | Name                           | Description   | R/W <sup>a</sup> |
|------------|--------------------------------|---------------|------------------|
| 29.11.15:0 | FEC uncorrected blocks counter | Error counter | RO, NR           |

<sup>a</sup>RO = Read only, NR = Non Roll-over

#### 45.2.11 Vendor specific MMD 1 registers

The assignment of registers in the vendor specific MMD 1 is shown in Table 45–422. A vendor specific MMD may have a device address of either 30 or 31. It is recommended that the device address is configurable and that the configuration is performed by some means other than via the MDIO.

**Table 45–422—Vendor specific MMD 1 registers**

| Register address        | Register name                            |
|-------------------------|------------------------------------------|
| 30.0, 30.1              | Vendor specific                          |
| 30.2, 30.3              | Vendor specific MMD 1 device identifier  |
| 30.4 through 30.7       | Vendor specific                          |
| 30.8                    | Vendor specific MMD 1 status register    |
| 30.9 through 30.13      | Vendor specific                          |
| 30.14, 30.15            | Vendor specific MMD 1 package identifier |
| 30.16 through 30.65 535 | Vendor specific                          |

##### 45.2.11.1 Vendor specific MMD 1 device identifier (Registers 30.2 and 30.3)

Registers 30.2 and 30.3 provide a 32-bit value, which may constitute a unique identifier for a particular type of vendor-specific device. The identifier shall be composed of the 3rd through 24th bits of the Organizationally Unique Identifier (OUI) assigned to the device manufacturer by the IEEE, plus a six-bit model number, plus a four-bit revision number. A vendor-specific device may return a value of zero in each of the 32 bits of the vendor specific MMD 1 device identifier.

The format of the vendor specific MMD 1 device identifier is specified in 22.2.4.3.1.

#### 45.2.11.2 Vendor specific MMD 1 status register (Register 30.8)

The assignment of bits in the vendor specific MMD 1 status register is shown in Table 45–423. All the bits in the vendor specific MMD 1 status register are read only; a write to the vendor specific MMD 1 status register shall have no effect.

**Table 45–423—Vendor specific MMD 1 status register bit definitions**

| Bit(s)     | Name           | Description                                                                                                                                                                                | R/W <sup>a</sup> |
|------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 30.8.15:14 | Device present | 15 14<br>1 0 = Device responding at this address<br>1 1 = No device responding at this address<br>0 1 = No device responding at this address<br>0 0 = No device responding at this address | RO               |
| 30.8.13:0  | Reserved       | Value always 0                                                                                                                                                                             | RO               |

<sup>a</sup>RO = Read only

##### 45.2.11.2.1 Device present (30.8.15:14)

When read as <10>, bits 30.8.15:14 indicate that a device is present and responding at this register address. When read as anything other than <10>, bits 30.8.15:14 indicate that no device is present at this register address or that the device is not functioning properly.

#### 45.2.11.3 Vendor specific MMD 1 package identifier (Registers 30.14 and 30.15)

Registers 30.14 and 30.15 provide a 32-bit value, which may constitute a unique identifier for a particular type of package that the vendor specific MMD 1 is instantiated within. The identifier shall be composed of the 3rd through 24th bits of the Organizationally Unique Identifier (OUI) assigned to the package manufacturer by the IEEE, plus a six-bit model number, plus a four-bit revision number. A vendor specific MMD 1 may return a value of zero in each of the 32 bits of the vendor specific MMD 1 package identifier.

A non-zero package identifier may be returned by one or more MMDs in the same package. The package identifier may be the same as the device identifier.

The format of the vendor specific MMD 1 package identifier is specified in 22.2.4.3.1.

#### 45.2.12 Vendor specific MMD 2 registers

The assignment of registers in the vendor specific MMD 2 is shown in Table 45–424. A vendor specific MMD may have a device address of either 30 or 31. It is recommended that the device address is configurable and that the configuration is performed by some means other than via the MDIO.

**Table 45–424—Vendor specific MMD 2 registers**

| Register address | Register name                           | Subclause |
|------------------|-----------------------------------------|-----------|
| 31.0, 31.1       | Vendor specific                         |           |
| 31.2, 31.3       | Vendor specific MMD 2 device identifier | 45.2.12.1 |

**Table 45–424—Vendor specific MMD 2 registers (*continued*)**

| Register address        | Register name                            | Subclause |
|-------------------------|------------------------------------------|-----------|
| 31.4 through 31.7       | Vendor specific                          |           |
| 31.8                    | Vendor specific MMD 2 status register    | 45.2.12.2 |
| 31.9 through 31.13      | Vendor specific                          |           |
| 31.14, 30.15            | Vendor specific MMD 2 package identifier | 45.2.12.3 |
| 31.16 through 31.65 535 | Vendor specific                          |           |

#### 45.2.12.1 Vendor specific MMD 2 device identifier (Registers 31.2 and 31.3)

Registers 31.2 and 31.3 provide a 32-bit value, which may constitute a unique identifier for a particular type of vendor-specific device. The identifier shall be composed of the 3rd through 24th bits of the Organizationally Unique Identifier (OUI) assigned to the device manufacturer by the IEEE, plus a six-bit model number, plus a four-bit revision number. A vendor-specific device may return a value of zero in each of the 32 bits of the vendor specific MMD 2 device identifier.

The format of the vendor specific MMD 2 device identifier is specified in 22.2.4.3.1.

#### 45.2.12.2 Vendor specific MMD 2 status register (Register 31.8)

The assignment of bits in the vendor specific MMD 2 status register is shown in Table 45–425. All the bits in the vendor specific MMD 2 status register are read only; a write to the vendor specific MMD status register shall have no effect.

**Table 45–425—Vendor specific MMD 2 status register bit definitions**

| Bit(s)     | Name           | Description                                                                                                                                                                                                                           | R/W <sup>a</sup> |
|------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| 31.8.15:14 | Device present | <div>15 14</div> <div>1 0 = Device responding at this address</div> <div>1 1 = No device responding at this address</div> <div>0 1 = No device responding at this address</div> <div>0 0 = No device responding at this address</div> | RO               |
| 31.8.13:0  | Reserved       | Value always 0                                                                                                                                                                                                                        | RO               |

<sup>a</sup>RO = Read only

##### 45.2.12.2.1 Device present (31.8.15:14)

When read as <10>, bits 31.8.15:14 indicate that a device is present and responding at this register address. When read as anything other than <10>, bits 31.8.15:14 indicate that no device is present at this register address or that the device is not functioning properly.

### 45.2.12.3 Vendor specific MMD 2 package identifier (Registers 31.14 and 31.15)

Registers 31.14 and 31.15 provide a 32-bit value, which may constitute a unique identifier for a particular type of package that the vendor specific MMD is instantiated within. The identifier shall be composed of the 3rd through 24th bits of the Organizationally Unique Identifier (OUI) assigned to the package manufacturer by the IEEE, plus a six-bit model number, plus a four-bit revision number. A vendor specific MMD may return a value of zero in each of the 32 bits of the package identifier.

A non-zero package identifier may be returned by one or more MMDs in the same package. The package identifier may be the same as the device identifier.

The format of the vendor specific MMD 2 package identifier is specified in 22.2.4.3.1.

## 45.3 Management frame structure

The MDIO interface frame structure is compatible with the one defined in 22.2.4.5 such that the two systems can co-exist on the same MDIO bus. The electrical specification for the MDIO interface is incompatible to that defined in 22.2.4.5; therefore, if the two systems are to co-exist on the same bus, a voltage translation device is required (see Annex 45A). The extensions that are used for MDIO indirect register accesses are specified in Table 45–426.

**Table 45–426—Extensions to management frame format for indirect access**

|                             | Management frame fields |    |    |       |       |    |                  |      |
|-----------------------------|-------------------------|----|----|-------|-------|----|------------------|------|
| Frame                       | PRE                     | ST | OP | PRTAD | DEVAD | TA | ADDRESS / DATA   | IDLE |
| Address                     | 1...1                   | 00 | 00 | PPPPP | EEEE  | 10 | AAAAAAAAAAAAAAAA | Z    |
| Write                       | 1...1                   | 00 | 01 | PPPPP | EEEE  | 10 | DDDDDDDDDDDDDDDD | Z    |
| Read                        | 1...1                   | 00 | 11 | PPPPP | EEEE  | Z0 | DDDDDDDDDDDDDDDD | Z    |
| Post-read-increment-address | 1...1                   | 00 | 10 | PPPPP | EEEE  | Z0 | DDDDDDDDDDDDDDDD | Z    |

Each MMD shall implement a sixteen bit address register that stores the address of the register to be accessed by data transaction frames. The address register shall be overwritten by address frames. At power up or device reset, the contents of the address register are undefined.

Write, read, and post-read-increment-address frames shall access the register whose address is stored in the address register. Write and read frames shall not modify the contents of the address register.

Upon receiving a post-read-increment-address frame and having completed the read operation, the MMD shall increment the address register by one. For the case where the MMD's address register contains 65 535, the MMD shall not increment the address register.

Implementations that incorporate several MMDs within a single component shall implement separate address registers so that the MMD's address registers operate independently of one another.

#### **45.3.1 IDLE (idle condition)**

The idle condition on MDIO is a high-impedance state. All three state drivers shall be disabled and the MMD's pull-up resistor will pull the MDIO line to a one.

#### **45.3.2 PRE (preamble)**

At the beginning of each transaction, the station management entity shall send a sequence of 32 contiguous one bits on MDIO with 32 corresponding cycles on MDC to provide the MMD with a pattern that it can use to establish synchronization. An MMD shall observe a sequence of 32 contiguous one bits on MDIO with 32 corresponding cycles on MDC before it responds to any transaction.

#### **45.3.3 ST (start of frame)**

The start of frame for indirect access cycles is indicated by the <00> pattern. This pattern assures a transition from the default one and identifies the frame as an indirect access. Frames that contain the ST=<01> pattern defined in Clause 22 shall be ignored by the devices specified in Clause 45.

#### **45.3.4 OP (operation code)**

The operation code field indicates the type of transaction being performed by the frame. A <00> pattern indicates that the frame payload contains the address of the register to access. A <01> pattern indicates that the frame payload contains data to be written to the register whose address was provided in the previous address frame. A <11> pattern indicates that the frame is read operation. A <10> pattern indicates that the frame is a post-read-increment-address operation.

#### **45.3.5 PRTAD (port address)**

The port address is five bits, allowing 32 unique port addresses. The first port address bit to be transmitted and received is the MSB of the address. A station management entity has to have a priori knowledge of the appropriate port address for each port to which it is attached, whether connected to a single port or to multiple ports.

#### **45.3.6 DEVAD (device address)**

The device address is five bits, allowing 32 unique MMDs per port. The first device address bit transmitted and received is the MSB of the address.

#### **45.3.7 TA (turnaround)**

The turnaround time is a 2 bit time spacing between the device address field and the data field of a management frame to avoid contention during a read transaction. For a read or post-read-increment-address transaction, both the STA and the MMD shall remain in a high-impedance state for the first bit time of the turnaround. The MMD shall drive a zero bit during the second bit time of the turnaround of a read or post-read-increment-address transaction. During a write or address transaction, the STA shall drive a one bit for the first bit time of the turnaround and a zero bit for the second bit time of the turnaround. Figure 22–15 shows the behavior of the MDIO signal during the turnaround field of a read or post-read-increment-address transaction.

#### **45.3.8 ADDRESS / DATA**

The address/data field is 16 bits. For an address cycle, it contains the address of the register to be accessed on the next cycle. For the data cycle of a write frame, the field contains the data to be written to the register.

For a read or post-read-increment-address frame, the field contains the contents of the register. The first bit transmitted and received shall be bit 15.

## 45.4 Electrical interface

### 45.4.1 Electrical specification

The electrical characteristics of the MDIO interface are shown in Table 45–427. The MDIO uses signal levels that are compatible with devices operating at a nominal supply voltage of 1.2 V. More information on the electrical interface is given in Annex 45A. Voltage translators between the Clause 22 electrical interface and the Clause 45 electrical interface are described in 45A.3 and 45A.4.

NOTE—It is possible to implement the MDIO electrical interface using open drain buffers and a resistive pull-up to a  $V_{DD}$  of 1.2 V (see 45A.1).

**Table 45–427—MDIO electrical interface characteristics**

| Symbol     | Parameter             | Condition                  | Min.   | Max.   |
|------------|-----------------------|----------------------------|--------|--------|
| $V_{IH}$   | Input high voltage    |                            | 0.84 V | 1.5 V  |
| $V_{IL}$   | Input low voltage     |                            | −0.3 V | 0.36 V |
| $V_{OH}$   | Output high voltage   | $I_{OH} = -100 \text{ uA}$ | 1.0 V  | 1.5 V  |
| $V_{OL}$   | Output low voltage    | $I_{OL} = 100 \text{ uA}$  | −0.3 V | 0.2 V  |
| $I_{OH}^a$ | Output high current   | $V_{OH} = 1.0 \text{ V}$   |        | −4 mA  |
| $I_{OL}$   | Output low current    | $V_{OL} = 0.2 \text{ V}$   | +4 mA  |        |
| $C_i$      | Input capacitance     |                            |        | 10 pF  |
| $C_L$      | Total capacitive load |                            |        | 470 pF |

<sup>a</sup> $I_{OH}$  parameter is not applicable to open drain drivers.

### 45.4.2 Timing specification

MDIO is a bidirectional signal that can be sourced by the Station Management Entity (STA) or the MMD. When the STA sources the MDIO signal, the STA shall provide a minimum of 10 ns of setup time and a minimum of 10 ns of hold time referenced to the rising edge of MDC, as shown in Figure 45–3, measured at the MMD.

When the MDIO signal is sourced by the MMD, it is sampled by the STA synchronously with respect to the beginning of the rising edge of MDC. The clock to output delay from the MMD, as measured at the STA, shall be a minimum of 0 ns, and a maximum of 300 ns, as shown in Figure 45–4.

The timing specification for the MDC signal is given in 22.2.2.13.

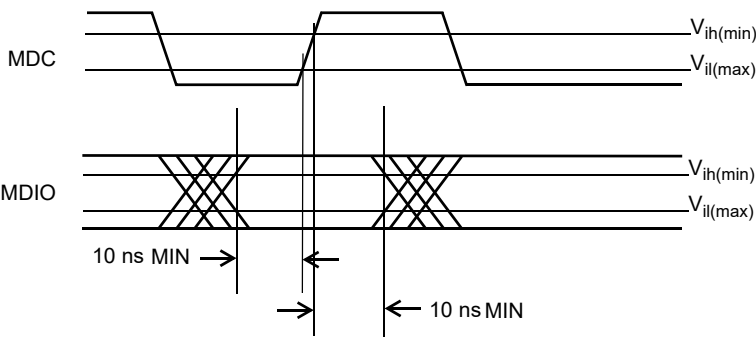


Figure 45-3—MDIO sourced by STA

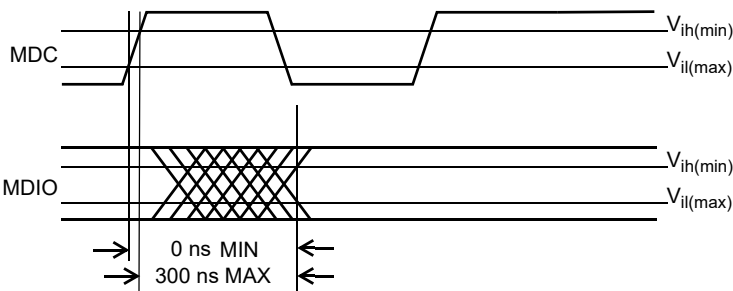


Figure 45-4—MDIO sourced by MMD

## 45.5 Protocol implementation conformance statement (PICS) proforma for Clause 45, Management Data Input/Output (MDIO) Interface<sup>87</sup>

### 45.5.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 45, MDIO interface, shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

### 45.5.2 Identification

#### 45.5.2.1 Implementation identification

|                                                                                                                                                     |  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|--|
| Supplier <sup>1</sup>                                                                                                                               |  |
| Contact point for inquiries about the PICS <sup>1</sup>                                                                                             |  |
| Implementation Name(s) and Version(s) <sup>1,3</sup>                                                                                                |  |
| Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) <sup>2</sup> |  |
| NOTE 1—Required for all implementations.                                                                                                            |  |
| NOTE 2—May be completed as appropriate in meeting the requirements for the identification.                                                          |  |
| NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).      |  |

#### 45.5.2.2 Protocol summary

|                                                                                                                                                                                                            |                                                                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Identification of protocol standard                                                                                                                                                                        | IEEE Std 802.3-2022, Clause 45, Management Data Input/Output (MDIO) Interface |
| Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS                                                                                            |                                                                               |
| Have any Exception items been required? No <input type="checkbox"/> Yes <input type="checkbox"/><br>(See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3-2022.) |                                                                               |

|                   |  |
|-------------------|--|
| Date of Statement |  |
|-------------------|--|

<sup>87</sup>Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.



### 45.5.2.3 Major capabilities/options

| Item  | Feature                                                                                 | Subclause    | Value/Comment | Status   | Support                      |
|-------|-----------------------------------------------------------------------------------------|--------------|---------------|----------|------------------------------|
| *PMA  | Implementation of PMA/PMD MMD                                                           | 45.2.1       |               | O        | Yes [ ]<br>No [ ]            |
| *WIS  | Implementation of WIS MMD                                                               | 45.2.2       |               | O        | Yes [ ]<br>No [ ]            |
| *PCS  | Implementation of PCS MMD                                                               | 45.2.3       |               | O        | Yes [ ]<br>No [ ]            |
| *PX   | Implementation of PHY XS MMD                                                            | 45.2.4       |               | O        | Yes [ ]<br>No [ ]            |
| *DX   | Implementation of DTE XS MMD                                                            | 45.2.5       |               | O        | Yes [ ]<br>No [ ]            |
| *VSA  | Implementation of Vendor specific MMD 1                                                 | 45.2.11      |               | O        | Yes [ ]<br>No [ ]            |
| *VSB  | Implementation of Vendor specific MMD 2                                                 | 45.2.12      |               | O        | Yes [ ]<br>No [ ]            |
| *TC   | Implementation of the TC MMD                                                            | 45.2.6       |               | 10P*2B:M | Yes [ ]<br>No [ ]            |
| *CTT  | Implementation of the Clause 22 extension MMD                                           | 45.2.10      |               | O        | Yes [ ]<br>No [ ]            |
| *ODB  | Open drain buffer                                                                       | 45.4.1       |               | O        | Yes [ ]<br>No [ ]            |
| *AN   | Implementation of Auto-Negotiation MMD                                                  | 45.2.7       |               | O        | Yes [ ]<br>No [ ]            |
| *BTC  | Implementation of BASE-T1 PMA/PMD                                                       | 45.2.1.214.1 |               | PMA:O    | Yes [ ]<br>No [ ]<br>N/A [ ] |
| *EO   | Implementation of OFDM MMD                                                              | 45.2.8       |               | O        | Yes [ ]<br>No [ ]            |
| *PODL | Implementation of Power over Data Lines (PoDL) of Single Balanced Twisted-Pair Ethernet | 45.2.9       |               | O        | Yes [ ]<br>No [ ]            |

### 45.5.3 PICS proforma tables for the Management Data Input Output (MDIO) Interface

#### 45.5.3.1 MDIO signal functional specifications

| Item | Feature                       | Subclause | Value/Comment | Status | Support |
|------|-------------------------------|-----------|---------------|--------|---------|
| SF1  | MDIO uses three-state drivers | 45.4.1    |               | M      | Yes [ ] |

### 45.5.3.2 PMA/PMD MMD options

| Item    | Feature                                        | Subclause   | Value/Comment | Status | Support                      |
|---------|------------------------------------------------|-------------|---------------|--------|------------------------------|
| *ALB    | Implementation of PMA local loopback function  | 45.2.1.1.5  |               | PMA:O  | Yes [ ]<br>No [ ]<br>N/A [ ] |
| *RLB    | Implementation of PMA remote loopback function | 45.2.1.1.4  |               | PMA:O  | Yes [ ]<br>No [ ]<br>N/A [ ] |
| *PLF    | Implementation of fault detection              | 45.2.1.7    |               | PMA:O  | Yes [ ]<br>No [ ]<br>N/A [ ] |
| *ALP    | Implementation of PMA/PMD low power ability    | 45.2.1.2.5  |               | PMA:O  | Yes [ ]<br>No [ ]<br>N/A [ ] |
| *PTD    | Implementation of transmit disable function    | 45.2.1.8    |               | PMA:O  | Yes [ ]<br>No [ ]<br>N/A [ ] |
| *2.5T   | Implementation of a 2.5GBASE-T PMA             | 45.2.1.19   |               | PMA:O  | Yes [ ]<br>No [ ]            |
| *2.5GKX | Implementation of 2.5GBASE-KX PMA/PMD          | 45.2.1.19.2 |               | PMA:O  | Yes [ ]<br>No [ ]            |
| *5GKR   | Implementation of 5GBASE-KR PMA/PMD            | 45.2.1.19.1 |               | PMA:O  | Yes [ ]<br>No [ ]            |
| *5T     | Implementation of a 5GBASE-T PMA               | 45.2.1.19   |               | PMA:O  | Yes [ ]<br>No [ ]            |
| *10T    | Implementation of the 10GBASE-T PMA            | 45.2.1.6    |               | PMA:O  | Yes [ ]<br>No [ ]            |
| *25T    | Implementation of the 25GBASE-T PMA            | 45.2.1.17   |               | PMA:O  | Yes [ ]<br>No [ ]            |
| *40T    | Implementation of the 40GBASE-T PMA            | 45.2.1.12   |               | PMA:O  | Yes [ ]<br>No [ ]            |
| *40G    | Implementation of 40 Gb/s PMA/PMD              | 45.2.1.4    |               | PMA:O  | Yes [ ]<br>No [ ]            |
| *100G   | Implementation of 100 Gb/s PMA/PMD             | 45.2.1.4    |               | PMA:O  | Yes [ ]<br>No [ ]            |
| *10M    | Implementation of 10 Mb/s PMA/PMD              | 45.2.1.4    |               | PMA:O  | Yes [ ]<br>No [ ]            |
| *100M   | Implementation of 100 Mb/s PMA/PMD             | 45.2.1.4    |               | PMA:O  | Yes [ ]<br>No [ ]            |
| *1G     | Implementation of 1000 Mb/s PMA/PMD            | 45.2.1.4    |               | PMA:O  | Yes [ ]<br>No [ ]            |
| *10P    | Implementation of the 10PASS-TS PMA/PMD        | 45.2.1.4    |               | PMA:O  | Yes [ ]<br>No [ ]            |
| *2B     | Implementation of the 2BASE-TL PMA/PMD         | 45.2.1.4    |               | PMA:O  | Yes [ ]<br>No [ ]            |

| Item    | Feature                                                       | Subclause  | Value/Comment                             | Status | Support           |
|---------|---------------------------------------------------------------|------------|-------------------------------------------|--------|-------------------|
| *2.5G   | Implementation of a 2.5 Gb/s PMA/PMD                          | 45.2.1.4   |                                           | PMA:O  | Yes [ ]<br>No [ ] |
| *5G     | Implementation of a 5 Gb/s PMA/PMD                            | 45.2.1.4   |                                           | PMA:O  | Yes [ ]<br>No [ ] |
| *10G    | Implementation of 10 Gb/s PMA/PMD                             | 45.2.1.4   |                                           | PMA:O  | Yes [ ]<br>No [ ] |
| 10XR    | Implementation of 10GPASS-XR PMA/PMD                          | 45.2.1.4   |                                           | PMA:O  | Yes [ ]<br>No [ ] |
| *KX     | Implementation of 1000BASE-KX PMA/PMD                         | 45.2.1.6   |                                           | PMA:O  | Yes [ ]<br>No [ ] |
| *KX4    | Implementation of 10GBASE-KX4 PMA/PMD                         | 45.2.1.6   |                                           | PMA:O  | Yes [ ]<br>No [ ] |
| *KR     | Implementation of 10GBASE-KR PMA/PMD                          | 45.2.1.6   |                                           | PMA:O  | Yes [ ]<br>No [ ] |
| *PXAR   | Implementation of PMA/PMD Extended Ability register           | 45.2.1.10  | Required for certain abilities            | PMA:O  | Yes [ ]<br>No [ ] |
| *40XAR  | Implementation of 40G/100G PMA/PMD Extended Ability register  | 45.2.1.12  | Required for certain abilities            | PMA:O  | Yes [ ]<br>No [ ] |
| 10XRAR  | Implementation of EPoC PMA/PMD ability register               | 45.2.1.15  |                                           | PMA:O  | Yes [ ]<br>No [ ] |
| *MMD8   | Implementation of separately addressable instance, address 8  | 45.2.1     | All register definitions as for address 1 | PMA:O  | Yes [ ]<br>No [ ] |
| *MMD9   | Implementation of separately addressable instance, address 9  | 45.2.1     | All register definitions as for address 1 | PMA:O  | Yes [ ]<br>No [ ] |
| *MMD10  | Implementation of separately addressable instance, address 10 | 45.2.1     | All register definitions as for address 1 | PMA:O  | Yes [ ]<br>No [ ] |
| *MMD11  | Implementation of separately addressable instance, address 11 | 45.2.1     | All register definitions as for address 1 | PMA:O  | Yes [ ]<br>No [ ] |
| *FEC-R  | Implementation of BASE-R FEC                                  | 45.2.1.107 |                                           | PMA:O  | Yes [ ]<br>No [ ] |
| *RS-FEC | Implementation of RS-FEC                                      | 45.2.1.117 |                                           | PMA:O  | Yes [ ]<br>No [ ] |
| *BH     | Implementation of a BASE-H PMA/PMD                            |            |                                           | PMA:O  | Yes [ ]<br>No [ ] |

### 45.5.3.3 PMA/PMD management functions

| Item | Feature                                                                                     | Subclause  | Value/Comment                                                                                            | Status | Support            |
|------|---------------------------------------------------------------------------------------------|------------|----------------------------------------------------------------------------------------------------------|--------|--------------------|
| MM1  | Device responds to all register addresses for that device                                   | 45.2       |                                                                                                          | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM2  | Writes to undefined and read-only registers have no effect                                  | 45.2       |                                                                                                          | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM3  | Operation is not affected by writes to reserved and unsupported bits.                       | 45.2       |                                                                                                          | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM4  | Reserved and unsupported bits return a value of zero                                        | 45.2       |                                                                                                          | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM5  | Latching low bits remain low until after they have been read via the management interface   | 45.2       |                                                                                                          | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM6  | Latching low bits assume correct value once read via the management interface               | 45.2       | Correct value is based upon current state                                                                | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM7  | Latching high bits remain high until after they have been read via the management interface | 45.2       |                                                                                                          | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM8  | Latching high bits assume correct value once read via the management interface              | 45.2       | Correct value is based upon current state                                                                | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM9  | Action on reset                                                                             | 45.2.1.1.1 | Reset the registers of the entire device to default values and set bit 15 of the Control register to one | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM10 | Return 1 until reset completed                                                              | 45.2.1.1.1 |                                                                                                          | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM11 | Control and management interfaces are restored to operation within 0.5 s of reset           | 45.2.1.1.1 |                                                                                                          | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM12 | Responds to reads of bit 15 and 1.8.15:14 during reset                                      | 45.2.1.1.1 |                                                                                                          | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM13 | Device responds to transactions necessary to exit low-power mode while in low-power state   | 45.2.1.1.2 |                                                                                                          | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM14 | Invalid writes to speed selection bits are ignored                                          | 45.2.1.1.3 |                                                                                                          | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM15 | PMA is set into local loopback mode when bit 0 is set to a one                              | 45.2.1.1.5 |                                                                                                          | ALB:M  | Yes [ ]<br>N/A [ ] |
| MM16 | PMA transmit data is returned on receive path when in local loopback                        | 45.2.1.1.5 |                                                                                                          | ALB:M  | Yes [ ]<br>N/A [ ] |

| Item | Feature                                                                                       | Subclause  | Value/Comment | Status     | Support            |
|------|-----------------------------------------------------------------------------------------------|------------|---------------|------------|--------------------|
| MM17 | PMA ignores writes to this bit if it does not support local loopback.                         | 45.2.1.1.5 |               | PMA*!ALB:M | Yes [ ]<br>N/A [ ] |
| MM18 | PMA returns a value of zero when read if it does not support local loopback.                  | 45.2.1.1.5 |               | PMA*!ALB:M | Yes [ ]<br>N/A [ ] |
| MM19 | PMA is set into remote loopback mode when bit 1 is set to a one                               | 45.2.1.1.4 |               | RLB:M      | Yes [ ]<br>N/A [ ] |
| MM20 | PMA receive data is returned on transmit path when in remote loopback                         | 45.2.1.1.4 |               | RLB:M      | Yes [ ]<br>N/A [ ] |
| MM21 | PMA ignores writes to this bit if it does not support remote loopback.                        | 45.2.1.1.4 |               | PMA*!RLB:M | Yes [ ]<br>N/A [ ] |
| MM22 | PMA returns a value of zero when read if it does not support remote loopback.                 | 45.2.1.1.4 |               | PMA*!RLB:M | Yes [ ]<br>N/A [ ] |
| MM23 | Writes to status 1 register have no effect                                                    | 45.2.1.2   |               | PMA:M      | Yes [ ]<br>N/A [ ] |
| MM24 | Receive link status implemented with latching low behavior                                    | 45.2.1.2.4 |               | PMA:M      | Yes [ ]<br>N/A [ ] |
| MM25 | Unique identifier is composed of 22 bits of OUI, model number and revision                    | 45.2.1.3   |               | PMA:M      | Yes [ ]<br>N/A [ ] |
| MM26 | PMA/PMD type is selected using bits 5:0                                                       | 45.2.1.6.3 |               | PMA:M      | Yes [ ]<br>N/A [ ] |
| MM27 | PMA/PMD ignores writes to type selection bits that select types that it has not advertised    | 45.2.1.6.3 |               | PMA:M      | Yes [ ]<br>N/A [ ] |
| MM28 | Writes to the status 2 register have no effect                                                | 45.2.1.7   |               | PMA:M      | Yes [ ]<br>N/A [ ] |
| MM29 | PMA/PMD returns a value of zero for transmit fault if it is unable to detect a transmit fault | 45.2.1.7.4 |               | PMA:M      | Yes [ ]<br>N/A [ ] |
| MM30 | Transmit fault is implemented using latching high behavior                                    | 45.2.1.7.4 |               | PMA*PLF:M  | Yes [ ]<br>N/A [ ] |
| MM31 | PMA/PMD returns a value of zero for receive fault if it is unable to detect a receive fault   | 45.2.1.7.5 |               | PMA*!PLF:M | Yes [ ]<br>N/A [ ] |
| MM32 | Receive fault is implemented using latching high behavior                                     | 45.2.1.7.5 |               | PMA*PLF:M  | Yes [ ]<br>N/A [ ] |
| MM33 | Writes to register 9 are ignored by device that does not implement transmit disable           | 45.2.1.8   |               | PMA*!PTD:M | Yes [ ]<br>N/A [ ] |
| MM34 | Single wavelength device uses bit 1.9.0 for transmit disable                                  | 45.2.1.8   |               | PMA*PTD:M  | Yes [ ]<br>N/A [ ] |

| Item | Feature                                                                                   | Subclause   | Value/Comment                               | Status       | Support                      |
|------|-------------------------------------------------------------------------------------------|-------------|---------------------------------------------|--------------|------------------------------|
| MM35 | Single wavelength device ignores writes to bits 10:1 and returns a value of zero for them | 45.2.1.8    |                                             | PTD:M        | Yes [ ]<br>N/A [ ]           |
| MM36 | Bits 1 to 10, set to one disables transmission on corresponding lane                      | 45.2.1.8    |                                             | PTD:M        | Yes [ ]<br>No [ ]<br>N/A [ ] |
| MM37 | Bits 1 to 10, set to zero enables transmission on corresponding lane                      | 45.2.1.8    |                                             | PTD:M        | Yes [ ]<br>No [ ]<br>N/A [ ] |
| MM38 | Setting bit 0 to a one disables transmission                                              | 45.2.1.8.7  |                                             | PMA*PTD:M    | Yes [ ]<br>No [ ]<br>N/A [ ] |
| MM39 | Setting bit 0 to a zero enables transmission                                              | 45.2.1.8.7  | Only if all lane transmit disables are zero | PMA*PTD:M    | Yes [ ]<br>No [ ]<br>N/A [ ] |
| MM40 | Receive signal detect register behaves as described                                       | 45.2.1.9    |                                             | PMA:M        | Yes [ ]<br>N/A [ ]           |
| MM41 | Writes to the extended ability register have no effect                                    | 45.2.1.10   |                                             | PMA:M        | Yes [ ]<br>N/A [ ]           |
| MM42 | Unique identifier is composed of 22 bits of OUI, model number and revision                | 45.2.1.11   |                                             | PMA:M        | Yes [ ]<br>N/A [ ]           |
| MM43 | Writes to the 40G/100G PMA/PMD extended ability register have no effect                   | 45.2.1.12   |                                             | PMA:M        | Yes [ ]<br>N/A [ ]           |
| MM44 | Bit indicates link down while initializing                                                | 45.2.1.2.4  |                                             | PMA*10P*2B:M | Yes [ ]<br>N/A [ ]           |
| MM45 | Bit remains a one and writing a one is ignored when link is up or initializing            | 45.2.1.11   |                                             | PMA*10P*2B:M | Yes [ ]<br>N/A [ ]           |
| MM46 | PRBS pattern testing implemented                                                          | 45.2.1.169  | Ability indicated in register 1.1500        | PMA:O        | Yes [ ]<br>N/A [ ]           |
| MM47 | Square wave testing implemented                                                           | 45.2.1.169  | Bit 1.1500.12 is set to one                 | PMA:O        | Yes [ ]<br>N/A [ ]           |
| MM48 | Counters corresponding to lanes that are not implemented return all zeros                 | 45.2.1.173  |                                             | MM46:M       | Yes [ ]<br>N/A [ ]           |
| MM49 | Counters are reset to zero by read or PMA reset                                           | 45.2.1.173  |                                             | MM46:M       | Yes [ ]<br>N/A [ ]           |
| MM50 | Counters are held at all ones in the case of overflow                                     | 45.2.1.173  |                                             | MM46:M       | Yes [ ]<br>N/A [ ]           |
| MM51 | Bit set to zero upon reset or upon link down                                              | 45.2.1.28.1 | -O subtypes only                            | PMA*10P*2B:M | Yes [ ]<br>N/A [ ]           |
| MM52 | Bit set to one upon reset or upon link down                                               | 45.2.1.28.1 | -R subtypes only                            | PMA*10P*2B:M | Yes [ ]<br>N/A [ ]           |
| MM53 | Handshake tones not sent while bit is set to zero                                         | 45.2.1.28.1 |                                             | PMA*10P*2B:M | Yes [ ]<br>N/A [ ]           |

| Item | Feature                                                                  | Subclause   | Value/Comment                       | Status       | Support            |
|------|--------------------------------------------------------------------------|-------------|-------------------------------------|--------------|--------------------|
| MM54 | Writes to set unsupported modes or when link is not down are ignored     | 45.2.1.28.4 |                                     | PMA*10P*2B:M | Yes [ ]<br>N/A [ ] |
| MM55 | Setting bit to one to one issues a clear-down command                    | 45.2.1.28.5 |                                     | PMA*10P*2B:M | Yes [ ]<br>N/A [ ] |
| MM56 | MMD clears bit to zero when clear-down command is issued or on reset     | 45.2.1.28.5 |                                     | PMA*10P*2B:M | Yes [ ]<br>N/A [ ] |
| MM57 | Writes ignored if link is not in “Link down (ready)” state               | 45.2.1.28.5 | Link state described in 45.2.1.29.4 | PMA*10P*2B:M | Yes [ ]<br>N/A [ ] |
| MM58 | PMA/PMD does not respond to handshake tones while bit is set to one      | 45.2.1.28.6 |                                     | PMA*10P*2B:M | Yes [ ]<br>N/A [ ] |
| MM59 | PMA/PMD responds to handshake tones properly when bit is set to zero     | 45.2.1.28.6 |                                     | PMA*10P*2B:M | Yes [ ]<br>N/A [ ] |
| MM60 | Bit set to zero upon MMD reset                                           | 45.2.1.28.6 |                                     | PMA*10P*2B:M | Yes [ ]<br>N/A [ ] |
| MM61 | Writes to set unsupported modes are ignored                              | 45.2.1.28.7 |                                     | PMA*10P*2B:M | Yes [ ]<br>N/A [ ] |
| MM62 | Bits zero when link is down or initializing                              | 45.2.1.29.1 |                                     | PMA*10P*2B:M | Yes [ ]<br>N/A [ ] |
| MM63 | Bits set indicate linked port type or link status                        | 45.2.1.29.4 |                                     | PMA*10P*2B:M | Yes [ ]<br>N/A [ ] |
| MM64 | Bits indicate 001 while link is initializing                             | 45.2.1.29.4 |                                     | PMA*10P*2B:M | Yes [ ]<br>N/A [ ] |
| MM65 | Bits indicate 000 when link is down and handshake tones are not detected | 45.2.1.29.4 |                                     | PMA*10P*2B:M | Yes [ ]<br>N/A [ ] |
| MM66 | Bits indicate 100 when link is down and handshake tones are detected     | 45.2.1.29.4 |                                     | PMA*10P*2B:M | Yes [ ]<br>N/A [ ] |
| MM67 | Bit held as one during operation, clears to zero after                   | 45.2.1.30.1 |                                     | PMA*10P*2B:M | Yes [ ]<br>N/A [ ] |
| MM68 | Result = failed after 10 second timeout                                  | 45.2.1.30.1 |                                     | PMA*10P*2B:M | Yes [ ]<br>N/A [ ] |
| MM69 | Writes to one while link is down are marked completed and failed         | 45.2.1.30.1 |                                     | PMA*10P*2B:M | Yes [ ]<br>N/A [ ] |
| MM70 | Bit held as one during operation, clears to zero after                   | 45.2.1.30.2 |                                     | PMA*10P*2B:M | Yes [ ]<br>N/A [ ] |
| MM71 | Result = failed after 10 second timeout                                  | 45.2.1.30.2 |                                     | PMA*10P*2B:M | Yes [ ]<br>N/A [ ] |
| MM72 | Writes to one while link is down are marked completed and failed         | 45.2.1.30.2 |                                     | PMA*10P*2B:M | Yes [ ]<br>N/A [ ] |

| Item | Feature                                                       | Subclause   | Value/Comment          | Status       | Support            |
|------|---------------------------------------------------------------|-------------|------------------------|--------------|--------------------|
| MM73 | Bit set to result of the “Get” operation                      | 45.2.1.31.1 |                        | PMA*10P*2B:M | Yes [ ]<br>N/A [ ] |
| MM74 | Bit set to zero on read or reset                              | 45.2.1.31.1 |                        | PMA*10P*2B:M | Yes [ ]<br>N/A [ ] |
| MM75 | Bit set to result of the “Send” operation                     | 45.2.1.31.2 |                        | PMA*10P*2B:M | Yes [ ]<br>N/A [ ] |
| MM76 | Bits are reset to zero when read or on reset                  | 45.2.1.34   |                        | PMA*10P*2B:M | Yes [ ]<br>N/A [ ] |
| MM77 | Bits are held to all ones upon counter overflow               | 45.2.1.34   |                        | PMA*10P*2B:M | Yes [ ]<br>N/A [ ] |
| MM78 | Bits are reset to zero when read or reset                     | 45.2.1.41   |                        | PMA*10P:M    | Yes [ ]<br>N/A [ ] |
| MM79 | Bits are reset to zero when read or reset                     | 45.2.1.42   |                        | PMA*10P:M    | Yes [ ]<br>N/A [ ] |
| MM80 | Bits are held at all ones when PHY cannot determine value     | 45.2.1.45.1 | ex: While link is down | PMA*10P:M    | Yes [ ]<br>N/A [ ] |
| MM81 | Bit remain as one while tones are being refreshed             | 45.2.1.55.1 |                        | PMA*10P:M    | Yes [ ]<br>N/A [ ] |
| MM82 | Bit set to zero when operation completes or upon reset        | 45.2.1.55.1 |                        | PMA*10P:M    | Yes [ ]<br>N/A [ ] |
| MM83 | Bit remain as one while tones are being activated/deactivated | 45.2.1.55.2 |                        | PMA*10P:M    | Yes [ ]<br>N/A [ ] |
| MM84 | Bit set to zero when operation completes or upon reset        | 45.2.1.55.2 |                        | PMA*10P:M    | Yes [ ]<br>N/A [ ] |
| MM85 | Bit remain as one while tone direction is being changed       | 45.2.1.55.3 |                        | PMA*10P:M    | Yes [ ]<br>N/A [ ] |
| MM86 | Bit set to zero when operation completes or upon reset        | 45.2.1.55.3 |                        | PMA*10P:M    | Yes [ ]<br>N/A [ ] |
| MM87 | Bit remain as one while SNR margins parameters are loaded     | 45.2.1.55.4 |                        | PMA*10P:M    | Yes [ ]<br>N/A [ ] |
| MM88 | Bit set to zero when operation completes or upon reset        | 45.2.1.55.4 |                        | PMA*10P:M    | Yes [ ]<br>N/A [ ] |
| MM89 | Bit remain as one while PSD level is set                      | 45.2.1.55.5 |                        | PMA*10P:M    | Yes [ ]<br>N/A [ ] |
| MM90 | Bit set to zero when operation completes or upon reset        | 45.2.1.55.5 |                        | PMA*10P:M    | Yes [ ]<br>N/A [ ] |
| MM91 | Bit remain as one while reference PSD level is set            | 45.2.1.55.6 |                        | PMA*10P:M    | Yes [ ]<br>N/A [ ] |
| MM92 | Bit set to zero when operation completes or upon reset        | 45.2.1.55.6 |                        | PMA*10P:M    | Yes [ ]<br>N/A [ ] |
| MM93 | Bits are reset to zero when read or upon reset                | 45.2.1.56.1 |                        | PMA*10P:M    | Yes [ ]<br>N/A [ ] |
| MM94 | Bits read as zero                                             | 45.2.1.57.6 |                        | PMA*10P:M    | Yes [ ]<br>N/A [ ] |



| Item  | Feature                                                                                                 | Subclause    | Value/Comment                          | Status                                                        | Support            |
|-------|---------------------------------------------------------------------------------------------------------|--------------|----------------------------------------|---------------------------------------------------------------|--------------------|
| MM95  | Bits read as zero                                                                                       | 45.2.1.57.7  |                                        | PMA*10P:M                                                     | Yes [ ]<br>N/A [ ] |
| MM96  | Writes to an invalid value are ignored                                                                  | 45.2.1.59    | Valid values are decimal 10, 20, or 40 | PMA*10P:M                                                     | Yes [ ]<br>N/A [ ] |
| MM97  | Bits set to default value on MMD reset                                                                  | 45.2.1.59    | Default value is decimal 20            | PMA*10P:M                                                     | Yes [ ]<br>N/A [ ] |
| MM98  | Writes to set an invalid value are ignored                                                              | 45.2.1.62.1  | Valid values are decimal 3 through 89  | PMA*10P:M                                                     | Yes [ ]<br>N/A [ ] |
| MM99  | Writes to set an invalid value are ignored                                                              | 45.2.1.62.2  | Valid values are decimal 3 through 89  | PMA*10P:M                                                     | Yes [ ]<br>N/A [ ] |
| MM100 | Writes to set an invalid value are ignored                                                              | 45.2.1.62.3  | Valid values are decimal 1 through 86  | PMA*10P:M                                                     | Yes [ ]<br>N/A [ ] |
| MM101 | Writes to set an invalid value are ignored                                                              | 45.2.1.62.5  | Invalid value is 11                    | PMA*10P:M                                                     | Yes [ ]<br>N/A [ ] |
| MM102 | Bits set to zero when read or reset                                                                     | 45.2.1.63    |                                        | PMA*2B:M                                                      | Yes [ ]<br>N/A [ ] |
| MM103 | Bits set to zero when read or reset                                                                     | 45.2.1.65    |                                        | PMA*2B:M                                                      | Yes [ ]<br>N/A [ ] |
| MM104 | Bits set to zero when read or reset                                                                     | 45.2.1.67    |                                        | PMA*2B:M                                                      | Yes [ ]<br>N/A [ ] |
| MM105 | Bits set to zero when read or reset                                                                     | 45.2.1.69    |                                        | PMA*2B:M                                                      | Yes [ ]<br>N/A [ ] |
| MM106 | Bits set to zero when read or reset                                                                     | 45.2.1.71    |                                        | PMA*2B:M                                                      | Yes [ ]<br>N/A [ ] |
| MM107 | Writes to set an invalid value are ignored                                                              | 45.2.1.76.1  | Valid values are decimal 3 through 89  | PMA*10P:M                                                     | Yes [ ]<br>N/A [ ] |
| MM108 | Writes to set an invalid value are ignored                                                              | 45.2.1.76.2  | Valid values are decimal 3 through 89  | PMA*10P:M                                                     | Yes [ ]<br>N/A [ ] |
| MM109 | Writes to set an invalid value are ignored                                                              | 45.2.1.76.3  | Valid values are decimal 1 through 86  | PMA*10P:M                                                     | Yes [ ]<br>N/A [ ] |
| MM110 | Writes to set an invalid value are ignored                                                              | 45.2.1.76.5  | Invalid value is 11                    | PMA*10P:M                                                     | Yes [ ]<br>N/A [ ] |
| MM111 | Bit set to zero if PMA link_status=FAIL                                                                 | 45.2.1.77.1  |                                        | PMA*2.5T:M<br>PMA*5T:M<br>PMA*10T:M<br>PMA*25T:M<br>PMA*40T:M | Yes [ ]<br>N/A [ ] |
| MM112 | Skew delay register update rate.                                                                        | 45.2.1.93    | At least once per second               | PMA*2.5T:M<br>PMA*5T:M<br>PMA*10T:M<br>PMA*25T:M<br>PMA*40T:M | Yes [ ]<br>N/A [ ] |
| MM113 | A PMA/PMD that is unable to detect a transmit fault returns a value of zero for the transmit fault bit. | 45.2.1.104.3 |                                        | KX:M                                                          | Yes [ ]<br>N/A [ ] |

| Item  | Feature                                                                                                                   | Subclause                                                                             | Value/Comment                                           | Status                                      | Support            |
|-------|---------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------|---------------------------------------------|--------------------|
| MM114 | The transmit fault bit is implemented with latching high behavior.                                                        | 45.2.1.104.3                                                                          |                                                         | KX:M                                        | Yes [ ]<br>N/A [ ] |
| MM115 | A PMA/PMD that is unable to detect a receive fault returns a value of zero for the receive fault bit.                     | 45.2.1.104.4                                                                          |                                                         | KX:M                                        | Yes [ ]<br>N/A [ ] |
| MM116 | The receive fault bit is implemented with latching high behavior.                                                         | 45.2.1.104.4                                                                          |                                                         | KX:M                                        | Yes [ ]<br>N/A [ ] |
| MM117 | FEC enable is set to zero upon execution of PHY reset                                                                     | 45.2.1.108.1                                                                          |                                                         | FEC-R:M                                     | Yes [ ]<br>N/A [ ] |
| MM118 | RS-FEC counters are reset when read or upon PHY reset.                                                                    | 45.2.1.118,<br>45.2.1.119,<br>45.2.1.120,<br>45.2.1.122,<br>45.2.1.123,<br>45.2.1.124 |                                                         | RS-FEC:M                                    | Yes [ ]<br>N/A [ ] |
| MM119 | RS-FEC counters are held at all ones in the case of overflow                                                              | 45.2.1.118,<br>45.2.1.119,<br>45.2.1.120,<br>45.2.1.122,<br>45.2.1.123,<br>45.2.1.124 |                                                         | RS-FEC:M                                    | Yes [ ]<br>N/A [ ] |
| MM120 | FEC corrected blocks counters are reset when read or upon PHY reset.                                                      | 45.2.1.109,<br>45.2.1.131                                                             |                                                         | FEC-R:M                                     | Yes [ ]<br>N/A [ ] |
| MM121 | FEC corrected blocks counters are held at all ones in the case of overflow                                                | 45.2.1.109,<br>45.2.1.131                                                             |                                                         | FEC-R:M                                     | Yes [ ]<br>N/A [ ] |
| MM122 | FEC uncorrected blocks counters are reset when read or upon PHY reset.                                                    | 45.2.1.110,<br>45.2.1.149                                                             |                                                         | FEC-R:M                                     | Yes [ ]<br>N/A [ ] |
| MM123 | FEC uncorrected blocks counters are held at all ones in the case of overflow                                              | 45.2.1.110,<br>45.2.1.149                                                             |                                                         | FEC-R:M                                     | Yes [ ]<br>N/A [ ] |
| MM124 | Backplane Ethernet AN ability bit is set to one for 1000BASE-KX, 2.5GBASE-KX, 5GBASE-KR, 10GBASE-KX4 and 10GBASE-KR PHYs. | 45.2.7.12.4                                                                           |                                                         | KX:M<br>2.5GKX:M<br>5GKR:M<br>KX4:M<br>KR:M | Yes [ ]<br>N/A [ ] |
| MM125 | Writes to this register have no effect                                                                                    | 45.2.1.11                                                                             |                                                         | PMA:M                                       | Yes [ ]<br>N/A [ ] |
| MM126 | Writes to the BASE-T1 PMA/PMD extended ability register have no effect                                                    | 45.2.1.16                                                                             |                                                         | PMA:M                                       | Yes [ ]<br>N/A [ ] |
| MM127 | BASE-T1 MASTER config                                                                                                     | 45.2.1.214.1                                                                          | PHY operates as MASTER when bit 1.2100.14 is set to one | BTC:M                                       | Yes [ ]<br>N/A [ ] |

| Item  | Feature                                                                                                                          | Subclause    | Value/Comment                                           | Status | Support            |
|-------|----------------------------------------------------------------------------------------------------------------------------------|--------------|---------------------------------------------------------|--------|--------------------|
| MM128 | BASE-T1 SLAVE config                                                                                                             | 45.2.1.214.1 | PHY operates as SLAVE when bit 1.2100.14 is set to zero | BTC:M  | Yes [ ]<br>N/A [ ] |
| MM129 | EEE deep sleep capability indicated for each port type                                                                           | 45.2.1.14    |                                                         | EEE:M  | Yes [ ]<br>N/A [ ] |
| MM130 | A reset sets all 1000BASE-T1 PMA/PMD registers to their default states                                                           | 45.2.1.237.1 |                                                         | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM131 | 1000BASE-T1 PMA/PMD returns a one in bit 1.2304.15 when a reset is in progress                                                   | 45.2.1.237.1 |                                                         | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM132 | 1000BASE-T1 PMA/PMD returns a zero in bit 1.2304.15 when a reset is complete                                                     | 45.2.1.237.1 |                                                         | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM133 | During a reset, the 1000BASE-T1 PMD/PMA responds to reads from register bits 1.2304.15, 1.8.15:14, and 1.0.15.                   | 45.2.1.237.1 |                                                         | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM134 | During a reset, the 1000BASE-T1 PMD/PMA register bits are ignored.                                                               | 45.2.1.237.1 |                                                         | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM135 | Setting either 1.2304.15 or 1.0.15 resets the 1000BASE-T1 PMA/PMD.                                                               | 45.2.1.237.1 |                                                         | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM136 | When bit 1.2304.14 is set to a one, the PMA disables output on the transmit path.                                                | 45.2.1.237.2 |                                                         | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM137 | When bit 1.2304.14 is set to a zero, the PMA enables output on the transmit path.                                                | 45.2.1.237.2 |                                                         | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM138 | Setting either bit 1.2304.14 or 1.9.0 disables the 1000BASE-T1 PMA transmit path.                                                | 45.2.1.237.2 |                                                         | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM139 | While in the low-power mode, the device, as a minimum, responds to management transactions necessary to exit the low-power mode. | 45.2.1.237.3 |                                                         | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM140 | Setting either 1.2304.11 or 1.0.11 puts the 1000BASE-T1 PMA/PMD in low-power mode.                                               | 45.2.1.237.3 |                                                         | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM141 | The 1000BASE-T1 PMA/PMD that is unable to detect a fault condition on the receive path returns a value of zero for bit 1.2305.1. | 45.2.1.238.6 |                                                         | PMA:M  | Yes [ ]<br>N/A [ ] |

| Item  | Feature                                                                                                                          | Subclause    | Value/Comment | Status | Support                      |
|-------|----------------------------------------------------------------------------------------------------------------------------------|--------------|---------------|--------|------------------------------|
| MM142 | Bit 1.2305.0 is implemented with latching low behavior.                                                                          | 45.2.1.238.6 |               | PMA:M  | Yes [ ]<br>N/A [ ]           |
| MM143 | Bit 1.2306.1 is set to zero if the 1000BASE-T1 PHY does not support OAM.                                                         | 45.2.1.239.2 |               | PMA:M  | Yes [ ]<br>N/A [ ]           |
| MM144 | Bit 1.2306.0 is set to zero if the 1000BASE-T1 PHY does not support EEE.                                                         | 45.2.1.239.3 |               | PMA:M  | Yes [ ]<br>N/A [ ]           |
| MM145 | OAM capability is enabled only when both the 1000BASE-T1 PHY and link partner are advertising OAM capability.                    | 45.2.1.240.2 |               | PMA:M  | Yes [ ]<br>N/A [ ]           |
| MM146 | EEE capability is enabled only when both the 1000BASE-T1 PHY and link partner are advertising EEE capability                     | 45.2.1.240.3 |               | PMA:M  | Yes [ ]<br>N/A [ ]           |
| MM147 | Bit 1.2100.14 is ignored when the Auto-Negotiation enable bit 7.512.12 is set to one.                                            | 45.2.1.214.1 |               | PMA:M  | Yes [ ]<br>N/A [ ]           |
| MM148 | Bits 1.2100.3:0 is ignored when the Auto-Negotiation enable bit 7.512.12 is set to one.                                          | 45.2.1.214.2 |               | PMA:M  | Yes [ ]<br>N/A [ ]           |
| MM149 | Writes to the register 1.22 have no effect                                                                                       | 45.2.1.20    |               | PMA:O  | Yes [ ]<br>No [ ]<br>N/A [ ] |
| MM150 | Writes of reserved values or modes of operation different of those advertised by register 1.22, to bits 1.900.3:0 have no effect | 45.2.1.158   |               | BH:M   | Yes [ ]<br>N/A [ ]           |
| MM151 | Bits 1.2100.3:0 are ignored when Auto-Negotiation enable bit 7.512.12 is set to one                                              | 45.2.1.214.2 |               | PMA:M  | Yes [ ]<br>N/A [ ]           |
| MM152 | A reset sets all 10BASE-T1L PMA registers to their default states                                                                | 45.2.1.231.1 |               | PMA:M  | Yes [ ]<br>N/A [ ]           |
| MM153 | 10BASE-T1L PMA returns a one in bit 1.2294.15 when a reset is in progress; otherwise, it returns a value of zero                 | 45.2.1.231.1 |               | PMA:M  | Yes [ ]<br>N/A [ ]           |
| MM154 | The control and management interface is restored to operation within 0.5 s from the setting of bit 1.2294.15                     | 45.2.1.231.1 |               | PMA:M  | Yes [ ]<br>N/A [ ]           |
| MM155 | During a reset, the 10BASE-T1L PMA responds to reads from register bits 1.2294.15, 1.8.15:14, and 1.0.15                         | 45.2.1.231.1 |               | PMA:M  | Yes [ ]<br>N/A [ ]           |

| Item  | Feature                                                                                                      | Subclause    | Value/Comment                                                                                                          | Status | Support            |
|-------|--------------------------------------------------------------------------------------------------------------|--------------|------------------------------------------------------------------------------------------------------------------------|--------|--------------------|
| MM156 | Setting either 1.2294.15 or 1.0.15 sets the other                                                            | 45.2.1.231.1 |                                                                                                                        | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM157 | Clearing either 1.2294.15 or 1.0.15 clears the other                                                         | 45.2.1.231.1 |                                                                                                                        | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM158 | Setting either 1.2294.15 or 1.0.15 resets the 10BASE-T1L PMA                                                 | 45.2.1.231.1 |                                                                                                                        | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM159 | When bit 1.2294.14 is set to one, the 10BASE-T1L PMA disables output on the transmit path                    | 45.2.1.231.2 |                                                                                                                        | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM160 | When bit 1.2294.14 is set to zero, the 10BASE-T1L PMA enables output on the transmit path                    | 45.2.1.231.2 |                                                                                                                        | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM161 | Setting either 1.2294.14 or 1.9.0 sets the other                                                             | 45.2.1.231.2 |                                                                                                                        | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM162 | Clearing either 1.2294.14 or 1.9.0 clears the other                                                          | 45.2.1.231.2 |                                                                                                                        | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM163 | Setting either 1.2294.14 or 1.9.0 disables the transmitter                                                   | 45.2.1.231.2 |                                                                                                                        | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM164 | 10BASE-T1L Transmit voltage amplitude control when Auto-Negotiation is not implemented or is not enabled     | 45.2.1.231.3 | 1 = 10BASE-T1L PMA transmits using 2.4 Vpp operating mode<br>0 = 10BASE-T1L PMA transmits using 1.0 Vpp operating mode | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM165 | When Auto-Negotiation is implemented and enabled, setting bit 1.2294.12 has no effect                        | 45.2.1.231.3 |                                                                                                                        | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM166 | While in low-power mode, the device responds to management transactions necessary to exit the low-power mode | 45.2.1.231.4 |                                                                                                                        | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM167 | Setting either 1.2294.11 or 1.0.11 sets the other                                                            | 45.2.1.231.4 |                                                                                                                        | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM168 | Clearing either 1.2294.11 or 1.0.11 clears the other                                                         | 45.2.1.231.4 |                                                                                                                        | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM169 | Setting either 1.2294.11 or 1.0.11 puts the 10BASE-T1L PMA in low-power mode                                 | 45.2.1.231.4 |                                                                                                                        | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM170 | 10BASE-T1L EEE functionality when Auto-Negotiation is not implemented or is not enabled                      | 45.2.1.231.5 | 1 = PHY operates with EEE enabled<br>0 = PHY operates with EEE disabled                                                | PMA:M  | Yes [ ]<br>N/A [ ] |

| Item  | Feature                                                                                                                                                                | Subclause    | Value/Comment | Status | Support            |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|---------------|--------|--------------------|
| MM171 | When Auto-Negotiation is implemented and enabled, setting bit 1.2294.10 has no effect                                                                                  | 45.2.1.231.5 |               | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM172 | When bit 1.2294.0 is set to one, the 10BASE-T1L PMA is placed into near-end loopback mode, and it accepts data on the transmit path and returns it on the receive path | 45.2.1.231.6 |               | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM173 | Setting either 1.2294.0 or 1.0.0 sets the other                                                                                                                        | 45.2.1.231.6 |               | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM174 | Clearing either 1.2294.0 or 1.0.0 clears the other                                                                                                                     | 45.2.1.231.6 |               | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM175 | Setting either 1.2294.0 or 1.0.0 enables loopback                                                                                                                      | 45.2.1.231.6 |               | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM176 | The 10BASE-T1L PMA that is unable to detect a fault condition on the receive path returns a value of zero for bit 1.2295.1                                             | 45.2.1.232.7 |               | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM177 | The 10BASE-T1L PMA receive fault bit is implemented with latching high behavior                                                                                        | 45.2.1.232.7 |               | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM178 | The 10BASE-T1L PMA receive link status bit is implemented with latching low behavior                                                                                   | 45.2.1.232.8 |               | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM179 | A reset sets all 10BASE-T1S PMA registers to their default states                                                                                                      | 45.2.1.234.1 |               | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM180 | 10BASE-T1S PMA returns a one in bit 1.2297.15 when a reset is in progress; otherwise, it returns a value of zero                                                       | 45.2.1.234.1 |               | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM181 | The control and management interface is restored to operation within 0.5 s from the setting of bit 1.2297.15                                                           | 45.2.1.234.1 |               | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM182 | During a reset, the 10BASE-T1S PMA responds to reads from register bits 1.2297.15, 1.8.15:14, and 1.0.15                                                               | 45.2.1.234.1 |               | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM183 | Setting either 1.2297.15 or 1.0.15 sets the other                                                                                                                      | 45.2.1.234.1 |               | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM184 | Clearing either 1.2297.15 or 1.0.15 clears the other                                                                                                                   | 45.2.1.234.1 |               | PMA:M  | Yes [ ]<br>N/A [ ] |

| Item  | Feature                                                                                                                                                                               | Subclause    | Value/Comment | Status | Support            |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|---------------|--------|--------------------|
| MM185 | Setting either 1.2297.15 or 1.0.15 resets the 10BASE-T1S PMA                                                                                                                          | 45.2.1.234.1 |               | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM186 | When bit 1.2297.14 is set to one, the 10BASE-T1S PMA disables output on the transmit path                                                                                             | 45.2.1.234.2 |               | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM187 | When bit 1.2297.14 is set to zero, the 10BASE-T1S PMA enables output on the transmit path                                                                                             | 45.2.1.234.2 |               | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM188 | Setting either 1.2297.14 or 1.0.14 sets the other                                                                                                                                     | 45.2.1.234.2 |               | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM189 | Clearing either 1.2297.14 or 1.0.14 clears the other                                                                                                                                  | 45.2.1.234.2 |               | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM190 | Setting either 1.2297.14 or 1.0.14 disables the transmitter                                                                                                                           | 45.2.1.234.2 |               | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM191 | While in low-power mode, the device responds to management transactions necessary to exit the low-power mode                                                                          | 45.2.1.234.3 |               | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM192 | Setting either 1.2297.11 or 1.0.11 sets the other                                                                                                                                     | 45.2.1.234.3 |               | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM193 | Clearing either 1.2297.11 or 1.0.11 clears the other                                                                                                                                  | 45.2.1.234.3 |               | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM194 | Setting either 1.2297.11 or 1.0.11 puts the 10BASE-T1S PMA in low-power mode                                                                                                          | 45.2.1.234.3 |               | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM195 | When bit 1.2297.10 is set to one, the 10BASE-T1S PMA is configured to operate in multidrop mode                                                                                       | 45.2.1.234.4 |               | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM196 | If multidrop mode is not supported according to bit 1.2298.10, writing to bit 1.2297.10 has no effect, and the PCS operates in half duplex mode with bits 3.2291.8 and 0.8 set to one | 45.2.1.234.4 |               | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM197 | When bit 1.2297.0 is set to one, the 10BASE-T1S PMA is placed into loopback mode, and it accepts data on the transmit path and returns it on the receive path                         | 45.2.1.234.5 |               | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM198 | Setting either 1.2297.0 or 1.0.0 sets the other                                                                                                                                       | 45.2.1.234.5 |               | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM199 | Clearing either 1.2297.0 or 1.0.0 clears the other                                                                                                                                    | 45.2.1.234.5 |               | PMA:M  | Yes [ ]<br>N/A [ ] |

| Item  | Feature                                                                                                                    | Subclause    | Value/Comment | Status | Support            |
|-------|----------------------------------------------------------------------------------------------------------------------------|--------------|---------------|--------|--------------------|
| MM200 | Setting either 1.2297.0 or 1.0.0 enables loopback                                                                          | 45.2.1.234.5 |               | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM201 | The 10BASE-T1S PMA that is unable to detect a fault condition on the receive path returns a value of zero for bit 1.2298.1 | 45.2.1.235.5 |               | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM202 | The 10BASE-T1S PMA receive fault bit is implemented with latching high behavior                                            | 45.2.1.235.5 |               | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM203 | Setting bit 1.2309.15 sets all MultiGBASE-T1 PMA/PMD registers to their default states.                                    | 45.2.1.242.1 |               | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM204 | The MultiGBASE-T1 PMA/PMD returns a value of one in bit 1.2309.15 when a reset is in progress.                             | 45.2.1.242.1 |               | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM205 | The MultiGBASE-T1 PMA/PMD returns a value of zero in bit 1.2309.15 when a reset is complete.                               | 45.2.1.242.1 |               | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM206 | During a reset, the MultiGBASE-T1 PMA/PMD responds to reads from register bits 1.2309.15, 1.8.15:14, and 1.0.15.           | 45.2.1.242.1 |               | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM207 | During a reset, all MultiGBASE-T1 PMA/PMD register bits are ignored.                                                       | 45.2.1.242.1 |               | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM208 | Setting or clearing either bit 1.2309.15 or 1.0.15 sets or clears the other bit.                                           | 45.2.1.242.1 |               | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM209 | Setting either bit 1.2309.15 or 1.0.15 resets the MultiGBASE-T1 PMA/PMD.                                                   | 45.2.1.242.1 |               | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM210 | When bit 1.2309.14 is set to a one, the PMA disables output on the transmit path.                                          | 45.2.1.242.2 |               | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM211 | When bit 1.2309.14 is zero, the PMA enables output on the transmit path.                                                   | 45.2.1.242.2 |               | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM212 | Setting or clearing either bit 1.2309.14 or 1.0.14 sets or clears the other bit.                                           | 45.2.1.242.2 |               | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM213 | Setting either bit 1.2309.14 or 1.0.14 disables the MultiGBASE-T1 transmit path.                                           | 45.2.1.242.2 |               | PMA:M  | Yes [ ]<br>N/A [ ] |



| Item  | Feature                                                                                                                          | Subclause    | Value/Comment                                                                                           | Status | Support            |
|-------|----------------------------------------------------------------------------------------------------------------------------------|--------------|---------------------------------------------------------------------------------------------------------|--------|--------------------|
| MM214 | While in low-power mode, the device, at a minimum, responds to management transactions necessary to exit the low-power mode.     | 45.2.1.242.3 |                                                                                                         | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM215 | Setting or clearing either bit 1.2309.11 or 1.0.11 sets or clears the other bit.                                                 | 45.2.1.242.3 |                                                                                                         | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM216 | Setting either bit 1.2309.11 or 1.0.11 puts the MultiGBASE-T1 PMA/PMD in low-power mode.                                         | 45.2.1.242.3 |                                                                                                         | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM217 | A MultiGBASE-T1 PMA/PMD that is unable to detect a fault condition on the receive path returns a value of zero for bit 1.2310.1. | 45.2.1.243.7 |                                                                                                         | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM218 | The receive fault bit (1.2310.1) is implemented with latching high behavior.                                                     | 45.2.1.243.7 |                                                                                                         | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM219 | The link status bit (1.2310.0) is implemented with latching low behavior.                                                        | 45.2.1.243.8 |                                                                                                         | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM220 | When bit 1.2311.5 is set to a one, the PHY uses 1.23.11.3:2 for the value of PrecodeSel.                                         | 45.2.1.244.2 |                                                                                                         | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM221 | When bit 1.2311.5 is set to a zero, the PHY controls the value of PrecodeSel.                                                    | 45.2.1.244.2 |                                                                                                         | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM222 | Advertisement of support for MultiGBASE-T1 OAM.                                                                                  | 45.2.1.244.5 | Support is advertised if bit 1.2311.1 is set to one, and not advertised if bit 1.2311.1 is set to zero. | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM223 | Advertisement of support for MultiGBASE-T1 EEE.                                                                                  | 45.2.1.244.6 | Support is advertised if bit 1.2311.0 is set to one, and not advertised if bit 1.2311.0 is set to zero. | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM224 | OAM capability is enabled only when both the local PHY and its link partner are advertising OAM capability.                      | 45.2.1.245.4 |                                                                                                         | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM225 | EEE capability is enabled only when both the local PHY and its link partner are advertising EEE capability.                      | 45.2.1.245.5 |                                                                                                         | PMA:M  | Yes [ ]<br>N/A [ ] |

| Item  | Feature                                                                                                                                                                                                 | Subclause    | Value/Comment                                                                                           | Status | Support            |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|---------------------------------------------------------------------------------------------------------|--------|--------------------|
| MM226 | When bit 1.2313.11 is set to one, the local transmitter's precoder shall be controlled by the value of bits 1.2313.10:9, and the precoder requested by the link partner in PrecodeSel shall be ignored. | 45.2.1.246.2 |                                                                                                         | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM227 | When bit 1.2313.11 is set to zero, the transmitter shall ignore bits 1.2313.10:9, and the precoder is set according to the value of PrecodeSel.                                                         | 45.2.1.246.2 |                                                                                                         | PMA:M  | Yes [ ]<br>N/A [ ] |
| MM228 | PMA/PMD type selection                                                                                                                                                                                  | 45.2.1.27.2  | Device ignore writes to the PMA/PMD type selection bits that select PMA/PMD types it has not advertised | M      | Yes [ ]            |

#### 45.5.3.4 WIS options

| Item | Feature                                  | Subclause | Value/Comment | Status | Support                      |
|------|------------------------------------------|-----------|---------------|--------|------------------------------|
| *WPT | Implementation of PRBS31 pattern testing | 45.2.2    |               | WIS:O  | Yes [ ]<br>No [ ]<br>N/A [ ] |

#### 45.5.3.5 WIS management functions

| Item | Feature                                                                                   | Subclause | Value/Comment                             | Status | Support            |
|------|-------------------------------------------------------------------------------------------|-----------|-------------------------------------------|--------|--------------------|
| WM1  | Device responds to all register addresses for that device                                 | 45.2      |                                           | WIS:M  | Yes [ ]<br>N/A [ ] |
| WM2  | Writes to undefined and read-only registers have no effect                                | 45.2      |                                           | WIS:M  | Yes [ ]<br>N/A [ ] |
| WM3  | Operation is not affected by writes to reserved and unsupported bits.                     | 45.2      |                                           | WIS:M  | Yes [ ]<br>N/A [ ] |
| WM4  | Reserved and unsupported bits return a value of zero                                      | 45.2      |                                           | WIS:M  | Yes [ ]<br>N/A [ ] |
| WM5  | Latching low bits remain low until after they have been read via the management interface | 45.2      |                                           | WIS:M  | Yes [ ]<br>N/A [ ] |
| WM6  | Latching low bits assume correct value once read via the management interface             | 45.2      | Correct value is based upon current state | WIS:M  | Yes [ ]<br>N/A [ ] |

| Item | Feature                                                                                                            | Subclause  | Value/Comment                                                                                            | Status     | Support            |
|------|--------------------------------------------------------------------------------------------------------------------|------------|----------------------------------------------------------------------------------------------------------|------------|--------------------|
| WM7  | Latching high bits remain high until after they have been read via the management interface                        | 45.2       |                                                                                                          | WIS:M      | Yes [ ]<br>N/A [ ] |
| WM8  | Latching high bits assume correct value once read via the management interface                                     | 45.2       | Correct value is based upon current state                                                                | WIS:M      | Yes [ ]<br>N/A [ ] |
| WM9  | Action on reset                                                                                                    | 45.2.2.1.1 | Reset the registers of the entire device to default values and set bit 15 of the Control register to one | WIS:M      | Yes [ ]<br>N/A [ ] |
| WM10 | Return 1 until reset completed                                                                                     | 45.2.2.1.1 |                                                                                                          | WIS:M      | Yes [ ]<br>N/A [ ] |
| WM11 | Reset completes within 0.5 s                                                                                       | 45.2.2.1.1 |                                                                                                          | WIS:M      | Yes [ ]<br>N/A [ ] |
| WM12 | Responds to reads of bits 2.0.15 and 2.8.15:14 during reset                                                        | 45.2.2.1.1 |                                                                                                          | WIS:M      | Yes [ ]<br>N/A [ ] |
| WM13 | Loopback mode                                                                                                      | 45.2.2.1.2 | Whenever bit 2.0.14 is set to a one                                                                      | WIS:M      | Yes [ ]<br>N/A [ ] |
| WM14 | Data received from PMA ignored during loopback                                                                     | 45.2.2.1.2 |                                                                                                          | WIS:M      | Yes [ ]<br>N/A [ ] |
| WM15 | Transmit data returned on receive path during loopback                                                             | 45.2.2.1.2 |                                                                                                          | WIS:M      | Yes [ ]<br>N/A [ ] |
| WM16 | Device responds to transactions necessary to exit low-power mode while in low-power state                          | 45.2.2.1.3 |                                                                                                          | WIS:M      | Yes [ ]<br>N/A [ ] |
| WM17 | Speed selection bits 13 and 6 are written as one                                                                   | 45.2.2.1.4 |                                                                                                          | WIS:M      | Yes [ ]<br>N/A [ ] |
| WM18 | Invalid writes to speed selection bits are ignored                                                                 | 45.2.2.1.4 |                                                                                                          | WIS:M      | Yes [ ]<br>N/A [ ] |
| WM19 | Writes to status 1 register have no effect                                                                         | 45.2.2.2   |                                                                                                          | WIS:M      | Yes [ ]<br>N/A [ ] |
| WM20 | Fault bit implemented using latching high behavior                                                                 | 45.2.2.2.1 |                                                                                                          | WIS:M      | Yes [ ]<br>N/A [ ] |
| WM21 | Link status bit implemented using latching low behavior                                                            | 45.2.2.2.2 |                                                                                                          | WIS:M      | Yes [ ]<br>N/A [ ] |
| WM22 | Unique identifier is composed of 22 bits of OUI, model number and revision                                         | 45.2.2.3   |                                                                                                          | WIS:M      | Yes [ ]<br>N/A [ ] |
| WM23 | Setting bit 2.7.5 to a one enables PRBS31 receive pattern testing if bit 2.8.1 is a one and bit 2.7.2 is not a one | 45.2.2.6.1 |                                                                                                          | WIS* WPT:M | Yes [ ]<br>N/A [ ] |
| WM24 | Setting bit 2.7.5 to a zero disables PRBS31 receive pattern testing                                                | 45.2.2.6.1 |                                                                                                          | WIS* WPT:M | Yes [ ]<br>N/A [ ] |

| Item | Feature                                                                                                             | Subclause   | Value/Comment | Status     | Support            |
|------|---------------------------------------------------------------------------------------------------------------------|-------------|---------------|------------|--------------------|
| WM25 | Setting bit 2.7.4 to a one enables PRBS31 transmit pattern testing if bit 2.8.1 is a one and bit 2.7.1 is not a one | 45.2.2.6.2  |               | WIS* WPT:M | Yes [ ]<br>N/A [ ] |
| WM26 | Setting bit 2.7.4 to a zero disables PRBS31 transmit pattern testing                                                | 45.2.2.6.2  |               | WIS* WPT:M | Yes [ ]<br>N/A [ ] |
| WM27 | Setting bit 3 to one selects the square wave test pattern                                                           | 45.2.2.6.3  |               | WIS:M      | Yes [ ]<br>N/A [ ] |
| WM28 | Setting bit 3 to zero selects the pseudo random test pattern                                                        | 45.2.2.6.3  |               | WIS:M      | Yes [ ]<br>N/A [ ] |
| WM29 | Setting bit 2 to one enables receive pattern testing                                                                | 45.2.2.6.4  |               | WIS:M      | Yes [ ]<br>N/A [ ] |
| WM30 | Setting bit 2 to zero disables receive pattern testing                                                              | 45.2.2.6.4  |               | WIS:M      | Yes [ ]<br>N/A [ ] |
| WM31 | Setting bit 1 to one enables transmit pattern testing                                                               | 45.2.2.6.5  |               | WIS:M      | Yes [ ]<br>N/A [ ] |
| WM32 | Setting bit 1 to zero disables transmit pattern testing                                                             | 45.2.2.6.5  |               | WIS:M      | Yes [ ]<br>N/A [ ] |
| WM33 | Setting bit 0 to a one enables 10GBASE-W logic and sets interface speed                                             | 45.2.2.6.6  |               | WIS:M      | Yes [ ]<br>N/A [ ] |
| WM34 | Setting bit 0 to a zero disables 10GBASE-W logic, sets interface speed, and bypasses data                           | 45.2.2.6.6  |               | WIS:O      | Yes [ ]<br>N/A [ ] |
| WM35 | Writes to bit are ignored by WIS not capable of supporting 10GBASE-R                                                | 45.2.2.6.6  |               | WIS:M      | Yes [ ]<br>N/A [ ] |
| WM36 | Bit returns one when read if WIS is not capable of supporting 10GBASE-R                                             | 45.2.2.6.6  |               | WIS:M      | Yes [ ]<br>N/A [ ] |
| WM37 | Writes to status 2 register have no effect                                                                          | 45.2.2.7    |               | WIS:M      | Yes [ ]<br>N/A [ ] |
| WM38 | Counter is reset to all zeros when read or reset                                                                    | 45.2.2.8    |               | WIS* WPT:M | Yes [ ]<br>N/A [ ] |
| WM39 | Counter is held at all ones at overflow                                                                             | 45.2.2.8    |               | WIS* WPT:M | Yes [ ]<br>N/A [ ] |
| WM40 | Unique identifier is composed of 22 bits of OUI, model number and revision                                          | 45.2.2.9    |               | WIS:M      | Yes [ ]<br>N/A [ ] |
| WM41 | Writes to Status 3 register have no effect                                                                          | 45.2.2.10   |               | WIS:M      | Yes [ ]<br>N/A [ ] |
| WM42 | SEF bit implemented using latching high behavior                                                                    | 45.2.2.10.1 |               | WIS:M      | Yes [ ]<br>N/A [ ] |

| Item | Feature                                                          | Subclause    | Value/Comment | Status | Support            |
|------|------------------------------------------------------------------|--------------|---------------|--------|--------------------|
| WM43 | Far end PLM-P/LCD-P bit implemented using latching high behavior | 45.2.2.10.2  |               | WIS:M  | Yes [ ]<br>N/A [ ] |
| WM44 | Far end AIS-P/LOP-P bit implemented using latching high behavior | 45.2.2.10.3  |               | WIS:M  | Yes [ ]<br>N/A [ ] |
| WM45 | LOF bit implemented using latching high behavior                 | 45.2.2.10.4  |               | WIS:M  | Yes [ ]<br>N/A [ ] |
| WM46 | LOS bit implemented using latching high behavior                 | 45.2.2.10.5  |               | WIS:M  | Yes [ ]<br>N/A [ ] |
| WM47 | RDI-L bit implemented using latching high behavior               | 45.2.2.10.6  |               | WIS:M  | Yes [ ]<br>N/A [ ] |
| WM48 | AIS-L bit implemented using latching high behavior               | 45.2.2.10.7  |               | WIS:M  | Yes [ ]<br>N/A [ ] |
| WM49 | LCD-P bit implemented using latching high behavior               | 45.2.2.10.8  |               | WIS:M  | Yes [ ]<br>N/A [ ] |
| WM50 | PLM-P bit implemented using latching high behavior               | 45.2.2.10.9  |               | WIS:M  | Yes [ ]<br>N/A [ ] |
| WM51 | AIS-P bit implemented using latching high behavior               | 45.2.2.10.10 |               | WIS:M  | Yes [ ]<br>N/A [ ] |
| WM52 | LOP-P bit implemented using latching high behavior               | 45.2.2.10.11 |               | WIS:M  | Yes [ ]<br>N/A [ ] |

#### 45.5.3.6 PCS options

| Item   | Feature                                                                                    | Subclause | Value/Comment | Status                      | Support                      |
|--------|--------------------------------------------------------------------------------------------|-----------|---------------|-----------------------------|------------------------------|
| *100CR | Implementation of 100GBASE-R PCS                                                           | 45.2.3.7  |               | PCS:O                       | Yes [ ]<br>No [ ]<br>N/A [ ] |
| *40CR  | Implementation of 40GBASE-R PCS                                                            | 45.2.3.7  |               | PCS:O                       | Yes [ ]<br>No [ ]<br>N/A [ ] |
| *CR    | Implementation of 10GBASE-R PCS                                                            | 45.2.3    |               | PCS:O                       | Yes [ ]<br>No [ ]<br>N/A [ ] |
| *XCR   | Implementation of 10/40/100GBASE-R PCS                                                     | 45.2.3.7  |               | 100CR:M<br>40CR:M<br>10CR:M | Yes [ ]<br>No [ ]<br>N/A [ ] |
| *CT    | Implementation of the 10GBASE-T PCS or the PCS of any other member of the MultiGBASE-T set | 45.2.3    |               | PCS:O                       | Yes [ ]<br>No [ ]<br>N/A [ ] |
| *CX    | Implementation of 10GBASE-X PCS                                                            | 45.2.3    |               | PCS:O                       | Yes [ ]<br>No [ ]<br>N/A [ ] |

| Item   | Feature                                           | Subclause               | Value/Comment | Status | Support                      |
|--------|---------------------------------------------------|-------------------------|---------------|--------|------------------------------|
| *XP    | Implementation of 10GBASE-X pattern testing       | 45.2.3                  |               | CX:O   | Yes [ ]<br>No [ ]<br>N/A [ ] |
| *PPT   | Implementation of PRBS31 pattern testing          | 45.2.3                  |               | PCS:O  | Yes [ ]<br>No [ ]<br>N/A [ ] |
| *PTT   | Implementation of PRBS9 pattern testing           | 45.2.3                  |               | PCS:O  | Yes [ ]<br>No [ ]<br>N/A [ ] |
| *EPC   | Implementation of the 10BASE-TS/2BASE-TL PCS      | 45.2.3.27               |               | PCS:O  | Yes [ ]<br>No [ ]<br>N/A [ ] |
| *PAF   | Implementation of the PME aggregation function    | 45.2.3.27               |               | EPC:O  | Yes [ ]<br>No [ ]<br>N/A [ ] |
| *CPR   | Implementation of 10GBASE-PR or 10/1GBASE-PRX PCS | 45.2.3                  |               | PCS:O  | Yes [ ]<br>No [ ]<br>N/A [ ] |
| *EEE   | Implementation of EEE                             |                         |               | O      | Yes [ ]<br>N/A [ ]           |
| *BHOAM | 1000BASE-H OAM channel implementation             | 45.2.3.51 and 45.2.3.52 |               | PCS:O  | Yes [ ]<br>No [ ]<br>N/A [ ] |
| 2.5GX  | Implementation of 2.5GBASE-X PCS                  | 45.2.3.8                |               | PCS:O  | Yes [ ]<br>No [ ]<br>N/A [ ] |
| 5GR    | Implementation of 5GBASE-R PCS                    | 45.2.3.8                |               | PCS:O  | Yes [ ]<br>No [ ]<br>N/A [ ] |

#### 45.5.3.7 PCS management functions

| Item | Feature                                                                                   | Subclause | Value/Comment | Status | Support            |
|------|-------------------------------------------------------------------------------------------|-----------|---------------|--------|--------------------|
| RM1  | Device responds to all register addresses for that device                                 | 45.2      |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM2  | Writes to undefined and read-only registers have no effect                                | 45.2      |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM3  | Operation is not affected by writes to reserved and unsupported bits                      | 45.2      |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM4  | Reserved and unsupported bits return a value of zero                                      | 45.2      |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM5  | Latching low bits remain low until after they have been read via the management interface | 45.2      |               | PCS:M  | Yes [ ]<br>N/A [ ] |

| Item | Feature                                                                                                     | Subclause  | Value/Comment                                                                                            | Status | Support            |
|------|-------------------------------------------------------------------------------------------------------------|------------|----------------------------------------------------------------------------------------------------------|--------|--------------------|
| RM6  | Latching low bits assume correct value once read via the management interface                               | 45.2       | Correct value is based upon current state                                                                | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM7  | Latching high bits remain high until after they have been read via the management interface                 | 45.2       |                                                                                                          | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM8  | Latching high bits assume correct value once read via the management interface                              | 45.2       | Correct value is based upon current state                                                                | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM9  | Action on reset                                                                                             | 45.2.3.1.1 | Reset the registers of the entire device to default values and set bit 15 of the Control register to one | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM10 | Return 1 until reset completed                                                                              | 45.2.3.1.1 |                                                                                                          | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM11 | Reset completes within 0.5 s                                                                                | 45.2.3.1.1 |                                                                                                          | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM12 | Device responds to reads of register bits 3.0.15 and 3.5.15:14 during reset                                 | 45.2.3.1.1 |                                                                                                          | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM13 | Loopback mode                                                                                               | 45.2.3.1.2 | Whenever bit 3.0.14 is set to a one                                                                      | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM14 | Transmit data is returned on the receive path during loopback                                               | 45.2.3.1.2 |                                                                                                          | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM15 | Writes to loopback bit are ignored with port type selections other than 100BASE-T1, 10GBASE-R, or 10GBASE-T | 45.2.3.1.2 |                                                                                                          | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM16 | Loopback bit returns zero with port type selections other than 100BASE-T1, 10GBASE-R, or 10GBASE-T          | 45.2.3.1.2 |                                                                                                          | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM17 | Device responds to transactions necessary to exit low-power mode while in low-power state                   | 45.2.3.1.3 |                                                                                                          | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM18 | Speed selection bits 13 and 6 are written as one                                                            | 45.2.3.1.5 |                                                                                                          | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM19 | Invalid writes to speed selection bits are ignored                                                          | 45.2.3.1.5 |                                                                                                          | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM20 | Writes to PCS status 1 register have no effect                                                              | 45.2.3.2   |                                                                                                          | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM21 | Receive link status implemented using latching low behavior                                                 | 45.2.3.2.7 |                                                                                                          | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM22 | Unique identifier is composed of 22 bits of OUI, model number and revision                                  | 45.2.3.3   |                                                                                                          | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM23 | PCS type is selected using bits 4 through 0                                                                 | 45.2.3.6.1 |                                                                                                          | PCS:M  | Yes [ ]<br>N/A [ ] |

| Item | Feature                                                                                                                                 | Subclause   | Value/Comment | Status        | Support            |
|------|-----------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------|---------------|--------------------|
| RM24 | Writes to the type selection bits that select types that have not been advertised are ignored                                           | 45.2.3.6.1  |               | PCS:M         | Yes [ ]<br>N/A [ ] |
| RM25 | Writes to PCS status 2 register have no effect                                                                                          | 45.2.3.7    |               | PCS:M         | Yes [ ]<br>N/A [ ] |
| RM26 | Transmit fault implemented with latching high behavior                                                                                  | 45.2.3.7.4  |               | PCS:M         | Yes [ ]<br>N/A [ ] |
| RM27 | Receive fault implemented with latching high behavior                                                                                   | 45.2.3.7.5  |               | PCS:M         | Yes [ ]<br>N/A [ ] |
| RM28 | Unique identifier is composed of 22 bits of OUI, model number and revision                                                              | 45.2.3.9    |               | PCS:M         | Yes [ ]<br>N/A [ ] |
| RM29 | Writes to 10GBASE-X PCS status register have no effect                                                                                  | 45.2.3.13   |               | PCS* CX:M     | Yes [ ]<br>N/A [ ] |
| RM30 | Register returns zero if the PCS does not implement the 10GBASE-X port type                                                             | 45.2.3.13   |               | PCS* !CX:M    | Yes [ ]<br>N/A [ ] |
| RM31 | EEE capability indicated for each port type                                                                                             | 45.2.3.10   |               | EEE:M         | Yes [ ]<br>N/A [ ] |
| RM32 | EEE wake error counter behavior as specified                                                                                            | 45.2.3.12   |               | EEE:M         | Yes [ ]<br>N/A [ ] |
| RM33 | Writes to bit are ignored and reads return a value of zero                                                                              | 45.2.3.14.1 |               | PCS* PX:M     | Yes [ ]<br>N/A [ ] |
| RM34 | Setting the bits to <10> selects the mixed frequency pattern                                                                            | 45.2.3.14.2 |               | PCS* PX:M     | Yes [ ]<br>N/A [ ] |
| RM35 | Setting the bits to <01> selects the low-frequency pattern                                                                              | 45.2.3.14.2 |               | PCS* PX:M     | Yes [ ]<br>N/A [ ] |
| RM36 | Setting the bits to <00> selects the high-frequency pattern                                                                             | 45.2.3.14.2 |               | PCS* PX:M     | Yes [ ]<br>N/A [ ] |
| RM37 | Writes to BASE-R and MultiGBASE-T PCS status 1 register have no effect                                                                  | 45.2.3.15   |               | CR:M<br>CT:M  | Yes [ ]<br>N/A [ ] |
| RM38 | Reads from BASE-R and MultiGBASE-T PCS status 1 register return zero for PCS that does not support 10GBASE-R                            | 45.2.3.15   |               | CR:M<br>CT:M  | Yes [ ]<br>N/A [ ] |
| RM39 | Writes to BASE-R and MultiGBASE-T PCS status 2 register have no effect                                                                  | 45.2.3.16   |               | XCR:M<br>CT:M | Yes [ ]<br>N/A [ ] |
| RM40 | Reads from BASE-R and MultiGBASE-T PCS status 2 register return zero for PCS that does not support 10/40/100GBASE-R or any MultiGBASE-T | 45.2.3.16   |               | XCR:M<br>CT:M | Yes [ ]<br>N/A [ ] |
| RM41 | Latched block lock implemented with latching low behavior                                                                               | 45.2.3.16.1 |               | XCR:M<br>CT:M | Yes [ ]<br>N/A [ ] |



| Item | Feature                                                                                                                | Subclause   | Value/Comment | Status                    | Support            |
|------|------------------------------------------------------------------------------------------------------------------------|-------------|---------------|---------------------------|--------------------|
| RM42 | Latched high BER implemented with latching high behavior                                                               | 45.2.3.16.2 |               | XCR:M<br>CT:M             | Yes [ ]<br>N/A [ ] |
| RM43 | BER counter resets to all zeros on read or reset                                                                       | 45.2.3.16.3 |               | XCR:M<br>CT:M             | Yes [ ]<br>N/A [ ] |
| RM44 | BER counter holds at all ones at overflow                                                                              | 45.2.3.16.3 |               | !RM53:M                   | Yes [ ]<br>N/A [ ] |
| RM45 | Errored blocks counter implemented as a non roll over counter                                                          | 45.2.3.16.4 |               | !RM57:M                   | Yes [ ]<br>N/A [ ] |
| RM46 | Errored blocks counter resets to all zeros on read                                                                     | 45.2.3.16.4 |               | XCR:M<br>CT:M             | Yes [ ]<br>N/A [ ] |
| RM47 | Setting bit 3.42.5 to a one enables PRBS31 receive pattern testing if bit 3.32.2 is a one and bit 3.42.2 is not a one  | 45.2.3.19.3 |               | PCS* PPT:M                | Yes [ ]<br>N/A [ ] |
| RM48 | Setting bit 3.42.5 to a zero disables PRBS31 receive pattern testing                                                   | 45.2.3.19.3 |               | PCS* PPT:M                | Yes [ ]<br>N/A [ ] |
| RM49 | Setting bit 3.42.4 to a one enables PRBS31 transmit pattern testing if bit 3.32.2 is a one and bit 3.42.3 is not a one | 45.2.3.19.4 |               | PCS* PPT:M                | Yes [ ]<br>N/A [ ] |
| RM50 | Setting bit 3.42.4 to a zero disables PRBS31 transmit pattern testing                                                  | 45.2.3.19.4 |               | PCS* PPT:M                | Yes [ ]<br>N/A [ ] |
| RM51 | Test-pattern error counter resets to all zeros on read or reset                                                        | 45.2.3.20   |               | XCR:M                     | Yes [ ]<br>N/A [ ] |
| RM52 | Test-pattern error counter holds at all ones at overflow                                                               | 45.2.3.20   |               | XCR:M                     | Yes [ ]<br>N/A [ ] |
| RM53 | BER high order counter implemented                                                                                     | 45.2.3.21   |               | CR:O<br>40CR:M<br>100CR:M | Yes [ ]<br>N/A [ ] |
| RM54 | High order bits latched on read to 3.33                                                                                | 45.2.3.21   |               | RM53:M                    | Yes [ ]<br>N/A [ ] |
| RM55 | Counter reset on read to 3.33 or PCS reset                                                                             | 45.2.3.21   |               | RM53:M                    | Yes [ ]<br>N/A [ ] |
| RM56 | Counter held at all ones in the case of an overflow                                                                    | 45.2.3.21   |               | RM53:M                    | Yes [ ]<br>N/A [ ] |
| RM57 | Errored blocks high order counter implemented                                                                          | 45.2.3.22   |               | CR:O<br>40CR:M<br>100CR:M | Yes [ ]<br>N/A [ ] |
| RM58 | Register bit 3.45.15 set to one                                                                                        | 45.2.3.22   |               | RM57:M                    | Yes [ ]<br>N/A [ ] |
| RM59 | High order bits latched on read to 3.33                                                                                | 45.2.3.22   |               | RM57:M                    | Yes [ ]<br>N/A [ ] |
| RM60 | Counter reset on read to 3.33 or PCS reset                                                                             | 45.2.3.22   |               | RM57:M                    | Yes [ ]<br>N/A [ ] |

| Item | Feature                                                                                   | Subclause   | Value/Comment           | Status    | Support                      |
|------|-------------------------------------------------------------------------------------------|-------------|-------------------------|-----------|------------------------------|
| RM61 | Counter held at all ones in the case of an overflow                                       | 45.2.3.22   |                         | RM57:M    | Yes [ ]<br>N/A [ ]           |
| RM62 | Bit indicates fault when any PCS register indicates fault                                 | 45.2.3.2.5  | If subtype is supported | PCS*EPC:O | Yes [ ]<br>No [ ]<br>N/A [ ] |
| RM63 | Writes to multi-lane BASE-R PCS alignment status 1 register have no effect                | 45.2.3.23   |                         | XCR:M     | Yes [ ]<br>N/A [ ]           |
| RM64 | Non multi-lane BASE-R device shall return all zeros                                       | 45.2.3.23   |                         | !XCR:M    | Yes [ ]<br>N/A [ ]           |
| RM65 | Device shall return a zero for lanes not required                                         | 45.2.3.23   |                         | XCR:M     | Yes [ ]<br>N/A [ ]           |
| RM66 | Writes to multi-lane BASE-R PCS alignment status 2 register have no effect                | 45.2.3.24   |                         | XCR:M     | Yes [ ]<br>N/A [ ]           |
| RM67 | Non multi-PCS lane BASE-R device shall return all zeros                                   | 45.2.3.24   |                         | !XCR:M    | Yes [ ]<br>N/A [ ]           |
| RM68 | Device shall return a zero for lanes not required                                         | 45.2.3.24   |                         | XCR:M     | Yes [ ]<br>N/A [ ]           |
| RM69 | Writes to multi-lane BASE-R PCS alignment status 3 register have no effect                | 45.2.3.25   |                         | XCR:M     | Yes [ ]<br>N/A [ ]           |
| RM70 | Non multi-PCS lane BASE-R device shall return all zeros                                   | 45.2.3.25   |                         | !XCR:M    | Yes [ ]<br>N/A [ ]           |
| RM71 | Device shall return a zero for lanes not required                                         | 45.2.3.25   |                         | XCR:M     | Yes [ ]<br>N/A [ ]           |
| RM72 | Writes to multi-lane BASE-R PCS alignment status 4 register have no effect                | 45.2.3.26   |                         | XCR:M     | Yes [ ]<br>N/A [ ]           |
| RM73 | Non multi-PCS lane BASE-R device shall return all zeros                                   | 45.2.3.26   |                         | !XCR:M    | Yes [ ]<br>N/A [ ]           |
| RM74 | Device shall return a zero for lanes not required                                         | 45.2.3.26   |                         | XCR:M     | Yes [ ]<br>N/A [ ]           |
| RM75 | Counters reset on read to 3.200 through 3.219 or PCS reset                                | 45.2.3.47   |                         | XCR:M     | Yes [ ]<br>N/A [ ]           |
| RM76 | Counters held at all ones in the case of an overflow                                      | 45.2.3.47   |                         | XCR:M     | Yes [ ]<br>N/A [ ]           |
| RM77 | Writing this bit to a one activates the PAF when link is established                      | 45.2.3.28.3 |                         | PCS*PAF:M | Yes [ ]<br>N/A [ ]           |
| RM78 | Writes to bit are ignored while link is active or initializing or if PAF is not supported | 45.2.3.28.3 |                         | PCS*PAF:M | Yes [ ]<br>N/A [ ]           |
| RM79 | Bits indicate device capabilities upon reset                                              | 45.2.3.29   |                         | PCS*PAF:M | Yes [ ]<br>N/A [ ]           |

| Item | Feature                                                                                                        | Subclause | Value/Comment | Status    | Support            |
|------|----------------------------------------------------------------------------------------------------------------|-----------|---------------|-----------|--------------------|
| RM80 | Writes to the register through any PCS MMD in the same package effect the register equally                     | 45.2.3.29 |               | PCS*PAF:M | Yes [ ]<br>N/A [ ] |
| RM81 | Single bit set to one and all others cleared to zero when device does not support aggregation of multiple PMEs | 45.2.3.29 |               | PCS:M     | Yes [ ]<br>N/A [ ] |
| RM82 | PME aggregation used if one or more bits set                                                                   | 45.2.3.30 |               | PCS*PAF:M | Yes [ ]<br>N/A [ ] |
| RM83 | Registers set to all zeros upon reset                                                                          | 45.2.3.30 |               | PCS*PAF:M | Yes [ ]<br>N/A [ ] |
| RM84 | Writes to the register through any PCS MMD in the same package effect the register equally                     | 45.2.3.30 |               | PCS*PAF:M | Yes [ ]<br>N/A [ ] |
| RM85 | Bits reset to zero when read or upon MMD reset                                                                 | 45.2.3.31 |               | PCS*PAF:M | Yes [ ]<br>N/A [ ] |
| RM86 | Bits held to one upon overflow                                                                                 | 45.2.3.31 |               | PCS*PAF:M | Yes [ ]<br>N/A [ ] |
| RM87 | Bits reset to zero when read or upon MMD reset                                                                 | 45.2.3.32 |               | PCS*PAF:M | Yes [ ]<br>N/A [ ] |
| RM88 | Bits held to one upon overflow                                                                                 | 45.2.3.32 |               | PCS*PAF:M | Yes [ ]<br>N/A [ ] |
| RM89 | Bits reset to zero when read or upon MMD reset                                                                 | 45.2.3.33 |               | PCS*PAF:M | Yes [ ]<br>N/A [ ] |
| RM90 | Bits held to one upon overflow                                                                                 | 45.2.3.33 |               | PCS*PAF:M | Yes [ ]<br>N/A [ ] |
| RM91 | Bits reset to zero when read or upon MMD reset                                                                 | 45.2.3.34 |               | PCS*PAF:M | Yes [ ]<br>N/A [ ] |
| RM92 | Bits held to one upon overflow                                                                                 | 45.2.3.34 |               | PCS*PAF:M | Yes [ ]<br>N/A [ ] |
| RM93 | Bits reset to zero when read or upon MMD reset                                                                 | 45.2.3.35 |               | PCS*PAF:M | Yes [ ]<br>N/A [ ] |
| RM94 | Bits held to one upon overflow                                                                                 | 45.2.3.35 |               | PCS*PAF:M | Yes [ ]<br>N/A [ ] |
| RM95 | Bits reset to zero when read or upon MMD reset                                                                 | 45.2.3.36 |               | PCS*PAF:M | Yes [ ]<br>N/A [ ] |
| RM96 | Bits held to one upon overflow                                                                                 | 45.2.3.36 |               | PCS*PAF:M | Yes [ ]<br>N/A [ ] |
| RM97 | Bits reset to zero when read or upon MMD reset                                                                 | 45.2.3.37 |               | PCS*PAF:M | Yes [ ]<br>N/A [ ] |
| RM98 | Bits held to one upon overflow                                                                                 | 45.2.3.37 |               | PCS*PAF:M | Yes [ ]<br>N/A [ ] |
| RM99 | Bits reset to zero when read or upon MMD reset                                                                 | 45.2.3.38 |               | PCS*PAF:M | Yes [ ]<br>N/A [ ] |

| Item  | Feature                                                                                                                                           | Subclause   | Value/Comment | Status     | Support                      |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------|------------|------------------------------|
| RM100 | Bits held to one upon overflow                                                                                                                    | 45.2.3.38   |               | PCS*PAF:M  | Yes [ ]<br>N/A [ ]           |
| RM101 | Setting bit 3.42.6 to a one enables PRBS9 transmit pattern testing if bit 3.32.3 is a one and bit 3.42.3 is not a one and bit 3.42.4 is not a one | 45.2.3.19.2 |               | PCS* PTT:M | Yes [ ]<br>N/A [ ]           |
| RM102 | Setting bit 3.42.6 to a zero disables PRBS9 transmit pattern testing                                                                              | 45.2.3.19.2 |               | PCS* PTT:M | Yes [ ]<br>N/A [ ]           |
| RM103 | Corrected FEC codeword counter is reset when read or upon PHY reset                                                                               | 45.2.3.41   |               | CPR:M      | Yes [ ]<br>No [ ]<br>N/A [ ] |
| RM104 | Corrected FEC codeword counter is held at all ones in the case of overflow                                                                        | 45.2.3.41   |               | CPR:M      | Yes [ ]<br>No [ ]<br>N/A [ ] |
| RM105 | Uncorrected FEC codeword counter is reset when read or upon PHY reset                                                                             | 45.2.3.42   |               | CPR:M      | Yes [ ]<br>No [ ]<br>N/A [ ] |
| RM106 | Uncorrected FEC codeword counter is held at all ones in the case of overflow                                                                      | 45.2.3.42   |               | CPR:M      | Yes [ ]<br>No [ ]<br>N/A [ ] |
| RM107 | This action sets all 1000BASE-T1 PCS registers to their default states.                                                                           | 45.2.3.76.1 |               | PCS:M      | Yes [ ]<br>N/A [ ]           |
| RM108 | Bit 3.2304.15 is self-clearing, and the 1000BASE-T1 PCS returns a value of one in bit 3.2304.15 when a reset is in progress.                      | 45.2.3.76.1 |               | PCS:M      | Yes [ ]<br>N/A [ ]           |
| RM109 | Otherwise, bit 3.2304.15 returns a value of zero.                                                                                                 | 45.2.3.76.1 |               | PCS:M      | Yes [ ]<br>N/A [ ]           |
| RM110 | During a reset, the 1000BASE-T1 PCS responds to reads from register bits 3.0.15, 3.8.15:14, and 3.2304.15.                                        | 45.2.3.76.1 |               | PCS:M      | Yes [ ]<br>N/A [ ]           |
| RM111 | All other register bits are ignored during a reset.                                                                                               | 45.2.3.76.1 |               | PCS:M      | Yes [ ]<br>N/A [ ]           |
| RM112 | Setting either bit 3.2304.15 or 3.0.15 resets the 1000BASE-T1 PCS.                                                                                | 45.2.3.76.1 |               | PCS:M      | Yes [ ]<br>N/A [ ]           |
| RM113 | The 1000BASE-T1 PCS is placed in a loopback mode of operation when bit 3.2304.14 is set to a one.                                                 | 45.2.3.76.2 |               | PCS:M      | Yes [ ]<br>N/A [ ]           |
| RM114 | When bit 3. 2304.14 is set to a one, the 1000BASE-T1 PCS accepts data on the transmit path and return it on the receive path.                     | 45.2.3.76.2 |               | PCS:M      | Yes [ ]<br>N/A [ ]           |

| Item  | Feature                                                                                                                                                                  | Subclause   | Value/Comment | Status | Support            |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------|--------|--------------------|
| RM115 | Setting either bit 3.2304.14 or 3.0.14 enables loopback.                                                                                                                 | 45.2.3.76.2 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM116 | All the bits in the 1000BASE-T1 PCS status 1 register are read only; a write to the 1000BASE-T1 PCS status 1 register has no effect.                                     | 45.2.3.77   |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM117 | Bit 3.2305.11 is implemented with latching high behavior.                                                                                                                | 45.2.3.77.1 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM118 | Bit 3.2305.10 is implemented with latching high behavior.                                                                                                                | 45.2.3.77.2 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM119 | All the bits in the 1000BASE-T1 PCS status 2 register are read only; a write to the 1000BASE-T1 PCS status 2 register has no effect.                                     | 45.2.3.78   |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM120 | Bit 3.2306.7 is implemented with latching high behavior.                                                                                                                 | 45.2.3.78.4 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM121 | Bit 3.2306.6 is implemented with latching low behavior.                                                                                                                  | 45.2.3.78.5 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM122 | Bits 3.2306.5:0 are reset to all zeros when the 1000BASE-T1 PCS status 2 register is read by the management function or upon execution of the 1000BASE-T1 PCS reset.     | 45.2.3.78.6 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM123 | Bits 3.2306.5:0 are held at all ones in the case of overflow.                                                                                                            | 45.2.3.78.6 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM124 | The state machine assigns a value alternating between zero and one to associate with the 8-octet OAM message transmitted by the BASE-T1 PHY.                             | 45.2.3.79.2 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM125 | Bit 3.2308.14 is read and recorded prior to setting 3.2308.15 to one.                                                                                                    | 45.2.3.79.2 |               | PCS:O  | Yes [ ]<br>No [ ]  |
| RM126 | Bit 3.2308.14 self clears on read.                                                                                                                                       | 45.2.3.79   |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM127 | Bit 3.2308.13 indicates whether the most recently transmitted OAM message with a toggle bit value in 3.2308.12 was received, read, and acknowledged by the link partner. | 45.2.3.79.3 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM128 | Bit 3.2308.13 clears on read.                                                                                                                                            | 45.2.3.79.3 |               | PCS:M  | Yes [ ]<br>N/A [ ] |

| Item  | Feature                                                                                                                                                                   | Subclause   | Value/Comment                                                                                                                                                               | Status          | Support            |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|--------------------|
| RM129 | Bit 3.2313.15 is set to one when the OAM message from the link partner is stored into registers 3.2314, 3.2315, 3.2316, and 3.2317 and the message number in 3.2313.11:8. | 45.2.3.81.1 |                                                                                                                                                                             | PCS:M           | Yes [ ]<br>N/A [ ] |
| RM130 | Bit 3.2313.15 self-clears when register 3.2317 is read.                                                                                                                   | 45.2.3.81.1 |                                                                                                                                                                             | PCS:M           | Yes [ ]<br>N/A [ ] |
| RM131 | 1000BASE-H OAM message format to be transmitted                                                                                                                           | 45.2.3.51.5 | TXO_DATA0[10:0] = OUI[23:12], TXO_DATA1[15:3] = OUI[11:0], and TXO_DATA1[3:0] = protocol number. Content of TXO_DATA2 to TXO_DATA8 is vendor specific per OUI or Company ID | PCS<br>*BHOAM:M | Yes [ ]<br>N/A [ ] |
| RM132 | Default operation mode                                                                                                                                                    | 45.2.3.53.1 | Default value of bits 3.518.15:13 is binary 000 (normal mode)                                                                                                               | PCS:M           | Yes [ ]<br>N/A [ ] |
| RM133 | Test operation modes                                                                                                                                                      | 45.2.3.53.1 | Values binary 001 through binary 110 of bits 3.518.15:13 select test modes of Table 45–280, with behavior per 115.5                                                         | PCS:M           | Yes [ ]<br>N/A [ ] |
| RM134 | Default loopback mode                                                                                                                                                     | 45.2.3.53.2 | Default value of bits 3.518.12:10 is binary 000 (no loopback operation)                                                                                                     | PCS:M           | Yes [ ]<br>N/A [ ] |
| RM135 | Loopback modes                                                                                                                                                            | 45.2.3.53.2 | Bits 3.518.12:10 select loopback modes with behavior per 115.10                                                                                                             | PCS:M           | Yes [ ]<br>N/A [ ] |
| RM136 | 1000BASE-H OAM conditional enable                                                                                                                                         | 45.2.3.53.3 | Write a one to bit 3.518.1 only produces effect if bit 3.519.1 is read as one                                                                                               | PCS<br>*BHOAM:M | Yes [ ]<br>N/A [ ] |
| RM137 | 1000BASE-H OAM disable                                                                                                                                                    | 45.2.3.53.3 | Write a zero to bit 3.518.1 prevents 1000BASE-H OAM channel establishment with link partner                                                                                 | PCS<br>*BHOAM:M | Yes [ ]<br>N/A [ ] |
| RM138 | 1000BASE-H OAM enable change effect                                                                                                                                       | 45.2.3.53.3 | Changes in the value of bit 3.518.1 only takes effect after PMA reset                                                                                                       | PCS<br>*OAM:M   | Yes [ ]<br>N/A [ ] |
| RM139 | EEE conditional enable                                                                                                                                                    | 45.2.3.53.4 | Write a one to bit 3.518.0 only produces effect if bit 3.519.0 is read as one                                                                                               | PCS<br>*EEE:M   | Yes [ ]<br>N/A [ ] |
| RM140 | EEE disable                                                                                                                                                               | 45.2.3.53.4 | Write a zero to bit 3.518.0 prevents EEE operation with link partner                                                                                                        | PCS<br>*EEE:M   | Yes [ ]<br>N/A [ ] |
| RM141 | EEE enable change effect                                                                                                                                                  | 45.2.3.53.4 | Changes in the value of bit 3.518.0 only takes effect after PMA reset                                                                                                       | PCS<br>*EEE:M   | Yes [ ]<br>N/A [ ] |

| Item  | Feature                                                                                                                                        | Subclause    | Value/Comment                                                                                                                               | Status          | Support            |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------|-----------------|--------------------|
| RM142 | Link status behavior                                                                                                                           | 45.2.3.54.3  | Bit 3.519.13 has latching low behavior                                                                                                      | PCS:M           | Yes [ ]<br>N/A [ ] |
| RM143 | Link status mirroring                                                                                                                          | 45.2.3.54.3  | Bit 1.1.2, bit 3.1.2, and bit 3.519.13 reflect same state variable link_status and a read of any of them releases the latch of all the bits | PCS:M           | Yes [ ]<br>N/A [ ] |
| RM144 | PHD lock status behavior                                                                                                                       | 45.2.3.54.6  | Bit 3.519.10 has latching low behavior                                                                                                      | PCS:M           | Yes [ ]<br>N/A [ ] |
| RM145 | Tx Assert LPI received behavior                                                                                                                | 45.2.3.54.8  | Bit 3.519.8 has latching high behavior                                                                                                      | PCS<br>*EEE:M   | Yes [ ]<br>N/A [ ] |
| RM146 | Tx Assert LPI received mirroring                                                                                                               | 45.2.3.54.8  | Bit 3.519.8 is a copy of bit 3.1.11 and a read of any of them releases the latch of all the bits                                            | PCS<br>*EEE:M   | Yes [ ]<br>N/A [ ] |
| RM147 | Rx Assert LPI generated behavior                                                                                                               | 45.2.3.54.9  | Bit 3.519.7 has latching high behavior                                                                                                      | PCS<br>*EEE:M   | Yes [ ]<br>N/A [ ] |
| RM148 | Rx Assert LPI generated mirroring                                                                                                              | 45.2.3.54.9  | Bit 3.519.7 is a copy of bit 3.1.10 and a read of any of them releases the latch of all the bits                                            | PCS<br>*EEE:M   | Yes [ ]<br>N/A [ ] |
| RM149 | 1000BASE-H OAM ability                                                                                                                         | 45.2.3.54.14 | Registers 3.500 through 3.517 are implemented and 3.519.1 reads as one                                                                      | PCS<br>*BHOAM:M | Yes [ ]<br>N/A [ ] |
| RM150 | BER test mode counter is reset to all zeros when read                                                                                          | 45.2.3.57    |                                                                                                                                             | PCS:M           | Yes [ ]<br>N/A [ ] |
| RM151 | BER test mode counter holds all ones in case of overflow                                                                                       | 45.2.3.57    |                                                                                                                                             | PCS:M           | Yes [ ]<br>N/A [ ] |
| RM152 | This action sets all 10BASE-T1L PCS registers to their default states                                                                          | 45.2.3.70.1  |                                                                                                                                             | PCS:M           | Yes [ ]<br>N/A [ ] |
| RM153 | The 10BASE-T1L PCS returns a value of one in bit 3.2278.15 when a reset is in progress; otherwise, returns a value of zero                     | 45.2.3.70.1  |                                                                                                                                             | PCS:M           | Yes [ ]<br>N/A [ ] |
| RM154 | The control and management interface is restored to operation within 0.5 s from the setting of bit 3.2278.15                                   | 45.2.3.70.1  |                                                                                                                                             | PCS:M           | Yes [ ]<br>N/A [ ] |
| RM155 | During a reset, the 10BASE-T1L PCS responds to reads from register bits 3.0.15, 3.8.15:14, and 3.2278.15; reads for all other bits are ignored | 45.2.3.70.1  |                                                                                                                                             | PCS:M           | Yes [ ]<br>N/A [ ] |
| RM156 | Setting either 3.2278.15 or 3.0.15 sets the other                                                                                              | 45.2.3.70.1  |                                                                                                                                             | PCS:M           | Yes [ ]<br>N/A [ ] |
| RM157 | Clearing either 3.2278.15 or 3.0.15 clears the other                                                                                           | 45.2.3.70.1  |                                                                                                                                             | PCS:M           | Yes [ ]<br>N/A [ ] |

| Item  | Feature                                                                                                                                                   | Subclause   | Value/Comment | Status | Support            |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------|--------|--------------------|
| RM158 | Setting either 3.2278.15 or 3.0.15 resets the 10BASE-T1L PCS                                                                                              | 45.2.3.70.1 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM159 | When bit 3.2278.14 is set to one, the 10BASE-T1L PCS is set to loopback mode, and it accepts data on the transmit path and returns it on the receive path | 45.2.3.70.2 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM160 | Setting either 3.2278.14 or 3.0.14 sets the other                                                                                                         | 45.2.3.70.2 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM161 | Clearing either 3.2278.14 or 3.0.14 clears the other                                                                                                      | 45.2.3.70.2 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM162 | Setting either 3.2278.14 or 3.0.14 enables loopback                                                                                                       | 45.2.3.70.2 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM163 | A write to the 10BASE-T1L PCS status 1 register has no effect                                                                                             | 45.2.3.71   |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM164 | Bit 3.2279.11 is implemented with latching high behavior                                                                                                  | 45.2.3.71.1 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM165 | Bit 3.2279.10 is implemented with latching high behavior                                                                                                  | 45.2.3.71.2 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM166 | Bit 3.2279.7 is implemented with latching high behavior                                                                                                   | 45.2.3.71.5 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM167 | Bit 3.2279.2 is implemented with latching low behavior                                                                                                    | 45.2.3.71.6 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM168 | This action sets all 10BASE-T1S PCS registers to their default states                                                                                     | 45.2.3.72.1 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM169 | The 10BASE-T1S PCS returns a value of one in bit 3.2291.15 when a reset is in progress; otherwise, returns a value of zero                                | 45.2.3.72.1 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM170 | The control and management interface is restored to operation within 0.5 s from the setting of bit 3.2291.15                                              | 45.2.3.72.1 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM171 | During a reset, the 10BASE-T1S PCS responds to reads from register bits 3.0.15, 3.8.15:14, and 3.2291.15; reads for all other bits are ignored            | 45.2.3.72.1 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM172 | Setting either 3.2291.15 or 3.0.15 sets the other                                                                                                         | 45.2.3.72.1 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM173 | Clearing either 3.2291.15 or 3.0.15 clears the other                                                                                                      | 45.2.3.72.1 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM174 | Setting either 3.2291.15 or 3.0.15 resets the 10BASE-T1S PCS                                                                                              | 45.2.3.72.1 |               | PCS:M  | Yes [ ]<br>N/A [ ] |



| Item  | Feature                                                                                                                                                   | Subclause   | Value/Comment | Status | Support            |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------|--------|--------------------|
| RM175 | When bit 3.2291.14 is set to one, the 10BASE-T1S PCS is set to loopback mode, and it accepts data on the transmit path and returns it on the receive path | 45.2.3.72.2 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM176 | Setting either 3.2291.14 or 3.0.14 sets the other                                                                                                         | 45.2.3.72.2 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM177 | Clearing either 3.2291.14 or 3.0.14 clears the other                                                                                                      | 45.2.3.72.2 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM178 | Setting either 3.2291.14 or 3.0.14 enables loopback                                                                                                       | 45.2.3.72.2 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM179 | Bit 3.2291.8 or 0.8 is ignored when the Auto-Negotiation enable bit 7.512.12 is set to one                                                                | 45.2.3.72.3 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM180 | Setting either 3.2291.8 or 0.8 sets the other                                                                                                             | 45.2.3.72.3 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM181 | Clearing either 3.2291.8 or 0.8 clears the other                                                                                                          | 45.2.3.72.3 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM182 | The 10BASE-T1S PCS fault bit is implemented with latching high behavior                                                                                   | 45.2.3.73.1 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM183 | A write to the 10BASE-T1S PCS status register has no effect                                                                                               | 45.2.3.73   |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM184 | A write to the 10BASE-T1S PCS diagnostic register has no effect                                                                                           | 45.2.3.74   |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM185 | Remote jabber count does not wrap                                                                                                                         | 45.2.3.74.1 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM186 | Writes to PCS diagnostic 2 register have no effect                                                                                                        | 45.2.3.75   |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM187 | Setting bit 3.2322.15 sets all MultiGBASE-T1 PCS registers to their default states.                                                                       | 45.2.3.85.1 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM188 | Bit 3.2322.15 is self-clearing, and the MultiGBASE-T1 PCS returns a value of one in bit 3.2322.15 when a reset is in progress.                            | 45.2.3.85.1 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM189 | Otherwise, bit 3.2322.15 returns a value of zero.                                                                                                         | 45.2.3.85.1 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM190 | During a reset, the MultiGBASE-T1 PCS responds to reads from register bits 3.0.15, 3.8.15:14, and 3.2322.15.                                              | 45.2.3.85.1 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM191 | All other register bits are ignored during a reset.                                                                                                       | 45.2.3.85.1 |               | PCS:M  | Yes [ ]<br>N/A [ ] |

| Item  | Feature                                                                                                                                                   | Subclause   | Value/Comment | Status | Support            |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------|--------|--------------------|
| RM192 | Setting or clearing either bit 3.2322.15 or 3.0.15 sets or clears the other bit.                                                                          | 45.2.3.85.1 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM193 | Setting either bit 3.2322.15 or 3.0.15 resets the MultiGBASE-T1 PCS.                                                                                      | 45.2.3.85.1 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM194 | The MultiGBASE-T1 PCS is placed in a loopback mode of operation when bit 3.2322.14 is set to a one.                                                       | 45.2.3.85.2 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM195 | When bit 3.2322.14 is set to a one, the MultiGBASE-T1 PCS accepts data on the transmit path and returns it on the receive path.                           | 45.2.3.85.2 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM196 | Setting or clearing either bit 3.2322.14 or 3.0.14 sets or clears the other bit.                                                                          | 45.2.3.85.2 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM197 | Setting either bit 3.2322.14 or 3.0.14 enables loopback.                                                                                                  | 45.2.3.85.2 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM198 | All bits in the MultiGBASE-T1 PCS status 1 register are read only; a write to the MultiGBASE-T1 PCS status 1 register has no effect.                      | 45.2.3.86   |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM199 | Bit 3.2323.11 is implemented with latching high behavior.                                                                                                 | 45.2.3.86.1 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM200 | Bit 3.2323.10 is implemented with latching high behavior.                                                                                                 | 45.2.3.86.2 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM201 | Bit 3.2323.2 is implemented with latching low behavior.                                                                                                   | 45.2.3.86.6 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM202 | All bits in the MultiGBASE-T1 PCS status 2 register are read only; a write to the MultiGBASE-T1 PCS status 2 register has no effect.                      | 45.2.3.87   |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM203 | Bit 3.2324.7 is implemented with latching high behavior.                                                                                                  | 45.2.3.87.4 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM204 | Bit 3.2324.6 is implemented with latching low behavior.                                                                                                   | 45.2.3.87.5 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM205 | Bits 3.2324.5:0 are reset to all zeros when the MultiGBASE-T1 PCS 2 register is read by management function or upon execution of the MultiGBASE-T1 reset. | 45.2.3.87.6 |               | PCS:M  | Yes [ ]<br>N/A [ ] |
| RM206 | Bits 3.2324.5:0 are held at all ones in case of overflow.                                                                                                 | 45.2.3.87.6 |               | PCS:M  | Yes [ ]<br>N/A [ ] |

#### 45.5.3.8 Auto-Negotiation options

| Item | Feature                                                                     | Subclause | Value/Comment | Status | Support                      |
|------|-----------------------------------------------------------------------------|-----------|---------------|--------|------------------------------|
| *AT  | Implementation of MultiGBASE-T Auto-Negotiation                             | 45.2.7    |               | AN:O   | Yes [ ]<br>No [ ]<br>N/A [ ] |
| *AB  | Implementation of Backplane Ethernet Auto-negotiation                       | 45.2.7    |               | AN:M   | Yes [ ]<br>No [ ]<br>N/A [ ] |
| *ABN | Implementation of Next Page support for Backplane Ethernet Auto-negotiation | 45.2.7    |               | AB:M   | Yes [ ]<br>No [ ]<br>N/A [ ] |
| *EEE | Implementation of EEE                                                       |           |               | O      | Yes [ ]<br>N/A [ ]           |

#### 45.5.3.9 Auto-Negotiation management functions

| Item | Feature                                                                                     | Subclause  | Value/Comment                                                                                            | Status | Support            |
|------|---------------------------------------------------------------------------------------------|------------|----------------------------------------------------------------------------------------------------------|--------|--------------------|
| AM1  | Device responds to all register addresses for that device                                   | 45.2       |                                                                                                          | AN:M   | Yes [ ]<br>N/A [ ] |
| AM2  | Writes to undefined and read-only registers have no effect                                  | 45.2       |                                                                                                          | AN:M   | Yes [ ]<br>N/A [ ] |
| AM3  | Operation is not affected by writes to reserved and unsupported bits                        | 45.2       |                                                                                                          | AN:M   | Yes [ ]<br>N/A [ ] |
| AM4  | Reserved and unsupported bits return a value of zero                                        | 45.2       |                                                                                                          | AN:M   | Yes [ ]<br>N/A [ ] |
| AM5  | Latching low bits remain low until after they have been read via the management interface   | 45.2       |                                                                                                          | AN:M   | Yes [ ]<br>N/A [ ] |
| AM6  | Latching low bits assume correct value once read via the management interface               | 45.2       | Correct value is based upon current state                                                                | AN:M   | Yes [ ]<br>N/A [ ] |
| AM7  | Latching high bits remain high until after they have been read via the management interface | 45.2       |                                                                                                          | AN:M   | Yes [ ]<br>N/A [ ] |
| AM8  | Latching high bits assume correct value once read via the management interface              | 45.2       | Correct value is based upon current state                                                                | AN:M   | Yes [ ]<br>N/A [ ] |
| AM9  | Action on reset                                                                             | 45.2.7.1.1 | Reset the registers of the entire device to default values and set bit 15 of the Control register to one | AN:M   | Yes [ ]<br>N/A [ ] |
| AM10 | Return 1 until reset completed                                                              | 45.2.7.1.1 |                                                                                                          | AN:M   | Yes [ ]<br>N/A [ ] |

| Item | Feature                                                                                                    | Subclause  | Value/Comment                             | Status | Support            |
|------|------------------------------------------------------------------------------------------------------------|------------|-------------------------------------------|--------|--------------------|
| AM11 | Reset completes within 0.5 s                                                                               | 45.2.7.1.1 |                                           | AN:M   | Yes [ ]<br>N/A [ ] |
| AM12 | Device responds to reads of register bit 7.0.15                                                            | 45.2.7.1.1 | All other register bits should be ignored | AN:M   | Yes [ ]<br>N/A [ ] |
| AM13 | Bit set enables Extended Next Page exchange                                                                | 45.2.7.1.2 |                                           | AN:M   | Yes [ ]<br>N/A [ ] |
| AM14 | Bit 7.0.13 has no effect unless bit 7.16.12 is set                                                         | 45.2.7.1.2 |                                           | AN:M   | Yes [ ]<br>N/A [ ] |
| AM15 | Auto-Negotiation enabled by setting bit 7.0.12                                                             | 45.2.7.1.3 |                                           | AN:M   | Yes [ ]<br>N/A [ ] |
| AM16 | Bits 1.0.13, 1.0.6, 1.0.5:2, and 1.7.3:0 have no effect and Auto-Negotiation determines link configuration | 45.2.7.1.3 |                                           | AN:M   | Yes [ ]<br>N/A [ ] |
| AM17 | Writes are ignored and bit is set to zero if Auto-Negotiation is disabled or unsupported                   | 45.2.7.1.4 |                                           | AN:M   | Yes [ ]<br>N/A [ ] |
| AM18 | Setting bit to one restarts Auto-Negotiation if supported and enabled                                      | 45.2.7.1.4 |                                           | AN:M   | Yes [ ]<br>N/A [ ] |
| AM19 | Bit set until Auto-Negotiation started then self cleared to zero                                           | 45.2.7.1.4 |                                           | AN:M   | Yes [ ]<br>N/A [ ] |
| AM20 | Setting this bit restarts Auto-Negotiation if previously completed                                         | 45.2.7.1.4 |                                           | AN:M   | Yes [ ]<br>N/A [ ] |
| AM21 | Setting the bit to zero has no effect on Auto-Negotiation process                                          | 45.2.7.1.4 |                                           | AN:M   | Yes [ ]<br>N/A [ ] |
| AM22 | Writing the bit to one is ignored if 7.1.3 = 0 or Auto-Negotiation is disabled                             | 45.2.7.1.4 |                                           | AN:M   | Yes [ ]<br>N/A [ ] |
| AM23 | Writes to register ignored                                                                                 | 45.2.7.2   |                                           | AN:M   | Yes [ ]<br>N/A [ ] |
| AM24 | Bit set to zero when bits 7.0.13 or 7.16.12 are set to zero                                                | 45.2.7.2.2 |                                           | AN:M   | Yes [ ]<br>N/A [ ] |
| AM25 | Extended Next Page not used when bit set to zero                                                           | 45.2.7.2.2 |                                           | AN:M   | Yes [ ]<br>N/A [ ] |
| AM26 | Bit set to one with link codeword received and stored                                                      | 45.2.7.2.3 |                                           | AN:M   | Yes [ ]<br>N/A [ ] |
| AM27 | Bit set to zero upon read of register 7.1                                                                  | 45.2.7.2.3 |                                           | AN:M   | Yes [ ]<br>N/A [ ] |
| AM28 | Bit set to zero when bit 7.0.12 is set to zero                                                             | 45.2.7.2.4 |                                           | AN:M   | Yes [ ]<br>N/A [ ] |
| AM29 | Bit set to zero if device does not support Auto-Negotiation                                                | 45.2.7.2.4 |                                           | AN:M   | Yes [ ]<br>N/A [ ] |

| Item | Feature                                                                        | Subclause        | Value/Comment | Status | Support            |
|------|--------------------------------------------------------------------------------|------------------|---------------|--------|--------------------|
| AM30 | Bit remains set until cleared                                                  | 45.2.7.2.5       |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM31 | Bit cleared to zero upon reset or read of register                             | 45.2.7.2.5       |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM32 | Bit remains set until cleared by register read                                 | 45.2.7.2.7       |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM33 | Bit cleared upon AN reset                                                      | 45.2.7.2.7       |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM34 | Bit set to one if link partner is able to participate in Auto-Negotiation      | 45.2.7.2.8       |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM35 | Bit set to zero if link partner is not able to participate in Auto-Negotiation | 45.2.7.2.8       |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM36 | Unique Identifier is composed of 22 bits of OUI, model number and revision     | 45.2.7.5         |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM37 | Prior to Auto-Negotiation bits 7.16.12:5 accurately reflect device capability  | 45.2.7.6         |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM38 | Write has no effect                                                            | 45.2.7.7         |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM39 | At power up or AN reset message code set to null message                       | 45.2.7.8         |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM40 | Write has no effect                                                            | 45.2.7.9         |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM41 | Auto-Negotiation used to determine MASTER or SLAVE when bit set to zero        | 45.2.7.10.1      |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM42 | Operates as MASTER when bit set to one                                         | 45.2.7.10.2      |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM43 | Operates as SLAVE when bit set to zero                                         | 45.2.7.10.2      |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM44 | Advertise 10GBASE-T PHY capability when bit is set to one                      | 45.2.7.10.4      |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM45 | 10GBASE-T PHY capability not advertised when bit is set to zero                | 45.2.7.10.4      |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM46 | Advertise loop timing capability when bit is set to one                        | 45.2.7.10.1<br>5 |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM47 | Loop timing capability not advertised when bit is set to zero                  | 45.2.7.10.1<br>5 |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM48 | Write has no effect                                                            | 45.2.7.11        |               | AN:M   | Yes [ ]<br>N/A [ ] |

| Item | Feature                                                                      | Subclause   | Value/Comment                                                                                                    | Status   | Support            |
|------|------------------------------------------------------------------------------|-------------|------------------------------------------------------------------------------------------------------------------|----------|--------------------|
| AM49 | Bit set if MASTER-SLAVE cannot be successfully concluded                     | 45.2.7.11.1 |                                                                                                                  | AN:M     | Yes [ ]<br>N/A [ ] |
| AM50 | Bit set to zero on read of register 7.33                                     | 45.2.7.11.1 |                                                                                                                  | AN:M     | Yes [ ]<br>N/A [ ] |
| AM51 | Bit set to zero by MultiGBASE-T PMA reset                                    | 45.2.7.11.1 |                                                                                                                  | AN:M     | Yes [ ]<br>N/A [ ] |
| AM52 | Bit set to zero when Auto-Negotiation is enabled                             | 45.2.7.11.1 |                                                                                                                  | AN:M     | Yes [ ]<br>N/A [ ] |
| AM53 | Bit set to one if local receiver is OK as defined in 55.2.2.7                | 45.2.7.11.3 |                                                                                                                  | AN:M     | Yes [ ]<br>N/A [ ] |
| AM54 | Bit is set to one if the remote receiver status is OK as defined in 55.2.2.8 | 45.2.7.11.4 |                                                                                                                  | AN:M     | Yes [ ]<br>N/A [ ] |
| AM55 | Parallel detection fault (7.1.9)                                             | 45.2.7.2.1  | ONE to indicate more than one Backplane Ethernet PMA indicates link_status of OK when autoneg_wait_timer expires | AB:M     | Yes [ ]<br>N/A [ ] |
| AM56 | Parallel detection fault clearing                                            | 45.2.7.2.1  | ZERO on read of the AN status register.                                                                          | AN:M     | Yes [ ]<br>N/A [ ] |
| AM57 | Bit 7.48.0 set to one                                                        | 45.2.7.12.4 | Set to one if KX, KX4, or KR PHY is implemented                                                                  | AB:M     | Yes [ ]<br>N/A [ ] |
| AM58 | EEE capability in advertisement register for each port type                  | 45.2.7.14   |                                                                                                                  | AN:EEE:M | Yes [ ]<br>N/A [ ] |
| AM59 | EEE LP advertisement register reflects link partner's capabilities           | 45.2.7.15   |                                                                                                                  | AN:EEE:M | Yes [ ]<br>N/A [ ] |
| AM60 | Writes to EEE LP advertisement register have no effect                       | 45.2.7.15   |                                                                                                                  | AN:EEE:M | Yes [ ]<br>N/A [ ] |
| AM61 | Advertise 40GBASE-T PHY capability when bit is set to one                    | 45.2.7.10.5 |                                                                                                                  | AN:M     | Yes [ ]<br>N/A [ ] |
| AM62 | 40GBASE-T PHY capability not advertised when bit is set to zero              | 45.2.7.10.5 |                                                                                                                  | AN:M     | Yes [ ]<br>N/A [ ] |
| AM63 | Advertise 25GBASE-T PHY capability when bit is set to one                    | 45.2.7.10.6 |                                                                                                                  | AN:M     | Yes [ ]<br>N/A [ ] |
| AM64 | 25GBASE-T PHY capability not advertised when bit is set to zero              | 45.2.7.10.6 |                                                                                                                  | AN:M     | Yes [ ]<br>N/A [ ] |
| AM65 | Advertise 5GBASE-T PHY capability when bit is set to one                     | 45.2.7.10.8 |                                                                                                                  | AN:M     | Yes [ ]<br>N/A [ ] |
| AM66 | 5GBASE-T PHY capability not advertised when bit is set to zero               | 45.2.7.10.8 |                                                                                                                  | AN:M     | Yes [ ]<br>N/A [ ] |
| AM67 | Advertise 2.5GBASE-T PHY capability when bit is set to one                   | 45.2.7.10.9 |                                                                                                                  | AN:M     | Yes [ ]<br>N/A [ ] |

| Item | Feature                                                                                                                                                                                                                                                           | Subclause   | Value/Comment | Status | Support            |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------|--------|--------------------|
| AM68 | 2.5GBASE-T PHY capability not advertised when bit is set to zero                                                                                                                                                                                                  | 45.2.7.10.9 |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM69 | Setting bit 7.512.15 to a one sets all BASE-T1 AN registers to their default states.                                                                                                                                                                              | 45.2.7.20.1 |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM70 | Bit 7.512.15 is self-clearing, and AN shall return a value of one in bit 7.512.15 when a reset is in progress and a value of zero otherwise.                                                                                                                      | 45.2.7.20.1 |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM71 | The reset process is completed within 0.5 s from the setting of bit 7.512.15.                                                                                                                                                                                     | 45.2.7.20.1 |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM72 | During an AN reset, AN responds to reads from register bit 7.512.15.                                                                                                                                                                                              | 45.2.7.20.1 |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM73 | All other register bits are ignored.                                                                                                                                                                                                                              | 45.2.7.20.1 |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM74 | The Auto-Negotiation function is enabled by setting bit 7.512.12 to a one.                                                                                                                                                                                        | 45.2.7.20.2 |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM75 | If bit 7.512.12 is set to one, then PHY type bits 1.2100.3:0 and MASTER-SLAVE bits 1.2100.14 has no effect on the link configuration, and the Auto-Negotiation process determines the link configuration.                                                         | 45.2.7.20.2 |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM76 | If the BASE-T1 PHY reports via bit 7.513.3 that it lacks the ability to perform Auto-Negotiation, then the value of bit 7.512.12 is zero.                                                                                                                         | 45.2.7.20.2 |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM77 | If the BASE-T1 PMA/PMD reports (via bit 7.513.3) that it lacks the ability to perform Auto-Negotiation, or if Auto-Negotiation is disabled, the BASE-T1 PMA/PMD returns a value of zero in bit 7.512.9 and any attempt to write a one to bit 7.512.9 are ignored. | 45.2.7.20.3 |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM78 | Otherwise, the Auto-Negotiation process is restarted by setting bit 7.512.9 to one.                                                                                                                                                                               | 45.2.7.20.3 |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM79 | Bit 7.512.9 is self-clearing, and the BASE-T1 PMA/PMD shall return a value of one in bit 7.512.9 until the Auto-Negotiation process has been initiated.                                                                                                           | 45.2.7.20.3 |               | AN:M   | Yes [ ]<br>N/A [ ] |

| Item | Feature                                                                                                                                                                                                                                          | Subclause   | Value/Comment | Status | Support            |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------|--------|--------------------|
| AM80 | If Auto-Negotiation was completed prior to this bit being set, the process is reinitialized.                                                                                                                                                     | 45.2.7.20.3 |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM81 | The Auto-Negotiation process is affected by clearing this bit to zero.                                                                                                                                                                           | 45.2.7.20.3 |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM82 | All the bits in the BASE-T1 AN status register are read only; therefore, a write to the BASE-T1 AN status register has no effect.                                                                                                                | 45.2.7.21   |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM83 | The Page received bit (7.513.6) is set to one to indicate that a new link codeword has been received and stored in the BASE-T1 AN LP Base Page ability registers 7.517 to 7.519 or the BASE-T1 AN LP Next Page ability registers 7.523 to 7.525. | 45.2.7.21.1 |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM84 | The Page received bit is reset to zero on a read of the BASE-T1 AN status register (register 7.513).                                                                                                                                             | 45.2.7.21.1 |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM85 | The BASE-T1 PMA/PMD returns a value of zero in bit 7.513.5 if Auto-Negotiation is disabled by clearing bit 7.512.12.                                                                                                                             | 45.2.7.21.2 |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM86 | The BASE-T1 PMA/PMD also returns a value of zero in bit 7.513.5 if it lacks the ability to perform Auto-Negotiation.                                                                                                                             | 45.2.7.21.2 |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM87 | The remote fault bit is implemented with a latching function, such that the occurrence of a remote fault causes the bit 7.513.4 to become set and remain set until it is cleared.                                                                | 45.2.7.21.3 |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM88 | Bit 7.513.4 is cleared each time register 7.513 is read via the management interface, and is also cleared by a AN reset.                                                                                                                         | 45.2.7.21.3 |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM89 | The link status bit is implemented with a latching function, such that the occurrence of a link_status equals FAIL condition causes the link status bit to become cleared and remain cleared until it is read via the management interface.      | 45.2.7.21.5 |               | AN:M   | Yes [ ]<br>N/A [ ] |



| Item  | Feature                                                                                                                                                          | Subclause   | Value/Comment | Status | Support            |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------|--------|--------------------|
| AM90  | Bit 7.513.2 is cleared upon AN reset.                                                                                                                            | 45.2.7.21.5 |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM91  | A write to the BASE-T1 AN LP Base Page ability register has no effect.                                                                                           | 45.2.7.23   |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM92  | On power-up or AN reset, registers 7.520, 7.521, and 7.522 contain the default value, which represents a Message Page with the message code set to Null Message. | 45.2.7.24   |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM93  | A write to the BASE-T1 AN LP Next Page ability register has no effect.                                                                                           | 45.2.7.25   |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM94  | When bit 7.526.15 is set to one, the PHY advertises 10BASE-T1L capability                                                                                        | 45.2.7.26.1 |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM95  | When bit 7.526.15 is set to zero, the PHY does not advertise 10BASE-T1L capability                                                                               | 45.2.7.26.1 |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM96  | If a 10BASE-T1L PHY supports EEE ability and desires to operate in EEE mode, bit 7.526.14 is set to one                                                          | 45.2.7.26.2 |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM97  | If a 10BASE-T1L PHY supports the 2.4 Vpp operating mode, bit 7.526.13 is set to one                                                                              | 45.2.7.26.3 |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM98  | When bit 7.526.12 is set to one, a request to operate the 10BASE-T1L PHY in increased transmit level mode is advertised                                          | 45.2.7.26.4 |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM99  | When bit 7.526.12 is set to zero, a request to operate the 10BASE-T1L PHY in increased transmit level mode is not advertised                                     | 45.2.7.26.4 |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM100 | When bit 7.526.7 is set to one, the PHY advertises 10BASE-T1S full duplex ability                                                                                | 45.2.7.26.5 |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM101 | When bit 7.526.7 is set to zero, the PHY does not advertise 10BASE-T1S full duplex ability                                                                       | 45.2.7.26.5 |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM102 | When bit 7.526.6 is set to one, the PHY advertises 10BASE-T1S half duplex capability                                                                             | 45.2.7.26.6 |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM103 | When bit 7.526.6 is set to zero, the PHY does not advertise 10BASE-T1S half duplex capability                                                                    | 45.2.7.26.6 |               | AN:M   | Yes [ ]<br>N/A [ ] |

| Item  | Feature                                                                                                                                     | Subclause | Value/Comment | Status | Support            |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|--------|--------------------|
| AM104 | Writing to the 10BASE-T1 AN status register has no effect                                                                                   | 45.2.7.27 |               | AN:M   | Yes [ ]<br>N/A [ ] |
| AM105 | When the AN process is complete, the 10BASE-T1 AN status register reflects the contents of the link partner's 10BASE-T1 AN control register | 45.2.7.27 |               | AN:M   | Yes [ ]<br>N/A [ ] |

#### 45.5.3.10 PHY XS options

| Item | Feature                           | Subclause | Value/Comment | Status | Support                      |
|------|-----------------------------------|-----------|---------------|--------|------------------------------|
| *PL  | Implementation of loopback        | 45.2.4    |               | PX:O   | Yes [ ]<br>No [ ]<br>N/A [ ] |
| *PT  | Implementation of pattern testing | 45.2.4    |               | PX:O   | Yes [ ]<br>No [ ]<br>N/A [ ] |

#### 45.5.3.11 PHY XS management functions

| Item | Feature                                                                                     | Subclause | Value/Comment                             | Status | Support            |
|------|---------------------------------------------------------------------------------------------|-----------|-------------------------------------------|--------|--------------------|
| PM1  | Device responds to all register addresses for that device                                   | 45.2      |                                           | PX:M   | Yes [ ]<br>N/A [ ] |
| PM2  | Writes to undefined and read-only registers have no effect                                  | 45.2      |                                           | PX:M   | Yes [ ]<br>N/A [ ] |
| PM3  | Operation is not affected by writes to reserved and unsupported bits                        | 45.2      |                                           | PX:M   | Yes [ ]<br>N/A [ ] |
| PM4  | Reserved and unsupported bits return a value of zero                                        | 45.2      |                                           | PX:M   | Yes [ ]<br>N/A [ ] |
| PM5  | Latching low bits remain low until after they have been read via the management interface   | 45.2      |                                           | PX:M   | Yes [ ]<br>N/A [ ] |
| PM6  | Latching low bits assume correct value once read via the management interface               | 45.2      | Correct value is based upon current state | PX:M   | Yes [ ]<br>N/A [ ] |
| PM7  | Latching high bits remain high until after they have been read via the management interface | 45.2      |                                           | PX:M   | Yes [ ]<br>N/A [ ] |
| PM8  | Latching high bits assume correct value once read via the management interface              | 45.2      | Correct value is based upon current state | PX:M   | Yes [ ]<br>N/A [ ] |

| Item | Feature                                                                                   | Subclause   | Value/Comment                                                                                            | Status   | Support            |
|------|-------------------------------------------------------------------------------------------|-------------|----------------------------------------------------------------------------------------------------------|----------|--------------------|
| PM9  | Action on reset                                                                           | 45.2.4.1.1  | Reset the registers of the entire device to default values and set bit 15 of the Control register to one | PX:M     | Yes [ ]<br>N/A [ ] |
| PM10 | Return 1 until reset completed                                                            | 45.2.4.1.1  |                                                                                                          | PX:M     | Yes [ ]<br>N/A [ ] |
| PM11 | Reset completes within 0.5 s                                                              | 45.2.4.1.1  |                                                                                                          | PX:M     | Yes [ ]<br>N/A [ ] |
| PM12 | Device responds to reads of bits 4.0.15 and 4.8.15:14 during reset                        | 45.2.4.1.1  |                                                                                                          | PX:M     | Yes [ ]<br>N/A [ ] |
| PM13 | Loopback mode                                                                             | 45.2.4.1.2  | Whenever bit 4.0.14 is set to a one                                                                      | PX*PL:M  | Yes [ ]<br>N/A [ ] |
| PM14 | Receive data is returned on transmit path during loopback                                 | 45.2.4.1.2  |                                                                                                          | PX*PL:M  | Yes [ ]<br>N/A [ ] |
| PM15 | Writes to loopback bit are ignored and reads return zero                                  | 45.2.4.1.2  |                                                                                                          | PX*!PL:M | Yes [ ]<br>N/A [ ] |
| PM16 | Device responds to transactions necessary to exit low-power mode while in low-power state | 45.2.4.1.3  |                                                                                                          | PX:M     | Yes [ ]<br>N/A [ ] |
| PM17 | Speed selection bits 13 and 6 are written as one                                          | 45.2.4.1.6  |                                                                                                          | PX:M     | Yes [ ]<br>N/A [ ] |
| PM18 | Invalid writes to speed selection bits are ignored                                        | 45.2.4.1.6  |                                                                                                          | PX:M     | Yes [ ]<br>N/A [ ] |
| PM19 | Writes to status 1 register have no effect                                                | 45.2.4.2    |                                                                                                          | PX:M     | Yes [ ]<br>N/A [ ] |
| PM20 | Transmit link status implemented using latching low behavior                              | 45.2.4.2.7  |                                                                                                          | PX:M     | Yes [ ]<br>N/A [ ] |
| PM21 | Unique identifier is composed of 22 bits of OUI, model number and revision                | 45.2.4.3    |                                                                                                          | PX:M     | Yes [ ]<br>N/A [ ] |
| PM22 | Writes to status 2 register have no effect                                                | 45.2.4.6    |                                                                                                          | PX:M     | Yes [ ]<br>N/A [ ] |
| PM23 | Transmit fault implemented with latching high behavior                                    | 45.2.4.6.2  |                                                                                                          | PX:M     | Yes [ ]<br>N/A [ ] |
| PM24 | Receive fault implemented with latching high behavior                                     | 45.2.4.6.3  |                                                                                                          | PX:M     | Yes [ ]<br>N/A [ ] |
| PM25 | Unique identifier is composed of 22 bits of OUI, model number and revision                | 45.2.4.7    |                                                                                                          | PX:M     | Yes [ ]<br>N/A [ ] |
| PM26 | Writes to 10G PHY XGXS Lane status register have no effect                                | 45.2.4.10   |                                                                                                          | PX:M     | Yes [ ]<br>N/A [ ] |
| PM27 | Writes to bit are ignored and reads return a value of zero                                | 45.2.4.11.1 |                                                                                                          | PX*!PT:M | Yes [ ]<br>N/A [ ] |

| Item | Feature                                                      | Subclause   | Value/Comment | Status  | Support            |
|------|--------------------------------------------------------------|-------------|---------------|---------|--------------------|
| PM28 | Setting the bits to <10> selects the mixed frequency pattern | 45.2.4.11.2 |               | PX*PT:M | Yes [ ]<br>N/A [ ] |
| PM29 | Setting the bits to <01> selects the low-frequency pattern   | 45.2.4.11.2 |               | PX*PT:M | Yes [ ]<br>N/A [ ] |
| PM30 | Setting the bits to <00> selects the high-frequency pattern  | 45.2.4.11.2 |               | PX*PT:M | Yes [ ]<br>N/A [ ] |

#### 45.5.3.12 DTE XS options

| Item | Feature                           | Subclause | Value/Comment | Status | Support                      |
|------|-----------------------------------|-----------|---------------|--------|------------------------------|
| *DT  | Implementation of pattern testing | 45.2.5    |               | DX:O   | Yes [ ]<br>No [ ]<br>N/A [ ] |

#### 45.5.3.13 DTE XS management functions

| Item | Feature                                                                                     | Subclause  | Value/Comment                                                                                            | Status | Support            |
|------|---------------------------------------------------------------------------------------------|------------|----------------------------------------------------------------------------------------------------------|--------|--------------------|
| DM1  | Device responds to all register addresses for that device                                   | 45.2       |                                                                                                          | DX:M   | Yes [ ]<br>N/A [ ] |
| DM2  | Writes to undefined and read-only registers have no effect                                  | 45.2       |                                                                                                          | DX:M   | Yes [ ]<br>N/A [ ] |
| DM3  | Operation is not affected by writes to reserved and unsupported bits                        | 45.2       |                                                                                                          | DX:M   | Yes [ ]<br>N/A [ ] |
| DM4  | Reserved and unsupported bits return a value of zero                                        | 45.2       |                                                                                                          | DX:M   | Yes [ ]<br>N/A [ ] |
| DM5  | Latching low bits remain low until after they have been read via the management interface   | 45.2       |                                                                                                          | DX:M   | Yes [ ]<br>N/A [ ] |
| DM6  | Latching low bits assume correct value once read via the management interface               | 45.2       | Correct value is based upon current state                                                                | DX:M   | Yes [ ]<br>N/A [ ] |
| DM7  | Latching high bits remain high until after they have been read via the management interface | 45.2       |                                                                                                          | DX:M   | Yes [ ]<br>N/A [ ] |
| DM8  | Latching high bits assume correct value once read via the management interface              | 45.2       | Correct value is based upon current state                                                                | DX:M   | Yes [ ]<br>N/A [ ] |
| DM9  | Action on reset                                                                             | 45.2.5.1.1 | Reset the registers of the entire device to default values and set bit 15 of the Control register to one | DX:M   | Yes [ ]<br>N/A [ ] |

| Item | Feature                                                                                   | Subclause   | Value/Comment                       | Status   | Support            |
|------|-------------------------------------------------------------------------------------------|-------------|-------------------------------------|----------|--------------------|
| DM10 | Return 1 until reset completed                                                            | 45.2.5.1.1  |                                     | DX:M     | Yes [ ]<br>N/A [ ] |
| DM11 | Reset completes within 0.5 s                                                              | 45.2.5.1.1  |                                     | DX:M     | Yes [ ]<br>N/A [ ] |
| DM12 | Device responds to reads of bits 5.0.15 and 5.8.15:14 during reset                        | 45.2.5.1.1  |                                     | DX:M     | Yes [ ]<br>N/A [ ] |
| DM13 | Loopback mode                                                                             | 45.2.5.1.2  | Whenever bit 5.0.14 is set to a one | DX:M     | Yes [ ]<br>N/A [ ] |
| DM14 | Transmit data is returned on receive path during loopback                                 | 45.2.5.1.2  |                                     | DX:M     | Yes [ ]<br>N/A [ ] |
| DM15 | Device responds to transactions necessary to exit low-power mode while in low-power state | 45.2.5.1.3  |                                     | DX:M     | Yes [ ]<br>N/A [ ] |
| DM16 | Speed selection bits 13 and 6 are written as one                                          | 45.2.5.1.6  |                                     | DX:M     | Yes [ ]<br>N/A [ ] |
| DM17 | Invalid writes to speed selection bits are ignored                                        | 45.2.5.1.6  |                                     | DX:M     | Yes [ ]<br>N/A [ ] |
| DM18 | Writes to status 1 register have no effect                                                | 45.2.5.2    |                                     | DX:M     | Yes [ ]<br>N/A [ ] |
| DM19 | Receive link status implemented using latching low behavior                               | 45.2.5.2.7  |                                     | DX:M     | Yes [ ]<br>N/A [ ] |
| DM20 | Unique identifier is composed of 22 bits of OUI, model number and revision                | 45.2.5.3    |                                     | DX:M     | Yes [ ]<br>N/A [ ] |
| DM21 | Writes to status 2 register have no effect                                                | 45.2.5.6    |                                     | DX:M     | Yes [ ]<br>N/A [ ] |
| DM22 | Transmit fault implemented with latching high behavior                                    | 45.2.5.6.2  |                                     | DX:M     | Yes [ ]<br>N/A [ ] |
| DM23 | Receive fault implemented with latching high behavior                                     | 45.2.5.6.3  |                                     | DX:M     | Yes [ ]<br>N/A [ ] |
| DM24 | Unique identifier is composed of 22 bits of OUI, model number and revision                | 45.2.5.7    |                                     | DX:M     | Yes [ ]<br>N/A [ ] |
| DM25 | Writes to 10G DTE XGXS Lane status register have no effect                                | 45.2.5.10   |                                     | DX:M     | Yes [ ]<br>N/A [ ] |
| DM26 | Writes to bit are ignored and reads return a value of zero                                | 45.2.5.11.1 |                                     | DX*!DT:M | Yes [ ]<br>N/A [ ] |
| DM27 | Setting the bits to <10> selects the mixed frequency pattern                              | 45.2.5.11.2 |                                     | DX*DT:M  | Yes [ ]<br>N/A [ ] |
| DM28 | Setting the bits to <01> selects the low-frequency pattern                                | 45.2.5.11.2 |                                     | DX*DT:M  | Yes [ ]<br>N/A [ ] |
| DM29 | Setting the bits to <00> selects the high-frequency pattern                               | 45.2.5.11.2 |                                     | DX*DT:M  | Yes [ ]<br>N/A [ ] |

#### 45.5.3.14 OFDM management functions

| Item | Feature                                                    | Subclause | Value/Comment | Status | Support            |
|------|------------------------------------------------------------|-----------|---------------|--------|--------------------|
| OFM1 | Implementation of 10GPASS-XR DS OFDM channel ID            | 45.2.8.1  |               | EO:M   | Yes [ ]<br>N/A [ ] |
| OFM2 | Implementation of 10GPASS-XR DS profile descriptor control | 45.2.8.2  |               | EO:M   | Yes [ ]<br>N/A [ ] |
| OFM3 | Implementation of 10GPASS-XR US profile descriptor control | 45.2.8.3  |               | EO:M   | Yes [ ]<br>N/A [ ] |
| OFM4 | Implementation of 10GPASS-XR US pre-equalizer coefficients | 45.2.8.4  |               | EO:M   | Yes [ ]<br>N/A [ ] |
| OFM5 | Implementation of 10GPASS-XR receive MER control           | 45.2.8.5  |               | EO:M   | Yes [ ]<br>N/A [ ] |
| OFM6 | Implementation of 10GPASS-XR receive MER measurement       | 45.2.8.6  |               | EO:M   | Yes [ ]<br>N/A [ ] |

#### 45.5.3.15 Power over Data Lines (PoDL) of Single Balanced Twisted-Pair Ethernet management functions

| Item   | Feature               | Subclause  | Value/Comment                                                                                                 | Status     | Support            |
|--------|-----------------------|------------|---------------------------------------------------------------------------------------------------------------|------------|--------------------|
| PODLM1 | PSE Enable bit        | 45.2.9.1.2 | PSE function enabled when set to one and disabled when set to zero                                            | PODL:<br>M | Yes [ ]<br>N/A [ ] |
| PODLM2 | Power Denied bit      | 45.2.9.2.1 | Implemented with latching high behavior and set to one when power_available transitions from true to false    | PODL:<br>M | Yes [ ]<br>N/A [ ] |
| PODLM3 | Valid Signature bit   | 45.2.9.2.2 | Implemented with latching high behavior and set to one when mr_valid_signature transitions from false to true | PODL:<br>M | Yes [ ]<br>N/A [ ] |
| PODLM4 | Invalid Signature bit | 45.2.9.2.3 | Implemented with latching high behavior and set to one when mr_valid_signature transitions from false to true | PODL:<br>M | Yes [ ]<br>N/A [ ] |
| PODLM5 | Class Timeout bit     | 45.2.9.2.4 | Implemented with latching high behavior and set to one when tclass_timer_done transitions from false to true  | PODL:<br>M | Yes [ ]<br>N/A [ ] |

| Item   | Feature           | Subclause  | Value/Comment                                                                                            | Status     | Support            |
|--------|-------------------|------------|----------------------------------------------------------------------------------------------------------|------------|--------------------|
| PODLM6 | Overload bit      | 45.2.9.2.5 | Implemented with latching high behavior and set to one when overload_held transitions from false to true | PODL:<br>M | Yes [ ]<br>N/A [ ] |
| PODLM7 | MFVS Absent bit   | 45.2.9.2.6 | Implemented with latching high behavior and set to one when mfvs_timeout transitions from false to true  | PODL:<br>M | Yes [ ]<br>N/A [ ] |
| PODLM8 | Invalid Class bit | 45.2.9.3.1 | Implemented with latching high behavior as defined in 45.2                                               | PODL:<br>M | Yes [ ]<br>N/A [ ] |

#### 45.5.3.16 Vendor specific MMD 1 management functions

| Item | Feature                                                                    | Subclause | Value/Comment | Status | Support            |
|------|----------------------------------------------------------------------------|-----------|---------------|--------|--------------------|
| VSA1 | Device responds to all register addresses for that device                  | 45.2      |               | VSA:M  | Yes [ ]<br>N/A [ ] |
| VSA2 | Writes to undefined and read-only registers have no effect                 | 45.2      |               | VSA:M  | Yes [ ]<br>N/A [ ] |
| VSA3 | Operation is not affected by writes to reserved and unsupported bits       | 45.2      |               | VSA:M  | Yes [ ]<br>N/A [ ] |
| VSA4 | Reserved and unsupported bits return a value of zero                       | 45.2      |               | VSA:M  | Yes [ ]<br>N/A [ ] |
| VSA5 | Unique identifier is composed of 22 bits of OUI, model number and revision | 45.2.11.1 |               | VSA:M  | Yes [ ]<br>N/A [ ] |
| VSA6 | Writes to status register have no effect                                   | 45.2.11.2 |               | VSA:M  | Yes [ ]<br>N/A [ ] |
| VSA7 | Unique identifier is composed of 22 bits of OUI, model number and revision | 45.2.11.3 |               | VSA:M  | Yes [ ]<br>N/A [ ] |

#### 45.5.3.17 Vendor specific MMD 2 management functions

| Item  | Feature                                                    | Subclause | Value/Comment | Status | Support            |
|-------|------------------------------------------------------------|-----------|---------------|--------|--------------------|
| VSBI  | Device responds to all register addresses for that device  | 45.2      |               | VSBI:M | Yes [ ]<br>N/A [ ] |
| VSBI2 | Writes to undefined and read-only registers have no effect | 45.2      |               | VSBI:M | Yes [ ]<br>N/A [ ] |

| Item | Feature                                                                    | Subclause | Value/Comment | Status | Support            |
|------|----------------------------------------------------------------------------|-----------|---------------|--------|--------------------|
| VSBB | Operation is not affected by writes to reserved and unsupported bits       | 45.2      |               | VSBB:M | Yes [ ]<br>N/A [ ] |
| VSBB | Reserved and unsupported bits return a value of zero                       | 45.2      |               | VSBB:M | Yes [ ]<br>N/A [ ] |
| VSBB | Unique identifier is composed of 22 bits of OUI, model number and revision | 45.2.12.1 |               | VSBB:M | Yes [ ]<br>N/A [ ] |
| VSBB | Writes to status register have no effect                                   | 45.2.12.2 |               | VSBB:M | Yes [ ]<br>N/A [ ] |
| VSBB | Unique identifier is composed of 22 bits of OUI, model number and revision | 45.2.12.3 |               | VSBB:M | Yes [ ]<br>N/A [ ] |

#### 45.5.3.18 Management frame structure

| Item | Feature                                                                                                               | Subclause | Value/Comment | Status | Support |
|------|-----------------------------------------------------------------------------------------------------------------------|-----------|---------------|--------|---------|
| MF1  | Device has implemented sixteen bit address register                                                                   | 45.3      |               | M      | Yes [ ] |
| MF2  | Address register is overwritten by address frames                                                                     | 45.3      |               | M      | Yes [ ] |
| MF3  | Write, read, and post-read-increment-address frames access the register whose address is held in the address register | 45.3      |               | M      | Yes [ ] |
| MF4  | Write and read frames do not modify the address register                                                              | 45.3      |               | M      | Yes [ ] |
| MF5  | Post-read-increment-address frames increment the address register by one unless the address register contains 65 535  | 45.3      |               | M      | Yes [ ] |
| MF6  | Components containing several MMDs implement separate address registers                                               | 45.3      |               | M      | Yes [ ] |
| MF7  | Tri state drivers are disabled during idle                                                                            | 45.3.1    |               | M      | Yes [ ] |
| MF8  | STA sources 32 contiguous ones at the beginning of each transaction                                                   | 45.3.2    |               | M      | Yes [ ] |
| MF9  | MMD observes 32 contiguous ones at the beginning of each transaction                                                  | 45.3.2    |               | M      | Yes [ ] |
| MF10 | Frames containing ST=<01> sequence are ignored                                                                        | 45.3.3    |               | M      | Yes [ ] |



| Item | Feature                                                                                                          | Subclause | Value/Comment | Status | Support |
|------|------------------------------------------------------------------------------------------------------------------|-----------|---------------|--------|---------|
| MF11 | STA tri state driver is high impedance during first bit of TA during read or post-read-increment-address frames  | 45.3.7    |               | M      | Yes [ ] |
| MF12 | MMD tri state driver is high impedance during first bit of TA during read or post-read-increment-address frames  | 45.3.7    |               | M      | Yes [ ] |
| MF13 | MMD tri state driver drives a zero bit during second bit of TA during read or post-read-increment-address frames | 45.3.7    |               | M      | Yes [ ] |
| MF14 | STA tri state driver drives a one bit followed by a zero bit for the TA during write or address frames           | 45.3.7    |               | M      | Yes [ ] |
| MF15 | First bit transmitted and received is bit 15                                                                     | 45.3.8    |               | M      | Yes [ ] |

#### 45.5.3.19 TC management functions

| Item | Feature                                                                            | Subclause  | Value/Comment | Status | Support            |
|------|------------------------------------------------------------------------------------|------------|---------------|--------|--------------------|
| TC1  | Device responds to all register addresses for that device                          | 45.2       |               | TC:M   | Yes [ ]<br>N/A [ ] |
| TC2  | Writes to undefined and read only register have no effect                          | 45.2       |               | TC:M   | Yes [ ]<br>N/A [ ] |
| TC3  | Operation is not affected by writes to reserved and unsupported bits               | 45.2       |               | TC:M   | Yes [ ]<br>N/A [ ] |
| TC4  | Reserved and unsupported bits return a value of zero                               | 45.2       |               | TC:M   | Yes [ ]<br>N/A [ ] |
| TC5  | Setting bit to a one sets all TC registers to their default states                 | 45.2.6.1.1 |               | TC:M   | Yes [ ]<br>N/A [ ] |
| TC6  | Bit reads one while reset is in progress otherwise reads zero                      | 45.2.6.1.1 |               | TC:M   | Yes [ ]<br>N/A [ ] |
| TC7  | Control and management interface is restored within 0.5s from setting bit to a one | 45.2.6.1.1 |               | TC:M   | Yes [ ]<br>N/A [ ] |
| TC8  | During reset, TC responds to reads from bit                                        | 45.2.6.1.1 |               | TC:M   | Yes [ ]<br>N/A [ ] |
| TC9  | Writes that would select an unsupported speed are ignored                          | 45.2.6.1.2 |               | TC:M   | Yes [ ]<br>N/A [ ] |
| TC10 | Identifier composed properly                                                       | 45.2.6.2   |               | TC:M   | Yes [ ]<br>N/A [ ] |

| Item | Feature                                                        | Subclause  | Value/Comment | Status | Support            |
|------|----------------------------------------------------------------|------------|---------------|--------|--------------------|
| TC11 | Identifier composed properly                                   | 45.2.6.5   |               | TC:M   | Yes [ ]<br>N/A [ ] |
| TC12 | Register is unique across all PCS MMDs in a package            | 45.2.6.6   |               | TC:M   | Yes [ ]<br>N/A [ ] |
| TC13 | Operation ignored when link is up or initializing              | 45.2.6.6.1 |               | TC:M   | Yes [ ]<br>N/A [ ] |
| TC14 | Bits indicate “Ready” when PAF is capable                      | 45.2.6.6.1 |               | TC:M   | Yes [ ]<br>N/A [ ] |
| TC15 | Writes ignored and “Ready” indicated if PAF is unsupported     | 45.2.6.6.1 |               | TC:M   | Yes [ ]<br>N/A [ ] |
| TC16 | Bits indicate “Ready” when operation is complete or upon reset | 45.2.6.6.1 |               | TC:M   | Yes [ ]<br>N/A [ ] |
| TC17 | Bits read as zero if PAF is unsupported                        | 45.2.6.7.1 |               | TC:M   | Yes [ ]<br>N/A [ ] |
| TC18 | Bits read as zero if PAF is unsupported                        | 45.2.6.7.2 |               | TC:M   | Yes [ ]<br>N/A [ ] |
| TC19 | Register is unique across all PCS MMDs in a package            | 45.2.6.8   |               | TC:M   | Yes [ ]<br>N/A [ ] |
| TC20 | Register is unique across all PCS MMDs in a package            | 45.2.6.9   |               | TC:M   | Yes [ ]<br>N/A [ ] |
| TC21 | Operation ignored when link is up or initializing              | 45.2.6.9.1 |               | TC:M   | Yes [ ]<br>N/A [ ] |
| TC22 | Bits indicate “Ready” when PAF is capable                      | 45.2.6.9.1 |               | TC:M   | Yes [ ]<br>N/A [ ] |
| TC23 | Writes ignored and “Ready” indicated if PAF is unsupported     | 45.2.6.9.1 |               | TC:M   | Yes [ ]<br>N/A [ ] |
| TC24 | Bits indicate “Ready” when operation is complete or on reset   | 45.2.6.9.1 |               | TC:M   | Yes [ ]<br>N/A [ ] |
| TC25 | Register is unique across all PCS MMDs in a package            | 45.2.6.10  |               | TC:M   | Yes [ ]<br>N/A [ ] |
| TC26 | Bits reset to zero when read or reset                          | 45.2.6.11  |               | TC:M   | Yes [ ]<br>N/A [ ] |
| TC27 | Bits held to one upon overflow                                 | 45.2.6.11  |               | TC:M   | Yes [ ]<br>N/A [ ] |
| TC28 | Bits reset to zero when read or reset                          | 45.2.6.12  |               | TC:M   | Yes [ ]<br>N/A [ ] |
| TC29 | Bits held to one upon overflow                                 | 45.2.6.12  |               | TC:M   | Yes [ ]<br>N/A [ ] |

#### 45.5.3.20 Clause 22 extension options

| Item | Feature                   | Subclause | Value/Comment | Status | Support                      |
|------|---------------------------|-----------|---------------|--------|------------------------------|
| *FEC | Implementation of PHY FEC | 65.4.4.6  |               | CTT:O  | Yes [ ]<br>No [ ]<br>N/A [ ] |

#### 45.5.3.21 Clause 22 extension management functions

| Item | Feature                                                                                         | Subclause   | Value/Comment | Status    | Support            |
|------|-------------------------------------------------------------------------------------------------|-------------|---------------|-----------|--------------------|
| CT1  | Device responds to all register addresses for that device                                       | 45.2        |               | CTT:M     | Yes [ ]<br>N/A [ ] |
| CT2  | Writes to undefined and read only register have no effect                                       | 45.2        |               | CTT:M     | Yes [ ]<br>N/A [ ] |
| CT3  | Operation is not affected by writes to reserved and unsupported bits                            | 45.2        |               | CTT:M     | Yes [ ]<br>N/A [ ] |
| CT4  | Reserved and unsupported bits return a value of zero                                            | 45.2        |               | CTT:M     | Yes [ ]<br>N/A [ ] |
| CT5  | Bits set to zero upon PHY reset                                                                 | 45.2.10.3.1 |               | CTT*FEC:M | Yes [ ]<br>N/A [ ] |
| CT6  | Bits reset to all zeros when the register is read by the management function or upon PHY reset. | 45.2.10.4   |               | CTT*FEC:M | Yes [ ]<br>N/A [ ] |
| CT7  | Bits held at all ones in the case of overflow                                                   | 45.2.10.4   |               | CTT*FEC:M | Yes [ ]<br>N/A [ ] |
| CT8  | Bits reset to all zeros when the register is read by the management function or upon PHY reset. | 45.2.10.5   |               | CTT*FEC:M | Yes [ ]<br>N/A [ ] |
| CT9  | Bits held at all ones in the case of overflow                                                   | 45.2.10.5   |               | CTT*FEC:M | Yes [ ]<br>N/A [ ] |
| CT10 | Bits reset to all zeros when the register is read by the management function or upon PHY reset. | 45.2.10.6   |               | CTT*FEC:M | Yes [ ]<br>N/A [ ] |
| CT11 | Bits held at all ones in the case of overflow                                                   | 45.2.10.6   |               | CTT*FEC:M | Yes [ ]<br>N/A [ ] |

#### 45.5.3.22 Signal timing characteristics

| Item | Feature                    | Subclause | Value/Comment                                       | Status | Support |
|------|----------------------------|-----------|-----------------------------------------------------|--------|---------|
| ST1  | MDIO setup and hold time   | 45.4.2    | Setup min = 10 ns; Hold min = 10 ns per Figure 45–3 | M      | Yes [ ] |
| ST2  | MDIO clock to output delay | 45.4.2    | Min = 0 ns; Max = 300 ns per Figure 45–4            | M      | Yes [ ] |
| ST3  | MDC min high/low time      | 45.4.2    | 160 ns                                              | M      | Yes [ ] |
| ST4  | MDC min period             | 45.4.2    | 400 ns                                              | M      | Yes [ ] |

#### 45.5.3.23 Electrical characteristics

| Item | Feature                    | Subclause | Value/Comment                                                                                              | Status | Support            |
|------|----------------------------|-----------|------------------------------------------------------------------------------------------------------------|--------|--------------------|
| EC1  | $V_{OH}$                   | 45.4.1    | $\geq 1.0 \text{ V}$ ( $I_{OH} = -100 \text{ uA}$ )<br>$\leq 1.5 \text{ V}$ ( $I_{OH} = -100 \text{ uA}$ ) | M      | Yes [ ]            |
| EC2  | $V_{OL}$                   | 45.4.1    | $\geq -0.3 \text{ V}$ ( $I_{OL} = 100 \text{ uA}$ )<br>$\leq 0.2 \text{ V}$ ( $I_{OL} = 100 \text{ uA}$ )  | M      | Yes [ ]            |
| EC3  | $V_{IH}$                   | 45.4.1    | $0.84 \text{ V} \leq V_{IH} \leq 1.5 \text{ V}$                                                            | M      | Yes [ ]            |
| EC4  | $V_{IL}$                   | 45.4.1    | $-0.3 \leq V_{IL} \leq 0.36 \text{ V}$                                                                     | M      | Yes [ ]            |
| EC5  | Input capacitance for MDIO | 45.4.1    | $\leq 10 \text{ pF}$                                                                                       | M      | Yes [ ]            |
| EC6  | Total capacitive load      | 45.4.1    | $\leq 470 \text{ pF}$                                                                                      | M      | Yes [ ]            |
| EC7  | $I_{OH}$                   | 45.4.1    | $\leq -4 \text{ mA}$ at $V_{OH} = 1.0 \text{ V}$                                                           | !ODB:M | Yes [ ]<br>N/A [ ] |
| EC8  | $I_{OL}$                   | 45.4.1    | $\geq +4 \text{ mA}$ at $V_{OL} = 0.2 \text{ V}$                                                           | M      | Yes [ ]            |